-**GZN**- pumice
-**GŢW**- igneous rock
-**GŢY**- sedimentary rock
-**GŢL**- metamorphic rock
-**ŢDŘ**- petrified wood

7.0 NATURAL PHENOMENA (BIOLOGICAL/ZOOLOGICAL)

-**LCP**- **'CELLULAR ORGANELLE / PLASMA MEMBRANE / CYTOPLASM'**		
BSC	**STEM 1:** (to be) an organelle within a biological cell	**STEM 2:** (to be) the plasma membrane / cell membrane of a biological cell
CTE	(to be) the physical body of an organelle within a biological cell	
CSV	(to be) the physical operation of an organelle within a biological cell	**STEM 3:** (to be) the cytoplasm (cytosol) within a biological cell
OBJ	(to be) function/purpose of an organelle within a biological cell	

The following stems follow the same Specification pattern as the above root:

-**PPW**- 1. cellular nucleus 2. nucleolus 3. chromatin
-**PPY**- 1. nuclear envelope 2. nuclear pore 3. endoplasmic reticulum
-**PPL**- 1. ribosome 2. centrosome 3. peroxisome
-**PPR**- 1. mitochondrion 2. lysosome 3. microtubule
-**PPŘ**- 1. Golgi apparatus 2. Golgi vesicle 3. secretory vesicle
-**PPH**- 1. microfilament 2. intermediate filament 3. vacuole

7.1 FOOD/EATING/INGESTION

-TX- 'EAT / DRINK / NUTRITIONAL CONSUMPTION'		
BSC	**STEM 1:** (to be) food and drink; to eat and drink	**STEM 2:** (to be) food, i.e., (semi-/quasi-)solid consumable/digestible material; to eat (semi-/quasi-)solid food **STEM 3:** (to be) a drink, i.e., a consumable/digestible liquid; to drink
CTE	(to be) (the biological process of consuming) food and drink as nourishment; eat/drink as nourishment	
CSV	(to be) the physical process of food/drink consumption, the physical process of eating/drinking (i.e., placement of consumable in mouth/chewing/swallowing/digestion).	
OBJ	(to be) the food and drink one consumes; the actual food/drink items one consumes	

-NTX- 'MASTICATION / SWALLOWING / BITING'		
BSC	**STEM 1:** (to be) an act of chewing/mastication on something; to chew something; to masticate on something	**STEM 2:** (to be) an act of swallowing something **STEM 3:** (to be) an act of biting (on) something; to bite (on) something [i.e., sink the forward half of one's teeth into an entity in order to either bite off a piece, deliver venom, or to keep it from escaping] [CPT = to bite off (a piece of) something]
CTE	(to be) the biological process of chewing/mastication; chewing as a biological process	
CSV	(to be) the physical process of chewing/mastication [use Phase categories if necessary to specify a single iteration or other pattern of chewing]	
OBJ	(to be) the entity/piece of food/object one chews on	

-SDR- 'STIR / BLENDING IN CIRCULAR PATTRN' (useful to describe food preparation but also applicable any other appropriate context)		
BSC	**STEM 1:** (to be) an act/instance of stirring something; to chew something; to masticate on something	**STEM 2:** (to be) an act/instance of "folding" something into something else as in baking; to "fold" as in baking (i.e., combining a lighter/thinner component into a heavier/thicker component while retaining as much air as possible) **STEM 3:** (to be) an act/instance of mechanically blending something via rotary action (as with a mechanical or electric-powered blender)
CTE	(to be) the state of something being stirred or undergoing stirring	
CSV	(to be) the physical process of stirring [use Phase categories if necessary to specify a single iteration or appropriate affixes or concatenation with shape/form roots to describe particular patterns of stirring]	
OBJ	(to be) the entity/material/substance/object one stirs	

The Specification pattern of the above root applies to the following roots as well:

-SDL- 1. mash 2. grind 3. pulverize / granulize

-SDŘ- 1. sprinkle onto 2. brush onto 3. baste

-SDW- 1. smear onto 2. spread with finger(s) CPT = coat by means of spreading 3. spread with spreading utensi CPT = coat by means of spreading

-SDY- 1. dip into CPT = coat by dipping into 2. melt / soften via heat 3. whip, i.e., soften or liquify via whipping with a utensil (as in cooking/baking)

-LKS- 'FOODSTUFF OR BEVERAGE' Associated affix: DFB			
	Stem 1	Stem 2	Stem 3
BSC	(to be) a state of there being a particular food or beverage available for consumption	(to be) a state/act of making/preparing/cooking or fermenting a particular food or beverage	(to be) a state/act of sourcing or raising or farming/growing/cultivating the animal/plant/mineral that will become a foodstuff
CTE	(to be) an amount/portion of a particular food/beverage (the entity/substance or liquid itself)	(to be) a state of a food or beverage in preparation/being cooked/being fermented	(to be) the state of an animal/plant/mineral being raised, grown, cultivated, or being a source for food
CSV	(to be) an act of eating/drinking a particular food or beverage; to eat/drink a particular food or beverage	(to be) the physical act of making/preparing/cooking or fermenting a particular food or beverage	(to be) a physical act of sourcing, raising, growing, cultivating an animal/plant/mineral for food
OBJ	(to be) a serving/dish/plate/glass/cup of a particular food or beverage; to serve (a dish/plate/glass/cup) of a particular food or beverage	(to be) the prepared/cooked/fermented food or beverage itself to be served	(to be) the plant/animal/mineral that will be (consumed as) food

-xč	**CK1 Cooking Methods 1**
1	sautéed; cook by sautéing X
2	stir-fried; cook by stir-frying X
3	pan-fried; cook by pan-frying X
4	seared; cook by searing X
5	deep-fried; cook by deep-frying X
6	flash-fried; cook by flash-frying X
7	grilled; cook by grilling X
8	broiled; cook by broiling X
9	roasted; cook by roasting X

-xc	**CK2 Cooking Methods 2**
1	steamed; cook by steaming X
2	blanched; heat/cook by blanching
3	poached; cook by poaching X
4	simmered; cook by simmering X
5	boiled; cook by boiling X
6	braised; cook by braising X
7	stewed; cook by stewing X
8	toasted; cook by toasting X
9	barbecued; cook by barbecuing

-xẓ	**CK3 Cooking Methods 3**
1	uncooked, raw; serve X raw/uncooked
2	smoked; cook by smoking X
3	cured; prepare/preserve by curing X
4	fermented; prepare by fermenting X
5	brewed; prepare by brewing X
6	pressure-cooked / pressure-steamed; cook by pressure-cooking X
7	slow-cooked in crockpot; cook by slow-cooking X in crockpot (e.g., underground or)
8	slow-cooked underground; cook by slow-cooking X underground)
9	baking; cook by baking X

NOTE: To express parboiling, parbaking, or parcooking, use the **PTT$_3$/4** affix with one of the affixes above.

The following roots follow the same Stem and Specification pattern as the root **-LKS-** 'FOODSTUFF OR BEVERAGE' above:

-**LKST**- 'DISH/RECIPE MADE FROM/WITH ANIMAL PRODUCTS / NON-VEGAN DISH'
-**LKSW**- 'DISH/RECIPE MADE FROM/WITH FISH/SEAFOOD'
-**LKSL**- 'DISH/RECIPE MADE FROM/WITH MAMMALIAN FLESH'
-**LKSR**- 'DISH/RECIPE MADE FROM/WITH POULTRY/AVIAN FLESH'
-**LKSŘ**- 'DISH/RECIPE MADE FROM/WITH CEREAL GRAINS'
-**LKSM**- 'DISH/RECIPE MADE FROM/WITH EGGS'
-**LKSN**- 'DISH/RECIPE MADE FROM/WITH REPTILIAN OR AMPHIBIAN FLESH'
-**LKSP**- 'DISH/RECIPE MADE FROM/WITH A MIXTURE OF ANIMAL/PLANT PRODUCTS/PARTS'
-**LKT**- 'VEGAN DISH/RECIPE'
-**LKTW**- 'DISH/RECIPE MADE FROM/WITH FRUIT'
-**LKTR**- 'DISH/RECIPE MADE FROM/WITH SEEDS/NUTS'
-**LKTL**- 'DISH/RECIPE MADE FROM/WITH BEANS/LEGUMES'
-**LKTY**- 'DISH/RECIPE MADE FROM/WITH VEGETABLES'
-**LKTŘ**- 'DISH/RECIPE MADE FROM/WITH PLANT LEAVES/SHOOTS'
-**LKSŇ**- 'DESSERT'
-**LKŠM**- 'JAM/JELLY/MARMALADE/CONFITURE MADE FROM FRUIT'
-**LKŠN**- 'FOOD EATEN AS A SNACK'

-**ŢMW**- 'WINE' (i.e., an alcoholic beverage made by the fermentation of a fruit, whether grapes, apples, berries, etc. - thus this stem includes the meaning of the English terms (alcoholic) *cider* and *perry*)
-**ŢDR**- 'BEER'
-**ŢGL**- 'FERMENTED ALCOHOLIC BEVERAGE' (other than fruit-based wine or beer, e.g., mead, vegetable-based liquor, nut-based liqueurs, etc.)
-**ŢGR**- 'FERMENTED FOODSTUFF'
-**ŢGW**- 'DECOCTION/INFUSION' (including teas and tisanes)

-**ČKW**- 'CEREAL GRAIN'
-**ČKY**- 'BAKED GOOD'
-**ČPW**- 'BREAD'
-**ČPY**- 'FLATBREAD'

-ḌNY- 'MILK FROM MAMMAL'
-**ḌNW-** '(LAIN) EGG FROM ANIMAL'
-ḌMW- 'ICE CREAM / SHERBET/GELATO'
-ḌMY- 'SORBET'

-**ČPL**- ‘CAKE’
-**ČPR**- ‘PASTRY’
-**ČPŘ**- ‘PIE’
-**ČFL**- ‘PASTA’
-**ČFR**- ‘NOODLE’

-**BVL**- ‘BROTH-BASED SOUP’
-**BVR**- ‘PUREE-BASED SOUP’
-**BVW**- ‘STEW’
-BVY- ‘GRUEL/PORRIDGE’
-**BVŘ**- ‘SAUCE’
-**BFL**- ‘VINEGAR’
-**BFR**- ‘GELATIN’

-ḌŇ- ‘CHEESE’
-ḌŇW- ‘BUTTER’
-ḌḌN- ‘CUSTARD/ PUDDING’
-**ŠBR**- ‘MOLASSES’
-**ŠBW**- ‘SUGAR [i.e., as foodstuff, not as chemical compound]
-**ŠBŘ**- ‘SUGAR SUBSTITUTE’
-**ḶTY**- ‘FRUIT PUNCH’
-**ḶTW**- ‘PROCESSED COMMERCIAL CARBONATED BEVERAGE / SODA’
-**ḶTŘ**- ‘FRUIT JUICE’
-**ḶTF**- ‘CANDY OR CONFECTION’
-**ḶTÇ**- ‘CARAMEL’
-**ḶPÇ**- ‘TOFFEE’
-**ḶPḶ**- ‘ICING/FROSTING

7.1.1 Common Food Plants and Fungi

The stems listed in this section are excerpted from the listings of lexical roots for plants in Sections 7.3.2 and 7.3.3 of this document. Each stem refers to the plant or fungus itself (i.e., the tree, shrub-tree, shrub, bush, liana, vine, grass stalk, herb, etc.). To create words for the edible portion of the plant, use an **EPP** affix or the **DFB** affix, both shown here for convenience.

-kc	EPP Edible or Autonomous Plant Parts/Components
1	seed/nut of X
2	(seed) pod of X
3	fruit of X
4	leaf of X
5	flower/bloom/mushroom of X
6	juice of X
7	sap of X
8	rhizome/rootstalk/shoot/stolon of X
9	tuber/bulb/cormis of X

-lks	DFB Derived Foodstuff/Beverage
1	non-sweet (dry) fermented alcoholic beverage made from X
2	sweet alcoholic beverage made from X
3	non-alcoholic beverage made from X
4	fermented X as foodstuff
5	recipe/dish made from/with X
6	dessert made from X
7	puree of X
8	processed/derived food product made from X (e.g., flour)
9	flavoring/seasoning made from/with X

Fruit plants (including squashes):

-BḌ- Stem 1: European cantaloupe plant **Stem 2:** North American cantaloupe plant **Stem 3:** muskmelon plant

-BḌN- Stem 1: watermelon **Stem 2:** winter melon / wax gourd / ash gourd **Stem 3:** tinda / Indian squash / round melon

-BḌW- Stem 1: honeydew melon plant **Stem 2:** Crenshaw melon plant **Stem 3:** casaba melon plant

-BḌY- Stem 1: cucumber vine **Stem 2:** cackrey / West Indian gherkin vine **Stem 3:** aardvark cucumber / aardvark pumpkin

-**CVR**- **Stem 1:** avocado, coyo

-**ḌKW**- **Stem 1:** pineapple

-**FMW**- **Stem 2:** elderberry

-**GḌM**- **Stem 1:** zucchini squash vine **Stem 2:** scallop summer squash **Stem 3:** crookneck squash vine

-**GḌN**- **Stem 1:** pumpkin vine **Stem 2:** acorn squash vine **Stem 3:** spaghetti squash vine

-**LČKL**- **Stem 1:** durian

-**LMSKW**- **Stem 2:** starfruit / carambola

-**LŠM**- **Stem 1:** redcurrant **Stem 2:** white currant **Stem 3:** blackcurrant

-**LŠN**- **Stem 1:** gooseberry **Stem 2:** jostaberry **Stem 3:** golden currant / clove currant / pruterberry / buffalo currant

-**MSFW**- **Stem 3:** santol tree / sentul tree / cotton fruit tree

-**NÇP**- **Stem 1:** banana, plantain **Stem 2:** enset **Stem 3:** Chinese dwarf banana / golden lotus banana

-ŇḌ- Stem 1: blueberry **Stem 2:** cranberry **Stem 3:** huckleberry, whortleberry

-**ŘDK**- **Stem 1:** guava / yellow guava / lemon guava, Costa Rican guava / cas, Jamaican guava, Cattley guava / strawberry guava / cherry guava, mountain guava, Galápagos guava, little guava, purple guava, araçá **Stem 3:** finger cherry / Cooktown loquat, native guava, rose myrtle

-**ŘDKW**- **Stem 1:** feijoa / pineapple guava / guavasteen **Stem 2:** ubajay

-ŇḌL- Stem 1: eastern teaberry [source of wintergreen flavoring] **Stem 2:** salal / shallon

-ŇḌR- Stem 1: lingonberry / partridge berry / cowberry / foxberry / redberry **Stem 2:** deerberry

-ŇḌW- Stem 1: kiwi **Stem 2:** kiwi berry / grape kiwi / dessert kiwi / cocktail kiwi / golden kiwi

-ŇḌY- Stem 1: Japanese persimmon **Stem 2:** date-plum **Stem 3:** American persimmon

-NḶ- Stem 1: sweet orange **Stem 2**: bitter orange **Stem 3:** key lime

-NḶK- Stem 3: mango tree

-**NḶML**- **Stem 1:** bael / Bengal quince / golden apple / Japanese bitter orange / stone apple / wood apple **Stem 2:** curd fruit / wood-apple / elephant-apple **Stem 3:** feroniella / ka sang / cra san / kawista-krikil / krasang / mak sang / canthan

-**NḶMW**- **Stem 1:** African cherry orange

-NḶN- Stem 1: lemon **Stem 2:** Meyer lemon **Stem 3:** grapefruit

-NḶŇ- Stem 1: lime **Stem 2:** mandarin orange **Stem 3:** citron

-NḶŇW- Stem 1: kumquat **Stem 2:** pomelo/shaddock **Stem 3:** other citrus fruit

-NḶNY- **Stem 1:** clementine **Stem 2:** tangelo **Stem 3:** other citrus hybrid

-**ŇŇP**- **Stem 1:** pomegranate

-NŽ- **Stem 1:** apple tree **Stem 2:** crabapple **Stem 3:** pear

-ŇZKR- **Stem 1:** chicle, ausubo / massaranduba / balatá, sapotilla, apota / sapodilla / naseberry **Stem 2:** eggfruit, abiu, lúcuma, mamey sapote, green sapote, black plum **Stem 3:** shea

-ŇZKŘ- **Stem 1:** miracle berry / sweet berry **Stem 2:** tar apple / star apple / milk fruit / cainito / Damson plum / wild star-apple / caimitillo

-NŽT- **Stem 1:** almond tree **Stem 2:** peach tree **Stem 3:** peach tree bearing nectarine fruit / "nectarine tree"

-NŽTW- **Stem 1:** plum tree **Stem 2:** apricot tree **Stem 3:** plum-apricot hybrid (pluot, plumcot, apriplum)

-NŽTY- **Stem 1:** cherry tree

-NŽW- **Stem 1:** quince **Stem 2:** Chinese quince **Stem 3:** Japanese quince

-ŘÇ- **Stem 1:** fig tree **Stem 2:** mulberry tree **Stem 3:** African breadfruit tree

-ŘÇW- **Stem 1:** breadfruit **Stem 2:** jackfruit **Stem 3:** monkey fruit / lakoocha

-RMV- **Stem 1:** red beet / beetroot **Stem 2:** sugar beet

-ŘT- **Stem 1:** raspberry plant **Stem 2:** blackberry plant **Stem 3:** dewberry plant

-ŘTL- **Stem 1:** marionberry plant **Stem 2:** youngberry plant **Stem 3:** olallieberry plant

-ŘTW- **Stem 1:** cloudberry plant **Stem 2:** salmonberry plant **Stem 3:** thimbleberry plant

-ŘTY- **Stem 1:** loganberry plant **Stem 2:** boysenberry plant **Stem 3:** tayberry plant

-ŘŽ- **Stem 2:** strawberry cultivar

-**RẔXM**- **Stem 1:** tamarind

-**ŢN**- **Stem 1:** (European *Vitus vinifera*) grape **Stem 2:** culinary/cultivated grape (other than *Vitus vinifera*)

-**VSTW**- **Stem 1:** lychee tree **Stem 2:** longan, alupag **Stem 3:** rambutan, korlan, pulasan

-**VSTY**- **Stem 1:** ackee fruit tree **Stem 2:** titberry, bastard taaibos, dune false crowberry / dune false currant **Stem 3:** native tamarind, small-leaved tamarind

-**ZP**- **Stem 2:** tomato **Stem 3:** eggplant / aubergine

-**ZPÇ**- **Stem 1:** tomatillo **Stem 2:** winter-cherry, Chinese/Japanese lantern **Stem 3:** groundcherry, Inca berry, golden strawberry, strawberry-tomato

-**ZPH**- **Stem 1:** habanero pepper **Stem 2:** aji chili pepper **Stem 3:** tabasco pepper / piri piri / Hawaiian pepper / xiaomila pepper, manzano pepper / rocoto pepper / locoto pepper

-**ZPL**- **Stem 1:** wild tomato, currant tomato **Stem 2:** Ethiopian eggplant **Stem 3:** naranjilla / lulo

-**ZPR**- **Stem 1:** Turkey berry / wild eggplant / pea eggplant / susumber **Stem 2:** pepino melon / melon pear **Stem 3:** tamarillo / tree tomato / blood fruit

-**ZPŘ**- **Stem 1:** wolf apple **Stem 2:** cocona **Stem 3:** kangaroo apple, desert raisin / bush raisn / bush saltana, bush tomato, potato bush, Australian wild tomato

-**ZPW**- **Stem 1:** bell pepper **Stem 2:** sweet Italian pepper, pepperoncino **Stem 3:** paprika

-**ZPY**- **Stem 1:** serrano pepper **Stem 2:** cayenne pepper **Stem 3:** jalapeño pepper

-ŽŽPŢ- Stem 1: papaya / pawpaw

Tuberous plants:

-BVK- Stem 2: ulluco
-**ḌKY**- **Stem 1:** chufa flatsedge [tuber used as food]
-**PŠMY- Stem 2:** duck potato / swamp potato / tule potato / wapato / arrowhead / watato / wapati
-**PSVL**- **Stem 1:** sweet potato
-**TXW**- **Stem 1:** yam **Stem 2:** batflower, Polynesian arrowroot / pia
-**ZP**- **Stem 1:** potato

Spice/condiment plants:

-**CVW**- **Stem 1:** laurel / bay laurel / sweet bay
-**CVY**- **Stem 1:** cinnamon tree **Stem 2:** sassafras
-**CFŘ**- **Stem 2:** grains of Selim / Kani pepper / Senegal pepper / Ethiopian pepper
-**FŠK**- **Stem 1:** common mint / spearmint **Stem 2:** corsican mint
-**FŠP**- **Stem 1:** common thyme **Stem 2:** lemon thyme **Stem 3:** citrus thyme
-**FŠPL**- **Stem 1:** oregano **Stem 2:** marjoram
-**FŠPŘ**- **Stem 1:** sage **Stem 2:** rosemary
-**FŠPR**- **Stem 1:** summer savory **Stem 2:** winter savory / mountain savory
-**FŠPW**- **Stem 1:** caraway thyme
-**FŠKL- Stem 1:** peppermint **Stem 2:** gingermint / redmint / Scotch spearmint **Stem 3:** Cuban mint / mojito mint / large apple mint
-**FŠPF**- **Stem 1:** basil
-**KBW**- **Stem 2:** cumin
-**KPW**- **Stem 2:** fennel **Stem 3:** parsley
-**KPY**- **Stem 1:** dill
-**KŠPḶ**- **Stem 2:** adjwain / ajowan / bishop's weed / carom, radhuni / wild celery
-**KŠPW**- **Stem 1:** anise, short-fruit pimpinella **Stem 2:** caraway, meridian fennel, Persian cumin **Stem 3:** coriander, cilantro / Chinese parsley
-**MŽDY**- **Stem 1:** stevia
-**NÇPW**- **Stem 1:** ginger **Stem 2:** turmeric **Stem 3:** cardamon
-**NÇPY**- **Stem 1:** galangal **Stem 2:** Ethiopian cardamom plant, melegueta pepper / alligator pepper / grains of paradise **Stem 3:** torch ginger
-**ŘDK**- **Stem 2:** allspice / myrtle pepper / pimenta / pimento, West Indian bay tree / bay rum tree / ciliment
-**ŘDKŢ**- **Stem 2:** clove tree

-**RDL**- **Stem 2:** sugarcane

-**RSPY**- **Stem 1:** vanilla bean

-**SSW**- **Stem 1:** pepper, pepper vine, pepperleaf, peppercorn, root bear plant, yerba santa, betel leaf vine, long pepper, ant plant / ant piper, kawakawa, cubeb, kava, pariparoba

Palms:

-**LBL**- **Stem 1:** areca nut palm [i.e., betel nut palm]

-**LBV**- **Stem 2:** forest coconut palm

-**LBW**- **Stem 1:** coconut palm **Stem 2:** date palm **Stem 3:** açaí palm

-**LBY**- **Stem 1:** jelly palm, yatay palm, pindo palm **Stem 2:** oil palm, macaw-fat **Stem 3:** peach palm

-**LBZ**- **Stem 2:** bertam / wild Bornean sago palm **Stem 3:** raffia palm

Grains/cereals/pseudo-cereals:

-**BK**- **Stem 1:** Asian rice **Stem 2:** African rice **Stem 3:** red rice / brownbeard rice

-**BKL**- **Stem 1:** teff

-**BKR**- **Stem 1:** finger millet

-**BKW**- **Stem 1:** wild rice / Canada rice / Indian rice / water oats

-**BT**- **Stem 1:** common wheat / bread wheat **Stem 2:** durum wheat / pasta wheat **Stem 3:** spelt / dinkel wheat

-**BTH**- **Stem 1:** oat **Stem 2:** naked oat / hulless oat **Stem 3:** Ethiopian oat

-**BTW**- **Stem 1:** barley **Stem 2:** rye **Stem 3:** triticale

-**BTY**- **Stem 1:** einkorn wheat **Stem 2:** emmer wheat **Stem 3:** khorasan wheat / kamut wheat

-**RDL**- **Stem 1:** corn / maize **Stem 3:** sorghum

-**RMD**- **Stem 2:** quinoa

-**RND**- **Stem 1:** amaranth

-RNTW- Stem 2: buckwheat

-**RẔPŠ**- **Stem 3:** alfalfa plant

-**ŢXL**- **Stem 1:** common millet **Stem 2:** pearl millet **Stem 3:** foxtail millet / Italian millet

-**ŢXR**- **Stem 1:** fonio / hungry rice **Stem 2:** raishan

Leafy vegetables:

-BVK- Stem 1: Malabar spinach / vine spinach / Ceylon spinach

-**RBZ**- **Stem 1:** (loose)leaf lettuce **Stem 2:** romaine/cos lettuce **Stem 3:** iceberg/cripshead lettuce

-**RBZL**- **Stem 1:** endive / escarole **Stem 2:** common chicory, radicchio, Belgian endive
-**RBZW**- **Stem 1:** butter(head) lettuce **Stem 2:** red leaf lettuce **Stem 3:** summercrisp lettuce
-**RMD**- **Stem 1:** spinach
-**RMDW**- **Stem 1:** saltbush / orache
-**RMDY**- **Stem 1:** Good-King-Henry / poor-man's asparagus / Lincolnshire spinach / markery **Stem 2:** salt-green / sowbane / Australian-spinach
-RMFR- Stem 1: sea beans / sea asparagus / samphire greens
-RMV- Stem 3: (Swiss) chard
-RMVW- Stem 1: Caucasian spinach
-**RND**- **Stem 2:** Lagos spinach
-RNTW- Stem 1: rhubarb **Stem 3:** sorrel, dock
-RNX- Stem 1: New Zealand spinach, dune spinach
-RẔH- Stem 1: salsify / oyster plant

Vegetables (including vegetable-like fruits, e.g., olive, okra):

-CK- Stem 1: tea
-**ÇX**- **Stem 1:** arabica coffee **Stem 2:** robusta coffee
-**FSPÇ**- **Stem 1:** safflower
-**ḌKL**- **Stem 2:** Chinese water chestnut
-**FSPR**- **Stem 1:** burdock
-**FSPW**- **Stem 1:** globe artichoke / French artichoke / green artichoke **Stem 2:** cardoon **Stem 3:** white artichoke
-**FSPY**- **Stem 1:** lesser wild artichoke
-**KBL**- **Stem 2:** shepherd's-needle / Stork's needle / Venus' comb
-**KBW**- **Stem 1:** carrot **Stem 3:** chervil
-**KPŘ**- **Stem 1:** arracacha / racacha
-**KPW**- **Stem 1:** celery
-**KŠP**- **Stem 1:** parsnip **Stem 2:** lovage **Stem 3:** licorice
-**LCKM**- **Stem 1:** okra
-**LZM**- **Stem 1:** olive
-ŇZKH- Stem 1: brazil nut **Stem 2:** paradise nut
-**ŘŽBL**- **Stem 1:** sesame
-**SFW**- **Stem 1:** asparagus **Stem 3:** agave
-**SŢL**- **Stem 1:** garlic **Stem 2:** chive **Stem 3:** scallion
-**SŢR**- **Stem 1:** onion **Stem 2:** shallot **Stem 3:** leek

-ŽGŘ- Stem 1: winter cress / yellow rocket, land cress / American cress / bank cress / Burmuda cress / herb barbara / rocketcress / winter rocket **Stem 2:** horseradish

-ŽK- Stem 1: broccoli **Stem 2:** cauliflower, romanesco, broccoflower **Stem 3:** Chinese broccoli / gai lan, broccolini

-ŽKÇ- Stem 1: Siberian kale **Stem 2:** tatsoi

-ŽKF- Stem 1: radish, daikon **Stem 2:** watercress **Stem 3:** arugula / rocket

-ŽKH- Stem 1: brown mustard, wrapped heart mustard cabbage **Stem 2:** white mustard **Stem 3:** black mustard

-ŽKL- Stem 1: savoy cabbage **Stem 2:** rutabaga

-ŽKR- Stem 1: turnip **Stem 2:** Napa cabbage **Stem 3:** bok choy

-ŽKŘ- Stem 1: (field) mustard **Stem 2:** rapini / broccoli rabe **Stem 3:** komatsuna / Japanese mustard spinach

-ŽKŢ- Stem 1: mizuna **Stem 2:** choy sum / flowering cabbage **Stem 3:** canola

-ŽKV- Stem 1: wasabi **Stem 2:** garden cress, field cress **Stem 3:** wild arugula / perennial wall-rocket

-ŽKW- Stem 1: cabbage **Stem 2:** brussel sprout **Stem 3:** kohlrabi

-ŽKY- Stem 1: kale **Stem 2:** collard **Stem 3:** spring greens

-ŽŽP- Stem 1: caper, karir **Stem 2:** shepherd tree, hanza **Stem 3:** swarstorm, cadaba

Edible Brown Alga (Kelp)

-**ŇŽG**- **Stem 1:** (*Laminaria*) kombu, haidai seaweed, oarweed **Stem 2:** (*Saccharina*) kombu **Stem 3:** (*Undaria*) wakame, hiromi

-**ŇŽGW**- **Stem 1:** (*Eisenia*) arame / sea oak **Stem 2:** (*Alaria*) winged kelp, badderlocks **Stem 3:** (*Ecklonia*) paddle weed / kajime / noro-kajime

-**ŇŽGY**- **Stem 1:** (*Durvillaea*) rimurapa / cochayuyo **Stem 2:** (*Postelsia*) sea palm / palm seaweed **Stem 3:** (*Nereocystis*) bull kelp / bullwhip kelp / ribbon kelp

-**ŇŽGL**- **Stem 1:** (*Kjellmaniella*) sugar kelp / sea belt / Devil's apron **Stem 2:** (*Fucus*) bladderwrack, spiral wrack **Stem 3:** (*Pelvetia*) channelled wrack

-**ŇŽGR**- **Stem 1:** (*Himanthalia*) thongweed **Stem 2:** (*Hizikia*) hiziki **Stem 3:** (*Cladosiphon*) mozuku

-**ŇŽGŘ**- **Stem 1:** (*Nemacystus*) ito-mozuku **Stem 2:** (*Tinocladia*) futo mozuku Stem 3: (*Sphaerotrichia*) ishi-mozuku

Legumes and Nuts:

-**JFW**- **Stem 1:** soybean / soya bean, wild soybean, glycine, love creeper **Stem 2:** jicama plant / yam bean, ahipa / Andean yam bean, goiteño / nupe / jacatupe / Amazonian yam bean **Stem 3:** kudzu / Japanese arrowroot, puero, East Asian arrowroot

-**JPL**- **Stem 1:** runner bean plant **Stem 2:** tepary bean plant **Stem 3:** year bean plant

-**JPŘ**- **Stem 1:** cowpea, black-eyed pea, crowderpea, Chinese long-bean / pea-bean **Stem 2:** adzuki bean **Stem 3:** mung bean / green gram / golden gram / mash bean / green soybean / celera-bean / Jerusalem-pea

-**JPR**- **Stem 1:** lima/butter bean plant **Stem 2:** spotted bean plant **Stem 3:** slimjim bean plant, other *Phaseolus* bean plant

-**JPW**- **Stem 1:** kidney/red bean plant **Stem 2:** white/navy bean plant **Stem 3:** cannellini bean plant

-**JPY**- **Stem 1:** pinto bean plant **Stem 2:** black turtle bean plant **Stem 3:** flageolet bean plant

-**NḌKN**- **Stem 1:** macadamia tree

-NḶK- Stem 1: pistachio plant **Stem 2:** cashew tree

-**ŘNTÇ**- **Stem 2:** pecan tree

-**ŘNTF**- **Stem 1:** walnut tree, butternut tree

-**ŘNTN**- **Stem 3:** hazel tree, filbert tree

-**ŘPP**- **Stem 1:** pea plant **Stem 2:** snap pea plant **Stem 3:** snow pea plant

-**ŘPPL**- **Stem 1:** peanut plant

-**ŘPPW**- **Stem 1:** fava bean plant **Stem 3:** lentil

-**ŘPPY**- **Stem 2:** chickpea/garbanzo bean plant

Mushrooms/Fungi:

-**GZNW**- **Stem 2:** (*Fistulina*) beefsteak mushroom / ox tongue mushroom

-**GZNY**- **Stem 1:** (*Hygrophorus*) woodwax / waxy cap, herald of winter

-**LVM**- Stem 1: (*Merasmius*) parachute mushroom, fairy ring mushroom, pinwheel mushroom

-**LVMW**- **Stem 1:** (*Lentinula*) shiitake mushroom

-**LVMY**- **Stem 1:** (*Flammulina*) enoki / golden needle / seafood mushroom / lily mushroom / winter mushroom / velvet foot / velvet shank / velvet stem / winter fungus **Stem 2:** (*Amillaria*) honey mushroom / honey fungus

-**NZGR**- **Stem 1:** *Pleurotus*) oyster mushroom, abalone mushroom, king tuber mushroom

-**NZGW**- **Stem 1:** (*Amanita caesaria*) Caesar's mushroom

-**NZGY**- **Stem 3:** (*Volvariella*) straw mushroom

-**PŠXL**- **Stem 1:** (*Hypsizygus*) beech mushroom, bunapi-shimeji, buna-shimeji **Stem 2:** (*Calocybe*) St. George's mushroom / marzolino

-**PŠXR**- **Stem 1:** (*Clitocybe*) wood blewit / blewitt muschroom

-**PṬP- Stem 1:** (*Agaricus*) common mushroom, agaricus mushroom, button mushroom, crimini mushroom, champignon, baby bella, portabella **Stem 2:** (*Coprinus*) shaggy ink cap / shaggy mane / lawyer's wig

-**PṬPF- Stem 1:** (*Stropharia*) winecap / burgundy / garden giant / king stropharia mushroom, roundhead mushroom **Stem 2:** (*Agrocybe*, a.k.a., *Cyclocybe*) pioppino / velvet pioppino / poplar mushroom / black poplar mushroom **Stem 3:** (*Pholiota*) nameko mushroom

-**PṬPL- Stem 1:** (*Macrolepiota*) parasol mushroom

-**PṬPW- Stem 2:** (*Calbovista*) sculptured puffball / sculptured giant puffball / warted giant puffball

-**RJBW**- **Stem 1:** (*Morchella*) morel

-**RJTF**- **Stem 1:** (*Gyromitra*) false morel

-**RJTḶ**- **Stem 1:** (*Tuber*) truffle

-**RṬK**- Stem 1: (*Cantherellus*) chanterelle / yellow chanterelle **Stem 2:** (*Craterellus*) tube chanterelle / yellowfoot chanterelle / yellow-leg, trompette de la mort / horn of plenty

-**RṬKL**- **Stem 1:** (*Laetiporus*) sulpher shelf / sulphur polypore / chicken-of-the-woods / crab-of-the-woods / lobster-of-the-woods **Stem 2:** (*Laccocephalum*, a.k.a. *Polyporus*) native bread / blackfellow's bread **Stem 3:** (*Cerioporus*) dryad's saddle / pheasant's back mushroom

-RṬKM- **Stem 2:** (*Lactarius*) milk-cap **Stem 3:** (*Lactifluus*) milky, milk-cap

-RṬKN- **Stem 1:** (*Hericium*) lion's mane / monkey head / bearded tooth / satyr's bear / bearded hedgehog / pom pom mushroom

-**RṬKŘ**- **Stem 1:** (*Auricularia*) wood ear / Judas' ear / jelly ear, cloud ear

-**RṬKR**- **Stem 1:** (*Grifola*) maitake mushroom / ram's head mushroom / sheep's head mushroom **Stem 2:** (*Sparassis*) cauliflower fungus / cauliflower mushroom

-**RṬKṬ**- **Stem 1:** (*Chroogomphus*) pine-spike / spike-cap

-**RṬKW**- Stem 1: (*Hydnum*) sweet tooth fungus / hedgehog mushroom / urchin of the woods, pied de mouton

-**RṬX**- **Stem 1:** (*Boletus*) cep mushroom / penny bun / porcini mushroom / bolete, old man of the woods **Stem 2:** (*Leccinum*) red-capped scaber stalk, northern roughstem, birch bolete, Aspen bolete, slate bolete, manzanita bolete **Stem 3:** (*Suillus*) weeping bolete / granulated bolete, Slippery Jack, woolly-capped suillus, Jersey cow mushroom / bovine bolete

-**Ẓ BV**- **Stem 1:** (*Tremella*) snow fungus / snow ear / silver ear fungus / white jelly mushroom

Miscellaneous Food Stems:

-**VST**- **Stem 1:** maple tree

-**PFTW**- **Stem 1:** spirulina **Stem 2:** aphanizomenon **Stem 3:** nostoc

-**KC**- **'PLANT PARTS I'** Associated Affix: EPP		
BSC	**Stem 1:** (to be) a seed or pit of a plant	**Stem 2:** (to be) a fruit or nut of a plant **Stem 3:** (to be) a bulb/tuber/corm of a plant
CTE	(to be) the potential for germination/sprouting/nourishment contained within a plant seed or pit	
CSV	(to be) the physical flesh/material of which a plant seed or pit is composed	
OBJ	(to be) a seed/pit of a plant as an energy source; to consume a plant seed/pit as energy source (e.g., by burning, eating, fermenting, etc.)	

The following two roots have the same Specification pattern as the root -**KC**- above:

-KŠW- ‘PLANT PARTS II’		
1. root	2. stem/stalk/shoot	3. branch/limb/twig

-KŠN- ‘PLANT PARTS III’		
1. leaf	2. flower	3. bud

-VTY- ‘PSYCHOACTIVE/INTOXICATING SUBSTANCE’		
BSC	**Stem 1:** (to be) an act/state of being intoxicated by using/ingesting a naturally derived psychoactive chemical substance (e.g., ethyl alcohol, cannabis, peyote, psilocybin, ayahuasca, etc.); to be(come) intoxicated due to use/ingestion of such a substance	**Stem 2:** (to be) a synthetic medicinal chemical used recreationally (e.g., a prescription drug) **Stem 3:** (to be) a chemical substance repurposed for recreational use (e.g., glue/solvents, etc.)
CTE	(to be) in a state of intoxication from a naturally derived chemical substance; to be/get drunk/high/intoxicated	
CSV	(to be) an act of using/ingesting a naturally derived chemical substance; to use/ingest/smoke/inject an intoxicating substance	
OBJ	(to be) a(n amount/serving/hit/drink/cigarette, etc. of a) particular intoxicating/psychoactive substance itself	

-STH- ‘OTHER METHODS OF BODILY INGESTION’		
BSC	**Stem 1:** (to be) an act/state of smoking of a gaseous or aerosolized substance (i.e., the oral intake/inhalation of fumes via an ingressive velaric airstream)	**Stem 2:** (to be) an act/state of nasal inhalation of a substance (e.g., snuff, cocaine, other powdery substance) **Stem 3:** (to be) an act/process of subcutaneous or intravenous injection (e.g., with a syringe) of a substance
CTE	(to be) the smoke/fumes of a burning substance which one takes in or inhales orally via smoking	
CSV	(to be) an act of smoking; to smoke, to intake/inhale orally by means of smoking	
OBJ	(to be) the apparatus utilized for an act of smoking	

7.2 THE BODY

-NṬ- 'BIRTH / BE BORN / GESTATION / CONCEPTION'		
BSC	**Stem 1:** (to be) a state/act of birth; to give birth or be born (i.e., the leaving of a child from their mother's womb to enter the world, with no implied perspective as to it being an act/state of the mother or the child)	**Stem 2:** (to be) a state of maternal gestation; to gestate, carry a developing fetus within one's womb **Stem 3:** (to be) an act of biological conception; to conceive a zygote
CTE	(to be) a state/act of being born [from the child's perspective]	
CSV	(to be) the physical act of giving birth [from the perspective of the mother]	
OBJ	(to be) the newborn infant	

-Š- SEX (i.e., genetically-based, the normally 2-way distinction in higher living organisms for reproductive/procreative purposes) Associated Affix: **SEX**		
BSC	**Stem 1:** (be/manifest) a particular sex of an organism that is normally one of two sexes (e.g., chordates) [both the existence and the clinical/biological determination/identification thereof]	**Stem 2:** (to be/manifest) the female sex of an organism that is normally one of two biological sexes **Stem 3:** (to be/manifest) the male sex of an organism that is normally one of two biological sexes
CTE	(be) a particular biological sex	
CSV	(be) an act/process/state of clinical/biological determination/identification of sex; to clinically/biologically determine a particular sex	
OBJ	(to be) that which biologically determines or by which one identifies a particular sex	

-ŽF- 'HEALTHINESS / PHYSICAL CONDITION / FITNESS / SANITY' Associated Affix: **HEA**		
BSC	**Stem 1:** (to be) a state of physical and mental healthiness (i.e., freedom from illness or malady)	**Stem 2:** (to be) one's state of mental health **Stem 3:** (to be) one's physical condition or state of health/fitness (i.e., the extent to which one is sufficiently healthy/fit to carry out specialized physical tasks or feats of physical prowess/stress)
CTE	(to be) the state of healthiness itself; good health	
CSV	(to be) the state of living and going about one's desired normal life due to good health; to live in a state of healthiness, be healthy	
OBJ	(to be) what one does to stay healthy; a healthy living praxis (e.g., diet, exercise, avoiding stress, prophylactic medication, healthy lifestyle, etc.)	

-MT- ‘ONE’S AGE / LIFE-STAGE OR PHASE’		
BSC	**Stem 1:** (to be) a state of aging; to age, get older	**Stem 2:** (to be) a (developmental) stage/phase in one’s life (i.e., a period of time during the life of an entity where it displays/manifests certain characteristics, activities, ideas, values, attributes, etc. specific to that period of time) **Stem 3:** (to be) a state/act of celebration/recognition/acknowledgement of one reaching a particular age; a birthday; to celebrate/recognize/acknowledge one’s birthday (i.e., anniversary of one’s birth)
CTE	(to be) a state of being/having a certain age	
CSV	(to be) the state/process of aging, getting older; to do/feel/notice/observe something that manifests/demonstrates one is aging/getting older	
OBJ	(to be) one’s age (i.e., the amount of elapsed time since an entity came into being)	

-ḌḌ- ‘SLEEP / UNCONSCIOUSNESS / COMA’		
BSC	**Stem 1:** (to be) a state of sleep, being asleep; (be) an act of putting to sleep; to sleep, be asleep	**Stem 2:** (to be) a state of unconsciousness, act of rendering an entity unconscious; to be/render unconscious **Stem 3:** (to be) a state of coma; act of putting someone into a coma; to be in a coma, to put into/induce a coma
CTE	(to be) the state of sleeping	
CSV	(to be) a state/act of falling asleep; to fall asleep, put to sleep	
OBJ	(to be) the quality/duration/description of one’s sleep; how one sleeps	

-ḶF- ‘WAKEFULNESS / CONSCIOUSNESS’		
BSC	**Stem 1:** (to be) a state of being awake, being fully conscious	**Stem 2:** (to be) a state/process of waking (up); to transition from sleep to wakefulness, to “come about”, to be waking up, to be regaining consciousness [CPT version = to regain consciousness, to come fully awake] **Stem 3:** (to be) a state of dozing or being half-awake; to doze, to be half-asleep, half-awake, to be in a state of half-sleep
CTE	(to be) the state of wakefulness/consciousness	
CSV	(to be) a state/act of doing something that demonstrates one is awake/conscious	
OBJ	(to be) the level of awareness of oneself and one’s surroundings one has while awake	

-MR- ‘DEATH’		
BSC	**Stem 1:** (to be) a state/act of dying; to die/perish	**Stem 2:** (to be) a state/process of approaching death, to be dying, to have a fatal condition, to be in a fatal state (i.e., to be in a terminal state where death is immanent, e.g., from a terminal illness, starvation, a mortal wound, etc.) **Stem 3:** (to be) a state of mortality; be mortal (i.e., certain that one will eventually die)
CTE	(to be) the state of being dead	
CSV	(to be) an act of dying; to undergo the act of passing from life to death	
OBJ	(to be) the cause/manner of one’s death	

-LM- **'DEGREE OF BODILY STRENGTH / ENERGY / VIGOR'** Affix: **STR** (use SUF/EXN/EXD, etc.)		
BSC	**STEM 1:** (to be/manifest a) degree of brute bodily strength; to manifest/apply a particular degree of strength (= available bodily dynamic energy) — [both the quality and the entity manifesting such]	**STEM 2:** (to be/manifest a) degree of bodily energy/vigor/vitality; "feel weak/strong" **STEM 3:** (to be/manifest a) degree of stamina
CTE	(to be) something with a degree of brute bodily strength	
CSV	(to be) a degree of brute bodily strength	
OBJ	(to be) an entity/party impacted/affected by one's degree of brute bodily strength	

-SM- **'DEGREE OF BODILY APPLIED ENERGY / POWER / FORCE'** Affix: **FRC** (use SUF/EXN/EXD, etc.)		
BSC	**STEM 1:** (to be/manifest a) degree of applied physical force — [both the quality and the entity manifesting such]	**STEM 2:** (to be/manifest a) degree of energy **STEM 3:** (to be/manifest a) degree of potency, "might" (= potential strength/power to cause or accomplish something)
CTE	(to be) something with a degree of applied physical force	
CSV	(to be) a degree of applied physical force	
OBJ	(to be) an entity/party impacted/affected by one's degree of applied physical force	

-ẔŇ- **'DEGREE OF PHYSICAL STAMINA / ENDURANCE / RESISTANCE TO PAIN'** (use with SUF/EXN affixes)		
BSC	**Stem 1:** (to be) a degree of brute bodily/physical stamina/endurance	**Stem 2:** (to be) a degree of resistance to physical pain; display/manifest a degree of resistance to physical pain, endure physical pain **Stem 3:** (to be) a degree of physical hardship; undergo physical hardship
CTE	(to be) one's physical stamina as a faculty	
CSV	(to be) an act of maintaining physical stamina	
OBJ	(to be) what one is/must have stamina/endurance against or for	

-JMW- **'HAND AS FLAT RIGID "BLADE"'**		
BSC	**Stem 1:** (to be) a hand as flat rigid "blade" for use of outside "edge" of rigidly-held hand (as in a karate chop)	**Stem 2:** (to be) a hand as flat rigid "blade" for use of fingers used "head-on" for pressing/jabbing/poking **Stem 3:** (to be) a hand held flat and rigid for use in covering something or stopping something from escaping (e.g., from a hole or tear)
CTE	(to be) the physical process of chopping with one's hand	
CSV	(to be) a closed fist used in chopping with one's hand	
OBJ	(to be) the blow/impact delivered from chopping with one's hand	

-JMY- 'CLOSED FIST'		
BSC	**Stem 1:** (to be) a closed fist for pounding; use outside surface of closed fist in hammer-like manner for pounding	**Stem 2:** (to be) a closed fist for hitting; use front of fist (i.e, the surface created by the dorsal side of one's fingers) in cudgel-like manner for hitting **Stem 3:** (to be) a closed fist for knocking; use knuckles of fist for knocking
CTE	(to be) the physical process of pounding with one's fist	
CSV	(to be) a closed fist used in pounding	
OBJ	(to be) the blow/impact delivered from a pounding fist	

-ŇP- 'COMMON HUMAN BODILY POSITION'		
BSC	**Stem 1:** (to be) a state/act of standing upright/erect; to (be) stand(ing) upright, stand erect	**Stem 2:** (to be) a state/act of sitting, being seated, being in a sitting/seated position; to sit, be seated, be in sitting/seated position, to be sitting down **Stem 3:** (to be) a state of lying prostrate; in prostrate postion [whether prone, supine, or laterally recumbent]; to lie prostrate, to be lying down
CTE	(to be) a state of being standing (upright/erect)	
CSV	(to be) a physical act of standing up; to stand up, to get up on one's feet	
OBJ	(to be) what one stands/is standing upon or within	

-ŇPY- 'RECUMBENT BODILY POSITION'		
BSC	**Stem 1:** (to be) a state/act of lying in laterally recumbent position; to lie with one's body in a laterally position position [i.e., on one's side]	**Stem 2:** (to be) a state/act of lying prone; to lie in a prone position [i.e., face-down on one's stomach] **Stem 3:** (to be) a state of lying in a supine position; to lie in a supine position [i.e., flat on one's back]
CTE	(to be) a state of being in a laterally recumbent position	
CSV	(to be) a physical act of lying down on one's side	
OBJ	(to be) what one is lying (down) on	

-ŇPW- 'LESS COMMON HUMAN BODILY POSITIONS'		
BSC	**Stem 1:** (to be) a state/act of crouching (down), being in a crouch(ed position); to be crouched (down), be in a crouched/crouching position	**Stem 2:** (to be) a state/act of (being in a) kneeling (position); to kneel, be in a kneeling position, be on one's knees **Stem 3:** (to be) a state,act of being curled up in to a (quasi-)fetal position; to (be) curl(ed) up into a (quasi-)fetal position
CTE	(to be) a state of being in a crouched/crouching position	
CSV	(to be) a physical act of crouching down; to crouch down, to get down into a crouched/crouching position	
OBJ	(to be) what one crouches upon or within	

7.2.1 Parts of the Body

The various roots/stems below for bodily parts/organs/tissues all have the following Specification pattern:

BSC	(to be) a particular bodily part/organ/tissue [both the material/physical aspect and the functional aspect thereof]
CTE	(to be) the function of a particular bodily part/organ/tissue
CSV	(to be) the physical/material make-up of a particular bodily part/organ/tissue
OBJ	(to be) the body to whom the particular bodily part/organ/tissue belongs or is from

-GW- ‘BODILY ORGAN / GLAND / TISSUE’	
STEM 1	bodily organ
STEM 2	gland
STEM 3	tissue (non-organ, non-gland)

-PN- ‘UPPER G.I. ORGANS’	
STEM 1	stomach
STEM 2	esophagus
STEM 3	pancreas

-ŠM- ‘MANIPULATIVE OR AMBULATORY APPENDAGE’	
STEM 1	limb
STEM 2	tentacle
STEM 3	pseudopod

-LŇ- ‘TAIL / CAUDAL STRUCTURE’	
STEM 1	tail
STEM 2	flagellum
STEM 3	tail-like body part or appendage; caudal structure

-ŠN- ‘LEG’	
STEM 1	leg/strut as support structure of animal or inanimate entity
STEM 2	leg as ambulatory appendage of animal or ambulatory entity
STEM 3	leg as functional “tool”/manipulator (e.g., with which to kick, push, press, apply force, etc.)

-CM- ‘ARM’	
STEM 1	arm as support(ing) structure of animal or inanimate entity
STEM 2	arm as appendage of animal (or anthropomorphic entity) for holding, carrying, lifting
STEM 3	arm as functional “tool”/manipulator by which to reach, hit, push, press, apply force, protect oneself, etc.

-RŠN- ‘PART OF THE LEG’	
STEM 1	thigh
STEM 2	calf [i.e., lower half of leg between the knee and ankle]
STEM 3	shin [i.e., front/ventral portion of the lower leg]

-RCM- ‘PART OF THE ARM’	
STEM 1	upper arm
STEM 2	forearm
STEM 3	inside/dorsal portion of forearm

-TY- ‘HEAD’	
STEM 1	head as a living being’s primary “interface” or “access point” for communication, ingestion, non-tactile sensory input, etc.
STEM 2	head as seat of one’s consciousness/personality/identity/mind/brain
STEM 3	head as “top” or “forward” extension of bodily form

-NL- ‘HAND’	
STEM 1	hand as an animal’s primary body part for fine-motor physical manipulation/handling of external entities
STEM 2	hand as holder, grasper, striker
STEM 3	hand as primary tactile-sensory interface, “feeler”, toucher

-PṬY- ‘PARTS OF THE HEAD’	
STEM 1	forehead
STEM 2	brow ridge
STEM 3	temple

-LPÇ- ‘OUTER EAR’	
STEM 1	helical, antihelical, and tragal structures of the outer ear
STEM 2	earlobe/lobule
STEM 3	concha, scapha, and triangular fossa depressions of the outer ear

Stem Zero = the outer (i.e., visible) ear

-NR- ‘FOOT / PAW’	
STEM 1	foot/paw as an entity’s primary contact/support point with ground/firmament when standing/ambulating
STEM 2	foot/paw as ambulatory appendage
STEM 3	foot/paw as functional “tool”/manipulator (e.g., with which to kick, push, press, apply force, etc.)

-MFR- ‘PART OF HAND OR FOOT’	
STEM 1	quasi-flat ventral mid-part of appendage (concatenate ‘hand’ or ‘foot’ to specify whether palm or underside of foot)
STEM 2	ball of the foot or upper palmar ridge of the hand (concatenate ‘hand’ or ‘foot’ to specify which)
STEM 3	butt of the hand or foot (concatenate ‘hand’ or ‘foot’ to specify whether butt of the palm or heel)

-MPG- 'HIP / BUTTOCK / HAUNCH'	
STEM 1	hip (area of flesh on the body, not just the bone-joint itself)
STEM 2	buttock
STEM 3	haunch / hindquarter

-TÇŇ- 'BODILY SURFACE INDENTATION'	
STEM 1	navel
STEM 2	armpit
STEM 3	eye socket

-NK- 'BODILY DIGIT'	
STEM 1	finger
STEM 2	thumb
STEM 3	toe

-ḌL- 'CARDIO-PULMONARY ORGANS'	
STEM 1	heart
STEM 2	lung
STEM 3	diaphragm

-NDY- 'BLOOD VESSEL'	
STEM 1	artery
STEM 2	vein
STEM 3	capillary

-VZ- 'CENTRAL NERVOUS SYSTEM ORGAN/TISSUE'	
STEM 1	brain
STEM 2	brain stem
STEM 3	spinal chord

-NGY- 'PERIPHERAL NERVOUS TISSUE'	
STEM 1	sensory (afferent) nerve tissue
STEM 2	motor (efferent) nerve tissue
STEM 3	somatic nerve tissue

-RDY- 'AUTONOMIC NERVOUS TISSUE'	
STEM 1	sympathetic nerve tissue
STEM 2	parasympathetic nerve tissue
STEM 3	enteric nerve tissue

-ŢŘ- 'BODILY LIMB-JOINT'	
STEM 1	mid-limb joint [concatenate stems for leg or arm to specify knee or elbow]
STEM 2	proximal limb joint [concatenate stems for leg or arm to specify hip-joint or shoulder-joint]
STEM 3	distal limb joint [concatenate stems for leg or arm to specify ankle or wrist]

-GŘ- 'BONE'	
STEM 1	quasi-cylindrical bone (e.g., tibia, ulna, phalanx, etc.)
STEM 2	plate-like or quasi-planar bone (e.g., skull-section, scapula, pelvis)
STEM 3	special-shaped bone (e.g., vertebra, rib, calcaneus, talus, antler, etc.

-ŢFR- 'JOINT STRUCTURAL COMPONENT / CARTILAGE'	
STEM 1	cartilage
STEM 2	joint cavity [containing synovial fluid]
STEM 3	articular capsule of a joint

-ZGR- 'MAJOR CONFIGURATIVE BONE SYSTEM OF BODY'	
STEM 1	skull
STEM 2	rib cage
STEM 3	spine/vertebral column (including cervical vertebrae and coccyx)

-PFL- 'DETOXIFICATION/FILTRATION/RECYCLING ORGAN/TISSUE'	
STEM 1	liver
STEM 2	kidney
STEM 3	spleen

-LȚ- 'INTEGUMENT / SKIN / PELLICLE'	
STEM 1	skin / hide / pellicle / integument
STEM 2	bark / integument of tree trunk/limb or plant main stem
STEM 3	peel / rind / husk / casing of seed or pod or fruit

-VMR- 'ENDOCRINE GLANDS OF THE BRAIN'	
STEM 1	hypothalamus
STEM 2	pituitary gland
STEM 3	pineal gland

-FSR- 'ENDOCRINE GLANDS OF THE BODY'	
STEM 1	thyroid gland
STEM 2	parathyroid gland
STEM 3	adrenal gland

-STY- 'ORIENTATIONAL BODY SURFACE'	
STEM 1	dorsal surface (i.e., the back [or topside] of a corporeal body)
STEM 2	ventral surface (i.e., the front [or underbelly] of a corporeal body)
STEM 3	lateral surface (i.e., the side/flank of a corporeal body)

-LŽ- 'FACE / VISAGE'	
STEM 1	face (as gestalt entity made up of the various facial quasi-planar surfaces plus the eyes, nose, lips, eyebrows, etc.)
STEM 2	face (as one's primary interface area for vocalization and display of emotion)
STEM 3	face/visage (as one's primary source/trait for recognition/identification)

-TÇM- 'EXCRETORY GLAND OR ORGAN'	
STEM 1	sweat gland; to sweat
STEM 2	bladder
STEM 3	sebaceous gland

-SKW- 'CENTRAL BODY SECTION/PART'	
STEM 1	chest, pectoral area
STEM 2	abdomen, midriff, waist, "stomach"/ "belly" / "tummy" area
STEM 3	trunk, thorax

-ḶP- ‘PRIMATE MAMMARY ORGAN/TISSUE’	
STEM 1	breast (of human or primate)
STEM 2	nipple
STEM 3	areola

-ḶPǍ- ‘NON-PRIMATE MAMMARY ORGAN/TISSUE’	
STEM 1	non-primate mammary organ (e.g., udder, dug)
STEM 2	nipple of non-primate mammary organ
STEM 3	areola or surrounding structure of nipple in non-primate mammary organ

-MḌ- ‘BODILY HAIR / FUR / FEATHER’	
STEM 1	a (single strand of) human body hair (other than thick scalp/facial/pubic/armpit hair)
STEM 2	a (single strand of) non-human bodily hair/fur/fleece
STEM 3	a(n avian) feather

-NḌ- ‘HUMAN-SPECIFIC HAIR’	
STEM 1	a (single strand of) human head hair (other than androgenic facial hair or eyebrows/eyelashes)
STEM 2	a (single strand of) human androgenic facial hair
STEM 3	a (single strand of) human armpit or pubic or eyebrow/eyelash hair (i.e,, thick body hair that stops growing at a certain length)

-ḌDR- ‘SPECIALTY HAIR-LIKE TISSUE’	
STEM 1	an eyelash hair [use appropriate configuration for “eyelashes”]
STEM 2	an eyebrow hair [use appropriate configuration for “eyebrow”]
STEM 3	a whisker/vibrissa

-NZG- ‘HARD KERATINOUS TISSUE’	
STEM 1	fingernail/toenail or claw
STEM 2	hoof
STEM 3	horn of animal

-ŇZM- ‘MUCOSAL TISSUE’	
STEM 1	inside surface of oral/naso-pharyngeal cavity
STEM 2	interior lining of gastro-intestinal tract
STEM 3	interior lining of bodily orifice (anus/rectum, vaginal canal, etc.)

-MZV- ‘MUSCLE / LIGAMENT / TENDON’	
STEM 1	muscle
STEM 2	ligament
STEM 3	tendon

-MZF- ‘OTHER BODILY TISSUE TYPE’	
STEM 1	fascia / membrane
STEM 2	aponeurosis
STEM 3	epicondyle

-ḌGḌ- ‘SMALL INTESTINE’	
STEM 1	jejunum
STEM 2	duodenum
STEM 3	ileum

-RGḌ- ‘LARGE INTESTINE / COLON’	
STEM 1	ascending colon
STEM 2	transverse colon
STEM 3	descending and sigmoid colon

-DŘ- ‘TOOTH / FANG / TUSK’	
STEM 1	tooth
STEM 2	fang
STEM 3	tusk

-GN- ‘NON-ORAL BODILY ORIFICE’	
STEM 1	nostril
STEM 2	ear canal
STEM 3	anus

-ŇW- ‘NECK’	
STEM 1	neck
STEM 2	neck (as bodily support/swivel point for head)
STEM 3	neck (as narrowest, most vulnerable body part)

-NXF- ‘ABDOMINAL TISSUE’	
STEM 1	peritoneum
STEM 2	mesentery
STEM 3	caecum

-DDV- ‘SPECIFIC TOOTH TYPE OTHER THAN FANG’	
STEM 1	incisor tooth
STEM 2	bicuspid
STEM 3	molar [concatenate this stem with -**PFW**- to render ‘premolar’]

-KKF- ‘HARDENED NON-LIVING BODILY TISSUE’	
STEM 1	callus
STEM 2	wart
STEM 3	corn

-LR- ‘WING’	
STEM 1	wing (avian/feathered)
STEM 2	wing (mammalian or reptilian, i.e., a membrane extending from arm-like appendages)
STEM 3	wing-like extension (mechanical, but still used for flying, e.g., wing of an airplane)

-PKL- ‘FIN / FLIPPER’	
STEM 1	fin (e.g., of a fish)
STEM 2	flipper (e.g., of a dolphin, seal, walrus, manatee, etc.)
STEM 3	fin- or flipper-like extension (mechanical)

-ŘC- ‘THROAT’	
STEM 1	throat (i.e., front/ventral portion of neck
STEM 2	upper portion of throat (i.e., the underside of the jaw)
STEM 3	lower portion of the throat (i.e., overlying the gullet/trachea/larynx)

-XBY- ‘PARTS OF A BODILY JOINT’	
STEM 1	(to be) the superior/dorsal side of a bodily joint [i.e., the “hard/bony” side of a bodily joint]*
STEM 2	(to be) the inferior/ventral side of a bodily joint [i.e., the soft, concave side of a bodily joint]*
STEM 3	(to be) the internal “swivel” mechanism of a bodily joint (i.e., the internal anatomy of a joint which allows an appendage to bend/twist*

*concatenate the stem of a specific limb or digit to specify ‘elbow’, ‘knee’, ‘knuckle (of finger)’, ‘knuckle (of toe), ‘shoulder’, etc.

-LDN- ‘GUM TISSUE / HARD PALATE	
STEM 1	gum tissue
STEM 2	alveolar ridge
STEM 3	hard palate [i.e., the area of hard flesh covering the roof of the mouth; for the underlying bone, see -**LTÇR**-]

-LPF- ‘INTERIOR TISSUES OF MOUTH’	
STEM 1	soft area between jaws underneath the front half of the tongue
STEM 2	soft palate
STEM 3	tonsil

-BF- ‘GENERIC REPRODUCTIVE ORGANS / CELLS’	
STEM 1	gonad (i.e., testicle, ovary)
STEM 2	gamete (i.e., ovum/egg cell, spermatozoon/sperm cell)
STEM 3	transport duct/tubule for gametes from gonad (i.e., Fallopian tube, vas deferens

-NM- ‘INTERNAL FEMALE REPRODUCTIVE ORGANS’	
STEM 1	uterus / womb
STEM 2	cervix
STEM 3	vagina

-GČ- ‘MALE REPRODUCTIVE ORGANS / TISSUES’	
STEM 1	penis
STEM 2	glans of penis
STEM 3	scrotum

-BDW- ‘EXTERNAL FEMALE REPRODUCTIVE ORGANS/TISSUES’	
STEM 1	clitoris
STEM 2	inner labium
STEM 3	outer labium

-LMW- PROBOSCIS / NOSE / SNOUT / BEAK	
STEM 1	nose [i.e., the visible external proboscis, not the nostrils or one's olfactory sense organ (see -**ŽG**-)]
STEM 2	snout, proboscis [mammalian, e.g., elephant trunk, tapir's snout, etc.]
STEM 3	beak

-JGŘ- 'ADIPOSE TISSUE (i.e., FAT)'	
STEM 1	adipose tissue / fat; have fat, be fat
STEM 2	blubber
STEM 3	cellulite

-NKW- 'TONGUE'	
STEM 1	tongue as organ for manipulation of food for eating and seat of taste buds
STEM 2	tongue as manipulative organ for the production of spoken language
STEM 3	proboscis [non-mammalian], i.e., tubular organ(s) for feeding/sensing as in insects, worms, molluscs, etc.)

-LLN- 'MEMBRANOUS TISSUE'	
STEM 1	eardrum, tympanic membrane
STEM 2	synovial membrane (interior lining of ariticular capsule of joint)
STEM 3	serous membrane (membrane lining a body cavity, e.g., peritoneal, pleural, pericardial membranes)

-MFP- 'LIP' *	
STEM 1	lip as aperture or "seal" to one's mouth
STEM 2	lip as manipulative organ for the production of spoken language
STEM 3	lip as body part used in puckering; to pucker

-JPH- 'CHEEK / CHIN / JAW'	
STEM 1	one's cheek (on face)
STEM 2	one's chin
STEM 3	one's jaw (i.e., lower portion of face that moves on hinge-like joint

* the stems of this root are almost always used with a DUPLEX configurations

-LPSL- 'MAJOR ARTERY' I	
STEM 1	aorta
STEM 2	coronary artery
STEM 3	pulmonary artery

-LPÇL- 'MAJOR ARTERY' II	
STEM 1	brachiocephalic artery
STEM 2	carotid artery
STEM 3	subclavian artery

-LPFL- 'MAJOR ARTERY' III	
STEM 1	iliac artery
STEM 2	obturator artery
STEM 3	genicular artery

-LBZL- 'MAJOR VEIN' I	
STEM 1	vena cava
STEM 2	hepatic vein
STEM 3	renal vein

-LBŽL- 'MAJOR VEIN' II	
STEM 1	brachiocephalic vein
STEM 2	jugular vein
STEM 3	azygos vein

-LĘL- 'MAJOR VEIN' III	
STEM 1	subclavian vein
STEM 2	axillary vein
STEM 3	cephalic vein

-LJL- 'MAJOR VEIN' IV	
STEM 1	iliac vein
STEM 2	adrenal vein
STEM 3	phrenic vein

-LDL- 'LYMPHATIC SYSTEM'	
STEM 1	lymph node
STEM 2	lymph vessel
STEM 3	thoracic duct

-FSGR- 'SALIVARY GLAND'	
STEM 1	parotid gland
STEM 2	sublingual gland
STEM 3	submandibular gland

-NKŢW- 'AVIAN DIGESTIVE ORGANS'	
STEM 1	crop
STEM 2	proventriculus
STEM 3	gizzard

-JTF- 'UROLOGICAL ORGAN'	
STEM 1	urethra
STEM 2	ureter
STEM 3	cloaca

-LPÇM- 'BLOOD CELL TYPE'	
STEM 1	erythrocyte
STEM 2	thrombocyte
STEM 3	leukocyte

-LPÇN- 'LEUKOCYTE TYPE'	
STEM 1	lymphocyte
STEM 2	granulocyte
STEM 3	monocyte

-CBR- 'SKIN ERUPTION'	
STEM 1	pimple, pustule
STEM 2	blister on skin
STEM 3	boil on skin

-GZDR- 'SKIN ANOMALY'	
STEM 1	a rash
STEM 2	mole, birthmark
STEM 3	scar

-JDL- 'LESION / ABRASION / WOUND'	
STEM 1	lesion
STEM 2	abrasion
STEM 3	wound

-JKŘ- 'ABNORMAL BODILY GROWTH'	
STEM 1	cyst
STEM 2	bodily nodule or amorphous malformation or placque
STEM 3	tumor

-PČ- 'MOUTH / GULLET / ORAL CAVITY'		
BSC	**Stem 1:** (to be) the mouth [of a chordate animal]	**Stem 2:** (to be) the gullet [of a non-chordate animal, plant, protist] **Stem 3:** (to be) the bodily apparatus/process by which an entity takes in external food
CTE	(to be) the oral cavity (i.e., the interior space of a mouth)	
CSV	(to be) the structural tissue(s) which form a mouth	
OBJ	(to be) that which enters or is placed within one's mouth	

-LCB- 'INNER EAR'	
STEM 1	cochlea
STEM 2	vestibule
STEM 3	semicircular canal

-LCBW- 'COCHLEAR DUCT MEMBRANES'	
STEM 1	basilar membrane
STEM 2	vestibular membrane
STEM 3	tectorial membrane

-LCBY- 'VESTIBULE OF EAR'	
STEM 1	otolith
STEM 2	saccule
STEM 3	utricle

-LCBR- 'STRUCTURES OF THE EAR'S SEMICIRCULAR CANAL'	
STEM 1	osseous ampula
STEM 2	ampullary cupola
STEM 3	hair cell

-PGW- 'THE FRONT OF THE EYE'	
STEM 1	cornea
STEM 2	lens
STEM 3	sclera

-PGY- 'UVEA OF THE EYE'	
STEM 1	iris
STEM 2	ciliary body
STEM 3	choroid

-PGL- 'PHOTORECEPTOR CELL OF THE EYE'	
STEM 1	cone cell
STEM 2	rod cell
STEM 3	intrinsically photosensitive retinal ganglion

-PGR- 'OTHER PARTS OF THE EYE'	
STEM 1	eyelid
STEM 2	lacrimal gland
STEM 3	lacrimal sac

-NXT- 'THE OUTER NOSE'	
STEM 1	ala (of the nostril area)
STEM 2	philtrum
STEM 3	nasal bridge

-NXN- 'NASAL CAVITY'	
STEM 1	nasal septum
STEM 2	nasal concha
STEM 3	nasal meatus

-RẔG- 'PARTS OF THE NASO-PHARYNGEAL CAVITY'	
STEM 1	sinus
STEM 2	tonsil
STEM 3	eustachian tube

-RẔGW- 'UPPER GASTRO-INTESTINAL TRACT'	
STEM 1	pharynx
STEM 2	epiglottis
STEM 3	esophagus

-RẔGY- 'LINGUISTIC STRUCTURES OF THE THROAT'	
STEM 1	larynx
STEM 2	vocal chord
STEM 3	glottis

-RẔGL- 'RESPIRATORY STRUCTURES'	
STEM 1	trachea
STEM 2	bronchus
STEM 3	pulmonary alveolus

-RẔGR- 'THE THROAT'	
STEM 1	hyoid bone
STEM 2	thyroid cartilage
STEM 3	cricoid cartilage

-LČP- 'AUDITORY OSSICLES'	
STEM 1	stapes
STEM 2	incus
STEM 3	malleus

7.2.1.1 Technical Terms for Various Bones, Muscles, and Tissue Types

-LTÇP- 'BONE TISSUE TYPE'	
STEM 1	(an amount/instance of) cortical bone tissue
STEM 2	(an amount/instance of) trabecular bone tissue
STEM 3	(an amount/instance of) myeloid bone tissue

-LTÇK- 'BONE CELL TYPE'	
STEM 1	osteocyte
STEM 2	chondrocyte
STEM 3	stromal cell

-LÇPW- 'TARSAL BONES' I	
STEM 1	calcaneus bone
STEM 2	talus bone
STEM 3	navicular bone

-LÇTW- 'TARSAL BONES' II	
STEM 1	metatarsal bone
STEM 2	cuneiform bone
STEM 3	cuboid bone

-LÇPY- 'PHALANGES OF FOOT OR HAND OR PAW'	
STEM 1	proximal phalanx
STEM 2	medial (or intermediate) phalanx
STEM 3	distal phalanx

-LÇTY- 'CARPAL BONES' I	
STEM 1	scaphoid bone
STEM 2	lunate bone
STEM 3	hamate bone

-LÇPL- 'LEG BONES'	
STEM 1	femur
STEM 2	tibia
STEM 3	fibula

-LÇTL- 'CARPAL BONES' II	
STEM 1	metacarpal bone
STEM 2	trapezium, trapezoid, or capitate bone
STEM 3	triquetum bone

-LÇPR- 'SESAMOID BONES'	
STEM 1	patella
STEM 2	fetlock
STEM 3	pisiform

-LÇTR- 'HIP STRUCTURES'	
STEM 1	acetabulum
STEM 2	labrum
STEM 3	pubic symphysis

-LÇPŘ- 'HIP BONES'	
STEM 1	ilium
STEM 2	ischium
STEM 3	pubis

-LÇTŘ- 'AVIAN HIP BONES'	
STEM 1	iliofemoralis
STEM 2	iliotibialis
STEM 3	iliofibularis

-LÇFW- ‘VERTEBRAE’	
STEM 1	thoracic vertebra
STEM 2	lumbar vertebra
STEM 3	cervical vertebra

-LÇFY- ‘TAIL BONES’	
STEM 1	sacrum
STEM 2	coccygial vertebra (=> coccyx)
STEM 3	pygostyle

-LÇFL- ‘THORACIC BONES’	
STEM 1	true rib
STEM 2	false rib
STEM 3	floating rib

-LÇFR- ‘SHOULDER BONES’	
STEM 1	clavicle
STEM 2	scapula plus acromion
STEM 3	caracoid

-LÇFŘ- ‘OTHER ABDOMINAL BONE’	
STEM 1	furcula/wishbone
STEM 2	keel/carina
STEM 3	cleithrum

-LÇFM- ‘STERNUM’	
STEM 1	body of the sternum
STEM 2	manubrium
STEM 3	xyphoid process

-LÇFN- ‘ARM BONES’	
STEM 1	humerus
STEM 2	radius
STEM 3	ulna

-LTÇG- ‘JAW BONES’ I	
STEM 1	mandible/dentary bone
STEM 2	angular bone
STEM 3	surangular bone

-LTÇB- ‘JAW BONES’ II	
STEM 1	articular bone
STEM 2	splenial bone
STEM 3	gular bone

-LTÇD- ‘CHEEK BONES’	
STEM 1	zygomatic/jugal bone
STEM 2	squamosal/pterotic bone
STEM 3	quadrate bone

-LTÇW- ‘SKULL BONES’ I	
STEM 1	temporal bone
STEM 2	sphenoid bone
STEM 3	ethmoid

-LTÇL- ‘SKULL BONES’ II	
STEM 1	vomer bone
STEM 2	palpebral
STEM 3	sclerotic ring

-LTÇR- ‘SKULL BONES’ III	
STEM 1	hard palate [i.e., the bone underlying the tissue covering the hard palate]
STEM 2	maxilla
STEM 3	palatine

-LTÇV- ‘SKULL BONES’ IV	
STEM 1	frontal bone
STEM 2	parietal bone
STEM 3	occipital bone

-LTÇN- ‘NOSE BONES’	
STEM 1	nasal bone
STEM 2	lacrimal bone
STEM 3	narial bone

-LTÇM- ‘CONNECTIVE TISSUE TYPE’	
STEM 1	(an amount/instance of) enthesis
STEM 2	bone fiber
STEM 3	periodontium

-MZN- ‘MUSCLE TISSUE TYPE’	
STEM 1	(an amount/instance of) skeletal muscle
STEM 2	(an amount/instance of) smooth muscle
STEM 3	(an amount/instance of) cardiac muscle

-MZŇ- ‘MUSCLE CELL TYPE’	
STEM 1	myocyte
STEM 2	sarcomere
STEM 3	sarcolemma

-MZNW- ‘MUSCLE FUNCTION TYPE’	
STEM 1	muscle as flexor type
STEM 2	muscle as extensor
STEM 3	muscle as abductor or adductor

-MZNY- ‘FORM OF MUSCLE FIBER’	
STEM 1	parallel muscle fiber
STEM 2	fusiform muscle fiber
STEM 3	pennate muscle fiber

-MZŇW- 'ADDITIONAL MUSCLE FIBER FORM PLUS RETINACULUM	
STEM 1	convergent muscle fiber
STEM 2	circular (i.e., sphincter-like) muscle fiber
STEM 3	retinaculum

-MZY- 'MUSCLE FASCICLE TISSUE'	
STEM 1	(an amount/instance of) epimysium
STEM 2	(an amount/instance of) perimysium
STEM 3	(an amount/instance of) endomysium

-GŽP- 'NEURAL TISSUE'	
STEM 1	(an amount/instance of) epineurium
STEM 2	(an amount/instance of) perineurium
STEM 3	(an amount/instance of) endoneurium

Muscles, Ligaments and Other Tissues/Structures

The following roots/stems name the various muscles and ligaments of the human and/or mammalian body. In a few instances, important tissues other than muscles and ligaments (e.g., fascia, retinacula, membranes, and tendons) have been assigned stems as well. For those non-bone or non-muscle tissues not assigned a stem, concatenate the stem for the particular tissue-type (e.g., tendon, fascia/membrane, retinaculum, enthesis) with the stem for the bone, muscle, or organ to which the tissue is connected or associated. DYN Function refers to muscle contraction or ligament tension/pulling.

-MZX- 'LATERAL FOOT MUSCLE'	
STEM 1	abductor digiti minimi
STEM 2	flexor digiti minimi brevis
STEM 3	opponens digiti minimi

-MZXW- 'MIDDLE FOOT MUSCLE'	
STEM 1	lumbrical muscle
STEM 2	plantar interossei
STEM 3	dorsal interossei

-MZXL- 'MEDIAL FOOT MUSCLE'	
STEM 1	abductor hallucis
STEM 2	flexor hallucis brevis
STEM 3	adductor hallucis

-MZXR- 'DORSAL FOOT MUSCLE'	
STEM 1	extensor hallucis brevis
STEM 2	extensor digitorum brevis
STEM 3	flexor digitorum brevis

-MZFW- 'MEDIAL HAND (VOLAR) MUSCLE'	
STEM 1	abductor digiti minimi
STEM 2	flexor digiti minimi brevis
STEM 3	opponens digiti minimi

-MZFY- 'INTERMEDIATE HAND (VOLAR) MUSCLE'	
STEM 1	lumbrical muscle
STEM 2	palmar interossei
STEM 3	dorsal interossei

-MZFL- 'LATERAL/THENAR HAND (VOLAR) MUSCLE'	
STEM 1	abductor pollicis brevis
STEM 2	flexor pollicis brevis
STEM 3	opponens pollicis

-MZFR- 'OTHER HAND OR FOOT MUSCLE'	
STEM 1	palmaris brevis
STEM 2	adductor pollicis
STEM 3	quadratus plantae

-MZTW- 'LATERAL CALF MUSCLE'	
STEM 1	peroneus longus
STEM 2	peroneus brevis
STEM 3	peroneus/fibularis tertius

-MZTY- 'ANTERIOR CALF MUSCLE'	
STEM 1	tibialis anterior
STEM 2	extensor digitorum longus
STEM 3	fibularis longus

-MZTL- 'OUTER POSTERIOR CALF MUSCLE'	
STEM 1	gastrocnemius
STEM 2	soleus
STEM 3	plantaris

-MZTR- 'INNER POSTERIOR CALF MUSCLE'	
STEM 1	tibialis posterior
STEM 2	flexor hallucis longus
STEM 3	flexor digitorum longus

-MZPW- 'FOREARM MUSCLE' I	
STEM 1	flexor carpi radialis
STEM 2	flexor pollicis longus
STEM 3	flexor digitorum profundus

-MZPY- 'FOREARM MUSCLE' II	
STEM 1	flexor carpi ulnaris
STEM 2	flexor digitorum superficialis
STEM 3	supinator

-MZPL- ‘FOREARM MUSCLE’ III	
STEM 1	palmaris longus
STEM 2	pronator teres
STEM 3	pronator quadratus

-MZPR- ‘FOREARM MUSCLE’ IV	
STEM 1	extensor carpi radialis longus
STEM 2	extensor carpi radialis brevis
STEM 3	extensor digitorum

-MZPF- ‘FOREARM MUSCLE’ V	
STEM 1	brachioradialis
STEM 2	extensor digiti minimi
STEM 3	extensor carpi ulnaris

-MZPȚ- ‘FOREARM MUSCLE’ VI	
STEM 1	extensor pollicis
STEM 2	abductor pollicis longus
STEM 3	extensor indicis

-MZKW- ‘ANTERIOR THIGH MUSCLE’ I	
STEM 1	vastus lateralis
STEM 2	vastus medialis
STEM 3	vastus intermedius

-MZKY- ‘ANTERIOR THIGH MUSCLE’ II	
STEM 1	rectus femoris
STEM 2	articularis genus / subcrureus
STEM 3	tensor vastus intermedius

-MZKL- ‘ANTERIOR THIGH MUSCLE’ III	
STEM 1	sartorius
STEM 2	tensor fascia latae
STEM 3	gracilis muscle

-MZKR- ‘INNER THIGH/GROIN MUSCLE’	
STEM 1	adductor brevis
STEM 2	adductor longus
STEM 3	adductor magnus

-MZKŘ- ‘UPPER THIGH MUSCLE’	
STEM 1	pectineus muscle
STEM 2	obturator externus
STEM 3	obturator internis

-MZKÇ- ‘LATERAL ROTATOR MUSCLE’	
STEM 1	piriformis/caudofemoralis
STEM 2	superior gemellus
STEM 3	inferior gemellus

-MZKH- 'OTHER THIGH AND HIP MUSCLE/TISSUE'	
STEM 1	quadratus femoris
STEM 2	adductor minimus
STEM 3	iliotibial tract / iliotibial band

-MZKF- 'POSTERIOR KNEE MUSCLE/LIGAMENT'	
STEM 1	popliteus muscle
STEM 2	poplitofibular ligament
STEM 3	patellar ligament (patellar tendon)

-MZKŢ- 'HAMSTRING'	
STEM 1	biceps femoris
STEM 2	semimembranosus
STEM 3	semitendinosus

-MVG- 'ARM MUSCLE' I	
STEM 1	biceps brachii
STEM 2	brachialis anticus
STEM 3	articularis cubiti

-MVGL- 'ARM MUSCLE' II	
STEM 1	triceps brachii (long head)
STEM 2	triceps brachii (medial head)
STEM 3	triceps brachii (lateral head)

-MVGR- 'ARM MUSCLE' III	
STEM 1	coracobrachialis
STEM 2	anconeus
STEM 3	epicondyle of humerus

-MVGW- 'OTHER ARM TISSUE'	
STEM 1	tendon of biceps brachii
STEM 2	bicipital aponeurosis
STEM 3	antebrachial fascia

-MŽKW- 'GLUTEAL MUSCLE'	
STEM 1	gluteus maximus
STEM 2	gluteus medius
STEM 3	gluteus minimus

-MŽKY- 'ADDITIONAL HIP-RELATED TISSUE'	
STEM 1	iliotibial tract
STEM 2	ciribriform fascia
STEM 3	tensor fascia of lata

-MŽKL- ‘ILIOPSOAS MUSCLE’	
STEM 1	psoas major
STEM 2	psoas minor
STEM 3	iliacus

-MŽKR- ‘PELVIC (LEVATOR ANI) MUSCLE’	
STEM 1	pubococcygeus
STEM 2	iliococcygeus
STEM 3	puborectalis

-MŽKŘ- ‘PERINEAL MUSCLE’ I	
STEM 1	ischiocavernosus muscle
STEM 2	bulbospongiosus muscle
STEM 3	transverse perineal muscle

-MŽKÇ- ‘PERINEAL MUSCLE’ II	
STEM 1	anal sphincter
STEM 2	urethral sphincter
STEM 3	coccygeus

-MŽKF- ‘ERECTOR SPINAE MUSCLE’	
STEM 1	spinalis
STEM 2	longissiumus
STEM 3	iliocostalis

-MŽKŢ- ‘SEMISPINALIS MUSCLE’	
STEM 1	semispinalis capitis
STEM 2	cervicis
STEM 3	thoracis

-MŽKH- ‘OTHER PELVIC/UROGENITAL TISSUE’	
STEM 1	cremaster muscle
STEM 2	rectovaginal/rectoprostatic fascia
STEM 3	pubovesical ligament

-LŽBW- ‘VERTABRAL CONNECTIVE TISSUE’	
STEM 1	interspinalis muscle
STEM 2	interspinous ligament
STEM 3	supraspinous ligament

-LŽBY- ‘TRANSVERSOSPINALIS MUSCLE’	
STEM 1	multifidus
STEM 2	rotatores
STEM 3	intertransversarius

-LŽBL- ‘NECK MUSCLES’	
STEM 1	splenius capitus
STEM 2	spenius cervicis
STEM 3	levator scapulae

-LŽBR- 'BACK MUSCLES' I	
STEM 1	trapezius
STEM 2	latissimus dorsi
STEM 3	rhomboid muscle

-LŽBŘ- 'BACK MUSCLES' II	
STEM 1	serratus posterior inferior
STEM 2	serratus posterior superior
STEM 3	serratus anterior

-LŽGW- 'THORACIC CAVITY'	
STEM 1	pectoralis major
STEM 2	pectoralis minor
STEM 3	subclavius

-LŽGY- 'INNER CHEST MUSCLE' I	
STEM 1	diaphragm
STEM 2	levatores costarum
STEM 3	quadratus lumborum

-LŽGL- 'INNER CHEST MUSCLE' II	
STEM 1	intercostal muscle
STEM 2	subcostalis
STEM 3	transversus thoracis

-LŽGR- 'ABDOMINAL MUSCLE'	
STEM 1	rectus abdominis
STEM 2	transversus abdominis
STEM 3	pyramidalis

-LŽGŘ- 'ABDOMINAL WALL'	
STEM 1	abdominal external oblique
STEM 2	internal oblique
STEM 3	transversalis

-LŽGV- 'THORACIC FASCIA'	
STEM 1	pectoral fascia
STEM 2	clavipectoral fascia
STEM 3	thoracolumbar fascia

-LŽDW- 'SHOULDER MUSCLE'	
STEM 1	deltoid
STEM 2	teres major
STEM 3	teres minor

-LŽDY- 'ROTATOR CUFF'	
STEM 1	supraspinatus
STEM 2	infraspinatus
STEM 3	subscapularis

-LŽDR- 'SHOULDER FASCIA'	
STEM 1	supraspinous fascia
STEM 2	infraspinus fascia
STEM 3	deltoid fascia

-LŽDŘ- 'UPPER ARM FASCIA'	
STEM 1	brachial fascia
STEM 2	antebrachial fascia
STEM 3	axillary fascia

-MŽT- 'NECK MUSCLE' I	
STEM 1	longus colli
STEM 2	longus capiti
STEM 3	rectus capitis anterior

-MŽTW- 'NECK MUSCLE' II	
STEM 1	scalene muscle
STEM 2	obliquus capitis
STEM 3	rectus capitis posterior

-MŽTY- 'NECK MUSCLE' III	
STEM 1	platysma
STEM 2	sternocleidomastoid
STEM 3	rectus capitis lateralis

-MŽTL- 'THROAT MUSCLE' I	
STEM 1	digastric
STEM 2	mylohyoid
STEM 3	geniohyoid

-MŽTR- 'THROAT MUSCLE' II	
STEM 1	omohyoid
STEM 2	thyrohyoid
STEM 3	stylohyoid

-MŽTŘ- 'THROAT MUSCLE' III	
STEM 1	sternohyoid
STEM 2	sternothyroid
STEM 3	cricothyroid

-MŽP- 'LARYNX'	
STEM 1	arytenoid
STEM 2	thyroarytenoid
STEM 3	cricoarytenoid

-MŽPW- 'PHARYNGEAL/TRACHEAL MUSCLE' I	
STEM 1	inferior pharyngeal constrictor
STEM 2	medial pharyngeal constrictor
STEM 3	superior pharyngeal constrictor

-MŽPY- 'PHARYNGEAL/TRACHEAL MUSCLE' II	
STEM 1	stylopharyngeus
STEM 2	salpingopharyngeus
STEM 3	trachealis

-MŽPL- 'TONGUE MUSCLE & TISSUE' I	
STEM 1	styloglossus
STEM 2	hyoglossus
STEM 3	genioglossus

-MŽPR- 'TONGUE MUSCLE& TISSUE' II	
STEM 1	superior longitudinal muscle
STEM 2	inferior longitudinal muscle
STEM 3	lingual mucosa

-MŽPŘ- 'TONGUE MUSCLE & TISSUE' III	
STEM 1	vertical muscle of the tongue
STEM 2	transverse muscle of the tongue
STEM 3	septum of the tongue

-MŽPF- 'SOFT PALATE' I	
STEM 1	tensor veli palatini
STEM 2	levator veli palatini
STEM 3	palatine aponeurosis

-MŽPŢ- 'SOFT PALATE' II	
STEM 1	musculus uvulae (uvula)
STEM 2	palatopharyngeus
STEM 3	palatoglossus

-MŽM- 'MASTICATORY MUSCLE'	
STEM 1	lateral pterygoid
STEM 2	medial pterygoid
STEM 3	masseter

-MŽMW- 'UPPER ORAL CAVITY'	
STEM 1	zygomaticus
STEM 2	levator labii superioris
STEM 3	levator anguli oris

-MŽMY- 'INNER ORAL CAVITY'	
STEM 1	orbicularis oris
STEM 2	buccinator
STEM 3	risorius

-MŽŇ- 'CHIN MUSCLE'	
STEM 1	mentalis
STEM 2	depressor labii inferioris / quadratus labii inferioris
STEM 3	depressor anguli oris

-MŽN- 'NASAL MUSCLE' I	
STEM 1	procerus
STEM 2	anomalous nasi
STEM 3	levator labii superior alaeque nasi

-MŽNW- 'NASAL MUSCLE' II	
STEM 1	alar nasalis
STEM 2	transverse nasalis
STEM 3	compressor narium minor

-MŽNY- 'NASAL MUSCLE' III	
STEM 1	dilator naris anterior
STEM 2	dilator naris posterior
STEM 3	depressor septi nasi

-MÇF- 'EYELID MUSCLE'	
STEM 1	corrugator supercilii
STEM 2	orbicularis oculi
STEM 3	superior tarsal

-MÇFW- 'EYEBALL MUSCLE' I	
STEM 1	superior rectus
STEM 2	inferior rectus
STEM 3	medial or lateral rectus

-MÇFY- 'EYEBALL MUSCLE' II	
STEM 1	levator palpebrae superioris
STEM 2	superior oblique
STEM 3	inferior oblique

-MÇFL- 'HEAD MUSCLE & TISSUE' I	
STEM 1	frontalis
STEM 2	occipitalis
STEM 3	temporalis

-MÇFR- 'HEAD MUSCLE & TISSUE' II	
STEM 1	temporal fascia
STEM 2	masseteric fascia
STEM 3	temporoparietalis

-MÇFŘ- 'EAR MUSCLE'	
STEM 1	auricularis superior
STEM 2	auricularis anterior
STEM 3	auricularis posterior

-PKW- 'FOOT RETINACULUM'	
STEM 1	superior extensor retinaculum
STEM 2	inferior extensor retinaculum
STEM 3	other retinaculum of foot

-PKHW- 'OTHER RETINACULA OF THE FOOT AND KNEE'	
STEM 1	fibular/peroneous retinaculum
STEM 2	lateral retinaculum
STEM 3	medial patellar retinaculum

-PKY- 'OTHER FOOT TISSUE'	
STEM 1	calcaneal (Achilles) tendon
STEM 2	plantar fascia
STEM 3	arch of foot

-VZKW- 'FOOT LIGAMENT' I	
STEM 1	plantar interphalangeal ligament
STEM 2	plantar metatarsophalangeal ligament
STEM 3	plantar tarsometatarsal ligament

-VZKY- 'FOOT LIGAMENT' II	
STEM 1	collateral interphalangeal ligament
STEM 2	collateral metatarsophalangeal ligament
STEM 3	collateral tarsometatarsal ligament

-VZKL- 'FOOT LIGAMENT' III	
STEM 1	plantar intermetatarsal/metatarsal ligament
STEM 2	superficial transverse intermetatarsal/metatarsal ligament
STEM 3	deep transverse intermetatarsal/metatarsal ligament

-VZKR- 'FOOT LIGAMENT' IV	
STEM 1	dorsal intermetatarsal/metatarsal ligament
STEM 2	interosseous intermetatarsal/metatarsal ligament
STEM 3	interosseous intercuneiform ligament

-VZKŘ- 'FOOT LIGAMENT' V	
STEM 1	plantar cuneonavicular ligament
STEM 2	plantar cuboideonavicular ligament
STEM 3	plantar intercueniform ligament

-VZKÇ- 'FOOT LIGAMENT' VI	
STEM 1	dorsal cuneonavicular ligament
STEM 2	dorsal cuboideonavicular ligament
STEM 3	dorsal intercueniform ligament

-VZKF- 'FOOT LIGAMENT' VII	
STEM 1	plantar calcaneonavicular/spring ligament
STEM 2	dorsal talonavicular ligament
STEM 3	bifurcated (calcaneonavicular) ligament

-VZKŢ- 'FOOT LIGAMENT' VIII	
STEM 1	plantar or long plantar calcaneocuboid ligament
STEM 2	dorsal calcaneocuboid ligament
STEM 3	bifurcated (calcaneocuboid) ligament

-VZKV- ‘FOOT LIGAMENT’ IX	
STEM 1	anterior talocalcaneal ligament
STEM 2	lateral talocalcaneal ligament
STEM 3	interosseous talocalcaneal ligament

-VZKM- ‘FOOT LIGAMENT’ X	
STEM 1	posterior talocalcaneal ligament
STEM 2	medial talocalcaneal ligament
STEM 3	tibiocalcaneal ligament

-VZKN- ‘FOOT LIGAMENT’ XI	
STEM 1	anterior tibiotalar ligament
STEM 2	posterior tibiotalar ligament
STEM 3	tibionavicular ligament

-VZKH- ‘FOOT LIGAMENT’ XII	
STEM 1	anterior talofibular ligament
STEM 2	posterior talofibular ligament
STEM 3	calcaneofibular ligament

-VZXM- ‘LOWER LEG LIGAMENT’	
STEM 1	anterior tibiofibular ligament
STEM 2	posterior tibiofibular ligament
STEM 3	interosseous membrane of the leg

-VZXN- ‘LOWER LEG AND KNEE LIGAMENT’	
STEM 1	anterior ligament of the head of the fibula
STEM 2	posterior ligament of the head of the fibula
STEM 3	articular capsule of the knee / capsular ligament

-KŢC- ‘KNEE LIGAMENT’ I	
STEM 1	anterior cruciate ligament / cranial cruciate ligament
STEM 2	medial meniscus
STEM 3	transverse/anterior meniscomeniscal ligament

-KŢCW- ‘KNEE LIGAMENT’ II	
STEM 1	posterior cruciate ligament / caudal cruciate ligament
STEM 2	lateral meniscus
STEM 3	anterolateral ligament

-KŢCY- ‘KNEE LIGAMENT’ III	
STEM 1	anterior meniscofemoral ligament
STEM 2	medial/tibial collateral ligament
STEM 3	oblique popliteal ligament

-KŢCM- ‘KNEE LIGAMENT’ IV	
STEM 1	posterior meniscofemoral ligament
STEM 2	fibular/lateral collateral ligament
STEM 3	arcuate popliteal ligament

-KŢCN- ‘OTHER KNEE TISSUE’	
STEM 1	bursa of the knee
STEM 2	synovial membrane of the knee
STEM 3	intrapatellar fat pad

-KSLW- ‘HIP LIGAMENT’ I	
STEM 1	iliofemoral ligament / Y-ligament
STEM 2	pubofemoral ligament
STEM 3	ischiofemoral ligament

-KSLY- ‘HIP LIGAMENT’ II	
STEM 1	head of the femur ligament
STEM 2	transverse acetabular ligament
STEM 3	acetabular labrum

-PŠLW- ‘OTHER HIP TISSUE’	
STEM 1	articular capsule (capsicular ligament) of the hip
STEM 2	zona orbicularis / annular ligament
STEM 3	acetabulum / cotyloid cavity / hip socket

-KSŢ- ‘GROIN LIGAMENT’	
STEM 1	inguinal ligament
STEM 2	lacunar ligament
STEM 3	pectineal ligament

-PŢL- ‘UTERO-OVARIAN LIGAMENT’	
STEM 1	ovarian ligament /utero-ovarian ligament / proper ovarian ligament
STEM 2	suspensory ligament of the ovary
STEM 3	round ligament of the uterus

- - ‘PELVIC LIGAMENT’	
STEM 1	anterior sacroiliac ligament
STEM 2	posterior sacroiliac ligament
STEM 3	interosseous sacroiliac ligament

- - ‘PELVIC LIGAMENT’	
STEM 1	superior pubic ligament
STEM 2	inferior pubic ligament
STEM 3	sacrotuberous ligaent

-PŠLY- ‘OTHER PELVIC TISSUE’	
STEM 1	sacrospinous ligament
STEM 2	obturator membrane
STEM 3	obturator canal

-BZLW- ‘THORACO-VERTEBRAL LIGAMENT’ I	
STEM 1	anterior sacrococcygeal ligament
STEM 2	posterior sacrococcygeal ligament
STEM 3	intra-articular ligament

-GZLW- 'THORACO-VERTEBRAL LIGAMENT' II	
STEM 1	costotransverse ligament
STEM 2	lumbocostal ligament
STEM 3	radiate ligament

-ẒLW- 'THORACO-VERTEBRAL LIGAMENT' III	
STEM 1	intra-articular sternocostal ligament
STEM 2	radiate sternocostal ligament
STEM 3	costoxiphoid ligament

-BZLY- 'VERTEBRAL LIGAMENT' I	
STEM 1	iliolumbar ligament
STEM 2	anterior atlanto-axial ligament
STEM 3	posterior atlanto-axial ligament

-GZLY- 'VERTEBRAL LIGAMENT' II	
STEM 1	cruciate ligament of atlas (transverse ligament of atlas)
STEM 2	alar ligament
STEM 3	apical ligament of dens

-ẒLY- 'VERTEBRAL LIGAMENT' III	
STEM 1	anterior longitudinal ligament
STEM 2	posterior longitudinal ligament
STEM 3	ligamenta flava

-JLY- 'VERTEBRAL LIGAMENT' IV	
STEM 1	supraspinous (nuchal) ligament
STEM 2	interspinous ligament
STEM 3	intertransverse ligament

-JLW- 'SPINAL (INTERVERTEBRAL) DISC'	
STEM 1	anulus fibrosus
STEM 2	nucleus pulposus
STEM 3	tectorial membrane of atlanto-axial joint

-FSLW- 'ATLANTO-OCCIPITAL JOINT LIGAMENT'	
STEM 1	atlanto-occipital articular capsule
STEM 2	anterior atlanto-occipital membrane
STEM 3	posterior atlanto-occipital membrane

-FSLY- 'TEMPOROMANDIBULAR JOINT TISSUE'	
STEM 1	capsule and articular disk of temporo-mandibular joint
STEM 2	lateral temporomandibular ligament
STEM 3	medial temporomandibular ligament (including sphenomandibular and stylomandibular ligament)

-KŢTW- 'HAND LIGAMENT' I	
STEM 1	collateral interphalangeal ligament
STEM 2	collateral metacarpophalangeal ligament
STEM 3	deep transverse metacarpal ligament

-KŢTY- 'HAND LIGAMENT' II	
STEM 1	palmar interphalangeal ligament
STEM 2	palmar metacarpophalangeal ligament
STEM 3	superficial transverse metacarpal ligament

-KŢTL- 'HAND LIGAMENT' III	
STEM 1	dorsal carpometacarpal ligament
STEM 2	radial collateral ligament
STEM 3	dorsal inercarpal ligament

-KŢTR- 'HAND LIGAMENT' IV	
STEM 1	palmar carpometacarpal ligament
STEM 2	ulnar collateral ligament
STEM 3	palmar intercarpal ligament

-KŢTŘ- 'HAND LIGAMENT' V	
STEM 1	radiate carpal ligament
STEM 2	interosseous intercarpal ligament
STEM 3	scapholunate ligament

-KŢTÇ- 'HAND LIGAMENT' VI	
STEM 1	dorsal radiocarpal ligament
STEM 2	dorso ulnocarpal ligament
STEM 3	volar plate

-KŢTH- 'OTHER HAND/WRIST TISSUE' I	
STEM 1	palmar radiocarpal ligament
STEM 2	palmar ulnocarpal ligament
STEM 3	pisometacarpal ligament

-PŢTW- 'OTHER HAND/WRIST TISSUE' II	
STEM 1	pisohamate ligament
STEM 2	flexor retinaculum of the hand
STEM 3	extensor retinaculum of the hand

-RŢTH- 'OTHER HAND/WRIST TISSUE' III	
STEM 1	carpal tunnel
STEM 2	ulnar canal / Guyon's canal
STEM 3	hook of the hamate bone

-PŢTY- 'FOREARM LIGAMENT'	
STEM 1	dorsal radio-ulnar ligament
STEM 2	palmar radio-ulnar ligament
STEM 3	interosseous membrane of the forearm

-PŢTL- 'ELBOW LIGAMENT' I	
STEM 1	anular ligament
STEM 2	oblique cord
STEM 3	quadrate ligament / ligament of Denucé

-PŢTR- 'ELBOW LIGAMENT' II	
STEM 1	radial collateral ligament
STEM 2	ulnar collateral ligament
STEM 3	capsule of elbow joint

-PŢTŘ- 'SHOULDER LIGAMENT' I	
STEM 1	anterior sternoclavicular ligament
STEM 2	posterior sternoclavicular ligament
STEM 3	interclavicular ligament

-PŢTḶ- 'SHOULDER LIGAMENT' II	
STEM 1	costoclavicular ligament
STEM 2	coraco-acromial ligament
STEM 3	acromioclavicular ligament

-PŢTÇ- 'SHOULDER LIGAMENT' III	
STEM 1	superior transverse scapular ligament
STEM 2	inferior transverse scapular ligament
STEM 3	coracoclavicular ligament

-PŢTH- 'SHOULDER LIGAMENT' IV	
STEM 1	superior glenohumeral ligament
STEM 2	middle glenohumeral ligament
STEM 3	inferior glenohumeral ligament

-PŢTHW- 'SHOULDER LIGAMENT' V	
STEM 1	coracohumeral ligament
STEM 2	transverse humeral ligament
STEM 3	capsule of shoulder joint (including glenoid labrum)

-PÇL- 'OTHER BODILY TISSUE' I	
STEM 1	tendinous intersction / tendinous inscription
STEM 2	linea alba
STEM 3	linea semilunaris

- PÇR- 'OTHER BODILY TISSUE' II	
STEM 1	xiphoid process / ensiform process / xiphisternum / metasternum
STEM 2	parametrium
STEM 3	perimetrium

7.2.2 Bodily Fluids, Secretions, Excretia

While many bodily fluids/secretions/excretia can be named by either utilizing the **ROC**/8 affix with an existing stem naming a bodily process (e.g., urination, ejaculation, etc.) or by concatenating -**TKL**- Stem 2 or -**RTKL**- Stem 1 into a stem naming a body part, it is nevertheless convenient to have lexical stems for the most common of such fluids/secretions/excretia.

Keep in mind that, while the English translations of the following stems are almost all non-count/mass nouns, the Ithkuil stems are countable, each meaning "(to be) an amount of" the particular substance/fluid in question. The stems below have the following Specification pattern.

BSC	**Stem 1:** (to be) a state/act/process of (the body) producing/having (a volume/amount of) the particular bodily fluid/secretion/excretia
CTE	(to be) an amount of the particular bodily fluid/secretion/excretia
CSV	(to be) a state/act/process of secreting/excreting an amount of the particular bodily fluid; to secrete/excrete an amount of the particular bodily fluid/secretion/excretia
OBJ	(to be) the organ or gland which secretes the particular bodily fluid/secretion/excretia

-**LPL**- 1. blood 2. blood plasma 3. lymph

-**TÇN**- 1. sweat 2. saliva 3. sebum

-**NGM**- 1. breast milk 2. vaginal fluid 3. amniotic fluid

-**ŇZR**- 1. earwax 2. perilymph 3. synovial fluid

-**GPH**- 1. feces 2. urine 3. pus

-ŇZL- 1. mucus 2. phlegm/sputum 3. rheum

-**NXM**- 1. tears 2. vitreous humor 3. aqueous humor

-**KŢČ**- 1. semen 2. bile/gall 3. gastric acid/stomach acid

-**FŢŘ**- 1. cerebrospinal fluid 2. peritoneal fluid 3. pericardial fluid

7.2.3 Disability/Disease/Sickness/Illness/Malady

-ÇK- 'ILLNESS / SICKNESS / DISEASE / MALADY / AILMENT / SYMPTOM'			
	Stem 1	**Stem 2**	**Stem 3**
BSC	(to be) a state of physical or mental illness/sickness malady; to be physically or mentally ill/sick, to suffer an illness/disease/ ailment/sickness/malady	(to be) a state of manifesting/displaying a symptom of an illness/disease/ailment/sickness/malady; to show/manifest/display a symptom of an illness/ailment/sickness/malady	(to be) a state of physical or mental incapacity or impairment or weakness or decrepitude due to an illness/disease/ailment/sickness/malady
CTE	(to be) a disease/illness/ailment/malady	(to be) a symptom an illness/disease/ailment/sickness/malady	(to be) the particular impairment(s) one suffers or experiences due to an illness/ailment/sickness/malady
CSV	(to be) the state of physically suffering from a disease/illness/ailment/malady; to suffer therefrom	(to be) the physical act of manifesting a symptom; for a symptom to physically occur	(to be) a state of having to deal/struggle with or adapt to a physical or mental impairment due to an illness/disease/ailment/sickness/malady
OBJ	(to be) a living being infected/suffering from an infectious illness/disease; entity/party that is sick	(to be) a living being suffering/manifesting/displaying a symptom of an illness/disease/ailment/sickness/malady	(to be) a living being experiencing/suffering from a physical or mental incapacity or impairment or weakness or decrepitude due to an illness/disease/ailment/sickness/malady

-KÇK- 'PHYSICAL DISABILITY DUE TO INJURY'		
BSC	**Stem 1:** (to be) a state of living with a static/permanent physical or mental disability (i.e., not disease-related, e.g., missing a limb, paralysis due to a war injury, blindness due to an accident, etc.)	**Stem 2:** (to be) a state of living with a temporary physical or mental disability (i.e., not disease-related, e.g., a broken leg, blindness while recovering from eye surgery, etc.) **Stem 3:** (to be) a state of using/living with a prosthetic or adaptive device or process or routine to compensate for a physical disability
CTE	(to be) a state of physical disability (i.e., not having all of one's poential abilities)	
CSV	(to be) a state of suffering/enduring/dealing with the effect(s) of a disability	
OBJ	(to be) the disability itself	

The following roots each denote a disease category or general disease type. Their stems and Specifications are modeled on the root -**ÇK**- above:

-**MŠK**- infectious (i.e., microbial) illness/disease

-**MŠKW**- disease/malady caused by exposure to adverse environmental element(s)

-**MŠKY**- disease /malady caused by exposure to toxic chemical substance/poison

-**MŠKL**- disease/malady caused by adverse personal habits (e.g., smoking, diet, sloth, etc.)

-**MŠKR**- metabolic disorder/malady

-**MŠKŘ**- neoplastic disorder/malady (i.e., cancer)

-**MŠKF**- genetic disorder/malady

-**MŠKŢ**- ideopathic disorder/malady

-**MŠKÇ**- autoimmune disorder

-**MŠKV**- malady caused by complications from an injury or injuries

-**MŠKM**- iatrogenic or nosocomial disorder/malady caused by medical/medicinal/surgical treatment or hospitalization

-**MŠKN**- prion-based disease

-**MŠKH**- congenital defect

-**MŠKHW**- disease/malady/disorder of unknown etiology, unrecognizable/unclassifiable disease/malady/disorder

The following roots refer to specific adverse physical or medical conditions/illnesses defined by a specific symptom or set of symptoms rather than a cause. Their Specification pattern is as follows:

BSC	(to be) a state of physical having/suffering from the particular symptom or set of symptoms; to experience/suffer from the particular symptom or set of symptoms
CTE	(to be) the underlying cause (i.e., disease/illness/ailment/malady) which gives rise to the particular symptom(s)
CSV	(to be) a state of physical manifesting the particular symptom or set of symptoms; to physically manifiest/display the particular symptom or set of symptoms
OBJ	(to be) the living being manifesting/experiencing/suffering from the particular symptom(s)

-**PÇW**- **Stem 1:** common cold **Stem 2:** nasal congestion **Stem 3:** pneumonia

-**NŢM**- **Stem 1:** inflammation, immune reaction **Stem 2:** sore throat **Stem 3:** fever

-**VMW**- **Stem 1:** nausea, stomach ache **Stem 2:** vomiting, to vomit **Stem 3:** diarrhea

-**DPH**- **Stem 1:** shortness of breath, difficulty breathing **Stem 2:** asthma **Stem 3:** sleep apnea

-**ŇMW**- **Stem 1:** anemia **Stem 2:** hemochromatosis **Stem 3:** hypercholesterolemia

-**ŇMY**- **Stem 1:** ischemia **Stem 2:** hemorrhage, bruising, hematoma **Stem 3:** blood clot, to clot (blood), ischemia

-**ŇNW**- **Stem 1:** hypertension **Stem 2:** aneurysm **Stem 3:** venous thromboembolism

-ŇNY- **Stem 1:** cardiac arrhythmia **Stem 2:** cardiac fibrillation, flutter **Stem 3:** palpitation

-ŠKM- **Stem 1:** syncope (fainting) **Stem 2:** altitude sickness **Stem 3:** nitrogen narcosis ("the bends")

-ŠKN- **Stem 1:** dehydration **Stem 2:** heat exhaustion **Stem 3:** heat stroke / sunstroke

-LCR- **Stem 1:** gastro-intestinal upset **Stem 2:** ulcer **Stem 3:** diverticulitis

-JDŘ- **Stem 1:** organ failure (i.e., insufficiency) **Stem 2:** organ arrest (i.e., complete shutdown) **Stem 3:** organ hyperactivity

-RCL- **Stem 1:** osteoporosis **Stem 2:** osteopenia **Stem 3:** osteomalacia

-ĻPL- **Stem 1:** spasm **Stem 2:** tremor **Stem 3:** convulsion => tonic-clonic seizure

-ĻPR- **Stem 1:** spell, trance **Stem 2:** non-convulsive seizure, absence seizure **Stem 3:** fugue state / dissociative state

-PSX- **Stem 1:** lethargy, torpor **Stem 2:** listlessness, inability to come to full alertness **Stem 3:** sluggishness

-LKSX- **Stem 1:** muscle weakness **Stem 2:** muscle cramping **Stem 3:** abdominal cramping

-RPSX- **Stem 1:** calcification, calciferous hardening **Stem 2:** ossification, hardening **Stem 3:** fibrosis (including cirrhosis)

-SLW- **Stem 1:** paresis, monoparesis **Stem 2:** hemiparesis, paraparesis **Stem 3:** quadriparesis

-SLY- **Stem 1:** paralysis, monoplegia **Stem 2:** hemiplegia, paraplegia **Stem 3:** quadriplegia

-ĻNW- **Stem 1:** paresthesia **Stem 2:** dyskinesia **Stem 3:** locked-in syndrome

-VPĻ- **Stem 1:** bone fracture **Stem 2:** concussion / traumatic brain injury **Stem 3:** locked retinal detachment

-KNW- **Stem 1:** delusion, deluded thinking **Stem 2:** uncontrolled/inappropriate behavior **Stem 3:** dementia

<table>
<tr><th colspan="3">-ŘÇK- 'CAUSE OF AN ILLNESS / SICKNESS / DISEASE / MALADY / AILMENT'</th></tr>
<tr><td>BSC</td><td>Stem 1: (to be) the pathogenic or toxicological cause of an illness/ disease/ailment/sickness/malady (e.g., microbial, toxic substance, venom)</td><td rowspan="4">Stem 2: (to be) the idiopathic/congenital cause of an illness/disease/ailment/sickness/ malady (e.g., the cause(s) of cancer, autoimmune disorder, mental illness, Type-1 diabetes)

Stem 3: (to be) the iatrogenic or lifestyle/behavioral/environmental cause of an illness/ disease/ailment/sickness/malady (e.g., smoking, gluttony, heatstroke, hypoxia due to living at high altitude, alcoholism, drug addiction, botched surgery, bodily injury)</td></tr>
<tr><td>CTE</td><td>(to be) a state of infection/exposure to a pathogen/poison/venom</td></tr>
<tr><td>CSV</td><td>(to be) an act of contracting or being exposed to a pathogen/poison/venom</td></tr>
<tr><td>OBJ</td><td>(to be) the microbe, poison, venom, toxic substance involved</td></tr>
</table>

7.2.4 Parasitic/Microbial Diseases

The following are lists of diseases along with the stems indicating their parasitic or microbial pathogens. These parasitic/microbial stems are excerpted from the taxonomic lists in Sections 7.3.1, 7.3.3, 7.3.4, 7.3.5, and 7.3.6 of this document. To name a particular pathogenic disease, either concatenate a pathogen stem below with an appropriate stem from one of the various applicable roots shown above, or add the new **ADI** affix below to the stem.

-lkř	**ADI Adverse Disease/Disorder/Illness**
1	temporarily asymptomatic despite having [contracted] condition/illness/disorder/pathogen X
2	temporarily mildly ill/symptomatic due to having [contracted] condition/illness/disorder/pathogen X
3	temporarily moderately ill/symptomatic/disabled due to having [contracted] condition/illness/disorder/pathogen X
4	temporarily severely ill/symptomatic/disabled due to having [contracted] condition/illness/disorder/pathogen X
5	terminally ill due to having [contracted] condition/illness/disorder/pathogen X [CPT Version = deceased due to the condition//disorder]
6	permanently or chronically severely ill/symptomatic/disabled due to having [contracted] condition/illness/disorder/pathogen X
7	permanently or chronically moderately ill/symptomatic/disabled due to having [contracted] condition/illness/disorder/pathogen X
8	permanently or chronically mildly ill/symptomatic due to having [contracted] condition/illness/disorder/pathogen X
9	permanently or chronically asymptomatic despite having [contracted] condition/illness/disorder/pathogen X

Eukaryotic Endoparasites

Disease	Pathogen
anisakiasis	-**LŢPW**- Stem 2: anisakis
ascariasis	-**LŢPY**- Stem 1: ascaris / large roundworm
cercarial dermatitis	-**LŢTL**- Stem 2: trichobilharzia
chlonorchiasis	-**LŢTR**- Stem 2: chlonorchis / Chinese liver fluke
diphyllobothriasis	-**LŢTW**- Stem 2: broad tapeworm / fish tapeworm / broad fish tapeworm
dircrocoeliasis	-**LŢTR**- Stem 3: dicrocoelium / lancet liver fluke
dirofilariasis	-**LŢPL**- Stem 1: dirofilaria
dracunculiasis	-**LŢPŘ**- Stem 1: guinea worm
echinococcosis	-**LŢTY**- Stem 2: echinococcus
elephantiasis	-**LŢPR**- Stem 2: brugia, wuchereria
enterobiasis	-**LŢPŘ**- Stem 3: pinworm / threadworm / seatworm
fascioliasis	-**LŢTR**- Stem 1: fasciola / common liver fluke
gnathostomiasis	-**LŢPW**- Stem 3: gnathostoma

gongylonemiasis	-**LṬPL**- Stem 3: gongylonema
helminthiasis	-**LṬPÇ**- Stem 1: Old World hookworm Stem 2: New world hookworm
hymenolepiasis	-**LṬTW**- Stem 3: dwarf tapeworm, rat tapeworm
loiasis / Calabar swelling	-**LṬPY**- Stem 2: loa
myiasis	-**LZVL**- Stem 1: [maggot of] dermatobia / [maggot of] human botfly
myiasis	-**LZVL**- Stem 2: [maggot of] cordylobia / [maggot of] tumbu fly / [maggot of] mango fly
myiasis	-**LZVL**- Stem 3: [maggot of] cochliomyia / [maggot of] screwworm)
onchocerciasis (river blindness)	-**LṬPY**- Stem 3: onchocerca
opisthorchiasis	-**LṬTŘ**- Stem 1: opisthorchis / Southeast Asian liver fluke
paragonimiasis	-**LṬTL**- Stem 3: paragonimus
schistosomiasis / bilharziosis / snail fever	-**LṬTL**- Stem 1: schistosoma / blood fluke
sparganosis	-**LṬTY**- Stem 1: cat tapeworm, dog tapeworm, raccoon tapeworm
streptocerciasis, mansonelliasis	-**LṬPR**- Stem 1: mansonella
strongyloidiasis	-**LṬPÇ**- Stem 3: threadworm
taeniasis, cysticercosis	-**LṬTW**- Stem 1: taenia, pork tapeworm, beef tapeworm, Asian tapeworm
thelaziasis	-**LṬPL**- Stem 2: thelazia
toxocariasis	-**LṬPŘ**- Stem 2: toxocara, cat roundworm, dog roundworm, fox roundworm
trichinosis / trichinellosis	-**LṬPW**- Stem 1: trichinella / trichina worm
trichuriasis, whipworm infection	-**LṬPR**- Stem 3: whipworm
tungiasis	-**ŘẔŇW**- Stem 3: tunga / Chigoe flea / jigger

Eukaryotic Ectoparasites

Disease	Pathogen
baker's itch	**-ẔFM-** Stem 1: acarus / flour mite
cheyletiellosis / cheyletiella dermatitis / walking dandruff	**-ẔFR-** Stem 3: cheyletiella mite
feather pillow dermatitis, oral mite anaphylaxis	**-ẔFL-** Stem 2: dermatophagoides mite / dust mite / house dust mite Stem 3: euroglyphus / Mayne's house dust mite
gamasoidosis	**-ẔFR-** dermanyssus (red mite)
grain itch	**-ẔFŘ-** Stem 1: cheyletus mite Stem 2: tryophagus / cheese mite Stem 3: pyemotes / oak leaf gall mite / itch mite
grocer's itch	**-ẔFM-** Stem 2: glyciphagus mite Stem 2: carpoglyphus mite
papular urticaria	**-ẔFY-** Stem 1: Ixidoidea hard tick

papular urticaria (bed bug-induced)	-**NḶC**- Stem 1: bed bug
pediculosis (head lice, body lice)	-**CXL**- Stem 2: pediculus / head louse, body louse
phthiriasis (crab lice)	-**CXL**- Stem 3: pthirus / crab louse / pubic louse
rodent mite dermatitis	**-ẔFN-** Stem 1: liponyssoides / house rat mite Stem 2: laelaps / spiny rat mite
rodent mite dermatitis, gamasoidosis	**-ẔFR-** Stem 2: ornithonyssus
scabies	**-ẔFL-** Stem 1: sarcoptes / scabies mite
trombiculosis	**-ẔFN-** Stem 3: trombiculid mite / chigger / red bug / harvest mite / scrub-itch mite, berry bug

Fungal/Yeast Pathogens [not including poisonous mushrooms]

Disease	Pathogen
allescheriasis, pseudallescheriasis	-**RJTḶ**- Stem 1: pseudallescheria
alternariasis	-**RJTW**- Stem 3: alternaria
aspergillosis, otomycosis	-**RJTR**- Stem 2: aspergillus
blastomycosis	-**RJTÇ**- Stem 2: blastomyces
candidiasis	-**RJXM**- Stem 2: candida yeast
coccidioidomycosis (valley fever)	-**RJTR**- Stem 1: coccidioides
cryptococcosis	-**ẔBV**- Stem 2: cryptococcus
eumycetoma	-**RJTW**- Stem 2: pyrenochaeta
fusariosis, hyalohyphomycosis	-**RJTH**- Stem 1: fusarium
histoplasmosis	-**RJTÇ**- Stem 1: histoplasma
hyalohyphomycosis	-**RJTH**- Stem 2: acremonium
maduromycosis	-**RJTH**- Stem 3: madurella
mucormycosis	-**RJGR**- Stem 1: mucormycosis-causing fungus
paracoccidioidomycosis	-**RJTÇ**- Stem 3: paracoccidioides
phaeohyphomycosis	-**RJTL**- Stem 1: cladophialophora Stem 2: rhinocladiella Stem 3: exophiala
pneumocystis carinii pneumonia (PCP)	-**RJXN**- Stem 2: pneumocystis yeast
ringworm / tinea / athlete's foot / dermatophytosis	-**RJTŘ**- Stem 1: trichophyton Stem 2: microsporum Stem 3: epidermophyton
sporithrichosis (rose handler's disease)	-**RJTḶ**- Stem 2: sporothrix
talaromycosis	-**RJTR**- Stem 3: talaromyces
tinea nigra / superficial phaeohyphomycosis	-**RJTW**- Stem 1: Hortea wernecki
white piedra	-**ẔBV**- Stem 3: trichosporon

Unicellular Protist Pathogens

Disease	Pathogen
acanthamoeba keratitis	-**ŇŠKR**- Stem 1: acanthamoeba
amoebiasis / amoebic dysentery	-**ŇŠKN**- Stem 1: entamoeba
babesiosis	-**NŢFL**- Stem 1: babesia
balantidiasis	-**NŢTH**- Stem 2: balantidium
cryptosporidiasis	-**NŢFW**- Stem 3: cryptosporidium
cyclosporiasis	-**NŢFY**- Stem 1: cyclospora
dientamoebiasis (traveller's diarrhea)	-**SKMW**- Stem 2: dientamoeba
endolimax infection with chronic diarrhea	-**ŇŠKM**- Stem 3: endolimax
equine piroplasmosis	-**NŢFL**- Stem 3: theileria
giardiasis	-**SKMW**- Stem 3: giardia
granulomatous amoebic encephalitis	-**ŇŠKR**- Stem 2: balamuthia
leishmaniasis	-**RSKÇ**- Stem 2: leishmania
malaria	-**NŢFW**- Stem 1: plasmodium
primary amoebic meningoencephalitis	-**RSKH**- Stem 1: naegleria
rhinosporidiosis	-**LMVR**- Stem 3: rhinosporidium
toxoplasmosis	-**NŢFW**- Stem 2: toxoplasma
trichomoniasis	-**SKMW**- Stem 1: trichomonas
trypanosomiasis (sleeping sickness), Chagas disease	-**RSKÇ**- Stem 1: trypanosoma

Bacterial Pathogens

Disease	Pathogen
actinomycetoma	-**PFTĻ**- Stem 2: actinomadura
actinomycosis	-**PFTL**- Stem 1: actinomyces
anthrax	-**PFP**- Stem 1: bacillus
bacteraemia	-**PFTÇ**- Stem 1: eggerthella
bacterial vaginosis	-**PFTR**- Stem 3: gardnerella

bacterial vaginosis	-**PFTR̆**- Stem 1: atopobium
bacterial vaginosis	-**PFTÇ**- Stem 3: bifidobacterium
bacterial vaginosis	-**PFTĻ**- Stem 1: mobiluncus
bacteroides infection	-**PFMW**- Sem 1: bacteroides
botulism	-**PFPL**- Stem 1: clostridium
brucellosis	-**KSKÇ**- Stem 1: brucella
campylobacter enteritis	-**KSKN**- Stem 2: campylobacter
catscratch disease, Carrion's disease, trench fever, bacillary angiomatosis	-**KSKÇ**- Stem 2: bartonella
chlamydia infection	-**VZXW**- Stem 1: chlamydia Stem 2: chlamydophila
cholera	-**KSKY**- Stem 3: vibrio
chromobacteriosis	-**KSKŢ**- Stem 1: chromobacterium
diarrhea-causing colstidioides infection	-**PFPL**- Stem 2: clostridioides
diphtheria	-**PFTL**- Stem 2: corynebacterium
E. coli infection, traveller's diarrhea, bloody diarrhea	-**KSKW**- Stem 3: escherichia
ehrlichiosis	-**KSKM**- Stem 3: ehrlichia
eikenella infection, infective endocarditis	-**KSKŢ**- Stem 2: eikenella
gonorrhea, neisseriasis, Waterhouse-Friderichsen syndrome	-**KSKF**- Stem 1: neisseria
haemophilus infection (bacteremia, pneumonia, meningitis, infective endocarditis)	-**KSKY**- Stem 2: haemophilus
infective endocarditis	-**KSKHW**- Stem 1: cardiobacterium
intestinal spirochaetosis	-**LMSY**- Stem 1: brachyspira
klebsiella pneumonia	-**KSKR**- Stem 3: klebsiella
legionnaire's disease, Pontiac fever	-**KSKR**- Stem 1: legionella
leptospirosis	-**LMSW**- Stem 2: leptospira
listeriasis	-**PFP**- Stem 2: listeria
Lyme disease	-**LMSW**- Stem 1: borrelia
melioidosis	-**KSKF**- Stem 3: burkholderia
mycoplasma pneumonia	-**PFPĻ**- Stem 1: mycoplasma
nocardiosis	-**PFTR**- Stem 1: nocardia
peptic ulcer	-**KSKN**- Stem 1: helicobacter
periodontal diseae, Lemierre's syndrome	-**LMS̆**- Stem 1: fusobacterium
periodontal disease	-**PFTR̆**- Stem 2: olsenella Stem 3: cryptobacterium
periodontal disease	-**PFMW**- Sem 2: prevotella

periodontal disease	-**PFMW**- Sem 3: porphyromonas
periodontal disease	-**PFMY**- Stem 1: tannerella
periodontal disease, infective endocarditis	-**KSKŘ**- Stem 1: aggregatibacter
plague	-**KSKY**- Stem 1: yersinia
pseudomonas infection	-**KSKL**- Stem 1: pseudomonas
Q-fever / query fever	-**KSKR**- Stem 2: coxiella
salmonellosis, typhoid fever	-**KSKW**- Stem 1: salmonella
scrub typhus	-**KSKM**- Stem 2: orientia
shigellosis / bacillary dysentery	-**KSKW**- Stem 2: shigella
staphylococcosis / "staph" infection	-**PFP**- Stem 3: staphylococcus
stenotrophomonas infection	-**KSKHW**- Stem 2: stenotrophomonas
"strep" infection, pneumococcal infection, scarlet fever, rheumatic fever	-**PFPW**- Stem 2: streptococcus
syphilis, yaws, bejel, pinta	-**LMSW**- Stem 3: treponema
tuberculosis, leprosy / Hanesen's disease	-**PFTL**- Stem 3: mycobacterium
tularemia / rabbit fever	-**KSKL**- Stem 2: francisella
typhus, spotted fever (Rocky Mountain, Helvetica, Mediterranean, Queensland)	-**KSKM**- Stem 1: rickettsia
urinary tract infection	-**PFPW**- Stem 1: enterococcus
whooping cough	-**KSKF**- Stem 2: bordetella

Viral/Viroid Pathogens

Disease	Pathogen
aichi viral infection	-**MÇPR**- Stem 1: aichi virus
Alkhurma virus infection	-**LFSC**- Stem 3: Alkhurma virus
astrovirus infection	-**ŇSŢR**- Stem 1: human astrovirus
Barmah Forest virus infection	-**LFSN**- Stem 2: Barmah Forest virus (BFV)
Bas-Congo virus infection, Ekpoma virus infection	-**NXPĻ**- Stem 1: Bas-Congo virus, Ekpoma virus 1 & 2
BK polyomavirus infection	-**RŠTL**- Stem 1: BK polyomavirus
California encephalitis, La Crosse virus infection, Jamestown Canyon virus infection, Cache Valley virus infection	-**NXPY**- Stem 2: California encephalitis virus, La Crosse virus (LACV), Jamestown Canyon virus, Cache Valley orthobunyavirus (CVV)
cardiovirus infection	-**MÇPL**- Stem 1: cardiovirus B, Saffold virus (SAFV)
Chandipura virus infection, vesicular stomatitis	-**NXPÇ**- Stem 3: Chandipura virus (CHPV), vesicular stomatitis virus

chicken pox, shingles	-**LFMW**- Stem 1: varicella zoster virus
Chikungunya virus infection	-**LFSŘ**- Stem 1: Chikungunya virus
CMV mononucleosis, cytomegalovirus infection	-**LFMY**- Stem 1: cytomegalovirus (CMV)
cold sores, oral herpes, genital herpes	-**LFMW**- Stem 2: herpes simplex 1, herpes simplex 2
Colorado tick fever, European Eyach infection	-**MSŢW**- Stem 1: Colorado tick fever virus, European Eyach virus
common cold, upper repiratory infection	-**ŠŠTW**- Stem 1: human coronavirus OC43, human coronavirus HKU1
common cold, upper repiratory infection	-**ŠŠTL**- Stem 1: human coronavirus 229E
common cold, upper repiratory infection	-**GZXL**- Stem 1: human mastadenovirus A/B/C/D/E/F/G
common cold, upper repiratory infection	-**MÇPY**- Stem 1: human rhinovirus A Stem 2: human rhinovirus B Stem 3: human rhinovirus C
COVID-19	-**ŠŠT**- Stem 3: severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2)
Crimean-Congo hemorhagic fever	-**NXPY**- Stem 3: Crimean-Congo hemorhagic fever orthonairovirus (CCHF)
dengue fever	-**LFSK**- Stem 2: dengue virus
ebola	-**NXPŘ**- Stem 1: ebolavirus
foot-and-mouth disease	-**MÇPF**- Stem 1: foot-and-mouth disease virus (FMDV)
hantavirus infection	-**NXPY**- Stem 1: hantavirus
hepatitis A	-**MÇPÇ**- Stem 1: hepatovirus A, hepatitis A virus
hepatitis B	-**RTÇP**- Stem 1: hepatitis B virus
hepatitis C	-**LFSW**- Stem 1: human hepatitis C virus
hepatitis E	-**LFSL**- Stem 2: orthohepevirus, human hepatitis E virus
HIV infection / AIDS	-**RTÇT**- Stem 1: human immunodeficiency virus 1 / HIV-1 Stem 2: human immunodeficiency virus 2 / HIV-2
HPV infection	-**RŠTY**- Stem 1: human papillomavirus
influenza / “the flu”	-**NXP**- Stem 1: influenza A virus Stem 2: influenza B virus Stem 3: influenza C virus
Japanese encephalitis	-**LFSČ**- Stem 1: Japanese encephalitis virus
JC polyomavirus infection	-**RŠTL**- Stem 2: JC polyomavirus
Kaposi’s sarcoma, primary effusion lymphoma, Castleman’s disease	-**LFML**- Stem 1: rhadinovirus
Kyansur Forest disease	-**LFSP**- Stem 2: Kyansur Forest disease virus
Lassa virus infection, lymphocytic choriomeningitis, Lujo virus infection	-**NXPL**- Stem 1: Lassa virus (LASV), lymphocytic choriomeningitis virus (LCMV), Lujo virus (LUJV)
Marburg virus infection	-**NXPŘ**- Stem 2: Marburg virus, Ravn virus
Mayaro virus infection, Una virus infection	-**LFSŘ**- Stem 2: Mayaro virus, Una virus
measles / rubeola	-**NXPŢ**- Stem 1: measles virus (MeV)

Merkel cell polyomavirus infection	-**RŠTL**- Stem 3: Merkel cell polyomavirus
MERS	-**ŠŠT**- Stem 1: Middle East respiratory syndrome-related coronavirus (MERS-CoV)
Middelburg virus infection	-**LFSN**- Stem 1: Middelburg virus (MIDV)
mononucleosis / Epstein-Barr infection	-**LFML**- Stem 2: lymphocryptovirus / Epstein-Barr virus
mumps	-**NXFW**- Stem1: mumps virus (MuV)
Murray Valley encephalitis	-**LFST**- Stem 1: Murray Valley encephalitis virus (MVEV)
norovirus infection (winter vomiting bug), Sapporo viral infection	-**MÇPĻ**- Stem 1: norovirus, winter vomiting bug 2. (genus *Sapovirus*) Sapporo virus
Omsk hemorrhagic fever	-**LFSP**- Stem 1: Omsk hemorrhagic fever virus
parechoviral infection	-**MÇPŘ**- Stem 1: parechovirus A /human parechovirus 1 / HPeV1, human parechovirus 2
picobirnavirus infection	-**NSŢR**- Stem 1: human picobirnavirus
poliomyelitis	-**MÇP**- Stem 1: poliovirus, enterovirus C
Powassan virus infection	-**LFSP**- Stem 3: Powassan virus (POWV)
quaranjavirus infection	-**NXPW**- Stem 1: quaranjavirus
rabies	-**NXPÇ**- Stem 1: rabies virus (RABV)
respiratory syncytial virus infection, metapneumovirus infection	-**NXFW**- Stem 3: human respiratory syncytial virus (HRSV), human metapneumovirus (HMPV)
respirovirus infection	-**NXPŢ**- Stem 3: human respirovirus 1 & 3
Rift Valley fever, Punta Toro phlebovirus infection, Sicilian phlebovirus infection	-**NXPL**- Stem 3: Rift Valley fever phlebovirus, Punta Toro phlebovirus, Sicilian phlebovirus
roseola / sixth diseasee	-**LFMY**- Stem 2: roseolovirus / human betaherpesvirus 6 or 7
Ross River virus infection	-**LFSŘ**- Stem 3: Ross River virus (RRV)
rotavirus infection	-**MSŢ**- Stem 1: rotavirus
rubella / German measles	-**LFSL**- Stem 1: rubella virus (RuV)
Saint Louis encephalitis	-**LFSČ**- Stem 2: Saint Louis encephalitis virus
SARS	-**ŠŠT**- Stem 2: severe acute respiratory syndrome coronavirus (SARS-CoV)
smallpox / variola	-**PSF**- Stem 1: smallpox virus / variola virus
T-cell lymphotropic virus infection	-**RTÇL**- Stem 1: human T-cell lymphotropic virus (HTLV-1, HTLV-2, HTLV-3, HTLV-4)
thogotovirus infection, dhori virus infection	-**NXPW**- Stem 2: thogotovirus (THOV), dhori virus (DHOV)
tick-borne encephalitis	-**LFST**- Stem 1: tick-borne encephalitis virus (TBEV)
West Nile virus infection, Kunjin virus infection	-**LFST**- Stem 1: West Nile virus, Kunjin virus
Whitewater Arroyo virus infection, Machupo virus infection, Sabiá virus infection, Chapare virus infection, Guanarito virus infection, Junín virus infection	-**NXPL**- Stem 2: Whitewater Arroyo virus (WWAV), Machupo virus (MACV), Sabiá virus (SBAV), Chapare virus (CHAPV), Guanarito virus (GTOV), Junín virus (JUNV)
yellow fever	-**LFSK**- Stem 1: yellow fever virus

zika virus infection	-**LFSK**- Stem 3: zika virus

7.2.5 Non-Genetic Metabolic Disorders/Diseases (e.g., Due to Environmental Factors, Improper Nutrition, etc.)

Disease	Root
nutrient deficiency, malnutrition	-**ŇZT**-
scurvy	-**ŇZTW**-
rickets	-**ŇZTY**-
diabetes Type 2	-**ŇZTL**-
electrolyte imbalance	-**ŇZTR**-
ADH imbalance	-**ŇZTŘ**-

7.2.6 Genetic Disorders/Diseases

The Stems and Specifications of the following roots follow the pattern of the root -**ÇK**- shown in Sec. 7.2.3 above. They may also be used with the **ADI** affix.

Disease	Root
1p36 deletion syndrome	-**BZZGL**-
18p deletion syndrome	-**GZZGL**-
21-hydroxylase deficiency	-**VZZGL**-
Alpha 1-antitrypsin deficiency	-**MZZGL**-
AAA syndrome (achalasia–addisonianism–alacrima syndrome)	-**NZZGL**-
Aarskog–Scott syndrome	-**ŇZZGL**-
ABCD syndrome	-**LZZGL**-
Aceruloplasminemia	-**RZZGL**-
Acheiropodia	-**ŘZZGL**-
Achondrogenesis type II	-**BZZGR**-
achondroplasia	-**GZZGR**-
Acute intermittent porphyria	-**VZZGR**-
adenylosuccinate lyase deficiency	-**MZZGR**-
Adrenoleukodystrophy	-**NZZGR**-

Alagille syndrome	-**ŇZZGR**-
ADULT syndrome	-**LZZGR**-
Aicardi–Goutières syndrome	-**RZZGR**-
Albinism	-**ŘZZGR**-
Alexander disease	-**BZZGŘ**-
alkaptonuria	-**GZZGŘ**-
Alport syndrome	-**VZZGŘ**-
Alternating hemiplegia of childhood	-**MZZGŘ**-
Amyotrophic lateral sclerosis – Frontotemporal dementia	-**NZZGŘ**-
Alström syndrome	-**ŇZZGŘ**-
Alzheimer's disease	-**LZZGŘ**-
Amelogenesis imperfecta	-**RZZGŘ**-
Aminolevulinic acid dehydratase deficiency porphyria	-**ŘZZGŘ**-
Androgen insensitivity syndrome	-**BZZGW**-
Angelman syndrome	-**GZZGW**-
Apert syndrome	-**VZZGW**-
Arthrogryposis–renal dysfunction–cholestasis syndrome	-**MZZGW**-
Ataxia telangiectasia	-**NZZGW**-
Axenfeld syndrome	-**ŇZZGW**-
Beare–Stevenson cutis gyrata syndrome	-**LZZGW**-
Beckwith–Wiedemann syndrome	-**RZZGW**-
Benjamin syndrome	-**ŘZZGW**-
biotinidase deficiency	-**BZZGY**-
Björnstad syndrome	-**GZZGY**-
Bloom syndrome	-**VZZGY**-
Birt–Hogg–Dubé syndrome	-**MZZGY**-
Brody myopathy	-**NZZGY**-
Brunner syndrome	-**ŇZZGY**-
CADASIL syndrome	-**LZZGY**-
CRASIL syndrome	-**RZZGY**-
Chronic granulomatous disorder	-**ŘZZGY**-
Campomelic dysplasia	-**BZZBL**-

Canavan disease	-**GZZBL**-
Carpenter Syndrome	-**VZZBL**-
Cerebral dysgenesis–neuropathy–ichthyosis–keratoderma syndrome (SEDNIK)	-**MZZBL**-
Cystic fibrosis	-**NZZBL**-
Charcot–Marie–Tooth disease	-**ŇZZBL**-
CHARGE syndrome	-**LZZBL**-
Chédiak–Higashi syndrome	-**RZZBL**-
Cleidocranial dysostosis	-**ŘZZBL**-
Cockayne syndrome	-**BZZBR**-
Coffin–Lowry syndrome	-**GZZBR**-
Cohen syndrome	-**VZZBR**-
collagenopathy, types II and XI	-**MZZBR**-
Congenital insensitivity to pain with anhidrosis (CIPA)	-**NZZBR**-
Congenital Muscular Dystrophy	-**ŇZZBR**-
Cornelia de Lange syndrome (CDLS)	-**LZZBR**-
Cowden syndrome	-**RZZBR**-
CPO deficiency (coproporphyria)	-**ŘZZBR**-
Cranio-lenticulo-sutural dysplasia	-**BZZBŘ**-
Cri du chat	-**GZZBŘ**-
Crohn's disease	-**VZZBŘ**-
Crouzon syndrome	-**MZZBŘ**-
Crouzonodermoskeletal syndrome (Crouzon syndrome with acanthosis nigricans)	-**NZZBŘ**-
Darier's disease	-**ŇZZBŘ**-
Dent's disease (Genetic hypercalciuria)	-**LZZBŘ**-
Denys–Drash syndrome	-**RZZBŘ**-
De Grouchy syndrome	-**ŘZZBŘ**-
Down Syndrome	-**BZZBW**-
Di George's syndrome	-**GZZBW**-
Distal hereditary motor neuropathy	-**VZZBW**-
Distal muscular dystrophy	-**MZZBW**-
Duchenne muscular dystrophy	-**NZZBW**-
Dravet syndrome	-**ŇZZBW**-

Edwards Syndrome	-**LZZBW**-
Ehlers–Danlos syndrome	-**RZZBW**-
Emery–Dreifuss syndrome	-**ŘZZBW**-
Epidermolysis bullosa	-**BZZBY**-
Erythropoietic protoporphyria	-**GZZBY**-
Fanconi anemia (FA)	-**VZZBY**-
Fabry disease	-**MZZBY**-
Factor V Leiden thrombophilia	-**NZZBY**-
Fatal familial insomnia	-**ŇZZBY**-
Familial adenomatous polyposis	-**LZZBY**-
Familial dysautonomia	-**RZZBY**-
Familial Creutzfeld–Jakob Disease	-**ŘZZBY**-
Feingold syndrome	-**BZZDL**-
FG syndrome	-**GZZDL**-
Fragile X syndrome	-**VZZDL**-
Friedreich's ataxia	-**MZZDL**-
G6PD deficiency	-**NZZDL**-
Galactosemia	-**ŇZZDL**-
Gaucher disease	-**LZZDL**-
Gerstmann–Sträussler–Scheinker syndrome	-**RZZDL**-
Gillespie syndrome	-**ŘZZDL**-
Glutaric aciduria, type I and type 2	-**BZZDR**-
GRACILE syndrome	-**GZZDR**-
Griscelli syndrome	-**VZZDR**-
Hailey–Hailey disease	-**MZZDR**-
Harlequin type ichthyosis	-**NZZDR**-
Hemochromatosis, hereditary	-**ŇZZDR**-
Hemophilia	-**LZZDR**-
Hepatoerythropoietic porphyria	-**RZZDR**-
Hereditary coproporphyria	-**ŘZZDR**-
Hereditary hemorrhagic telangiectasia (Osler–Weber–Rendu syndrome)	-**BZZDŘ**-
Hereditary inclusion body myopathy	-**GZZDŘ**-

Hereditary multiple exostoses	-**VZZDŘ**-
Hereditary spastic paraplegia (infantile-onset ascending hereditary spastic paralysis)	-**MZZDŘ**-
Hermansky–Pudlak syndrome	-**NZZDŘ**-
Hereditary neuropathy with liability to pressure palsies (HNPP)	-**ŇZZDŘ**-
Heterotaxy	-**LZZDŘ**-
Homocystinuria	-**RZZDŘ**-
Huntington's disease	-**ŘZZDŘ**-
Hunter syndrome	-**BZZDW**-
Hurler syndrome	-**GZZDW**-
Hutchinson–Gilford progeria syndrome	-**VZZDW**-
Hyperlysinemia	-**MZZDW**-
Hyperoxaluria, primary	-**NZZDW**-
Hyperphenylalaninemia	-**ŇZZDW**-
Hypoalphalipoproteinemia (Tangier disease)	-**LZZDW**-
Hypochondrogenesis	-**RZZDW**-
Hypochondroplasia	-**ŘZZDW**-
Immunodeficiency–centromeric instability–facial anomalies syndrome (ICF syndrome)	-**BZZDY**-
Incontinentia pigmenti	-**GZZDY**-
Ischiopatellar dysplasia	-**VZZDY**-
Isodicentric 15	-**MZZDY**-
Jackson–Weiss syndrome	-**NZZDY**-
Joubert syndrome	-**ŇZZDY**-
Juvenile primary lateral sclerosis (JPLS)	-**LZZDY**-
Keloid disorder	-**RZZDY**-
Kniest dysplasia	-**ŘZZDY**-
Kosaki overgrowth syndrome	-**BZZFL**-
Krabbe disease	-**GZZFL**-
Kufor–Rakeb syndrome	-**VZZFL**-
LCAT deficiency	-**MZZFL**-
Lesch–Nyhan syndrome	-**NZZFL**-
Li–Fraumeni syndrome	-**ŇZZFL**-
Limb-Girdle Muscular Dystrophy	-**LZZFL**-

Lynch syndrome	-**RZZFL**-
lipoprotein lipase deficiency	-**ŘZZFL**-
Malignant hyperthermia	-**BZZFR**-
Maple syrup urine disease	-**GZZFR**-
Marfan syndrome	-**VZZFR**-
Maroteaux–Lamy syndrome	-**MZZFR**-
McCune–Albright syndrome	-**NZZFR**-
McLeod syndrome	-**ŇZZFR**-
MEDNIK syndrome	-**LZZFR**-
Mediterranean fever, familial	-**RZZFR**-
Menkes disease	-**ŘZZFR**-
Metachromatic leukodystrophy	-**BZZFŘ**-
Methemoglobinemia	-**GZZFŘ**-
Methylmalonic acidemia	-**VZZFŘ**-
Micro syndrome	-**MZZFŘ**-
Microcephaly	-**NZZFŘ**-
Morquio syndrome	-**ŇZZFŘ**-
Mowat–Wilson syndrome	-**LZZFŘ**-
Muenke syndrome	-**RZZFŘ**-
Multiple endocrine neoplasia type 1 (Wermer's syndrome)	-**ŘZZFŘ**-
Multiple endocrine neoplasia type 2	-**BZZFW**-
Muscular dystrophy	-**GZZFW**-
Muscular dystrophy, Duchenne and Becker type	-**VZZFW**-
Myostatin-related muscle hypertrophy	-**MZZFW**-
myotonic dystrophy	-**NZZFW**-
Natowicz syndrome	-**ŇZZFW**-
Neurofibromatosis type I	-**LZZFW**-
Neurofibromatosis type II	-**RZZFW**-
Niemann–Pick disease	-**ŘZZFW**-
Nonketotic hyperglycinemia	-**BZZFY**-
Nonsyndromic deafness	-**GZZFY**-
Noonan syndrome	-**VZZFY**-

Norman–Roberts syndrome	-**MZZFY**-
Ogden syndrome	-**NZZFY**-
Omenn syndrome	-**ŇZZFY**-
Osteogenesis imperfecta	-**LZZFY**-
Pantothenate kinase-associated neurodegeneration	-**RZZFY**-
Patau syndrome (Trisomy 13)	-**ŘZZFY**-
PCC deficiency (propionic acidemia)	-**BZZVL**-
Porphyria cutanea tarda (PCT)	-**GZZVL**-
Pendred syndrome	-**VZZVL**-
Peutz–Jeghers syndrome	-**MZZVL**-
Pfeiffer syndrome	-**NZZVL**-
Phenylketonuria	-**ŇZZVL**-
Pipecolic acidemia	-**LZZVL**-
Pitt–Hopkins syndrome	-**RZZVL**-
Polycystic kidney disease	-**ŘZZVL**-
Polycystic ovary syndrome (PCOS)	-**BZZVR**-
Porphyria	-**GZZVR**-
Prader–Willi syndrome	-**VZZVR**-
Primary ciliary dyskinesia (PCD)	-**MZZVR**-
Primary pulmonary hypertension	-**NZZVR**-
Protein C deficiency	-**ŇZZVR**-
Protein S deficiency	-**LZZVR**-
Pseudo-Gaucher disease	-**RZZVR**-
Pseudoxanthoma elasticum	-**ŘZZVR**-
Retinitis pigmentosa	-**BZZVŘ**-
Rett syndrome	-**GZZVŘ**-
Roberts syndrome	-**VZZVŘ**-
Rubinstein–Taybi syndrome (RSTS)	-**MZZVŘ**-
Sandhoff disease	-**NZZVŘ**-
Sanfilippo syndrome	-**ŇZZVŘ**-
Schwartz–Jampel syndrome	-**LZZVŘ**-
Sjogren-Larsson syndrome	-**RZZVŘ**-

Spondyloepiphyseal dysplasia congenita (SED)	-**ŘZZVŘ**-
Shprintzen–Goldberg syndrome	-**BZZVW**-
Sickle cell anemia	-**GZZVW**-
Siderius X-linked mental retardation syndrome	-**VZZVW**-
Sideroblastic anemia	-**MZZVW**-
Sly syndrome	-**NZZVW**-
Smith–Lemli–Opitz syndrome	-**ŇZZVW**-
Smith–Magenis syndrome	-**LZZVW**-
Snyder–Robinson syndrome	-**RZZVW**-
Spinal muscular atrophy	-**ŘZZVW**-
Spinocerebellar ataxia (types 1–29)	-**BZZVY**-
SSB syndrome (SADDAN)	-**GZZVY**-
Stargardt disease (macular degeneration)	-**VZZVY**-
Stickler syndrome (multiple forms)	-**MZZVY**-
Strudwick syndrome (spondyloepimetaphyseal dysplasia, Strudwick type)	-**NZZVY**-
Tay–Sachs disease	-**ŇZZVY**-
Tetrahydrobiopterin deficiency	-**LZZVY**-
Thanatophoric dysplasia	-**RZZVY**-
Treacher Collins syndrome	-**ŘZZVY**-
Tuberous sclerosis complex (TSC)	-**BZZMW**-
Tumor necrosis factor receptor-associated periodic syndrome (TRAPS)	-**GZZMW**-
Turner syndrome	-**VZZMW**-
Usher syndrome	-**MZZMW**-
Variegate porphyria	-**NZZMW**-
von Hippel–Lindau disease	-**ŇZZMW**-
von Willebrand disease	-**LZZMW**-
Waardenburg syndrome	-**RZZMW**-
Weissenbacher–Zweymüller syndrome	-**ŘZZMW**-
Williams syndrome	-**BZZMY**-
Wilson disease	-**GZZMY**-
Woodhouse–Sakati syndrome	-**VZZMY**-
Wolf–Hirschhorn syndrome	-**MZZMY**-

Xeroderma pigmentosum	-**NZZMY**-
X-linked intellectual disability and macroorchidism (fragile X syndrome)	-**ŇZZMY**-
X-linked spinal-bulbar muscle atrophy (spinal and bulbar muscular atrophy)	-**LZZMY**-
Xp11.2 duplication syndrome	-**RZZMY**-
X-linked severe combined immunodeficiency (X-SCID)	-**ŘZZMY**-
X-linked sideroblastic anemia (XLSA)	-**BZZNW**-
47,XXX (triple X syndrome)	-**GZZNW**-
XXXX syndrome (48, XXXX)	-**VZZNW**-
XXXXX syndrome (49, XXXXX)	-**MZZNW**-
XYY syndrome (47,XYY)	-**NZZNW**-
Zellweger syndrome	-**ŇZZNW**-

7.2.7 Autoimmune Diseases/Disorders

The Stems and Specifications of the following roots follow the pattern of the root -**ÇK**- shown in Sec. 7.2.3 above. They may also be used with the new **ADI** affix.

autoimmune myocarditis, autoimmune cardiomyopathy, Coxsackie myocarditis	-**MMSPW**-
postmyocardial infarction syndrome / Dressler's syndrome	-**MMSPY**-
postpericardiotomy syndrome	-**MMSPL**-
anti-glomerular basement membrane nephritis / Goodpastures syndrome, glomerulonephritis Type 1	-**MMSPR**-
Autoimmune hepatitis / Lupoid hepatitis	-**MMSKW**-
Primary biliary cholangitis (PBC)	-**MMSKY**-
Primary sclerosing cholangitis	-**MMSKL**-
Churg-Strauss syndrome / Eosinophilic pneumonia	-**MMSKR**-
Antisynthetase syndrome	-**MMSTW**-
Alopecia Areata	-**MMSTY**-
Autoimmune Angioedema	-**MMSTL**-

Autoimmune progesterone dermatitis	-**MMSTR**-
Autoimmune urticaria	-**MMSTŘ**-
Bullous pemphigoid	-**MMSTH**-
Cicatricial pemphigoid / Benign Mucosal Pemphigoid / Ocular cicatricial pemphigoid	-**MMŠPW**-
Dermatitis herpetiformis	-**MMŠPY**-
Discoid lupus erythematosus	-**MMŠPL**-
Epidermolysis bullosa acquisita	-**MMŠPR**-
Erythema nodosum	-**MMŠPŘ**-
Gestational pemphigoid	-**MMŠPH**-
Hidradenitis suppurativa	-**MMŠKW**-
Lichen planus	-**MMŠKY**-
Lichen sclerosus	-**MMŠKL**-
Linear IgA disease (LAD)	-**MMŠKR**-
Morphea	-**MMŠKŘ**-
Pemphigus vulgaris	-**MMŠKH**-
Pityriasis lichenoides et varioliformis acuta	-**MMŠTW**-
Mucha–Habermann disease	-**MMŠTY**-
Psoriasis	-**MMŠTL**-
Systemic scleroderma	-**MMŠTR**-
Vitiligo	-**MMŠTŘ**-
Addisons Disease / Autoimmune polyendocrine syndrome (APS) type 1 / Whitaker's Syndrome / Autoimmune polyendocrinopathy-candidiasis-ectodermal dystrophy (APECED) / Polyglandular Autoimmune Syndrome 1 (PGAS-1)	-**MMÇPW**-
Autoimmune polyendocrine syndrome (APS) type 2 / Schmidt syndrome / Polyglandular Autoimmune Syndrome 2 (PGAS-2)	-**MMÇPY**-
Autoimmune polyendocrine syndrome (APS) type 3	-**MMÇPL**-
Autoimmune pancreatitis (AIP)	-**MMÇPR**-
Diabetes mellitus type 1	-**MMÇPŘ**-
Autoimmune thyroiditis / Chronic lymphocytic thyroiditis / Hashimoto's thyroiditis	-**MMÇPH**-
Ord's thyroiditis	-**MMÇKW**-

Graves' disease	-**MMÇKY**-
Autoimmune oophoritis	-**MMÇKL**-
Endometriosis	-**MMÇKR**-
Autoimmune orchitis	-**MMÇKŘ**-
Sjögren syndrome	-**MMÇKH**-
Autoimmune enteropathy	-**MMÇTW**-
Coeliac disease	-**MMÇTY**-
Crohn's disease	-**MMÇTL**-
Esophageal achalasia	-**MMÇTR**-
Microscopic colitis	-**MMÇTŘ**-
Ulcerative colitis	-**MMÇTH**-
Antiphospholipid syndrome (APS, APLS) / Hughes syndrome	-**MMFPW**-
Aplastic anemia	-**MMFPY**-
Autoimmune hemolytic anemia	-**MMFPL**-
Autoimmune lymphoproliferative syndrome / Canale-Smith syndrome	-**MMFPR**-
Autoimmune neutropenia	-**MMFPŘ**-
Autoimmune thrombocytopenic purpura / Idiopathic Thrombocytopenic Purpura (ITP)	-**MMFPH**-
Cold agglutinin disease / Autoimmune hemolytic anemia	-**MMFKW**-
Essential mixed cryoglobulinemia	-**MMFKY**-
Evans syndrome	-**MMFKL**-
Pernicious anemia	-**MMFKR**-
Pure red cell aplasia	-**MMFKŘ**-
Thrombocytopenia / Neonatal thrombocytopenia	-**MMFKH**-
Adiposis dolorosa / Dercum's disease	-**MMFTW**-
Adult-onset Still's disease	-**MMFTY**-
Ankylosing spondylitis	-**MMFTL**-
CREST syndrome	-**MMFTR**-

Drug-induced lupus	-**MMFTŘ**-
Enthesitis-related arthritis	-**MMFTH**-
Eosinophilic fasciitis / Shulman's syndrome	-**MMŢPW**-
Felty syndrome	-**MMŢPY**-
IgG4-related autoimmune disease / IgG4-related systemic sclerosing disease / IgG4-associated multifocal systemic fibrosis / IgG4 syndrome / Hyper-IgG4 disease / Systemic IgG4-related plasmacytic syndrome	-**MMŢPL**-
Juvenile arthritis / Juvenile rheumatoid arthritis / Juvenile idiopathic arthritis	-**MMŢPR**-
Lyme disease (chronic)	-**MMŢPŘ**-
Mixed connective tissue disease (MCTD)	-**MMŢKW**-
Palindromic rheumatism / Hench-Rosenberg syndrome	-**MMŢKY**-
Parry–Romberg syndrome	-**MMŢKL**-
Parsonage–Turner syndrome	-**MMŢKR**-
Psoriatic arthritis	-**MMŢKŘ**-
Reactive arthritis / Reiter's syndrome	-**MMŢTW**-
Relapsing polychondritis / Atrophic polychondritis / systemic chondromalacia / chronic atrophic polychondritis / Meyenburg-Altherr-Uehlinger syndrome / generalized chondromalacia / systemic chondromalacia	-**MMŢTY**-
Retroperitoneal fibrosis	-**MMŢTL**-
Rheumatic fever	-**MMŢTR**-
Rheumatoid arthritis	-**MMŢTŘ**-
Sarcoidosis	-**MMZPW**-
Schnitzler syndrome	-**MMZPY**-
Systemic lupus erythematosus (SLE) / Lupus	-**MMZPL**-
Undifferentiated connective tissue disease (UCTD) / Latent lupus, incomplete lupus	-**MMZPR**-
Dermatomyositis / Juvenile dermatomyositis	-**MMZKW**-
Fibromyalgia	-**MMZKY**-
Inclusion body myositis	-**MMZKL**-
Myositis	-**MMZKR**-
Myasthenia gravis	-**MMZTW**-
Neuromyotonia / Isaacs' syndrome	-**MMZTY**-

Paraneoplastic cerebellar degeneration	-**MMZTL**-
Polymyositis	-**MMZTR**-
Acute disseminated encephalomyelitis (ADEM) / Perivenous encephalomyelitis / Acute hemorrhagic leukoencephalitis (AHL, AHLE) / Acute necrotizing encephalopathy (ANE) / Acute hemorrhagic encephalomyelitis (AHEM) / Acute necrotizing hemorrhagic leukoencephalitis (ANHLE) / Weston-Hurst syndrome /Hurst's disease	-**NNSPW**-
Acute motor axonal neuropathy	-**NNSPY**-
Anti-N-Methyl-D-Aspartate (Anti-NMDA) Receptor Encephalitis	-**NNSPL**-
Balo concentric sclerosis / Balo disease / Schilders disease	-**NNSPR**-
Bickerstaff's encephalitis	-**NNSPŘ**-
Chronic inflammatory demyelinating polyneuropathy / Relapsing polyneuropathy (CRP) / chronic inflammatory demyelinating polyradiculoneuropathy / Chronic inflammatory demyelinating polyneuritis	-**NNSPH**-
Guillain–Barré syndrome / Miller–Fisher syndrome / Landry's paralysis	-**NNSKW**-
Hashimoto's encephalopathy / Steroid-responsive encephalopathy associated with autoimmune thyroiditis (SREAT) / Nonvasculitic autoimmune meningoencephalitis (NAIM) / Encephalopathy Associated with Autoimmune Thyroid Disease (EAATD)	-**NNSKY**-
Idiopathic inflammatory demyelinating diseases	-**NNSKL**-
Lambert–Eaton myasthenic syndrome	-**NNSKR**-
Multiple sclerosis, pattern II / Primary progressive multiple sclerosis / Relapsing-remitting multiple sclerosis / disseminated sclerosis / encephalomyelitis disseminata	-**NNSKŘ**-
Oshtoran syndrome	-**NNSKH**-
Pediatric Autoimmune Neuropsychiatric Disorder Associated with Streptococcus (PANDAS)	-**NNSTW**-
Progressive inflammatory neuropathy	-**NNSTY**-
Restless legs syndrome	-**NNSTL**-
Stiff-person syndrome	-**NNSTR**-
Sydenham's chorea	-**NNSTŘ**-
Transverse myelitis	-**NNSTH**-
Autoimmune retinopathy	-**NNŠPW**-
Autoimmune uveitis	-**NNŠPY**-
Cogan syndrome	-**NNŠPL**-
Glaucoma	-**NNŠPR**-

Graves' ophthalmopathy	-**NNŠPŘ**-
Intermediate uveitis / Pars planitis / peripheral uveitis	-**NNŠKW**-
Ligneous conjunctivitis	-**NNŠKY**-
Mooren's ulcer	-**NNŠKL**-
Neuromyelitis optica / Devic's disease	-**NNŠKR**-
Opsoclonus myoclonus syndrome	-**NNŠKŘ**-
Optic neuritis	-**NNŠTW**-
Scleritis	-**NNŠTY**-
Susac's syndrome / Retinocochleocerebral vasculopathy	-**NNŠTL**-
Sympathetic ophthalmia	-**NNŠTR**-
Tolosa–Hunt syndrome	-**NNŠTŘ**-
Autoimmune inner ear disease (AIED)	-**NNŠPH**-
Ménière's disease	-**NNŠKH**-
Behçet's disease, Hughes–Stovin syndrome	-**NNÇPW**-
Eosinophilic granulomatosis with polyangiitis (EGPA)	-**NNÇPY**-
Giant cell arteritis / Cranial arteritis / temporal arteritis	-**NNÇPL**-
Granulomatosis with polyangiitis (GPA)	-**NNÇPR**-
IgA vasculitis (IgAV) / Anaphylactoid purpura / Henoch-Schonlein purpura / purpura rheumatica / Schönlein–Henoch purpura	-**NNÇPŘ**-
Kawasaki disease / Kawasaki syndrome / lymph node syndrome / mucocutaneous lymph node syndrome	-**NNÇKW**-
Leukocytoclastic vasculitis	-**NNÇKY**-
Lupus vasculitis	-**NNÇKL**-
Rheumatoid vasculitis	-**NNÇKR**-
Microscopic polyangiitis (MPA) / Microscopic polyarteritis / microscopic polyarteritis nodosa	-**NNÇKŘ**-
Podoconiosis / non-filarial elephantiasis	-**NNÇTW**-
Polyarteritis nodosa (PAN) / Panarteritis nodosa / periarteritis nodosa / Kussmaul disease / Kussmaul–Maier disease	-**NNÇTY**-
Polymyalgia rheumatica	-**NNÇTL**-
Urticarial vasculitis	-**NNÇTR**-

Vasculitis	-**NNÇTR̆**-
Chronic fatigue syndrome	-**NNŢPW**-
Complex regional pain syndrome / Amplified Musculoskeletal Pain Syndrome / Reflex Neurovascular Dystrophy / Reflex sympathetic dystrophy	-**NNŢPY**-
Eosinophilic esophagitis	-**NNŢPL**-
Gastritis	-**NNŢPR**-
Interstitial lung disease	-**NNŢPR̆**-
POEMS syndrome	-**NNŢKW**-
Raynaud's phenomenon	-**NNŢKY**-
Primary immunodeficiency	-**NNŢKL**-
Pyoderma gangrenosum	-**NNŢKR**-
IPEX syndrome / X-linked polyendocrinopathy / immunodeficiency and diarrhea-syndrome (XLAAD)	-**NNŢKR̆**-

7.2.8 Idiopathic Diseases/Disorders, Non-Genetic Congenital Disorders, or Diseases/Disorders of Unknown/Uncertain Etiology

The Stems and Specifications of the following roots follow the pattern of the root -**ÇK**- shown in Sec. 7.2.3 above. They may also be used with the new **ADI** affix.

Agammaglobulinemia	-**N̆N̆SPW**-
Amyloidosis	-**N̆N̆SPY**-
Amyotrophic lateral sclerosis (ALS) / Lou Gehrig's disease / Motor Neuron Disease	-**N̆N̆SPL**-
Anti-tubular basement membrane nephritis	-**N̆N̆SPR**-
Arterio-venous malformation	-**N̆N̆SPR̆**-
Atherosclerosis	-**N̆N̆SPH**-
Atopic allergy	-**N̆N̆SKW**-
Atopic dermatitis	-**N̆N̆SKY**-
Autoimmune peripheral neuropathy	-**N̆N̆SKL**-
Blau syndrome	-**N̆N̆SKR**-
Castleman's disease	-**N̆N̆SKR̆**-

Central hypoventilation syndrome, Congenital central hypoventilation syndrome, Acquired central hypoventilation syndrome / Ondine’s curse	-**N̆N̆SKH**-
Chronic obstructive pulmonary disease (including emphysema, chronic bronchitis)	-**N̆N̆STW**-
Chronic recurrent multifocal osteomyelitis / Majeed syndrome	-**N̆N̆STY**-
Complement component 2 deficiency	-**N̆N̆STL**-
Congenital heart block	-**N̆N̆STR**-
Contact dermatitis	-**N̆N̆STR̆**-
Cushing's syndrome	-**N̆N̆STH**-
Cutaneous leukocytoclastic angiitis	-**N̆N̆SPW**-
Dego's disease	-**N̆N̆SPY**-
Cerebral palsy	-**N̆N̆SPL**-
Eczema	-**N̆N̆SPR**-
Eosinophilic gastroenteritis	-**N̆N̆SPR̆**-
Erythroblastosis fetalis	-**N̆N̆SPH**-
Fibrodysplasia ossificans progressiva	-**N̆N̆SKW**-
Gastrointestinal pemphigoid	-**N̆N̆SKY**-
Gout	-**N̆N̆SKL**-
Hypogammaglobulinemia	-**N̆N̆SKR**-
Idiopathic giant-cell myocarditis / Giant cell myocarditis	-**N̆N̆SKR̆**-
Idiopathic pulmonary fibrosis / Fibrosing alveolitis	-**N̆N̆SKH**-
IgA nephropathy / IgA nephrits / Berger's disease / Synpharyngitic Glomerulonephritis	-**N̆N̆STW**-
Irritable bowel syndrome	-**N̆N̆STY**-
Ligneous conjunctivitis	-**N̆N̆STL**-
Narcolepsy	-**N̆N̆STR**-
Pellucid marginal degeneration (PMD)	-**N̆N̆STR̆**-
Rapid-onset obesity with hypthalamic dysregulation, hypoventilation, and autonomic dysregulation (ROHHAD)	-**N̆N̆STH**-
Rasmussen's encephalitis	-**N̆N̆ZPW**-
Serum sickness	-**N̆N̆ZPY**-
Spina bifida	-**N̆N̆ZPL**-
Spondyloarthropathy	-**N̆N̆ZPR**-

Sweet's syndrome	-**N̆N̆ZPŘ**-
Takayasu's arteritis	-**N̆N̆ZPH**-

7.2.9 Psychological/Behavioral Disorders

The Stems and Specifications of the following roots follow the pattern of the root -**ÇK**- shown in Sec. 7.2.3 above. They may also be used with the new **ADI** affix.

Disabilities relating to a deficit in a particular behavior or skill can be expressed by a use of the Type-2 INB/7 or INB/8 affix or other applicable affixes (e.g., speech communication disorders, stuttering, etc., hand-eye coordination, insomnia, clumsiness, etc.). Phobias can be expressed either by use of the Type-2 ENG/1 or ENG/2 affix or by concatenating an appropriate stem with Stem 3 of he root -**Ẓ**GR-.

intellectual disability / intellectual developmental disorder / global development delay	-**MMZBW**-
autism spectrum disorder	-**MMZBY**-
attention-deficit hyperactivity disorder	-**MMZBL**-
bipolar disorder	-**MMZBR**-
mania	-**MMZBŘ**-
monopolar depression, depressive disorder	-**MMZGW**-
generalized anxiety disorder	-**MMZGY**-
social anxiety disorder	-**MMZGL**-
panic disorder	-**MMZGR**-
separation anxiety disorder	-**MMZGŘ**-
acute stress disorder	-**MMZDW**-
adjustment disorder	-**MMZDY**-
PTSD	-**MMZDL**-
reactive adjustment disorder	-**MMZDR**-
dissociative amnesia	-**MMZDŘ**-
dissociative identity disorder	-**NNZBW**-
depersonalization/derealization disorder	-**NNZBY**-

somatic symptom disorder	-**NNZBL**-
conversion disorder	-**NNZBR**-
factitious disorder (e.g., Munchausen syndrome)	-**NNZBŘ**-
anorexia nervosa	-**NNZGW**-
bulimia	-**NNZGY**-
rumination disorder	-**NNZGL**-
pica	-**NNZGR**-
binge-eating disorder	-**NNZGŘ**-
prasomnia (e.g., sleepwalking, night terrors)	-**NNZDW**-
kleptomania	-**NNZDY**-
pyromania	-**NNZDL**-
intermittent explosive disorder	-**NNZDR**-
conduct disorder	-**NNZDŘ**-
oppositional defiant disorder	-**ŇŇZBW**-
substance addiction	-**ŇŇZBY**-
gambling disorder	-**ŇŇZBL**-
sex addiction	-**ŇŇZBR**-
delirium	-**ŇŇZBŘ**-
neurocognitive disorder	-**ŇŇZGW**-
schizophrenia	-**ŇŇZGY**-
catatonia	-**ŇŇZGL**-
obsessive-compulsive disorder (OCD)	-**ŇŇZGR**-
body-dysmorphic disorder	-**ŇŇZGŘ**-
hoarding disorder	-**ŇŇZDW**-
excoriation disorder	-**ŇŇZDY**-
personality disorder	-**ŇŇZDL**-
antisocial personality disorder	-**ŇŇZDR**-
avoidant personality disorder	-**ŇŇZDŘ**-
borderline personality disorder	-**MMZVL**-
dependent personality disorder	-**MMZVR**-

histrionic personality disorder	-**MMZVŘ**-
narcissistic personality disorder	-**NNZVL**-
obsessive-compulsive personality disorder	-**NNZVR**-
paranoid personality disorder	-**NNZVŘ**-
schizoid personality disorder	-**N̆N̆ZVL**-
schizotypal personality disorder	-**N̆N̆ZVR**-
Tourette syndrome	-**N̆N̆ZVŘ**-

7.2.10 Miscellaneous Roots Related to Illness

-ŘD- 'REMEDY / HEAL / RECOVER / RECUPERATE / RESTORE'		
BSC	**Stem 1:** (to be) a state/act of medically treating, remedying something or someone; to treat [medically], to (try to) remedy something	**Stem 2:** (to be) a state/act of recuperation/healing/recovery; to recuperate, to heal, to recover
CTE	(to be) a state of [medical] treatment	
CSV	(to be) a physical act of treating/remedying	**Stem 3:** (to be) a state/act of restoration; to restore
OBJ	(to be) a remedy/treatment	

-ŘDV- 'MEDICATION/MEDICINE'		
BSC	**Stem 1:** (to be) an act/state of administering a medicine/medicaton orally in solid form (e.g., pill, tablet, capsule, etc.); to administer a (solid) oral medication	**Stem 2:** (to be) an act/state of administering a medicine/medication orally in liquid form (e.g., cough syrup, chemical dissolved in liquid, etc.); to administer a liquid medication
CTE	(to be) a state of being on a solid oral medication	
CSV	(to be) an act of swallowing a (solid) oral medication; to swallow a pill/tablet/capsule	**Stem 3:** (to be) to be) an act/state of administering a medicine/medication via injection; to inject a medication
OBJ	(to be) the (pill/tablet/capsule of) medication itself	

-ŠKL- 'PSYCHIATRIC TREATMENT / THERAPY'		
BSC	**STEM 1:** (to be) an act of psychiatric treatment or therapeutic psychological therapy	**STEM 2:** (to be) an act of subliminal psychological manipulation **STEM 3:** (to be) an act of brainwashing [i.e., psychological manipulation to the extent that a person's personality/motivations/values/memories/beliefs, etc. are changed for covert or nefarious purposes]
CTE	(to be) a state of being under or subject to psychiatric/psychological treatment/therapy	
CSV	(to be) a physical act of administering psychiatric/psychological treatment/therapy; to treat a patient by means of psychiatry/psychological therapy	
OBJ	(to be) the nature/specifics of a (course of) psychiatric/psychological therapy	

-DGW- 'BANDAGE / DRESSING / PATCH'		
BSC	**STEM 1:** (to be) an act/instance of bandaging a wound; to cover a wound with a bandage, to bandage a wound	**STEM 2:** (to be) an act/instance of using a pad/compress/dressing to absorb/staunch loss of blood or fluid **STEM 3:** (to be) an act/instance of using a piece or section of material as a patch to cover, seal, and protect a wound or area of damage to something; to patch something
CTE	(to be) a state of protecting a wound by having a bandage on it; to have a bandage on	
CSV	(to be) a physical act of bandaging, to cover with a bandage	
OBJ	(to be) a bandage	

The Specification pattern of the above root applies to the following root as well:

-DGY- 'MATERIAL USED AS BANDAGE or PATCH'
1. (adhesive) tape 2. rubber(-like) patch 3. film(-like material)

7.3 BIOLOGICAL LIFE-FORMS

In addition to roots and stems associated with formalized, hierarchically arranged biological taxonomies (which are beginning in Sec. 7.3.1-+), the following roots and stems are available for informal use when talking about biological life-forms in general:

-ŠW- 'LIFE-FORM, LIVING THING'		
BSC	**STEM 1:** (to be) a living being; a life-form; a form of life; to be something that is alive	**STEM2:** (to be) something vital; manifest vitality; activity consistent with being alive **STEM 3:** (to be) something reproduced or brought to life; to reproduce [biologically], bring to life
CTE	(to be) the individual essence/soul, set of instincts, or mind of a life-form (that distinguishes it from other individuals)	
CSV	(to be) the physical body or corporeal aspect of a living thing	
OBJ	(to be) an entity/party impacted/affected by the presence of, or interaction with a living being	

As will be seen beginning in Sec. 7.3.1 in the quasi-taxonomic listings of animals, numerous animal classes are divided up into several roots due to the diversity of similar taxonomic/phylogenic orders, sub-orders, infra-orders, families, super-families, etc. An example of this is the myriad number of moth types. However, it becomes necessary for laypersons to have a simple generic term for 'moth' irrespective of the particular type of moth. While the **RES**, **TXN**, and **HSR** affixes listed later below are available to form generic terms, these generic notions of animal types are semantically salient enough to warrant their own "shortcut" roots/stems for non-scientific, non-taxonomic lexical categorization. Such roots/stems are provided here; they follow the same Specification pattern as the root **-ŠW-** above:

-FST- 'PROKARYOTIC, NON-VISIBLE LIFE FORMS'
1. bacterium
2. archaeon
3. virus

-SV- 'EUKARYOTIC LIFE FORM'
1. plant
2. animal
3. non-plant / non-animal eukaryote, i.e., protist, fungus, or chromist)

-KST- 'NON-PLANT & NON-ANIMAL EUKARYOTIC LIFE-FORM'
1. protist (i.e., protozoa, amoeboids, protophytic algae, slime molds)
2. fungus
3. chromist (brown algae, diatoms, water molds)

-LḌ- 'TREE / SHRUB / BUSH'
1. tree
2. shrub
3. bush

-LḌR- 'GENERAL TREE TYPE'
1. conifer or conifer-like evergreen tree
2. non-conifer evergreen tree
3. deciduous tree

-LḌL- 'NON-VASCULAR (i.e., PRIMITIVE) PLANT FORM'
1. alga
2. moss
3. hornwort, liverwort, or lichen

-ZHW- 'GRASS-LIKE, STALK-LIKE PLANT FORM'
1. grass
2. stalk-like plant
3. cereal grain or pseudo-cereal grain (e.g., wheat, rice, buckwheat, amaranth, etc.)

-ZTÇ- 'FERN-LIKE PLANT'
1. fern
2. eusporangiate fern (e.g., whisk fern, grapefern, horsetail, adder's tongue)
3. lycopod (e.g., club moss, firmoss, spike moss, quillwort)

-LMY- 'FLOWERING PLANT'
1. flowering plant/tree/bush
2. herb
3. vine / liana

-VN- 'BASIC ANIMAL TYPES'
1. animal of land or air ("terroid")
2. waterlife-animal, animal living in water ("aquoid")
3. amphibian

-TN- 'MAMMAL'
1. placental mammal
2. marsupial
3. monotreme (egg-laying mammal)

-L- 'HUMAN BEING'
1. adult human being
2. human child
3. adolescent human being, "teenager"

-ŢL- 'ANIMALS OF THE AIR' (i.e., capable of self-powered flight)
1. bird
2. flying insect
3. bat

-ZZ- '(SMALLER) REPTILE'
1. lizard-like reptile (e.g., lizard, skink, gecko, monitor)
2. snake or legless lizard
3. turtle/tortoise

-JXW- 'AMPHIBIAN'
1. Salientian amphibian (i.e., frog, toad)
2. Caudatan amphibian (i.e., salamander, newt)
3. Gymnophione (caecilian) amphibian (i.e., snake-like or worm-like amphibian)

-FŇ- 'NON-PLANT WATERLIFE FORM'
1. motile, free-swimming waterlife form, e.g., fish, eels, whales, octopus, etc.
2. crawling waterlife form, e.g., crustaceans, crabs
3. non-motile/sessile waterlife form, e.g., shellfish, sea urchin, starfish, etc.

-PFT- ‘TYPES OF FISH’
1. skate / ray
2. shark
3. [lobe-finned or ray-finned] fish

-NGL- ‘MOLLUSC [other than cephalopod]’
1. snail or slug
2. bivalve (e.g., clam, mussel, scallop, oyster, cockle, etc.)
3. other non-cephalopod mollusc (e.g., nautilus, chiton, solenogaster, etc.)

-NGW- ‘WORM-LIKE INVERTEBRATE’
1. worm (i.e., annelid worm or nematode worm)
2. flatworm or tapeworm
3. fluke or leech

-NGŘ- ‘OTHER INVERTBRATE’
1. cephalopod (e.g., octopus, squid, cuttlefish, argonaut, etc.)
2. sponge (i.e., poriferan)
3. cnidarian, ctenophore, or placozoan (e.g., jellyfish, coral, sea anemone, hydra, comb jelly, etc.)

-ŽJ- ‘ARTHROPOD’
1. hexapod or chelicerate (i.e., insect or arachnid, including springtail, conehead, dipluran, horseshoe crab, and sea spider)
2. crustacean
3. myriapod (i.e., millipede or centipede)

-ZT- ‘ARACHNID’
1. spider or harvester (a.k.a. harvestman or daddy longlegs)
2. scorpion or solifuge (a.k.a. wind scorpion, camel spider, sun spider)
3. tick or mite

-ḌP- ‘LARGE DECAPODIAN CRUSTACEAN’
1. lobster / crayfish
2. crab
3. shrimp / prawn

-GLL- ‘BUG-LIKE INSECT’
1. beetle or beetle-like insect (in terms of appearance/behavior)
2. ant or ant-like insect (in terms of behavior/appearance)
3. rounded “bug”-like insect other than beetle (e.g., bed bug, stink bug, junebug, etc.)

-ZZC- ‘STINGING FLYING INSECT’
1. bee, bumblebee or bee-like insect (in terms of appearance/behavior)
2. wasp, hornet, sawfly, or wasp-like insect (in terms of appearance/behavior)
3. gnat / midge / mosquito or similar insect (in terms of appearance/behavior)

-NNZ- 'FLY-LIKE INSECT'
1. fly or fly-like insect (in terms of appearance/behavior)
2. louse, aphid, thrip or similar insect (in terms of behavior/appearance)
3. lacewing or lacewing-like insect (in terms of appearance/behavior), e.g., stonefly, alderfly, fishfly, antlion, owlfly, etc.

-NV- 'BUTTERFLY / MOTH'
1. butterfly
2. moth
3. caterpillar

-KSTW- 'FUNGUS FUNCTIONAL TYPE / MUSHROOM'
1. mushroom (edible)
2. non-harmful / non-parasitic fungus
3. harmful (i.e., parasitic, pathogenic or poisonous) fungus/mushroom

-KSTY- 'OTHER FUNGUS-RELATED LIFEFORM
1. a yeast
2. a mold
3. a lichen

-KSTF- 'PROTIST-TYPE'
1. non-amoeboid protozoan
2. amoeboid life-form
3. protophytic alga or slime mold

-VŽG- 'GAMETOPHYTE / SPOROPHYTE / ZYGOTE'
1. gametophyte
2. sporophyte
3. zygote

-ŽŇ- 'NON-MAMMALIAN LIFE STAGE'			
BSC	**STEM 1:** (to be) a newborn, newly hatched, newly sprouted, first-stage life-form (e.g., sprout, larva, tadpole, caterpillar, gametophyte, sporophyte, etc.)	**STEM 2:** (to be) the pupal stage of a metamorphic life-form	**STEM 3:** (to be) the "birth" or hatching of a life-form from a cocoon, chrysalis, external (shell-bearing) egg, or other interim metamorphic life-stage; to hatch, emerge from a cocoon, emerge from a marsupial pouch, etc.
CTE	(to be) the individual essence/soul, set of instincts, or mind of a first-stage life-form (that distinguishes it from other individuals)	(to be) the state of metamorphosis undergone by a pupa	(to be) the state of having completed a metamorphic life-stage and becoming one's final corporeal form
CSV	(to be) the physical body or corporeal aspect of a first-stage life-form	(to be) a pupa, i.e., a life-form in a pupal stage of development	(to be) the process of hatching/emerging itself
OBJ	(to be) an entity/party impacted/affected by the presence of, or interaction with a first-stage life-form	(to be) a cocoon or chrysalis	(to be) the discarded cocoon, egg-casing, bud, etc. emerged from

7.3.0.1 Preliminary Note on Quasi-Taxonomic Roots/Stems Denoting Biological Life-forms

Beginning in Section 7.3.1 below, the roots and stems or specific life-forms are presented. The general rule regarding roots associated with biological life-forms is the use of a simple two- or three- (sometimes four-) consonant root to designate the "base level" (i.e., most common or most representative) genus and/or species of a particular class/family/order/sub-order of animals, e.g., common domesticated animals, common insects, common birds, common reptiles, etc. Each of these roots then serves as the basis for expansion of the root by a single consonant (e.g., the adding of -**y**, -**w**, -**l**, -**r**, -**ř**, -**m**, -**n**, -**f**, -**ţ**, -**v**, etc. and/or the prefixing of **l**-, **r**- or **ř**-) to designate less common genera in the same class/family/order/sub-order or to designate corollary or associated roots.

When the distinction between similar life-forms is a common one, stems are assigned to individual species within a genus. Otherwise, the designation of species and sub-species is to be accomplished via use of a Type-2 V_XC_S affix or via Type-2 stem concatenation, the specific affixes and concatenated roots as-yet-to-be-determined. As an example, Stem No. 3 of the root -**RRN**- 'OCELOT-LIKE FELINES' designates any less common species of the genus *Leopardus* other than the ocelot itself (*L. pardalis*) (designated by Stem No. 1) or the margay (*L. wiedii*) (designated by Stem No. 2). Therefore, Stem No. 3 refers to an animal that could be either an oncilla (*L. tigrina*), an Andean mountain cat (*L. jacobita*), a Pampas cat (*L. colocola*), a kodkod (*L. guigna*), a southern tigrina (*L. guttula)*, or Geoffroy's cat (*L. geoffroyi*). To distinguish these individual six species, either an appropriate Type-2 V_XC_S affix would be used or a Type- 2 concatenated stem. At this time the author has not yet assigned the specific affixes or stems to be used due to time constraints.

NOTE: The three stems of many of these roots are structured to reflect quasi-trinary divisions within the scientific taxonomies of each animal-type. Often, these quasi-trinary distinctions are of little practical importance to laypersons, in which case the use of Stem "Zero" may be of the most practical value when referring to the particular animal. As an example, the three stems of the root -**MVY**- refer to the three species (or sub-species) of bison; however, most speakers will likely be uninterested in (or ignorant about) distinguishing between a plains bison versus a wood bison versus a wisent; thus, use of Stem "Zero" to refer simply to a bison is convenient. (Alternately, use of Stem 1 with additional morphology such as a "fuzzy" configuration, or RPV Essence, or the REA/7 affix may be used.)

7.3.0.2 Affixes Used with Taxonomic Stems for Genus or Species or Sub-Species Identification

In addition to Type-2 stem-concatenation, the following affixes are available to use with biological life-form stems to help differentiate (or, if used as a Type-2 affix, to formally name) the species or sub-species of a particular genus, or in cases where a stem refers to an entire family or class of life-forms (as is the case for less familiar or less differentiated life-forms such as annelid worms), to identify or name a particular genus. (Note that the use of affixes to identify species of a genus are certainly not limited to the following affixes; these affixes are presented here only because they have been specifically designed for use with such stems. Conversely, these affixes may also be used with any relevant/applicable stem, and are not reserved solely for use with stems denoting biological/taxonomic life-forms.)

-ẓč	PG1 Physical Features Based on Geography 1 for Species or Sub-Species Differentiation
1	having physical features characteristic of mainland Asian origin or environment
2	having physical features characteristic of Southeast Asian/Polynesian origin or environment
3	having physical features characteristic of Central Asian origin or environment
4	having physical features characteristic of the Indian sub-continent in origin or environment
5	having physical features characteristic of Western Asian/Mesopotamian origin or environment
6	having physical features characteristic of Western European/Nordic origin or environment
7	having physical features characteristic of Eastern European/Balkans origin or environment
8	having physical features characteristic of sub-Saharan African origin or environment
9	having physical features characteristic of North African/Mediterranean origin or environment

-ẓj	PG2 Physical Features Based on Geography 2 for Species or Sub-Species Differentiation
1	having physical features characteristic of Arctic origin or environment
2	having physical features characteristic of Mediterranean Sea origin or environment
3	having physical features characteristic of Pacific Ocean or environment
4	having physical features characteristic of North Atlantic origin or environment
5	having physical features characteristic of South Atlantic origin or environment
6	having physical features characteristic of South Pacific oceanic origin or environment
7	having physical features characteristic of Indian Ocean origin or environment
8	having physical features characteristic of Antarctic origin or environment
9	having physical features characteristic of New World / American origin or environment

-ẓg	Biogeographic Realm/Ecozone
1	inhabiting or originating in the Palearctic ecozone
2	inhabiting or originating in the Nearctic ecozone
3	inhabiting or originating in the Holarctic ecozone
4	inhabiting or originating in the Afrotropic ecozone
5	inhabiting or originating in the Neotropic ecozone
6	inhabiting or originating in the Australasia ecozone
7	inhabiting or originating in the Indomalaya ecozone
8	inhabiting or originating in the Oceania ecozone
9	inhabiting or originating in the Antarctic ecozone

-šb	GEO Geographic or Environmental Niche for Species or Sub-Species Differentiation
1	inhabiting or associated with high mountains
2	inhabiting or associated with the desert
3	inhabiting or associated with rocky terrain
4	inhabiting or associated with the tundra
5	inhabiting or associated with brushland, scrubland, bushy terrain
6	inhabiting or associated with swampy/boggy/marshy terrain
7	inhabiting or associated with savannah/grassland
8	inhabiting or associated with trees / arboreal
9	inhabiting or associated with the rainforest

-šd	WBN Water-based Environmental Niche for Species or Sub-Species Differentiation
1	inhabiting or associated with islands
2	inhabiting or associated with offshore / shallow ocean dwelling
3	inhabiting or associated with the seashore / littoral
4	inhabiting or associated with estuaries/lagoons
5	inhabiting or associated with saltwater marshes
6	inhabiting or associated with freshwater lakes
7	inhabiting or associated with freshwater ponds
8	inhabiting or associated with rivers / riparian
9	inhabiting or associated with streams or brooks

-šg	OBN Ocean-based Environmental Niche for Species or Sub-Species Differentiation
1	inhabiting or associated with the intertidal zone
2	inhabiting or associated with a coral reef
3	inhabiting or associated with the neritic zone
4	inhabiting or associated with the pelagic zone
5	inhabiting or associated with seamount
6	inhabiting or associated with hydrothermal vent
7	inhabiting or associated with a cold seep
8	inhabiting or associated with the demersal zone
9	inhabiting or associated with the benthic zone

-zf	ENS Environmental Niche for Species or Sub-Species Differentiation
1	active at twilight / crepuscular
2	active at night / nocturnal
3	active around dawn
4	active during the morning
5	active during the day / diurnal
6	sessile, not motile -- adhering to a substrate by direct attachment (not via a stalk/stipe/pedicel/connecting medium)
7	attached to a substrate via a stalk/stipe/pedicel/connecting medium
8	motile in reaction to heat
9	motile in reaction to light

-ţč	DBM Distinctive Bodily/Fur Markings for Species or Sub-Species Differentiation
1	having rows of lighter-colored spots
2	having rows of darker-colored spots
3	having lighter-colored facial band(s)
4	having darker-colored facial band(s)
5	having a distinctive pattern of fur/bodily markings colored differently from the rest of the body/fur
6	having lighter-colored dorsal band(s) or stripe(s)
7	having darker-colored dorsal band(s) or stripe(s)
8	having lighter-colored lateral band(s) or stripe(s)
9	having darker-colored lateral band(s) or stripe(s)

-vž	MB1 Morphological Bodily Distinctions for Species or Sub-Species Differentiation 1
1	having a longer-than-average tail
2	having a larger-than-average-mouth/beak/gullet
3	having larger-than-average eyes
4	having larger/longer-than-average wings/wingspan
5	having larger/longer-than-average nose/snout/proboscis
6	having larger/longer-than-average limbs
7	having a larger or more robust body than average
8	being longer/taller than average
9	being more rotund/fatter/thicker than average

-ţj	MB2 Morphological Bodily Distinctions for Species or Sub-Species Differentiation 2
1	having a shorter-than-average tail
2	having a smaller-than-average-mouth/beak/gullet
3	having smaller-than-average eyes
4	having smaller/shorter-than-average wings/wingspan
5	having smaller/shorter-than-average nose/snout/proboscis
6	having smaller/shorter-than-average limbs
7	having a smaller body than average
8	being shorter than average
9	being more slender/streamlined than average

-cd	RES Resemblances to Other Species
1	having behavioral characteristics similar to X
2	having eating/foraging/hunting habits similar to X
3	having a voice/call/cry/whistle, etc. similar to X
4	inhabiting an environmental niche similar to X
5	having various factors of appearance and behavior similar to X
6	having a facial resemblance to X
7	having a bodily form similar to X
8	having bodily markings similar to X
9	having nesting/parenting/burrowing habits similar to X

Species differentiation can also be designated using the PTY affix.

-zv	SYM Symbiotic Relationship
1	in an obligate [i.e., required/mandatory/necessary] relationship with another party to mutual benefit, i.e., obligate mutualistic relationship
2	in an obligate [i.e., required/mandatory/necessary] relationship to the benefit of oneself but neutral to the other party, i.e., obligate commensal relationship
3	in an obligate [i.e., required/mandatory/necessary] relationship to the benefit of oneself but harmful to the other party, i.e., obligate parasitic relationship
4	in a facultative [i.e., optional] relationship with another party to mutual benefit, i.e., optionally mutualistic relationship
5	in a facultative [i.e., optional] relationship to the benefit of oneself but neutral to the other party, i.e., optional commensal relationship
6	in a facultative [i.e., optional] relationship to the benefit of oneself but harmful to the other party, i.e., optional parasitic relationship
7	in an obligate or facultative relationship with another party of no benefit to oneself but advantageous/beneficial to the other party
8	in an obligate or facultative relationship with another party harmful to oneself but advantageous/beneficial to the other party
9	in a competitive relationship with another party [potentially] harmful to both parties

The eleven **Areal/Cultural Association** affixes may also be used with taxonomic genus/species stems (see the Affixes document)

7.3.0.3 Affixes for Grouping Biological Life-Form Roots/Stems into Classificatory Schemata/Taxonomies

-bv	**TXN Taxonomic Classification Hierarchy (to be added to the Base Root forms for various groupings)**	Type-2 **TXN** affixes are used specifically for biological taxonomic classification, as listed below:
1	X as representative of a 3rd-level sub-grouping of the basic-level grouping	the sub-species or strain of which X is a member
2	X as representative of a sub-sub-group or sub-sub-section of X's base-level group	the species of which X is a member
3	X as representative of a sub-group or sub-section of X's base-level group	the genus of which X is a member
4	X as the basic or common hierarchical level representative of a group; X as member, X as representative	the tribe of which X is a member
5	the "tribe", group, club, section of which X is a member	the family of which X is a member
6	the "family" of which X is a member; the grouping to which X's "tribe"/club/group/section belongs	the order of which X is a member
7	3rd-order hierarchical super-group of which X is a member; the grouping to which the grouping containing X's group/section belongs	the class of which X is a member
8	4th-order hierarchical super-group of which X is a member	the phylum of which X is a member
9	5th-order hierarchical super-group of which X is a member	the kingdom of which X is a member

-dv	**HSR Hierarchical Sub- and Super-Rankings**
1	sub-sub- (two degree of ranking below)
2	sub- (one degrees of ranking below)
3	comprising some of a section; some of the sub-sections of a section
4	comprising most of a section but not including all sub-sections; most of the sub-sections of a section
5	comprising more than one sub-section
6	comprising more than one section
7	comprising at least one section and at least one subsection of another section
8	super- (one degree of ranking above)
9	super-super (two degrees of ranking above)

7.3.0.4 Affixes for Describing Extinct Genera

With few exceptions, the list of quasi-taxonomic roots for life-forms below do not include extinct genera. To signify/name an extinct genus, choose an extant genus in the same family/order, etc., apply the appropriate RES affix (shown above), then add the new PEB affix in Degree-9 shown below, along with one of the additional affixes shown below indicating a particular era or epoch (if signifying a fossil species).

-rç-	PEB Overall Presence/Participation Within an Environmental Niche or Biosphere
1	dominant component/feature/element/life-form/factor
2	near-dominant
3	thriving
4	up-and-coming, close to thriving
5	newfound / newly arrived / newly present
6	declining
7	threatened
8	near-extinct
9	extinct

-xn-	PZE Paleozoic Era
1	of/from/during the Paleozoic Era
2	of/from/during the Early (or Lower) or Middle Cambrian Period
3	of/from/during the Late (Upper) Cambrian epoch or Furongian period
4	of/from/during the Ordovician period
5	of/from/during the Silurian period
6	of/from/during the Devonian period
7	of/from/during the Mississippian period of the Carboniferous period
8	of/from/during the Pennsylvanian period of the Carboniferous period
9	of/from/during the Permian period

-pn-	MZE Mesozoic Era
1	of/from/during the Mesozoic Era
2	of/from/during the Early (or Lower) Triassic period
3	of/from/during the Middle Triassic period
4	of/from/during the Late (or Upper) Triassic period
5	of/from/during the Early (or Lower) Jurassic period
6	of/from/during the Middle Jurassic period
7	of/from/during the Late (or Upper) Jurassic period
8	of/from/during the Early (or Lower) Cretaceous period
9	of/from/during the Late (or Upper) Cretaceous period

-gn-	CZE Cenozoic Era
1	of/from/during the Cenozoic Era
2	of/from/during the Paleocene epoch
3	of/from/during the Eocene epoch
4	of/from/during the Oligocene epoch
5	of/from/during the Miocene epoch
6	of/from/during the Pliocene epoch
7	of/from/during the Pleistocene epoch
8	of/from/during the Holocene epoch
9	of/from/during the Anthropocene epoch

7.3.1 Animals

All of the following stems referring to living animals have the following Specification structure:

BSC	(to be) an animal as a holistic entity, including its physical/corporeal body and its and mental identity and living essence; to live/be alive
CTE	(to be) that which gives a particular animal its individual identity; the living essence or mental identity of an animal
CSV	(to be) the physical body of an animal; the corporeal aspect of an animal
OBJ	(to be) an activity engaged in by an animal; what an animal is doing; to act (as a particular animal species does)

The listing of roots and stems below are grouped by more or less familiar taxonomic classification schemes, with pragmatic factors such as degree of familiarity/utility among laypersons taken into account. For example, animal orders/classes familiar to laypersons such as mammals have many stems at the species level, while unfamiliar orders/classes such as Annelids forego taxonomic lexicalization entirely and instead lexicalize common-sense, superficial morphological distinctions, i.e., 'earthworm' vs. 'leech', vs. 'other ringed/segmented worm'.

7.3.1.1 Mammals

Feliforms

-**RR**- 'CAT' 1. (*Felis catus*) cat (domestic) 2. (genus *Felis*, other than *catus* species) wildcat 3. (genera *Prionailurus* or *Otocolobus*) leopard cat, fishing cat, flat-headed cat, rusty-spotted cat, Pallas's cat/manul

-**RRW**- 'LARGE FELINE' (genus *Panthera*) ' 1. tiger 2. lion 3. jaguar

-**RRY**- 'LEOPARD' (genera *Panthera* and *Neofelis*) 1. leopard 2. snow leopard 3. clouded leopard

-**RRM**- 'LYNX-LIKE ANIMAL' (genus *Lynx*) 1. (*L. rufus*) bobcat 2. (*L. lynx, L. pardinus*) Eurasian or Iberian lynx 3. (*L. canadensis*) Canada lynx

-**RRN**- 'OCELOT-LIKE FELINE' (genus *Leopardus*)' 1. ocelot 2. margay 3. other ocelot-like feline (e.g., oncilla, pampas cat, kodkod, tigrina, etc.)

-**RRL**- 'PUMA-LIKE FELINE' 1. (genus *Acinonyx*) cheetah 2. (genus *Puma*) cougar (a.k.a. puma/panther/catamount) 3. (genus *Herpailurus*) jaguarundi

-**RRF**- 'BAY CAT FELINE' (genera *Catopuma, Pardofelis*)' 1. bay cat 2. Asian golden cat 3. marbled cat

-**RRṬ**- 'CARACAL FELINE' (genera *Caracal, Leptailurus*)' 1. caracal 2. African golden cat 3. serval

-**RRC**- 'OTHER FELIFORM' 1. (genus *Nandinia*) African palm civet 2. (genus *Prionodon*) banded linsang 3. (genus *Prionodon*) spotted linsang

Canines

-**ZV**- 'COMMON CANINE' 1. (*Canis lupus familiaris, C.lupus dingo*) domestic dog, dingo 2. (*Canis lupus*) gray wolf 3. (*Canis latrans*) coyote (= American jackal)

-**ZVY**- ' OTHER CANINE' 1. (genus *Canis*; species *anthus, simensis, aureus, mesomelas, adustus*) African wolf, Asian wolf, jackal 2. (genus *Cuon*) dhole, Asiatic wild dog 3. (genus *Lycaon*) African wild dog/painted wolf

-**ZVW** - 'FOX' 1. (*Vulpes vulpes*) red fox 2. (other *Vulpes* species: *zerda, cana, chama, rueppellii, corsac, ferrilata, macrotis, lagopus*) fox 3. (genus *Otocyon*) bat-eared fox

-**ZVL**- 'SOUTH AMERICAN CERDOCYONINE FOX/DOG/WOLF' 1. (genus *Lycalopex*; species *vetulus, sechurae, fulvipes, gymnocercus, griseus, culpaeus)* [South American] fox 2. (genus *Speothos*) bush dog 3. other South American cerdocyonine canine (e.g., genera *Chrysocyon, Dusicyon, Cerdocyon, Atelocynus*)

-**ZVR**- 'ADDITIONAL CANINE GENUS' 1. (genus *Urocyon*) gray fox 2. (genus *Urocyon*) island fox 3. (genus *Nyctereutes*) raccoon dog

Ovids

-**NÇ**- 'GOAT-LIKE ANIMAL' 1. (*Capra aegagrus hircus*) domestic goat 2. ibex 3. (genera *Capra, Hemitragus, Arabitragus, Ammotragus*) wild goat, tahr, tur, markhor

-**NÇW**- 'SHEEP-LIKE ANIMAL' 1. (*Ovis aries*) domestic sheep 2. (genus *Ovis*, other than *aries* species) wild sheep, mouflon, argali, urial, dall 3. (genera *Nilgritragus, Pseudois*) bharal, dwarf blue sheep, Nilgiri tahr

-**NÇL**- 'OTHER CAPRINOID ANIMAL' 1. (genus *Capricornis*) serow 2. (genus *Nemorhaedus*) goral 3. (genera *Oreamnos, Rupicapra*) mountain goat, chamois

-**NÇR**- 'OVIBOVINE' 1. (genus *Ovibos*) musk-ox 2. (genus *Budorcas*) takin 3. golden takin

-**NÇŘ**- 'ANTELOPE' 1. (genus *Hippotragus*) antelope 2. (genus *Oryx*) oryx, gemsbok 3. (genus *Addax*) addax

-**NÇF**- 'ALCELAPHINE ANTELOPE' 1. (genus *Alcelaphus*) hartebeest 2. (genus *Connochaetes*) wildebeest 3. (genera *Damaliscus, Beatragus*) topi, bontebok, blesbok, tsessebe, korrigum, hirola

-**NÇŢ**- 'ANTELOPE-LIKE ANIMAL' 1. (genus *Aepyceros*) impala 2. (genera *Gazella, Eudorcas, Nanger, Procapra, Antidorcas*) gazelle, springbok 3. (genera *Litocranius, Antilope, Ammodorcas*) East-African or Asian antelope (e.g., dibatag, gerenuk, blackbuck)

-**NÇV**- 'DWARF ANTELOPE' 1. (genus *Madoqua*) dik-dik 2. (genera *Neotragus, Nesotragus*) Bates' pygmy antelope, suni, royal antelope 3. (genera *Dorcatragus, Oreotragus, Ourebia, Raphicerus*) beira, klipspringer, oribi, steenbok, grysbok

-**NÇM**- 'DUIKER' 1. (genus *Cephalophus*) duiker 2. (genus *Philantomba*) blue duiker, Maxwells's duiker, Walter's duiker 3. (genus *Sylvicapra*) common duiker

-**NÇN**- 'REDUNCINE ANTELOPE' 1. (genus *Redunca*) reedbuck 2. (*Kobus ellipsiprymnus*) waterbuck 3. (genus *Kobus* other than *ellipsyrymnus* species) kob, puku, lechwe

Bovines

-**MV**- 'CATTLE' 1. (*Bos primigenius*) cow/bull, cattle, auroch 2. (subgenus *Bibos*) gaur, gayal/mithun, banteng 3. (subgenera *Poephagus*, *Novibos*) yak, kouprey

-**MVY**- 'BISON' 1. (*Bison bison bison*) American plains bison 2. (*Bison bison athabascae*) American wood bison 3. (*Bison bonasus*) European bison/wisent

-**MVW**- 'WATER BUFFALO' 1. (genus *Bubalus*) Asian water buffalo 2. (genus *Bubalus*) anoa, tamaraw 3. (genus *Syncerus*) African water buffalo

-**MVL**- 'TRAGELAPHINE BOVINE' 1. (genus *Taurotragus*) common eland, giant eland 2. (genus *Tragelaphus*) greater and lesser kudu, nyala, mountain nyala 3. (genus *Tragelaphus*) bongo, sitatunga, kéwel, imbabala/bushbuck

-**MVR**- 'BOVINE-RELATED ANIMAL' 1. (genus *Giraffa*) giraffe 2. (genus *Okapia*) okapi 3. (genus *Pseudoryx*) saola

-**MVM**- 'BOVINE HYBRID' 1. dzo/yakow (cow/bull + yak) 2. "beefalo"/ "zubron" (cow + bison) 3. "yakalo" (bison + yak)

-**MVŘ**- 'OTHER BOVINE-LIKE ANIMAL' I 1. (genus *Antilocapra*) pronghorn 2. (genera *Hyemoschus, Moschiola*) chevrotain 3. (genus *Tragulus*) mouse-deer

-**MVN**- 'OTHER BOVINE-LIKE ANIMAL' II 1. (genus *Boselaphus*) nilgai 2. (genera *Tetracerus*) four-horned antelope / chousingha 3. other bovine-like or ovid-like animal

Cervines

-**LV**- 'CERVINE (OLD WORLD DEER) (genera *Cervus, Rusa*)' 1. [Old World] deer 2. [American] elk, wapiti 3. (genera other than *Cervus* or *Rusa*) hog deer, fallow deer, chital, swamp deer, thamin

-**LVW**- 'LARGER CAPREOLINE (NEW WORLD DEER) 1. (genus *Alces*) moose / Eurasian elk 2. (genus *Rangifer*) reindeer/caribou 3. (genus *Blastocerus*) marsh deer

-**LVY**- 'SMALLER CAPREOLINE (NEW WORLD DEER)' 1. (genus *Mazama*) brocket 2. (genus *Hippocamelus*) taruca, huemul 3. (genera *Capreolus, Ozotoceros*, *Pudu, Odocoileus*) roe deer, pudu, pampas deer, white-tailed deer, mule deer

-**LVR**- 'OTHER DEER(-LIKE ANIMAL)' 1. (genus *Muntiacus*) muntjac 2. (genus *Elaphodus*) tufted deer 3. (genus *Moschus*) musk deer

Ursids

-**RH**- 'BEAR (Genus *Ursus*)' 1. Black Bear (species *americanus*) 2. Grizzly Bear (species *horribilis*) 3. Big Brown Bear (species *arctos*)

-**RHM**- 'OTHER URSUS BEAR SPECIES' (Genus *Ursus* [previously *Thalarctos*]) 1. Polar Bear (species *maritimus*) 2. Sun bear (species *malayanus*) 3. Sloth bear (species *ursinus*)

-**RHN**- 'OTHER BEAR GENUS' 1. giant panda (genus *Ailuropoda*) 2. Qinling panda (genus *Ailuropoda*) 3. spectacled bear (genus *Tremarctos*)

Musteloids

-**ČT**- 'PROCYONINE' 1. (genus *Procyon*) raccoon 2. (genera *Nasua, Nasuella*) coati 3. (genus *Bassariscus*) ringtail cat, cacomistle

-**ČTW**- 'OTHER PROCYONID' 1. kinkajou 2. olingo 3. olinguito

-**ČTY**- 'MUSTELID' 1. (genus *Mustela*) weasel/polecat/stoat/European mink 2. ferret (= genus *Mustela*; domesticated weasel/polecat) 3. (genus *Neovison*) American mink

-**ČTL**- 'GULONINE' 1. (genera *Martes, Pekania*) marten/sable/fisher 2. (genus *Gulo*) wolverine 3. (genus *Eira*) tayra

-**ČTR**- 'OTTER' 1. (genus *Enhydra*) sea otter 2. (genus *Lontra*) river otter 3. other otters (genera *Aonyx, Lutra, Hydrictis, Lutrogale, Pteronura*)

-**ČTŘ**- 'BADGER' 1. (genus *Taxidea*) American badger 2. (genus *Meles*) European/Asian/Japanese badger 3. (genus *Arctonyx*) hog badger

-**ČTF**- 'SKUNK' 1. (genera *Mephitis, Spilogale*) skunk 2. (genus *Conepatus*) hognose skunk 3. (genus *Mydaus*) stink badger

-**ČTÇ**- 'ICTONYCHINEA' 1. (genus *Galictis*) grison 2. (genera *Ictonyx, Vormela*) striped polecat, marbled polecat 3. (genus *Lyncodon, Poecilogale*) Patagonian weasel, African striped weasel

-**ČTH**- 'OTHER MUSTELOID' 1. (genus *Ailurus*) red panda 2. (genus *Melogale*) ferret-badger, 3. (genus *Mellivora*) honey badger

Suids

-**PX**- 'SUID' 1. (*Sus scrofa domestica*) domestic pig 2. (*Sus scrofa*) wild boar 3. (other *Sus* species) warty pig, bearded pig

-**PXW**- 'OTHER SUID' 1. (genus *Porcula*) pygmy hog 2. (genus *Potamochoerus*) bushpig, red river hog 3. (genus *Babyrousa*) babirusa

-**PXL**- 'SUID-LIKE ANIMAL' 1. (genus *Phacochoerus*) warthog 2. (genus *Hylochoerus*) giant forest hog 3. (genus *Tayassu*) peccary/javelina/skunk pig

Hippopotamus

-**VḌ**- HIPPOPOTAMUS 1. (genus *Hippopotamus*) hippopotamus 2. (genus *Choeropsis*) pygmy hippo 3. [stem unassigned]

Cetaceans

-**BŽ-** 'CETACEAN' 1. baleen whale 2. toothed whale 3. delphinoid (includes oceanic dolphins, river dolphins, porpoises, and narwhals)

-**BŽW**- 'SPERM WHALE' 1. (genus *Physeter*) sperm whale 2. (*Kogia breviceps*) pygmy sperm whale 3. (*Kogia sima*) dwarf sperm whale

-**BŽY**- 'RIVER DOLPHIN' 1. (genus *Platanista*) Asian river dolphin 2. (genus *Inia*) South American river dolphin 3. (genus *Pontoporia*) La Plata dolphin/franciscana

-**BŽL**- 'BEAKED WHALE' 1. (genus *Berardius*) giant beaked whale 2. (genus *Hyperoodon*) bottlenose whale 3. (genera *Indopacetus, Mesoplodon, Tasmacetus, Ziphius*) small beaked whale

-**BŽR**- 'OCEANIC DOLPHIN' 1. (genera *Delphinus, Tursiops, Lissodelphis, Sotalia, Sousa, Stenella, Cephalorhyncus, Grampus, Lagenodelphis, Lagenorhyncus*) dolphin 2. (genus *Orcinus*) killer whale 3. (genera *Feresa, Pseudorca*) pygmy killer whale, false killer whale

-**BŽŘ**- 'OTHER OCEANIC DOLPHIN' 1. (genus *Globicephala*) pilot whale 2. (genus *Peponocephala*) melon-headed whale 3. (genus *Orcaella*) snubfin dolphin

-**BŽM**- 'PORPOISE' 1. (genus *Phocoena*) porpoise 2. (genus *Neophocoena*) finless porpoise 3. (genus *Phocoenoides*) Dall's porpoise

-**BŽN**- 'MONODONT DELPHINOID' 1. (genus *Delphinapterus*) beluga whale, white whale 2. (genus *Monodon*) narwhal 3. [stem unassigned]

Camelids

-**ŇĻW**- 'CAMEL (genus *Camelis*)' 1. dromedary camel 2. domesticated bactrian camel 3. wild bactrian camel

-**ŇĻY**- 'LAMINID' 1. (genus *Lama*) llama, guanaco 2. (*Vicugna pacos*) alpaca 3. (*Vicugna vicugna*) vicuña

Rhinoceros

-**PẔ**- 'RHINOCEROS 1. (genus *Rhinoceros*) Indian & Javan rhinoceros 2. (genus *Dicerorhinus*) Sumatran rhinoceros 3. (genera *Ceratotherium, Diceros*) African white and black rhinoceros

Tapirs

-**XJ**- 'TAPIR' (genus *Tapirus*) 1. Central American tapir 2. South American and Mountain tapir 3. Asian tapir

Equines

-**XC**- 'EQUINE (genus *Equus*)' 1. (subgenus *ferus*) horse 2. (subgenus *asinus*) ass, kiang, onager, kulan, donkey 3. (subgenus *hippotigris*) zebra

-**XCW**- 'EQUINE HYBRID' 1. mule 2. hinny 3. "zebroid" (including "zorse", "zeedonk", "zony")

Viverroids

-**ZB**- 'VIVERRID' 1. (Viverrines and Hemigalines: genera *Viverra, Viverricula, Civettictis, Hemigalus, Cynogale, Diplogale, Chrotogale*) civet 2. (Paradoxurines: genera *Paradoxurus, Arctictis, Paguma, Macrogalidia, Arctogalidia*) [Asian] civet, binturong/bearcat 3. (Genettines: genera *Genetta, Poiana*) genet, oyan or African linsang

-**ZBY**- 'HYAENA' 1. (genus *Crocuta*) spotted hyaena 2. (genera *Hyaena*, *Parahyaena*) hyaena (striped, brown) 3. (genus *Proteles*) aardwolf

-**ZBW**- 'MONGOOSE' 1. (Herpestine subfamily: genera *Herpestes, Atilax, Cynictis, Ichneumia, Bdeogale, Galerella, Rhynchogale, Paracynictis*) mongoose 2. (Mongotine subfamily: genera *Mungos, Suricata, Crossarchus, Helogale, Dolgale, Liberiictis*) mongoose, meerkat, kusimanse, dwarf mongoose 3. (Euplerine and Galidiine subfamilies: *genera Cryptoprocta, Eupleres, Fossa, Galidia,Galidictus, Mungotictis, Salanoia*) fossa, falanouc, vontsira, Malagasy mongoose, Malagasy civet

Eulipotyphla (Hedgehogs, Gymnures, Shrews, Moles, Shrew-Moles, Desmans, Solenodons)

-**DX**- 'HEDGEHOG' 1. (genus *Erinaceus*) common hedgehog 2. (genus *Atelerix*) African hedgehog 3. (genera *Hemiechinus, Mesechinus, Paraechinus*) Asian hedgehog

-**DXW**- 'MOONRAT/GYMNURE' 1. (genus *Echinosorex*) moonrat 2. (genera *Hylomys, Neohylomys, Podogymnura*) gymnure 3. (genus *Neotetracus*) shrew gymnure, shrew hedgehog

-**DXL**- 'SHREW' 1. (Crocidurines; genera *Crocidura, Diplomesodon, Feroculus, Palawanosorex, Paracrocidura, Ruwenzorisorex, Scutisorex, Solisorex, Suncus, Sylvisorex*) white-toothed shrew, large-headed shrew, armored shrew / hero shrew, forest shrew 2. (Myosoricines; genera *Myosorex, Congosorex, Surdisorex*) African shrew 3. (Soricines; genera *Anourosorex, Blarinella, Blarina, Cryptotis, Chimarrogale, Chodsigoa, Episoriculus, Nectogale, Neomys, Soriculus, Megasorex, Notiosorex, Sorex*) red-toothed shrew

-**DXR**- 'MOLE/SHREW-MOLE/DESMAN/SOLENODON' 1. (genera *Condylura, Parascalops, Scalopus, Scapanulus, Scapanus, Uropsilus, Euroscaptor, Mogera, Parascaptor, Scaptochirus, Talpa, Scaptonyx, Dymecodon, Urotrichus, Neurotrichus*) mole, shrew-mole 2. (genera *Desmana, Galemys*) desman 3. (genus *Solenodon*) solenodon

Leporiforms And Rodents

-**BZ-** 'RABBIT/HARE/PIKA' 1. (genera *Nesolagus, Poelagus, Pronolagus, Romerolagus, Sylvilagus, Brachylagus, Caprolagus, Oryctolagus, Bunolagus, Pentalagus*) rabbit/bunny/coney 2. (genus *Lepus*) hare, jackrabbit 3. (genus *Ochotona*) pika

-**BZW**- 'BEAVER' 1. (*Castor canadensis*) North American beaver 2. (*Castor fiber*) Eurasian beaver 3. (genus *Aplodontia*) mountain beaver

-**BZY**- 'CHIPMUNK/MARMOT/PRAIRIE DOG' 1. (genera *Tamias, Eutamias, Neotamias*) chipmunk 2. (genus *Marmota*) marmot 3. (genus *Cynomys*) prairie dog

-**BZL**- 'SQUIRREL' 1. (genera *Spermophilus, Urocitellus, Notocitellus, Ammospermophilus, Otospermophilus, Callospermophilus, Xerospermophilus, Ictidomys, Atlantoxerus, Xerus, Spermophilopsis, Epixerus, Funisciurus, Heliosiurus, Myosiurus, Paraxerus, Protoxerus, Callosciurus, Dremomys, Exilisciurus, Glyphotes, Hyosciurus, Lariscus, Menetes, Nannosciurus, Prosciurillus, Rhinosciurus, Rubrisciuris, Sundasciurus, Tamiops, Funambulus, Ratufa, Poliocitellus, Sciurotamias*) ground squirrel, Asian ornate squirrel, Oriental giant squirrel 2. (genera *Sciurus, Rheithrosciurus, Microsciurus, Sciurillus, Tamiasciurus, Syntheosciurus*) tree squirrel, neotropical pygmy squirrel 3. (genera *Eoglacomys, Glaucomys, Hylopetes, Iomys, Petaurillus, Petinomys, Aeretes, Aeromys, Belomys, Biswamoyopterus, Eupetaurus, Petaurista, Pteromys, Pteromyscus, Trogopterus*) flying squirrel

-**BZR**- 'VOLE/LEMMING/MUSKRAT' 1. (genera *Arvicola, Blanfordimys, Chionomys, Lasiopodomys, Lemmiscus, Microtus, Neodon, Phaiomys, Proedromys, Volemys, Ellobius, Alticola, Caryomys, Eothenomys, Hyperacrius, Myodes, Dinaromys, Prometheomys, Arborimus, Phenacomys*) vole, water vole, mole-vole, snow vole, tree vole, heather vole 2. (genera *Lemmus, Lagurus, Eolagurus, Dicrostonyx, Myopus, Synaptomys*) lemming, steppe lemming, bog lemming, collared lemming 3. (genera *Ondatra, Neofiber*) muskrat

-**BZŘ**- 'HAMSTER' 1. (genera *Cricetus, Allocricetulus, Cansumys, Tscherskia*) hamster 2. (genus *Mesocricetus*) golden hamster 3. (genera *Cricetulus, Phodopus*) dwarf hamster

-**BZM**- 'RAT' 1. (Murid/Old world rats; Murines, Lophiomyines, Leimacomyines; numerous genera) rat 2. (Sigmodontine and Tylomyine New World/South American rats; numerous genera) cotton rat, rice rat, marsh rat, fish-eating rat, water rat, forest rat, swamp rat, bunny rat, conyrat, tapecua, giant rat, vesper rat, climbing rat 3. (Neotomine, Nyctomyine, Nesomyid and Deomyine rats; numerous genera) woodrat, isthmus rat, hamster-rat, pouched rat, giant pouched rat, antsangy, tufted-tail rat, short-tailed rat, brush-furred rat, link rat

-**BZN**- 'MOUSE' 1. (Murid/Old world mice; numerous genera) mouse 2. (Sigmodontine/New World/South American mice; numerous genera) South American water mouse, bristly mouse, spiny mouse, cane mouse, montaine mouse, arboreal mouse, climbing mouse, Oldfield mouse, grass mouse, long-clawed mouse, brucie, bolo mouse,

hocidudo, cerrado mouse, chaco mouse, vesper mouse, big-eared mouse, chincilla mouse, gerbil mouse, leaf-eared mouse 3. (Neotomine , Nesomyid, and Deomyine mice; numerous genera) pygmy mouse, deer mouse, harvest mouse, grasshopper mouse, pouched mouse, climbing mouse, African fat mouse, big-footed mouse, island mouse, voalavo, rock mouse, pygmy rock mouse, brush-furred mouse, spiny mouse

-**BZK**- 'DORMOUSE' 1. (genera *Glirulus, Glis*) dormouse 2. (genus *Graphiurus*) African dormouse 3. (genera *Chaetocauda, Dryomys, Eliomys, Muscardinus, Myomimus, Selevinia*) Asian dormouse

-**BZKW**- 'PORCUPINE' 1. (genera *Hystrix, Atherurus, Trychis*) Old World Porcupine 2. (genus *Erethizon*) North American porcupine 3. (genera *Coendou, Chaetomys*) South American prehensile-tailed porcupine, Brazilian bristle-spined porcupine

-**BZKY**- 'GOPHER' 1. (genera *Cratogeomys, Pappogeomys, Thomomys, Zygogeomys*) western & Mexican pocket gopher 2. (genus *Geomys*) eastern pocket gopher 3. (genus *Orthogeomys*) giant pocket gopher, taltuza

-**BZKL**- 'HETEROMYID' 1. (genus *Heteromys*) spiny pocket mouse 2. (genera *Dipodomys, Microdipodops*) kangaroo rat, kangaroo mouse 3. (genera *Perognathus, Chaetodipus*) pocket mouse

-**BZKR**- 'DIPODID' 1. (genus *Sicista*) birch mouse 2. (genera *Eozapus, Napaeozepus, Zapus*) jumping mouse 3. (genera *Allactaga, Allactodipus, Pygeretmus, Cardiocranius, Salpingotulus, Salpingotus, Dipus, Eremodipus, Jaculus, Stylodipus, Paradipus, Euchoreutes*) jerboa

-**BZKŘ**- 'PLATACANTHOMYID' 1. (genus *Platacanthomys*) spiny dormouse 2. (genus *Typhlomys*) Chinese pygmy dormouse

-**BZKF**- 'SPALACID' 1. (genera *Myospalax, Eospalax*) zokor 2. (genera *Rhizomys, Cannomys, Tachyoryctes*) bamboo rat (a.k.a., root rat), African mole-rat 3. (genus *Spalax*) blind mole-rat

-**BZKÇ**- 'OTHER RODENT' 1. (genus *Calomyscus*) mouse-like hamster 2. (genus *Pedetes*) springhare 3. (genus *Petromus*) dassie rat

-**BZKŢ**- 'GERBIL / JIRDS / TATERIL 1. (genera *Dipodillus, Pachyuromys, Desmodilliscus, Ammodillus, Gerbillus, Microdillus, Desmodillus, Gerbilliscus, Gerbillurus*) gerbil, pygmy gerbil 2. (genera *Brachionis, Meriones, Psammomys, Rhombomys, Sekeetamys*) jird, sand rat 3. (genus *Taterillus*) tateril

-**BZKH**- 'ANOMALURINE SCALY-TAILED FLYING SQUIRREL' 1. (genus *Anomalurus*) scaly-tailed flying squirrel 2. (genus *Idiurus*) long-eared flying mouse, pygmy scaly-tailed flying squirrel 3. (genus *Zenkerella*) Cameroon scaly-tail

-**BZKHW**- 'NICHE AFRICAN RODENT' 1. (genera *Ctenodactylus, Felovia, Massoutiera, Pectinator*) gundi 2. (genus *Thryonomys*) cane rat 3. (genera *Georychus, Cryptomys, Fukomys, Heliophobius, Bathyergus*) blesmol / mole-rat

-**BZD**- 'CHINCHILLA/VISCACHA/PACARANA 1. (genus *Chinchilla*) chinchilla 2. (genera *Lagidium, Lagostomus*) viscacha 3. (genus *Dinomys*) pacarana

-**BZDL**- 'AGOUTI / ACOUCHI / PACA' 1. (genus *Dasyprocta*) agouti 2. (genus *Myoprocta*) acouchi 3. (genus *Cuniculus*) paca

-**BZDR**- 'GUINEA PIG / CAVY / CAPYBARA' 1. (genera *Cavia, Galea, Microcavia*) guinea pig / cavy 2. (genus *Dolichotis*) Patagonian cavy (a.k.a. mara) 3. (genera *Hydrochoerus, Kerodon*) capybara, rock cavy

-**BZDŘ**- OCTODONTID RODENT 1. (genus *Aconaemys*) Andean rock rat 2. (genera *Octodon, Octodontomys*,) degu 3. (genera *Octomys, Pipanacoctomys, Tympanoctomys*) viscacha rat

-**BZDW**- 'ECHIMYID RODENT' 1. (numerous genera) spiny rat, spiny tree-rat, punaré, armored rat, guiara 2. (genera *Capromys, Geocapromys, Mesocapromys, Mysateles, Plagiodontia*) hutia 3. (genus *Myocastor*) nutria/coypu

-**BZDY**- 'OTHER SOUTH AMERICAN RODENT' 1. (genus *Ctenomys*) tuco-tuco 2. (genera *Abrocoma, Cuscomys*) chinchilla-rat 3. (genus *Spalocopus*) coruro

Bats

-**BH**- 'BAT' 1. (Microchiroptera; numerous genera) [micro]bat 2. (Megachiroptera; numerous genera) [mega]bat / fruit-bat [other than flying fox] 3. (genera *Pteropus, Acerodon*) flying fox

Treeshrews

-**JŇW**- 'TREESHREW' 1. (genus *Anathana*) Indian treeshrew 2. (genera *Dendrogale*, *Tupaia*, *Urogale*) Asian treeshrew 3. (genus *Ptilocercus*) pen-tailed treeshrew

Apes and Monkeys [for genus *Homo*, use the root -**L**- given earlier)

-**LGW**- 'APE' 1. (genus *Pan*) chimpanzee, bonobo 2. (genus *Gorilla*) gorilla 3. (genus *Pongo*) orangutan

-**LGY**- 'GIBBON' 1. (genera *Hylobates, Nomascus*) dwarf gibbon, crested gibbon 2. (genus *Hoolock*) hoolock gibbon 3. (genus *Symphalangus*) siamang

-**LGL**- 'LEMUR' 1. (genera *Lemur, Eulemur, Varecia, Hapalemur, Prolemur, Lepilemur, Indri, Avahi, Propithecus, Daubentonia*) lemur, indri, sifaka, aye-aye 2. (genera *Cheirogaleus, Microcebus, Mirza, Allocebus, Phaner*) dwarf lemur, mouse lemur 3. (genera *Cynocephalus, Gaelopterus*) colugo [flying lemur]

-**LGR**- 'BABOON/MACAQUE/MANGABEY 1. (genus *Papio*) baboon 2. (genus *Macaca*) macaque 3. (genera *Lophocebus, Cercocebus*) mangabey

-**LGŘ**- 'OTHER BABOON-LIKE MONKEY' 1. (genus *Mandrillus*) drill, mandrill 2. (genus *Theropithecus*) gelada 3. (genus *Rungwecebus*) kipunji

-**LGZ**- 'ALLOUATTINE AND AOTID MONKEY' 1. (Alouattineae; genus *alouatta*) howler monkey 2. (Aotidae; genus *Aotus, lemurinus* variety) gray-necked night monkey / owl monkey / douroucoulis 3. (Aotidae; genus *Aotus, azarae* variety) red-necked night monkey / owl monkey / douroucoulis

-**LGZB**- 'PITHECIID MONKEY' 1. (genera *Plecturocebus, Callicebus, Cheracebus*) titi 2. (genus *Cacajao*) uakari 3. (genera *Pithecia, Chiropotes*) saki, bearded saki

-**LGZV**- 'ATELINE MONKEY' 1. (genus *Ateles*) spider monkey 2. (genus *Brachyteles*) muriqui / woolly spider monkey 3. (genus *Laothrix*) woolly monkey

-**LGZW**- 'COLOBUS MONKEY' 1. (genus *Colobus*) black-and-white colobus 2. (genus *Piliocolobus*) red colobus 3. (genus *Procolobus*) olive colobus

-**LGZY** 'LANGUR/LUTUNG/ SURILLI' 1. (genera *Trachypithecus, Simias*) lutung/langur/leaf monkey, pig-tailed langur 2. (genus *Semnopithecus*) gray langur 3. (genus *Presbytis*) surilli

-**LGZL**- 'ODD-NOSED MONKEY' 1. (genus *Pygathrix*) douc 2. (genus *Rhinopithecus*) snub-nosed monkey 3. (genus *Nasalis*) proboscis monkey

-**LGZR**- 'CAPUCHIN/SQUIRREL MONKEY' 1. (genus *Cebus*) [gracile] capuchin monkey 2. (genus *Sapajus*) [robust] capuchin monkey 3. (genus *Saimiri*) squirrel monkey

-**LGZŘ**- 'CERCOPITHECINE MONKEY' 1. (genus *Cercopithecus*) guenon, mona monkey 2. (genus *Miopithecus*) talapoin 3. (genera *Chlorocebus, Allenopithecus, Erythrocebus*) grivet, vervet, malbrouck, swamp monkey, patas monkey

-**LGZM**- 'LORIS/POTTO/ANGWANTIBO' 1. (genera *Loris*, *Nycticebus*) loris 2. (genera *Perodicticus, Pseudopotto*) potto, false potto 3. (genus *Arctocebus*) angwantibo

-**LGZN**- 'GALAGO/BUSHBABY' 1. (genera *Galago, Galagoides*) galago/bushbaby 2. (genus *Euoticus*) needle-clawed bushbaby 3. (genera *Otolemur, Sciurocheirus*) greater galago / thick-tailed bushbaby / squirrel galago

-**LGZG**- 'TARSIER' 1. (genus *Tarsius*) [South-East Asian] tarsier 2. (genus *Cephalopachus*) western tarsier 3. (genus *Carlito*) Philippine tarsier

-**LGZD**- 'MARMOSET/TAMARIN' 1. (genera *Cebuella, Callibella, Mico, Callithrix, Callimico*) marmoset 2. (genus *Saguinus*) tamarin 3. (genus *Leontopithecus*) lion tamarin

Seals, Sea Lions & Walrus

-**VXL**- 'PHOCID ("EARLESS"/ "TRUE") SEAL' 1. (Arctic/Northern seals: genera *Phoca, Pusa, Halichoerus, Histriophoca, Pagophilus, Erignathus, Cystophora*) seal 2. (Antarctic seals: genera *Mirounga, Ommatophoca, Hydrurga, Leptonychotes, Lobodon*) [Antarctic] seal 3. (tropical seals: genera *Monachus, Neomonachus*) monk seal

-**VXR**- 'OTARIID ("EARED") SEAL' 1. (genera *Arctocephalus, Callorhinus*) fur seal 2. (genera *Neophoca, Phocarctos, Zalophus, Otaria, Eumetopias*) sea lion 3. (genus *Odobenus*) walrus

Pangolins

-**ZḌW**- 'PANGOLIN (Scaly Anteaters)' 1. (genus *Manis*) Asian scaly anteater 2. (genus *Phataginus*) African tree pangolin and long-tailed pangolin 3. (genus *Smutsia*) African giant pangolin and ground pangolin

Elephant Shrews, Sengis

-**ZKṬ**- 'ELEPHANT SHREW / SENGI' 1. (genus *Elephantulus*) 'elephant shrew' 2. (genera *Macroscelides, Petrodomus, Petrosaltator*) sengi 3. (genus *Rhynchocyon*) checkered elephant shrew / giant sengi

Tenrecs

-**ZKṬW**- 'TENREC' 1. (genus *Tenrec*) common tenrec 2. (genera *Microgale, Nesogale*) shrew tenrec 3. (genera *Geogale, Setifer, Hemicentetes, Echinops, Oryzorictes*) large-eared tenrec, hedghog tenrec, streaked tenrec, rice tenrec

-**ZKṬY**- 'AFRICAN INSECTIVORE (other than tenrecs)' 1. (genus *Orycteropus*) aardvark 2. (genera *Potamogale, Micropotamogale*) otter shrew 3. (Chrysochloridae; numerous genera) golden mole

Elephants

-**GM**- 'ELEPHANT/MAMMOTH' 1. (genus *Loxodonta*) African elephant 2. (genus *Elephas*) Asian elephant 3. (genus *Mammuthus*) mammoth

-**GMW**- 'HYRAX' 1. (genus *Procavia*) rock hyrax 2. (genus *Heterohyrax*) bush hyrax 3. (genus *Dendrohyrax*) tree hyrax

-**GMY**- 'MANATEE/DUGONG' 1. (genus *Trichechus*) manatee / sea cow 2. (genus *Dugong*) dugong 3. (genus *Hydrodamalis*) Steller's sea cow

Armadillos

-**VCW**- 'ARMADILLO' 1. (Tolypeutines and Dasypodides; genera *Cabassous, Priodontes, Tolypeutes, Dasypus*) naked-tailed armadillo, giant armadillo, three-banded armadillo, long-nosed armadillo 2. (Euphractines; genera *Eupractus, Zaedyus, Chaetophractus*) hairy armadillo, six-banded armadillo, pichi 3. (Clamyphorines; genera *Calyptophractus, Chlamyphorus*) fairy armadillo

(New World) Anteaters and Sloths

-**ŇDW**- '(NEW WORLD) ANTEATER' 1. (genus *Myrmecophaga*) giant anteater 2. (genus *Cyclopes*) silky anteater 3. (genus *Tamandua*) tamandua

-**ŇDY**- 'SLOTH' 1. (*genus Bradypus*) three-toed sloth 2. (*Choloepus didactylus*) Linnaeus' two-toed sloth / unau 3. (*Choloepus hoffmanni*) Hoffmann's two-toed sloth

Opossums

-**MZTḶ**- 'OPOSSUM' 1. (numerous sub-families and genera) opossum 2. (genera *Caenolestes, Lestoros, Rhyncholestes*) shrew opossum (a.k.a. rat opossum, caenolestid) 3. (genus *Dromiciops*) colocolo opossum (a.k.a. monito del monte)

Marsupials

-**RDK** - 'DASYURINE MARSUPIAL' 1. (genus *Dasyurus*) quoll 2. (genus *Myoictis*) dasyure 3. (genus *Neophascogale*) speckled dasyure

-**RDKW**- 'OTHER DASYURINE MARSUPIAL' 1. (genus *Dasycercus*) mulgara 2. (genus *Dasykaluta*) kaluta 3. (genus *Dasyuroides*) kowari

-**RDKY**- 'MORE DASYURINE MARSUPIAL' 1. (genus *Pseudantechinus*) false antechinus 2. (genus *Phascolosorex*) marsupial shrew 3. (genus *Parantechinus*) dibbler

-**RDKL**- 'PHASCOGALINE MARSUPIAL' 1. (genus *Antechinus*) antechinus 2. (genus *Phascogale*) phascogale / wambenger 3. (genera *Micromurexia, Murexechinus, Murexia, Paramurexia, Phascomurexia*) Habbema dasyure, black-tailed dasyure, short-furred dasyure, broad-striped dasyure, long-nosed dasyure

-**RDKR**- 'SMINTHOPSINE MARSUPIAL' 1. (genus *Sminthopis*) dunnart 2. (genus *Planigale*) planigale 3. (genera *Antechinomys, Ningaui*) kultarr, ningaui

-**RDKŘ**- 'OTHER DASYUROMORPHS' 1. (genus *Sarcophilus*) Tasmanian devil 2. (genus *Myrmecobius*) numbat 3. (genus *Thylacinus*) thylacine [Tasmanian tiger]

-**RDKF** - 'PHALANGERIFORM POSSUM' 1. (Phalagerides; numerous genera) cuscus 2. (Trichosurines; genera *Trichosurus, Wyulda*) brushtail possum, scaly-tailed possum 3. (Burramyides; genera *Burramys, Cercartetus*) pygmy possum

-**RDKŢ**- 'PETAUROID POSSUM' 1. (Pseudocheirides; numerous genera) ring-tailed possum 2. (Petaurides; genera *Dactylopsila, Gymnobelideus, Petaurus*) triok, glider 3. (Tarsipedides and Acrobatides; genera *Tarsipes, Acrobates, Distoechurus*) honey possum, feathertail glider, feather-tailed possum

Bandicoots and Bilbies

-**KŢXW**- 'PERAMELEMORPH & NOTORYCTEMORPH' 1. (Peramelines; numerous genera) bandicoot 2. (genus *Macrotis*) bilby 3. (genus *Notoryctes*) marsupial mole

Wombats and Koalas

-**LJXW**- 'WOMBAT/KOALA' 1. (genus *Vombatus*) common wombat 2. (genus *Lasiorhinus*) hairy-nosed wombat 3. (genus *Phascolarctos*) koala

Kangaroos, Wallaroos, Wallabies

-**LMKW**- 'KANGAROO/WALLAROO/WALLABY (genera *Macropus, Onychogalea, Wallabia*)' 1. kangaroo 2. wallaroo 3. wallaby

-**LMKY**- 'POTOROIFORM' 1. (genera *Aepyprymnus, Hypsiprymnodon*) rat-kangaroo, musky rat-kangaroo 2. (genus *Bettongia*) bettong 3. (genus *Potorous*) potoroo

-**LMKL**- 'OTHER MACROPODID 1. (genus *Dendrolagus*,) tree-kangaroo, dingiso, tenkile 2. (genera *Dorcopsis, Dorcopsulus, Lagorchestes*) dorcopsis, hare-wallaby 3. (genus *Lagostrophus*) banded hare-wallaby/mernine/munning

-**LMKR**- 'OTHER MACROPODINE' 1. (Genus *Petrogale*) rock-wallaby, monjon, nabarlek 2. (Genus *Setonix*) quokka 3. (Genus *Thylogale*) pademelon

Monotremes

-**LCX**- 'MONOTREME' 1. (genus *Ornithorhyncus*) platypus 2. (genus *Tachyglossus*) short-beaked echidna 3. (genus *Zaglossus*) long-beaked echidna

7.3.1.2 Reptiles

Turtles

-**ZZC**- 'TURTLE/TORTOISE' 1. (Pleurodira turtles; numerous genera) snake-necked, side-necked, twist-necked, swamp, snapping, big-headed, helmeted turtle 2. (Polycryptodira turtles [except tortoises]; numerous genera) [pond, box, water, river, leaf, roofed, mud, sea, leatherback, softshell, pignose] turtle 3. (Testudinides; numerous genera) tortoise

Lizard-Like Animals [NOTE: To refer to "lizard" as a generic or catch-all term, use Stem 1 of the root -**ZZ**- shown in Sec. 7.3 above.]

-**ZZW**- 'PLEURODONTINE LIZARD' 1. (iguanide, liolaemid, and oplurine lizards; numerous genera) iguana, chuckwalla 2. (crotaphytine, hoplocercine, phrynosomatine, polychrotine, and leiosaurid lizards; numerous genera) [collared, leopard, wood, clubtail, earless, spiny, tree, side-blotched, horned] lizard 3. (tropidurides; numerous genera) neotropical ground lizard, lava lizard, Pacific iguana, whorltail iguana, thornytail iguana

-**ZZY**- 'DISTINCTIVE PLEURODONTINE LIZARD' 1. (corytophanides; genera *Basiliscus, Corytophanes, Laemanctus*) basilisk, helmeted iguana, casquehead iguana 2. (dactyloids and polychrotides; numerous genera) anole 3. (genus *Leiocephalus*) curly tailed lizard

-**ZZL**- 'AGAMIDE LIZARD' 1. (agamide and uromasticine lizards; numerous genera) frilled lizard, agama, mastigure, dabb lizard 2. (amphibolurine, draconine, and hydrosaurine lizards; numerous genera) dragon lizard, fan-throated lizard, lyreshead lizard, gliding lizard, sailfin lizard 3. (leiolepidines; numerous genera) butterfly lizards

-**ZZR**- 'OTHER IGUANA-LIKE ANIMAL'- 1. (chamaeleonide lizards; numerous genera) chameleon 2. (genus *Sphenodon*) tuatara 3. [stem unassigned]

-**ZZŘ**- 'PALEOANGUIMORPH' 1. (genera *Veranus, Lanthanotus* [other than *V. veranus komodoensis*]) monitor lizard 2. (*V. veranus komodoensis*) komodo dragon 3. (genus *Shinisaurus*) Chinese crocodile lizard

-**ZZM**- 'NON-ANGUIOID NEOANGUIMORPH' 1. (genus Heloderma [other than species H. suspectum]) beaded lizard 2. (Heloderma suspectum) gila monster 3. (genus) Xenosaurus) knob-scaled lizard

-**ZZN**- 'ANGUIOID' 1. (genera *Celestus, Diploglossus*) galliwasp 2. (anguines and anniellids; several genera) slowworm, glass lizard, scheltopusik, American legless lizard 3. (genera *Gerrhonotus, Abronia, Elgaria*) alligator lizard

-**ZZF**- 'TEIFORM LIZARD' 1. (Teiine lizards; numerous genera) jungle-runner, whiptail lizard, desert tegu 2. (Tupinambine lizards; several genera) false monitor, Caiman lizard, tegu 3. (gymnophthalmids; numerous genera) spectacled lizard / microteiid

-**ZZV**- 'LACERETIBAENIAN LIZARD' 1. (Lacertid lizards; numerous genera) wall lizard, true lizard, lacerta 2. (Amphisbaenian legless lizards; numerous genera) worm lizard 3. (genus *Bipes*) Mexican mole lizard

Skinks

-**ZZŇ**- 'SCINCIFORM LIZARD' 1. (Scincids; numerous genera) skink 2. (Gerrhosaurids and Cordylides; several genera) plated lizard, girdled lizards/spinytail lizards 3. (Xantusiids; genera *Cricosaura, Lepidophyma, Xantusia*) night lizard

Geckos & Close Relatives

-**ZZH**- 'GECKO (and genetically-related creatures)' 1. (carphodactylids; numerous genera) gecko 2. (pygopodides; several genera) legless lizard, snake-lizard, flap-footed lizard 3. (dibamides; genera *Dibamus, Anelytropsis*) blind skink

Snakes [NOTE: To refer to "snake" as a generic or "catch-all" term, use Stem 2 of the root -**ZZ**- shown earlier.

-**ZZG**- 'PYTHON-LIKE SNAKE' 1. (Pythonids; several genera) python 2. (genus *Xenopeltis*) sunbeam snake 3. (genus *Loxocemus*) Mexican python

-**ZZB**- 'BOA-LIKE SNAKE' 1. (Boids; numerous genera) boa, anaconda 2. (genus Xenophidion) spinejaw snake 3. (genera *Bolyeria, Casarea*) splitjaw snake

-**ZZD**- 'VIPER' 1. (Viperines; numerous genera) [true or pitless] viper, asp 2. (Crotalines; numerous genera) pit viper, lancehead, rattlesnake, moccasin, bushmaster, temple viper 3. (genus *Azemiops*) Fea's viper

-**ZZT**- 'LESSER-KNOWN PROTEROGLYPH SNAKE' 1. (Homalopsids; numerous genera) Indo-Australian water snake, mudsnake, bockadam 2. (Colubrids; numerous genera) colubrid snake 3. (Lamprophiids; numerous genera) lamprophiid snake

-**ZZP**- 'RARE CAENOPHID SNAKE' 1. (genus *Acrochordus*) wart snake/ file snake/ elephant trunk snake/ dogface snake 2. (Xenodermids; several genera) xenodermid snake) 3. (Pareids; several genera) pareid snake

-**ZZḌ**- 'NON CAENOPHID and NON-CONSTRICTING BOOIDE SNAKE' 1. (Scolecophids and Anomalepidids; numerous genera) blind snake, thread snake, dawn blind snake 2. (Cylindrophiids, Anomochiids, and Aniliids; numerous genera) pipe snake 3. (Uropeltids; several genera) shield-tailed snake

-**ZZK**- 'ELAPID SNAKE' I 1. (genera *Naja, Pseudohaje, Walterinnesia, Boulengerina, Aspidelaps*) cobra 2. (genus *Ophiophagus*) king cobra 3. (genus *Hemachatus*) ring-necked spitting cobra /rinkhal

-**ZZKW**- 'ELAPID SNAKE' II 1. (genus *Dendroaspis*) mamba 2. (genus *Bungarus*) krait 3. (genus *Elapsoidea*) African garter snake / venemous garter snake

-**ZZKY**- 'ELAPID SNAKE' III 1. (genera *Calliophis, Hemibungaris, Sinomicrurus*) Old World coral snake 2. (genera *Leptomicrurus, Micruroides, Micrurus*) New World coral snake 3. (genera *Salomonelaps, Simoselaps*) Soloman coral snake, Australian coral snake

-**ZZKL**- 'ELAPID SNAKE' IV 1. (genus *Laticauda*) sea krait 2. (genus *Notechis*) tiger snake 3. (genus *Antaioserpens*) burrowing snake

-**ZZKR**- 'ELAPID SNAKE' V 1. (genera *Ephalophis, Hydrelaps*) mudsnake 2. (genus *Hydrophis*) sea snake 3. (genera *Acalyptophis, Aipysurus, Astrotia, Emydocephalus, Enhydrina, Kerilia, Kolpophis, Lapemis, Parahydrophis, Pelamis, Praescutata, Rhinoplocephalus, Thalassophis*) other sea snake

-**ZZKŘ**- 'ELAPID SNAKE' VI 1. (genus *Oxyuranus*) taipan 2. (genus *Ogmodon*) Fiji snake / bolo snake / bola 3. (genus *Pseudechis*) black snake / king brown snake

-**ZZKF**- 'ELAPID SNAKE' VII 1. (genus *Acanthophis*) death adder 2. (genus *Aspidomorphus*) collared adder 3. (genus *Austrelaps*) copperhead

-**ZZKŢ**- 'ELAPID SNAKE' VIII 1. (genus *Cacophis*) rainforest crowned snake 2. (genus *Brachyurophis*) shovel-nosed snake 3. (genus *Demansia*) whipsnake

-**ZZKÇ**- 'ELAPID SNAKE' IX 1. (genus *Echiopsis*) bardick 2. (genera *Drysdalia, Elapognathus*) grass snake 3. (genus *Denisonia*) ornamental snake

-**ZZKS**- 'ELAPID SNAKE' X 1. (genus *Homoroselaps*) harlequin snake 2. (genus *Toxicocalamus*) New Guinea forest snake 3. (genus *Parapistocalamus*) Hediger's snake

-**ZZKŠ**- 'ELAPID SNAKE' XI 1. (genera *Hemiaspis*, *Hoplocephalus, Paroplocephalus, Pseudonaja, Tropidechis, Vermicella*) Australian venemous snake (various): swamp snake, broad-headed snake, pale-headed snake, Lake Cronin snake, dugite, brown snake, rough-scaled snake, bandy-bandy / hoop snake 2. (genus *Suta*) hooded snake, monk snake, black-backed snake, curl snake, myall snake, spotted snake, copper snake, short-tailed snake, whip snake, Dwyer's snake, Rosens's snake 3. (genera *Loveridgelaps, Microphechis*) small-eyed snake

Crocodilians

-**GZW**- 'CROCODILIAN' 1. (genera *Crocodylus, Osteolaemus, Mecistops*) crocodile 2. (genera *Alligator, Caiman, Paleosuchus*) alligator, caiman 3. (genera *Gavialis, Tomistoma*) gharial, false gharial

Extinct Sauropsids (including Dinosaurs)

-**GZY**- 'SAUROPSID' 1. (Avemetarsalians other than birds) dinosaur, pterosaur, aphanosaur 2. (all extinct Eureptilia other than Avemetarsalia) extinct non-dinsosaurian reptile 3. parareptile (extinct non-reptilian sauropsid, i.e., mesosaur, millerettid, procolophonid, pareiassaur)

7.3.1.3 Amphibians

-**NXL**- 'FROG/TOAD' 1. (Neobatrachians [other than Bufonids] and Mesobatrachians; numerous genera) frog 2. (Bufonids; numerous genera) toad 3. (Archaeobatrachians; numerous genera) small primitive frog/toad

-**NXR**- 'SALAMANDER/NEWT/SIREN' 1. (Salamandrines, Cryptobranchids and Hynobiids; numerous genera) [mole, lungless, torrent, giant, Asiatic] salamander, mudpuppy/olm 2. (Pleurodelines; numerous genera) newt 3. (genera *Siren*, *Pseudobranchus*) siren, dwarf siren

7.3.1.4 Birds

-**NSD**- ‘PALEOGNATH’ I 1.(Struthionids; genus *Struthio*) ostrich 2. (Rheids; genus *Rhea*) rhea 3. (Dromaiids; genus *Dromaius*) emu

-**NSDW**- PALEOGNATH’ II 1. (Casuariids; genus *Casuarius*) cassowary 2. (Apterygids; genus Apteryx) kiwi 3. (Tinamids; genus Tinamus) tinamou

-**NSDY**- ‘MEGAPODIID (BRUSHTURKEY/SCRUBFOWL)’ 1. (Alecturines; genera *Talegalla, Alectura, Aepypodius*) brushturkey 2. (genus *Leipoa*) malleefowl 2. (Megapodiins; genera *Macrocephalon, Eulipoa, Megapodius*) scrubfowl, maleo, megapode

-**NSDL**- ‘CRACID (GUAN & CURASSOW)’ 1. (Cracida - Penelopines and Ortalidines; several genera) guan, chachalaca 2. (Oreophasines; genus *Oreophasis*) horned guan 3. (Cracinii; several genera) curassow

-**NSDR**- ‘ODONTOPHORID & NUMIDID (NEW WORLD QUAIL & GUINEAFOWL)’ 1. (Odontophorines; numerous genera) New World quail 2. (Ptilopachines; genus Ptilopachus) stone partridge, Nahan’s partridge 3. (Numidids; several genera) guineafowl

-**NSP**- **‘**CHICKEN & TURKEY’ 1. (*Gallus gallus domesticus*) chicken 2. (genus *Gallus* other than *G. g. domesticus*) junglefowl 3. (genus *Meleagris*) turkey, wild turkey, ocellated turkey

-**NSPW**- ‘TETRAOGALLINE (PARTRIDGE / QUAIL)’ 1. (genera *Coturnix, Anurophasis, Synoicus, Excalfactoria, Ophrysia, Perdicula*) quail 2. (genera *Ammoperdix, Margaroperdix, Alectoris*) [Old World] partridge 3. (genus *Tetraogallus*) snowcock

-**NSPY**- ‘OTHER PARTRIDGE-LIKE BIRD’ 1. (Rollulines; several genera) Asiatic partridge / jungle partridge / wood partridge 2. (Tetraogallines and Perdicines - genera *Pternistis, Dendroperdix, Francolinus, Galloperdix,Peliperdix, Scleroptila*) francolin, spurfowl 3. (other Perdicines; genera *Arborophila, Bambusicola, Haematortyx, Lerwa, Perdix, Tetraophasis, Tropicoperdix*) partridge, monal-partridge

-**NSPL**- ‘PHASIANINE (PHEASANT)’ I 1. (genus *Phasianus*) [common] pheasant 2. (genus *Tragopan*) horned pheasant, tragopan 3. (genus *Ithaginis*) blood pheasant

-**NSPR**- ‘PHASIANINE (PHEASANT)’ II 1. (Pavonii; genera *Pavo, Afropavo*) peafowl (inluding peacock and peahen) 2. (Polyplectronini; genus *Polyplectron*) peacock-pheasant 3. (genus *Syrmaticus*) long-tailed pheasant

-**NSPŘ**- ‘PHASIANINE (PHEASANT)’ III 1. (genus *Argusianus*) great argus 2. (genus *Rheinardia*) crested argus 3. (genus *Catreus*) cheer pheasant / Wallich’s pheasant

-**NSPF**- ‘PHASIANINE (PHEASANT)’ IV 1. (genus *Chrysolophus*) ruffed pheasant, golden pheasant 2. (genus *Crossoptilon*) eared pheasant 3. (genus *Lophophorus*) monal

-**NSPV**- ‘PHASIANINE (PHEASANT)’ V 1. (genus *Lophura*) gallopheasant, fireback 2. (genus *Pucrasia*) koklass 3. [stem unassigned]

-**NSPÇ**- ‘TETRAONINE (GROUSE)’ 1. (genera *Falcipennis, Dendragapus, Tetrao, Tetrastes, Bonasa, Centrocercus*) grouse, capercaillie 2. (genus *Lagopus*) ptarmigan 3. (genus *Tympanuchus*) prairie grouse, prairie chicken

-**NSG**- **‘**DUCK’ I 1. (Anatines; numerous genera) dabbling duck, mallard, teal, shoveler, pintail, widgeon, gadwall, Asian/Madagascar/East African/Pacific duck 2. (Merginines; numerous genera) seaduck, merganser, smew, eider, scoter, goldeneye, bufflehead, long-tailed duck, harlequin duck 3. (Oxyurini; genera Heteronetta, Nomonyx, Oxyura) black-headed duck, masked duck, stiff-tailed duck

-**NSGW**- ‘DUCK’ II 1. (Dendrocynines; genera Dendrocygna, Thalassornis) whistling duck, white-backed duck 2. (Stictonettines; genus Stictonetta) freckled duck 3. (genus Tachyeres) steamer duck

-**NSGL**- ‘TADORNINE (SHELDUCK / SHELDGOOSE)’ 1. (genera *Tadorna, Radjah, Hymenolaimus, Merganetta, Malacorhynchus, Aix, Carina, callonetta, Chenonetta, Salvadorina, Sarkidiornis*) shelduck, blue duck, torrent duck, pink-eared duck, Mandarin duck, wood duck, Muscovy duck, ringed teal, maned duck, Salvadorini’s teal, knob-

billed duck 2. (genera *Alopochen, Neochen, Chloephaga, Cyanochen, Plectropterus*) sheldgoose, Egyptian goose, Orinoco goose, Andean goose, blue-winged goose, spur-winged goose 3. (genera *Aythya, Netta, Marmaronetta*) diving duck, pochard, scaup, canvasback, marbled teal

-**NSGY**- 'GOOSE' 1. (genus *Anser*) goose (including grey and white geese) 2. (genus *Branta*) black goose, Canada goose, barnacle goose, cackling goose, nene, brant 3. (genus *Cereopsis*) Cape Barren goose

-**NSGR**- 'SWAN' 1. (genus *Cygnus*) swan 2. (genus *Olor*) black-and-yellow-billed swan 3. (genus *Coscoroba*) coscoroba swan

-**NSGŘ**- 'ANSERIFORM' 1. (Anhimids; genera *Anhima, Chauna*) screamer 2. (Anseranatids; genus *Anseranas*) magpie goose 3. (genus *Nettapus*) pygmy goose

-**NSB**- **'**DOVE' 1. (Turturini,Claravinae and Phabini except *Ocyphaps, Petrophassa*, *Leucosarcia*; several genera) dove 2. (Zenaidini; several genera) quail-dove, zenaida dove 3. (genera *Reinwardtoena, Turacoena, Macropygia*) cuckoo-dove

-**NSBW**- 'PIGEON' 1. (Raphini, Treronini, Ptilinopini; numerous genera) pigeon 2. (genus *Patagioenas*) American pigeon 3. (genera *Ocyphaps, Petrophassa*, *Leucosarcia*) phabine pigeon

-**NSBY**- 'OTHER COLUMBIFORM BIRD' 1. (Mesitornithids; genera *Monias, Mesitornis*) mesite 2. (Pteroclids; several genera) sandgrouse 3. (Raphinae; genus *Rhaphus*) dodo bird

-**NSBL**- 'GRUIFORM' 1. (Gruids; genera *Grus, Belearica*) crane 2. (Aramids; genus *Aramus*) limpkin / carrao / courlan / crying bird 3. (Psophiids; genus *Psophia*) trumpeter

-**NSBR**- 'RALLID (RAIL)' I 1. (genera *Rallus, Lewinia, Gallirallus, Rallicula, Dryolimnas, Gymnocrex, Hypotaenidia, Pardirallus, Aramides, Himantornis, Megacrex, Coturnicops*) rail, weka 2. (genus *Crex, Anurolimnas, Mustelirallus, Amaurolimnas, Porzana, Aenigmatolimnus, Amaurornis, Gallicrex, Rallina, Myrcropygia, Laterallus, Mundia, Hapalocrex*) crake, sora, bush-hen, watercock 3. (genera *Gallinula, Paragallinula, Porphyrio, Tribonyx*) moorhen, marsh hen, gallinule, nativehen, waterhen / swamp chicken

-**NSBŘ**- 'RALLID (RAIL)' II 1. (genus *Canirallus*) grey-throated rail 2. (genus *Sarothrura*) flufftail 3. (genus *Rougetius*) Rouget's rail

-**NSBV**- 'HELIORNITHID (FINFOOT)' 1. (genus *Heliopais*) masked finfoot / Asian finfoot 2. (genus *Podica*) African finfoot 3. (genus *Heliornis*) sungrebe

-**ŇSBV**- 'OTIDIMORPH (CUCKOO / BUSTARD)' 1. (Cuculids; numerous genera) cuckoo, koel, malkoha, coua, yellowbill, roadrunner, coucal, ani 2. (Musophagids; several genera) turaco, go-away-bird, plantain-eater 3. (Otidids; numerous genera) bustard, florican, korhaan

-**ŇSDV**- 'CAPRIMULGIFORM (NIGHTJAR / NIGHTHAWK)' I 1. (Caprimulgines and Eurostopodines; numerous genera) nightjar, poorwill, whip-poor-will, pauraque 2. (Chordeilines; genera *Nyctiprogne, Chordeiles, Lurocalis*) nighthawk 3. (Aegothelids; genus *Aegotheles*) owlet-nightjar

-**ŇSGV**- 'CAPRIMULGIFORM (NIGHTJAR / NIGHTHAWK)' II 1. (Podargids; genera *Podgarus, Batrachostomus, Rigidipenna*) frogmouth 2. (Nyctibiids; genus *Nyctibius*) potoo 3. (Steatornithids; genus *Steatornis*) oilbird

-**NSGV**- 'APODID (SWIFT / TREESWIFT)' 1. (Cypseloidini, Apodini, Collocalliini; several genera) swift, palm swift, swallow-tail swift, swiftlet 2. (Chaeturini; several genera) needletail, spinetail, spine-tailed swift 3. (Hemiprocnids; genus Hemiprocne) treeswift / crested swift

-**NSDV**- 'TROCHILID (HUMMINGBIRD)' 1. (Trochilines; numerous genera) hummingbird 2. (Phaethornithines; several genera) hermit, sicklebill, barbthroat 3. [stem unassigned]

-**NŠT**- 'CHARADRIIFORM' I 1. (Charadriids; genera *Pluvialis, Pluvianus, Charadrius, Thinornis, Phegornis*) plover 2. (Ibidorhynchids; genus *Ibidorhyncha*) ibisbill 3. (Charadriines, Anarhynchines, and Vanellines; genera *Orepholus, Eurodromias, Peltohyas, Erythrogonus, Anarhynchus, Vanellus*) dotterel, wrybill, lapwing

-**NŠTW**- 'CHARADRIIFORM' II 1. (Haematopodids; genus *Haematopus*) oystercatcher 2. (Recurvirostrids; genus *Recurvirostra*) avocet 3. (Recurvirostrids; genera *Cladorhynchus, Himantopus*) stilt

-**NŠTY**- ‘CHIONIDIFORM’ 1. (Burhinids; genera *Burhinus, Esacus*) stone-curlew, thick-knee 2. (Chionids; genus *Chionis*) sheathbill / paddy 3. (Pluvianellids; genus *Pluvianellus*) Magellanic plover

-**NŠTÇ**- ‘JACANID’ 1. (several genera except *Irediparra*) jacana 2. (genus *Irediparra*) lotusbird / lilytrotter / comb-crested jacana 3. [stem unassigned]

-**NŠTF**- ‘THINOCOROID’ 1. (Rostratulids; genera *Rostratula, Nycticryphes*) painted-snipe 2. (Thinocorids; genera *Attagis, Thinocorus*) seedsnipe 3. (Pedionomids; genus *Pedionomus*) plains-wanderer

-**NŠTL**- ‘SCOLOPACID (SANDPIPER)’ I 1. (genera *Calidris/Eriola/Ereunetes, Bartramia, Actitis, Xenus, Prosobonia*) sandpiper, stint / peep, knot, sanderling, dunlin, ruff, surfbird 2. (genus *Numenius*) curlew, whimbrel 3. (genus *Limosa*) godwit

-**NŠTR**- ‘SCOLOPACID (SANDPIPER)’ II 1. (genus *Arenaria*) turnstone 2. (genus *Phalaropus*) phalarope 3. (genus *Tringa*) tattler, yellowlegs, shank

-**NŠTŘ**- ‘SCOLOPACINE (SNIPE)’ 1. (genera *Gallinago, Coenocorypha, Lymnocryptes*) snipe, jacksnipe 2. (genus *Limnodromus*) dowitcher 3. (genus *Scolopax*) woodcock

-**NŠTV**- ‘TURNICID & GLAREOLOID’ 1. (Turnicids; genera *Turnix, Ortyxelos*) buttonquail, quail-plover 2. (Dromadids; genus *Dromas*) crab-plover 3. (Glareolids; genera *Glareola, Stiltia, Rhinoptilus, Cursorius*) pratincole, courser

-**NŠTM**- ‘ALCOID (SKUA / PUFFIN / AUKLET)’ 1. (Stercorariids; genus *Stercorarius*) skua, jaeger 2. (Fraterculines; genera *Fratercula, Cerorhinca*) puffin, rhinoceros puffin 3. (Fraterculines; genera *Aethia, Ptychoramphus*) auklet

-**NŠTN**- ‘ALCINE (MURRE / GUILLEMOT / AUK)’ 1. (genera *Brachyramphus, Synthliboramphus*) murrelet 2. (genera *Cepphus, Uria*) guillemot, murre / turr 3. (genera *Alle*, *Alca*) dovekie / little auk, razorbill / lesser auk

-**NSDŘ**- ‘LAROID (GULL / TERN / NODDY / SKIMMER)’ 1. (Larines; numerous genera) gull / seagull, kittiwake 2. (Sternines and Gygines; numerous genera) tern 3. (Anoines and Rynchopines; genera *Anous, Rynchops*) noddy, skimmer

-**NŠK**- **‘**MIRADORNITHID’ 1. (Phoenicopterids; several genera) flamingo 2. (Podicipedids; several genera) grebe 3. [stem unassigned]

-**NŠKW**- ‘EURYPYGIMORPH’ 1. (Phaetontids; genus *Phaethon*) tropicbird 2. (Eurypygids; genus *Eurypyga*) sunbittern 3. (Rhynochetids; genus *Rhynochetos*) kagu

-**NŠKY**- ‘PROCELLARIFORM (TUBENOSE)’ 1. (Gaviids; genus *Gavia*) loon / diver 2. (Diomedeids; several genera) albatross, mollymawk 3. (Hydrobatids and Oceanitids; numerous genera) storm petrel

-**NŠKL**- ‘PROCELLARIID (PETREL / PRION / SHEARWATER)’ 1. (genera *Macronectes, Fulmarus, thalassoica, Daption, Pagodroma, Pterodroma*) petrel, fulmar 2. (genera *Pachyptila, Halobaena*) prion 3. (genera Calonectris, Puffinus, Ardenna, Bulweria, Pseudobulweria, Lugensa, Procellaria) shearwater

-**NŠKR**- ‘SPHENISCID (PENGUIN)’ 1. (genus *Aptenodytes*) king penguin, emperor penguin 2. (genus *Eudyptes*) crested penguin, rockhopper penguin, macaroni penguin, royal penguin, fiordland penguin 3. (genera *Spheniscus, Pygoscelis, Megadyptes, Eudyptula*) penguin, Megellanic penguin, Humboldt penguin, Galápagos penguin, African penguin, banded penguin, fairy penguin, gentoo penguin, Adélie penguin, chinstrap penguin, rump-legged penguin / brush-tailed penguin, yellow-eyed penguin, little penguin

-**NŠKŘ**- ‘CICONIID (STORK)’ 1. (genera *Mycteria*, Ciconia, Ephippiorhynchus, Leptoptilos) stork, adjutant 2. (genus *Anastomus*) openbill 3. (genus *Jabiru*) jabiru

-**NŠKM**- ‘SULIFORM (BOOBY / GANNET / FRIGATEBIRD)’ I 1. (Sulids; genera *Sula, Papasula*) booby 2. (Sulids; genus *Morus*) gannet 3. (Fregatids; genus *Fregata*) frigatebird

-**NŠKN**- ‘SULIFORM (CORMORANT / DARTER)’ II 1. (Phalacrocoracids; genera *Phalacrocorax, Microcarbo*) cormorant, shag 2. (Phalacrocoracids; genus *Leucocarbo*) blue-eyed shag 3. (Anhingids; genus *Anhinga*) darter, snakebird, water turkey

-**NŠKÇ**- ‘PELECANIFORM’ 1. (Pelecanids; genus *Pelecanus*) pelican 2. (Scopids; genus *Scopus*) hamerkop 3. (Balaenicipitids; genus *Balaeniceps*) shoebill / whalehead

-**NŠKF**- ‘ARDEID (HERON / EGRET / BITTERN)’ 1. (Ardeines and Tigriornithines; numerous genera) heron 2. (Ardeines; genera *Egretta, Bubulcus*) egret, cattle egret 3. (Botaurines; several genera) bittern

-**NŠKŢ**- ‘THRESKORNITHID (IBIS / SPOONBILL)’ 1. (Threskornithines; numerous genera) ibis 2. (Plataleines; genus Platalea) spoonbill 3. [stem unassigned]

-**ŇSD**- ‘VULTURE / CONDOR’ 1. (Catharids; genera *Cathartes, Coragyps, Sarcoramphus*) [New World] vulture 2. (Catharids; genera *Gymnogyps, Vultur*) condor 3. (Aegypiines and Gypaetines; several genera) [Old World] vulture

-**ŇSDW**- ‘ACCIPITRIFORM & OPISTHOCOMID’ 1. (Pandionids; genus *Pandion*) osprey 2. (Sagittariids; genus *Sagittarius*) secretarybird 3. (Opisthocomids; genus *Opisthocomus*) hoatzin

-**ŇSDY**- ‘ACCIPITRIID’ I 1. (Aquilines; numerous genera) eagle, hawk-eagle 2. (Haliaeetines; genus *Haliaeetus*) sea eagle, ern, fish eagle, bald eagle 3. (Circaetines; several genera) snake-eagle, serpent-eagle, bateleur

-**ŇSDL**- ‘ACCIPITRIID’ II 1. (Milvines and Elanines; numerous genera) kite 2. (Pernines; several genera) honey buzzard, kite 3. (Harpiines; genera *Harpia, Morphnus, Harpyopsis*) harpy eagle, crested eagle

-**ŇSDR**- ‘ACCIPITRIID’ III 1. (Buteonines; numerous genera) hawk, buzzard, buzzard-eagle 2. (Acciptrines and Melieraxines; several genera) hawk, goshawk, sparrowhawk 3. (Circines and Polyoroidines; genera *Circus, Polyboroides*) harrier, harrier-hawk

-**MSKW**- ‘OWL’ 1. (Strigids; numerous genera) owl (including screech owls, eared owls, horned owls, eagle-owls) 2. (Tytonines; genus *Tyto*) barn owl, grass owl, masked owl 3. (Tytonines; genus *Phodilus*) bay owl

-**MSKY**- ‘CORACIIMORPH’ 1. (Coliids; genera *Colius, Urocolius*) mousebird 2. (Leptosomids; genus *Leptosomus*) cuckoo roller / courol 3. (Trogonids; several genera) trogon, quetzal

-**MSKL**- ‘BUCEROTIFORM’ 1. (Bucerotids and Bucorvids; numerous genera) hornbill, ground hornbill 2. (Phoeniculids [except Rhinopomastus] and Upupids; genera Phoenuculus, Upupa) hoopoe, wood hoopoe 3. (Pheoniculids; genus Rhinopomastus) scimitarbill

-**ŇSP**- **‘**PICIDS (WOODPECKER)’ 1. (Picines; numerous genera) woodpecker 2. (Picumnines and Nesoctitines; genera Picumnus, Sasia, Nesoctites) piculet 3. (Jyngines; genus Jynx) wryneck

-**ŇSPW**- ‘PICIFORM’ I 1. (Indicatorids; several genera) honeyguide, honeybird 2. (Megalaimids, Lybiids, Captonids, Semnornithids; numerous genera) barbet, toucan-barbet 3. (Ramphastids; several genera) toucan, toucanet, araçari

-**ŇSPY**- ‘PICIFORM’ II 1. (Galbulids; several genera) jacamar 2. (Bucconids; genera *Notharchus, Bucco, Nystalus, Hypnelus, Malacoptila, Chelidoptera*) puffbird 3. (Bucconids; genera *Monasa, Hapaloptila, Nonnula, Micromonacha*) nunbird, nunlet, monklet

-**ŇSPL**- ‘CORACIIFORM’ I 1. (Coraciids; genera *Coracias, Eurystomus*) roller, dollarbird 2. (Brachypteraciids; several genera) ground roller 3. (Meropids; genera *Merops*, *Meropogon, Nyctyornis*) bee-eater

-**ŇSPR**- ‘CORACIIFORM’ II 1. (Alcedinids; numerous genera) kingfisher, kookaburra 2. (Todids; genus *Todus*) tody 3. (Momotids; several genera) motmot

-**ŇSPŘ**- ‘FALCONIFORM & CARIAMIFORM’ 1. (Falconines; several genera) falcon, falconet, kestrel, hobby, gyrfalcon 2. (Polyborines; several genera) caracara 3. (Cariamids; genera *Cariama, Chunga*) seriema

-**ŇSPF**- ‘PSITTACOID (PARROT)’ 1. (Psittacoids; numerous genera) parrot, parakeet, macaw, amazon, rosella, bluebonnet, racket-tail 2. (Loriines; numerous genera) lory, lorikeet 3. (Melopsittacines; genus *Melopsittacus*) budgerigar / budgie

-**ŇSPV**- ‘CACATUID (COCKATOO)’ 1. (Cacatuines; several genera) cockatoo, corella, galah 2. (Calyptorhynchines; genus *Calyptorhynchus*) black cockatoo 3. (Nymphicines; genus *Nymphicus*) cockatiel / weiro / quarrion

-**ŇSPÇ**- ‘STRIGOPOID (NEW ZEALAND PARROT)’ 1. (genus *Nestor*) kaka 2. (*N. notabilis*) kea 3. (genus *Strigops*) kakapo

-**ŇSGW**- ‘SAPAYOID & ACANTHISITTID BIRD’ 1. (Sapayoids; genus *Sapayoa*) sapayoa 2. (Acanthisittids; genus *Acanthisitta*) rifleman 3. (Acanthisittids; genus *Xenicus*) New Zealand rock wren

-**ŇSGY**- ‘EURYLAIMOID BIRD’ 1. (Eurylaimids and Calyptomenids; several genera) broadbill 2. (Pittids; genera *Pitta, Hydrornis, Erythropitta*) pitta 3. (Philepittids; genera *Philepitta*, *Neodrepanis*) asity, sunbird-asity

-**ŇSGL**- ‘TYRANNIOID BIRD’ I 1. (Piprids; numerous genera) manakin, tyrant-manakin, piprites 2. (Cotingids; numerous genera) cotinga, plantcutter, berryeater, fruiteater, cock-of-the-rock, bellbird, piha, fruitcrow, capuchinbird, umbrellabird 3. (Tityrids; numerous genera) sharpbill, flycatcher, mylobius, tityra, schiffornis, mourner, purpletuft, laniisoma, xenopsaris, becard

-**ŇSGR**- ‘TYRANNIOID BIRD’ II 1. (Tyrannids; numerous genera) flycatcher, tyrant, tyrannulet, kiskadee 2. (Platyrinchids; genus *Platyrinchus*) spadebill 3. (Rhynchocyclids; genera *Rhynchocyclus, Tolmomyias,Ramphotrigon*) flatbill

-**ŇSBW**- ‘FORMICAROID BIRD’ I 1. (Melanopareiids; genus *Malanopareia*) crescentchest 2. (Thamnophilids; numerous genera) antbird, antshrike, antwren, antvireo 3. (Conopophagids; genera *Conopophaga, Pittasoma*) gnateater

-**ŇSBY**- ‘FORMICAROID BIRD’ II 1. (Grallariids; several genera) antpitta 2. (Rhynocryptids; numerous genera) tapaculo, huet-huet, turca, gallito, bamboowren, bristlefront 3. (Formicariids; genera Formicarius, Chamaeza) antthrush

-**ŇSBL**- ‘FURNARIID (NEOTROPICAL OVENBIRD)’ I 1. (Furnariini; numerous genera) neotropical ovenbird, hornero, reedhaunter, rushbird, streamcreeper, earthcreeper, barbtail, tuftedcheeks 2. (Pygarrhichnini plus *Xenops* and *Berlepschia*; several genera) treerunner, xenops, earthcreeper, palmcreeper 3. (Philydorini; numerous genera) foliage-gleaner, hookbill, treehunter, great xenops

-**ŇSBR**- ‘FURNARIID (NEOTROPICAL OVENBIRD)’ II 1. (Synallaxini; numerous genera) spintail, treerunner, rayadito, wiretail, tit-spinetail, thornbird, firweood-gatherer, brushrunner, cacholote, false canastero, wren-spinetail, prickletail, plushcrown, graveteiro, softtail, barbtail, reedhaunter 2. (Sclerurines; genera *Sclerurus, Geositta*) leaftosser, miner 3. (Dendrocolaptines; numerous genera) woodcreeper

-**ŇST**- ‘MENUROID BIRD’ 1. (Menurids; genus *Menura*) lyrebird 2. (Atrichornithids; genus *Atrichornis*) scrubbird 3. [stem unassigned]

-**ŇSTW**- ‘CLIMACTERIOID BIRD’ 1. (Ptilonorhynchids; genera *Archboldia, Amblyomis, Prionodura, Sericulus, Ptilonorhynchus, Chlamydera*) bowerbird 2. (Ptilonorhynchids; genera *Ailuroedus, Scenopoeetes*) catbird 3. (Climacterids; genera *Climacteris, Cormobates*) Australian treecreeper

-**ŇSTY**- ‘MALURID (AUSTRALIAN WREN)’ 1. (Malurini; several genera) fairywren 2. (Stipiturini; genus *Stipiturus*) emu-wren 3. (Amytornithines; genus *Amytornis*) grasswren

-**ŇSTL**- ‘ACANTHIZID (AUSTRALIAN WARBLER)’ I 1. (genera *Sericornis, Aethomyias, Neosericornis*) scrubwren 2. (genus *Orescopus*) fernwren 3. (genera, *Hylacola, Calamanthus*) fieldwren, heathwren

-**ŇSTR**- ‘ACANTHIZID (AUSTRALIAN WARBLER)’ II 1. (genus *Acanthiza*) thornbill 2. (genus *Aphelocephala*) whiteface 3. (genus *Pachycare*) goldenface

-**ŇSTŘ**- ‘ACANTHIZID (AUSTRALIAN WARBLER)’ III 1. (genus *Gerygone*) gerygone 2. (genus *Acanthornis*) scrubtit 3. (genus *Smicornis*) weebill

-**ŇSTÇ**- ‘ACANTHIZID (AUSTRALIAN WARBLER)’ IV 1. (genus *Pyncoptilus*) pilotbird 2. (genus *Pyrrholaemus*) redthroat, speckled warbler 3. (genus *Origma*) rockwarbler, mouse-warbler

-**ŇSTM**- ‘MELIPHAGOID BIRD’ 1. (Dasyornithids; genus *Dasyornis*) bristlebird 2. (Pardalotes; genus *Pardalotes*) pardalote / peep-wren 3. (Meliphagids; numerous genera) honeyeater, myzomela, chat, friarbird, kikau, straightbill, wattlebird, melidectes, myza

-**ŇSTN**- ‘ORTHONYCHOID BIRD’ 1. (Orthonychids; Genus *Orthonyx*) logrunner 2. (*O. spaldingii*) chowchilla 3. (Pomatostomids; genus *Pomatostomus*) Australasian babbler / pseudo-babbler

-**ŇSTF**- ‘CORVIFORM BIRD’ 1. (Cinclosomatids; genera *Cinclosoma, Ptilorrhoa*) quail-thrush, jewel-babbler 2. (Campephagids; numerous genera) cuckooshrike, triller, minivet 3. (Neosittids; genus *Daphoenositta*) sitella

-**ŇSTV**- ‘MOHOUID (genus *Mohoua*)’ 1. (*M. albicilla*) whitehead 2. (*M. novaeseelandiae*) pipipi / brown creeper 3. (*M. ochrocephala*) yellowhead

-**ŇSK**- ‘ORIOLID’ 1. (genus *Oriolus*) oriole 2. (genus *Sphecotheres*) figbird 3. (genus *Pitohui*) pitohui

-**ŇSKW**- ‘ORIOLOID BIRD’ I 1. (Pachycephalids; several genera) whistler, strike-thrush 2. (Oreocids; several genera) Australo-Papuan bellbird 3. (Falcunculids; genus *Falcunculus*) crested shriketit

-**ŇSKY**- ‘ORIOLOID BIRD’ II 1. (Psophodids; genera *Psophodes, Androphobus*) whipbird, wedgebill 2. (Paramythiids; genera *Paramythia, Oreocharis*) painted berrypecker 3. (Eulacestomids; genus *Eulacestoma*) wattled ploughbill

-**ŇSKL**- ‘VIREONID’ I 1. (genus *Vireo*) vireo 2. (genus *Vireolanius*) shrike-vireo 3. (genera *Hylophilus, Tunchiornis, Pachysylvia*) greenlet

-**ŇSKR**- ‘VIREONID’ II 1. (genus *Cyclarhis*) peppershrike 2. (genus *Pteruthius*) shrike-babbler 3. (genus *Epornis*) white-bellied epornis

-**ŇSKŘ**- ‘ARTAMID’ 1. (Artamines; genus *Artamus*) woodswallow 2. (Peltopsines; genus *Peltops*) peltops 3. (Cracticines; genera *Melloria, Cracticus, Strepara, Gymnorhina*) butcherbird, currawong, Australian magpie

-**ŇSKM**- ‘MALACONOTOID BIRD’ I 1. (Machaerirhynchids; genus *Machaerirhynchus*) boatbill 2. (Rhagologids; genus *Rhagologus*) mottled berryhunter / mottled whistler 3. (Aegithinids; genus *Aegithina*) iora

-**ŇSKN**- ‘MALACONOTOID BIRD’ II 1. (Pityriaseids; genus *Pityriasis*) bristled shrike / bald-headed wood-shrike / Bornean bristlehead 2. (Malaconotids; genera *Malaconotus, Laniarius, Telophorus, Chlorophoneus, Rhodophoneus*) bush-shrike, boubou, gonolek 3. (Malaconotids; genera *Nilaus, Dryoscopus, Bocagia, Tchagra*) brubru, puffback, tchagra

-**ŇSKÇ**- ‘PLATYSTEIRID’ 1. (genus *Platysteira*) wattle-eye 2. (genus *Batis*) batis 3. (genus *Lanioturdus*) white-tailed shrike

-**ŇSKF**- ‘VANGID’ I 1. (genera *Vanga, Calicalicus, Schetba, Xenopirostris, Falculea, Artamella, Leptopterus, Cyanolanius, Oriolia, Tylas, Hypositta*) vanga 2. (genus *Newtonia*) newtonia 3. (*genusTephrodornis*) woodshrike

-**ŇSKV**- ‘VANGID’ II 1. (genus *Prionops*) helmetshrike 2. (genus *Philentoma*) philentoma 3. (genera *Bias, Pseudobias, Hemipus, Megabyas, Mystacornis*) flycatcher, flycatcher-shrike, shrike-flycatcher, Crossley’s babbler

-**NST**- ‘CORVINE (CROW)’ 1. (genus *Corvus*) crow, raven, rook 2. (genus *Coloeus*) jackdaw 3. (genus *Nucifraga*) nutcracker

-**NSTW**- ‘CORVID’ I 1. (genus *Pica, Cyanopica, Cissa, Urocissa*) magpie 3. (genus *Garrulus*) [Old World] jay 3. (Crypsirinines; genera *Dendrocitta, Crypsirina, Temnurus, Platysmurus*) treepie

-**NSTY**- ‘CORVID’ II 1. (genus *Podoces*) ground jay / ground chough 2. (genus *Ptilostomus*) piacpiac 3. (genus *Zavattariornis*) bush crow

-**NSTL**- ‘CORVID’ III 1. (genera *Cyanocorax, Cyanocitta, Cyanolyca*) jay, blue-jay, Steller’s jay 2. (genus *Psilorhinus*) brown jay 3. (genus *Calocitta*) magpie-jay

-**NSTR**- ‘CORVID’ IV 1. (genera *Aphelocoma, Gymnorhinus*) scrub-jay, pinyon jay 2. (genus *Perisoreus*) Canada jay, Siberian jay, Sichuan jay 3. (genus *Pyrrhocorax*) chough

-**NSTM**- ‘CORVOID’ I 1. (Laniids; several genera) shrike 2. (Terpsiphonines; several genera) monarch, crested flycatcher, paradise flycatcher 3. (Monarchines; numerous genera) monarch, shrikebill, magpie-lark, torrent-lark, broad-billed flycatcher, ‘elepaio

-**NSTN**- ‘CORVOID’ II 1. (Dicrurids; genus *Dicrurus*) drongo, balicassiao 2. (Ifritids; genus *Ifrita*) blue-caped ifrit 3. (Melampittids; genus *Melampitta*) melampitta

-**NSTŘ**- ‘RHIPIDURID’ 1. (genus *Rhipidura*) fantail, wagtail 2. (genus *Lamprolia*) silktail 3. (genus *Chaetorhynchus*) drongo fantail / pygmy drongo

-**NSTŇ**- ‘CORCORACID (MUDNESTER)’ 1. (genus *Corcorax*) white-winged chough 2. (genus *Struthidea*) apostlebird 3. [stem unassigned]

-**NSTÇ**- ‘PARADISAEID (BIRD-OF-PARADISE)’ I 1. (genera *Paradisaea, Pteridophora, Semioptera, Seleucidis, Cicinnurus, Paradisornis*) bird-of-paradise 2. (genera *Epimachus, Drepanornis*) sicklebill 3. (genera *Lophorina*) riflebird, superb bird-of-paradise

-**NSTF**- ‘PARADISAEID (BIRD-OF-PARADISE)’ II 1. (genera Manucodia, *Phonygammus*) manucode 2. (genus *Astrapia*) astrapia 3. (genus *Parotia*) parotia

-**NSTV**- ‘PARADISAEID (BIRD-OF-PARADISE)’ III 1. (genus *Lycocorax*) paradise-crow 2. (genus *Paradigalla*) paradigalla 3. [stem unassigned]

-**MST**- ‘PASSERID’ I 1. (Melanocharitids; genera *Melanocharis, Rhamphocharis*) berrypecker 2. (Melanocharitids; genera *Toxorhamphus, Oedistoma*) longbill 3. (Cnemophilids; genera *Cnemophilus, Loboparadisea*) satinbird

-**MSTW**- ‘PASSERID’ II 1. (Notiomystids; genus *Notiomystis*) stitchbird / hihi 2. (Petroicids; numerous genera) Australasian robin, flyrobin 3. (Callaeids; genera *Callaeas, Philestumus, Heteralocha*) kokako, saddleback, huia

-**MSTY**- ‘EUPETOID BIRD’ 1. (Picathartids; genus *Picathartes*) rockfowl / bald crow 2. (Chaetopids; genus *Chaetops*) rockjumper 3. (Eupetids; genus *Eupetes*) rail-babbler

-MSTL- ‘PAROID BIRD’ 1. (Stenostirids; several genera) fairy fly-catcher, crested flycatcher, canary-flycatcher 2. (Hyliotids; genus Hyliota) hyliota 3. (Remizids; several genera) penduline tit, verdin

-**MSTR**- ‘PARID (TIT / TITMOUSE / CHICKADEE)’ 1. (genus *Parus, Melaniparus, Sittiparus, Pardaliparus, Periparus, Sylviparus, Cephalopyrus, Machlolophus, Melanochlora, Pseudopodoces, Cyanistes, Poecile*) tit, chickadee 2. (genus *Baeolophus*) titmouse 3. (genus *Lophophanus*) crested tit

-**MSTŘ**- ‘MACROSPHENID (AFRICAN WARBLER)’ 1. (genus *Macrosphenus*) longbill 2. (genus *Sylvietta*) crombec 3. (genera *Achaetops, Sphenoeacus, Cryptillas, Melocichla*) rockrunner, grass warbler, grassbird

-**MSP**- ‘SYLVIOID BIRD’ I 1. (Alaudids; numerous genera) lark, skylark, sparrow-lark, hoopoe-lark 2. (Nicatorids; genus *Nicator*) nicator 3. (Panurids; genus *Panurus*) bearded reedling / bearded parrotbill / bearded tit

-**MSPW**- ‘SYLVIOID BIRD’ II 1. (Pnoepygids; genus *Pnoepyga*) cupwing / wren-babbler 2. (Acrocephalids; genus *Acrocephalus*) reed warbler, swamp warbler, millerbird 3. (Locustellids; numerous genera) grass warbler, grassbird, bush warbler, grasshopper warbler, malia

-**MSPY**- ‘SYLVIOID BIRD’ III 1. (Donacobiids; genus *Donacobius*) black-capped donacobius 2. (Bernierids; several genera) Madagascar warbler, yellowbrow, bernieria, wedge-tailed jery 3. (Phylloscopids; genus *Phylloscopus*) leaf warbler, chiffchaff

-**MSPL**- ‘SYLVIOID BIRD’ IV 1. (Pycnonotids; numerous genera) bulbul, greenbul, bristlebill, brownbul, leaflove 2. (Cettiids; several genera) bush warbler, stubtail, tesia, broad-billed warbler 3. (Hyliids; genera *Hylia, Pholidornis*) hylia

-**MSPR**- ‘SYLVIOID BIRD’ V 1. (Sylviids; genus *Sylvia*) [Old World] warbler, blackcap, whitethroat, parisoma 2. (Sylviids; genera *Myzornis*, *Parphasma*) myzornis, Abyssian catbird 3. (Aegithalids; several genera) bushtit, tit-warbler

-**MSPŘ**- ‘SYLVIOID BIRD’ VI 1. (Sylviids; genera *Myzornis, Pseudoalcippe, Horizorhinus, Lioptilus, Fulvetta, Chrysomma, Moupinia,* Rhopophilus) babbler, thrush-babbler, fulvetta 2. (Sylviids; genera *Conostoma, Cholornis, Suthora, Sinosuthora, Neosuthora, Chleuasicus, Psittiparus, Paradoxornis*) parrotbill 3. (Sylviids; genus *Chamaea*) wrentit

-**MSPÇ**- ‘SYLVIOID BIRD’ VII 1. (Timaliids; numerous genera) [Old World] babbler, wren-babbler, scimitar-babbler, tit-babbler, tawny-bellied babbler, chestnut-capped babbler, wedge-billed babbler, dark-fronted babbler 2. (Pellorneids; numerous genera) babbler, grass babbler, fulvetta, wren-babbler, iliadopsis, grassbird 3. (Leiothrichids; numerous genera) laughingthrush, babbler, cutia, crocia, barwing, minla, sibia, liocichla, babax

-**MSPŢ**- ‘SYLVIOID BIRD’ VIII 1. (Scotocercids; genus *Scotocerca*) streaked scrub warbler 2. (Erythrocercids; genus *Erythrocercus*) little yellow flycatcher, Livingstone’s flycatcher, chestnut-capped flycatcher 3. other sylvioid bird

-**MSPF**- 'HIRUNDID (SWALLOW / MARTIN)' 1. (genera *Hirundo, Cecropsis, Alopochelidon, Neochelidon, Atticora, Haplocheledon, Notiochelidon, Tachycineta, Cheramoeca, Petrochelidon*) swallow, cliff swallow 2. (genera *Delichon, Progne, Ptyonoprogne, Riparia, Phedina, Pseudochelidon*) martin, river martin, crag martin, house martin 3. (genus *Psalidoprocne*) saw-wing

-**ŇSB**- **'**CISTICOLID (AFRICAN & ASIAN WARBLER)' I 1. (genus *Cisticola*) cisticola 2. (genus *Neomixis*) jery 3. (genera *Oreophilais, Incana, Drymocichla, Spiloptila, Phyllolais, Malcorus, Hypergerus, Eminia, Euryptila, Bathmocercus, Sceopmycter, Artisomus, Poliolais*) [African / Asian] warbler

-**ŇSBŘ**- 'CISTICOLID (AFRICAN & ASIAN WARBLER)' II 1. (genera *Prinia, Schistolais, Phragmacia*) prinia 2. (genera *Apalis, Oreolais*) apalis 3. (genus *Orthotomus*) tailorbird

-**ŇSG**- **'**CISTICOLID (AFRICAN & ASIAN WARBLER)' III 1. (genus *Micromacronus*) miniature babbler 2. (genus *Urolais*) green longtail 3. (genus *Calamonastes*) wren-warbler

-**ŇSGŘ**- 'CISTICOLID (AFRICAN & ASIAN WARBLER)' IV 1. (genus *Camaroptera*) camaroptera 2. (genus *Eremomela*) eremomela 3. [stem unassigned]

-**NSK**- **'**TURDID (THRUSH / BLACKBIRD)' 1. (genus *Turdus*) thrush, [Old World] blackbird, ouzel, fieldfare, redwing, American robin 2. (genus *Sialia*) bluebird 3. (genera other than *Turdus* and *Sialia*) grandala, solitaire, cochoa, fruithunter

-**NSKW**- 'ERITHACINE' I 1. (genus *Erithacus*) European robin / robin redbreast 2. (genus *Chamaetylas*) alethe 3. (genera *Cossypha, Cossyphicula*) robin-chat, cave chat, ground robin

-**NSKY**- 'ERITHACINE' II 1. (genera *Swynnertonia, Pogonocichla, Stiphrornis*) forest robin, robin-chat 2. (genus *Sheppardia*) akalat 3. (genus *Cichladusa*) palm thrush

-**NSKL**- 'SAXICOLINE' I 1. (genus *Luscinia*) nightingale, thrust nightingale, redstart, bluethroat 2. (genera *Heinrichia, Heteroxenicus, Brachypteryx*) shortwing 3. (genera *Irania, Larvivora, Vauriella, Myiomela, Tarsiger, Cinclidium*) blue robin, bush robin, bluetail, blue-fronted robin

-**NSKR**- 'SAXICOLINE' II 1. (genera *Saxicola, Campicoloides, Pinarochroa, Thamnolaea, Emarginata,Myrmecocichla, Pinarornis, Namibornis*) chat, stonechat, cliff chat, moorland chat 2. (genus *Phoenicurus*) redstart 3. (genus *Oenanthe*) wheatear

-**NSKŘ**- 'SAXICOLINE' III 1. (genus *Calliope*) rubythroat, firethroat, blackthroat 2. (genus *Enicurus*) forktail 3. (genera *Ficedula, Muscicapella, Humblotia*) flycatcher, pygmy flycatcher

-**NSKF**- 'SAXICOLINE' IV 1. (genus *Myophonus*) whistling thrush 2. (genus *Monticola*) rock thrush 3. [stem unassigned]

-**RNSK**- **'**MUSCICAPOID BIRD' I 1. (Troglodytids; numerous genera) wren 2. (Certhiids; genera *Certhia, Salpornis*) treecreeper, spotted creeper 3. (Polioptilids; genera *Polioptila, Microbates, Ramphocaenus*) gnatcatcher, gnatwren

-**RNSKW**- 'MUSCICAPOID BIRD 'II 1. (Tichodromadids; genus *Tichodroma*) wallcreeper 2. (Sittids; genus *Sitta*) nuthatch 3. (Regulids; genus *Regulus*) kinglet / crest

-**RNSKY**- 'MUSCICAPOID BIRD' III 1. (Elachurids; genus *Elachura*) spotted elachura / spotted wren-babbler 2. (Ptiliogonatids; several genera) silky-flycatcher, phainoptila, phainopepla 3. (Cinclids; genus *Cinclus*) dipper

-**RNSKL**- 'MUSCICAPOID BIRD' IV 1. (Bombycillids; genus *Bombycilla*) waxwing 2. (Dulids; genus *Dulus*) palmchat 3. (Hypocoliids; genus *Hypocolius*) grey hypocolus

-**RNSKR**- 'MUSCICAPOID BIRD' V 1. (Mimids; numerous genera) mockingbird, catbird, thrasher 2. (Sturnids; numerous genera) starling, myna, Philippine creeper 3. (Buphagids; genus *Buphagus*) oxpecker

-**RNSKŘ**- 'MUSCICAPID' 1. (Muscicapini and Niltavines; several genera) [Old World] flycatcher, forest flycatcher, tit-flycatcher, silverbird 2. (Muscicapinae; genera *Copsychus, Cercotrichas*) magpie-robin, shama, scrub robin / bush-chat 3. (Niltavines; genus *Sholicola*) sholakili

-**RNSKF**- 'ZOSTEROPID (WHITE-EYE)' 1. (numerous genera other than *Yuhina, Sterrhoptilus* and *Heleia*) white-eye, striped babbler, pygmy babbler 2. (genera *Yuhina, Sterrhoptilus*) yuhina 3. (genus *Heleia*) heleia

-**ŇŠT**- ‘FRINGILLID (FINCH)’ 1. (Fringillines; genus *Fringilla*) chaffinch, brambling 2. (Carduellines; numerous genera) grosbeak, hawfinch, bullfinch, rosefinch, drepanid / honeycreeper, greenfinch, twite, linnet, redpoll, crossbill, serin, mountain finch, desert finch, African canary, goldfinch, siskin 3. (Euphoniines; genera *Euphonia*, *Chlorophonia*) euphonia, chlorophonia

-**ŇŠTW**- ‘ ESTRILDOID BIRD’ 1. (Ploceids; numerous genera) weaver , bishop, widowbird, fody, quelea, malimbe 2. (Viduids; genera *Vidua, Anomalospiza*) indigobird, whydah, cuckoo-finch 3. (Estrildids; numerous genera) waxbill, mannikin, munia, silverbill, twinspot, firefinch, quailfinch, antpecker, nigrita, oliveback, pytilia, crimsonwing, seedcracker, bluebill, cordon-bleu, purple grenadier, avadavat, estrildid finch

-**ŇŠTY**- ‘PASSERIFORM BIRD’ I 1. (Promeropids; genus *Promerops*) sugarbird 2. (Modulatricids; genera *Modulatrix, Arcanator, Kakamega*) spot-throat, dapple-throat, grey-chested babbler 3. (Dicaeids; genera *Dicaeum, Prionochilus*) flowerpecker

-**ŇŠTL**- ‘PASSERIFORM BIRD’ II 1. (Nectariniids; numerous genera) sunbird, spiderhunter 2. (Urocynchramids; genus *Urocynchramus*) Przewalski’s finch / Przewalksi’s pinktail 3. (Irenids; genus *Irena*) fairy-bluebird

-**ŇŠTR**- ‘PASSERIFORM BIRD’ III 1. (Chloropseids; genus *Chloropsis*) leafbird 2. (Peucedramids; genus *Peucedramus*) olive warbler 3. (Prunellids; genus *Prunella*) accentor, dunnock, hedge sparrow

-**ŇŠTŘ**- ‘PASSAROID BIRD’ 1. (Passerids; numerous genera) [Old World] sparrow, rockfinch, snowfinch, bush sparrow, ibon 2. (Motacillids; several genera) wagtail, longclaw, pipit, shorttail, madanga 3. (Calcariids; genera *Calcarius, Plectrophenax, Rhynchophanes*) longspur, bunting

-**ŇŠTÇ**- ‘ICTERID’ I 1. (genera *Agelaius, Xanthspar, Agelasticus, Chrysomus, Nesopsar, Xanthocephalus, Dives, Euphagus, Gymnomystax, Amblyramphus, Curaeus, Anumara, Gnorimopsar, Oreopsar*) [New World] blackbird 2. (genera *Quiscalus, Hypopyrrhus, Lampropsar, Macroagelaius*) grackle 3. (genus *Icterus*) [New World] oriole, troupial

-**ŇŠTF**- ‘ICTERID’ II 1. (genera *Molothrus, Agelaioides*) cowbird, baywing 2. (genus *Sturnella, Leistes*) meadowlark 3. (genus *Pseudoleistes*) marshbird

-**ŇŠTV**- ‘ICTERID’ III 1. (genus *Psarocolius*) oropendola 2. (genera *Cacicus, Cassiculus, Amblycercus*) cacique 3. (genus *Dolichonyx*) bobolink

-**NŠP**- **‘**EMBERIZOID BIRD’ I 1. (Passerellids; numerous genera) [American] sparrow, towhee, lark bunting, ground sparrow, junco 2. (Emberizids; genus *Emberiza*) bunting, reed bunting, yellowhammer 3. (Rhodinochichlids; genus *Rhondinocichla*) rosy thrush-tanager

-**NŠPW**- ‘EMBERIZOID BIRD’ II 1. (Phaenicophilids; genera *Phaenicophilus, Xenoligea, Microligea*) tanager, white-winged warbler, green-tailed warbler 2. (Zeledoniids; genus *Zeledonia*) wrenthrush 3. (Teretistrids; genus *Teretistris*) Cuban warbler

-**NŠPY**- ‘EMBERIZOID BIRD’ III 1. (Parulids; numerous genera) [New World] warbler, wood warbler, oven bird, waterthrush, yellowthroat, whitestart, parula, redstart 2. (Icteriids; genus *Icteria*) yellow-breasted chat 3. (Spindalids; genus *Spindalis*) spindalis

-**NŠPL**- ‘EMBERIZOID BIRD’ IV 1. (Microspingids; genera *Mitrospingus, Orthogonys, Lamprospiza*) micropspingid tanager 2. (Calyptophilids; genus *Calyptophilus*) chat-tanager 3. (Nesospingids; genus *Nesospingus*) Puerto Rican tanager

-**NŠPR**- ‘THRAUPID (TANAGER)’ I 1. (numerous genera) tanager, tanager-finch, conebill, flowerpiercer 2. (Tholospiza [finch-like tanagers]; numerous genera) bullfinch, orangequit, grassquit, warbler-finch, ground finch, tree finch, bananaquit 3. (genera *Cyanicterus, Bangsia, Buthraupis, Cnemathraupis, Chloromis, Wetmorethraupis, Anisognathus, Dubusia, Pseudosaltator*) mountain tanager

-**NŠPŘ**- ‘THRAUPID (TANAGER)’ II 1. (genera *Tersina, Cyanerpes, Chlorophanes, Iridophanes*) honeycreeper 2. (genus *Dacnis*) dacnis 3. (genera *Sporophila, Dolospingus, Oryzoburus*) seedeater, seed-finch

-**NŠPF**- **‘**THRAUPID (TANAGER)’ III 1. (genera Emberizoides, *Embernagra*) grass-finch, pampa-finch 2. (genus *Hemispingus*) hemispingus 3. (genus *Nephelornis*) pardusco

-**NŠPV**- ‘THRAUPID (TANAGER)’ IV 1. (genus *Catamblyrhynchus*) plushcap 2. (genus *Incaspiza*) Inca finch 3. (genera *Saltator, Saltatricula*) saltator, many-colored Chaco finch

-**MSTF**- ‘CARDINALID (CARDINAL)’ I 1. (genera *Cardinalis, Gubernatrix*) cardinal 2. (genera *Periporphyrus, Caryothraustes, Rhodothraupis, Cyanocompsa, Cyanoloxia, Pheucticus*) grosbeak 3. (genera *Chlorothraupis, Habia, Piranga*) [cardinalid] tanager

-**MSTV**- ‘CARDINALID (CARDINAL)’ II 1. (genus *Passerina*) [North American] bunting 2. (genera *Spiza, Amaurospiza*) dickcissel, seedeater 3. (genus *Granatellus*) chat

7.3.1.5 Fish, Tunicates and Lancelets

Hagfish, Lancelets, Tunicates, Lampreys, Skates, Sawfish, & Rays

-**PSGW**- ‘HAGFISH / LANCELET’ 1. (genera *Rubicundus, Eptatretus, Myxine, Nemamyxine, Neomyxine, Notomyxine*) hagfish 2. (Ampioxiforms; several genera) lancelet

-**PSGY**- ‘TUNICATE’ 1. (Phlebobranchians and Aplousobranchians; numerous genera) sea squirt 2. (Thaliaceans; numerous genera) salp, pyrosome, doliolid tunicate 3. (Stolidobranchians and Larvaceans; several genera) other tunicate

-**PSGL**- ‘LAMPREY’ 1. (genus *Geotria*) pouched lamprey 2. (genus *Mordacia*) Southern topeyed lamprey 3. (Petromyzontids; numerous genera) Northern lamprey

-**PSGR**- SKATE’ 1. (Rajids and Gurgesiellids; numerous genera) skate 2. (Anacanthobatids; several genera) smooth skate / leg skate 3. (Arhynchobatids; numerous genera) softnose skate

-**PSGŘ**- ‘SAWFISH’ (genera *Pristis, Anoxypristis*) 1. (*P. pristis*) [largetooth, common, wide, freshwater, river, northern] sawfish / carpenter shark 2. (Indo-Pacific species; *A. cuspidata, P. clavata, P. zijsron*) [knifetooth, pointed, Queensland, green, longcomb, narrowsnout, olive] sawfish 3. (Atlantic species; *P. pectinata*) smalltooth sawfish

-**PSGḌ**- ‘ELECTRIC RAY’ 1. (genera Torpedo, Tetronarce) electric ray 2. (Narcinenes and Hypnines; several genera) numbfish, coffin ray 3. (Narkines, several genera) sleeper ray

-**PSGV**- ‘STING RAY’ 1. (Myliobatoids; numerous genera) stingray, [cownose, eagle, manta, devil, butterfly, river, whiptail, deepwater, sixgill] ray 2. (genus *Zanobatis*) panray 3. (genera *Platyrhina, Platyrhinoidis*) thornback

Sharks and Chimaeras [To signify a ‘shark’ in general, use the appropriate stem from the general ‘TYPES OF FISH’ root -**PFT**- shown in Sec. 7.3 above]

-**PSDW**- ‘COW SHARK / FRILLED SHARK’ (Hexanchids; genera *Heptranchias, Notorhynchus*) sevengill cow shark 2. (genus *Hexanchus*) sixgill cow shark 3. (genus *Chlamydoselachus*) frilled shark

-**PSDY**- ‘SQUALIFORM SHARK’ I 1. (Squalids; genera *Cirrhigaleus, Squalus*) dogfish shark 2. (Dalatiids; several genera) kitefin shark 3. (Somniosids several genera) sleeper shark

-**PSDL**- ‘SQUALIFORM SHARK’ II 1. (Etmopterids; several genera) lantern shark 2. (genera *Centrophorus, Deania*) gulper shark 3. (genus *Oxynotus*) rough shark

-**PSDR**- ‘SQUATINOFORM SHARK’ 1. (genus *Squatinus*) angel shark 2. (genera *Pliotrema, Pristiophorus*) sawshark 3. (genus *Echinorhinus*) bramble shark, prickly shark

-**PSDŘ**- ‘GROUND SHARK’ 1. (Carcharhinids; numerous genera) requiem shark 2. (Hemigaleids; several genera) weasel shark 3. (genera *Eusphyra, Sphyrna*) hammerhead shark, winghead shark, bonnethead, scoophead

-**PSSDW**- ‘HOUNDSHARK / BULLHEAD SHARK’ 1. (Triakines; genera *Musteles, Scylliogaleus, Triakis, Leptocharius*) houndshark, barbeled houndshark 2. (Galeorhinines; several genera) whiskery shark, school shark, sailback houndshark, blacktip tope 3. (genus *Heterodontus*) bullhead shark

-**PSSDY**- ‘MACKEREL SHARK (LAMNID)’ 1. (genus *Carcharodon*) great white shark 2. (genus *Isurus*) mako 3. (genus *Lamna*) salmon shark, porbeagle

-**PSBW**- 'LAMNIFORM SHARK' I 1. (genus *Alopius*) thresher shark 2. (genus *Cetorhinus*) basking shark 3. (genus *Megachasma*) megamouth shark

-**PSBY**- 'LAMNIFORM SHARK' II 1. (genus *Mitsukurina*) goblin shark 2. (Odontaspidids; genera *Carcharias, Odontaspis*) sand shark / sand tiger shark / grey nurse shark / ragged tooth shark 3. (genus *Pseudocarcharias*) crocodile shark

-**PSBL**- 'CARPET SHARK' I 1. (Ginglymostomatids and Brachaelurids; numerous genera) nurse shark, blind shark 2. (Orectobids; several genera) wobbbegong shark 3. Hemiscylliids; genera *Chyloscyllium, Hemiscyllium*) bamboo shark

-**PSBR**- 'CARPET SHARK' II 1. (genus *Rhincodon*) whale shark 2. (genus *Stegostoma*) zebra shark 3. (genera *Cirrhoscyllium, Parascyllium*) collared carpet shark

-**PSBǍ**- 'CATSHARK' 1. (Scyliorhinids; numerous genera) catshark 2. (Proscylliids, several genera) finback catshark (including ribbontail catshark, harlequin catshark, graceful catshark, magnificent catshark) 3. (Pseudotriakids; genera *Pseudotriakis, Planonasus, Gollum*) false catshark, gollumshark

-**PSDV**- 'CHIMAERA' 1. (Chimaerids; genera *Chimaera, Hydrolagus*) chimaera, rabbitfish, ghostshark, ratfish 2. (Callorhinchids; genus *Callorhinchus*) ploughnose chimaera, Cape elephantfish, Australian ghost shark 3. (Rhinochimaerids; genera *Harriotta, Neoharriotta, Rhinochimaera*) spookfish, sicklefin chimaera, paddle-nose chimaera, longnose chimaera

Lobe-Finned and Ray-Finned Fishes

-**PSSGW**- 'ARCHAIC FISH' 1. (Coelacanths; genus Latimeria) coelacanth 2. (Ceratodontids; Lepidosirenids; several genera) lungfish 3. (Polypterids; genera *Erpetoichthys, Polypterus*) reedfish, bichir

-**PSSGY**- 'STURGEON / PADDLEFISH' 1. (Acipenserids; several genera) sturgeon 2. (genus *Polyodon*) American paddlefish 3. (genus *Psephurus*) Chinese paddlefish

-**PSSGL**- 'BOWFIN / GAR' 1. (genus *Amia*) bowfin 2. (genus *Atractosteus*) alligator gar, Cuban gar, tropical gar 3. (genus *Lepisosteus*) spotted gar, longnose gar, shortnose gar, Florida gar

-**PSSGR**- 'EEL & EEL-LIKE FISH' 1. (Anguilliforms; numerous genera) eel, moray, conger 2. (Notacanthids; genera *Lipogenys, Notacanthus, Polyacanthonotus*) deep-sea spiny eel 3. (Halosaurids; genera *Aldrovandia, Halosaurus, Halosaropsis*) halosaur

-**PSSGǍ**- 'ELOPIFORM & ALBULIFORM FISH' 1.(genus *Elops*) ladyfish, skipjack, jack-rash, tenpounder 2. (genus *Megalops*) tarpon 3. (genera *Nemoossis, Pterothrissus, Albula*) bonefish

-**PSSGV**- **'**OSTEOGLOSSOID FISH' I 1. (Arapaimids & Osteoglossids; genera *Heterotis, Arapaima, Scleropages, Osteoglossum*) bonytongue, arowana 2. (Pantodontids; genus *Pantodon*) butterflyfish 3. (Hiodontiforms; genus *Hiodon*) mooneye, goldeye

-**PSSGḌ**- 'OSTEOGLOSSOID FISH' II 1. (Notopterids; several genera) knifefish, featherback 2. (Gymnarchids; genus *Gymnarchus*) aba aba / freshwater rat-tail, 3. (Mormyrids; numerous genera) elephant fish

-**PST**- 'CLUPEIFORM FISH 1. (Engraulids; numerous genera) anchovy 2. (Clupeids; numerous genera) herring, sprat, sardine, shad 3. (Clupeiforms other than Clupeids and Engraulids) denticle herring, longfin herring, wolf herring, round herring

-**PSTW**- 'GONORYNCHID FISH 1. (genus *Chanos*) milkfish 2. (genus *Gonorynchus*) beaked salmon / beaked sandfish 3. (Knerioids; several genera) shellear, hingemouth/snake mudhead

-**PSTY**- 'CYPRININE AND ALEPOCEPHALID FISH 1. (*Carassius auratus*) goldfish 2. (Cyprinines other than *Carassius*) carp, Crucian carp, Prussian carp, ginbuna, koi 3. (Alepocephalids; numerous genera) smooth-head, slick-head marine smelt

-**PSTL**- 'LEUCISCINE FISH' I 1. (small Leuciscines; numerous genera) minnow 2. (genera *Condoma, Coreoleucisccus, Cyprinella, Luxilus, Lythrurus, Notemigonus, Notropis, Pteronotropis*) shiner 3. (genera *Hesperoleucus, Rutilus, Samarutilus, Leucos*) roach, pigo, kutum, other roach-like fish

-**PSTR**- 'LEUCISCINE FISH' II 1. (genera *Abramis, Ballerus, Blicca*) bream 2. (genera *Chrosomus, Clinostomus, Leuciscus, Margariscus, Leucalburnus, Telestes*) dace 3. (genera *Chondrostoma, Protochondrostoma, Pseudochondrostoma*) nase, sneep

-**PSTŘ**- 'LEUCISCINE FISH' III 1. (genus *Tica*) tench, doctor fish 2. (genus *Campostoma*) stoneroller 3. (genus *Acrocheilus*) chiselmouth

-**PSTÇ**- 'LEUCISCINE FISH' IV 1. (genus *Mylopharadon*) hardhead 2. (genus *Orthodon*) Sacramento blackfish 3. (genus *Mylocheilus*) peamouth

-**PSTF**- 'LEUCISCINE FISH' V 1. (genus *Leucaspius*) sunbleak, belica, moderlieschen 2. (genus *Pelecus*) sabre carp, sabrefish, ziege, sichel 3. (genus *Vimba*) vimba

-**PSTĻ**- 'LEUCISCINE FISH' VI 1. (genera *Hybopsis, Semotilus, Squalius, Hemitremia, Platygobio, Nocomis, Couesius, Iotichthys, Snyderichthys, Oregonichthys, Petroleuciscus, Erimystax, Gila, Siphateles, Algansea*) chub 2. (genus *Scardinius*) rudd 3. (numerous genera) other leuciscine fish

-**RPST**- 'CYPRINID FISH (Other than Cyprinines and Leuciscines)' I 1. (Acheilognathines; several genera) bitterling 2. (Alburnines; several genera) bleak 3. (Gobionines; numerous genera) gobionine fish (including gudeons, steeds)

-**RPSTW**- 'CYPRINID FISH (Other than Cyprinines and Leuciscines)' II 1. (Barbines and Leptobarbines; numerous genera) barbine fish (including barbs, barbels, snowtrouts) and *Leptobarbus* 2. (Labeonines; numerous genera) labeonine fish (including mud carps, labeos, yellowfish) 3. (Cultrines; numerous genera) cultrine fish (including sharpbellies)

-**RPSTY**- 'CYPRINID FISH (Other than Cyprinines and Leuciscines)' III 1. (Danionines; numerous genera) danionine fish (including carplets, rasboras, flying barbs, razorbelly minnows) 2. (Squaliobarbines & Xenocyprines; several genera) squaliobarbine and xenocyprine fish (including grass carp, black carp, bighead carp) 3. (Psylorhynchids and unclassified Cyprinids; numerous genera) other cyprinid fish [not otherwise indicated] (including giant salmon carp, blind barbs, mahseers, cavefish, catla, rohtees)

-**RPSTL**- 'CYPRINOID FISH (Other than Cyprinids)' 1. (Catastomids; numerous genera) sucker fish 2. (genus *Psilorhynchus*) psylorhynchus 3. (Cobitoids and Gyrinocheilids; numerous genera) loach, algae-eater

-**RPSTR**- 'CHARACIFORM FISH' 1. (Distichodontids; numerous genera) distichodontid fish 2. (Citharinids; genera Citharidium, Citharinops, Citharinus) lutefish 3. (Crenuchids; numerous genera) South American darter, sailfin tetra

-**RPSTŘ**- 'ALESTIOID FISH' 1. (Hepsetids; genus *Hepsetus*) African pike characin 2. (Alestiids; numerous genera) African tetra 3. [stem unassigned]

-**RPSTH**- 'SERRASALMID FISH' 1. (genera *Catoprion, Pristobrycon, Pygocentrus, Pygopristis, Serrasalmus*) piranha 2. (numerous genera) pacu 3. (numerous genera) silver dollar

-**ŘPSM**- 'ERYTHINOID FISH' I 1. (Tarumaniids; genus *Tarumania*) tarumania 2. (Erythrinids; genera *Erythrinus, Hoplerythrinus, Hoplias*) trahira/ tararira 3 (Cynodontids; several genera) dogtooth characin / vampire tetra / sabretooth tetra

-**ŘPSN**- 'ERYTHINOID FISH' II 1. (Hemiodontids; several genera) hemiodontid fish 2. (Paradontids; several genera) pardontid fish 3. (Prochilodontids; several genera) flannel-mouthed characin

-**ŘPSŇ**- 'ERYTHINOID FISH' III 1. (Chilodontids; genera *Caenotropus, Chilodus*) headstander 2. (Curimatids; several genera) toothless characin 3.(Anostomids; numerous genera) anostomid fish

-**ŘPSMW**- 'CHARACID FISH' I (including TETRA) 1. (Characins; several genera) characin fish 2. (Stevardiins and Characines other than Characins) characin-like fish 3. (Spintherobolines and Stethaprionines; numerous genera) other characin-like fish

-**ŘPSNW**- 'CHARACID FISH' II 1. (Ctenoluciids; genera *Boulengerella, Ctenolucius*) pike-characin 2. (Lebiasinids; several genera) pencil fish (including voladora) 3. (genus *Salminus*) dorado

-**PSL**- 'CATFISH' I 1. (Silurids; numerous genera) catfish 2. (Loricaroids; numerous genera) armored catfish, pencil catfish, parasitic catfish 3. (Diplomystids; numerous genera) velvet catfish

-**PSLW**- 'CATFISH' II 1. (Pangasiids, Mochokids, Claroteids; numerous genera) shark catfish, African catfish, squeaker 2. (Plotosids, Ictalurids, Clariids, Ailiids, Sisorids, Bagrids; numerous genera) eeltail catfish, ictalurid, airbreathing catfish, naked catfish 3. (Doradoids and Pimelodoids; numerous genera) banjo catfish, thorny catfish/talking catfish, driftwood catfish, three-barbeled catfish, bumblebee/dwarf marbled catfish, long-whiskered catfish

-**PSLY**- 'KNIFEFISH' 1. (Gymnotoids; numerous genera) banded knifefish 2. (Rhamphichthyoids; numerous genera) sand knifefish, bluntnose/grass/leaf knifefish 3. (Apteronotoids; numerous genera) glass knifefish, rattail knifefish, ghost knifefish

-**PSTH**- 'OTHER EUTELEOSTEOMORPHIC FISH' 1. (genus *Lepidogalaxias*) salamanderfish 2. (Argentinoids; numerous genera) smelt, barreleye 3. (Galaxiids; several genera) galaxia

-**PSK**- 'SALMONINE FISH' 1. (genus *Oncorhynchus*) Pacific salmon (i.e., Chinook, Coho, Pink, Sockeye, Chum), Pacific trout (i.e., Cutthroat, Rainbow/Steelhead) 2. (genus *Salmo*) Atlantic salmon, Brown trout 3. (genera *Salvelinus, Salvethymus*) char (including brook trout, lake trout, bull trout, Dolly Varden trout), long-finned char

-**PSKW**- 'SALMONID FISH' (other than salmon/trout/char) 1. Eurasian Salmonids; genera *Hucho, Parahucho, Brachymystax*) hucho, taimen/huchen, lenok 2. (genera *Coregonus*, *Prosopium, Stenodus*) whitefish 3. (genus *Thymallus*) grayling

-**PSKY**- 'ESOCIFORM FISH' 1. (genus *Esox*) pike, pickerel 2. (*E. masquinongy*) muskellunge 3. (Umbrids; several genera) mudminnow

-**PSKL**- 'OSMERIFORM FISH' 1. (Retropinnids and Osmerids; several genera) smelt, Australian-New Zealand grayling 2. (genus *Plecoglossus*) ayu / sweetfish 3. (Salangids; numerous genera) noodlefish / icefish

-**PSKR**- 'STERNOPTYCHID FISH' 1. (Maurolicines other than *Valenciennellus*; several genera) bottlelight, pearlside 2. (genus *Valenciennellus*) constellationfish 3. (Sternoptychines; genera *Argyropelecus, Polyipnus, Sternoptyx*) marine hatchetfish

-**PSKŘ**- 'STOMIIFORM' FISH (other than Sternoptychids) 1. (Gonostomatids; several genera) bristlemouth, anglemouth, fangjaw 2. (Stomiids; numerous genera) barbeled dragonfish, loosejaw, stareater, snaggletooth, viperfish 3. (Phosichthyids; several genera) lightfish

-**RPSKW**- 'ALEPISAUROID FISH' I 1. (genus *Alepisaurus*) lancetfish 2. (genus *Anotopterus*) daggertooth 3. (genus *Omosudis*) hammerjaw

-**RPSKY**- 'ALEPISAUROID FISH' II 1. (Evermannellids; several genera) sabertooth 2. (Paralepidids; numerous genera) barracudina 3. (Scopelarchids; several genera) pearleye

-**RPSKL**- 'AULOPIFORM FISH' I 1. (genera *Bathysauropsis, Bathysauroides*) grinner 2. (Chlorophthalmids; genera *Chlorophthalmus, Parasudis*) greeneye 3. (Ipnopids; several genera) tripod fish

-**RPSKR**- 'AULOPIFORM FISH' II 1. (Notosudids; several genera) waryfish 2. (genus *Bathysaurus*) deepsea lizardfish 3. (genus *Gigantura*) telescopefish

-**RPSKŘ**- 'SYNODONTOID FISH' 1. (Synodontids; several genera, and *Pseudotrichonotus*) lizardfish, Bombay duck / bummalo, sand-diving lizardfish 2. (Aulopids; several genera) flagfin 3. (genus *Paraulopus*) cucumberfish

-**PSSM**- 'MYCTOPHIFORM FISH & ATELEOPOD FISH' 1. (Myctophids; numerous genera) lanternfish 2. (Neoscopelids; several genera) blackchin 3. (Ateleopodids; several genera) jellynose fish / tadpole fish

-PSSN- 'LAMPRIFORM FISH' I 1. (Veliferids; genera *Velifer, Metavelifer*) sailfin moonfish 2. (genus *Lampris*) opah / moonfish / sunfish / kingfish 3. (genus *Radiicephalus*) tapertail

-**PSSŇ**- 'LAMPRIFORM FISH' II 1. (Lophotids; genera *Lophotus, Eumecichthys*) crestfish 2. (Trachipterids; genera *Desmodema, Trachipterus, Zu*) ribbonfish 3. (Regalacids; genera *Agrostichthys, Regalecus*) oarfish

-**PSKÇ**- 'PERCOPSIFORM FISH' 1. (genus *Percopsis*) trout-perch 2. (genus *Aphredotus*) pirate-perch 3. (Amblyopsids; several species) cavefish, swampfish, ricefish

-**PSKF**- 'ZEIFORM FISH' I 1. (Zeids; genera *Zeus, Zenopsis*) dory 2. (Cyttids; genus *Cyttus*) lookdown dory, silver dory, king dory, silver dory 3. (Zeniontids; genera *Zenion, Capromimus, Cyttomimus*) zeniontid, capro dory

-**PSKV**- 'ZEIFORM FISH' II 1. (Oreosomatids; several genera) oreo 2. (Parazenids; several genera) parazen 3. (Grammicolepidids; several genera) tinselfish

-**PSP**- 'GADID FISH' I 1. (genus *Gadus*) [true] cod 2. (genus *Melanogrammus*) haddock 3. (genus *Merlangius*) whiting / merling

-**PSPW**- **'**GADID FISH' II 1. (genus *Microgadus*) tomcod 2. (genus *Micromesistius*) blue whiting 3. (genus *Pollachius*) pollock

-**PSPY**- 'GADID FISH' III 1. (genera *Arctogadus, Boreogadus*) arctic/polar cod 2. (genera *Eleginus, Trisopterus*) saffron cod, navaga, poor cod, pout, pouting 3. (genus *Raniceps*) tadpole fish

-**PSPL**- 'LOTID FISH' I 1. (genus *Lota*) burbot / bubbot / lingcod / mariah / eelpout / coneyfish 2. (genus *Brosme*) cusk / brismak / brosmius / torsk / moonfish 3. (genus *Molva*) ling

-**PSPR**- 'LOTID FISH' II 1. (genus *Gaidropsaurus*) rockling 2. (genus *Enchelyopus*) fourbeard rockling 3. (genus *Ciliata*) ciliata fish

-**PSPŘ**- 'GADIFORM FISH' 1. (Bregmacerotids; genus *Bregmaceros*) codlet 2. (Euclichthyids; genus *Euclichthys*) eucla cod 3. (Muraenolepidids; genera *Muraenolepis, Notomuraenobathys*) eel cod

-**PSPÇ**- 'MORID FISH' 1. (genus *Mora*) common mora, googly-eyed cod 2. (genus *Auchenoceros*) ahuru 3. (other Morids; numerous genera) codling, hakeling, mora, beardie, skulpin, grenadier cod

-**PSPF**- 'OTHER PARACANTHOPTERYGIAN FISH' 1. (Macrourids; numerous genera) rattail / grenadier 2 (Melanonids; genus *Melanonus*) pelagic cod, arrowtail 3. (Stylephorids; genus *Stylephorus*) tube-eye / thread-tail

-**PSPH**- 'MERLUCCIID FISH' 1. (Merlucciids; several genera) hake 2. (genus *Maruronus*) blue grenadier 3. (Phycids; genera *Phycis, Urophycis*) phycid hake

-**PSC**- 'BERYCIFORM FISH' I 1. (Berycids; genus *Beryx*) alfonsino 2. (Berycids; genus *Centroberyx*) nannygal 3. (Ostracoberycids; genus *Ostracoberyx*) shellskin alfonsino

-**PSČ**- 'BERYCIFORM FISH' II 1. (Cetomimids; numerous genera) flabby whalefish, tapetail, hairyfish 2. (Rondeletiids; genus *Rondeletia*) redmouth whalefish 3. (Barbourisiids; genus *Barbourisia*) velvet whalefish

-**PŠČ**- 'BERYCIFORM FISH' III 1. (Melamphaids; several genera) ridgehead / bigscale 2. (Gibberichthyids; genus *Gibberichthys*) gibberfish 3. (Stephanoberycids and Hispidoberycids; several genera) pricklefish

-**PSCW**- 'TRACHICHTHYFORM FISH' I 1. (Trachichthyids; several genera except *Paratrychichthys*) roughy, slimehead, redfish, sawbelly 2. (genus *Paratrychichthys*) sandpaper fish 3. (Monocentrids; genera *Cleidopus, Monocentris*) pineapplefish / pinecone fish

-**PSČW**- 'TRACHICHTHYFORM FISH' II 1. (Anoplogastrids; genus *Anoplogaster*) fangtooth 2. (Diretmids; genera *Diretmus, Diretmoides, Diretmichthys*) spinyfin 3. (Anomalopids; several genera) lanternfish / flashlight fish, eyelight fish

-**PŠČW**- 'HOLOCENTRIFORM & POLYMIXIIFORM FISH' 1. (Holocentrines; genera *Holocentrus, Neomiphon, Sargocentron*) squirrelfish 2. (Myripristines; several genera) soldierfish 3. (Polymixiiforms; genus *Polymixia*) beardfish

-**LPŠT**- 'OPHIDIIFORM FISH' 1. (Carapids; several genera) pearlfish 2. (Ophidiids, Aphyonids and Parabrotulids; numerous genera) brotula, cusk-eel, false brotula, aphyonid 3. (Bythitids; numerous genera) viviparous brotula

-**LPŠTW**- 'KURTIFORM FISH & BATRACHOIDID FISH' 1. (Kurtids; genus Kurtus) nurseryfish 2. (Apogonids; numerous genera) cardinalfish 3. (Batrachoidids; numerous genera) toadfish / frogfish

-**LPŠTY**- ‘GEMPYLID FISH’ 1. (genus *Thyrsites, Tyrsitoides*) snoek 2. (genera *Tongaichthys, Rexichthys, Promethichthys, Lepidocybium, Diplospinus*) escolar 3. (genera *Epinnula, Gempylus, Nealotus, Neoepinnula, Nesiarchus, Paradiplospinus, Rexea, Ruvettus, Thyrsitops*) snake mackerel, domine, oilfish, sackfish, black gemfish

-**LPŠTL**- ‘GOBIOID FISH’ I 1. (Gobiines, Benthophilines, Gobionellines, Sicydiines; numerous genera) goby 2. (Amblyopines; numerous genera) eel goby / worm goby 3. (Oxucerdids; several genera) mudskipper

-**LPŠTR**- ‘GOBIOID FISH’ II 1. (Ptereleotrines; numerous genera) dartfish 2. (Kraemeriids; genera *Kraemeria, Gobitrichinotus*) sand darter 3. (Microdesmines; several genera) wormfish

-**LPŠTŘ**- ‘GOBIOID FISH’ III 1. (Thalasseleotridids; genera *Thalasseleotris, Grahamichthys*) gudgeon 2. (Eleotrids, Butids and Odontobutids; numerous genera) sleeper goby / sleeper 3. (Milyeringids; several genera) blind cave goby

-**LPŠTÇ**- ‘GOBIOID FISH’ IV 1. (Rhyacichthyids; genera Protogobius, Rhyacichthys) loach-goby 2. (Schindleriids; genus *Schindleria*) infantfish 3. (Xenisthmids; several genera) wriggler / collared wriggler, flathead wriggler

-**KŢS**- ‘SCOMBRID FISH’ 1. (Thunnines; several genera) tuna 2. (Scombrines, Scoberomorines; Gasterochismatines; several genera) mackerel, Spanish mackerel, butterfly kingfish 3. (Sardini; genera *Sarda, Cybiosarda, Gymnosarda, Orcynopsis*) bonito

-**KŢSP**- ‘SCOMBRIFORM FISH’ I 1. (Bramids; several genera) pomfret, fanfish 2. (Stromateids; genera *Pampus, Peprilus, Stromateus*) butterfish, silver pomfret, harvestfish, Pacific pompano 3. (Nomeids and Ariommatids; genera *Cubiceps, Nomeus, Psenes, Ariomma*) driftfish, fathead, cubehead, cigarfish, man-of-war, bluebottle, ariomma, ariommid

-**KŢST**- ‘ SCOMBRIFORM FISH’ II 1. (Scombrolabracids; genus *Scombrolabrax*) longfin escolar / black mackerel 2. (Amarsipids; genus *Amarsipus*) amarsipa 3. (Tetragonurids; genus *Tetragonurus*) squaretail

-**KŢSK**- ‘SCOMBRIFORM FISH’ III 1. (Centrolophids; several genera) medusafish, ruff, rudderfish, blackfish, barrelfish, butterfish, warehou / trevalla 2. (Icosteids; genus *Icosteus*) ragfish 3. (Scombropids; genus *Scombrops*) gnomefish

-**KŢSB**- ‘SCOMBRIFORM FISH’ IV 1. (genus *Sphyraena*) barracuda 2. (genus *Xiphias*) swordfish / broadbill 3. (Istiophorids; several genera) marlin / billfish

-**KŢSD**- ‘TRICHIURID FISH (CUTLASSFISH)’ 1. (genera *Aphanopus, Assurger, Evoxymetopon, Lepidopus*) scabbardfish 2. (genus *Benthodesmus*) frostfish 3. (genera *Eupleurogrammus, Demissolinea, Lepturacanthus, Tentoriceps, Thrichiurus*) hairtail

-**KŢSG**- ‘SYNGNATHIFORM FISH’ I 1. (genus *Hippocampus*) seahorse 2. (Syngnathines; numerous genera) pipefish, pipehorse, seadragon 3. (Solenostomids; genus *Solenostomus*) ghost pipefish

-**KŢSF**- ‘SYNGNATHIFORM FISH’ II 1. (Pegasids; genus *Pegasus, Euypegasus*) seamoth 2. (genus *Aulostomus*) trumpetfish 3. (genus *Fistularia*) cornetfish / flutemouth

-**KŢSV**- ‘SYNGNATHIFORM FISH’ III 1. (Mullids; several genera) goatfish / red mullet 2. (Callionymids; numerous genera) dragonet 3. (Draconettids; genera *Draconetta, Centrodraco*) slope dragonet

-**KŢSM**- ‘SYNGNATHIFORM FISH’ IV 1. (Macroramphosids; genus *Macroramphosus*) snipefish / bellowfish 2. (Centrisciids; genera *Aeoliscus, Centriscus*) shrimpfish 3. (Dactylopterids; genera *Dactyloptena, Dactylopterus*) flying gurnard / helmet gurnard

-**KŢSN**- ‘SYNBRANCHIFORM FISH’ 1. (Synbranchids; numerous genera) swamp eel 2. (Chaudhuriids; several genera) earthworm eel 3. (Mastacembelids; numerous - genera) spiny eel, fire eel

-**KŢSŇ**- ‘ANABANTOID FISH (GOURAMI)’ 1. (Osphronemids; numerous genera) gourami 2. (Helostomids; genus Helostoma) kissing gourami, kisser 3. (Anabantids; several genera) climbing gourami / climbing perch

-**KŢSH**- ‘ANABANTIFORM FISH’ 1. (Channids; several genera) snakefish 2. (Badids; genera *Badis, Dario*) chameleonfish 3. (Nandids and Pristolepidids; several genera) Asian leaffish

-**PŢS**- ‘CARANGID FISH’ I 1. (genus *Lichia*) leerfish / garrick 2. (genus *Trachinotus*) pompano 3. (genus *Scomberoides*) queenfish

-**PŢSP**- **‘**CARANGID FISH’ II 1. (genera *Oligoplites, Parona*) leatherjacket 2. (genus *Campogramma*) vadigo 3. (genus *Elagatis*) rainbow runner / rainbow yellowtail, Spanish jack / Hawaiian salmon

-**PŢST**- ‘CARANGID FISH’ III 1. (genus *Naucrates*) pilot fish 2. (genus *Seriola*) amberjack 3. (genus *Seriolina*) black-banded trevally

-**PŢSK**- ‘CARANGID FISH’ IV 1. (genus *Alectis*) threadfish, diamond trevally 2. (genus *Alepes*) scad 3. (genus *Atropus*) cleftbelly trevally

-**PŢSB**- ‘CARANGID FISH’ V 1. (genera *Caranx, Carangoides, Hemicaranx*) jack, trevally, bludger 2. (genus *Atule*) yellowtail scad 3. (genus *Chloroscombrus*) bumper / bumperfish

-**PŢSD**- ‘CARANGID FISH’ VI 1. (genus *Decapterus*) mackerel scad, round scad, roughear scad, Indian scad, redtail scad, Japanese scad, shortfin scad, koheru 2. (genus *Gnathodon*) golden trevally / golden kingfish / king trevally 3. (genus *Megalespis*) torpedo scad / hardtail scad / finny scad / finletted mackerel scad / cordyla scad

-**PŢSG**- ‘CARANGID FISH’ VII 1. (genus *Pantolebus*) fringefin trevally / round-finned trevally / reef herring 2. (genus *Parastromateus*) black pomfret 3. (genus *Pseudocaranx*) white trevally, skipjack trevally

-**PŢSF**- ‘CARANGID FISH’ VIII 1. (genus *Selar*) oxeye scad, bigeye scad 2. (genus *Selaroides*) yellowstripe scad / yellowstripe trevally / yellow-banded trevally / smooth-tailed trevally / slender-scaled trevally 3. (genus *Selene*) lookdown, moonfish

-**PŢSV**- ‘CARANGID FISH’ IX 1. (genus *Trachurus*) saurel / jack mackerel 2. (genus *Ulua*) silvermouth trevally, longrakered trevally 3. (genus *Uraspis*) whitetongue jack, cottonmouth jack, whitemouth jack

-**PŢSM**- ‘CARANGIFORM FISH’ I 1. (genus *Nematistius*) roosterfish 2. (genus *Coryphaena*) dolphinfish 3. (genus *Rachycentron*) cobia / black kingish / black bonito / lemonfish

-**PŢSN**- ‘CARANGIFORM FISH’ II 1. (Echeneids; several genera) remora / suckerfish, lousefish 2. (Menids; genus *Mene*) moonfish 3. (Centropomids; genus *Centropomus*) snook

-**PŢSŇ**- ‘CARANGIFORM FISH’ III 1. (Polynemids; several genera) threadfin, bobo, barbu, paradise fish 2. (Leptobramids; genus *Leptobrama*) beachsalmon 3. (Toxotids; genus *Toxotes*) archerfish

-**PŠK**- ‘PLEURONECTID FISH (RIGHTEYE FLOUNDER)’ I 1. (*Hippoglossus hippoglossus*) Atlantic halibut 2. (*Hippoglossus stenolepis*) Pacific halibut 3. (genus *Reinhardtius*) Greenland halibut / Greenland turbot

-**PŠKW**- ‘PLEURONECTID FISH (RIGHTEYE FLOUNDER)’ II 1. (genus *Hippoglossoides*) American plaice, flathead flounder, flathead sole, Bering flounder 2. (genus *Pleuronectes*) European plaice, Alaska plaice, American smooth flounder 3. (genus *Acanthopsetta*) scale-eye plaice

-**PŠKY**- ‘PLEURONECTID FISH (RIGHTEYE FLOUNDER)’ III 1. (genus *Microstomus*) lemon sole, Pacific Dover sole, slime flounder 2. (genus *Eopsetta*) petrale sole, shotted halibut 3. (genus *Parophrys*) English sole

-**PŠKL**- ‘PLEURONECTID FISH (RIGHTEYE FLOUNDER)’ IV 1. (genus *Embassichthys*) deepsea sole 2. (genus *Psettichthys*) Pacific sand sole 3. (genus *Lyopsetta*) slender sole

-**PŠKR**- ‘PLEURONECTID FISH (RIGHTEYE FLOUNDER)’ V 1. (genus *Lepidopsetta*) rocksole, dusky sole, Northern rock sole 2. (genus *Isopsetta*) butter sole 3. (genus *Clidoderma*) roughscale sole

-**PŠKŘ**- ‘PLEURONECTID FISH (RIGHTEYE FLOUNDER)’ VI 1. (*Glyptocephalus zachirus*) rex sole 2. (*Glyptocephalus cynoglossus*) witch flounder 3. (*Glyptocephalus stelleri*) blackfin flounder

-**PŠKÇ**- ‘PLEURONECTID FISH (RIGHTEYE FLOUNDER)’ VII 1. (genus *Pleuronichthys*) curlfin sole, C-O sole, ridge-eyed flounder, ocellated turbot, spotted turbot, horny-head turbot 2. (genus *Limanda*) yellowfin sole, Sakhalin sole, common dab, longhead dab, yellowtail flounder, speckled flounder 3. (genus *Hypsopsetta*) diamond turbot

-**PŠKF**- 'PLEURONECTID FISH (RIGHTEYE FLOUNDER)' VIII 1. (genus *Kereius*) stone flounder 2. (genus *Liopsetta*) Arctic flounder, Far Eastern smooth flounder 3. (genus *Platichthys*) European flounder, starry flounder, Baltic flounder

-**PŠKH**- 'PLEURONECTID FISH (RIGHTEYE FLOUNDER)' IX 1. (genus *Pseudopleuronectes*) winter flounder, yellow-striped flounder, cresthead flounder, marbled flounder 2. (genus *Tenakius*) willowy flounder 3. (genus *Verasper*) barfin flounder, spotted halibut

-**PŠKŢ**- 'PLEURONECTID FISH (RIGHTEYE FLOUNDER)' X 1. (genus *Paralichthodes*) peppered flounder / measles flounder 2. (genus *Atheresthes*) Kamchatka flounder, arrowtooth flounder 3. (genera *Cleisthenes, Dexistes*) other righteye flounder

-**PŠKM-** 'RHOMBOSOLEID FISH (SOUTHERN HEMISPHERE RIGHTEYE FLOUNDER)' I 1. (genus *Ammotretis*) shortfin flounder, elongate flounder, Tudor's flounder, longsnout flounder 2. (genus *Azygopus*) banded-fin flounder / spotted flounder 3. (genus *Oncopterus*) Remo flounder

-**PŠKN**- 'RHOMBOSOLEID FISH (SOUTHERN HEMISPHERE RIGHTEYE FLOUNDER)' II 1. (genus *Colistium*) New Zealand brill, New Zealand turbot 2 (genus *Peltorhamphus*) speckled sole, New Zealand sole 3. (genus *Pelotretis*) southern lemon sole / New Zealand lemon sole

-**PŠKV**- 'RHOMBOSOLEID FISH (SOUTHERN HEMISPHERE RIGHTEYE FLOUNDER)' III 1. (genus *Rhombosolea*) yellowbelly flounder, sand flounder, black flounder, greenback flounder 2. (genus *Psammodiscus*) Indonesian ocellated flounder 3. (genus *Taratretis*) Derwent flounder

-**LPSK**- BOTHID FISH (LEFTEYE FLOUNDER) I 1. (genus *Arnoglossus*) scaldfish 2. (genus *Bothus*) Pacific eyed flounder, Guinean flounder, Pacific leopard flounder, plate fish, mottled flounder, flowery flounder, St. Helena flounder, Indo-Pacific oval flounder, eyed flounder, leopard flounder, wide-eyed flounder, twospot flounder 3. (genus *Crossorhombus*) blue flounder, Kobe flounder, broadbrow flounder, Lord Howe Island flounder

-**LPSB**- 'BOTHID FISH (LEFTEYE FLOUNDER)' II 1. (genus *Engyophrys*) speckled-tail flounder, American spiny flounder 2. (genus *Grammatobothus*) Krempf's flounder, Pennant flounder, threespot flounder 3. (genus *Engyprosopon*) fringelip dwar flounder, Bellona flounder, Bleeker's flounder, Largescale flounder, Hureau's flounder, long pectoral fin flounder, olive wide-eyed flounder, Natal flounder, Regan's flatfish, long snout flounder, seven pelvic ray flounder

-**LPSG**- 'BOTHID FISH (LEFTEYE FLOUNDER)' III 1. (genus *Kamoharaia*) wide-mouthed flounder 2. (genus *Laeops*) clear fin-base flounder, Philippine slender flounder, Günther's flounder, khaki flounder, blackspotted flounder, small headed flounder, Longarm flounder 3. (genus *Monolene*) slim flounder, Asaedae flounder, Acapulco flounder, Pacific deepwater flounder, Merten's moonflounder, smalltooth moonflounder, deepwater flounder

-**LPSD**- 'BOTHID FISH (LEFTEYE FLOUNDER)' IV 1. (genus *Lophonectes*) crested flounder 2. (genus *Perissias*) striped-fin flounder 3. (genera *Asterorhombus, Chascanopsetta, Japonolaeops, Neolaeops, Parabothus, Psettina, Taeniopsetta, Tosarhombus, Tricopsetta*) other lefteye flounder

-**LPSC**- 'PARALYCHTHYID FISH (LARGE-TOOTH FLOUNDER)' I 1. (genus *Ancylopsetta*) Cyclope founder, three-spot flounder, three-eye flounder, four-eyed flounder, Gulf of Mexico ocellated flounder 2. (genus *Citharichthys*) megrim / whiff, sanddab 3. (genus *Cyclopsetta*) Mexican flounder, spotfin flounder, God's flounder, toothed flounder

-**LPSČ**- 'PARALYCHTHYID FISH (LARGE-TOOTH FLOUNDER)' II 1. (genus *Etropus*) fringed flounder, shelf flounder, smallmouth flounder, Peruvian flounder, gray flounder, sole flounder, Delsman's flounder 2. (genus *Gastropsetta*) shrimp flounder 3. (genus *Hippoglossina*) bigmouth flounder, fourspot flounder, American fourspot flounder

-**LPSY**- 'PARALYCHTHYID FISH (LARGE-TOOTH FLOUNDER)' III 1. (genus *Paralichthys*) fine flounder, Cortez flounder, gulf flounder, Brasilian flounder, California flounder, summer flounder, fluke, bastard halibut, Patagonian flounder, broad flounder, tropical flounder, speckled flounder, olive flounder 2. (genus *Pseudorhombus*) ringed flounder, peacock flounder, largetooth flounder, cinnamon flounder, four twin-spot flounder, ocellated flounder, deep flounder, Javan flounder, small-toothed flounder, Malayan flounder, roughscale flounder, fivespot flounder, five-eyed flounder, spiny flounder, three spotted flounder 3. (genus *Syacium*) Papillose flounder, shoal flounder, beach flounder, longfin flounder, clearspot flounder, channel flounder, oval flounder, dusky flounder

-**LPSŘ**- 'PARALYCHTHYID FISH (LARGE-TOOTH FLOUNDER)' IV 1. (genus *Tephrinectes*) Chinese brill 2. (genus *Xystreurys*) fantail flounder 3. (genera *Cephalopsetta, Tarphops, Thysanopsetta*) other large-tooth flounder

-**LPSŢ**- 'CITHARID FISH (LARGESCALE FLOUNDER)' 1. (genus *Brachypleura*) yellow-dabbled flounder 2. (genus *Lepidoblepharon*) scale-eyed flounder 3. (genera *Citharus, Citharoides*) branched ray flounder, twospot largescale flounder, spotted flounder

-**LPSX**- ‘POECILOPSETTINE FISH (BIGEYE FLOUNDER)’ 1. (genus *Marleyella*) comb flounder 2. (genus *Nematops*) large-scale righteye flounder, narrow-body righteye flounder, small-mouth righteye flounder 3. (genus *Poecilopsetta*) deepwater dab, colored righteye flounder, African righteye flounder, tile-colored righteye flounder, Alcock’s narrow-body righteye flounder

-**LPSV**- **‘**ACHIROPSETTID FISH (SOUTHERN FLOUNDER)’ 1. (genera Achiropsetta, Neoarchiropsetta) finless flounder / prickly flounder, armless flounder 2. (genus Mancopsetta) Antarctic armless flounder 3. (genus Pseudomancopsetta) pygmy flounder

-**LPSŇ**- ‘SAMARID FISH (CRESTED FLOUNDER)’ 1. (genus *Samaris, Samaretta*) crested flounder, cockattoo righteye flounder, large-scale crested righteye flounder 2. (genus *Samariscus*) coralline-red flounder, Huysman’s righteye flounder, deep-body righteye flounder, longfinned flounder, Luzon righteye flounder, large-mouth righteye flounder, spotted righteye flounder, New Caledonian righteye flounder, Nielsen’s righteye flounder, Sunier’s righteye flounder, three-spot righteye flounder 3. (genus *Plagiopsetta*) tongue flatfish

-**PŠT**- ‘ACHIRID FISH (AMERICAN SOLE)’ 1. (genus *Achirus*) drab sole, plainfin sole, brown sole, lined sole, Mazatlan sole, American sole, network sole 2. (genus *Trinectes*) fringed sole, freshwater sole, spotted-in sole, scrawled sole, hogchoker, spotted-cheek sole, slipper sole 3. (genera *Apionichthys, Catathyridium, Gymnachirus, Hypoclinemus*, *Pnictes*) other American sole fish

-**PŠTW**- ‘SOLEID FISH (TRUE SOLE)’ I 1. (*Solea solea*) common sole 2. (genus *Solea* [other than *S. solea*]) Egyptian sole, elongated sole, ovate sole, Senegalese sole, Stanaland’s sole 3. (genera *Achiroides, Barnardichthys, Bathysolea, Brachirus, Dagetichthys, Liachirus, Paradicula, Phyllichthys, Rendahlia, Rhinosolea, Synapturichthys, Synclidopus, Typhlachirus*) other true sole fish

-**PŠTY**- ‘SOLEID FISH (TRUE SOLE)’ II 1. (genus *Buglossidium*) yellow sole / solonette 2. (genus *Dexillus*) tufted sole 3. (genus *Dicologlossa*) wedge sole

-**PŠTL**- ‘SOLEID FISH (TRUE SOLE)’ III 1. (genus *Heteromycteris*) Cape sole, hook-nosed sole, bamboo sole, true sole 2. (genus *Leptachirus*) Allen’s sole, Bensbach River sole, Darwin sole, Kikori River sole, tailed sole, Lorentz River sole, manyscale sole, Robert’s sole, three-line sole 3. (genus *Microchirus*) bastard sole, Lusitanian sole, Frechkop’s sole, foureyed sole, thickback sole, banded sole

-**PŠTR**- ‘SOLEID FISH (TRUE SOLE)’ IV 1. (genus *Monochirus*) whiskered sole 2. (genus *Pardachirus*) piebald sole, peacock sole, southern peacock sole, finless sole, Persian carpet sole, estuary sole 3. (genus *Pegusa*) Cadenat’s sole, Adriatic sole, sand sole, blackhand sole, Cyclope sole

-**PŠTŘ**- ‘SOLEID FISH (TRUE SOLE)’ V 1. (genera *Aesopia, Pseudoaesopia, Zebria*) zebra sole, unicorn sole, banded sole, thickray sole, wavyband sole 2. (genus *Aseraggodes*) poisonous sole 3. (genus *Austroglossus*) west coast sole, mud sole

-**PŠTÇ**- ‘SOLEID FISH (TRUE SOLE)’ VI 1. (genus *Soleichthys*) whiteblotched sole, small-head sole, banded-eye sole, snakeskin sole 2. (genus *Synaptura*) Kaup’s sole, Guinean sole, Commerson’s sole, Portuguese sole, white-margined sole, saltpan sole 3. (genus *Vanstraelenia*) African solonette

-**PŠTF**- ‘SCOPHTALMID FISH (TURBOT / BRILL)’ 1. (genus *Scophthalmus*) turbot, brill, Black Sea turbot, windowpane flounder 2. (genus *Lepidorhombus*) megrim / whiff 3. (genera *Phrynorhombus, Zeugopterus*) topknot

-**PŠTH**- ‘OTHER PLEURONECTIFORM FISH’ 1. (Psettodids; genus *Psettodes*) spiny turbot, spottail spiny turbot, Indian halibut 2. (Cynoglossids; several genera) tonguefish 3. [stem unassigned]

-**PSSK**- ‘CICHLIFORM FISH’ (Pholidichthyids; genus *Pholidichthys*) convict or engineer blenny / convict or engineer globy 2. (Tilapioids [Coelotilapines, Coptodonines, Heterotilapines, Oreochromines, Pelmatolapiines, Tilapiines]; numerous genera) tilapia 3. (other Cichlids; numerous genera) cichlid

-**PSSKW**- ‘AQUARIUM CICHLID FISH’ 1. (genus *Pterophyllum*) angelfish 2. (genus Archocentrus) convict cichlid / zebra cichlid 3. (genus *Symphysodon*) discus, pompadour fish

-**PSSKY**- ‘SPORT-FISHING CICHLID FISH’ 1. (genus Cichla) peacock bass 2. (genus *Astronotus*) oscar, tiger oscar, velvet cichlid, marble cichlid 3. (genus Mayaheros) Mayan cichlid / Mexican mojarra

-**PSSKL**- ‘BELONIFORM FISH’ I 1. (Adrianichthyids; several genera) ricefish, medaka 2. (Belonids; numerous genera) needlefish, long tom 3. (Exocoetids; several genera) flying fish / flying cod

-**PSSKR**- 'BELONIFORM FISH' II 1. (Scomberesocids; genera Cololabis, Scomberesox) saury 2. (Hemiramphids; several genera) halfbeak / spipefish, ballyhoo 3. (Zenarchopterids; several genera) viviparous halfbeak

-**PSSKŘ**- 'CYPRINODONTIFORM FISH' 1. (Aplocheiloids, Profundulids, Fundulids, Valenciids; numerous genera) killifish, topminnow 2. (Cyprinodonids; numerous species) pupfish 3. (Goodeids; numerous genera) springfish, poolfish, goodeid, splitfin, characodon, skiffia

-**LPSKW**- 'ANABLEPID FISH' 1. (genus *Anableps*) four-eyed fish 2. (genus *Jenynsia*) onesided livebearer 3. (genus *Oxyzygonectes*) white-eye

-**LPSKY**- 'POECILIID FISH' 1. (Procatopodines; numerous genera) lampeye 2. (Aplocheilichthines; genus *Aplocheilichthys*) banded lampeye 3. (Poeciliines; numerous genera) livebearer

-**LPSKL**- 'ATHERINOPSID FISH (NEOTROPICAL SILVERSIDE)' 1. (genus *Atherinops*) topsmelt, topsmelt silverside 2. (genus *Atherinopsis*) jacksmelt, jack silverside 3. (genus *Leuresthes*) grunion

-**LPSKR**- 'ATHERINOFORM FISH' 1. (other Atherinopsids, Notocheirids; genera *Odontesthes, Basilichthys, Colpichthys, Notocheirus*) other neotropical silverside fish 2. (Menidiines; numerous genera) menidiine silverside 3. (Atherinids; numerous genera) Old World silverside

-**LPSKŘ**- 'ATHERINOID FISH' 1. (Isonids; genus *Iso*) surf sardine, surf sprite, flower of the wave 2. (Phallostethids; several genera) priapium fish 3. (Atherionids and Dentatherinids; genera *Atherion, Dentatherina*) pricklenose silverside, bearded silverside, McCulloch's hardyhead, tusked silverside

-**LPST**- 'MELANOTAENIID FISH (RAINBOWFISH)' 1. (Melanotaenines; several genera) rainbowfish 2. (Bedotiines and Telmatherinines; several genera) Madagascar rainbowfish, Celebes rainbowfish 3. (Pseudomugilines; genera *Kiunga, Pseudomugil, Scaturiginichthys*) blue-eye

-**LPSTW**- 'AMBASSID FISH' 1. (genera *Ambassis, Parambassis, Gymnochanda, Paradoxodacna, Pseudoambassis, Chanda*) glassy, perchlet, glass(y) perchlet, glassfish 2. (genus *Tetracentrum*) four-spined glass perchlet 3. (genus *Denariusa*) pennyfish

-**LPSTY**- 'MUGILID, EMBIOTOCID, & GRAMMATID FISH' 1. (Mugilids; numerous genera) grey mullet 2. (Embiotocids; numerous genera) surfperch / seaperch 3. (Grammatids; genera *Gramma, Lipogramma*) basslet

-**LPSTL**- 'PLESIOPID FISH (LONGFIN / ROUNDHEAD)' I 1. (genus *Acanthoclinus*) rockfish 2. (genus *Acanthoplesiops*) spiny basslet, scottie, longfin 3. (genus *Beliops*) Batan longfin, Southern longfin

-**LPSTR**- 'PLESIOPID FISH (LONGFIN / ROUNDHEAD)' II 1. (genus *Belonepterygion*) barred spiny basslet 2. (genus *Notograptus*) shark-tailed eel-blenny, spotted eel-blenny 3. (genus *Assessor*) yellow devilfish, blue devilfish, Randall's devilfish

-**LPSTŘ**- 'PLESIOPID FISH (LONGFIN / ROUNDHEAD)' III 1. (genus *Calloplesiops*) comet / marine betta 2. (genus *Fraudella*) carp prettyfin 3. (genus *Paraplesiops*) blue devil

-**LPSTÇ**- 'PLESIOPID FISH (LONGFIN / ROUNDHEAD)' IV 1. (genus *Plesiops*) longfin, prettyfin 2. (genus *Steeneichthys*) Christmas longfin, Steene's prettyfin 3. (genus *Trachinops*) hulafish

-**LPSP**- 'PSEUDOCHROMID FISH (DOTTYBACK)' 1. (Congrogadines; numerous genera) eel-blenny 2. (Pseudochromines and Anisochromines; numerous genera) dottyback 3. (Pseudoplesiopines; several genera) dottyback

-**LPSPW**- 'POMACENTRID FISH' 1. (Amphiprionines; genera *Amphiprion, Premnas*) clownfish / anemonefish 2. (Chromines [other than *Dascyllus*] and Lepidozygines; several genera) damselfish 3. (genus *Dascyllus*) dascyllus, humbug

-**LPSPY**- 'POMACENTRINE FISH' I 1. (genus *Abudefduf*) sergeant, sergeant-major, nightsergeant 2. (genus *Hypsypops*) garibaldi 3. (genus *Neopomacentrus*) demoiselle / lyretail damselfish

-**LPSPL**- 'POMACENTRINE FISH' II 1. (genus *Parma*) scalyfin, New Zealand black angelfish 2. (genus *Stegastes*) gregory 3. (genera *Chrysiptera, Amblyglyphidodon, Neopomacentrus, Cheiloprion, Dischistodus, Hemiglyphidodon, Teixeirichthys, Similiparma, Pomachromis, Plectroglyphidodon, Nexilosus, Microspathodon, Mecaenichthys*) damsel

-**PSSP**- 'BLENNIOID FISH' I 1. (Blenniids and Labrisomids; numerous genera) blenny, rockskipper, combtooth blenny 2. (Chaenopsids; numerous genera) pike-blenny, tube-blenny, flagblenny 3. (Dactyloscopids; numerous genera) sand stargazer

-**PSSPW**- **'**BLENNIOID FISH' II 1. (Clinids; numerous genera) klipfish, cline, weedfish, kelpfish, eel blenny 2. (Tripterygiids; numerous genera) triplefin / threefin, triplefin blenny

-**PSSPY**- 'BLENNIMORPHIC FISH' 1. (Opistognathids; several genera) jawfish 2. (Gobiesocids; numerous genera) clingfish 3. [stem unassigned]

-**PSSPL**- 'LABRID FISH (WRASSE)' I 1. (genus *Halichoeres* and numerous other genera) wrasse 2. (genera *Bodianus, Lachnolaimus*) hogfish 3. (genus *Labroides*) cleaner wrasse

-**PSSPR**- 'LABRID FISH (WRASSE)' II 1. (genus *Xyrichtys*) razorfish 2. (genus *Coris*) rainbow wrasse, coris 3. (genus *Choerodon*) tuskfish

-**PSSPŘ**- 'LABRID FISH (WRASSE)' III 1. (genus *Tautoga*) tautog, blackfish 2. (genus *Tautogolabrus*) bergall 3. (genus *Achoerodus*) blue groper

-**PSSPÇ**- 'LABRID FISH (WRASSE)' IV 1. (genus *Labrichthys*) tubelip wrasse 2. (genus *Diproctacanthus*) yellowtail tubelip 3. [stem unassigned]

-**PSSPF**- 'LABRIFORM FISH 1. (Odacids; several genera) cale, weed whiting 2. (Scarine labrids; numerous genera) parrotfish 3. [stem unassigned]

-**PSSPŢ**- 'TRACHINIFORM FISH' I 1. (Trachinids; genera *Trachinus, Echichthys*) weever / weeverfish 2. (Creediids; several genera) sandburrower / burrower, sand diver 3. (Ammodytids; several genera) sandlance / sand lance, sand eel, launce

-**PSSPḶ**- 'TRACHINIFORM FISH' II 1. (Chiasmodontids; several genera) snaketooth fish / swallower 2. (Champsodontid; genus *Champsodon*) crocodile toothfish, sabre-gill, gaper 3. (Cheimarrichthyids; genus *Cheimarrichthys*) torrentfish

-**PSSPV**- **'**TRACHINIFORM FISH' III 1. (Leptoscopids; genera *Crapatalus, Leptoscopus, Lesueuina*) southern sandfish, flathead pygmy-stargazer, estuary stargazer 2. (Percophids; numerous genera) duckbill 3. (Pinguipedids; several genera) sandperch, grubfish

-**PSSPH**- 'TRACHINIFORM FISH' IV 1. (Trichodontids; genera *Trichodon, Arctoscopus*) sandfish 2. (Trichonotids; genus *Trichonotus*) sand-diver 3. (Uranoscopids; several genera) stargazer

-**RPSP**- 'CENTRARCHID FISH' I 1. (genus Lepomis) sunfish 2. (genus *Micropterus*) black bass 3. (genus *Pomoxis*) crappie

-**RPSPW**- **'**CENTRARCHID FISH' II 1. (genus *Enneacanthus*) banded sunfish 2. (genus *Centrarchus*) flier 3. (genus *Archoplites*) Sacramento perch

-**RPSPY**- 'CENTRARCHID FISH' III 1. (genus *Ambloplites*) rock bass, shadow bass, Roanoke bass, Ozark bass 2. (genus *Acantharchus*) mud sunfish 3. [stem unassigned]

-**RPSPL**- 'CENTRARCHIFORM FISH' I: 1. (Aplodactylids; genus *Aplodactylus*) marblefish, sea carp, rock cale 2. (Latrids and Cheilodactylids; several genera) trumpeter, morwong / butterfish / fingerfin / jackassfish / sea carp / snapper / moki 3. (Chironemids; genus *Chironemus*) kelpfish, silver spot

-**RPSPR**- ' CENTRARCHIFORM FISH' II: 1. (Cirrhitids; numerous genera) hawkfish 2. (Dichistiids; genus *Dichistius*) galjoen 3. (Elassomatids; genus *Elassoma*) pygmy sunfish

-**RPSPŘ**- 'CENTRARCHIFORM FISH' III: 1. (Enoplosids; genus Enoplosus) old wife 2. (Girellines; genera *Girella, Graus*) nibbler, drummer, opaleye, parore, blackfish, stripey bream 3. (Kuhliids; genus Kuhlia) flagtail

-**RPSPH**- 'CENTRARCHIFORM FISH' IV: 1. (Kyphosines; several genera) sea chub 2. (Microcantines; several genera) footballer, moonlighter 3. (Scorpindines; several genera) halfmoon, grey knifefish, blue knifefish, sweep

-**RPSB**- 'CENTRARCHIFORM FISH' V: 1. (Oplegnathids; genus *Oplegnatus*) knifejaw 2. (Percichtyids, Percalatids, Perciliids, and Sinipercids; numerous genera) temperate perch, nightfish, pygmy perch, perchlet, mandarin fish, oriental perch 3. (Terapontids; numerous genera) grunter, tigerperch, terapon, trumpeter

-**PSSC**- ‘TETRADONTIFORM FISH’ I: 1. (Tetradontids and Triodontids; numerous genera) puffer, pufferfish, blowfish, globefish, balloonfish, blowie, bubble fish, swellfish, toadfish, toady, honey toad, sugar toad, sea squab, toby, blaasop 2. (Diodontids; several genera) porcupine fish / blowfish, burrfish, swelltoad 3. (Triacanthodids; numerous genera) spikefish, trumpetsnout

-**PSSČ**- ‘TETRADONTIFORM FISH’ II: 1. (Balistids; numerous genera) triggerfish, picasso fish 2. (Triacanthids; seveal genera) triplespine / tripodfish 3. (Monacanthids; numerous genera) filefish, foolfish, shingle, leatherjacket, scraper

-**PŠŠČ**- ‘TETRADONTIFORM FISH’ III: 1. (Molids; genera *Mola, Mastrurus, Ranzania*) sunfish / mola 2. (Ostraciids and Aracanids; several genera) boxfish, cowfish, trunkfish, turretfish, basketfish 3. [stem unassigned]

-**PSST**- ‘SCORPAENIFORM FISH’ I: 1. (Scorpaenids; numerous genera) scorpionfish, rock fish, red rock cod 2. (genus *Taenionotus*) paperfish / leaf scorpionfish 3. (genus *Centrogenys*) false scorpionfish

-**PSSTW**- ‘SCORPAENIFORM FISH’ II: 1. (genus *Dendrochirus*) lionfish, firefish, turkey fish 2. (genus *Ebosia*) lionfish 3. (genus *Pterois*) lionfish, zebrafish, firefish, turkeyfish / tastyfish / butterfly-cod

-**PSSTY**- ‘ SCORPAENIFORM FISH’ III: 1. (Apistids and Tetrarogids; numerous genera) wasp scorpionfish, waspfish 2. (genus *Centropogon*) fortescue 3. (genera *Glyptauchen*, *Neovespicula*) goblinfish, leaf goblinfish

-**PSSTL**- ‘SCORPAENIFORM FISH’ IV: 1. (genus *Gymnapistes*) cobbler / estuary cobbler 2. (genus *Notesthes*) bullrout / freshwater stonefish / kroki 3. (genus *Tetraroge*) bearded roguefish

-**PSSTR**- ‘ SCORPAENIFORM FISH’ V: 1. (Agonids, Bathyagonines and Bothragonines; several genera) poacher, snailfish, starsnout, rockhead 2. (genus *Agonus*) hooknose / pogge / armed bullhead 3. (Anoplagonines; several genera) alligatorfish

-**PSSTŘ**- ‘SCORPAENIFORM FISH’ VI: 1. (Anoplopomatids; genera *Anoplopoma, Erilepsis*) sablefish / butterfish /black cod / blue cod /bluefish / candlefish /coal cod / coalfish / beshow, skilfish 2. (Hexagrammids; several genera) greenling, ling cod / buffalo cod 3. (Normanichthyids; genus *Normanichthys*) mole sculpin

-**PSSTÇ**- ‘SCORPAENIFORM FISH’ VII: 1. (Sebastids; several genera) rockfish, rock perch, ocean perch, sea perch, thornyhead, sea ruffe, rockcod 2. (Neosebastids; genera *Neosebastes, Maxillicosta*) gurnard perch, gurnard scorpionfish 3. (Triglids; numerous genera) sea robin, gurnard, latchet, piper gurnard

-**PSSTF**- ‘SCORPAENIFORM FISH’ VIII: 1. (Synanceids; numerous genera) stonefish, stinger, stingfish, ghoul, devilfish 2. (Congiopodids and Perryenids; several genera) pigfish, horsefish, racehorse 3. (Pataecids; several genera) Australian prowfish, red Indianfish

-**PSSTH**- ‘SCORPAENIFORM FISH’ IX: 1. (Eschmeyerids; genus *Eschmeyer*) cofish 2. (Plectrogenids; genus *Plectrogenium*) stinger flathead 3. (Setarchids; several genera) deap-sea bristly scorpionfish

-**PSSTḶ**- ‘SCORPAENIFORM FISH’ X: 1. (Aploactinids and Gnathanacanthids; numerous genera) velvetfish, red velvetfish 2. (genus *Adventor*) visitor 3. (Caracanthids; genus *Caracanthus*) coral croucher, orbicular velvetfish

-**PSSDL**- ‘COTTOID FISH (SCULPIN)’ I: 1. (Cottids; numerous species) sculpin, bullhead, cabezon 2. (Bathylutichthyids; genus Bathylutichthys) Antarctic sculpin 3. (Hemitripterids; several genera) sea raven

-**PSSDR**- ‘COTTOID FISH (SCULPIN)’ II: 1. (Icelids; genus *Icelus*) scaled sculpin 2. (Comephorids; genus *Comephorus*) Baikal oilfish 3. (Cottocomephorids; several genera) Baikal sculpin / bighead sculpin

-**PSSDŘ**- ‘COTTOID FISH (SCULPIN)’ III: 1. (Ereuniids; genera *Ereunias, Marukawichthys*) deepwater bullhead sculpin 2. (Psychrolutids; numerous genera) blobfish, toadfish, flathead sculpin, tadpole sculpin, fathead sculpin 3. (Rhamphocottids; genus *Rhamphocottus*) grunt-fish / grunt sculpin

-**PSSDV**- ‘COTTOID FISH (SCULPIN)’ IV: 1. (Cyclopterids; several genera) lumpsucker / lumpfish 2. (Liparids; numerous genera) snailfish / sea snail 3. (Abyssocottids; several genera) deep-water sculpin

-**PSBV**- ‘PLATYCEPHALOID (FLATHEAD) FISH’ 1. (Platycephalids, Bembrids, Parabembrids, and Hoplichthyids; numerous genera) flathead, deep-water flathead, ghost flathead / armoured flathead 2. (genus *Papilloculiceps*) crocodilefish 3. (Peristediids; several genera) armored searobin / armored gurnard

-**PSPŢ**- ‘GASTEROSTEOID FISH’ 1. (Gasterosteids; several genera) stickleback 2. (Aulorhynchids; genus *Aulorhynchus*) tube-snout 3. (Hypoptychids; genera *Hypoptychus, Aulichthys*) sand-eel, tubenose, sandlance

-**PSSBW**- ‘LOPHIIFORM FISH (ANGLERFISH)’ I 1. (Lophiids; several genera) angler, goosefish, monkfish, sea-devil 2. (Brachionichthyids; several genera) handfish / warty angler 3. (Chaunacids; genera *Chaunax*, *Chaunacops*) sea toad / coffinfish, frogmouth

-**PSSBY**- ‘LOPHIIFORM FISH (ANGLERFISH)’ II 1. (Antennariines and Lophichthyids; several genera) frogfish 2. (Histiophrynines; several genra) anglerfish 3. (Tetrabrachiids; genera *Tetrabrachium, Dibrachichthys*) four-armed frogfish

-**PSSBL**- ‘LOPHIIFORM FISH (ANGLERFISH)’ III 1. (Ogcocephalids; numerous genera) batfish, seabat 2. (Himantolophids; genus *Himantolophus*) footballfish 3. (Centrophrynids; genus *Centrophryne*) horned lantern fish / prickly seadevil

-**PSSBR**- ‘LOPHIIFORM FISH (ANGLERFISH)’ IV 1. (Ceratiids; genera *Ceratias, Cryptopsaras*) warty seadevil 2. (Diceratiids; genera *Diceratias, Bufoceratias*) double angler / doublespine seadevil 3. (Neoceratids; genus *Neoceratias*) toothed seadevil / spiny seadevil / netbeard seadevil

-**PSSBŘ**- ‘LOPHIIFORM FISH (ANGLERFISH)’ V 1. (Caulophrynids; genera *Caulophryne*, *Robia*) fanfin / hairy anglerfish / fanfin seadevil 2. (Melanocetids; genus *Melanocetus*) black seadevil 3. (Thaumatichtyids; genera *Thaumatichthys, Lasiognathus*) wolftrap angler

-**PSSBV**- ‘LOPHIIFORM FISH (ANGLERFISH)’ VI: 1. (Oneirodids; numerous genera) dreamer, dreamarm, tyrant devil 2. (Gigantoactinids; genera *Gigantactis, Rhynchactis*) whipnose / whipnose angler / whipnose seadevil 3. (Linophrynids; several genera) leftvent / bearded seadevil, netdevil

-**PŠP**- **‘**PERCID FISH’ 1. (Percines; several genera) perch, ruffe, schraetzer, percarina 2. (Etheostomatines; several genera) darter 3. (Luciopercines; several genera) sculpin-perch, asprete, walleye, pike-perch, Zander, Sauger, streber, zingel

-**PŠPW**- ‘LUTJANIFORM FISH’ 1. (Lutjanids; numerous genera) snapper, jobfish, wenchman, pinjalo, beeliner, Chinamanfish 2. (Haemulines and Plectorhynchines; several genera) grunt, sargo, porkfish, margate, porgy, bonnetmouth, boga, dara, grunter, piggy, salema, sweetlips, rubberlip, slatey, thicklip, hotlips 3. (Caesionids; several genera) fusilier, banana fish

-**PŠPY**- ‘LOBOTIFORM FISH’ 1. (Datnioidids; genus *Datnioides*) tigerfish, tiger perch, freshwater tripletail 2. (Lobotids; genus *Lobotes*) tripletail 3. (Hapalogenyids; genus *Hapalogenys*) barbeled grunter, velvetchin, javelinfish

-**PŠPL**- ‘ACANTHUROID FISH’ I 1. (genera *Acanthuris, Ctenochaetus, Prionurus, Paracanthurus*) surgeonfish, sawtail, doctorfish 2. (genus *Naso*) unicorn fish 3. (genus *Zebrasoma*) tang

-**PŠPR**- ‘ACANTHUROID FISH’ II: 1. (Siganids; genus *Siganus*) rabbitfish / spinefoot 2. (Scatophagids; genera *Scatophagus, Selenotoca*) scat 3. (Luvarids; genus *Luvaris*) louvar / luvar

-**PŠPŘ**- ‘ACANTHUROID FISH’ III: 1. (Ephippids; several genera) spadefish, batfish 2. (Drepaneids; genus *Drepane*) sicklefish, concertina fish 3. (Zanclids; genus *Zanclus*) Moorish idol

-**PŠPÇ**- ‘PRIACANTHIFORM & CAPROIFORM FISH’ 1. (Priacanthids; several genera) bigeye, bulleye, bullseye, catalufa, glasseye 2. (Cepolids; several genera) bandfish 3. (Caproids; genera *Capros, Antigonia*) boarfish

-**PŠPF**- ‘SPARIFORM FISH’ 1. (Sparids; numerous species) sea bream, porgy, picarel, musselcracker, pinfish, oblade, scup, salema, goldline 2. (Nemipterids; several genera) whiptail bream, false snapper, thread-fin bream, monocle bream, whiptail 3. (Lethrinids; several genera) emperor, emperor bream, big-eye-bream, pigface bream, large-eye bream

-**PŠPH**- ‘SERRANID FISH (SEA BASS & GROUPER)’ 1. (Anthiadines, Liopropomatines, & Serranines; numerous genera) sea bass, wirrah, koester, seaperch, splitfin, swallowtail, jewelfish, harlequin fish, perchlet, anthias, basslet, goldie, hamlet, comber 2. (Epinephelines; numerous genera) grouper, coral grouper, lyretail, hind, mutton hamlet 3. (Grammistines; several genera) soapfish, podge, skunkfish

-**PSKH**- ‘ZOARCOID FISH’ I 1. (Anarhichadids; genera *Anarhichas, Anarrhyichthys*) wolffish / sea wolf, wolf eel 2. (Bathymasterids; genera *Bathymaster, Rathbunella, Ronquilus*) ronquil / ronchil, searcher 3. (Zoarcids; numerous genera) eelpout, pout, fish doctor

-**PSKM**- ‘ZOARCOID FISH’ II 1. (Cryptacanthodids; genus *Cryptacanthodes*) wrymouth 2. (Scytalinids; genus *Scytalina*) graveldiver 3. (Ptilichthyids; genus *Ptilichthys*) quillfish

-**PSKŢ**- ‘ZOARCOID FISH’ III 1. (Pholids; several genera) gunnel 2. (Stichaeids; numerous genera) prickleback / shanny 3. (Zaprorids; genus *Zaprora*) prowfish

-**PŠTĻ**- ‘NOTOTHENIOID FISH’ I 1. (Nototheniids; numerous genera) cod icefish / notothen, icedevil, toothfish, notie, rockcod, Antarctic silverfish, southern cod, scalyhead 2. (Bathydraconids; numerous genera) Antarctic dragonfish, ploughfish 3. (Channichthyids; numerous genera) crocodile icefish / white-blooded fish, icefish

-**PŠTV**- ‘NOTOTHENIOID FISH’ II 1. (Pseudaphritids; genus Pseudaphritis) congoli / tupong, catadromous icefish 2. (Eleginopsids; genus Eleginops) Patagonian blenny / Falkland’s mullet 3. (Bovichtids; several genera) thornfish, bull blenny

-**PŠPĻ**- ‘NOTOTHENIOID FISH’ III 1. (Harpagiferids; genus Harpagifer) spiny plunderfish / plunderfish 2. (Artedidraconids; several genera) barbeled plunderfish 3. [stem unassigned]

-**PŠTHW**- ‘PEMPHERIFORM FISH’ I 1. (Phempherids; genera *Pempheris, Parapriacanthus*) sweeper, bullseye 2. (Symphsanodontids; genus *Symphysanodon*) slopefish 3. (Banjosids; genus *Banjos*) banjofish

-**PŠKHW**- ‘PEMPHERIFORM FISH’ II 1. (Epigonids; several genera) deepwater cardinalfish 2. (Pentacerotids; several genera) armorhead, boarfish 3. (Glaucosomatids; genus *Glaucosoma*) pearl perch, jewfish

-**PŠPHW**- ‘PEMPHERIFORM FISH’ III 1. (Howellids; several genera) oceanic basslet 2. (Bathyclupeids; genera *Bathyclupea, Neobathyclupea*) deep-sea herring 3. (Polyprionids; genera *Polyprion, Stereolepis*) wreckfish, giant sea bass, bass grouper

-**RPŠK**- ‘MISCELLANEOUS PERCOMORPHARIAN FISH’ I 1. (Gerreids; several genera) mojarra 2. (Chaetodontids; several genera) butterflyfish, coralfish bannerfish, pennantfish, talma, barberfish 3. (Leiognathids; numerous genera) ponyfish, slipmouth, toothpony, slimy

-**RPŠKW**- ‘MISCELLANEOUS PERCOMORPHARIAN FISH’ II 1. (Lateolabracids, genus Lateolabrax) Asian seabass 2. (Dinopercids; genera *Dinoperca, Centrarchopos*) cavebass, barred seabass, lampfish 3. (Callanthiids; genera *Callantias, Grammatonotus*) splendid perch, groppo

-**RPŠKY**- ‘MISCELLANEOUS PERCOMORPHARIAN FISH’ III 1. (Moronids; genera *Morone, Dicentrarchus*) temperate bass, white perch 2. (Acropomatids; several genera) lanternbelly / temperate ocean bass, rosy seabass / blackthroat seaperch, seabass, splitfin 3. (Latids; several genera) lates perch, spikey bass, Waigieu seaperch, Waigeo barramundi

-**RPŠKL**- ‘MISCELLANEOUS PERCOMORPHARIAN FISH’ IV 1. (Caristiids; several genera) manefish 2. (Polycentrids; genera *Monocirrhus, Polycentrus*) leaffish 3. (Pomacanthids; several genera) marine angelfish

-**RPŠKR**- ‘MISCELLANEOUS PERCOMORPHARIAN FISH’ V 1. (Arripids; genus *Arripis*) ruff / Australian herring, Australian salmon, kahawai 2. (Malacanthids; several genera) tilefish, Quakerfish, blanquillo 3. (Zaniolepidids; genus Zaniolepis) combfish

-**RPŠKŘ**- ‘MISCELLANEOUS PERCOMORPHARIAN FISH’ VI 1. (Dinolestids; genus *Dinolestes*) long-finned pike / yellowfin pike 2. (Parascorpidids; genus *Parascorpis*) jutjaw 3. (Sillaginids; several genera) whiting, smelt-whiting, sand borer, sand-smelt, sillago

-**RPŠKH**- 'MISCELLANEOUS PERCOMORPHARIAN FISH' VII 1. (Sciaenids; numerous genera) drum, croaker, sheephead, wuss fish, shepherd's pie, gou, Gasper goo, grinder, meagre, kob, mulloway, weakfish, bahaba, corvina, spot, king fish, curbinata, pacora, drummer, red, queenfish, stardrum, totoaba / totuava 2. (Monodactylids; genera *Monodactylus*, *Schuettea*) moony, mono / Malayan angel, moonyfish, fingerfish, pomfret 3. (Emmelichthyids; several genera) rover, redbait, rubyfish, kali kali

-**RPŠG**- 'MISCELLANEOUS PERCOMORPHARIAN FISH' VIII 1. (Pomatomids; genus *Pomatomus*) bluefish, tailor, elf 2. (Lactariids; genus *Lactarius*) false trevally 3. [stem unassigned]

7.3.1.6 Echinoderms, Hemichordates, and Xenacoelomorphs

-**LMZW**- 'STARFISH / BRITTLE STAR / BASKET STAR' 1. (Asteroidea; numerous genera) starfish, sea star 2. (Ophiurids; numerous genera) brittle star 3. (Euryalids; several genera) basket star

-**LMZY**- 'SEA URCHIN / SAND DOLLAR' 1. (all Echinoids except Clypeasteroids and Spatangoids) sea urchin 2. (Clypeasteroids; numerous genera) sand dollar 3. (Spantagoids; several genera) heart urchin

-**LMZL**- 'SEA CUCUMBER' 1. (Holothuroids other than Molpadiids, Dendrochirotids, and Elasipodids) sea cucumber 2. (Molpadiids; several genera) molpadiid sea cucumber 3. (Dendrochirotids and Elasipodids; numerous genera) tentacled sea cucumber

-**LMZR**- 'SEA FLOWER' 1. (all Crinoids except Comatulids; several genera) sea lily 2. (Comatulids; several genera) feather star 3. (genus *Xyloplax*) sea daisy

-**LMZŘ**- 'HEMICHORDATE' 1. (Enteropneusta; numerous genera) acorn worm 2. (genus *Cephalodiscus*) cephalodiscus 3. (genus *Rhabdopleura*) rhabdopleura

-**LMZV**- 'XENACOELOMORPH' 1. (genus *Xenoturbella*) xenoturbella 2. (Acoela; numerous genera) acoel 3. (Nemertodermatids; several genera) nemertodermatid acoel

7.3.1.7 Protostomes (excluding Arthropods)

Annelids & Related Animals

-**LṬP**- 'ANNELID' 1. earthworm 2. leech 3. other annelid / ringed worm / segmented worm

-**LṬPW**- 'DISEASE-CAUSING NEMATODE' I 1. (genus *Trichinella*) trichinella / trichina worm 2. (genus *Anisakis*) anisakis 3. (genus *Gnathostoma*) gnathostoma

-**LṬPY**- 'DISEASE-CAUSING NEMATODE' II 1. (genus *Ascaris*) ascaris / large roundworm 2. (genus *Loa*) loa 3. (genus *Onchocerca*) onchocerca

-**LṬPL**- 'DISEASE-CAUSING NEMATODE' III 1. (genus *Dirofilaria*) dirofilaria 2. (genus *Thelazia*) thelazia, eyeworm 3. (genus *Gongylonema*) gongylonema

-**LṬPR** 'DISEASE-CAUSING NEMATODE' IV 1. (genus *Mansonella*) mansonella 2. (genera *Brugia*, *Wuchereria*) brugia, wuchereria 3. (genus *Trichuris*) whipworm

-**LṬPŘ**- - 'DISEASE-CAUSING NEMATODE' V 1. (genus *Dracunculus*) guinea worm 2. (genus *Toxocara*) toxocara, cat roundworm, dog roundworm, fox roundworm 3. (genus *Enterobius*) pinworm / threadworm / seatworm

-**LṬPÇ**- 'DISEASE-CAUSING NEMATODE' VI 1. (genus *Ancylostoma*) Old World hookworm 2. (genus *Necator*) New world hookworm 3. (genus *Strongyloides*) threadworm

-**LṬPḶ**- - 'NEMATODE(-LIKE ANIMAL)' 1. (Nematoda other than named genera in the six roots above; numerous genera) other nematode/roundworm 2. (Nematomorphs; numerous genera) horsehair worm / Gordian worm 3. (Onychophora; numerous genera) velvet worm

-**LṬB**- 'MICROSCOPIC ECDYSOZOAN' 1. (Tardigrada; numerous genera) tardigrade / "water bear" 2. (Loricefera; numerous genera) loriciferan 3. (Kinorhyncha; numerous genera) mud dragon / spiny crown worm

-**LṬBW**- ‘PRIAPULID WORM’ 1. (Priapulids; genera *Priapulus, Priapulosis*) priapulid worm (a.k.a. penis worm) 2. (genus *Tubilucus*) tubiluchide penis worm 3. (genus *Halicryptus*) giant penis worm

-**LṬBY**- ‘ROTIFER’ 1. (Eurotatoria; numerous genera) rotifer 2. (Monogononta; numerous genera) monogonant rotifer 3. (Seisonids; genera *Seison, Paraseison*) seisonid rotifer

-**LṬBL**- ‘GNATHIFER (OTHER THAN ROTIFER)’ 1. (Acanthocephala; numerous genera) thorny-headed worm, spiny-headed worm 2. (Chaetognatha; numerous genera) arrow worm 3. (Cycliophora; genus *Symbion*) symbion

-**LṬBR**- ‘OTHER SPIRALIA’ 1. (Gnathostomulids; numerous genera) jaw worm 2. (Micrognathozoa; genus *Limnognathia*) limnognathia 3. (Gastroticha; numerous genera) gastrotrich / hairyback

-**LṬTW**- ‘CESTODA (TAPEWORM)’ I 1.(genus *Taenia*) taenia, pork tapeworm, beef tapeworm, Asian tapeworm 2. (genus *Diphyllobothrium*) broad tapeworm / fish tapeworm / broad fish tapeworm 3. (genus *Hymenolepis*) dwarf tapeworm, rat tapeworm

-**LṬTY**- ‘CESTODA (TAPEWORM)’ II 1.(genus *Spirometra*) cat tapeworm, dog tapeworm, raccoon tapeworm 2. (genus *Echinococcus*) echinococcus 3. (Cestoda other than named genera; numerous genera) other tapeworm

-**LṬTL**- TREMATODE (FLUKE)’ I 1. (genus *Schistosoma*) schistosoma / blood fluke 2. (genus Trichobilharzia) trichobilharzia 3. (genus *Paragonimus*) paragonimus

-**LṬTR**- ‘TREMATODE (FLUKE)’ II 1. (genus *Fasciola*) fasciola / common liver fluke 2. (genus *Chlonorchis*) chlonorchis / Chinese liver fluke 3. (genus *Dicrocoelium*) dicrocoelium / lancet liver fluke

-**LṬTŘ**- ‘TREMATODE & OTHER FLATWORM’ 1. (genus *Opisthorchis*) opisthorchis / Southeast Asian liver fluke 2. (Trematoda other than named genera; numerous genera) other fluke 3. (non-cestoid, non trematoid platyhelminths; numerous genera) other flatworm, planarian

-**LṬTÇ**- ‘BRYOZOAN / ENTOPROCT’ 1. (Bryozoa/Ectoprocta; numerous genera) moss animal 2. (Entoprocta/Kamptozoa; numerous genera) entoproct / kamptozoan
3. [stem unassigned]

-**LṬTḶ**- ‘ANNELID-RELATED ANIMAL’ I 1. (Phoronida; genera Phononis, Actinotrocha) horseshoe worm 2. (Brachiopoda; numerous genera) brachiopod 3. (Sipuncula; numerous genera) peanut worm

-**LṬTH**- ‘ANNELID-RELATED ANIMAL’ II 1. (Nemertea; numerous genera) ribbon worm / proboscis worm 2. (Myzostomida; numerous genera) myzostomid 3. (Dicyemida/Rhombozoa; several genera) dicyemid/rhombozoan

Molluscs

Gastropods [Note: recent genetics-based studies show that gastropod phylogeny and taxonomy bears little relation to the distinctive morphological traits by which the gastropods were traditionally classified. Since laypersons are likely to be familiar only with practical distinctive morphology in distinguishing these creatures, the classificatory scheme below utilizes morphology as its lexico-semantic basis]

-**ṬPW**- ‘SNAIL’ 1. terrestrial snail 2. sea snail / whelk 3. limpet

-**ṬPY**- ‘DISTINCTIVE SEA SNAIL’ 1. abalone 2. conch [i.e., sea snail with pointed/spired/horned shell] 3. (Thecosomes; numerous genera) sea butterfly

-**ṬPL**- ‘SLUG’ 1. slug 2. sea slug / nudibranch 3. (Anaspids; several genera) sea hare

Bivalves

-**ŢPR**- ‘CLAM’ 1. clam 2. cockle 3. (Limids; several genera) file shell

-**ŢPŘ**- ‘MYID / HIATELLID’ 1. (Myids; several genera) soft-shell clam 2. (Hiatellids; several genera) geoduck 3. (genus *Teredo*) shipworm

-**ŢPÇ**- ‘OYSTER’ 1. (Ostreoids; numerous genera) [true] oyster, foam oyster / honeycomb oyster 2. (Pterioids; numerous genera) hammer oyster, pearl oyster, tree oyster, winged oyster, feather oyster, pulvinitid 3. (Dimyids; several genera) dimyarian oyster

-**ŢPF**- ‘MUSSEL / SCALLOP’ 1. (Unionoids; numerous genera) freshwater mussel 2. (Mytiloids; numerous genera) saltwater mussel 3. (Pectinids, Propeamussids, Entoliids; numerous genera) scallop, glass scallop, mud scallop

-**ŢPS**- ‘PTERIOMORPH’ I 1. (Anomiids; several genera) jingle shell / saddle oyster 2. (Placunids; windowpane shell / windowpane oyster 3. (Pinnids; several genera) pen shell

-ŢPŠ- ‘PTERIOMORPH’ II 1. (Arcids; numerous genera) ark shell / ark clam / bittersweet 2. (genus *Plicatula*) kittenpaw 3. (genus *Spondylus*) spiny oyster / thorny oyster

Cephalopods

-**ŢPŢ**- ‘OCTOPOD’ 1. (Octopoda; numerous genera) octopus 2. (genus *Vampyroteuthis*) vampire squid 3. (genus *Argonauta*) argonaut / paper nautilus

-**ŢPĻ**- ‘SQUID / CUTTLEFISH’ 1. (Myopsida, Oegopsida; numerous genera) squid 2. (Sepiolida, Spirulida; numerous genera) bobtail squid, ram’s horn squid 3. (Sepiadariids; numerous genera) cuttlefish

Other Molluscs

-**ŢPSS**- ‘CHITON / NAUTILUS / SCAPHOPOD’ 1. (genera *Allonautilus, Nautilus*) [chambered] nautilus 2. (Polyplacophorans; numerous genera) chiton / sea cradle 3. (Scaphopods; several genera) tusk shell / tooth shell

-**ŢPŠŠ**- ‘OTHER MOLLUSC’ 1. (Solenogastres/Neomeniomorphs; numerous genera) solenogaster 2. (Caudofoveata/Chaetodermomorpha; numerous genera) caudofoveate 3. (Monoplacophorans; several genera) monoplacophoran

7.3.1.8 Porifera, Ctenophora, Placozoa, and Cnidaria

Porifera

-**KFW**- ‘PORIFERA (SPONGE)’ I 1. (Calcarea; numerous genera) calcareous sponge 2. (Hexactinellids; numerous genera) glass sponge 3. (Demosponges; numerous genera) demosponge / coralline sponge

-**KFY**- ‘PORIFERA (SPONGE)’ II 1. (Plakinids; several genera) spiculate sponge 2. (Oscarellids; genera *Oscarella, Pseudocorticium*) slime sponge, aspiculate sponge 3. [stem unassigned]

Ctenophora and Placozoa

-**KFM**- 'CTENOPHORE (COMB JELLY) and PLACOZOAN' 1. (Tentaculates; numerous genera) comb jelly, Venus girdle 2. (Beroids; genera *Beröe, Neis*) beroid / cigar comb jelly 3. (Placozoa; genera *Trichoplax, Hoilungia, Polyplacotoma*) placozoan

Cnidaria

-**KFL**- 'SEA ANEMONE' 1. (Actiniaria; numerous genera) sea anemone 2. (Zoanthids; numerous genera) zoanthid / zoanthid coral 3. (Penicillaria and Spirularia; numerous genera) tube-dwelling anemone

-**KFR**- 'HEXACORALLINE CORAL' 1. (Scleractinia; numerous genera) stony coral / hard coral 2. (Corallimorpharia; numerous genera) mushroom coral, elephant ear anemone 3. (Antipatharia; numerous genera) black coral / thorn coral

-**KFŘ**- 'OCTOCORALLINE CORAL' 1. (Pennatulacea; numerous genera) sea pen, sea pansy 2. (Alcyonaea; numerous genera) soft coral, sea fan, sea whip, gorgonian 3. (Heliopora; genera Heliopora, Nanipora) soft coral

-**KFT**- 'JELLYFISH' 1. (Coronamedusae and Discomedusae; numerous genera) jellyfish 2. (Cubozoa; numerous genera) box jellyfish 3. (Stauromedusae; several genera) stalked jellyfish

-**KFP**- 'HYDROZOAN' 1. (Hydrozoa in hydroid or polypoid stage; numerous genera) hydroid, polyp 2. (Hydrozoa in medusoid stage; numerous genera) hydrozoan jelly 3. (Siphonophorae; numerous genera) siphonophore [colonial hydrozoan] (including Portuguese man o'war)

-**KFN**- 'PARASITIC CNIDARIA' 1. (Polypodiids; genus Polypodium) polypodium 2. (Myxosporea; numerous genera) myxosporean myxozoan 3. (Malacosporea-Saccosporids; genera *Buddenbrockia, Tetracapsuloides*) saccosporid myxozoan

7.3.1.9 Arthropods (including Arachnids, Crustaceans, Myriapods, and Insects)

Extinct Arthropods

-ZTF- 1. (Artiopods) trilobite 2. (Megacheira) great appendage arthropod 3. (Dinocaridids) other extinct Arthropod

Chelicerates

Chelicerates > Eucheliceratа > Arachnida > Araneae > Mesothelae (containing single family Liphistiidae)

-ZŇW- 'LIPHISTIID SPIDER (ASIAN TRAPDOOR SPIDER)' 1. (genera *Heptathela, Ryuthela*) Japanese and Ryuku Island trapdoor spider, including kimura spider 2. (genera *Ganthela, Sinothela, Songthela, Vinathela, Qiongthela*) Chinese and Vietnamese trapdoor spider 3. (genus *Liphistius*) Thai/Malaysian/Burmese/Laotian trapdoor spider

Chelicerates > Eucheliceratа > Arachnida > Araneae > Mygalomorphae

Chelicerates > Eucheliceratа > Arachnida > Araneae > Mygalomorphae > Atypoidea

-**ZMŘ**- 'MYGALOMORPHIC ATYPOID SPIDER' 1. (Antrodiaetids; genera *Aliatypus, Antrodiaetus, Atypoides*) folding trapdoor spider 2. (Atypids; genera *Atypus, Calommata, Sphodros*) purseweb spider / atypical tarantula 3. (Mecicobothrids; genera *Hexura, Hexurella, Mecicobothrium, Megahexura*) sheet funnel-web spider / dwarf tarantula

Chelicerates > Euchelicerata > Arachnida > Araneae > Mygalomorphae > Avicularioidea

-**ZMW**- 'AVICULARIOID SPIDER' I 1. (Actinopodids; genera *Actinopus, Missulena, Plesiolena*) Actinopod spider, including mouse spider 2. (Atracids; genera *Atrax, Hadronyche, Illawarra*) atracid / Australian funnel-web spider 3. (Hexathelids, Macrothelids, Porrhothelids; several genera) funnel-web tarantula

-**ZMY**- 'AVICULARIOID SPIDER' II 1. (Diplurids; numerous genera) curtain-web spider 2. (genus *Macrothele*) macrothelid funnel-web spider 3. (genus *Porrhothele*) porrhothelid curtain-web spider

-**ZML** - 'AVICULARIOID SPIDER' III 1. (Ctenizids; genera *Cteniza, Cyrtocarenum, Stasimopus*) ctenizid / cork-lid trapdoor spider 2. (Halonoproctids; several genera) halonoproctid trapdoor spider 3. (Nemsiids; numerous genera) nemesiid / funnel-web tarantula

-**ZMR**- 'AVICULARIOID SPIDER' IV 1. (Cyrtaucheniids; numerous genera) wafer trapdoor spider 2. (Euctenizids; several genera) euctenizid trapdoor spider 3. (Idiopids; numerous genera) armoured trapdoor spider

-**ZNW** - 'AVICULARIOID SPIDER' V 1. (Barychelid; numerous genera) barychelid / brushed trapdoor spider / trapdoor baboon spider 2. (Migids; numerous genera) tree trapdoor spider 3. (Microstigmatids; several genera) microstigmatid spider

-**ZNY** - 'AVICULARIOID SPIDER' VI 1. (Therephosids; numerous genera) tarantula 2. (Paratropidines; genera *Anisaspis, Anisaspoides, Paratropis, Stormtropis*) [paratropidine] baldlegged spider 3. (genus *Melloina*) [glabropelmatine] baldlegged spider

Chelicerates > Euchelicerata > Arachnida > Araneae > Mygalomorphae > Araneomorphae > Hypochilidae, Filistatids, and Synspermiata

-BZPW- 'PRIMITIVE ARANEOMORPHIC SPIDER' I 1. (Hypochilids; genera Ectatosticta, Hypochilus) lampshade spider 2. (Filistatids; numerous genera) crevice weaver spider 3. (Austrochilids; genera *Austrochilus, Thaida*) austrochilid spider

-BZPY- 'PRIMITIVE ARANEOMORPHIC SPIDER' II 1. (Gradungulids; several genera) large-clawed spider 2. (genus *Hickmania*) Tasmanian cave spider 3. (Leptonetids; numerous genera) leptonetid spider

-BZPL- 'SYNSPERMIATIC (HAPLOGYNE) SPIDER' I 1. (Diguetiids; genera *Diguetia, Segestrioides*) coneweb spider 2. (Plectreurids; genera *Kibramoa, Plectreurys*) plectreurid spider 3. (Paculiids; several genera) paculiid spider

-BZPR- 'SYNSPERMIATIC (HAPLOGYNE) SPIDER' II 1. (Tetrablemmids; numerous genera) armoured spider 2. (Pholcids; numerous genera) daddy long-legs / cellar spider / carpenter spider / vibrating spider / skull spider 3. (Caponiids; numerous genera) caponiid spider

-BZPŘ- 'SYNSPERMIATIC (HAPLOGYNE) SPIDER' III 1. (Dysderids; numerous genera) woodlouse hunter / cell spider / sowbug-eating spider 2. (Segestriids; several genera) tube-dwelling spider / tubeweb spider 3. (Orsolobids; numerous genera) orsolobid spider

-BZPF- 'SYNSPERMIATIC (HAPLOGYNE) SPIDER' IV 1. (Oonopids; numerous genera) goblin spider / dwarf hunting spider 2. (Ochyroceratids; numerous genera) midget ground weaver spider 3. (Trogloraptorids; genus *Trogloraptor*) tragloraptor spider

-BZPĻ- 'SYNSPERMIATIC (HAPLOGYNE) SPIDER' V 1. (Sicariids; genera *Loxosceles, Hexophthalma, Sicarius*) recluse spider, violin spider, sand spider 2. (Drymusids; genera *Drymusa, Izithunzi*) false violin spider 3. (Periegopids; genus *Periogops*) periogops spider

-BZPÇ- 'SYNSPERMIATIC (HAPLOGYNE) SPIDER' VI 1. (Scytodids; several genera) spitting spider 2. (Telemids; numerous genera) long-legged cave spider 3. other synspemiatic (haplogyne) spider

Chelicerates > Euchelicerata > Arachnida > Araneae > Mygalomorphae > Araneomorphae > Palpimanoidea

-BZPH- 'PALPIMANOID SPIDER' I 1. (Archaeids; several genera) assassin spider / pelican spider 2. (Stenochilids; genera Colopea, Stenochilus) stenochild spider 3. (Palpimanids; numerous genera) palp-foot spider

-BZPHW- 'PALPIMANOID SPIDER' II 1. (Mecysmaucheniids; several genera) mecysmaucheniid spider 2. (genus Huttonia) huttonia spider 3. [stem unassigned]

Chelicerates > Euchelicerata > Arachnida > Araneae > Mygalomorphae > Araneomorphae > Entelegynae

-SPW- 'ENTELEGYNE SPIDER' I 1. (Theridiids; numerous genera) tangle-web spider / cobweb spider / comb-footed spider, common house spider, widow 2. (Araneids; numerous genera) orb-weaver spider 3. (Tetragnathids; numerous genera) long-jawed orb weaver / long jawed spider

-SPY- 'ENTELEGYNE SPIDER' II 1. (Mysmenids; numerous genera) spurred orb-weaver spider 2. (Symphytognathids; several genera) dwarf orb-weaver 3. (Arkyids; genera *Arkys, Demadiana*) arkyid orb-weaver, triangular spider / ambush spider

-SPŘ- 'ENTELEGYNE SPIDER' III 1. (Anapids; numerous genera) anapid spider 2. (Nesticids; numerous genera) scaffold web spider 3. (Linyphiids; numerous genera) sheet weaver spider / money spider / dwarf spider

-SPF- 'ENTELEGYNE SPIDER' IV 1. (Theridiosomatids; numerous genera) ray spider 2. (Cyatholipids; numerous genera) cyatholipid spider 3. (Pimoids; several genera) pimoid spider

-SPŢ- 'ENTELEGYNE SPIDER' V 1. (Synaphrids; several genera) synaphrid spider 2. (Synotaxids; genus *Synotaxus*) synotaxus spider 3. (Physoglenids; numerous genera) physoglenid spider

-SPV- 'ENTELEGYNE SPIDER' VI 1. (Agelenids; numerous genera) funnel weaver spider 2. (Toxopids; numerous genera) toxopid spider 3. (Amaurobiids; numerous genera) tangled nest spider

-SPĻ- 'ENTELEGYNE SPIDER' VII 1. (Corinnids; numerous genera) corinnid sac spider / dark sac spider 2. (Liocranids; numerous genera) liocranid sac spider 3. (Clubionids; numerous genera) sac spider

-SPÇ- 'ENTELEGYNE SPIDER' VIII 1. (Gnaphosids; numerous genera) ground spider / flat-bellied ground spider / long-spinneret ground spider 2. (Lamponids; numerous genera) lamponid spider 3. (Trochanteriids; numerous genera) flat spider

-SPH- 'ENTELEGYNE SPIDER' IX 1. (Ammoxenids; several genera) termite hunter spider 2. (Cithaeronids; genera *Cithaeron, Inthaeron*) cithaeronid spider 3. (Gallieniellids; several genera) gallieniellid spider

-SPFW- 'ENTELEGYNE SPIDER' X 1. (Salticids; numerous genera) jumping spider 2. (Selenopids; several genera) wall spider / wall crab spider / flatty 3. (Anyphaenids; numerous genera) anyphaenid sac spider

-SPFY- 'ENTELEGYNE SPIDER' XI 1. (Thomisids; numerous genera) crab spider, flower spider / flower crab spider 2. (Sparassids; numerous genera) huntsman spider / giant crab spider, wood spider, rain spider, wheel spider 3. (Philodromids; numerous genera) running crab spider

-SPFL- 'ENTELEGYNE SPIDER' XII 1. (Desids; numerous genera) intertidal spider 2. (Phrurolithids; numerous genera) phrurolithid spider 3. (Trachelids; numerous genera) trachelid spider

-SPFR- 'ENTELEGYNE SPIDER' XIII 1. (Dictynids; numerous genera) dictynid spider 2. (Cybaeids; numerous genera) cybaeid spider 3. (Hahniids; numerous genera) dwarf sheet spider

-SPFŘ- 'ENTELEGYNE SPIDER' XIV 1. (Eresids; numerous genera) velvet spider 2. (Hersiliids; numerous genera) tree trunk spider 3. (Oecobiids; several genera) disc web spider

-SPHW- 'ENTELEGYNE SPIDER' XV 1. (Lycosids; numerous genera) wolf spider 2. (Ctenids; numerous genera) wandering spider / tropical wolf spider 3. (Viridasiids; genera *Viridasius, Vulsor*) viridasiid spider

-SPÇW- 'ENTELEGYNE SPIDER' XVI 1. (Oxyopids; several genera) lynx spider 2. (Pisaurids; numerous genera) nursery web spider 3. (Stiphidiids; numerous genera) sheetweb spider

-SPÇÇ- 'ENTELEGYNE SPIDER' XVII 1. (Psechrids; genera *Fecenia, Psechrus*) psechrid spider 2. (Senoculids; genus Senoculus) senoculus spider 3. (Trechaleids; numerous genera) trechaleid spider

-SPŢW- 'ENTELEGYNE SPIDER' XVIII 1. (Zoropsids; numerous genera) false wolf spider 2. (Ududbids; several genera) ududbid spider 3. (Penestomids; genus Penestomus) penestomus spider

-SPŢY- 'ENTELEGYNE SPIDER' XIX 1. (Malkarids; numerous genera) shield spider 2. (Mimetids; numerous genera) pirate spider 3. (Nicodamids; several genera) nicodamid spider

-SPŢL- 'ENTELEGYNE SPIDER' XX 1. (Phyxelidids; numerous genera) lace web spider 2. (Titanoecids; several genera) titanoecid spider 3. (Megadictynids; genera *Forstertyna,Megadictyna*) megadictynid spider

-SPŢR- 'ENTELEGYNE SPIDER' XXI 1. (Uloborids; numerous genera) hackled orb weaver 2. (Deinopids; genera *Deinopis, Menneus*) net-casting spider 3. (Zodariids; numerous genera) ant spider

-SPŢŘ- 'ENTELEGYNE SPIDER' XXII 1. (Miturgids; numerous genera) long-legged sac spider 2. (Cheiracanthiids; numerous genera) cheiracanthiid spider 3. (Xenoctenids; several genera) xenoctenid sac spider

-SPĻW- 'ENTELEGYNE SPIDER' XXIII 1. (Cycloctenids; several genera) cycloctenid spider 2. (Homalonychids; genus Homalonychus) homalonychus spider 3. (Entelegyne spider other than from named family) other entelegyne spider

Chelicerates > Eucheliceratа > Arachnida > Scorpiones

-**GGZ**- 'SCORPION' I 1. (Buthida; numerous genera) thick-tailed scorpion, fat-tailed scorpion, bark scorpion 2. (Hemiscorpiidae; genus *Hemiscorpius*) rock scorpion / creeping scorpion / tree scorpion 3.(Euscorpiidae; several genera) wood-scorpion

-**GGV**- 'SCORPION' II 1. (Scorpionidae; numerous genera) giant forest scorpion, emperor scorpion, burrowing scorpion / hissing scorpion / serkets, pale-legged scorpion 2. (Vaejovidae; numerous genera) stripetailed scorpion, sand scorpion, slendertailed scorpion, northern scorpion, dune scorpion, unstriped scorpion, devil scorpion 3. (Caraboctonidae; genus *Hadrurus*) giant desert hairy scorpion, black-back scorpion

-**GGĐ**- 'SCORPION' III 1. (Pseudochactida; three genera) pseudochactid cave-dwelling scorpion 2. (Chaerilida, Bothriuridae, Iuridae, Superstitioniidae, Chactidae; numerous genera) other scorpion 3. [stem unassigned]

Chelicerates > Eucheliceratа > Arachnida > remaining families

-ẒFW- 'DISEASE-CARRYING TICK' 1. (genus *Ixodes*) deer tick 2. (genus *Dermacentor*) wood tick 3. (genus *Amblyomma*) Cayenne tick, lone star tick

-ẒFY- 'METASTIGMA (TICK)' 1. (Ixodids; numerous genera) hard tick 2. (Argasids; numerous genera) soft tick 3. (Nuttalliellids; genus Nuttalliella) nuttalliella tick

-ẒFL- 'PARASITIC MITE' I 1. (genus *Sarcoptes*) itch mite / scabies mite 2. (genus *Dermatophagoides*) dust mite / house dust mite 3. (genus *Euroglyphus*) Mayne's house dust mite

-ẒFR- 'PARASITIC MITE' II 1. (genus *Dermanyssus*) red mite 2. (genus *Ornithonyssus*) tropical rat mite, tropical fowl mite, northern fowl mite, bird mite 3. (genus *Cheyletiella*) cheyletiella mite

-ẒFŘ- 'PARASITIC MITE' III 1. (genus *Cheyletus*) cheyletus mite 2. (genus *Tryophagus*) cheese mite 3. (genus *Pyemotes*) oak leaf gall mite, itch mite

-ẒFM- 'PARASITIC MITE' IV 1. (genus *Acarus*) flour mite 2. (genus *Glyciphagus*) glyciphagus mite 3. (genus *Carpoglyphus*) fruit mite

-ẒFN- 'PARASITIC MITE' V 1. (genus *Liponyssoides*) house rat mite 2. (genus *Laelaps*) spiny rat mite 3. (Trombiculidae; numerous genera) chigger / red bug / harvest mite / scrub-itch mite, berry bug

-ẒFŇ- 'OTHER MITE' 1. (Sarcoptiformes; numerous genera) sarcoptiform mite (i.e., astigmatine, endeostigmatid, or oribatid mite) 2. (Trombidiformes; numerous genera) trombidiform mite (i.e., prostigmatid or sphaerolichid mite), chigger 3. (Holothyrids, Mesostigma, and Opilioacariformes; numerous genera) parasitiform mite

-ẒNW- 'OTHER ARACHNID' I 1. (Opiliones; numerous genera) harvestman / harvester / shepherd spider 2. (Pseudoscorpiones; numerous genera) pseudoscorpion / false scorpion / book scorpion 3. (Solifugae; numerous genera) solifuge / sun spider / camel spider / wind scorpion

-ẒNY- 'OTHER ARACHNID' II 1. (Amblypygids; numerous genera) whip spider / tailless whip scorpion 2. (Shizomids; numerous genera) shorttailed whipscorpion 3. (Thelyphonids; numerous genera) whip scorpion / vinegaroon

-ẒŇW- ‘OTHER ARACHNID’ III 1. (Palpigradids; numerous genera) palpigrade / microwhip scorpion 2. (Ricinuleids; genera *Cryptocellus, Pseudocellus, Ricinoides*) hooded tickspider 3. [stem unassigned]

Chelicerates > non-arachnids

-SPḶḶ- ‘NON-ARACHNID CHELICERATE’ 1. (Xiphosura; genera *Tachypleus, Limulus*, *Carcinoscorpious*) horseshoe crab 2. (Pycnogonid; numerous genera) sea spider 3. [stem unassigned]

Crustaceans

Crustaceans > branchiopods (brine shrimp)

-**KṬKW**- ‘BRANCHIOPOD’ I 1. (Anostraca; numerous genera) fairy shrimp, brine shrimp 2. (Notostraca; genera Triops, Lepidurus) tadpole shrimp / shield shrimp 3. (Cladocera; numerous genera) water flea

-**KṬKY**- ‘BRANCHIOPOD’ II (CLAM SHRIMP) 1. (Laevicaudata; genera Lynceiopsis, Lynceus, Paralimnetes) [laevidaudate] clam shrimp 2. (Cyclestherida; genera Cyclestheria, Paracyclesthereia) [cyclestherid] clam shrimp 3. (Spinicaudata; numerous genera) [spinicaudate] clam shrimp

-**KṬKÇ**- ‘OTHER SMALL CRUSTACEAN’ 1. (Cephalocarids; several genera) horseshoe shrimp 2. (Ostracods; numerous genera) seed shrimp 3. (Remipedia; several genera) remipede

Crustaceans > maxillopods (barnacles, copepods, fish lice, etc.)

-**KṬKL**- ‘COPEPOD & MYSTACOCARID’ 1. (Calanoids, Cyclopoids, Poecilostomatoids, Monstrilloids, Mormonillids, Platycopiids, Siphonostomatoids; numerous genera) [primarily] planktonic marine or freshwater copepod 2. (Harpacticoids, Speleophriids; numerous genera, plus Gellyellids; genus *Gelyella*) sea-floor -dwelling or cave-dwelling copepod 3. (Mystacocarids; genera *Derocheilocaris, Ctenocheilocharis*) mystacocarid

-**KṬKR**- ‘THECOSTRACA (BARNACLE & BARNACLE-LIKE CREATURE)’ 1. (Thoracica; numerous genera) [common] barnacle 2. (Rhizocephala; numerous genera) parasitic barnacle 3. (Acrothoracica, Ascothoracids, Facetotecta; several genera) acrothoracic (i.e.,calcareous material-boring) barnacle, ascothoracid parasite, facetotectid / Hansenocaris

-**KṬKŘ**- ‘SMALL PARASITIC MAXILLOPOD CRUSTACEAN’ 1. (Branchiura; several genera) fish louse, carp louse 2. (Pentastomids; numerous genera) tongue worm 3. (Tantulocarids; numerous genera) tantulocarid

Crustaceans > Malacostraca (lobsters, crabs, shrimp, etc.)

Crustaceans > Malacostraca > Eumalacostraca > Peracarida

-**KṬMW**- ‘AMPHIPOD (SCUD / SIDESWIMMER)’ 1. (Gammarids and Senticaudates; numerous genera) shrimp-like amphipod / scud / sideswimmer 2. (Ingolfiellids; several genera) worm-like amphipod 3. (Hyperiids; numerous genera) marine parasitic/predatory amphipod

-**KṬMY**- ‘ISOPOD (WOOD LOUSE / PILL BUG & SIMILAR)’ 1. (Oniscidea; numerous genera) wood louse / pill bug / armadillo bug / roly-poly / potato bug / sow bug 2. (Asselotes, Calabazoids, Cymothoids, Limnoriids, Phoratopids, Phreatoicids, Sphaeromatids, Tainisopids, Valviferids; numerous genera) marine or freshwater isopod 3. (Microcerberids; several genera) tiny worm-like isopod

-**KṬNW**- ‘PERACARID (BROOD-POUCH BEARING SHRIMP)’ I 1. (Cumacea; numerous genera) hooded shrimp / comma shrimp 2. (Mysida; numerous genera) opossum shrimp 3. (Tanaidacea; numerous genera) tanaid

-**KŢNY**- 'PERACARID (BROOD-POUCH BEARING SHRIMP)' II 1. (Mictacea, Lophogastrids; several genera) mictacean or lophogastrid crustacean 2. (Spelaeogriphacea; several genera) spelaeogriphacean (cave-dwelling) crustacean 3. (Thermosbaenacea; several genera) thermosbaenacean (thermal spring-dwelling) crustacean

Crustaceans > Malacostraca > Eumalacostraca > Eucarida

Crustaceans > Malacostraca > Eumalacostraca > Eucarida > Euphausiacea

-**KŢPW**- 'NON-DECAPOD EUCARID' 1. (Euphausiids; several genera) krill 2. (Bentheuphausiids; genus *Bentheuphausia*) deep sea krill 3. (Amphionidacea; genus *Amphionides*) amphionides

Crustaceans > Malacostraca > Eumalacostraca > Eucarida > Decapoda

Crustaceans > Malacostraca > Eumalacostraca > Eucarida > Decapoda > Dendrobranchiata

-**KŢPY**- 'DENDROBRANCHIATE (PRAWN)' 1. (Aristeids; several genera) gamba prawn / deep-sea shrimp 2. (Penaeoids other than Aristeids; numerous genera) [benthesicymid, penaeid, sicyonid, or solenocerid] prawn 3. (Sergestoids; several genera) [luciferid or sergestid] prawn

Crustaceans > Malacostraca > Eumalacostraca > Eucarida > Decapoda > Caridea

-**KŢPL**- 'CARIDEAN SHRIMP' I 1. (Alpheoids; numerous genera) snapping shrimp / pistol shrimp / alpheid shrimp, cleaner shrimp, broken-back shrimp / anemone shrimp 2. (Crangonoids; numerous genera) crangonoid shrimp 3. (Pandalids; numerous genera) pandalid shrimp

-**KŢPR**- 'CARIDEAN SHRIMP' II 1. (Atyids; numerous genera) atyid shrimp 2. (Campylonotoids; several genera) compylonotoid shrimp 3. (Palaemonoids; numerous genera) palaemonoid shrimp

-**KŢPŘ**- 'CARIDEAN SHRIMP' III 1. (Nematocarcinoids; several genera) nematocarcinoid shrimp 2. (Pasiphaeids; several genera) pasiphaeid shrimp 3. (Processoids; several genera) processoid shrimp

-**KŢPH**- 'CARIDEAN SHRIMP' IV 1. (Stylodactyloids; several genera) stylodactyloid shrimp 2. (Oplophorids; several genera) oplophorid shrimp 3. (Bresilioids; numerous genera) bresilioid shrimp

-**KŢPÇ**- 'RARE CARIDEAN SHRIMP' 1. (Psalidopodoids; genus *Psalidopus*) psalidopus shrimp 2. (Physetocaridoids; genus *Physetocaris*) physetocaris shrimp 3. (Galatheacaridoids; genus *Galatheacaris*) galatheacaris shrimp

-**KŢL**- 'LOBSTER' 1. (genus *Nephrops*) Norway lobster, Dublin Bay prawn, langoustine, scampi 2. (genus *Homarus*) American lobster / Maine lobster / Atlantic lobster / true lobster, European lobster / common lobster 3. (all other genera) Japanese lobster, China lobster, Australian scampi, New Zealand scampi, Bight lobster, Caribbean lobster, Cape lobster, etc.

-**KŢR**- 'ACHELATE LOBSTER' 1. (Palinurids; several genera) spiny lobster 2. (Synaxids; genera *Palinurellus, Palibythus*) furry lobster, mole lobster 3. (Scyllarids; numerous genera) slipper lobster / shovel-nose lobster / locust lobster

-**KŢŘ**- 'LOBSTER-LIKE CREATURE' 1. (Glypheidea; genera *Neoglyphea, Laurentaeglyphea*) glypheid lobster 2. (Polychelids; several genera) polychelid lobster 3. (Enoplometopids; genus *Enoplometopus*) reef lobster

-**KŢF**- 'CRAYFISH' 1. (Astacids; genera *Astacus, Austropotamobius, Pacifastacus*) European or Western N. American crayfish 2. (Cambarids; numerous genera) Eastern N. American & Central American crayfish 3. (Parastacids; numerous genera) Southern Hemisphere crayfish

-**KṬṬ**- ‘CRAB’ 1. (Eubrachyura; numerous genera) crab 2. (Raninids; several genera) frog crab 3. (Dromiaceans and Cyclodorippoids; numerous genera) podotremate crab

-**KṬṬW**- ‘CRAB-LIKE CREATURE’ I 1. (Paguroids; numerous genera) hermit crab, coconut crab / robber crab 2. (Lithodoids; numerous genera) king crab 3. (Hippoids; several genera) sand crab / mole crab

-**KṬṬY**- ‘CRAB-LIKE CREATURE’ II 1. (Galatheoids, Chirostyloids; numerous genera) squat lobster, porcelain crab 2. (Aeglids; genus Aegla) aegla crab 3. (Lomisids; genus *Lomis*) hairy stone crab

-**KṬN̆**- ‘OTHER DECAPOD’ 1. (Stenopodidea; numerous genera) stenopod crustacean, glass sponge shrimp 2. (Procaridídea; genera *Procaris, Vetericaris*) procarid shrimp 3. (Axiids and Gebiids; numerous genera) mud shrimp / mud lobster, ghost shrimp, burrowing shrimp

-**KṬN̆W**- ‘OTHER CRUSTACEAN’ 1. (Stomatopods; numerous genera) mantis shrimp 2. (Leptostraca; several genera) leptostracan 3. (Syncarids; numerous genera) syncarid crustacean, anaspid crustacean

Myriapods

-**BḌL**- ‘CHILOPOD (CENTIPEDE)' 1. (Scutigeromorphs; numerous genera) house centipede 2. (Lithobiomorphs; numerous genera) stone centipede 3. (Geophilomorphs, Scolopendromorphs, Craterostigmomorphs; numerous genera) soil centipede, tropical centipede, craterostigmus centipede

Myriapods > Diplopods (millipedes)

-**BḌR**- ‘DIPLOPOD (MILLIPEDE)’ 1. (Helminthomorphs; numerous genera) worm-like millipede 2. (Pentazonids; numerous genera) short-bodied pill millipede 3. (Penicillates; numerous genera) bristle millipede / bristly millipede / pincushion millipede

-**BḌŘ**- ‘OTHER MYRIAPOD’ 1. (Pauropod; numerous genera) pauropod 2. (Symphyla; several genera) symphylan / garden centipede / pseudocentipede 3. [stem unassigned]

Hexapods

Hexapods > Entognatha

-**CDW**- ‘ENTOGNATHID’ 1. (Collembola; numerous genera) springtail 2. (Diplura; numerous genera) dipluran / two-pronged bristletail 3. (Protrura; numerous genera) protruran / conehead

Hexapods > Insects

-**CDY**- ‘SILVERFISH / JUMPING BRISTLETAIL’ 1. (Zygentoma; numerous genera except *Thermobia domestica*) silverfish / fishmoth 2. (*Thermobia domestica*) firebrat 3. (Archaeognatha; numerous genera) jumping bristletail

-**CDL**- ‘DRAGONFLY’ 1. (Aeshnoids and Epiophlebiids; numerous genera) aeshnid / hawker / darner dragonfly, petaltail dragonfly, clubtail dragonfly, Himalayan relict dragonfly 2. (Cordulegastrids; numerous genera) spiketail / biddie / flying addie dragonfly, neopetalia 3. (Libelluloids; numerous genera) emerald, skimmer / percher, pennant, dasher, sylph, dragonlet, pondhawk, setwing, pintail, forestwatcher, rainpool glider, amberwing, meadowhawk / darter, saddlebag, whiteface, dropwing dragonfly

-**CDR**- ‘DAMSELFLY’ 1. (Hemiphlebiids and Coenagrionoids; numerous genera) ancient greenling, pond, narrow-wing, white-legged, shadowdamsel, threadtail, forest giant damselfly 2. (Lestoids; numerous genera) spreadwing, bluestrak, flatwing, shortwing, sylph damselfly 3. (Calopterygoids; numerous genera) relict, demoiselle, jewel, bareleg, odalisque, bannerwing damselfly

-**CDǍ**- ‘MAYFLY (a.k.a. SHADFLY / FISHFLY / UP-WINGED FLY)’ 1. (Pannota; numerous genera) pannotic mayfly, including spiny crawler mayfly, armored mayfly 2. (Schistonota; numerous genera) schistonotic mayfly, including brushlegged mayfly, minnow mayfly, flat-headed mayfly / stream mayfly, spiny-headed burrowing mayfly 3. (Leptophlebiids; numerous genera) leptophlebiid / prong-gilled mayfly

-**CDV**- ‘DERMAPTERAN / ZORAPTERAN’ 1. (Dermaptera [except Labiids and Spongiphorids]; numerous genera) earwig 2. (Labiids, Spongiphorids; numerous genera) little earwig 3. (Zoraptera; numerous genera) angel insect

Hexapods > Insects > Orthoptera (crickets, grasshoppers, katydids, etc.)

-**XFW**- ‘CRICKET’ 1. (Grylloids; numerous genera) cricket, scaly cricket 2. (Gryllotalpids; several genera) mole cricket 3. (Myrmecophilids; several genera) ant cricket / ant-loving cricket

-**XFY**- ‘CRICKET-LIKE CREATURE’ I 1. (Tettigoniids; numerous genera) katydid / bush cricket 2. (Hagloids; several genera) grig 3. (Stenopelmatids; several genera) stenopelmatid, Jerusalem cricket

-**XFL**- ‘CRICKET-LIKE CREATURE’ II 1. (Anostostomatids; numerous genera) weta [except cave weta] / king cricket 2. (Gryllacridids; numerous genera) leaf-rolling cricket / raspy cricket 3. (Cooloolids; genus *Cooloola*) Cooloola monster, dingo monster, Pearson’s monster, sugarcane monster

-**XFR**- ‘CRICKET-LIKE CREATURE’ III 1. (Rhaphidophorids; numerous genera) cave weta / cave cricket, camleback cricket / camel cricket, spider cricket, sand treader 2. (Schizodactylids; genera *Comicus, Schizodactylus*) dune cricket / splay-footed cricket 3. [stem unassigned]

-**XFM**- ‘GRASSHOPPER’ 1. (Acridoidea, Pyrgomorphoidea; numerous genera) grasshopper 2. (Pneumorids; several genera) bladder grasshopper 3. (Trigonopterygoidea; several genera) leaf grasshopper

-**XFN**- ‘GRASSHOPPER-LIKE CREATURE 1. (Eumastacids; numerous genera) monkey grasshopper / matchstick grasshopper 2. (Proscopiids; numerous genera) stick grasshopper / jumping stick 3. (Tetrigids; several genera) groundhopper / pygmy grasshopper / pygmy devil

-**XFǍ**- ‘TRIDACTYLOID’ 1. (Cylindrachetids; several genera) sandgroper 2. (Ripipterygids; genera *Mirhipipteryx, Ripipteryx*) mud cricket 3. (Tridactylids; several genera) pygmy mole cricket

-**XȚW** - ‘NOTOPTERA & EMBIOPTERA’ 1. (Grylloblattids; several genera) ice bug, ice-crawler 2. (Mantophasmatids; numerous genera) gladiator / rock crawler / heelwalker 3. (Embioptera; numerous genera) webspinner

-**XȚY**- ‘PHASMATODE (WALKING STICK / WALKING LEAF) 1. (Verophasmatodes [other than Phylliids]and Timematids; numerous genera) walking stick / stick-bug / bug stick 2. (Phylliids; several genera) walking leaf / leaf insect 3. (Agathemerids; genus Agathemera) agathemera

Hexapods > Insects > Mantodea (mantises)

-**ḌPḶ**- ‘MANTIS’ 1. (Mantids; numerous genera) praying mantis 2. ([various genera and species]) flower mantis 3. ([numerous genera]) mantis (other than Mantids and flower mantises)

Hexapods > Insects > Blattodea (cockroaches and termites)

-**ČŇ**- ‘COCKROACH’ 1. (Blattoids; numerous genera) Oriental, American, Australian, Brown, Smokbrown, Florida woods, Botany Bay, common shining, wood roach / brown-hooded cockroach 2. (Blaberoids; numerous genera) blaberid, giant cockroach, death’s head, false death’s head / drummer / discoid, speckled / lobster, greenhouse / Surinam, hissing, Dubia / orange-spotted, giant burrowing, Asian, brown-banded, European native, fulvous wood, German, Pennsylvania wood, Virginia wood, small yellow cockroach 3. (Corydioids; numerous genera) sand cockroach, cave cockroach

-**ČŇW**- ‘TERMITE’ 1. (Neoisopterans; numerous genera) [higher/newer] termite 2. (Euisopterans; numerous genera) dampwood termite, harvester termite, drywood termite 3. (Mastotermitids; *Mastotermes darwiniensis*) giant northern termite / Darwin termite

-**CXM**- ‘APHID’ 1. (Aphidids; numerous genera) aphid 2. (Adelgids; several genera) pine aphid, spruce aphid, wooly aphid 3. (Phylloxerids; several genera) phylloxeran

-**CXL**- ‘LOUSE’ 1. (Phthiraptera; numerous genera) louse 2. (genus *Pediculus*) head louse, body louse 3. (genus *Phthirus*) crab louse / public louse

-**CXR**- ‘PARANEOPTERAN BUG’ 1. (Thysanoptera; numerous genera) thrip 2. (Liposcelididae; numerous genera) booklouse 3. (Psocoptera other than Liposcelididae; numerous genera) barklouse / barkfly / psocid

Hexapods > Insects > Hemiptera (true bug)

-**NL̦T**- ‘CICADOMORPHIC BUG’ 1. (Cicadoids; numerous genera) cicada 2. (Cercopoids; numerous genera) froghopper / spittlebug 3. (Membracoids; numerous genera) treehopper / thorn bug

-**NL̦TW**- ‘REDUVIID (ASSASSIN BUG)’ I 1. (Triatomines; numerous genera) kissing bug / conenose bug / cone-headed bug / vampire bug 2. (Peiratines; numerous genera) corsair 3. (remaining Reduviid families or sub-families) assassin bug, wheel bug

-**NL̦TY**- ‘REDUVIID (ASSASSIN BUG)’ II 1. (Phymatines; numerous genera) ambush bug 2. (Emesines; numerous genera) three-legged bug 3. (Holoptilines; several genera) feather-legged bug / ant wolf

-**NL̦TL**- ‘PENTATOMOMORPHIC BUG’ I 1. (Pentatomoids; numerous genera) shield bug, stink bug, burrowing bug, jewel bug, ebony bug, chust bug, giant shield bug 2. (Lygaeoids; numerous genera) seed bug, stilt bug, big-eyed bug, dirt-colored seed bug, ash-gray leaf bug 3. (Pyrrhocoroids; numerous genera) red bug, cotton stainer, bordered plant bug

-**NL̦TR**- ‘PENTATOMOMORPHIC BUG’ II 1. (Hyocephalids; genera *Hyocephalus, Maevius*) hyocephalid bug 2. (Stenocephalids; genus *Dicranocephalus*) dicranocephalus 3. (Termitaphidid; Termitaphis, Termitaradus) termite bug

-**NL̦TŘ**- ‘STERNORRHYNCHAN BUG’ 1. (Psylloids; numerous genera) jumping plant lice 2. (Aleyrodids; numerous genera) whitefly 3. (Coccoids; numerous genera) scale insect, mealybug, cochineal, ground pearl

-**NL̦TF**- ‘COREOID BUG’ 1. (Alydids; numerous genera) broad-headed bug 2. (Coreids; numerous genera) leaf-footed bug, squash bug 3. (Rhopalids; numerous genera) scentless plant bug, soapberry bug

-**NL̦TÇ**- ‘FULGOROID BUG’ 1. (Fulgoridae; numerous genera) fulgorid planthopper 2. (Fulgoroidea other than Fulgoridae; numerous genera) other planthopper 3. [stem unasssigned]

-**NL̦C**- ‘CIMICOMORPHIC BUG’ I 1. (Cimicids; numerous genera) cimicid, bed bug 2. (Anthocorids, Lasiochilids, Lyctocorids; numerous genera) minute pirate bug / flower bug 3. (Microphysids; several genera) minute bladder bug

-**NL̦CW**- ‘CIMICOMORPHIC BUG’ II 1. (Mirids; numerous genera) capsid bug / mirid bug / plant bug / leaf bug / grass bug, lygus bug, apple dimpling bug, mosquito bug, honelylocust plant bug, green mind, potato mind 2. (Nabids; numerous genera) damsel bug 3. (Aradids; numerous genera) flat bug

-**NL̦Č**- ‘CIMICOMORPHIC BUG’ III 1. (Tingids; numerous genera) lace bug 2. (Thaumastocorids; several genera) royal palm bug 3. (Polyctenid; several genera) Old World bat bug / polyctenid bat bug

-**NL̦ČW**- ‘CIMICOMORPHIC BUG’ IV 1. (Velocipedids; several genera) velocipedid bug 2. (Joppeicids; genus *Joppeicus*) joppeicus bug 3. (Curaliids; geneus *Curalium*) curalium bug

-**NL̦TV**- ‘CIMICOMORPHIC BUG’ V 1. (Plokiophilids; several genera) plokiophilid bug 2. (Medocostids; genus *Medocostes*) medocostes bug 3. (Pachynomids; several genera) pachynomid bug

-**NL̦F**- ‘GERROID (SEMI-AQUATIC) BUG’ 1. (Gerrids; numerous genera) water strider / water skeeter / pond skater / water skipper / Jesus bug 2. (Veliids; numerous genera) riffle bug / broad-shouldered water strider 3. (Hermatobatids; genus *Hermatobates*) coral-treader

-**NḶFW**- ‘GERROMORPHIC (SEMI-AQUATIC) BUG’ 1. (Hydrometroids; several genera) water measurer / marsh treader 2. (Hebroids; several genera) velvet water bug 3. (Mesovelioids; numerous genera) water treader

-**NḶFY**- ‘DIPSOCOROMORPHIC BUG (LITTER BUG)’ 1. (Ceratocombids; several genera) litter bug 2. (Schizopterids and Hypsipterygids; numerous genera) jumping soil bug 3. (Stemmocryptid; genus *Stemmocrypta*) stemmocrypta bug

-NḶFL- ‘NEPOMORPHIC BUG’ I 1.(Belostomatids; numerous genera) giant water bug / toe-biter / electric-light bug / alligator tick / alligator flea 2. (Nepids; numerous genera) water scorpion, needle bug, water stick insect 3. (Corixids; numerous genera) water boatman

-**NḶFR**- ‘NEPOMORPHIC BUG’ II 1. (Gelastocorids; genera Gelastocoris, Nerthra) toad bug 2. (Ochterids; several genera) velvety shore bug 3. (Naucorids; numerous genera) creeping water bug / saucer bug

-**NḶFŘ**- ‘NEPOMORPHIC BUG’ III 1. (Notonectids and Pleoids; numerous genera) backswimmer, pygmy backswimmer 2. (Aphelochairids; genus Aphelocheirus) aphelocheirus bug 3. (Potamocorids; genera *Potamocoris, Coleopterocoris*) potamocorid bug

-**NḶV**- ‘HETEROPTERAN BUG’ 1. (Peloridiids; numerous genera) moss bug / beetle bug 2. (Enicocephalomorphs; numerous genera) gnat bug, unique-headed bug, aenictophecheid 3. (Leptopodomorphs; numerous genera) shore bug, spiny shore bug

Hexapods > Insects > Hymenoptera

Hexapods > Insects > Hymenoptera >Apoidea (bees and sphecoid wasp)

-**BZZ**- ‘APID BEE (CORBICULATE)’ 1. (Apini [genus *Apis*] and Meliponini; numerous genera) honey bee, stingless honey bee 2. (Bombini; genus *Bombus*) bumblebee 3. (Euglossini; several genera) orchid bee

-**BZZW**- ‘APID BEE (NON-CORBICULATE)’ 1. (Non-corbiculate Apines; numerous genera) digger bee, long-horned bee, non-corbiculate apine bee 2. (Nomadines; numerous genera) cuckoo bee 3. (Xylocopines; numerous genera) carpenter bee

-**BZZY**- ‘NON-APID BEE’ 1. (Megachilids; numerous genera) mason bee, leafcutter bee, carder bee 2. (Colletids; numerous genera) plasterer bee / polyester bee, yellow-faced bee 3. (Andrenids, Halictids, Melittids, Stenotritids; numerous genera) non-apid bee, sweat bee, mining bee

-**BZZL**- ‘SPHECID (THREAD-WAISTED) WASP’ 1. (Sceliphines; several genera) mud wasp, mud dauber 2. (Chloriontines; genus Chlorion) cricket-hunter wasp 3. (Ammophilines and Sphecines) thread-waisted wasp, sand wasp

-**BZZR**- ‘SPHECOID WASP’ 1. (Ampulicids;) cockroach wasp 2. (Philanthines; several genera) predatory wasp, bee-hunter, bee-killer wasp, ant-queen kidnapping wasp 3. (Pemphredonids, Psenids, Ammoplanids; numerous genera) aphid wasp

-**BZZŘ**- ‘BEMBICINE WASP’ 1. (Bembicini; numerous genera) sand wasp 2. (Nyssonini; numerous genera) kleptoparasitic bembicine wasp 3. (Astatines and Heterogynaids; several genera) bembicine wasp

-**BZZV**- ‘CABRONID WASP’ 1. (Cabronines; numerous genera) digger wasp, square-headed wasp 2. (Mellinines; genera *Mellinus, Xenosphex*) mellinine wasp 3. [stem unassigned]

Hexapods > Insects > Hymenoptera > Vespoidea & Chrysidoidea

-**BZZX**- ‘Genus VESPA (HORNET)’ 1. (*Vespa crabro*) common European hornet 2. (*V. mandarinia*) Asian giant hornet / murder hornet 3. (all other *Vespa* species) other hornet

-**BZZM**- ‘VESPINE WASP ’ 1. (genera *Vespula*) common wasp, red cuckoo wasp, red wasp, oriental hornet 2. (genus *Dolichovespula*) yellowjacket, tree wasp, Saxon wasp, Norwegian wasp, bald-face hornet 3. (genus *Provespa*) night wasp / night hornet

-**BZZN**- ‘VESPID WASP’ 1. (Eumenines and Zethines; numerous genera) potter wasp 2. (Polistines; numerous genra) paper wasp 3. (Stenogastrines; several genera) hover wasp

-**BZZŇ**- ‘CHRYSIDOID / VESPOID WASP’ I 1. (Chrysidids; numerous genera) cuckoo wasp, emerald wasp, jewel wasp, gold wasp, ruby wasp 2. (Chrysidoids other than Chrysidids [i.e., Bethylids, Dryinids, Embolemids, Plumariids, Sclerogibbids, Scolebythids]; numerous genera) chrysidoid wasp 3. (Massarines, Euparagiines, Gayellines; numerous genera) pollen wasp

-**BZZḌ**- ‘CHRYSIDOID / VESPOID WASP’ II 1. (Mutillids; numerous genera) velvet wasp / velvet ant 2. (Pompilids; numerous genera) spider wasp / spider-hunting wasp, tarantula hawk 3. (Vespoids other than Mutillids, Pompilids, and Vespids [i.e., Bradynobaenids, Rhopalosomatids, Sapygids, Scoliids, Sierolomorphids, Tiphiids]; numerous genera) other vespoid wasp

Hexapods > Insects > Hymenoptera > parasitoid [i.e., non-aculeate] wasps

-**BZZB**- ‘CHALCID WASP’ 1. (Agaonids; numerous genera) fig wasp 2. (Chalcidids; numerous genera) chalcid wasp 3. (Eurytomids; numerous genera) seed chalcid wasp

-**BZZD**- ‘PARASITOID [NON-ACULEATE] WASP’ I 1. (Mymarids; numerous genera) fairy wasp / fairyfly 2. (Aphelinids, Encyrtids, Eucharitids; Eulophids, Eupelmids, Leucospids, Ormyrids, Perilampids, Pteromalids, Rotoitids, Signiphorids, Tanaostigmatids, Tetracampids, Torymids, Trichogrammatids; numerous genera) chalcidoid wasp 3. (Ceraphronids and Megaspilids; numerous genera) ceraphronoid wasp

-**BZZG**- ‘PARASITOID [NON-ACULEATE] WASP’ II 1. (Cynipids; numerous genera) gall wasp / gallfly 2. (Figitids, Ibaliids, Lipterids, Austrocynipids) cynipoid wasp 3. (Diapriids, Maamingids, Monomachids, Austroniids) diaprioid wasp

-**BZZP**- ‘PARASITOID [NON-ACULEATE] WASP’ III 1. (Evaniids; numerous genera) ensign wasp / nightshade wasp / hatchet wasp 2. (Aulacids, Gasteruptiids, Trigonaloids; numerous genera) evanioid wasp, trigonaloid wasp 3. (Megalyrids; several genera) megalyrid wasp

-**BZZT**- ‘PARASITOID [NON-ACULEATE] WASP’ IV 1. (Ichneumonids; numerous genera) ichneumon wasp, scorpion wasp 2. (Braconids; numerous genera) braconid wasp [including the mummy-wasp] 3. (Mymarommatids; several genera) false fairy wasp

-**BZZK**- ‘PARASITOID [NON-ACULEATE] WASP’ V 1. (Sephanids; several genera) crown wasp 2. (Helorids, Plelcinids, Peradeniids, Proctorenyxids, Proctotrupids, Roproniids, Vanhorniids; numerous genera) proctotrupoid wasp 3. (Platygastrids, Scelionids; numerous genera) platygastroid wasp

Hexapods > Insects > Hymenoptera > Formicidae (ants)

-**MŢK**- ‘FORMICINE ANT’ I 1. (genus *Formica*) wood ant, mound ant, thatching ant, field ant , meadow ant, horse ant 2. (genus *Cataglyphis*) desert ant 3. (genus *Dinomyrmex*) giant forest ant

-**MŢKW**- ‘FORMICINE ANT’ II 1. (genus *Camponotus*) carpenter ant 2. (genus *Oecophylla*) weaver ant 3. (genus *Polyergus*) Amazon ant

-**MŢKY**- ‘FORMICINE ANT’ III 1. (genus *Brachymyrmex*) rover ant 2. (genus *Gigantiops*) Gigantiops jumping ant 3. (all other Formicine genera) formicine ant

-**MŢKL**- ‘MYRMICINE ANT’ I 1. (genus *Solenopsis*) fire ant, ginger ant, tropical fire ant 2. (genus *Monomorium*) monomorium ant, flower ant, pharaoh ant 3. (genus *Pogonomyrmex*) harvester ant, Maricopan harvester ant

-**MŢKR**- ‘MYRMICINE ANT’ II 1. (genus *Cremtogaster*) Saint Valentine ant / cocktail ant / acrobat ant 2. (genus *Cephalotes*) turtle ant 3. (all other Myrmicine genera) myrmicine ant

-**MŢKŘ**- ‘MYRMECIINE ANT’ 1. (genus *Myrmecia*) jack jumper ant / bulldog ant / bull ant 2. (genus *Nothomyrmecia*) dinosaur ant / dawn ant 3. (Pseudomyrmecines; genera *Pseudomyrmex, Myrcridis, Tetraponera*) pseudomyrmecine ant, slender ant

-**MŢKF**- ‘DOLICHODERINE ANT’ I 1. (genus *Linepithema*) Argentine ant 2. (genus *Iridomyrmex*) rainbow ant 3. (genus *Dorymyrex*) cone ant

-**MŢKŢ**- ‘DOLICHODERINE ANT’ II 1. (*Tapinoma erraticum*) erratic ant 2. (*Tapinoma sessile*) stink ant / odorous house ant / sugar ant / coconut ant 3. (all other Dolichoderine genera) dolichoderine ant

-**MŢKV**- ‘OTHER ANT’ 1. (Ectatommines, Heteroponerines, Amblyoponines, Paraponerines, Ponerines, Proceratiines); numerous genera) poneromorphic ant 2. (Aneuretines, Martialines, Agroecomyrmecines; several genera) other non-poneromorphic ant 3. [stem unassigned]

Hexapods > Insects > Hymenoptera > Symphyta (sawflies)

-**ZZVW**- 'SYMPHYTE (SAWFLY)' I 1. (Tenthredinids; numerous genera) common sawfly 2. (Diprionids; numerous genera) conifer sawfly / pine sawfly 3. (Argids, Blasticotomids, Cimbicids, Pergids; numerous genera) tenthredinoid sawfly

-**ZZVY**- 'SYMPHYTE (SAWFLY)' II 1. (Xyelids; several genera) xyelid sawfly 2. (Pamphiliids; several genera) leaf-rolling sawfly / web-spinning sawfly 3. (Megalodontesids; several genera) megalondontesid sawfly

-**ZZVL**- 'SYMPHYTE (SAWFLY)' III 1. (Cephoids; numerous genera) stem sawfly 2. (Siricids; several genera) horntail 3. (Anaxyelids; genus *Syntexis*) cedar wood wasp

-**ZZVR**- 'WOOD WASP' 1. (Xyphydriids; numerous genera) wood wasp 2. (Orussids; numerous genera) parasitic wood wasp 3. [stem unassigned]

Hexapods > Insects > Neuropterids (net-winged insects: lacewings, mantidflies, antlions, etc.)

-**Ẓ.GW**- 'PRIMITIVE FLY' 1. (Plecoptera; numerous genera) stonefly 2. (Raphidoptera; numerous genera) snakefly 3. (Nevrorthids; genera *Nevrorthus, Austroneurorthus, Nipponeurorthus*) nevrorthid lacewing

-**Ẓ.GY**- 'MEGALOPTERA' 1. (Sialids; numerous genera) alderfly 2. (Chauliodines; numerous genera) fishfly 3. (Corydalines; several genera) dobsonfly

-**Ẓ.BW**- 'NEUROPTERID (LACEWING)' I 1. (Osmylids; numerous genera) giant lacewing 2. (Ithonids; several genera) moth lacewing 3. (Chrysopids; numerous genera) green lacewing, stinkfly

-**Ẓ.BY**- 'NEUROPTERID (LACEWING)' II 1. (Hemerobiids; numerous genera) brown lacewing 2. (Coniopterygids; numerous genera) dustywing 3. (Dilarids; several genera) pleasing lacewing

-**Ẓ.BL**- 'NEUROPTERID (LACEWING)' III 1. (Mantispids; numerous genera) mantidfly / mantisfly 2. (Rhachiberothids; several genera) thorny lacewing 3. (Berothids; several genera) beaded lacewing

-**Ẓ.BR**- 'NEUROPTERID (LACEWING)' IV 1. (Nymphids; several genera) split-footed lacewing 2. (Nemopterids; numerous genera) spoonwing, thread-winged 3. (Sisyrids; several genera) spongillafly

-**Ẓ.BŘ**- 'NEUROPTERID (LACEWING)' V 1. (Psychopsids; several genera) silky lacewing 2. (Myrmeleontids; numerous genera) antlion 3. (Ascalaphids; numerous genera) owlfly

Hexapods > Insects > Coleoptera (beetles)

Hexapods > Insects > Coleoptera > Adephaga

-**FBW**- 'ADEGPHAGAN BEETLE' I 1. (Carabids; numerous genera) ground beetle, bombardier beetle, ant nest beetle / paussine, sand beetle, tiger beetle, harp beetle, violin beetle 2. (Rhysodids; numerous genera) wrinkled bark beetle 3. (Trachypachids; genera *Trachypachus, Systolosoma*) false ground beetle

-**FBY**- 'ADEGPHAGAN BEETLE' II 1. (Dystiscids; numerous genera) predacious diving beetle 2. (Haliplids; several genera) crawling water beetle, haliplid 3. (Noterids; numerous genera) burrowing water beetle

-**FBL**- 'ADEGPHAGAN BEETLE' III 1. (Amphizoans; genus *Amphizoa*) troutstream beetle 2. (Gyrinids; numerous genera) whirligig beetle 3. (Hygrobiids; genus Hygrobia) squeak beetle / screech-beetle

-**FBR**- 'ADEGPHAGAN BEETLE' IV 1. (Aspidytids; genus *Aspidytes*) aspidytes beetle 2. (Meruids; genus *Meru*) meru beetle 3. [stem unassigned]

Hexapods > Insects > Coleoptera > Polyphaga

Hexapods > Insects > Coleoptera > Polyphaga > Cucujiformia

-**RÇK**- 'CUCUJOID BEETLE' I 1. (Coccinelids; numerous genera) ladybug / ladybird / lady beetle 2. (Discolomatids; numerous genera) Mexican hat beetle 3. (Phalacrids; numerous genera) shining flower beetle

-**RÇKW**- 'CUCUJOID BEETLE' II 1. (Endomychids; numerous genera) handsome fungus beetle 2. (Erotylids; numerous genera) pleasing fungus beetle 3. (Cryptophagids; numerous genera) silken fungus beetle

-**RÇKY**- 'CUCUJOID BEETLE' III 1. (Cucujids; several genera) flat bark beetle 2. (Bothriderids; numerous genera) dry bark beetle / cocoon-forming beetle 3. (Cerylonids; numerous genera) minute bark beetle

-**RÇKL**- 'CUCUJOID BEETLE' IV 1. (Sylvanids; numerous genera) sylvan flat bark beetle 2. (Laemophloeids; numerous genera) lined flat bark beetle 3. (Passandrids; several genera) parasitic flat bark beetle

-**RÇKR**- 'CUCUJOID BEETLE' V 1. (Nitidulids; numerous genera) sap beetle, picnic beetle / beer bug, small hive beetle 2. (Kateritids; several genera) short-winged flower beetle 3. (Latridiids; numerous genera) minute brown scavenger beetle / fungus beetle

-**RÇKŘ**- 'CUCUJOID BEETLE' VI 1. (Monotomids; numerous genera) root-eating beetle 2. (Corylophids; numerous genera) minute hooded beetle / minute fungus beetle 3. (Smicripids; genus *Smicrips*) palmetto beetle

-**RÇKÇ**- 'CUCUJOID BEETLE' VII 1. (Sphindids; several genera) cryptic slime mold beetle 2. (Phloeostichids, Priasilphids, Propalticids, Protocucujids, Lamingtoniids, Helotids, Boganiids, Alexiids, Akalyptoischion; numerous genera) other cucujoid beetle 3. (Myraboliids, Tasmosalpingids, Hobartiids, Cyclaxyrids, Agapythids; several genera) Australasian cucujoid beetle

-**RÇKF**- 'CURCULIONOID BEETLE (i.e., WEEVIL) I 1. (Curculionids; numous genera) true weevil / snout beetle, flower weevil, acorn weevil, nut weevil, broad-nosed weevil, ambrosia beetle / pinhole borer, bark beetle 2. (Brentids; numerous genera) straight-snouted weevil 3. (Attelabids; several genera) leaf-rolling weevil

-**RÇKŢ**- 'CURCULIONOID BEETLE (i.e., WEEVIL) II 1. (Nemonychids; numerous genera) pine flower weevil 2. (Anthribids; numerous genera) fungus weevil 3. (Belids and Carids; numerous genera) primitive weevil, belid, cycad weevil, carid weevil

-**RÇKH**- 'CLEROID BEETLE' I 1. (Clerids; numerous genera) checkered beetle, ant beetle, steely blue beetle, ham beetle, yellow-horned clerid 2. (Melyrids; numerous genera) soft-winged flower beetle 3. (Biphyllids; several genera) false skin beetle

-**RÇKHW**- 'CLEROID BEETLE' II 1. (Byturids; several genera) fruitworm beetle 2. (Trogossitids; numerous genera) bark-gnawing beetle 3. (Chaetosomatids, Lophocaterids, Acanthocnemids, Mauroniscids, Peltids, Phloiophilids, Phycosecids, Prionocerids, Thaneroclerids, Protopeltids, Rentoniids, Rhadalids, Thymalids; numerous genera) other claroid beetle

-**RÇKM**- 'CHRYSOMELOID & LYMEXILID BEETLE' 1. (Chrysomelids, Megalopodids, Orsodacnids; numerous genera) leaf beetle, Colorado potato beetle / ten-striped spearman 2. (Cerambycids, Disteniids, Oxypeltids, Vesperids; numerous genera) long-horned beetle / longicorn, titan beetle, red milkweed beetle, huhu beetle, harlequin beetle 3. (Lymexilids; several genera) ship-timber beetle, timberworm beetle

-**RÇG**- 'TENEBRIONOID BEETLE' I 1. (Aderids; numerous genera) ant-like leaf beetle 2. (Ciids; numerous genera) minute tree-fungus beetle 3. (Mycetophagids; several genera) hairy fungus beetle

-**RÇGW**- 'TENEBRIONOID BEETLE' II 1. (Mordellids; numerous genera) tumbling flower beetle / pintail beetle 2. (Anthicids; numerous genera) ant-like flower beetle 3. (Scraptiids; numerous genera) false flower beetle

-**RÇGY**- 'TENEBRIONOID BEETLE' III 1. (Zopherids other than Colydiines; several genera) ironclad beetle 2. (Colydiines; numerous genera) cylindrical bark beetle 3. (Salpingids; numerous genera) narrow-waisted bark beetle

-**RÇGL**- 'TENEBRIONOID BEETLE' IV 1. (Meloids; numerous genera) blister beetle 2. (Oedemerids; numerous genera) false blister beetle / pollen-feeding beetle 3. (Pyrochroids; numerous genera) fire-colored beetle

-**RÇGR**- ‘TENEBRIONOID BEETLE’ V 1. (Stenotrachelids; several genera) false long-horned beetle 2. (Prostomids; two genera) jugular-horned beetle 3. (Ripiphorids; numerous genera) wedge-shaped beetle

-**RÇGŘ**- ‘TENEBRIONOID BEETLE’ VI 1. (Tenebrionids; numerous genera) darkling beetle, flour beetle 2. (Melandryids; numerous genera) false darkling beetle 3. (Archeocrypticids, Borids, Chalcodryids, Mycterids, Perimylopids, Pterogeniids, Pythids, Synchroids, Tetratomids, Trachelostenids, Trictenotomids, Ulodids; numerous genera) other tenebrionoid beetle

Hexapods > Insects > Coleoptera > Polyphaga > Scarabaeoidea

-**RBK**- ‘SCARABAEID (SCARAB) BEETLE’ I 1. (Scarabaeines; numerous genera) true dung beetle 2. (Aphodiines;, numerous genera) small dung beetle 3. (Cetoniines; numerous genera) flower chafer, flower scarab

-**RBKW**- ‘SCARABAEID (SCARAB) BEETLE’ II 1. (Dynastines; numerous genera) rhinoceros beetle / unicorn beetle / horn beetle, Atlas beetle, Hercules beetle, Neptune beetle, Caucasus beetle, rabbit beetle, Siamese beetle, Mars beetle, elephant beetle, actaeon beetle, ox beetle 2. (Rutelines; numerous genera) shining leaf chafer, Japanese beetle 3. (Melolonthines; numerous genera) May beetle, May bug, Junebug, cockchafer, monkey beetle, white-grub beetle

-**RBKY**- ‘SCARABAEID (SCARAB) BEETLE’ III 1. (Euchirines; several genera) long-armed scarab 2. (Aclopines, Aegialiines, Allidiostomatines, Chironines, Dynamopodines, Aulonocnemines, Eremazines, Phaenomeridines, Orphnines, Temitotrogines; numerous genera) other scarab beetle 3. [stem unassigned]

-**RBKL**- ‘SCARABAEOID BEETLE’ I 1. (Hybosorids; numerous genera) scavenger scarab beetle 2. (Ceratocanthids; numerous genera) pill scarab beetle 3. (Ochodaeids; numerous genera) sand-loving scarab beetle

-**RBKR**- ‘SCARABAEOID BEETLE’ II 1. (Lucanids; numerous genera) stag beetle 2. (Diphyllostomatids; genus *Diphyllostoma*) false stage beetle 3. (Passalids; numerous genera) bess beetle / bessbug / betsy beetle / horned passalus beetle

-**RBKŘ**- ‘SCARABAEOID BEETLE’ III 1. (Pleocomids; genus Pleocoma) rain beetle 2. (Trogids and; several genera) hide beetle 3. (Glareresids; genus *Glaresis*) enigmatic scarab beetle

-**RBKÇ**- ‘SCARABAEOID BEETLE’ IV 1. (Geotrupids; numerous genera) earth-boring dung beetle 2. (Glaphyrids; several genera) bumble bee scarab beetle 3. (Belohinids; genus *Belohina*) belohina beetle

Hexapods > Insects > Coleoptera > Polyphaga > Staphyliniformia

-**RBKF**- ‘STAPHYLINOID BEETLE’ I 1. (Silphids; numerous genera) carrion beetle / burying beetle 2. (Agyrtids; numerous genera) primitive carrion beetle 3. (Staphylinids; numerous genera) rove beetle, ant-like stone beetle, crab-like rove beetle

-**RBKV**- ‘STAPHYLINOID BEETLE’ II 1. (Leiodids; numerous genera) round fungus beetle, mammal-nest beetle 2. (Ptiliids; numerous genera) featherwing beetle 3. (Hydraenids; numerous genera) minute moss beetle

-**RBKŢ**- ‘STAPHYLINIFORM BEETLE’ 1. (Hydrophiloids; numerous genera) water scavenger beetle, minute mud-loving beetle 2. (Histerids and Sphaeritids; numerous genera) clown beetle, false clown beetle 3. (Synteliids; genus *Syntelia*) syntelia beetle

Hexapods > Insects > Coleoptera > Polyphaga > Elateriformia

Hexapods > Insects > Coleoptera > Polyphaga > Elateriformia>Elateroidea

-**RBD**- ‘ELATEROID BEETLE’ I 1. (Elaterids; numerous genera) click beetle / snapping beetle / spring beetle / elater / skipjack, false firefly 2. (Eucnemids; numerous genera) false click beetle 3. (Cerophytids; several genera) rare click beetle

-**RBDW**- ‘ELATEROID BEETLE’ II 1. (Cantharids; numerous genera) soldier beetle / leatherwing 2. (Omethids; several genera) false soldier beetles, long-lipped beetle 3. (Lycids; numerous genera) net-winged beetle

-**RBDY**- ‘ELATEROID BEETLE’ III 1. (Lampyrids; numerous genera) firefly / lightning bug 2. (Phengodids, Rhagophthalmids, Osmalisids; numerous genera) glowworm beetle 3. (Throscids; several genera) false metallic wood-boring beetle

-**RBDL**- ‘ELATEROID BEETLE’ IV 1. (Brachypsectrids; genus *Brachypsectra*) Texas beetle 2. (Artematopodids; several genera) soft-bodied plant beetle 3. (Jurasaids, Podabrocephalids, Rhinorhipids; several genera) other elateriform beetle

Hexapods > Insects > Coleoptera > Polyphaga > Elateriformia>Buprestoidea

-**RBDR**- ‘BUPRESTOID BEETLE’ 1. (Buprestids; numerous genera) jewel beetle / metallic wood-boring beetle 2. (Schizopodids; several genera) false jewel beetle 3. [stem unassigned]

Hexapods > Insects > Coleoptera > Polyphaga > Elateriformia>Dascilloidea

-**RBDV**- ‘DASCILLOID BEETLE’ 1. (Dascillids; several genera) soft-bodied plant beetle 2. (Rhipicerids; several genera) cicada parasite beetle 3. [stem unassigned]

Hexapods > Insects > Coleoptera > Polyphaga > Elateriformia>Scirtoidea

-**RBDŘ**- ‘SCIRTOID BEETLE’ 1. (Scirtids; numerous genera) marsh beetle 2. (Clambids; several genera) minute beetle / fringe-winged beetle 3. (Eucinetids and Decliniids; several genera) plate-thigh beetle, declinia beetle

Hexapods > Insects > Coleoptera > Polyphaga > Elateriformia>Byrrhoidea

-**RBC**- ‘BYRRHOID BEETLE’ I 1. (Byrrhids; numerous genera) pill beetle 2. (Callirhipids; several genera) cedar beetle 3. (Chelonariids; several genera) turtle beetle

-**RBČ**- ‘BYRRHOID BEETLE’ II 1. (Limnichids; numerous genera) minute mud beetle / minute marsh-loving beetle 2. (Heterocerids; numerous genera) variegated mud-loving beetle 3. (Psephenids; numerous genera) water-penny beetle

-**RBẔ**- ‘BYRRHOID BEETLE’ III 1. (Eulichadids; several genera) forest stream beetle 2. (Elmids; numerous genera) riffle beetle 3. (Dryopids; numerous genera) long-toed water beetle

-**RBJ**- ‘BYRRHOID BEETLE’ IV 1. (Ptilodactylids; several genera) toe-winged beetle 2. (Lutrochids; genus *Lutrochus*) travertine beetle 3. (Cneoglossids; genus *Cneoglossa*) cneoglossa beetle

Hexapods > Insects > Coleoptera > Polyphaga > Bostrichiformia

-**RBVM**- ‘BOSTRICHIFORM BEETLE’ I 1. (Bostrichids; numerous genera) horned powderpost beetle, false powderpost beetle, auger beetle 2. (Dermestids; numerous genera) carpet beetle, skin beetle, larder beetle, hide beetle / leather beetle, museum beetle, warehouse beetle, cabinet beetle, Khapra beetle / buscuit beetle 3. (Ptinids; numerous genera) death-watch beetle, spider beetle, common furniture beetle

-**RBVN**- ‘BOSTRICHIFORM BEETLE’ II 1. (Endecatomids; genus *Endecatomus*) endecatomus beetle 2. (Derontids; several genera) tooth-necked fungus beetle 3. (Nosedendrids, Jacobsoniids; several genera) wounded-tree beetle, slime flux beetle, Jacobson’s beetle

Hexapods > Insects > Coleoptera > remaining families

-**RBKH**- 'OTHER COLEOPTEROID' 1. (Myxophagans; several genera) myxophagan beetle, skiff beetle 2. (Archostematans; several genera) reticulated beetle, telephone-pole beetle, other archostematid beetle 3. (Strepsiptera; numerous genera) twisted-winged parasite / twisted-wing insect / stylops

Hexapods > Insects > Amphiesmenoptera (butterflies, moths, and caddisflies)

Hexapods > Insects > Amphiesmenoptera > Lepidoptera (butterfly, moth)

-**LSP**- 'PRIMITIVE MOTH' I 1. (Aglossata [Agathiphagids]; genus *Agathiphaga*) kauri moth 2. (Heterobathmiids; genus Heterobathmia) heterobathmiid moth, southern beech moth 3. (Zeugloptera [Micropterigids]; numerous genera) mandibulate archaic moth / small primitive metallic moth

-**LSPW**- 'PRIMITIVE MOTH' II 1. (Dacnonypha [Eriocraniids]; several genera) eriocraniid metallic moth 2. (Acanthopteroctetids; genera *Acanthopteroctetes, Catapterix*) archaic sun moth 3. (Lophocoronids; genus *Lophocorona*) lophocorona moth

-**LSPY**- 'PRIMITIVE MOTH' III 1. (Neopseustids; several genera) archaic bell moth 2. (Aenigmatineids; genus *Aenigmatinea*) enigma moth 3. (Exoporia; several genera) swift moth, ghost moth, New Zealand primitive moth

Hexapods > Insects > Amphiesmenoptera > Lepidoptera > Monotrysia (monotrysiic moths)

-**LSPL**- 'ADELOID MOTH' 1. (Adelids; several genera) fairy longhorn moth 2. (Prodoxids; several genera) prodoxid moth, currant shoot borer, yucca moth, bogus yucca moth 3. (Cecidosids, Heliozelids,Incurvariids; numerous genera) adeloid leaf-miner moth

-**LSPR**- 'NEPTICULOID MOTH' 1. (Nepticulines; several genera) pigmy moth / midget moth 2. (Pectinvalvines; genera Pectinivalva, Roscidotoga) Australian pigmy/midget moth 3. (Opostegids; several genera) white eyecap moth

-**LSPŘ**- 'MONOTRYSIAN MOTH' 1. (Andesianids; genus *Andesiana*) Andean endemic moth 2. (Palaephatids; several genera) Gondwanaland moth 3. (Tischerioids; several genera) trumpet leaf miner moth

Hexapods > Insects > Amphiesmenoptera > Lepidoptera > Dytrysia

-**LSPF**- 'TINEOID MOTH' 1. (Tineids; numerous genera) clothes moth, carpet moth, fungus moth, tineid moth 2. (Psychids; numerous genera) bagworm moth / bagworm / bagmoth / case moth 3. (Acrolophids; several genera) burrowing webworm moth

-**LSPV**- 'OTHER TINEOID MOTH & MILLIERIID MOTH' 1. (Eriocottids; several genera) Old World spiny-winged moth 2. (Arrhenophanids; several genera) arrhenophanid moth 3. (Millieriids; genera *Millieria, Phormoestes, Nyx*) millieriid moth

-**LSPĻ**- 'GRACILLARIOID MOTH' 1. (Bucculatricids; genus *Buculatrix*) ribbed cocoon maker / bucculatricid moth 2. (genus *Ogmograptis*) scribbly gum moth 3. (Gracillariids and Roeslerstammids; numerous genera) gracillarioid moth

-**LSPÇ**- 'YPONOMEUTOID MOTH' I 1. (Yponomeutids; numerous genera) ermine moth, 2. (Argyresthiids; several genera) shiny head-standing moth 3. (Praydids; several genera) false ermine moth

-**LSPH**- 'YPONOMEUTOID MOTH' II 1. (Plutellids; numerous genera) diamondback moth 2. (Acrolepiids; several genera) false diamondback moth 3. (Glyphipterigids; numerous genera) sedge moth

-**LSPÇW**- 'YPONOMEUTOID MOTH' III 1. (Heliodinids; numerous genera) sun moth 2. (Bedelliids; genus Bedellia) bedelliid moth, sweet potato leaf miner moth 3. (Lyonetiids; numerous genera) lyonetiid moth

-**LSPHW**- 'YPONOMEUTOID MOTH' IV 1. (Ypsolophids seveal genera) ypsolophid moth 2. (Attevids; genus Atteva) atteva moth 3. [stem unassigned]

Hexapods > Insects > Amphiesmenoptera > Lepidoptera > Apoditrysia

-**LFP**- 'PTEROPHORID & ALUCITOID MOTH' 1. (Pterophorids; numerous genera) plume moth 2. (Alucitids; several genera) many-plumed moth 3. (Tineodids; numerous genera) false plume moth

-**LFPW**- 'TORTRICID (a.k.a. TORTRIX or LEAFROLLER) MOTH' I 1. (Tortricids; numerous genera) leafroller moth / tortrix moth 2. ((genus *Cydia*) cydia tortrix, rusty oak moth, slash pine seedworm moth, hickory shuckworm moth, filbertworm moth, tea flush worm moth, alfalfa moth, larch gall moth, pea moth, jumping bean moth, spruce bark tortrix, ponderosa pine seedworm moth, codling moth, pear fruit moth / pear fruit tortricid, chestnut tortrix / acorn moth, spruce seed moth, beech moth, longleaf seedworm moth, nut fruit tortrix, larch bark moth, eastern pine seedworm moth 3. (genus *Grapholita*) grapholita tortrix, hemp moth / hemp borer, oriental fruit moth, cherry fruitworm moth, clover seed moth, hawthorn leafroller, peach moth, appleseed moth / small fruit tortrix, three-lined grapholita moth, grapholitha roseticolana / deep-brown piercer / rose-hip tortricid / briar moth

-**LFPY**- 'TORTRICID (a.k.a. TORTRIX or LEAFROLLER) MOTH' II 1. (genus *Archips*) fruit-tree leafroller moth, ugly-nest caterpillar moth, brown oak tortrix, oak webworm moth, baldcypress leafroller, black shield leafroller / gray archips moth, larger boxelder leafroller, large fruit tree tortrix, rose tortrix / rose leaf roller, oak leafroller, apple leafroller, variegated golden tortrix, striated tortrix moth / striated leafroller, southern ugly-nest caterpillar moth, spring spruce needle moth / spruce needleworm moth, omnivorous leafroller, dusky-back leafroller, smoked leafroller / hickory webworm moth, Georgia archips moth, exotic leafroller moth / apple tortrix, boldly-marked archips moth, Asiatic leafroller, groundnut leafroller 2. (genus *Tortrix*) green oak tortrix / green oak moth / European oak leafroller, Japanese oak leafroller 3. (genus *Sparganothis*) sparganothis fruitworm moth / blueberry leafroller, aproned sparganothis / beautiful sparganothis, aproned ceopis moth, white-tailed fruitworm moth, distinct sparganothis moth, chokecherry leafroller moth, spring dead-leaf roller, gray sparganothis moth, three-streaked sparganothis moth, one-lined sparganothis moth, mosaic sparganothis moth, long-palped tortrix / vine leaf roller

-**LFPL**- 'TORTRICID (a.k.a. TORTRIX or LEAFROLLER) MOTH' III 1. (genus *Choristoneura*) broken-banded leafroller /dark-banded fireworm moth, large aspen tortrix, two-year cycle budworm moth, spruce budworm moth, western spruce budworm moth, eastern spruce budworm moth, mountain-ash tortricid, strawberry leafroller, sugar pine tortrix, parallel-banded leafroller moth, jack pine budworm moth, oblique banded leaf roller / rosaceous leaf roller, zapulata moth 2. (genus *Argyrotaenia*) white-spotted leafroller, pondapple leafroller moth, orange tortrix / apple skinworm moth, Ivana leafroller moth, hickory leafroller moth, Kimball's leafroller moth, fall spruce needle moth, gray-banded leafroller moth, pine tube moth, yellow-winged oak leafroller moth, four-lined leafroller / four-banded leafroller / lesser all-green leafroller, jack pine tube moth / lodgpole needletier moth, red-banded leafroller moth 3. (genus *Pandemis*) chequered fruit-tree tortrix / hazel tortix moth / filbert tortricid / barred fruit tree moth, barred fruit-tree tortrix, green aspen leaftier, dark fruit-tree tortrix / apple brown tortrix, woodgrain leafroller moth, three-lined leafroller, apple pandemis / pandemis leafroller moth

-**LFPR**- 'TORTRICID (a.k.a. TORTRIX or LEAFROLLER) MOTH' IV 1. (genus *Spilonata*) bud moth, white fruit moth / larger apple fruit moth / eye-spotted bud moth, larch leafroller, apple fruit licker 2. (genus *Thaumatotibia*) cacao husk borer, false codling moth / orange moth / citrus codling moth / orange codling moth 3. (genus *Spilonata*) white fruit moth / larger apple fruit moth / eye-spotted bud moth, larch leafroller, apple fruit licker, bud moth

-**LFPŘ**- 'TORTRICID (a.k.a. TORTRIX or LEAFROLLER) MOTH' V 1. (genus *Adoxophyes*) appleleaf-curling moth, summer fruit tortrix moth, shimmering adoxophyes moth, bell moth / orange tip moth 2. (genus *Lobesia*) European grapevine moth 3. (genus *Paralobesia*) tulip-tree leaftier moth, grape berry moth

-**LFPF**- 'TORTRICID (a.k.a. TORTRIX or LEAFROLLER) MOTH' VI 1. (genus *Taniva*) spruce needleminer moth 2. (genus *Synnoma*) rabbitbrush webbing moth 3. (genus *Pammene*) chestnut leafroller, pygmy piercer moth

-**LFPV**- 'COSSOID MOTH' 1. (Cossids, Dudgeoneids, Metarbelids; numerous genera) carpenter moth / carpenter miller / cossid miller / goat moth / wood moth 2. (genus *Coryphodema*) quince borer / sad goat moth / apple-trunk borer moth 3. (genus *Prionoxystus*) carpenterworm moth / locust borer moth

-**LFPŢ**- 'SESSIOID MOTH' 1. (Brachodids; numerous genera) little bear moth 3. (Castniids; numerous genera) giant butterfly-moth, sun moth 3. (Sessiids; numerous genera) clearwing moth

-**LFPS**- 'ZYGAENOID MOTH' I 1. (Zygaenids; numerous genera) burnet moth, forester moth, smoky moth, skeltonizer moth, vine bud moth 2. (Megalopygids; numerous genera) flannel moth / crinkled flannel moth 3. (Epipyropids and Cyclotornids; numerous genera) planthopper parasite moth

-**LFPŠ**- 'ZYGAENOID MOTH' II 1. (Dalcerids, Limacodids; numerous genera) slug moth / cup moth, jewel caterpillar moth 2. (Heterogynids, Lacturids, Phaudids, Aidids, Anomoeotids, Somabrachyids, Himantopterids; numerous genera) other zygaenoid moth 3. [stem unassigned]

-**LFPĻ**- ‘GELECHIOID MOTH’ I 1. (Gelechioids other than Gelechiids; numerous genera) gelechioid moth / curved-horn moth 2. (Coleophorids; numerous genera) case-bearer moth / casebearing moth / case moth 3. (Cosmopterigids; numerous genera) cosmet moth

-**LFPĻW**- ‘GELECHIOID MOTH’ II 1. (Elachistids; numerous genera) grass-miner moth 2. (Argonoxenines and Blastodacnids; numerous genera) palm moth 3. (Lecithocerids; numerous genera) long-horned moth

-**LFPH**- ‘GELECHIOID MOTH’ III 1. (Momphids, numerous genera) mompha moth 2. (Oecophorids; numerous genera) concealer moth 3. (Schistonoeids; genus Schistonoea) scavenger moth

-**LFPHW**- ‘GELECHIOID MOTH’ IV 1. (Scythridids; numerous genera) flower moth 2. (Xyloryctids; numerous genera) hermit moth / timber moth 3. (Austostichids, Batrachedrids, Blastobasids, Chimabachids, Pterolonchids, Epimarptids, Lypusids, Depressariids, Peleopodids, Stathmopodids, Syringopaids; numerous genera) other gelechioid moth

-**LFPŠW**- ‘GELECHIID (TWIRLER) MOTH’ 1. (Gelechiids; numerous genera) twirler moth 2. (genera *Anarsia, Aproaerema, Keiferia, Phthorimaea, Sitotraga, Tecia, Tuta*) gelechiid twig, groundnut, tuber, grain, or tomato pest 3. (genus *Pectinophora*) pink bollworm moth

-**LFPÇ**- ‘APODITRYSIAN MOTH’ I 1. (Carposinids; numerous genera) fruitworm moth 2. (Copromorphids; numerous genera) tropical fruitworm moth 3. (Immids; numerous genera) immid moth

-**LFPÇW**- ‘APODITRYSIAN MOTH’ II 1. (Epermeniids; numerous genera) fringe-tufted moth 2. (Schreckensteiniids; several genera) bristle-legged moth 3. (Simaethistids; genera Simaethistis, Metaprotus) simaethistid moth

-**LFPSW**- ‘APODITRYSIAN MOTH’ III 1. (Choreutids; numerous genera) metalmark moth 2. (Urodids; several genera) false burnet moth 3. (Galacticids; several genera) galacticid moth

-**LFPSY**- ‘APODITRYSIAN MOTH’ IV 1. (Prodidactid; genus *Prodidactis*) prodidactis moth 2. (Douglasiids; genera *Tinagma, Klimeschia, Protonyctia*) Douglas moth 3. [stem unassigned]

Hexapods > Insects > Amphiesmenoptera > Lepidoptera > Obtectomera

-**LKW**- ‘PAPILIONID (SWALLOWTAIL) BUTTERFLY’ I 1. (Parnassiines; several genera) snow Apollo, mountain Apollo, false Apollo, souther festoon, eastern festoon, Spanish festoon, Bhutan glory, luehdorfina 2. (Papilionines; numerous genera) swallowtail, swordtail, kite swallowtail, bluebottle, triangle, jay, graphium, white-lady, zebra, dragontail, gorgon, Kaiser-i-hind, golden Kaiser-i-hind, batwing, windmill, clubtail, rose, clearwing swallowtail / big greasy, birdwing, cattleheart 3.(Baronines; genus Baronia) short-horned baronia

-**LKY**- ‘HESPERIID (SKIPPER) BUTTERFLY’ I 1. (Coeliadines; several genera) awl, awlet, policeman, African giant skipper 2.(Euschemonines; genus Euschemon) regent skipper 3. (Eudamines; numerous genera) dicot skipper

-**LKL**- ‘HESPERIID (SKIPPER) BUTTERFLY’ II 1. (Pyrigines; numerous genera) spread-winged skipper, firetail skipper 2. (Heteropterines; numerous genera) skipperling 3. (Hesperines; numerous genera) grass skipper / banded skipper

-**LKR**- ‘HESPERIID (SKIPPER) BUTTERFLY’ III 1. (Megathymines; several genera) giant skipper 2. (Trapezitines; numerous genera) Australian/New Guinea skipper 3. [stem unassigned]

-**LKŘ**- ‘PIERID BUTTERFLY’ I 1. (Pierines; numerous genera) white butterfly, orange tip, marble, yellow tip, black-tip, sulphur, sawtooth, dotted border, Jezebel, gull, puffin, albatross, blackvein, caper white, Arab, vagrant 2. (Coliadines; numerous genera) yellow butterfly, dogface, sulphur, emigrant 3. (Dismorphiines and Pseudopontiines; several genera) mimic sulphur, mimic white butterfly, pseudopontia

-**LKÇ**- ‘LYCAENID (GOSSAMER-WINGED) BUTTERFLY’ I 1. (Lycaenines; numerous genera) copper butterfly, sapphire butterfly 2. (Curetines; genus *Curetis*) sunbeam butterfly 3. (Polyommatines; numerous genera) blue, hairtail, Pierrot, argus, cerulean, azure, tailed-blue, cupid butterfly

-**LKF**- ‘LYCAENID (GOSSAMER-WINGED) BUTTERFLY’ II 1. (Miletines; numerous genera) harvester, wooly legs, moth butterfly 2. (Theclines; numerous genera) hairstreak, elfin, posy, imperial, oakblue, Atala 3. (Aphnaeines; numerous genera) opal, Roger’s gem, silver spot, silver-spotted copper, highflyer, scarce scarlet / golden flash, daisy copper, king copper

-**LKŢ**- ‘RIODINID & OTHER LYCAENID BUTTERFLY’ 1. (Riodinids; numerous genera) metalmark butterfly, Duke of Burgundy, orange harlequin, columbine, punch, punchinello 2. (Poritiines; numerous genera) poritiine, zulu, pentila, buff, glasswing / white mimic, tiger mimic, acraea mimic, harlequin, dot, acraea blue, epitola 3. [stem unassigned]

-**LKM**- ‘NYMPHALID (FOUR-FOOTED / BRUSH-FOOTED) BUTTERFLY’ I 1. (Nymphalines; numerous genera) tortoiseshell, admiral, mapwing, anglewing, comma, jester, painted lady, tiger beauty, pirate, eggfly, diadem, oakleaf, leaf, malachite, peacock, Fatima, pansy, commodore, blue beauty, buckeye, meadow argus, northern argus, mother-of-pearl butterfly 2. (Melitaeines; numerous genera) checkerspot, patch, crescent / crescentspot, fritillary, European peacock butterfly 3. (other Nymphalids; genera *Rhinopalpa, Kallimoides, Vanessula*) wizard, African leaf, black and orange / lady’s maid butterfly

-**LKN**- ‘NYMPHALID (FOUR-FOOTED / BRUSH-FOOTED) BUTTERFLY’ II 1. (Cyrestines; genera *Cyrestis, Chersonesia, Marpesia*) map, maplet, daggerwing butterfly 2. (Biblidines; numerous genera) tropical brushfoot, eighty-eight, numberwing, Pericloud, sailor, greenwing, asterope, glory, banner, cracker, calico, beauty, red flasher, firewing, catone, shoemaker, olivewing, bluewing, tree nymph, purplewing, red rim / crimson-banded black, castor, piper, sailor, Drury’s delight, joker, mestra, azeca banner 3. (Apaturines; numerous genera) emperor, empress, hackberry, courtesan, pasha, sorcerer, circe, siren, prince, courtier

-**LKV**- ‘NYMPHALID (FOUR-FOOTED / BRUSH-FOOTED) BUTTERFLY’ III 1. (Heliconiines; numerous generera) acraea, legionnaire, tawny coster, actinote, altinote, bematistes, lacewing, gulf fritillary / passion butterfly, longwing, postman, banded orange / orange tiger, Julia / flame / flambeau, Juliette, scarce bamboo page / longwing dido, leopard, spotted rustic, blotched leopard, cruiser, yeoman, royal Assyrian, rustic, vagrant, fritillary, polka dot 2. (Limenitidines; numerous genera) clipper, marquise, duke, archduke, count, earl, viscount, forest-glade nymph, pathfinder, brilliant nymph, forester, figeater, commander, guineafowl, false fritillary, false sergeant, sister, admiral, commodore, lascar, aeroplane, glider, knight, false acraea, blue sailer / blue sergeant 3. (Libytheines; genera Libythea, Libytheana) beak / snout butterfly

-**LKB**- ‘NYMPHALID (FOUR-FOOTED / BRUSH-FOOTED) BUTTERFLY’ IV 1. (Danaini; several genera) cleric, monarch, queen, tiger, glassy tiger, tree-nymph, wood-nymph, Schneider’s surprise, blue tiger, crow, paper, mimic queen, magpie butterfly 2. (Ithomiines; numerous genera) clearwing, tigerwing, clapping ticlear, glasswing butterfly 3. (Tellervini; genus *Tellervo*) tellervo butterfly

-**LKH**- ‘NYMPHALID (FOUR-FOOTED / BRUSH-FOOTED) BUTTERFLY’ V 1. (Calinagines; genus *Calinaga*) freak butterfly 2. (Charaxines; numerous genera) leafwing, nawab, rajah, pasha, begum, palla, forest queen, prepona, Nemesis 3. (Morphines; numerous genera) dryad, palmking, koh-i-noor, duffer, caliph jungle queen, jungleking, jungle glory, owl, giant owl, morpho

-**LKŠ**- ‘SATYRID BUTTERFLY’ 1. (Satyrini; numerous genera) brown, satyrid, satyr, grayling, ringlet, false grayling, wood-nymph, meadowbrown, xenica, alpine, widow, heath 2. (Elymniini; numerous genera) palmfly, diqadem, white owl, pearly-eye, wall, wallbrown, treebrown, woodbrown, bushbrown, labyrinth, goldenfork, silverstripe, lilacfork, silverfork, mystic 3. (Melanitini, Eritini, Haeterini, Ragadiini; numerous genera) mountain pride / Table Mountain beauty, bush beauty / forest pride, brown

-**LKHW**- ‘BUTTERFLY- MOTH / MOTH-BUTTERFLY’ 1. (Callidulids; several genera) Old World butterfly-moth 2. (Hedylids; genus *Macrosoma*) American moth-butterfly 3. [stem unassigned]

-**LKÇW**- OBTECTOMERAN MOTH’ I 1. (Thyridids; numerous genera) picture-winged leaf moth 2. (Hyblaeids; genera *Hyblaea, Erythrochrus*) teak moth 3. (Whalleyanids; genus *Whalleyana*) whalleyana moth

-**LKŠW**- ‘OBTECTOMERAN MOTH’ II 1. (Crambids; numerous genera) grass moth, European corn boere / European corn worm moth / European high-flyer, sod grass webworm moth, rice stem borer 2. (Pyralids; numerous genera) snout moth / pyralid moth, Indian meal moth / weevil moth / pantry moth / flour moth / grain moth, coconut spike moth / oil palm bunch moth, cacao moth / tobacco moth / warehouse moth, Mediterranean flour moth / mill moth, wax moth, rice moth 3. (Mimallonids; numerous genera) mimallonid moth / sack-bearer moth

Hexapods > Insects > Amphiesmenoptera > Lepidoptera > Macroheterocera

-**ŘZB**- ‘BOMBYCOID MOTH’ I 1. (Bombycids; numerousgenera) silk moth, emperor moth, sphinx moth, silkworm moth, Brahmin moth, true silkmoth / mulberry silkmoth 2. (Drepanoids; numerous genera) hook-tip moth, false owlet, gold moth, doid moth, drepanid moth 3. (Lasiocampids; numerous genera) eggar, snout moth, lappet moth, tent caterpillar moth

-**ŘZBW**- 'BOMBYCOID MOTH' II 1. (Saturnids; numerous genera) oakworm moth, regal moth, pine-devil moth, splendid royal moth, rosy maple moth, imperial moth, io moth, buck moth, emperor moth, moon moth, tussar moth, comet moth, Polyphemus moth, mopane moth, cecropia math, silkmoth, Hercules moth, Atlas moth 2. (Sphingids; numerous genera) sphinx moth, hawk- moth 3. (Apatelodids; numerous genera) American silkworm moth, shag-carpet caterpillar moth

-**ŘZBY**- 'BOMBYCOID MOTH' III 1. (Eupterotids, Phiditiids, Endromids, Carthaeids; numerous genera) bombycoid moth, dryandra moth 2. (Brahmaeids; several genera) brahmin moth, owl moth 3. (Anthelids; several genera) anthelid moth, rose anthelid

-**ŘZBL**- 'GEOMETRID (GEOMETER) MOTH' 1. (Geometrids; numerous genera) geometer, spinach, phoenix, barred straw, carpet, delicate, rivulet, seraphim, pug, oblique striped, mallow, chimney sweeeper, vestal, blood-vein, zachera, wave, lace border, scallop, mocha, maiden's blush, emerald, diminutive, purple-barred yellow moth 2. (Epicopeia; several genera) oriental swallowtail moth 3. (Sematurids and Pseudobistonids, several genera) sematurid moth, pseudobistonid moth

-**ŘZBR**- 'URANIID MOTH' 1. (Uraniines; numerous genera) urania, green page, tropical swallowtail, Laos brown butterfly, sunset moth, uraniine moth 2. (Epiplemines; numerous genera) scoopwing moth 3. (Auzeines, Microniines; several genera) auzeine moth, microniine moth

-**ŘZV**- 'EREBID MOTH' I 1. (Erebines; numerous genera) underwing moth, witch moth 2. (Arctiines; numerous genera) tiger, lichen, footman, wasp moth 3. (Calpinesand Scoliopterygines; several genera) fruit-piercing, fruit-sucking, piercing moth, cotton leaf worm moth

-**ŘZVW**- 'EREBID MOTH' II 1. (Herminiines; numerous genera) litter moth 2. (Hypenines; numerous genera) snout moth, bomolocha, hypena, hop looper / hop vine, lantana defoliator, cloverworm moth 3. (Lymantriines; numerous genera) tussock moth, gypsy moth

-**ŘZVY**- 'EREBID MOTH' III 1. (Aganaines; several genera) aganaine, tiger moth 2. (Scolecocampines; several genera) scolecocampine, algibelle, sigela, dead-wood borer moth 3. (Anobines, Boletobiines, Eulepidotines, Hypcalines, Pangraptines, Rivulines, Tinoliines, Toxocampines; numerous genera) other erebid moth

-**ŘZVL**- 'NOCTUID (OWLET) MOTH' 1. (Noctuines; numerous genera) owlet moth, cutworm moth, dart, clay, square-spot, Hebrew character, rustic 2. (Acronictines, Agaristines, Amphipyrines, Bagisarids, Bryophilines, Condicines; numerous genera) dagger, sycamore, marsh, slowpoke, Canadian giant, rusty shoulder knot moth, amphipyra, copper underwing, mouse moth, mountain forester, day flying moth, bagisara, marbled green, marbled gray, red groundling / pied groundling moth 3. (Cuculliines, Dilobines, Eustrotiines, Hadenines, Heliothines, Pantheines, Plusiines, Raphiines; numerous genera) star-wort, woodworm moth, asteroid, merveille du jour, sallow, catabena, toadflax, brocade, figure of eight, armyworm moth, cabbage moth, arch, zebra caterpillar moth, glaucous shears, feathered ear, white-point, wainscot, bright-eye, double line, white-speck, drab, Quaker, corn earworm moth, bollworm moth, pease blossum, yellowhorn, looper, burnished brass, pitcher plant, alpine beauty, silver Y, Bröther moth

-**ŘZVR**- 'NOCTUOID (OWLET) MOTH' 1. (Notodontids; numerous genera) prominent, puss moth, buff-tip, lobster, poplar kitten, California oakworm, beech caterpillar, oakleaf caterpillar, yellownecked caterpillar, walnut caterpillar 2. (Nolids; numerous genera) nolid, nola, hieroglyphic moth, green pea moth 3. (Oenosandrids, Doids, Thaumetopoeids; numerous genera) stub, Boisduval's autumn, epicoma, black spot, long-tailed bombyx / Sparshall's moth, notodontid moth

Hexapods > Insects > Amphiesmenoptera > Trichoptera

-**ŘZŇ**- 'TRICHOPTERAN (CADDISFLY / SEDGE-FLY / RAIL-FLY)' 1. (Annulipalpians; numerous genera) net-spinning caddisfly, tube-making caddisfly, trumpet-net caddisfly, finger-net caddisfly 2. (Integripalpians; numerous genera) portable case-maker caddisfly, snail-case caddisfly, stonecase caddisfly, tubecase caddisfly 3. (Spicipalpians; numerous genera) spicipalpian, purse-case caddisfly / microcaddisfly

Hexapods > Insects > Antliophora

Hexapods > Insects > Antliophora > Siphonaptera (flea)

-**ŘZŇW**- 'SIPHONAPTERAN (FLEA)' 1. (Pulicids; numerous genera) common flea / cat flea 2. (Pulicioids other than Pulicids; numerous genera) pulicioid flea 3. (Hectopsyllids; genera *Tunga, Hectopsylla*) jigger / chigoe flea, hectopsyllid flea

Hexapods > Insects > Antliophora > Mecoptera (scorpionflies)

-**ŘJŇ**- 'MECOPTERAN (SCORPIONFLY)' I 1. (Panorpids; several genera) common scorpionfly 2. (Panorpodids; genera *Panorpodes, Brachypanorpa*) short-faced scorpionfly 3. (Boreids; several genera) snow scorpionfly

-**ŘJŇW**- 'MECOPTERAN (SCORPIONFLY)' II 1. (Bittacids; numerous genera) hangingfly / hanging scorpionfly 2. (Meropeids; genera *Merope, Austromerope*) earwigfly / forcepfly 3. (Eomeropids, Apteropanorpids, Choristids, Nannochoristids; several genera) notiothauma scorpionfly, apteropanorpa scorpionfly, choristid scorpionfly, nannochoristid scorpionfly

Hexapods > Insects > Antliophora > Diptera (true fly)

Hexapods > Insects > Antliophora > Diptera > Nematocera > Ptychopteromorpha (phantom and primitive crane-flies)

-**LZGV**- 'TUPILOMORPHIC FLY (CRANE FLY)' I 1. (Tanyderids; several genera) primitive crane fly 2. (Ptychopterids; several genera) phantom crane fly 3. (Trichocerids; three genera) winter crane fly, winter gnat

-**LZGḌ**- 'TUPILOMORPHIC FLY (CRANE FLY)' I I 1. (Tipulids and Limoniines; numerous genra) crane fly, limoniid crane fly, snow fly 2. (Cylindrotomids; several genera) long-bodied cranefly 3. (Pediciids; several genera) hairy-eyed cranefly

Hexapods > Insects > Antliophora > Diptera > Nematocera > Culicomorpha (mosquitoes and biting midges)

-**LZG**- 'CULICOMORPH' I 1. (Culicids; numerous genera) mosquito 2. (Dixids; numerous genera) meniscus midge 3. (Chaoborids; numerous genera) phantom midge / glassworm midge

-**LZGW**- 'CULICOMORPH' II 1. (Simuliids; numerous genera) black fly / buffalo gnat / turkey gnat / white socks 2. (Ceratopogonids; numerous genera) biting midge / no-see-um 3. (Corethrellids; two genera) frog-biting midge

-**LZGY**- 'CULICOMORPH' III 1. (Chironomids; numerous genera) nonbiting midge / lake fly, harlequin fly 2. (Thaumaleids; several genera) solitary midge / trickle midge 3. [stem unassigned]

Hexapods > Insects > Antliophora > Diptera > Nematocera > Blephariceromorpha (net-winged midges, etc.)

-**LZGŘ**- 'BLEPHARICEROMORPH' 1. (Blepharicerids; several genera) net-winged midge, torrent midge 2. (Deuterophlebiomorph; genus Deuterophlebia) mountain midge 3. (Nymphomyiids; genus *Nymphomyia*) nymphomyiid fly / nymphomyiid midge

Hexapods > Insects > Antliophora > Diptera > Nematocera > Bibionomorpha (march fly, midges, gnats)

-**LZGL**- 'BIBIONOMORPH' I 1. (Bibionids; several genera) march fly / St. Mark's fly, love bug / honeymoon fly / double-headed bug 2. (Anisopodids; several genera) wood gnat / window-gnat 3. (Cedidomyiids; numerous genera) gall midge / gall gnat, wood midge

-**LZGR**- 'BIBIONOMORPH' II 1. (Mycetophilids; numerous genera) fungus gnat 2. (Lygistorrhinids; several genera) long-beaked fungus gnat 3. (Sciarids; numerous genera) dark-winged fungus gnat

-**LZGZ**- 'BIBIONOMORPH' III 1. (Rangomaramids; several genera) long-winged fungus gnat 2. (Keroplatids; numerous genera) predatory fungus gnat 3. (Ditomyids, Diadocidiids, Bolitophilids, Hesperinids; several genera) sciaroid gnat, bibionomorphic gnat

-**LZGŽ**- 'OTHER NEMATOCERAN FLY' '1. (Pachyneurids; several genera) pachyneurid gnat, pachyneurid fly, pachyneurid midge 2. (Axymyiids; three genera) axymyiid fly 3. [stem unassigned]

Hexapods > Insects > Antliophora > Diptera > Nematocera > Psychodomorpha

-**LZBḌ**- 'PSYCHODOMORPH' (Psychodids; numerous genera) drain fly / sink fly / filter fly / sewer gnat / moth fly 2. (Scatopsids; numerous genera) dung midge / minute black scavenger fly 3. (Cnathyloscelids and Perissommatids; several genera) other scatopsoid fly

Hexapods > Insects > Antliophora > Diptera > Brachycera > Tabanomorpha, Xylophagomorpha, and Stratiomyomorpha

-**LZB**- 'TABANID FLY' 1. (Tabanines; numerous genera) horse-fly, cleg 2. (Chrysopsines; numerous genera) deer fly / banded horse-fly 3. (Pangoniines; numerous genera) long-tongued horse-fly

-**LZBW**- 'TABANOMORPHOID FLY' 1. (Rhagionids, Pelecorhychids, Bolbomyiids, Austroleptids, Oreoleptids; numerous genera) snipe fly 2. (Athericids; numerous genera) ibis fly / water snipe fly 3. (Xylophagids; several genera) awl-fly

-**LZBY**- 'STRATIOMORPHIC FLY' 1. (Stratiomyids; numerous genera) soldier fly 2. (Xylomyids; several genera) wood soldier fly 3. (Pantophthalmids; 2 genera) timber fly

Hexapods > Insects > Antliophora > Diptera > Brachycera >Asilomorpha > Asiloidea

-**LZBL**- 'ASILOID FLY' I 1. (Asilids; numerous genera) robber fly / assassin fly 2. (Bombyliids; numerous genera) bee fly / humblefly 3. (Mydids and Apiocerids; numerous genera) Mydas fly, flower-loving fly

-**LZBR**- 'ASILOID FLY' II 1. (Therevids; numerous genera) stiletto fly 2. (Scenopinids; numerous genera) window fly 3. (Apsilocephalids, Evocoids; several genera) other theravoid fly

-**LZBŘ**- 'ASILOID FLY' III 1. (Mythicomyiids; numerous genera) mythicomyiid fly 2. (Hilarimorphids; genus *Hilarimorpha*) hilarimorphid fly 3. (Apystromyiids; genus Apystomyia) apystomyiid fly

Hexapods > Insects > Antliophora > Diptera > Brachycera >Asilomorpha >Empidoidea

-**LZBZ**- 'EMPIDOID FLY' I 1. (Empidids; numerous genera) dagger fly / balloon fly 2. (Hybotids; numerous genera) dance fly 3. (Dolichopodids; numerous genera) long-legged fly

-**LZBŽ**- 'EMPIDOID FLY FLY' II 1. (Ragadids; seveal genera) ragadid fly 2. (Atelestids; several genera) atelestid fly 3. [stem unassigned]

Hexapods > Insects > Antliophora > Diptera > Brachycera >Asilomorpha >Nemistrinoidea & Vermileonomorpha

-**LZBV**- 'NEMESTRINOID & VERMILEONID FLY' 1. (Acrocerids; numerous genera) small-headed fly / hunch-back fly / spider fly 2. (Nemestrinids; numerous genera) tangle-veined fly 3. (Vermileonids; several genera) wormlion fly

Hexapods > Insects > Antliophora > Diptera > Brachycera > Muscomorpha

Hexapods > Insects > Antliophora > Diptera > Brachycera > Muscomorpha > Aschiza

-**LZKW**- 'PLATYPEZOID FLY' 1. (Phorids; numerous genera) coffin fly, scuttle fly, ant-decapitating fly 2. (Lonchopterids; several genera) spear-winged fly / pointed-wing fly 3. (Platypezids and Opetiids; numerous genera) flat-footed fly

-**LZKY**- 'SYRPHOID & OTHER ASCHIZATE FLY' 1. (Syrphids; numerous genera) hoverfly / flower fly / syrphid fly, drone fly / "H-bee" 2. (Pipunculids; numerousgenera) big-headed fly 3. (Ironomyiids; genus *Ironomyia*) ironic fly

Hexapods > Insects > Antliophora > Diptera > Brachycera > Muscomorpha> Schizophora

Hexapods > Insects > Antliophora > Diptera > Brachycera > Muscomorpha> Schizophora > Acalyptrata

-**LZK**- 'TEPHRITOID FLY' I 1. (Tephritids; numerous genera) [tephritid] fruit fly / small fruit fly / peacock fly 2. (Platystomatids; numerous genera) signal fly 3. (Ulidiids; numerous genera) picture-winged fly

-**LZKL**- 'TEPHRITOID FLY' II 1. (Pallopterids; numerous genera) flutter-wing fly / trembling-wing fly / waving-wing fly 2. (Piophilids; numerous genera) cheese fly 3. (Pyrgotids, Ctenostylids, Richardiids; numerous genera) other tephritoid fly

-**LZKR**- 'NERIOID FLY' 1. (Neriids; numerous genera) cactus fly, banana stalk fly 2. (Micropezids; numerous genera) stilt-legged fly 3. (Cypselosomatids, several genera) cypselosomatid fly

-**LZKŘ**- 'DIOPSOID FLY' 1. (Diopsids; several genera) stalk-eyed fly 2. (Psilids; several genera) rust fly, carrot fly 3. (Gobryids, Megamerinids, Nothybids, Somatiids, Stongylophthalmyiids, Syringogastrids, Tanypezids; numerous genera) other diopsoid fly, stretched-foot fly

-**LZKF**- 'SCIOMYZOID FLY' I 1. (Sciomyzids; numerous genera) marsh fly, snail-killing fly 2. (Sepsids; numerous genera) black scavenger fly / ensign fly 3. (Coelopids; several genera) kelp fly / seaweed fly

-**LZKV**- 'SCIOMYZOID FLY' II 1. (Heterocheilids; genus *Heterochela*) half-bridge fly 2. (Dryomyzids, Helcomyzids, Helosciomyzids, Ropalomerids; numerous genera) other sciomyzoid fly

-**LZKM**- 'SPHAEROCEROID FLY' 1. (Sphaerocerids; numerousgenera) small dung fly / lesser dung fly / lesser corpse fly 2. (Chyromyids, Heleomyzids, Nannodastiids; numerous genera) other sphaeroceroid fly 3. [stem unassigned]

-**LZKÇ**- 'LAUXANIOID FLY' 1. (Chamaemyiids; numerous genera) aphid fly 2. (Celyphids; several genera) beetle fly / beetle-backed fly 3. (Lauxaniids, Cremifaniids; numerous genera) other lauxanioid fly

-**LZKH**- 'OPOMYZOID FLY' I 1. (Agromyzids; numerous genera) leaf-miner fly 2. (Aulacigastrids; several genera) sap fly 3. (Clusiids; numerous genera) druid fly / lekking fly

-**LZKHW**- 'OPOMYZOID FLY' II 1. (Opomyzids; several genera) opomyzid fly, cereal fly, grass fly 2. (Neurochaetids; two genera) upside-down fly 3. (Anthomyzids, Asteiids, Odiniids, Periscelidids, Teratomyzids, Xenasteiids; numerous genera) other opomyzoid fly

-**LZKFW**- 'EPHYDROID FLY' I 1. (Ephydrids; numerous genera) shore fly, brine fly 2. (Diastatids; several genera) bog fly 3. (Drosophilids; numerous genera) fruit fly / vinegar fly

-**LZKFY**- 'EPHYDROID FLY' II 1. (Curtonotids; several genera) quasimodo fly 2. (Camillids; several genera) camillid fly 3. (Braulids; two genera) bee louse

-**LZKFL**- 'CARNOID FLY' I 1. (Carnids; several genera) bird fly / filth fly 2. (Canacids; numerous genera) beach fly / surf fly / surge fly 3. (Milichiids; numerous genera) freloader fly / jackal fly

-**LZKFR**- 'CARNOID FLY' II 1. (Chloropids; numerous genera) frit fly / grass fly, eye gnat / eye fly 2. (Inbiomyiids; genus *Inbiomyia*) inbiomyiid fly 3. (Australimyzids; genus *Australimyza*) australimyzid fly

-**LZKŢ**- 'OTHER ACALYPTRATE FLY' I 1. (Conopids; numerous genera) thick-headed fly 2. (Lonchaeids; several genera) lance fly 3. (Cryptochetids; several genera) scale parasite fly

-**LZKŢW**- 'OTHER ACALYPTRATE FLY' II 1. (Paraleucophids; several genera) paraleucophid fly 2. (Mormotomyiids; genus *Mormotomyia*) frightful hairy fly / terible hairy fly 3. [stem unassigned]

Hexapods > Insects > Antliophora > Diptera > Brachycera > Muscomorpha> Schizophora > Calyptrata

-**LZL**- 'MUSCID FLY' 1. (genus *Musca*) housefly, face fly / autumn housefly, bazaar fly / eye-seeking fly, Australian bush fly 2. (ectoparasitic Stomoxyines; numerous genera including *Haematobia, Haematobosca, Stomoxys*) horn fly, buffalo fly, moose fly, stable fly / barn fly / dog fly / biting house fly / power mower fly 3. (Muscines [other than *Musca*], Atherigonines, Azeliines, Coenosiines, Cyrtoneurinines, Mydaeines, Phaoniines; numerous genera) other muscid fly

-**LZŘ**- 'MUSCOID FLY' 1. (Fanniids; several genera) little housefly 2. (Anthomyiids; numerous genera) root-maggot fly, cabbage fly, turnip root fly / summer cabbage fly, raspberry cane fly, beet leafminer, spinach leafminer, bean seed fly, onion fly, wheat bulb fly, willow catkin fly, knotting gall fly, seed fly 3. (Scathophagids; numerous genera) dung fly

-**LZV**- 'PARASITIC OESTROID FLY' 1. (genus *Dermatobia*) human bot fly 2. (genus *Cordylobia*) tumbu fly / mango fly / putzi fly / skin maggot fly 3. (genera *Cochliomyia, Chrysomya*) screw-worm fly

-**LZVL**- 'OESTROID FLY' I 1. (Oestrids; numerous genera) botfly / heel fly / warble fly / gadfly 2. (Calliphorids; numerous genera) bow fly / carrion fly / bluebottle / greenbottle / cluster fly, 3. (Sarcophagids; numerousgenera) flesh fly

-**LZVW**- 'OESTROID FLY' II 1. (Rhinophorids; numerous genera) woodlouse fly 2. (Mystacinobiids; genus Mystacinobia) New Zealand bat fly 3. (Tachinids, Rhiniids, Ulurumyiids, Mesembrinellids; numerous genera) other oestroid fly

-**LZVY**- 'HIPPOBOSCOID FLY' 1. (Glossinids; genus Glossina) tsetse fly / tik-tik fly 2. (Hippoboscids; numerous genera) louse fly / ked fly 3. (Nycteribiids and Streblids; numerous genera) bat fly

7.3.2 Plants

IMPORTANT NOTE: When utilizing lexical stems relating to plants, care should be taken to remember that each of these stems refers to the entire plant as a whole. If the context is referencing a particular part of a pant (e.g., fruit, seed, leaf, stem, root, tuber, flower, nut, pod, etc.), an appropriate EPP affix should be utilized or the stem should be concatenated with another stem naming the relevant part of the plant.

All of the following stems referring to plants have the following Specification structure:

BSC	(to be) a plant as a holistic entity, including its physical/corporeal body and its living essence; to live/be alive (as a plant)
CTE	(to be) that which gives a particular plant its individual identity; the living essence of a plant
CSV	(to be) the physical body of a plant; the corporeal aspect of a plant
OBJ	(to be) a state or activity being undergone or engaged in by a plant, most likely associated with a particular stage of a plant's life-cycle (e.g., dormancy as seed, sproutling, budding, in bloom, withering, winter dormancy, etc.)

7.3.2.1 Bryophytes (non-vascular land plants)

-**FXW**- 'GREEN ALGA' 1. (Chlorophytes; numerous genera) chlorophyte, chlorophytic green alga 2. (Charophyta [other than Embryophyta]; numerous genera) charophytic green alga 3. (Mesostigmatophyceae, Spirotaenia, Chlorokybophyceae; several genera) other green alga

-**FXM**- 'EDIBLE GREEN ALGA' 1. (genera *Monostroma*) aonori / hitoegusa-nori / green laver seaweed 2. (genus *Caulerpa*) sea grapes / green caviar 3. (genus Ulva) sea lettuce, green nori, gutweed

-**FXX**- 'NON-VASCULAR LAND PLANT' 1.(Marchantiophyta; numerous genera) liverwort / hepatic 2. (Anthocerotophyta; several genera) hornwort 3. (Bryophyta; numerous genera) moss

-**FXL**- 'MOSS (by branching type)' 1. acrocarpous moss 2. pleurocarpous moss 3. cladocarpous moss

-**FXR**- ‘MOSS (by taxonomic class)’ 1. (Spagnoposida; numerous genera) peat moss 2. (Bryopsida; numerous genera) bryopsid moss 3. (Takakiopsids, Andreaeopsids, Andreaeobyopsids, Oedipodiopsids, Polytrichopsids, Tetraphidopsids; numerous genera) other moss

7.3.2.2 Tracheophytes (Vascular Plants)

-**CN̆W**- ‘LYCOPOD’ 1. (Lycopodiaceae; numerous genera) clubmoss 2. (Selaginellaceae; genus *Selaginella*) spikemoss, lesser clubmoss 3. (Isoetaceae; genus *Isoetes*) quillwort

-**CNW**- ‘POLYPODIOPSID (FERN)’ I 1. (Equisetid; genus *Equisetum*) horsetail / horsetail fern 2. (Ophioglossids; several genera) moonwort, grape-fern, goblin fern, rattlesnake fern, hand fern, adder’s-tongue fern, whisk fern 3. (Marattiids; several genera) marattiid fern, giant fern, king fern, horse shoe fern

-**CNY**- ‘POLYPODIOPSID (FERN)’ II 1. (Polypodiales; numerous genera) wood fern / male fern / buckler fern, macho fern / swordfern, deersfoot fern / hare’s foot fern /rabbit foot fern / ball fern / shinobu fern, mahogany maidenhair, shieldfern, beech fern, bog fern / Massachusetts fern, maiden fern, swamp shield-fern, New York fern, Sierra marsh fern / Nevada march fern, lady-fern, twinsorus fern, silvery glade fern, spleenwort, Japanese lady fern, cliff fern, ostrich fern, sensitive fern, chain fern, waterfall fern, hard fern, water fern, rasp fern / amaumau fern, deer fern, half-net fern, bladderfern / fragile fern, oak fern, glade fern, hayscented fern / cup fern, batwing fern, beadfern, lace fern, bracken, brakenfern / eagle fern, maidenhair fern, cloak fern, false cloak fern, lip fern, brake, cliff brake, rock brake, leather fern, gold fern, shoestring fern, soralpouch fern, necklace fern 2. (Cyatheales; numerous genera) tree fern, scaly tree fern, manfern 3. (Salviniales; several genera) watermoss, water fern / fairy moss / duckweed fern / mosquito fern, pillwort, water clover

-**CMW**- ‘POLYPODIOPSID (FERN)’ III 1. (Schizaeaceae; several genera) ray fern, curlygrass fern / comb fern 2. (Lygodioids; genus *Lygodium*) climbing fern 3. (Anemioids; genus *Anemia*) pine fern, hemlock fern, flowering fern

-**CMY**- ‘POLYPODIOPSID (FERN)’ IV 1. (Gleicheniales; several genera) forked fern, coral fern, tangle fern, umbrella fern, fan fern 2. (Hymenophyllaceae; several genera) filmy fern, bristle fern 3. (Osumdaceae; several genera) royal fern, interrupted fern, lancea flowering fern, cinnamon fern

Tracheophytes > Spermatophytes (seed plants)

-**PSN̆**- ‘CYCAD & GINGKO’ 1.(Cycads [other than genus *Ecephalartos*; several genera) cycad, king sago, queen sago, Dukour sago 2. (genus *Encephalartos*) bread tree / bread palm / kaffir bread tree 3. (Gingkophyta; genus *Gingko*) gingko tree

-**PSN̆W**- ‘GNETOPHYTE’ 1. (Gnetaceae; genus *Genetum*) gnetum tree, gnetum shrub, gnetum liana 2. (Welwitschiaceae; genus *Welwitschia*) welwitschia / tree tumbo 3. (Ephedraceae; genus *Ephedra*) joint-pine / jointfir / Mormon-tea plant / Brigham tea plant

Tracheophytes > Spermatophytes > Pinophyta (conifers)

-**LSK**- ‘PINE TREE’ I (genus *Pinus*, subgenus *Pinus*) 1. (*P.P.* Subsection *Ponderosae*; numerous species) ponderosa, Jeffrey / yellow / black, Torrey, gray / foothill / digger pine, Durango, Coulter / big-cone, Arizona, various Mexican and Central American pines 2. (*P.P.* Subsections *Australes* and *Contortae*; numerous species) lodgpole, jack, sand, Virginia, knobcone, bishop, Monterey, Caribbean, Cuban, shortleaf, slash, spruce pine, egg-cone, longleaf, paula, pitch, pond, loblolly, ocote, various Mexican and Central American pines 3. (*P.P.* Subsection *Pinus* and *Pinaster*; numerous species) mountain pine, red, Scots, tropical, various Eurasian pines

-**LSKW**- ‘PINE TREE’ II (genus *Pinus*, subgenus *Strobus*) 1. (*P.S.* Subsections *Strobus, Krempfianae,* and *Gerardianae*; numerous species) whitebark, sugar, white, blue, limber, Chiapas, lacebark, chilgoza, various Eurasian white pines 2. (*P.S.* Subsection *Balfourianae*; several species) bristlecone pine, foxtail pine 3. (*P.S.* Subsections *Cembroides* and *Nelsonianae*; numerous species) pinyon pine

-**LSKY**- ‘LARCH TREE’ (genus *Larix*) 1. ([North American larches] *L. laricina, L. lyallii, L. occidentalis*) western larch, tamarack, subalpline larch 2. ([Northern Eurasian larches] *L. decidua, L. sibirica, L. gmelinii, L. kaempferi, L. czekanowskii*) European, Siberian, Dahurian, Japanese larch 3. ([Southern Eurasian larches] *L. ptaninii, L. mastersiana, L. griffithii*) chinese larch, Masters’ larch, Himalayan larch

-**LSKL**- ‘PINACEAE’ I 1. (genus *Picea*) spruce tree 2. (genus *Pseudotsuga*) Douglas fir / Oregon pine 3. (genus *Cathaya*) cathaya

-**LSKR**- ‘PINACEAE’ II 1. (genus *Abies*) fir tree 2. (genus *Cedrus*) cedar tree 3. (genus *Tsuga*) hemlock tree

-**LSKŘ**- ‘PINACEAE’ III 1. (genus *Nothotsuga*) bristlecone hemlock tree 2. (genera *Keteleeria*) keteleeria tree 3. (genus *Pseudolarix*) golden larch

-**LSKÇ**- ‘PINOPSID’ 1. (Araucariaceae; three genera) araucarian, kauri / dammara, wollemia 2. (Podocarpaceae; numerous genera) silver pine / Westland pine / white silver pine, creeping pine, celery pine, Huon pine / Macquarie pine, Asian bayberry, podocarp 3. (Sciadopityaceae; genus *Sciadopitys*) Japanese umbrella-pine

-**LSKH**- ‘CUPRESSOID’ 1. (genus *Cupressus*) cypress tree 2. (genus *Juniperus*) juniper 3. (Cupressoids other than Cupressus and Juniperus; several genera) thuja / arborvitae, hiba, false cypress, Fujian cypress, incense-cedar, Chinese arborvitae, sandarac / Barbary thuja, Siberian carpet-cypress / Russian arbor-vitae, Nootka cypress, Vietnamese golden cypress

-**LSKŢ**- ‘SEQUOIOID’ 1. (genus *Sequoia*) coast redwood 2. (genus *Sequoiadendron*) giant sequoia 3. (genus *Metasequoia*) dawn redwood

-**LSKF**- ‘CUPRESSACEAE’ 1. (Callitroids; several genera) cypress-pine, alerce / Patagonian cypress, callidtroid 2. (Taxodioids; several genera) bald cypress, swamp cypress, sugi 3. (Athrotaxidoids, Taiwanioids, Cunninghamioids; several genera) Tasmanian cedar, taiwania, cunninghamia

-**LSKM**- ‘TAXACEAE (YEW)’ I 1. (genus *Taxus*) common yew 2. (genus *Pseudotaxus*) whiteberry yew 3. (genus *Austrotaxus*) southern yew / New Caledonia yew

-**LSKN**- ‘TAXACEAE (YEW)’ II 1. (genus *Cephalotaxus*) plum-yew 2. (genus *Torreya*) nutmeg yew, torreya, gopher wood 3. (genus *Amentotaxus*) catkin-yew

Tracheophytes > Spermatophytes > Angiosperms (flowering plants)

-**CML**- ‘NYMPHAEALES’ 1. (Nymphaeaceae; several genera) water lily / pond-lily / bonnet lily / alligator-bonnet / spatterdock, blue lotus, Miami rose, Egyptian lotus, prickly waterlily / foxnut / gorgon nut / makhana 2. (Cobombaceae; genera *Brasenia, Cabomba*) water-shield, fanwort, cabomba 3. (Hydatellaceae; genus *Trithuria*) trithuria

-**CMR**- ‘PRIMITIVE ANGIOSPERM’ I 1. (Austrobreileyaceae; genus *Austrobaileya*) austrobaileya 2. (Trimeniaceae & Schisandrids; genera *Trimenia, Schisandra, Kadsura, Illicium*) trimenia, schisandrid, star anise / badian 3. (Amborellaceae; genus *Amborella*) amborella

-**CMŘ**- ‘PRIMITIVE ANGIOSPERM’ II 1. (Chloranthaceae; genera *Chloranthus, Sarcandra, Ascarina, Hedyosmum*) chloranthus, sacandra herb, hutu, ascarina 2. (Ceratophyllaceae; genus *Ceratophyllum*) coontail / hornwort 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > monocots

Tracheophytes > Spermatophytes > Angiosperms > monocots > alismatid monocots

-**PŠM**- ‘ARACEAE (ARUM)’ I 1. (Orontioids; three genera) skunk cabbage, golden club 2. (Gymnostachydoideae; genus *Gymnostachys*) settler’s twine / boorgay 3. (Lemnoids; several genera) duckweed / water lentil / water lens, bogmat / mud-midget, watermeal, dotted duckmeat

-**PŠMW**- ‘ARACEAE (ARUM)’ II 1. (Pothoids; several genera) anthurium / tailflower / flamingo flower 2. (Aroids; numerous genera) arum lily, aroid, corpse flower, elephant yam, elephant ear / heart of Jesus / angel wing, calla / bog arum / marsh calla / wild calla / squaw claw / water-arum, Chinese evergreen, green dragon, water cabbage / water lettuce / shellflower / Nile cabbage, philodendron, rascagarganta, vilevine, arrow arum, arum lilly / calla lily, dead horse arum lily 3. (Lasioids, Zamioculcadoids; numerous genera) other aroid

-**PŠMY**- ‘ALISMATACEAE’ 1. (genus *Damasonium*] star-fruit / thrumwort, star water-plantain 2 (genus *Sagittaria*] arrowhead / duck potato / katniss / swamp potato / tule potato / wapato / wapatoo /wapati / watato 3. (Alismataceae other than Stems 1 and 2; numerous genera) water-plantain, burhead / Amazon sword plant, yellow velvetleaf / sawah flower rush / sawah lettuce

-**PŠML**- ‘ALISMATID’ I 1. (Butomaceae; genus *Butomus*) flowering rush / grass rush 2. (Hydrocharitaceae; numerous genera) waterweed, tape-grass, waterthyme, European frogbit / water-poppy, oxygen weed, spongeplant / American frogbit, eelgrass / vallis, water nymph / naiad, guppy grass / najas grass, duck lettuce, swamp lily, water soldier, turtlegrass 3. (Tofieldiaceae; several genera) rush featherling, tofieldia, Scottish asphodel / Scotch false asphodel, false asphodel, Harper’s beauty

-**PŠMR**- ‘ALISMATID’ II 1. (Aponogetonaceae; genus *Aponogeton*) cape-pondweed / water hawthorn, laceleaf / lattice leaf / lace plant 2. (Scheuchzeriaceae; genus *Scheuchzeria*) rannoch-rush / pod grass 3. (Jungcaginaceae and Maundiaceae; several genera) arrowgrass, flowering quillwort / awl-leaf lilaea, southern water-ribbon / dwarf water-ribbon

-**PŠMŘ**- ‘ALISMATID’ III 1. (Posidoniaceae; genus *Posidonia*) Neptune grass / Mediterranean tapeweed, fibreball weed, posidonia seagrass 2. (Ruppiaceae; genus *Ruppia*) ruppia / widgeonweed / ditch grass / widgeon grass, tasselweed, tassel pondweed 3. (Cymodoceaceae; several genera) manatee-grass, little Neptune grass, narrowleaf seagrass, shoalweed / shoal grass, wire weed, sickle-leaved cymodocea

-**PŠŇ**- ‘ALISMATID’ IV 1. (Potamogetonaceae; several genera) pondweed, horned pondweed 2. (Zosteraceae; genera *Zostera, Phyllospadix*) marine eelgrass, surfgrass 3. (Acoraceae; genus *Acorus*) calamus / sweet flag

-**PŠŇW**- ‘MONSTEROID’ 1. (genus *Epipremnum*) satin pothos / silver vine, golden pothos / money plant / hunter’s robe / devil’s vine / devil’s ivy / marble queen / Ceylon creeper 2. (genus *Monstera*) monsteroid, shingle plant, Adanson’s monstera / five holes plant, ceriman / monster fruit plant / fruit salad plant / Swiss cheese plant 3. (Monsteroids other than *Epipremnum* and *Monstera*; several genera) spath / peace lily, centipede tongavine, other monsteroid

Tracheophytes > Spermatophytes > Angiosperms > monocots > lilioid monocots > Dioscoreales

-**TXW**- ‘DIOSCOREACEAE’ 1. (genera *Dioscorea, Tamus*) yam, purple yam / greater yam / ube, Chinese yam / Chinese potato / cinnamon-vine / nagaimo, Indian three-leaved yam, Japanese mountain yam / yamaimo, Zanzibar yam, white yam / guinea yam / African yam, black bryony / lady’s seat / black bindweed, barbasco, Mexican yam, wild yam, lesser yam, pencil yam, Indian yam / cush-cush / yampee, fourleaf yam / devil’s bones / rheumatism root, fiveleaf yam 2. (genus *Tacca*) batflower, Polynesian arrowroot / pia 3. (genera *Borderea, Epipetrum, Rajania, Stenomeris, Avetra, Trichopus*) other dioscorean plant

- **TXL**- ‘NARTHECIACEAE’ 1. (genus *Aletris*) colicroot / colicweed / crow corn / unicorn root 2. (genus *Lophiola*) goldencrest 3. (genus *Narthecium*) yellow asphodel / bog asphodel

- **TXR**- ‘OTHER DIOSCOREALES’ I 1. (Burmanniaceae; numerous genera) nodding-nixie, northern bluethread, yellowseed, thismia, fairy lantern, burmanniate 2. (Nartheciaceae; genera *Nietneria, Metanarthecium*) other nartheciacean 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > monocots > lilioid monocots > Liliales

-**LSW**- ‘LILIACEAE’ I 1. (genus *Lilium*) [true] lily 2. (genus *Fritillaria*) fritillary, spotted missionbells / spotted mountainbells, chocolate lily, kaiser’s crown / crown imperial, adobe lily, yellowbells, brown bells, checkered daffodil / frog-cup 3. (genus *Cardiocrinum*) giant lily

-**LSY**- ‘LILIACEAE’ II 1. (genera *Tulipa, Amana*) tulip 2. (genus *Gagea*) gagea, star-of-Bethlehem, Radnor lily, mountain spiderwort / Snowdon lily / common alplily, yellow star-of-Cypress 3. (genus *Erythronium*) fawn lily, trout lily, dog’s-tooth violet, avalanche-lily, adder’s tongue

-**LSL**- ‘LILIACEAE’ III 1. (genus *Clintonia*) bead liluy / bride’s bonnet / queen’s cup, speckled wood lily / Clinton’s lily, blue-bead lily / corn lily / snakeberry / dogberry / straw lily, clintonia 2. (genus *Medeola*) Indian cucumber-root / Cuke root 3. (genus *Calochortus*) globelily, mariposa lily, beavertail grass, star tulip, sego lily, pussy ears, cat’s ear lily

-**LSR**- ‘LILIACEAE’ IV 1. (genus *Streptotus*) twistedstalk / rosybells / rose mandarin / scootberry / liverberry / rose-bellwort 2. (genus *Posartes*) fairybells, drops of gold, yellow mandarin, spotted mandarin, Siskiyou bells 3. (genus *Scoliopus*) fetid adderstongue, slinkpod / brownies

-**LSŘ**- ‘LILIACEAE’ IV 1. (genus *Tricyrtis*) toad lily, hairy toad lily, tricyrtis 2. (genera *Nomocharis, Notholirion*) other liliate 3. [stem unassigned]

-**LSX**- ‘PARIDS’ 1. (genus *Trillum*), trillium / wakerobin / birthroot / birthwort / tri flower 2. (genus *Paris*) paris, herb-paris / true lover’s knot, bunchflower 3. (genus *Pseudotrillium*) brook wakerobin

-**LSF**- ‘MELANTHIDS’ I 1. (genus *Amianthium*) fly poison 2. (genus *Anticlea*) elegant camas / alkali grass, deathcamas 3. (genus *Melanthium*) slender bunchflower / cripsed bunchflower, Ozark bunch-flower / Wood’s bunchflower

-**LSV**- 'MELANTHIDS' II 1. (genus *Toxicoscordion*) deathcamas, star zigadene, sand-corn 2. (genus *Zigadenus*) sandbog death camas 3. (genus *Stenathium*) featherbell, Osceola's plume / black snakeroot / crow poison, pinebarren deathcamas

-**LSŢ**- 'MELANTHIDS' III 1. (genus *Veratrum*) false hellebore / corn-lily 2. (genus *Schoenocaulon*) feathershank, sabadilla 3. [stem unassigned]

-**LSḶ**- 'MELANTHIACEAE' 1. (Heloniads; genera *Helonias, Heloniopsis, Ypsilandra*) swamp pink 2. (Chionographids; genera *Chamaelirium, Chionographis*) blazing-star / devil's bit / false unicorn / fairy wand 3. (Xerophylleae; genus *Xerophyllum*) turkey beard / beartongue / grass-leaved helonias

-**LSM**- 'COLCHICAEAE' 1. (Burchardideae; genus *Burchardia*) milkmaid 2. (Uvularieae; genera *Uvularia, Disporum*) bellwort / bellflower / merrybells, fairy bells 3. (Tripladenieae; three genera) tripladene, lilac lily

-**LSN**- 'COLCHICACEAE' I 1. (genus *Colchicum* [including *Androcymbium* and *Merendera*]) autumn crocus / meadow saffron / naked lady 2. (genus *Gloriosa*) flame lily / fire lily / gloriosa lily / glory lily / superb lily / climbing lily / creeping lily 3. (genus *Hexacyrtis*) Namib lily

-**LSŇ**- 'COLCHICACEAE' II 1. (genus *Sandersonia*) Christmas bells / golden lily of the valley / Chinese lantern lily / chinese lantern bulb 2. (Anguillarieae; genera *Baeometra, Wurmbea*) beetle lily, wurmbea, early Nancy, eight Nancy 3. (genera *Ornithoglossum, Iphigenia, Camptorrhiza*) other colchicate plant

-**LSS**- 'LILIALES' I 1. (Smilacaceae; genera *Smilax, Heterosmilax*) greenbrier / catbrier / prickly-ivy 2. (Ripogonaceae; genus *Ripogonum*) supplejack 3. (Philesiaceae; genera *Lapageria, Philesia*) Chilean bellflower / copihue, austral bellflower

-**LSH**- 'LILIALES' II 1. (Alstroemeriaceae; genera *Alstroemeria, Bomarea*) Inca-lily, Peruvian lily / lily of the Incas, bomarea, salsilla 2. (Luzuriagaceae; genera *Drymophila, Luzuriaga*) turquoise berry / native Solomons seal, orange berry, quilineja, almond flower 3. (Corsiaceae, Campynemataceae, Petermanniaceae; several genera) other liliales plant

Tracheophytes > Spermatophytes > Angiosperms > monocots > lilioid monocots > Asparagales

Tracheophytes > Spermatophytes > Angiosperms > monocots > lilioid monocots > Asparagales > Asparagaceae

-**SFW**- 'ASPARAGOIDEAE & AGAVOIDEAE' I 1. (genus *Asparagus*) asparagus plant, large forest sicklethorn, katbos, bridal creeper, florist's fern 2. (genera *Yucca, Hesperoyucca*) yucca plant, Grand Canyon Quixote plant / Newberry's yucca, Quixote yucca / foothill yucca / Spanish bayonet 3. (genus *Agave*) agave / century plant, tuberose

-**SFY**- 'ASPARAGOIDEAE & AGAVOIDEAE' II 1. (genus *Camassia*) camas / quamash / Indian hyacinth / wild hyacinth 2. (genus *Chlorogalum*) soap plant / soaproot / amole 3. (genus *Eremocrinum*) Utah solitaire lily / desert lily / sand lily / lonely lily / intermountain false-wheatgrass

-**SFL**- 'ASPARAGOIDEAE & AGAVOIDEAE' III 1. (genus *Hastingsia*) rushlily 2. (genus *Hosta*) hostas, plantain lily 3. (genus *Leucocrinum*) common starlily, mountain lily

-**SFR**- 'ASPARAGOIDEAE & AGAVOIDEAE' IV 1. (genus *Shoenolirion*) rush-lily / sunnybell 2. (genus *Hemiphylacus*) hemiphylacus 3. (Aphyllanthoideae; genus *Aphyllantes*) aphyllantes

-**LSFW**- 'NOLINOIDS' I 1. (genus *Nolina*) beargrass, Bigelow's nolina, palmilla 2. (genus *Convallaria*) lily-of-the-valley 3. (genus *Danae*) Alexandrian laurel / poet's laurel

-**LSFY**- 'NOLINOIDS' II 1. (genus *Maianthemum*) false Solomon's seal, wild lily-of-the-valley, false lily-of-the-valley / Canada mayflower, false mayflower, May lily 2. (genus *Liriope*) monkey grass, creeping lilyturf, spider grass, liriope 3. (genus *Ophiopogon*) lilyturf

-**LSFL**- 'NOLINOIDS' III 1. (genus *Dracaena*) dragon tree, shrubby dracaena, snake plant, African spear, sansevieria, golden hala pepe, "lucky bamboo", Sait George's sword / mother-in-law's tongue / viper's bowstring hemp 2. (genus *Polygonatum*) King Solomon's seal 3. (genus *Rohdea*) Nippon lily / sacred lily

-**LSFR**- 'NOLINOIDS' IV 1. (genus *Ruscus*) butcher's broom 2. (genera *Aspidistra, Beaucarnea, Calibanus, Campylandra, Comospermum, Dasylirion, Disporopsis, Eriospermum, Hereopolygonatum, Peliosantes, Pleomele, Reineckea, Semele, Speirantha, Theropogon, Tupistra*) other nolinoid 3. [stem unassigned]

-**RSFW**- 'LOMANDROIDS' I 1. (genus *Lomandra*) fish bones, mat-rush, basket grass 2. (genus *Chamaescilla*) blue stars / blue squill / mudrurt 3. (genus *Cordyline*) cabbage tree

-**RSFY**- 'LOMANDROIDS' II 1. (genus *Dichopogon*) chocolate lily 2. (genus *Eustrephus*) wombat berry vine 3. (genus *Laxmannia*) paperlily

-**RSFL**- 'LOMANDROIDS' III 1. (genus *Arthropodium*) renga lily / New Zealand rock lily, vanilla lily 2. (genus *Sowerbaea*) purple tassels, vanilla lily, chocolate flower, many-stemmed lily 3. (genus *Thysanotus*) fringe-lily

-**RSFR**- 'LOMANDROIDS' IV 1. (genus *Xerolirion*) Basil's asparagus 2. (genera *Acanthocarpus, Chamaexeros, Murchisonia, Romnalda, Trichopetalum*) other lomandroid 3. [stem unassigned]

-**ŘSFW**- 'SCILLOIDEAE' I 1. (genus *Hyacinthus*) hyacinth 2. (genus *Muscari*) grape hyacinth 3. (genus *Hyacinthoides*) bluebells

-**ŘSFY**- 'SCILLOIDEAE' II 1. (genus *Scilla*) squill, glory-of-the-snow, Dalmation scilla 2. (genus *Drimia*) drimia, sea squill / sea onion 3. (genus *Prospero*) autumn squill

-**ŘSFL**- 'SCILLOIDEAE' III 1. (genus *Ledebouria*) silver squill / wood hyacinth, ledebouria 2. (genus *Leopoldia*) tassel hyacinth / tassel grape hyacinth 3. (genus *Ornithogalum*) star-of-Bethlehem, sun star, chincherinchee, snake flower, Bath asparagus / Prussian asparagus

-**ŘSFR**- 'SCILLOIDEAE' IV 1. (genus *Eucomis*) pineapple flower, pineapple lily 2. (genus *Drimiopsis*) little white soldiers / lepoard plant / false hosta 3. (genus *Massonia*) hedgehog lily, blistered massonia

-**ŘSFŘ**- 'OTHER SCILLOIDEAE & BRODIAEOIDEAE' 1. (genus *Albuca*) slime lily 2. (Scilloideae other than previously named genera) other scilloid plant 3. (Brodiaeoideae; numerous genera) cluster-lily, Mexican star, triplet lily, Baja lily / blue sand lily, brodiaeoid

Tracheophytes > Spermatophytes > Angiosperms > monocots > lilioid monocots > Asparagales > Amaryllidaceae

-**SŢR**- 'ALLIEAE (ONION)' I (genus *Allium*) 1. (*A. cepa*) onion plant 2. (*A. oschaninii*) shallot plant 3. (*A. ampeloprasum*) leek plant

-**SŢL**- 'ALLIEAE (ONION)' II (genus *Allium*) 1. (*A. sativum*) garlic 2. (*A. schoenoprasum*) chive plant 3. (various other *Allium* species) scallion plant

-**SŢW**- 'ALLIOIDEAE' 1. (Leucocoryneae; several genera) glory-of-the-sun, starflower / spring starflower, springstar, crowpoison / false garlic, honeybells / onion weed 2. (Tulbaghieae; genera *Tulbaghia*, *Prototulbaghia*) wild garlic / society garlic 3. (Gilliesieae; several genera) South American allioid / gilliesid

-**LSŢW**- 'AMARYLLIDACEAE' I 1. (Amaryllids; several genera) amarylid, swamplily, poisonbulb / Queen Emma lily, seven sisters 2. (Cyrtantheae; genus *Cyrtanthus*) fire lily, Knysna lily, falcate fire lily 3. (Haemantheae; several genera) Natal lily / bush lily / Kaffir lily, green-tip forest lily, paintbrush, blood flower / blood lily, haemanthus, Inyanga fireball, kukumakranka

-**LSŢY**- 'AMARYLLIDACEAE' II 1. (Calostemmateae; genera *Proiphys, Calostemma*) Cardwell lily, Brisbane lily / Moreton Bay lily, wilcannia lily, garland lily 2. (Lycorideae; genera *Lycoris, Ungernia*) hurricane lily, cluster amaryllis, red spider lily / hell flower / red magic lily / equinox flower, resurrection lily / surprise lily / magic lily / naked lady, ungernia 3. (Griffineae; genera *Griffinia, Worsleya*) griffinid, Empress-of-Brazil

-**LSŢL**- 'AMARYLLIDACEAE' III 1. (Narcisseae; genera *Narcissus, Sternbergia, Lapiedra*) daffodil, winter daffodil / autumn daffodil / lily-of-the-field / yellow autumn crocus 2. (Pancractieae; genera *Pancratium, Vagaria*) pancratid, sea daffofil 3. (Galantheae; genera *Acis, Galanthus, Leucojum*) snowflake, snowdrop

-**LSŢR**- 'AMARYLLIDACEAE' IV 1. (Hippeastreae; several genera) copperlily, pink fairy lily / pink rain lily, Barbados snowdrop, lily-of-the-palace, "amaryllis", Aztec lily / Jacobean lily, rainflower, zephyr lily, rain lily, Atamasco lily / wild Easter lily, traubiine 2. (Eustephieae; four genera) eustephid, fire lily / flame lily 3. (Eucharideae; several genera) eucharid, Amazon lily, giant Peruvian daffodil

-**SŢY**- 'AGAPANTHOIDEAE & OTHER AMARYLLIDACEAE' 1. (Agapanthoideae; genus *Agapanthus*) lily of the Nile, African lily, African tulip, African bluebell / African blue lily, agapanthus 2. (Hymenocallideae; three genera) spider lily, cahaba lily, Peruvian daffodil, amancay 3. (Clinantheae; three genera) clinanthus, clinanthid

Tracheophytes > Spermatophytes > Angiosperms > monocots > lilioid monocots > Asparagales > Asphodelaceae

-**RSŢW**- 'ASPHODELOIDEAE' I 1. (genus *Aloe*) true aloe, aloe vera 2. (genus *Aristaloe*) guinea-fowl aloe / lace aloe 3. (genus *Aloiampelos*) climbing-aloe

-**RSŢY**- 'ASPHODELOIDEAE' II 1. (genus *Eremurus*) foxtail lily / desert candle 2. (genus *Gonialoe*) partridge aloe 3. (genus *Asphodelus*) asphodel

-**RSŢL**- ‘ASPHODELOIDEAE’ III 1. (genus *Haworthia*) haworthia 2. (genus *Haworthiopsis*) haworthiopsis 3. (genus *Gasteria*) gasteria

-**RSŢR**- ‘ASPHODELOIDEAE’ IV 1. (genus *Kniphofia*) tritoma, red hot poker, torch lily, poker plant, “Bees’ Sunset”, “Brimstone Bloom”, “Buttercup” 2. (genus *Kumara*) fan-aloe, kumara 3. (Asphodeloideae other than previously named genera) other asphodeloid plant

-**ŘSŢW**- ‘HEMEROCALLIDOIDEAE’ I 1. (genus *Hemerocallis*) daylily, lemon lily, tiger lily / ditch lily 2. (genus *Simethis*) Kerry lily 3. (genus *Ticoryine*) yellow rush-lilly

-**ŘSŢY**- ‘HEMEROCALLIDOIDEAE’ II 1. (genus *Caesia*) grass lily 2. (genus *Johnsonia*) pipe lily 3. (genera *Corynotheca, Arnocrinum, Hensmania, Stawellia*) other johnsoniid hemerocallid

-**ŘSŢL**- ‘HEMEROCALLIDOIDEAE’ III 1. (genus *Phormium*) New Zealand flax 2. (genus *Geitonoplesium*) scrambling lily 3. (genus *Stypandra*) nodding blue lily / blind grass

-**ŘSŢR**- ‘HEMEROCALLIDOIDEAE’ IV 1. (genus *Dianella*) flax lily, blueberry lily / paroo lily, inkberry lily 2. (genus *Thelionema*) tufted blue-lily 3. (genus *Herpolirion*) sky lily

-**ŘSŢŘ**- ‘XANTHORRHOEOIDEAE & OTHER HEMEROCALLIDOIDEAE’ 1. (Xanthorrhoeoideae; genus *Xanthorrhoea*) grass tree 2. (genera *Excremis, Pasithea, Agrostocrinum*) other phormid hemerocallidoid 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > monocots > lilioid monocots > Asparagales > Iridaceae

-**RSP**- ‘IRIDEAE’ 1. (genus *Iris*) iris, juno 2. (genus *Moraea*) Cape tulip, Barbary nut 3. (genera *Dietes, Bobartia, Ferraria*) wood iris / fortnight iris / butterfly iris / Japanese iris / African iris, other irid

-**RSPF**- ‘IRIDOIDEAE’ 1. (Sisyrinchieae; several genera) blue-eyed grass, morning iris, purple-eyed grass / satin flower / grass-widow / olsynium, pretty grass-flag 2. (Tigridieae; numerous genera) peacock flower / tiger-flower / shell flower, peatleaf, propeller flower, lágrimas de la virgen, tigrid 3. (Trimezieae; several genera) walking iris, apostle’s iris, apostle plant, forenoon yellow flag

-**RSPL**- ‘CROCEAE’ I 1. (genus *Crocus*) crocus 2. (genus *Babiana*) baboon-root 3. (genus *Ixia*) corn lily

-**RSPR**- ‘CROCEAE’ II 1. (genus *Dierama*) hairbells, angel’s fishing rods / grassy bells / fairybell / wedding bell, wandflower 2. (genus *Sparaxis*) harlequin flower, wandflower, sparaxis 3. (genera *Chasmanthe, Duthieastrum, Geissorhiza, Hesperantha, Radinosiphon, Syringodea, Tritonia, Tritoniopsis*) other crocid

-**RSPŘ**- ‘CROCOIDEAE’ 1. (Gladiolaeae; genera *Gladiolus, Melasphaerula*) gladiolus, cornflag, painted lady, cliff lily, Afrikaner, sword lily, gladiola 2. (Watsonieae; several genera) bugle lily, watsoniid 3. (Freesieae; genera *Freesia, Xenoscapa, Crocosmia, Devia*) freesia, montbretia

-**RSPŢ**- ‘IRIDACEAE’ 1. (Aristeoideae; genus *Aristea*) blue flies / blue stars / blue-eyed iris, blue corn-lily 2. (Patersonioideae; genus *Patersonia*) native flag / purple flag / native iris, swamp iris 3. (Isophysidoideae, Geosiridoideae, Nivenioideae; several genera) other crocioid

Tracheophytes > Spermatophytes > Angiosperms > monocots > lilioid monocots > Asparagales > Orchidaceae

-**RSPW**- ‘ORCHIDACEAE’ 1. (Orchidoideae; numerous genera) orchidoid orchid 2. (Epidendroideae; numerous genera) epidendroid orchid 3. (Apostasioideae and Cypripendioideae; numerous genera) apostasioid or cypripendioid orchid

-**RSPY**- ‘VANILLOIDEAE’ 1. (Vanilleae; *Vanilla planifolia*) flat-leaved vanilla orchid plant [source of vanilla bean extract) 2. (Vanilleae; several genera/species other than *V.planifolia*) vanillid orchid plant 3. (Pogonieae; several genera) pogonia, snakemouth orchid, little five leaves, pogonid orchid

Tracheophytes > Spermatophytes > Angiosperms > monocots > lilioid monocots > Asparagales > remaining families

-**RSPH**- ‘OTHER ASPARAGALES’ I 1. (Doryanthaceae; genus *Doryanthes*) gymnea lily, giant spear lily 2. (Tecophilaeaceae; several genera) Chilean blue crocus, Hartweg’s doll’s lily, tecophilaeid 3. (Ixioliriaceae; genus *Ixiolirion*) lavender mountain lily / Siberian lily

-**RSPĻ**- 'OTHER ASPARAGALES' II 1. (Asteliaceae; several genera) pineapple grass / perching lily, silver spear, bush flax / bush lily, astelia, asteliid 2. (Hypoxidaceae; several genera) hypoxidiid, weevil-wort, curculigo 3. (Lanariaceae; genus *Lanaria*) lambtails / Cape edelweiss

-**RSPÇ**- 'OTHER ASPARAGALES' III 1. (Blandfordiaceae; genus *Blandfordia*) Christmas bells 2. (Boryaceae; genera *Boryna, Alania*) pincushion, borya, alania 3. (Xeronemataceae; genus *Xeronema*) Poor Knights lily

Tracheophytes > Spermatophytes > Angiosperms > monocots > lilioid monocots > remaining orders

-**RSPHW**- 'OTHER LILLIOID MONOCOT' 1. (Pandanales; numerous genera) pandan, screw-pine / screw palm / pandanus palm 2. (Petrosaviales; genera *Petrosavia, Japonolirion*) petrosaviate 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > monocots > commelinid monocots

Tracheophytes > Spermatophytes > Angiosperms > monocots > commelinid monocots > Commelinales

-**NÇPF**- 'COMMELINACEAE' 1. (Commelineae; several genera) dayflower, widow's tears, tropical spiderwort / wandering Jew, scurvy weed, scurvy grass, rat's ear, marble berry, dewflower 2. (Tradescantieae; numerous genera) spiderwort, Indian paint, boatlily / oyster plant, wild crocus, false dayflower, blue ginger 3. (Cartonemateae; genus *Cartonema*) cartonema, wiridjagu

-**NÇPH**- 'COMMELINALES' 1. (Haemodoraceae; numerous genera) bloodwort, kangaroo paw, winter bell, bloodroot, coneflower, redroot 2. (Pontederiaceae; several genera) mud plantain, water hyacinth, water stargrass, ducksalad, pickerel weed, false pickerelweed / oval-leafed pondweed 3. (Philydraceae & Hanguanaceae; genera *Philydrum, Philydrella, Helmholtzia, Hanguana*) stream lily, frogsmouth, woolly waterlily, hanguana

Tracheophytes > Spermatophytes > Angiosperms > monocots > commelinid monocots > Zingiberales

-**NÇP**- 'MUSACEAE' 1. (genus *Musa*) banana, plantain 2. (genus *Ensete*) enset, false banana 3. (genus *Musella*) Chinese dwarf banana / golden lotus banana

-**NÇPW**- 'ZINGIBERACEAE' I 1. (genus *Zingiber*) ginger plant, myoga [Japanese ginger] 2. (genus *Curcuma*) turmeric plant, zedoary / white turmeric, Java ginger / Javanese turmeric, Cape York lily, mango ginger, Siam tulip / summer tulip, East Indian arrowroot / narrow-leaved turmeric, mango-ginger 3. (genera *Amomum*, *Elettaria*) cardamom plant

-**NÇPY**- 'ZINGIBERACEAE' II 1. (genus *Alpinia*) galangal plant, shellflower / shellplant, alpinia, Chinese ginger, shell ginger, round Chinese cardamom 2. (genus *Aframomum*) false cardamom / Ethiopian cardamom plant, melegueta pepper / alligator pepper / grains of paradise 3. (genera *Etlingera*) torch ginger / torch lily / Philippine wax flower / Indonesian tall ginger / porcelain rose, Malay rose

-**NÇPL**- 'ZINGIBERACEAE' III 1. (genus *Hornstedtia*) Scott's ginger / jiddo / native cardamom 2. (genus *Hedychium*) garland flower / ginger lily / kahili ginger 3. (Zingiberaceae other than already named; numerous genera) other zingiberid

-**NÇPÇ**- 'ZINGIBERACEAE' IV 1. (genus *Lanxangia*) cao guo 2. (genus *Kaempferia*) kencur / lesser galangal / aromatic ginger / sand ginger / cutcherry / resurrection lily / cekur, Thai ginseng / Thai black ginger, silver spot 3. (genus *Boesenbergia*) Chinese keys / fingerroot / Chinese ginger / lesser galangal

-**NÇPR**- 'ZINGIBERALES' I 1. (Cannaceae; genus *Canna*) canna lily 2. (Heliconiaceae; genus *Heliconia*) lobster-claws / toucan beak / wild plantain / false bird-of-paradise 3. (Lowlaceae; genus *Orchidantha*) orchidantha

-**NÇPŘ**- 'ZINGIBERALES' II 1. (Strelitziaceae; three genera) strelitzia, bird of paradise, crane lily, traveller's tree / traveller's palm 2. (Costaceae; several genera) costus, orange tulip ginger, spiral ginger,spiralflag ginger / Indian head ginger, yellow trumpet, spiral flag, crêpe ginger, wax ginger 3. (Marantaceae; numerous genera) arrowroot, prayer plant, calathea, myrosma, alligator-flag, thalia, hardy canna, fire-flag

Tracheophytes > Spermatophytes > Angiosperms > monocots > commelinid monocots > Arecales

-**LBW**- 'COMMON FOOD PALM (ARECACEAE)' I 1. (genus *Cocos*) coconut tree 2. (genus *Phoenix*) date palm tree 3. (genus *Euterpe*) açaí palm tree

-**LBY**- 'COMMON FOOD PALM (ARECACEAE)' II 1. (genus *Butia*) jelly palm, yatay palm, pindo palm 2. (genus *Elaeis*) oil palm, macaw-fat 3. (genus *Bactris*) peach palm

-**LBL**- 'ARECEAE' I 1. (genus *Areca*) areca nut palm [betel nut palm] 2. (genus *Howea*) thatch palm / Kentia palm 3. (genus *Normanbya*) black palm

-**LBR**- 'ARECEAE' II 1. (genus *Nenga*) pinang palm 2. (genus *Heterospathe*) sagisi palm 3. (genus *Hedyscepe*) big mountain palm / umbrella palm

-**LBǍ**- 'ARECEAE' III 1. (genus *Acanthophoenix*) barbel palm / palmiste rouge 2. (genus *Wodyetia*) foxtail palm 3. (genus *Phoenicophorium*) thief palm / latanier palm

-**LBV**- 'COCOSEAE' 1. (genus *Jubaea*) Chilean wine palm 2. (genus *Jubaeopsis*) Pondoland palm 3. (genus *Voanioala*) forest coconut palm

-**LBḌ**- 'ARACOIDEAE' 1. Leopoldinieae; genus *Leopoldinia*) pissava palm / jará palm 2. (Roystoneae; genus *Roystonea*) royal palm, palmiste / imperial palm / cabbabe palm, yagua / cabiche 3. (Geonomeae, Iriarteeae, Manicarieae, Oranieae, Pelagodoxeae, Podococceae, Reinhardtieae, Sclerospermeae, Areceae other than named genera above; numerous genera) other aracoid palm

-**LBZ**- 'CALAMOIDEAE' 1. (Calameae; numerous genera) rattan palm 2. (Eugeissoneae; genus *Eugeissona*) bertam / wild Bornean sago palm 3. (Lepidocaryeae; several genera) raffia palm, butiti, poltamui

-**LBŽ**- 'CORYPHOIDEAE' 1. (Trachycarpeae; numerous genera) needle palm, lady palm, saw palmetto, carnauba wax palm, European fan palm / Mediterranean dwarf palm 2. (Crysophileae; several genera) zombie palm, Key thatch palm, silver pimeto 3. (Sabaleae, Borasseae, Caryoteae; Chuniophoeniceae, Chorypheae; numerous genera) fan-palm, palmetto, sea coconut palm / double coconut palm, fishtail palm, Hainan fan palm, white backed palm, Mazari palm, tahina palm

-**LBH**- 'OTHER ARECALES' 1. (Ceroxyloideae; several genera) ivory palm / ivory-nut palm / tagua palm, chonta palm, mastodon palm / fiber palm 2. (Nypoideae; genus *Nypa*) nipa / mangrove palm 3. (Dasypogonaceae; four genera) pineapple bush, tinsel lily, kingia / bullanock, baxteria

Tracheophytes > Spermatophytes > Angiosperms > monocots > commelinid monocots > Poales (grasses)

Tracheophytes > Spermatophytes > Angiosperms > monocots > commelinid monocots > Poales > Graminids

Tracheophytes > Spermatophytes > Angiosperms > monocots > commelinid monocots > Poales > Graminids > Poaceae

Tracheophytes > Spermatophytes > Angiosperms > monocots > commelinid monocots > Poales > Graminids > Poaceae > Avenine grasses

-BT- 'TRITICUM (WHEAT)' I 1. (*T. aestivum*) common wheat / bread wheat grass 2. (*T. durum*) durum wheat / pasta wheat grass 3. (*T. spelta*) spelt grass / dinkel wheat grass

-BTY- 'TRITICUM (WHEAT)' II 1. (*T. monococcum, T. boeoticum*) einkorn wheat grass 2. (*T. turgidum dicoccum, T. turgidum dicoccoides)* emmer wheat grass 3. (*T. turanicum*) khorasan wheat grass, kamut grass

-**BTW**- 'TRITICEAE (WHEAT-RELATED DOMESTICATED CEREAL GRAINS)' 1. (genus *Hordeum*) barley grass 2. (genus *Secale*) rye grass 3. (*Triticum*+*Secale* hybrid) triticale grass

-**BTL**- 'OTHER TRITICEAE' I 1. (genus *Agropyron*) wheatgrass 2. (genus *Aegilops*) goatgrass 3. (genus *Elymus*) couch grass, wildrye, bottlebrush grass, lyme grass, dune grass, strand grass / strand-wheat, squirreltail

-**BTR**- 'OTHER TRITICEAE' II 1. (genus *Hordelymus*) wood-barley 2. (genus *Leymus*) wildrye, lyme grass 3. (genus *Thinopyrum*) Eurasian quackgrass, rush wheatgrass, tall wheatgrass

-**BTǍ**- 'OTHER TRITICEAE' III 1. (genus *Triticum* other than already named species) club wheat, Persian wheat, Armenian wild emmer, red wild einkorn wheat, Ethiopian wheat, Zanduri wheat, Polish wheat 2. (Triticeae other than previously named genera) other wild wheatgrass 3. [stem unassigned]

-**BTH**- 'AVENA (OAT)' 1. (*A. sativa*) common oat grass 2. (*A. nuda*) naked oat / hulless oat grass 3. (*A. abyssinica*) Ethiopian oat grass

-**BTÇ**- 'POOIDEAE' I 1. (Aveninae-type grasses; numerous genera) oatgrass, junegrass, vernal grass / hornwort, bentgrass, hairgrass, button-grass, plumegrass, hedgehog grass, false mannagrass, swamp wallaby grass, beard grass / rabbitsfoot grass, nit grass, Ozark grass, shining grass, silkybent grass, windgrass 2. (Poinae-type grasses; numerous genera) meadow-grass, bluegrass, tussock, snow-grass, sussock-grass, muttonbird poa, quaking grass, pendant grass, hardgrass, alkali grass, salt grass, saltmarsh-grass, fescue, pinegrass, blue bunchgrass, pseudovina, English bluegrass, fescue tussock, ryegrass, cock's-foot grass, orchard grass, hair grass, tussock grass, sandgrass, common rivergrass, giant mountain grass, timothy grass, sloughgrass, woodreed, foxtail grass, hooded grass 3. (Bromeae, Littledaleae, Brachypodieae; genera *Bromus, Littledalea, Brachypodium*) brome, chess grass, cheat grass, quake grass, bull grass, rescue grass / Australian oat, false brome, tor-grass

-**BTF**- 'POOIDEAE' II 1. (Nardeae; genus *Nardus*) matgrass 2. (Lygeeae; genus *Lygeum*) esparto grass / cord grass / albardine 3. (Duthieeae; several genera) oat speargrass

-**BTV**- 'POOIDEAE' III 1. (Meliceae; several genera) melic grass, semaphore grass, false semaphore grass, false mesic 2. (Stipeae; numerous genera) needle grass, spear grass, silver spike grass, ricegrass, sleepygrass, achnatherum, needle-and-thread grass, feather grass, false needle grass 3. (Ampelodesmeae; genus *Ampelodesmos*) stramma / rope grass / Mauritania grass / dis grass

-**BTĻ**- 'POOIDEAE' IV 1. (Diarrheneae; genera *Diarrhena, Neomolinia*) breakgrain / break grass 2. (genus *Ammophila*) beach grass, marram grass 3. (Brachyelytreae, Phaenospermateae, Brylkinieae; genera *Brachyelytrum, Phaenosperma, Brylkinia*) other pooid grass

Tracheophytes > Spermatophytes > Angiosperms > monocots > commelinid monocots > Poales > Graminids > Poaceae >Bambusoideae

-**BĐD**- 'BAMBUSOIDEAE (BAMBOO)' 1. (Bambuseae; numerous genera) tropical woody bamboo 2. (Olyreae; numerous genera) tropical herbaceous bamboo 3. (Arundinarieae; numerous genera) temperate woody bamboo, hill cane, switchcane, river cane / giant cane

Tracheophytes > Spermatophytes > Angiosperms > monocots > commelinid monocots > Poales > Graminids > Poaceae >Oryzoideae

-**BK**- 'ORYZA (RICE)' 1. (*O. sativa*) Asian rice grass 2. (*O. glaberrima*) African rice grass 3. (*O. rufipogon*) red rice / brownbeard rice grass

-**BKW**- 'ORYZEAE' I 1. (genus *Zizania*) wild rice, Canada rice , Indian rice, water oats 2. (genus *Zizaniopsis*) giant cutgrass / water millet / southern wildrice 3. (genus *Oryza* other than previously named species) wild rice grass

-**BKY**- 'ORYZEAE' II 1. (genus *Leersia*) cutgrass, whitegrass, swamp rice grass 2. (genus *Potamophila*) Hastings River reed 3. (genera *Luziola, Hyrgroryza*) watergrass

-**BKÇ**- 'ORYZOIDEAE' 1. (Ehrharteae; four genera) weeping grass / weeping rice grass / weeping meadow grass, veldtgrass, zotovia 2. (Phyllorachideae, Streptogyneae) other oryzoid grass 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > monocots > commelinid monocots > Poales > Graminids > Poaceae > Chlororidoideae

-**BKL**- 'ERAGROSTIDEAE' 1. (*Eragrostis tef*) teff grass 2. (genus *Eragrostis* other than *E. tef*) lovegrass, canegrass, cause grass 3. (Eragrostids other than *Eragrostis*; several genera) robies cocksfoot, thimble grass / Zulu fescue, other eragrostid grass

-**BKR**- 'ELEUSININAE' 1. (*Eleusine coracana*) finger millet grass 2.(genus *Eleusine* other than *E. coracana*) goosegrass, wiregrass, crowfootgrass, yard-grass 3. (Eleusininae other than *Eleusine*; numerous genera) caterpillar grass, windmill grass, Mitchell grass, Bermuda grass / dog's tooth grass

-**BKF**- 'CYNODONTEAE' I 1. (Triodiinae; genus *Triodia*) spinifex, porcupine grass 2. (Dactylocteniinae; several genera) goose grass, crowfoot grass 3. (Muhlenbergiinae; genus *Muhlenbergia*) muhly, scratchgrass, deergrass, bullgrass, aparejograss

-**BKŢ**- 'CYNODONTEAE' II 1. (Scleropogoninae; several genera) eyelashgrass, desertgrass, desert fluff-grass, woollygrass, burrograss, Eureka dunegrass, false buffalograss 2. (Orcuttiinae; genera *Orcuttia, Neostapfia*) Colusa grass, Orcutt grass 3. (Cynodonteae other than Stems 1 and 2; numerous genera) fiveminute grass, grama grass, pappusgrass, burr grass, carrot-seed grass, Texasgrass, sandgrass, skeletongrass, curly mesquite, other cynodontid grass

-**BKŘ**- 'ZOYSIEAE' 1. (genus *Zoysia*) zoysia / zoysiagrass, mascarene grass / Korean velvet grass, Korean lawngrass, Manila grass 2. (genus *Sporobolus*) dropseed, sacaton grass, fairy grass / yakka grass, smut grass, giant rat's-tail grass, poverty grass, salt grass 3. (genera *Psilolemma, Urochondra*) other zoysiid grass

-**BKM**- 'CHLORIDOIDEAE' 1. (Centropodieae; two genera) centropodiid grass 2. (Triraphideae; three genera) triraphid grass 3. (genera *Grossweilerochola, Indopoa, Lepturopetium, Myriostachya, Pogonochloa, Pseudozoysia, Silentvalleya*) other chloridoid grass

Tracheophytes > Spermatophytes > Angiosperms > monocots > commelinid monocots > Poales > Graminids > Poaceae > Panicoideae > Andropogoneae

-**RDL**- 'ANDROPOGONEAE' I 1. (genus *Zea*) maize / corn plant 2. (genus *Saccharum*) sugarcane plant, ravenna grass, elephant grass 3. (genus *Sorghum*) sorghum grass

-**RDR**- 'ANDROPOGONEAE' II 1. (genus *Ischaemum*) murainagrass, duck-beak, paddle grass, lucuntu grass 2. (genus *Tripsacum*) gamagrass 3. (genus *Hemarthria*) jointgrass / limpograss / halt grass / quick grass

-**RDŘ**- 'ANDROPOGONEAE' III 1. (genus *Arthraxon*) carpet grass 2. (genus *Chrysopogon*) vetiver, amorseco / grama-amorosa / lesser spear grass 3. (Andropogoneae other than previously named genera) browntop, other andropogonid grass

-**RDM**- 'ANDROPOGONINES & SACCHARINES' 1. (genus *Cymbopogon*) lemongrass, citronella grass, palmarosa / Indian geranium, barbed wire grass, silky head, oily head, Cochin grass, Malabar grass 2. (Andropogonines other than *Cymbopogon*; numerous genera) broomsedge, whiskygrass, bluestem, Tambuki grass / gamba grass, beard grass, sweetpitted grass, leg grass / red grass / redleg, Flinders grass, tanglehead, scented-top grass 3. (Saccharines other than *Saccharum*) cogongrass / kunai grass / blady grass / satintail / lalang grass / cotton wool grass, crinkleawn grass, silvergrass

-**RDŇ**- 'ANDROPOGONOID PANICOIDEAE' I 1. (Paspaleae; numerous genera) carpet grass, vine mesquite, bedgrass, panicgrass, bahiagrass, paspalum, koda millet / kodra millet, hilo grass / carabao grass, dallis grass, crowngrass, marsh grass 2. (Gynerieae; genus *Gynerium*) wildcane, arrow cane 3. (Arundinelleae, Lecomtelleae; several genera) other andropogonoid panicoid grass

Tracheophytes > Spermatophytes > Angiosperms > monocots > commelinid monocots > Poales > Graminids > Poaceae > Danthonioideae

-**RDF**- 'DANTHONIOIDEAE' 1. (genus *Danthonia*) oatgrass / heathgrass / wallaby grass 2. (genus *Schismus*) Mediterranean grass 3. (Danthonioideae other than Stems 1 and 2; numerous genera) danthonioid grass, snowgrass

Tracheophytes > Spermatophytes > Angiosperms > monocots > commelinid monocots > Poales > Graminids > Poaceae > Panicoideae > Paniceae

-**ŢXL**- 'MILLET' I 1. (*Panicum milleaceum*) common millet grass [a.k.a. proso millet, broomcorn millet, hog millet, red millet, white millet] 2. (*Pennisetum glaucum*) pearl millet grass 3. (*Setaria italica*) foxtail millet / Italian millet

-**ŢXR**- 'MILLET' II 1. (*Digitaria exilis, D. iburua*) fonio / hungry rice 2. (*D. compacta*) raishan 3. (genus *Digitaria* [other than *D. exilis, D. iburua* and *D. compacta*) crabgrass, finger-grass, summer grass, witchgrass, cottontop, umbrella grass, blue couch, pangolagrass, sourgrass, Polish millet

-**ŢXM**- 'CENCHRINAE' 1. (genus *Cenchrus*) buffelgrass, sandbur, sand spur 2. (genus *Pennisetum* [other than *P. glaucum*]) fountaingrass, swamp-foxtail, bristlegrass, feather grass / bedding grass, mission grass, Napier grass, elephant grass / barner grass, feathertop / white foxtail 3. (genus *Setaria* [other than *S. italica*] pigeon grass, bristle grass, highland pitpit, grama negra, yellow foxtail, bristly foxtail, corn grass / Mary grass

-**ŢXN**- 'PANICEAE' 1. (other Cenchrinae; numerous genera) mudgrass, false bristlegrass, flat-stem grass, sandhill canegrass 2. (Boivinellinae; numerous genera) basketgrass, barnyard grass / cockspur grass, boivinelline grass 3. (other Paniceae [including *Panicum* other than *P. milleaceum*]; numerous genera) mulga grass, rosette grass, cupscale grass, signalgrass, panicum, tumbleweed, maidencane, switchgrass, panic, Guinea grass / buffalo grass, panic grass

Tracheophytes > Spermatophytes > Angiosperms > monocots > commelinid monocots > Poales > Graminids > Poaceae > Panicoideae > Centothecoid clade

-**ŢXW**- 'CENTOTHECOID' 1. (Chasmanthieae; genera *Chasmanthium, Bromuniola*) woodoats 2. (Thysanolaeneae; genus *Thysanolaena*) tiger grass / broom stick / Nepalese broom grass 3. (Zeugiteae, Steyermarkochloeae, Tristachyideae, Centotheceae, Cyperochloeae; several genera) other centothecoid grass

Tracheophytes > Spermatophytes > Angiosperms > monocots > commelinid monocots > Poales > Graminids > Poaceae > Micrairoideae

-**ŢXX**- 'MICRAIROIDEAE' 1. (genus *Isachne*) bloodgrass 2. (genus *Eriachne*) Wanderrie grass 3. (Mcrairoideae other than Stems 1 and 2; several genera) macrairoid grass

Tracheophytes > Spermatophytes > Angiosperms > monocots > commelinid monocots > Poales > Graminids > Poaceae > Arundinoideae

-**ŢTĻ**- 'ARUNDINOIDEAE' I 1. (genus *Phragmites*) common reed 2. (genus *Arundo*) giant cane / Spanish cane, Pliny's reed 3. (genus *Amphipogon*) greybeard grass

-**ŢTŘ**- 'ARUNDINOIDEAE' II 1. (genus *Molinia*) moor grass 2. (genus *Monachather*) mulga oats 3. (Arundinoideae other than previously named genera; several genera) other arundinoid grass or reed

Tracheophytes > Spermatophytes > Angiosperms > monocots > commelinid monocots > Poales > Graminids > Poaceae > Puelioideae, Pharoideae, and Anomochlooideae

-**ŢTY**- 'OTHER POACEAE' 1. (Puelioideae; genera *Puelia, Guaduella*) puelioid grass 2. (Pharoideae; genera *Pharus*, *Leptaspis*) stalkgrass, leptaspis 3. (Anomochlooideae; genera *Anomochloa, Streptochaeta*) anomochlooid grass

Tracheophytes > Spermatophytes > Angiosperms > monocots > commelinid monocots > Poales > Graminids > Ecdeiocoleaceae, Joinvilleaceae, and Flagellariaceae

-**ŢTH**- 'OTHER GRAMMINID GRASS FAMILY' 1. (Ecdeiocoleaceae; genera *Ecdeiocolea, Georgeantha*) ecdeiocoleid plant 2. (Joinvilleaceae; genus *Joinvillea*) joinvillea 3. (Flagellariaceae; genus *Flagellaria*) whip vine / hell tail / supplejack / false rattan / bush cane

Tracheophytes > Spermatophytes > Angiosperms > monocots > commelinid monocots > Poales > Bromeliads > Bromeliaceae and Typhaceae

-**ĎKW**- 'BROMELIAD' 1. (genus *Ananas*) pineapple 2. (Bromeliaceae other than *Ananas*; numerous genera) bromeliad, airplant, ball moss, Spanish moss 3. (Typhaceae; genera *Typha, Sparganium*) cattail / punk / reed / bulrush / reedmace, bur-reed

Tracheophytes > Spermatophytes > Angiosperms > monocots > commelinid monocots > Poales > Cyperid clade

-**ĎKY**- 'CYPERUS' 1. (*C. esulentus*) chufa flatsedge [tuber used as food] 2. (*C. papyrus*) papyrush sedge 3. (other *Cyperus* species) flatsedge, nutsedge, umbrella-sedge

-**ĎKL**- 'CYPERACEAE' I 1. (genus *Carex*) [true] sedge 2. (*Eleocharis dulcis*) Chinese water chestnut 3. (other *Eleocharis* species) spikerush / hairgrass, spikesedge, scallion grass

-**ĎKR**- 'CYPERACEAE' II 1. (genus *Scirpus*) club-rush, bulrush, beakrush 2. (genus *Fimbristylis*) fimbry / fringe-rush / fimbristyle 3. (Cyperaceae other than previously named genera; numerous genera) bulrush, hairsedge, twigrush, bogrush, nutrush, sedge

-**ĎKF**- 'JUNCACEAE' 1. (genus *Juncus*) rush, blackgrass 2. (genus *Luzula*) woodrush, Good Friday grass / sweep's brush 3. (Juncaceae other than Stems 1 and 2; several genera) other rush-like plant

-**ĎKŢ**- 'OTHER CYPERID' 1. (Thurniacieae; genera *Thurnia, Prionium*) palmiet, thurnia 2. (Rapateaceae; numerous genera) rapateaceous plant 3. (Mayacaceae; genus Mayaca) mayaca plant

Tracheophytes > Spermatophytes > Angiosperms > monocots > commelinid monocots > Poales > Restiid and Xyrid clades

-**ḌKŘ**- 'POALES' 1. (Restionaceae [including Anarthriaceae, Centrolepidaceae, and Lyginiaceae]; numerous genera) wire rush, bamboo rush, silver reed, Cape thatching reed, rope-rush, buttonrush, cord-rush, tassel rush, plume rush, centrolepis, Manypeaks rush, Australian reed, restiad, restio 2. (Eriocaulaceae; several genera) pipewort, hat pin, bog button 3. (Xyridaceae; several genera) yellow-eyed grass

Tracheophytes > Spermatophytes > Angiosperms > Eudicots

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > Buxales

-**KMW**- 'BUXACEAE' 1. (genus *Buxus*) box, boxwood 2. (genus *Pachysandra*) Allegheny spurge, carpet box / Japanese spurge, pachysandra 3. (genus *Srcococca*) sweet box, Christmas box

-**KMY**- 'OTHER BUXALES' 1. (Didymelaceae; genus *Didymeles*) didymeles 2. (Haptanthaceae; genus *Haptanthus*) haptanthus 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > Proteales

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > Proteales > Proteaceae

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > Proteales > Proteaceae > Proteoideae

-**NḌPW**- 'PROTEOIDEAE' I 1. (genus *Protea*) sugarbush, protea, honey flower, creeping beauty 2. (genus *Petrophile*) petrophile, conebush, conesticks, pixie mops 3. (genus *Aulax*) featherbush

-**NḌPY**- 'PROTEOIDEAE' II 1. (genus *Conospermum*) smokebush, blue brother, blue lace 2. (genus *Synaphea*) synaphea 3. (genus *Stirlingia*) blueboy

-**NḌPL**- 'PROTEOIDEAE' III 1. (genus *Isopogon*) coneflower, drumsticks, wild Irishman / horny cone bush, pincushion rose bush, isopogon 2. (genus *Adenanthos*) jugflower, glandflower, basket flower, woollybush, flame bush / bridle bush / sweat bush, adenanthos 3. (genus *Cenarrhenes*) Port Arthur plum / native plum

-**NḌPR**- 'PROTEOIDEAE' IV 1. (genus *Leucadendron*) cone bush 2. (genus *Serruria*) spiderhead, blushing bride / pride of Franschhoek 3. (genus *Paranomus*) sceptre, exploding baked apple

-**NḌPŘ**- 'PROTEOIDEAE' V 1. (genus *Vexatorella*) vexator 2. (genus *Sorocephalus*) powderpuff 3. (genus *Spatalla*) spoon plant

-**NḌPF**- 'PROTEOIDEAE' VI 1. (genus *Leucospermum*) pincushion plant 2. (genus *Mimetes*) pagoda plant 3. (genus *Diastella*) silkypuff

-**NḌPḶ**- 'PROTEOIDEAE' VII 1. (genus *Orothamnus*) marsh rose 2. (Proteoideae other than named genera; several genera) other proteoid plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > Proteales > Proteaceae > Grevilleoideae

-**NḌKW**- 'GREVILLEOIDEAE' I 1. (genus *Grevillea*) spider flower, silky oak, toothbrush plant, grevillea 2. (genus *Opisthiolepis*) blush silky oak / pink silky oak / brown silky oak / drunk rabbit 3. (genus *Buckinghamia*) ivory curl flower

-**NḌKY**- 'GREVILLEOIDEAE' II 1. (genus *Hakea*) hakea, mulloway needlebush, needlewood, corkbark tree, red pokers, bootlace tree / corkwood, cork tree, needles and corks, dungyn, yandena, walukara, needle tree, woody peach, kerosine bush / two-leaf, pin bush / water tree / booldoobah, emu bush, corkwood oak, honey bush 2. (genus Stenocarpus) scrub beefwood, firewheel tree, stenocarpus 3. (genus Lomatia) crinkle bush / parsely fern, guitarplant, lomatia, native holly, radal

-**NḌKL**- 'GREVILLEOIDEAE' III 1. (genus *Embothrium*) Chilean firebush / notro / ciruelillo / foforito 2. (genus *Telopea*) waratah 3. (genus *Alloxylon*) Queensland tree waratah / red silky oak, Dorrigo waratah

-**NḌKR**- 'GREVILLEOIDEAE' IV 1. (genus *Banksia*) banksia, parrot bush, dryandra, kerosene bush, couch honeypot 2. (genus *Musgravea*) crater silky oak, briar silky oak 3. (genus *Austromuellera*) Mueller's silky oak

-**NḌKŘ**- 'GREVILLEOIDEAE' V 1. (genus *Orites*) orites, mountain silky oak / prickly ash / white beefwood, yellow bush 2. (genus *Neorites*) fishtail oak / fishtail silky oak 3. (genus *Triunia*) spice bush, red nut

-**NḌKF**- ‘GREVILLEOIDEAE’ VI 1. (genus *Xylomelum*) woody pear 2. (genus *Lambertia*) wild honeysuckle 3. (genus *Hollandaea*) Sayer’s silky oak, hollandaea

-**NḌKM**- ‘GREVILLEOIDEAE’ VII 1. (genus *Helicia*) helicia, silky oak, pale oak / brown oak, Austral oak, hairy weiner / rusty oak 2. (genus *Sphalmium*) mystery oak / Mt. Lewis oak / poorman’s fishtail oak / buff silky oak / satin silky oak 3. (genus *Carnarvonia*) red oak / red silky oak

-**NḌKN**- ‘GREVILLEOIDEAE’ VIII 1. (genus *Macadamia*) macadamia nut tree, bauple nut / Queensland nut / bush nut / Hawaii nut 2. (genus *Brabejum*) wild almond / bitter almond 3. (genus *Catalepidia*) Hey’s nut oak

-**NḌKS**- ‘GREVILLEOIDEAE’ IX 1. (genus *Athertonia*) Atherton oak 2. (genus *Cordwellia*) northern silky oak / bull oak / golden spanglewood / lacewood / congaary 3. (genus *Gevuina*) Chilean hazelnut tree

-**NḌKŠ**- ‘GREVILLEOIDEAE’ X 1. (genus *Hicksbeachia*) red bopple nut / monkey nut / red nut /beef nut / rose nut / ivory silky oak 2. (Grevilleoideae other than named genera) other grevilleoid plant/tree 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > Proteales > Proteaceae > remaining sub-families: Bellendonoideae, Persoonioideae, and Symphionematoideae

-**NḌKH**- ‘OTHER PROTEACEAE’ I 1. (genus *Bellendena*) mountain rocket 2. (genus *Placospermum*) rose silky oak / plate-seeded oak 3. (genus *Persoonia*) geebung, snottybobble, persoonia, wild pear

-**NḌKHW**- ‘OTHER PROTEACEAE’ II 1. (genus *Agastachys*) white waratah 2. (Proteaceae other than named genera) other proteaceous plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > Proteales > remaining families: Nelumbonaceae, Platanaceae, and Sabiaceae

-**NḌPH**- ‘OTHER PROTEALES’ 1. (Nelumbonaceae; genus *Nelumbo*) lotus flower / sacred lotus / Indian lotus / rose of India / sacred water lilly, American lotus 2. (Platanaceae; genus *Platanus*) plane tree, American sycamore / buttonwood / water beech / occidental plane tree, California sycamore / western sycamore / aliso, Arizona sycamore 3. (Sabiaceae; genera *Sabia, Ophiocaryon, Meliosma*) sabiaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > Ranunculales

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > Ranunculales > Ranunculaceae

-**VZMW**- ‘RANUNCULACEAE’ I 1. (genus *Ranunculus*) buttercup, spearwort, crowfoot, water-crowfoot 2. (genus *Ceratocephala*) bur buttercup / curveseed butterwort 3. (genus *Ficaria*) fig buttercup / lesser celandine / pilewort

-**VZMY**- ‘RANUNCULACEAE’ II 1. (genus *Myosurus*) mousetail 2. (genus *Trautvetteria*) Carolina bugbane / false bugbane / tassel-rue 3. (genus *Caltha*) marsh marigold, caltha

-**VZML**- ‘RANUNCULACEAE’ III 1. (genus *Anemone*) anemone, Spanish marigold / windflower, thimbleweed 2. (genus *Clematis*) clematis, leather flower, pine hyacinth, traveller’s joy, satin curls, virgin’s bower, headache vine, vasevine, curlyheads, bluebill, devil’s darning needles, old man’s beard 3. (genus *Hepatica*) liverleaf / liverwort / hepatica

-**VZMR**- ‘RANUNCULACEAE’ IV 1. (genus *Pulsatilla*) pasque flower, windflower, prairie crocus / cutleaf anemone 2. (genus *Helleborus*) hellebore, setterwort, Christmas rose, Lenten rose, bear’s foot 3. (genus *Nigella*) love-in-a-mist / ragged lady / devil in the bush

-**VZMŘ**- ‘RANUNCULACEAE’ V 1. (genus *Actaea*) baneberry, bugbane, cohosh, herb crhistopher, doll’s eyes 2. (genus *Anemonopsis*) false anemone 3. (genus *Eranthis*) white aconite

-**VZVW**- ‘RANUNCULACEAE’ VI 1. (genus *Aconitum*) aconite, monkshood, wolfsbane, iron hammer, leopard’s bane, mousebane, devil’s helmet, queen of poisons, blue rocket 2. (genus *Consolida*) larkspur 3. (genus *Delphinium*) larkspur, delphinium

-**VZVY**- ‘RANUNCULACEAE’ VII 1. (genus *Adonis*) pheasant’s-eye, blooddrops, adonis 2. (genus *Trollius*) globeflower 3. (genus *Glaucidium*) Japanese wood poppy

-**VZVL**- ‘RANUNCULACEAE’ VIII 1. (genus *Hydrastis*) goldenseal / orangeroot / yellow puccoon 2. (genus *Coptis*) goldthread 3. (genus *Xanthorhiza*) yellowroot

-**VZVR**- ‘RANUNCULACEAE’ IX 1. (genus *Thalictrum*) meadow rue, rue-anemone 2. (genus *Isopyrum*) false rue-anemone 3. (genus *Enemion*) false rue-anemone

-**VZVŘ**- ‘RANUNCULACEAE’ X 1. (genus *Aquilegia*) columbine / granny’s bonnet / granny’s nightcap 2. (genus *Paraquilegia*) spurless columbine 3. (Ranunculaceae other than named genera; several genera) other ranunculaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > Ranunculales > Berberidaceae

-**VZBW**- ‘BERBERIDACEAE’ I 1. (genus *Berberis*) barberry 2. (genus *Mahonia*) Oregon grape, Beale’s barberry, Cascade barberry, currant-of-Texas / wild currant / chaparral berry, mahonia 3. (genus *Achlys*) vanilla leaf / deer’s foot / sweet after death

-**VZBY**- ‘BERBERIDACEAE’ II 1. (genus *Podophyllum*) mayapple / American mandrake / wild mandrake / ground lemon 2. (genus *Sinopodophyllum*) Himalayan may apple / Indian may apple 3. (genus *Diphylleia*) umbrella leaf

-**VZBL**- ‘BERBERIDACEAE’ III 1. (genus *Epimedium*) barrenwort, bishop’s hat, fairy wings, horny goat weed 2. (genus *Jeffersonia* [including *Plagiorhegma*]) twinleaf / rheumatism root, Asian twinleaf 3. (genus *Vancouveria*) inside-out flower

-**VZBR**- ‘BERBERIDACEAE’ IV 1. (genus *Nandina*) heavenly bamboo / sacred bamboo / nandina 2. (genus *Caulophyllum*) blue cohosh / squaw root / papoose root 3. (Berberidaceae other than named genera; several genera) other berberidaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > Ranunculales > Menispermaceae

-**VZGW**- ‘MENISPERMACEAE’ I 1. (genus *Menispermum*) moonseed, ghost grape, common parilla 2. (genus *Legnephora*) round-leaf vine 3. (genus *Sarcopetalum*) pearl vine

-**VZGY**- ‘MENISPERMACEAE’ II 1. (genus *Cocculus*) red-berried moonseed / Carolina moonseed, correhuela, broom creeper, laurel-leaved snail tree, queen coralbead 2. (genus *Cissampelos*) velvet leaf, abuta 3. (genus *Stephania*) snake vine

-**VZGL**- ‘MENISPERMACEAE’ III 1. (genus *Calycocarpum*) cupseed 2. (genus *Dioscoreophyllum*) serendipity berry 3. (genus *Jateorhiza*) calumba

-**VZGR**- ‘MENISPERMACEAE’ IV 1. (genus *Anamirta*) Indian berry / fishberry / Levant nut 2. (Menispermaceae other than named genera; numerous genera) other menispermaceous plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > Ranunculales > Lardizabalaceae

-**VZBŘ**- ‘LARDIZABALACEAE’ I 1. (genus *Lardizabala*) zabala fruit 2. (genus *Boquila*) chameleon vine 3. (genus *Decaisnea*) dead man’s fingers

-**VZBV**- ‘LARDIZABALACEAE’ II 1. (genus *Holboellia*) blue china vine 2. (genus *Akebia*) chocolate vine, akebia 3. (Lardizabalaceae other than named genera; several genera) other lardizabalaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > Ranunculales > Papaveraceae

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > Ranunculales > Papaveraceae > Papaveroideae

-**VZDW**- ‘PAPAVEROIDEAE’ I 1. (genus *Papaver*) poppy, blindeyes 2. (genus *Arctomecon*) bear poppy / bear-paw poppy 3. (genus *Argemone*) prickly poppy

-**VZDY**- ‘PAPAVEROIDEAE’ II 1. (genus *Canbya*) pygmy poppy 2. (genus *Meconopsis*) blue poppy, autumn poppy, Nepal poppy / satin poppy, Himalayan woodland-poppy 3. (genus *Romneya*) matilija poppy, California tree poppy

-**VZDL**- ‘PAPAVEROIDEAE’ III 1. (genus *Stylomecon*) windpoppy 2. (genus *Meconella*) fairypoppy 3. (genus *Platystemon*) creamcups

-**VZDR**- ‘PAPAVEROIDEAE’ IV 1. (genus *Eschscholzia*) California poppy, tufted poppy / foothill poppy, frying pans, desert gold(en) poppy / Mojave poppy, San Benito poppy, Lemmon’s poppy, pygmy poppy, Parish’s poppy, Channel Islands poppy 2. (genus *Dendromecon*) bush poppy, Channel Islands tree poppy 3. (genus *Hunnemannia*) Mexican tulip poppy

-**VZDŘ**- ‘PAPAVEROIDEAE’ V 1. (genus *Chelidonium*) celandine / greater celandine / nipplewort / swallowwort / tetterwort 2. (genus *Bocconia*) plume poppy / tree poppy / tree celandine / parrotweed / sea oxeye daisy / John Crow bush 3. (genus *Dicranostigma*) eastern horned poppy

-**VZDV**- ‘PAPAVEROIDEAE’ VI 1. (genus *Eomecon*) snow-poppy / dawn-poppy 2. (genus *Glaucium*) yellow horned poppy, sea poppy 3. (genus *Hylomecum*) forest poppy

-**VZZ̦**- ‘PAPAVEROIDEAE’ VII 1. (genus Macleaya) plume poppy 2. (genus Sanguinaria) bloodroot / bloodwort / redroot / red puccoon / Canada puccoon / pauson 3. (genus Stylophorum) celandine-poppy / woods-poppy

-**VZJ**- ‘PAPAVEROIDEAE’ VIII 1. (genus *Hesperomecon*) narrow-leaved meconella 2. (*Roemeria*) spotted Asian poppy 3. (Papaveroideae other than named genera) other papaveroid plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > Ranunculales > Papaveraceae > Fumarioideae

-**VZGV**- ‘FUMARIOIDEAE’ I 1. (genus *Fumaria*) fumitory, ramping-fumitory, earth smoke 2. (genus *Pseudofumaria*) pale corydalis / white coridalys 3. (genus *Corydalis*) fumewort, corydalis, scrambled eggs / golden smoke

-**VZGZ**- ‘FUMARIOIDEAE’ II 1. (genus *Capnoides*) harlequin corydalis / pale corydalis / pink corydalis 2. (genus *Dicentra*) bleeding-heart, steer’s-head, turkey-corn, squirrel-corn, Dutchman’s-breeches 3. (genus *Ichtyoselmis*) large-flowered dicentra

-**VZGŽ**- ‘FUMARIOIDEAE’ III 1. (genus *Ehrendorferia*) eardrops, yellow bleeding-heart 2. (genus Lamprocapnos) Asian bleeding-heart 3. (Fumarioideae other than named genera; seveal genera) other fumarioid plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > Ranunculales > remaining families: Eupteleaceae and Circaeasteraceae

-**VZGŘ**- ‘EUPTELEACEAE & CIRCAEASTERACEAE’ 1. (Eupteleaceae; genus *Euptelea*) euptelia 2. (Circaesteraceae; genera *Circaeaster, Kingdonia*) circaeasteraceous plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > Trochodendrales

-**VZḌ**- ‘TROCHODENDRACEAE’ 1. (genus *Trochodendron*) wheel tree 2. (genus *Tetracentron*) tetracentron 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > Gunnerales

-**VZNW**- ‘GUNNERALES’ 1.(Gunneraceae; genus *Gunnera*) giant rhubarb / Chilean rhubarb, gunnera, poor man’s umbrella 2. (Myrothamnaceae; genus *Myrothamnus*) resurrection plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > Dilleniales

-**VZNY**- ‘DILLENIACEAE’ 1. (genus *Hibbertia*) guinea flower, yellow buttercups, hibbertia 2. (genus *Dillenia*) red beech / golden guinea tree, elephant apple tree 3. (Delleniaceae other than Stems 1 and 2; several genera) other dilleniaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Campanulids

-**FTL**- 'AQUIFOLIALES' 1. (Aquifoliaceae; genus *Ilex*) holly 2. (Cardiopteridaceae, Helwingiaceae, Phylonomaceae, Stemonuraceae; several genera) other aquifoliate plant 3. [stem unassigned]

-**FTÇ**- 'ESCALLONIACEAE' 1. (genus *Escallonia*) cloven gum box, redclaws, escallonia 2. (genus *Anopterus*) native laurel / Tasmanian laurel, queensland laurel / Macleay laurel 3. (Escalloniaceae other than *Escallonia* and *Anopterus*) other escalloniaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Campanulids > Dipsacales

-**FMW**- 'ADOXACEAE' 1. (genus *Adoxa*) moschatel / hollowroot / muskroot / five-faced bishop / townhall clock / tuberous crowfoot 2. (genus *Sambucus*) elderberry 3. (genus *Viburnum* and other several other Axodoceae genera) viburnum / arrowwood, wild raisin / Appalachian tea, squashberry / mooseberry / highbush cranberry / lowbush cranberry, blackhaw, stag bush, Guelder-rose, Japanese snowball, nannyberry / sheepberry, hobble-bush, wayfaring tree, hoarwithy, laurustinus / laurustine

-**FMY**- 'CAPRIFOLIOIDEAE' 1. (genus *Lonicera*) honeysuckle 2. (genus *Symphoricarpos*) snowberry / waxberry / ghostberry / coralberry 3. (genus *Triosteum*) horse-gentian / feverwort

-**FNW**- 'VALERIANOIDEAE' I 1. (genus *Valeriana*) valerian 2. (genus *Valerianella*) cornsalad 3. (genus *Plectritis*) seablush, plectritis

-**FNY**- 'VALERIANOIDEAE' II 1. (genus *Nardostachys*) spikenard plant 2. (genus *Centranthus*) narrow-leaved valerian, long-spurred valerian, red balerian 3. (genus *Patrinia*) patrinia

-**FŇW**- 'CAPRIFOLIACEAE' 1. (genus *Diervilla*) bush honeysuckle 2. (genus *Linnaea*) twinflower, beautybush, abelia 3. (Caprifoliaceae other than named genera; several genera) other caprifoliate plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Campanulids > Apiales

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Campanulids > Apiales > Apiaceae

-**KPW**- 'APIEAE' I 1. (genus *Apium*) celery plant, marshwort 2. (genus *Foeniculum*) fennel plant 3. (genus *Petroselinum*) parsley plant

-**KPY**- 'APIEAE' II 1. (genus *Anethum*) dill plant 2. (genus *Ammi*) bishop's weed / bullwort / lady's face / laceflower, toothpick-plant / toothpickweed / bisnaga / khella 3. (Apieae other than named genera; several genera) other apiid plant

-**KPL**- 'SELINEAE' I 1. (genus *Selinum*) milk-parsley 2. (genus *Aethus*) fool's parsley / fool's cicely / poison parsley 3. (genus *Tauschia*) umbrellawort

-**KPR**- 'SELINEAE' II 1. (genus *Angelica*) angelica 2. (genus *Ammoselinum*) sandparsley 3. (genus *Apiastrum*) mock parsley

-**KPŘ**- 'SELINEAE' III 1. (genus *Arracacia*) arracacha / racacha 2. (genus *Cnidium*) cnidium [source of coumarins] 3. (genus *Cymopterus*) springparsley, Indian parsnip, wavewing

-**KPF**- 'SELINEAE' IV 1. (genus *Glehnia*) beach silvertop / American silvertop 2. (genus *Lomatium*) buscuitroot / Indian parsley / desert parsley 3. (genus *Oreonana*) mountainparsley

-**KPŢ**- 'SELINEAE' V 1. (genus *Orogenia*) Indian potato 2. (genus *Podistera*) podistera, woodroot 3. (genus *Saposhnikovia* [a.k.a. *Ledebouriella*]) siler

-**KPÇ**- 'SELINEAE' VI 1. (genus *Seseli*) moon carrot, stone-parsley 2. (genus *Sphenosciadium*) button parsley / woollyhead parsnip / ranger's buttons / button parsley / swamp white heads 3. (genus *Taenidia*) pimpernel

-**KPĻ**- 'SELINEAE' VII 1. (genus *Thaspium*) meadow-parsnip 2. (genus *Zizia*) alexanders, zizia 3. (Selineae other than named genera; several genera) other selenid parsley-like plant

-**KPH**- 'ACIPHYLLEAE' 1. (genus *Aciphylla*) golden speargrass / golden Spaniard, giant speargrass, mountain celery / snow aciphyll 2. (genus *Anisotome*) Campbell Island carrot, mountain celery 3. (Aciphylleae other than Stems 1 and 2; several genera) other acyphill plant

-**KBW**- ‘SCANDICEAE’ I 1. (genus *Daucus*) carrot plant 2. (genus *Cuminum*) cumin plant 3. (genus *Anthriscus*) chervil

-**KBY**- ‘SCANDICEAE’ II 1. (genus *Ferula*) galbanum plant, asafoetida plant, giant fennel 2. (genus *Chaerophyllum*) hairyfruit chervil / Southern chervil, bulbous chervil / parsnip chervil 3. (genus *Myrrhis*) myrrh / cicely / sweet chervil

-**KBL**- ‘SCANDICEAE’ III 1. (genus *Osmorhiza*) sweet cicely / sweetroot 2. (genus *Scandix*) shepherd’s-needle / Stork’s needle / Venus’ comb 3. (Scandiceae other than named genera; several genera) hedge parsley, false carrot

-**KBR**- ‘SANICULEAE’ I 1. (genus *Sanicula*) sanicle, black snakeroot, turkey pea 2. (genus *Eryngium*) eryngo, sea holly, rattlesnake master, beethistle, coyote thistle, snakeroot, blue devil, button-celery, prickfoot 3. (genus *Astrantia*) masterwort

-**KBŘ**- ‘SANICULEAE’ II 1. (genus *Hacquetia*) hacquetia 2. (Sanculeae other than named genera; several genera) other sanicle-like plant 3. [stem unassigned]

-**KDW**- ‘OENANTHEAE’ I 1. (genus *Berula*) water parsnip 2. (genus *Cicuta*) water hemlock 3. (genus *Cryptotaenia*) honewort

-**KDY**- ‘OENANTHEAE’ II 1. (genus *Ptilimnium*) harperella, bishopweed, laceflower 2. (genus *Lilaeopsis*) grasswort, mudflat quillplant 3. (genus *Oenanthe*) water dropwort, Chinese celery / Indian pennywort / Japanese parsley / water celery

-**KDL**- ‘OENANTHEAE’ III 1. (genus *Perideridia*) yampa / yampah, squaw potato, Sierra Queen Anne’s lace 2. (genus *Sium*) water parsnip 3. (Oenantheae other than named genera) other oenanthid plant

-**KDR**- ‘MACKINLAYOIDEAE’ I 1. (genus *Platysace*) platysace, karna, karno, tapeworm plant, Ravensthorphe radish, carrot tops 2. (genus *Actinotus*) flannel flower 3. (genus *Centella*) Asiatic pennywort / Indian pennywort

-**KDŘ**- ‘MACKINLAYOIDEAE’ II 1. (genus *Xanthosia*) xanthosia, southern cross 2. (Mackinlayoideae other than named genera) other mackinlayoid plant

-**KŠP**- ‘APIOIDEAE’ I 1. (genus *Pastinaca*) parsnip plant 2. (genus *Levisticum*) lovage 3. (genus *Ligusticum*) licorice-root

-**KŠPW** ‘APIOIDEAE’ II 1. (genus *Pimpinella*) anise, short-fruit pimpinella / chamnamul, burnet-saxifrage 2. (genus *Carum*) caraway / meridian fennel / Persian cumin 3. (genus *Coriandrum*) coriander / Chinese parsley / cilantro

-**KŠPY**- ‘APIOIDEAE’ III 1. (genus *Bunium*) great pignut / earthnut, black cumin / blackseed / black caraway 2. (genus *Crithmum*) rock samphire / sea fennel 3. (genus *Cyclospermum*) marsh parsley / fir-leafed celery

-**KŠPL**- ‘APIOIDEAE’ IV 1. (genus *Heracleum*) hogweed, giant hogweed, cow parsnip 2. (genus *Smyrnium*) alexanders / black-lovage / horse-parsley 3. (genus *Tordylium*) hartwort

-**KŠPR**- ‘APIOIDEAE’ V 1. (genus *Bupleurum*) hare’s ear, thorow-wax, thoroughwax 2. (genus *Falcaria*) sickleweed / longleaf 3. (genus *Erigenia*) harbinger of spring, pepper-and-salt

-**KŠPŘ**- ‘APIOIDEAE’ VI 1. (genus *Conium*) hemlock 2. (genus *Conioselinum*) hemlock-parsley 3. (genus *Opopanax*) Hercules-all-heal

-**KŠPḶ**- ‘APIOIDEAE’ VII 1. (genus *Silaum*) pepper-saxifrage 2. (genus *Trachyspermum*) adjwain / ajowan / bishop’s weed / carom, radhuni / wild celery 3. (Apioideae other than named genera; numerous genera) other apioid plant

-**KŠPH**- ‘OTHER APIACEAE’ 1. (Azorelloideae, numerous genera) azorella, cushion plant, llareta / yareta 2. (Apaiaceae other than named genera; numerous genera) other apaiaceous plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Campanulids > Apiales > Araliaceae

-**KSBW**- ‘ARALIACEAE’ I 1. (genus *Hedera*) ivy 2. (genus *Panax*) ginseng 3. (genus *Aralia*) spikenard, aralia, sarsaparilla, devil’s walking stick / Hercules’ club / angelica-tree / prickly ash / prickly elder

-**KSBY**- ‘ARALIACEAE’ II 1. (genus *Asrotricha*) star-hair, flannel leaf 2. (genus *Kalopanax*) castor aralia / tree aralia / prickly castor oil tree 3. (genus *Oplopanax*) devil’s club / Alaskan ginseng

-**KSBL**- ‘ARALIACEAE’ III 1. (genus *Polyscias*) parasol tree, Ming aralia, island pine, elderberry panax / small basswood, shield arala / plum aralia, pencil cedar, celery wood 2. (genus *Plerandra*) false aralia 3. (genus *Seemannaralia*) wild-maple / mock carrot tree

-**KSBR**- ‘ARALIACEAE’ IV 1. (genus *Tetrapanax*) rice-paper plant 2. (genus *Trachymene*) wild parsnip, white parsnip, sponge-fruit, blue lace flower, purple parsnip 3. (genus *Hydrocotyle*) water pennywort, marsh pennywort, waxweed, floating pennyroyal, whorled pennyroyal

-**KSBŘ**- ‘ARALIACEAE’ V 1. (genus *Stilbocarpa*) Macquarie Island cabbage 2. (Araliaceae other than named genera) other araliaceous plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Campanulids > Apiales > remaining families

-**KSBV**- ‘OTHER APIALES 1. (Pittosporaceae; several genera) pittosporum, billardiera, pronaya, Australian bluebell, appleberry, blackthorn, finger-flower, cheesewood, petroleum nut, lemonwood, native frangipani, marianthus, white marianth 2. (Torricelliaceae, Griseliniaceae, Myodocarpaceae, Pennantiaceae; numerous genera) other apiales plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Campanulids > Asterales

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Campanulids > Asterales > Asteraceae

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Campanulids > Asterales > Asteraceae > Asteroideae

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Campanulids > Asterales > Asteraceae > Asteroideae > Asterodae

-**RŽB**- ‘ASTEREAE’ I 1. (genus *Bellis*) daisy 2. (genus *Aphanostephus*) lazydaisy 3. (genus *Astranthium*) western-daisy

-**RŽBW**- ‘ASTEREAE’ II 1. (genus *Aster*) aster 2. (genus *Bradburia*) goldenaster 3. (genus *Sericocarpus*) white-topped aster

-**RŽBY**- ‘ASTEREAE’ III 1. (genus *Boltonia*) doll’s-daisy 2. (genus *Egletes*) tropic daisy 3. (genus *Pentachaeta*) pygmydaisy

-**RŽBL**- ‘ASTEREAE’ IV 1. (genera *Machaeranthera*, *Psilactis, Rayjacksonia*) tansyaster 2. (genus *Arida*) desert tansy-aster 3. (genus *Doelingeria*) tall flat-topped aster

-**RŽBR**- ‘ASTEREAE’ V 1. (genus *Xanthisma*) sleepydaisy 2. (genus *Corethrogyne*) sandaster 3. (genus *Xylorhiza*) woody-aster

-**RŽBŘ**- ‘ASTEREAE’ VI 1. (genus *Solidago*) goldenrod 2. (genus *Bigelowia*) rayless-goldenrod 3. (genus *Cuniculotinus*) rock goldenrod

-**RŽBV**- ‘ASTEREAE’ VII 1. (genus *Petradoria*) rock goldenrod 2. (genus *Xylothamia*) desert goldenrod 3. (genus *Brintonia*) rayless mock goldenrod

-**RŽD**- ‘ASTEREAE’ VIII 1. (genus *Almutaster*) alkali marsh aster 2. (genus *Ampelaster*) climbing aster 3. (genus *Callistephus*) China aster, annual aster

-**RŽDW**- ‘ASTEREAE’ IX 1. (genus *Brachyscome*) yellow-tongue daisy, varable daisy, grassland daisy, rock daisy, tufted daisy, mountain daisy 2. (genus *Calotis*) burr-daisy 3. (genus *Celmisia*) New Zealand aster, New Zealand daisy, cotton daisy, snow-daisy

-**RŽDY**- ‘ASTEREAE’ X 1. (genus *Eastwoodia*) yellow aster, yellow mock aster 2. (genus *Dieteria*) hoary-aster, hoary tansyaster 3. (genus *Eucephalus*) elegant aster, Olympic Mountain aster, Siskiyou aster, Klickitat aster, Cascade aster, rayless aster, wayside aster

-**RŽDL**- ‘ASTEREAE’ XI 1. (genus *Euthamia*) goldentop, grass-leaved goldenrod 2. (genus *Gundlachia*) goldenshrub 3. (genus *Pyrrocoma*) goldenweed

-**RŽDR**- ‘ASTEREAE’ XII 1. (genus *Herrickia*) Wasatch aster, horrid herrickia, King’s serpentweed / King’s aster, gray aster 2. (genus *Oclemena*) bog aster / bog nodding aster, whorled wood aster 3. (genus *Eurybia*) mountain aster / mountain wood aster / Applalachian heartleaf aster, slender aster, western showy aster, white-wood aster, thistleleaf aster / coyote-thistle aster, forked aster, southern prairie aster, thickstem aster, bigleaf aster / largeleaf aster, subalpine aster / arctic aster, bouquest aster / dwarf aster, southern swamp aster, rough wood aster / low rough aster, roughleaf aster, nettle-leaved Michaelmas-daisy, Siberian aster, purple wood aster / showy aster, pinewoods aster, creeping aster

-**RŽDŘ**- ‘ASTEREAE’ XIII 1. (genus *Stenotus*) goldenweed, mock goldenweed 2. (genus *Nestotus*) goldenweed, mock goldenweed 3. (genus *Oonopsis*) false goldenweed

-**RŽDV**- ‘ASTEREAE’ XIV 1. (genus *Oreostemma*) Peirson’s aster / Peirson’s mountaincrown, tundra aster / Anderson’s mountain crown, Plumas alpine aster 2. (genus *Ionactis*) ankle-aster 3. (genus *Symphyotrichum*) Pacific aster, meadow aster, western aster / intermountain aster, blue wood aster, serpentine aster, heath aster, alkali aster, smooth aster / glaucous aster, calico aster, aromatic aster, frost aster / hairy aster, sky-blue aster / azure aster, crooked-stem aster, purplestem aster / red-stemmed aster, silky aster, bushy aster / rice-button aster, New England aster, panicled aster / tall white aster

-**RŽG**- ‘ASTEREAE’ XV 1. (genus *Olearia*) daisy-bush 2. (genus *Townsendia*) Townsend daisy 3. (genus *Vittadinia*) New Holland daisy

-**RŽGW**- ‘ASTEREAE’ XVI 1. (genus *Croptilon*) scratchdaisy 2. (genus *Chaetopappa*) leastdaisy 3. (genus *Chrysopsis*) golden aster

-**RŽGY**- ‘ASTEREAE’ XVII 1. (genus *Tonestus*) serpentweed 2. (genus *Toiyabea*) alpine serpentweed 3. (genus *Gutierrezia*) snakeweed, matchweed

-**RŽGL**- ‘ASTEREAE’ XVIII 1. (genus *Ericameria*) goldenrush 2. (genus *Neonesomia*) goldenshrub 3. (genus *Isocoma*) jimmyweed, goldenweed

-**RŽGR**- ‘ASTEREAE’ XIX 1. (genus *Benitoa*) benitoa 2. (genus *Tracyina*) Indian headdress / beaked tracyina 3. (genus *Amphipappus*) chaffbush, eytelia

-**RŽGŘ**- ‘ASTEREAE’ XX 1. (genus *Erigeron*) fleabane 2. (genus *Grindelia*) gum-plant, resin-weed 3. (genus *Heterotheca*) goldenaster, telegraphweed, Mexican arnica, Zion golden aster, camphorweed, false golden aster, monarch golden golden aster

-**RŽGV**- ‘ASTEREAE’ XXI 1. (genus *Chrysothamnus*) rabbitbrush 2. (genus *Lorandersonia*) rabbitbush 3. (genus *Monoptilon*) desertstar

-**RŽGḌ**- ‘ASTEREAE’ XXII 1. (genus *Hazardia*) bristleweed 2. (genus *Rigiopappus*) wireweed 3. (genus *Amphiachyris*) broomweed

-**RŽM**- ‘ASTEREAE’ XXIII 1. (genus *Kippistia*) fleshy minuria 2. (genus *Gymnosperma*) gumhead, sticky selloa 3. (genus *Tetramolopium*) pamakani, tetramolopium

-**RŽMW**- ‘ASTEREAE’ XXIV 1. (genus *Triniteurybia*) Idaho goldenweed 2. (genus *Laennecia*) horseweed 3. (genus *Lessingia*) lessingia

-**RŽMY**- ‘ASTEREAE’ XXV 1. (genus *Miyamayomena*) Korean starwort 2. (genus *Lagenophora*) lagenophora, bottle-daisy 3. (Astereae other than named genera; numerous genera) other aster-like or daisy-like plant

-**RŽT**- ‘ANTHEMIDEAE’ I 1. (genus *Chrysanthemum*) chrysanthemum, mum, chrysanth 2. (genus *Chamaemelum*) chamomile, dogfennel 3. (genus *Anthemis*) chamomile, dog-fennel, mayweed

-**RŽTW**- ‘ANTHEMIDEAE’ II 1. (genus *Achillea*) yarrow, milfoil / allheal / thousand-leaf / bloodwort / carpenter’s grass / cammock / nosebleed / green adder’s mouth / soldier’s woundwort / dog daisy / old-man’s pepper, sweet Nancy / English mace, sneezewort /sneezeweed 2. (genus *Anacyclus*) anacyclus, white buttons, pellitory, Spanish chamomile / Mount Atlas daisy 3. (genus *Artemisia*) mugwort, wormwood, terragon, estragon, sagebrush, budsage

-**RŽTY**- ‘ANTHEMIDEAE’ III 1. (genus *Nipponanthemum*) Montauk daisy / Nippon daisy 2. (genus *Otanthus*) cottonweed plant 3. (genus *Leucanthemum*) ox-eye daisy / dog daisy, Shasta daisy, creeping daisy / mini marguerite, max chrysanthemum

-**RŽTL**- ‘ANTHEMIDEAE’ IV 1. (genus *Cotula*) water buttons / buttonweeds 2. (genus *Heteranthemis*) oxeye, sticky oxeye 3. (genus *Santolina*) cotton lavender / lavender-cotton

-**RŽTR**- ‘ANTHEMIDEAE’ V 1. (genus *Tanacetum*) tansy, golden-buttons, dusty-miller / silver-lace, feverfew, pyrethrum, painted daisy / Persian insect-flower, costmary 2. (genus *Soliva*) burr-weed 3. (genus *Tripleurospermum*) mayweed

-**RŽTŘ**- ‘ANTHEMIDEAE’ VI 1. (genus *Ismelia*) tricolour chrysanthemum / tricolor daisy / annual chrysanthemum 2. (genus *Rhodanthemum*) Moroccan daisy 3. (Anthemideae other than named genera; numerous genera) other anthemid plant

-**RŽTF**- ‘CALENDULEAE’ I 1. (genus *Calendula*) marigold, ruddle 2. (genus *Osteospermum*) daisybush, African daisy, Cape marguerite, Stinking Roger 3. (genus *Dimorphoteca*) Cape marigold / Namaqualand daisy, Cape daisy / rain daisy, rain flower / white bietou

-**RŽTÇ**- ‘CALENDULEAE’ II 1. (genus *Chrysantemoides*) bitou bush / bietou / tick berry 2. (Calenduleae other than named genera; several genera) other calendulid plant 3. [stem unassigned]

-**RŽTĻ**- ‘GNAPHALIEAE’ I 1. (genus *Gnaphalium*) cudweed 2. (genus *Gamochaeta*) everlasting, featherweed, cudweed 3. (genus *Antennaria*) catsfoot, pussytoes, everlasting

-**RŽČ**- ‘GNAPHALIEAE’ II 1. (genus *Anaphalis*) pearl, pearly everlasting 2. (gensus *Craspedia*) billy button / woollyhead 3. (genus *Pycnosorus*) drumstick / billy button

-**RŽTH**- ‘GNAPHALIEAE’ III 1. (genus *Rhodanthe*) sunray, pink paper daisy 2. (genus *Cassinia*) cassinia, dogwood / dolly bush, chinese scrub, cough bush / dead finish, bully bush / killmoke 3. (genus *Ozothamnus*) cottonwood / tauhinu, kerosene bush, rice flower / white dogwood / pill flower / sago bush, everlasting

-**RŽTHW**- ‘GNAPHALIEAE’ IV 1. (genus *Euchiton*) creeping cudweed, star cudweed / commond cudweed 2. (genus *Helichrysum*) curry plant, everlasting, strawflower, Cape gold, helichrysum, gold-and-silver 3. (Gnaphalieae other than named genera; numerous genera) other gnaphaliid plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Campanulids > Asterales > Asteraceae > Asteroideae > Senecionodae

-**RZB**- ‘SENECIONEAE’ I 1. (genus *Senecio*) ragwort, groundsel, squaw-weed, senecio, marsh fleabane, butterweed, dolphin necklace / flying dolphins / dolphin plant, wax ivy, Cape ivy / German ivy, string of bananas, string of pearls, old-man-in-the-spring 2. (genus *Arnoglossum*) Indian plantain 3. (genus *Barkleyanthus*) willow ragwort / willow groundsel

-**RZBW**- ‘SENECIONEAE’ II 1. (genus *Blennosperma*) stickyseed 2. (genus *Cacalioposis*) silvercrown 3. (genus *Crassocephalum*) ragleaf, thickhead

-**RZBY**- ‘SENECIONEAE’ III 1. (genus *Crocidium*) spring gold 2. (genus *Doronicum*) leopard’s bane 3. (genus *Emilia*) tasselflower

-**RZBL**- ‘SENECIONEAE’ IV 1. (genus *Erechtites*) fireweed, burnweed 2. (genus *Gynura*) purple passion 3. (genus *Lepidospartum*) broomsage, scalebroom

-**RZBR**- ‘SENECIONEAE’ V 1. (genus *Luina*) silverback 2. (genus *Petasites*) butterbur, coltsfoot 3. (genus *Psacalium*) Indian bush

-**RZBŘ**- ‘SENECIONEAE’ VI 1. (genus *Roldana*) groundsel 2. (genus *Rugelia*) Rugel’s ragwort / Rugel’s Indian plantain 3. (genus *Sinosenecio*) butterweed

-**RZBV**- ‘SENECIONEAE’ VII 1. (genus *Tetradymia*) horsebrush 2. (genus *Tussilago*) coltsfoot 3. (genus *Yermo*) desert yellowhead

-**RZBĐ**- ‘SENECIONEAE’ VIII 1. (genus *Bedfordia*) blanket leaf 2. (genus *Farfugium*) leopard plant / green leopard plant 3. (Senecioneae other than named genera; numerous genera) other senecionid plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Campanulids > Asterales > Asteraceae > Asteroideae > Helianthodae

-**MZB**- ‘COREOPSIDEAE’ I 1. (genus *Dahlia*) dahlia 2. (genus *Cosmos*) cosmos 3. (genus *Coreopsis*) calliopsis, tickseed

-**MZBW**- ‘COREOPSIDEAE’ II 1. (genus *Bidens*) beggarticks, tickseed sunflower, water-marigold, Spanish needles, bur-marigold, devil’s pitchfork / devil’s bootjack / pitchfork weed, sticktight, Meskel flower, black jack 2. (genus *Thelesperma*) greenthread, Navajo tea, Hopi tea 3. (Coreopsideae other than named genera; numerous genera) other coreopsid plant

-**MZBY**- ‘MADIEAE’ I 1. (genus *Madia*) madia, tarweed 2. (genera *Hemizonia, Hemizonella, Deinandra, Centromadia, Holcarpha*) tarweed, spikeweed 3. (genus *Blepharizonia*) big tarweed, big tarplant

-**MZBL**- ‘MADIEAE’ II 1. (genus *Argyroxiphium*) silversword, greensword 2. (genus *Wilkesia*) iliau 3. (genus *Arnica*) arnica, leopardbane, mountain tobacco

-**MZBR**- ‘MADIEAE’ III 1. (genus *Blepharipappus*) rough eyelash / rough eyelashweed 2. (genus *Calycadenia*) western rosinweed 3. (genus *Constancea*) Nevin’s woolly sunflower

-**MZBŘ**- ‘MADIEAE’ IV 1. (genus *Eatonella*) white false tickhead 2. (genus *Eriophyllum*) woolly sunflower 3. (genus *Guardiola*) Apache plant

-**MZBV**- ‘MADIEAE’ V 1. (genus *Harmonia*) harmonia, serpentine tarweed, nodding madia 2. (genus *Holozonia*) whitecrown 3. (genus *Hulsea*) alpinegold

-**MZD**- ‘MADIEAE’ VI 1. (genus *Kyhosia*) kyhosia, Bolander’s madia 2. (genus *Lagophylla*) hareleaf 3. (genus *Lasthenia*) goldfields

-**MZDW**- ‘MADIEAE’ VII 1. (genus *Layia*) tidytips 2. (genus *Monolopia*) monolopia, San Joaquin woollythread 3. (genus *Osmadenia*) false rosinweed

-**MZDY**- 'MADIEAE' VIII 1. (genus *Pseudobahia*) sunburst 2. (genus *Raillardella*) raillardella 3. (genus *Syntrichopappus*) Frémont's-gold, xerasid

-**MZDL**- 'MADIEAE' IX 1. (genus *Venegasia*) canyon sunflower 2. (genus *Achyrachaena*) blow-wives 3. (Madieae other than named genera; several genera) other madiid plant

-**MZDR**- 'INULEAE' I 1. (genus *Inula*) elecampane / horse-heal / elfdock, golden samphire, ploughman's-spikenard, yellowhead, inula, Irish fleabane, giant fleabane 2. (genus *Caesulia*) pink node flower 3. (genus *Pallenis*) starwort, golden star, rose of Jericho / dinosaur plant

-**MZDŘ**- 'INULEAE' II 1. (genus *Pechuel-loeschea*) stinkbush 2. (genus *Pulicaria*) false fleabane 3. (Inuleae other than Stems 1 and 2) other inulid plant

-**MZDV**- 'MILLERIEAE' I 1. (genus *Sigesbeckia*) St. Paul's wort 2. (genus *Smallanthus*) leafcup, bear's foot, yacón 3. (genus *Tetragonotheca*) nerveray

-**MZG**- 'MILLERIEAE' II 1. (genus *Bebbia*) sweetbush 2. (genus *Galinsoga*) galinsoga, gallant soldier / quickweed / potato weed, shaggy soldier / Peruvian daisy 3. (genus *Guizotia*) inga seed, black seed, ramtil, niger seed

-**MZGW**- 'MILLERIEAE' III 1. (genus *Melampodium*) blackfoot 2. (genus *Tridax*) tridax daisy / coatbuttons 3. (Millerieae other than named genera; several genera) other milleriid plant

-**MZGY**- 'TAGETEAE' I 1. (genus *Tagetes*) marigold 2. (genus *Adenophyllum*) dogweed 3. (genus *Dyssodia*) fetid marigold

-**MZGL**- 'TAGETEAE' II 1. (genus *Dysodiopsis*) false dogfennel 2. (genus *Haploesthes*) false broomweed 3. (genus *Jamesianthus*) Alabama warbonnet

-**MZGR**- 'TAGETEAE' III 1. (genus *Jaumea*) jaumea 2. (genus *Nicolletia*) hole-in-the-sand plant 3. (genus *Pectis*) cinchweed, chinchweed, donkeyweed

-**MZGŘ**- 'TAGETEAE' IV 1. (genus *Porophyllum*) poreleaf, pápalo / Bolivian coriander, jarilla 2. (genus *Pseudoclappia*) false clapdaisy 3. (genus *Sartwellia*) glowwort

-**MZGV**- 'TAGETEAE' V 1. (genus *Thymophylla*) pricklyleaf, Texas dogweed, ashy dogweed 2. (Tageteae other than named genera) other tagetid plant 3. [stem unassigned]

-**MŽB**- 'HELENIEAE' I 1. (genus *Helenium*) sneezeweed, rosilla, cabezona 2. (genus *Gaillardia*) blanket flower, gaillardia, bandanna daisy, firewheel, perfumeballs 3. (genus *Balduina*) honeycombhead

-**MŽBW**- 'HELENIEAE' II 1. (genus *Baileya*) desert marigold, laxflower 2. (genus *Hymenoxys*) rubberweed, bitterweed, four-nerve daisy, old-man-of-the-mountain, owl claws, prairie dawn 3. (genus *Marshallia*) Barbara's buttons

-**MŽBY**- 'HELENIEAE' III 1. (genus *Psathyrotes*) turtleback, brittlestem, fanleaf, velvet rosette, psathyrotes 2. (genus *Psilostrophe*) paperflower 3. [stem unassigned]

-**MŽBL**- 'HELENIEAE' IV 1. (genus *Schkuhria*) false threadleaf, canchalagua, dwarf Mexican marigold 2. (Helenieae other than named genera; several genera) other heleniid plant

-**MŽBR**- 'EUPATORIEAE' I 1. (genus *Eupatorium*) thoroughwort, dog-fennel, Yankeeweed, false fennel, justiceweed, boneset, hemp-agrimony 2. (genus *Eutrochium*) Joe-Pye weed 3. (genus *Isocarpha*) pearlhead

-**MŽBŘ**- 'EUPATORIEAE' II 1. (genus *Liatris*) blazing star, Cahaba torch, gayfeather, colicroot, prairie-pine / button snakewort, devil's bite 2. (genus *Carphephorus*) chaffhead, vanillaleaf 3. (genus *Ageratina*) snakeroot, richweed / white sanicle, mistflower, ageratina, Western eupatorium, Shasta eupatorium, creeping cottonwood

-**MŽBV**- 'EUPATORIEAE' III 1. (genus *Ageratum*) whiteweed, flossflower / bluemink / blueweed / pussy foot / Mexican paintbrush 2. (genus *Conoclinium*) mistflower, palmleaf thoroughwort 3. (genus *Fleischmannia*) slender-thoroughwort, pink thoroughwort

-**MŽD**- 'EUPATORIEAE' IV 1. (genus *Brickellia*) brickellbush, Flyr's nemesis, false boneset 2. (genus *Flyriella*) brickelbush 3. (genus *Asanthus*) Mule Mountain false brickellbush

-**MŽDW**- 'EUPATORIEAE' V 1. (genus *Trichocoronis*) bugheal 2. (genus *Shinnersia*) Rio Grande bugheal / Mexican oak leaf plant 3. (genus *Carphochaete*) bristlehead

-**MŽDY**- 'EUPATORIEAE' VI 1. (genus *Stevia*) stevia 2. (genus *Garberia*) garberia, Garber's scrub start 3. (genus *Hartwrightia*) Florida hartwrightia

-**MŽDL**- ‘EUPATORIEAE’ VII 1. (genus *Malperia*) brown turban 2. (genus *Pleurocoronis*) bush arrowleaf 3. (genus *Sclerolepis*) pink bogbutton

-**MŽDR**- ‘EUPATORIEAE’ VIII 1. (genus *Carminatia*) plumeweed 2. (Eupatorieae other than named genera; numerous genera) other eupatoriid plant

-**MŽDŘ**- ‘BAHIEAE’ I 1. (genus *Bahia*) bahia 2. (genus *Picradeniopsis*) many-flowere bahia, oppositeleaf bahia 3. (genus *Chamaechaenactis*) fullstem

-**MŽDV**- ‘BAHIEAE’ II 1. (genus *Palafoxia*) palafox, Spanish needles 2. (genus *Florestina*) florestina, sticky florestina 3. (genus *Hypenopappus*) woollywhite, oldplainsman, collegeflower, hymenopappus

-**MŽG**- ‘BAHIEAE’ III 1. (genus *Hymenothrix*) thimblehead 2. (genus *Peucephyllum*) pygmy cedar / desert fir /desert pine 3. (genus *Platyschkuhria*) basindaisy

-**MŽGW**- ‘BAHIEAE’ IV 1. (genus *Bartlettia*) Bartlett daisy 2. (genus *Amauriopsis*) ragleaf bahia, yellow ragweed 3. (Bahieae other than named genera) other bahiid plant

-**MŽGY**- ‘PERITYLEAE’ 1. (genus *Perityle*) rock daisy 2. (genus *Pericome*) mountain tall-leaf 3. (Perityleae other than Stems 1 and 2; several genera) other peritylid plant

-**MŽGL**- ‘PLUCHEEAE’ 1. (genus *Pluchea*) camphorweed, pluchea, cure-for-all, sweetscent / saltmarsh fleabane, arrowweed 2. (genus *Pterocaulon*) blackroot, applebush / fruit-salad plant 3. (Plucheeae other than Stems 1 and 2; several genera) other plucheid plant

-**MŽGR**- ‘CHAENACTIDEAE’ 1. (genus *Chaenactis*) dustymaiden, pincushion 2. (genus *Dimeresia*) doublet 3. (genus *Orochaenactis*) California mountain pincushion

-**MŽGŘ**- ‘NEUROLAENEAE & POLYMNIEAE’ 1. (genus *Calea*) bitter-grass / Mexican calea / dream herb 2. (Neurolaeneae other than Stem 1; several genera) other neuroleanid plant 3. (Polymnieae; genus *Polymnia*) leafcup

-**MŽGV**- ‘ATHROISMEAE & FEDDEEAE’ 1. (genus *Centipeda*) old man weed / scent weed 2. (Athroismeae other than Stem 1; several genera) other athroismid plant 3. (Feddeeae; genus *Feddea*) feddea

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Campanulids > Asterales > Asteraceae > Asteroideae > Helianthodae > Heliantheae

-**ŇZV**- ‘HELIANTHINAE’ I 1. (genus *Helianthus*) sunflower, Jerusalem artichoke / sunchoke / earth-apple 2. (genus *Viguiera*) goldeneye, yellow streamer 3. (genus *Heliomeris*) false goldeneye, showy goldeneye, hairy goldeneye, paria sunflower

-**ŇZVW**- ‘HELIANTHINAE’ II 1. (genus *Phoebanthus*) false sunflower 2. (genus *Sclerocarpus*) bonebract 3. (genus *Simsia*) bush sunflower

-**ŇZVY**- ‘HELIANTHINAE’ III 1. (genus *Tithonia*) tree marigold / Mexican tournesol / Mexican sunflower / Japanese sunflower / Nitobe chrysanthemum, red sunflower 2. (genus *Bahiopsis*) San Diego County sunflower / San Diego viguiera / tornleaf goldeneye 3. (Helianthinae other than named genera; several genera) other sunflower-like plant

-**ŇZB**- ‘ZINNIINAE’ I 1. (genus *Zinnia*) zinnia 2. (genus *Echinacea*) echinacea, coneflower 3. (genus *Heliopsis*) smooth oxeye, mountain oxeye, gold root, pinewoods oxeye, rough oxeye

-**ŇZBW**- ‘ZINNIINAE’ II 1. (genus *Sanvitalia*) creeping zinnia 2. (Zinniinae other than named genera; several genera) other zinniid plant 3. [stem unassigned]

-**ŇZBY**- ‘ECLIPTINAE’ I 1. (genus *Melanthera*) squarestem 2. (genus *Eclipta*) false daisy 3. (genus *Lipochaeta*) nehe

-**ŇZBL**- ‘ECLIPTINAE’ II 1. (genus *Sphagneticola*) creeping-oxeye 2. (genus *Wedelia*) wedelia, creeping oxeyes 3. (genus *Synedrella*) nodeweed

-**ŇZBR**- ‘ECLIPTINAE’ III 1. (genus *Baltimora*) beautyhead 2. (genus *Jefea*) jefea 3. (genus *Calyptocarpus*) straggler daisy / horseherb / lawnflower / creeping Cinderella-weed

-**ŇZBŘ**- ‘SPILANTHINAE’ 1. (genus *Acmella*) paracress / toothache plant / tingflowers / electric daisy / buzz buttons / Sichuan buttons, Spilanthes, spotflower 2. (Spilanthinae other than Stem 1; several genera) other spilanthine plant 3. [stem unassigned]

-**ŇZBV**- ‘AMBROSIINAE’ 1. (genus *Ambrosia*) ragweed, bursage, marko, altamisa, Roman wormwood, beach-bur, ambrosia, burrobush, lagoonweed 2. (genus *Parthenium*) feverfew, wild quinine, whitetop weed, guayule 3. (Ambrosiinae other than Stems 1 and 2; several genera) other ambrosiine plant

-**ŇZG**- ‘ENCELIINAE’ I 1. (genus *Encelia*) brittlebrush, encelia, noddinghead / nodding sunray 2. (genus *Enceliopsis*) sunray, Panamint daisy 3. (genus *Geraea*) desert sunflower, desert gold, geraea

-**ŇZGW**- ‘ENCELIINAE’ II 1. (genus *Flourensia*) tarbush, tarwort 2. (genus *Helianthella*) little sunflower 3. (other Enceliinae genera) other enceliine plant

-**ŇZGY**- ‘ENGELMANNIINAE’ I 1. (genus *Engelmannia*) Engelmann’s daisy / cutleaf daisy 2. (genus *Berlandiera*) greeneyes 3. (genus *Lindheimera*) Texas yellow star / Texas star

-**ŇZGL**- ‘ENGELMANNIINAE’ II 1. (genus *Silphium*) rosinweed 2. (genus *Wyethia*) mule’s ears, black sunflower, California compassplant, wyethia 3. (genus *Balsamorhiza*) balsamroot

-**ŇZGR**- ‘ENGELMANNIINAE’ III 1. (genus *Chrysogonum*) golden-knee / green-and-gold / goldenstar 2. (genus *Borrichia*) seaside tansy, sea-oxeye 3. (genus *Vigethia*) Mexican green-eyed sunflower

-**ŇZGŘ**- ‘VERBESININAE’ 1. (genus *Verbesina*) crownbeard, wingstem / yellow ironweed, goldweed / butter daisy / American dogweed, gravelweed, stick weed, frostweed 2. (Verbesininae other than Stem 1; several genera) other verbesinine plant 3. [stem unassigned]

-**ŇZGV**- ‘HELIANTHEAE’ 1. (genus *Rudbeckia*) black-eyed-susan 2. (genus *Rojasianthe*) rojasianthe 3. (Heliantheae other than named genera) other helianthid plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Campanulids > Asterales > Asteraceae > Cichorioideae > Cichorieae

-RBZ - ‘LACTUCA SATIVA (LETTUCE CULTIVAR)’ I 1. (loose)leaf lettuce 2. romaine/cos lettuce 3. iceberg/crisphead lettuce

-RBZW - ‘LACTUCA SATIVA (LETTUCE CULTIVAR)’ II 1. butter(head)/bibb lettuce 2. red leaf lettuce 3. summercrisp lettuce

-RBZY- ‘LACTUCINAE’ 1. (genus *Lactuca* other than *L. sativa*) wild lettuce, bitter lettuce, pliant lettuce, blue lettuce prickly lettuce / compassplant / scarole / milk thistle, willow lettuce / least lettuce, mountain lettuce, downy lettuce / hairy lettuce, grassleaf lettuce, woodland lettuce, tail lettuce, sow thistle 2. (genus *Cicerbita*) blue sow thistle 3. (genus *Notoseris*) notoseris

-RBZL- ‘CICHORUM’ 1. (*C. endivia*) endive plant, escarole 2. (*C. intybus*) common chicory, leaf chicory, radicchio, Belgian endive 3. (*C. pumilum* and several other species) wild endive

-RBZR- ‘CICHORIINAE’ 1. 2. (genus *Tolpis*) European umbrella milkwort 3. (genus *Arnoseris*) dwarf nipplewort / lamb-succory / lamb’s succory 3. (genera *Phalacroseris*, *Erythroseris*, *Rothmaleria*) mock dandelion, other cichoriine plant

-RBZŘ- ‘MICROSERIDINAE’ I 1. (genera *Microseris, Uropappus*) silverpuffs, microseris, sylvan scorzonella 2. (genus *Agoseris*) false dandelion, mountain dandelion, agoseris 3. (genus *Nothocalais*) false dandelion, false agoseris

-RBZV- ‘MICROSERIDINAE’ II 1. (genus *Malacothrix*) desert dandelion 2. (genus *Atrichoseris*) tobacco weed / parachute plant / gravel ghost 3. (genus *Munzothamnus*) Blair’s wirelettuce / Blair’s munzothamnus

-RBZM- ‘MICROSERIDINAE’ III 1. (genus *Stephanomeria*) wirelettuce 2. (genus *Rafinesquia*) plumeseed, California chicory, desert chicory 3. (genus *Glyptopleura*) carveseed / keysia / crustleaf, holy dandelion

-RBZN- ‘MICROSERIDINAE’ IV 1. (genus *Pleiacanthus*) thorn(y) skeletonweed 2. (genus *Prenanthella*) brightwhite 3. (genus *Krigia*) dwarf dandelion

-RBZŇ- ‘MICROSERIDINAE’ V 1. (genus *Pinaropappus*) rock lettuce 2. (genus *Pyrrhopappus*) desert-chicory 3. (genus *Shinneroseris*) beaked skeletonweed

-RBZH- ‘MICROSERIDINAE’ VI 1. (genus *Lygodesmia*) skeletonplant, rushpink 2. (genus *Chaetadelpha*)Wheeler’s skeletonweed 3. (Microseridiinae other than named genera; several genera) other microserid(ine) plant

-RBZZ- ‘HIERACIINAE’ 1. (genus *Hieracium*) hawkweed, Sicilian sparviere 2. (genus *Pilosella*) fox-and-cubs / orange hawk bit / devil’s paintbrush / grim-the-collier, white hawkweed, yellow hawkweed, king devil, yellow devil 3. (Hieraciinae other than Stem 1; several genera) other hieraciine plant

-RẐW- ‘HYOSERIDINAE’ 1. (genus *Sonchus*) sow thistle, hare thistle, hare lettuce, milk thistle 2. (genus *Reichardia*) brighteyes, false sowthistle 3. (Hyoseridinae other than Stems 1 and 2; genera *Hyoseris, Aposeris, Launaea*) other hyoserid(ine) plant

-RẔY- ‘CREPIDINAE’ I 1. (genus *Crepis*) hawksbeard 2. (genus *Lapsana*) nipplewort 3. (genus *Askellia*) elegant hawksbeard, dwarf alpine hawksbeard

-RẔL- ‘CREPIDINAE’ II 1. (genus *Teraxacum*) dandelion 2. (genus *Ixeris*) creeping lettuce 3. (genus *Youngia*) Japanese hawkweed, Oriental false hawkweed

-RẔR- ‘CREPIDINAE’ III 1. (genus *Nabalus*) white lettuce, rattllesnakeroot 2. (Crepidinae other than named genera; several genera) other crepid(ine) plant 3. [stem unassigned]

-RẔŘ- ‘HYPOCHAERIDINAE’ I 1. (genus *Hypochaeris*) cat’s ear, flatweed / false dandelion, chikku-chikku 2. (genus *Scorzoneroides*) autumn hawbit 3. (genus *Prenanthes*) rattlesnake root

-RẔV- ‘HYPOCHAERIDINAE’ II 1. (genus *Urospermum*) goldenfleece 2. (genus *Picris*) ox-tongue 3. (genus *Helminthotheca*) bristly ox-tongue

-RẔM- ‘HYPOCHAERIDINAE’ III 1. (genus *Leontodon*) hawkbit 2. (genus *Hedypnois*) scaly hawkbit 3. (Hypochaeridinae other than named genera) other hypochaeridine plant

-RẔN- ‘CHONDRILLINAE’ 1. (genus *Chondrilla*) rush skeletonweed, smallflower desert-chicory / Texas false dandelion 2. (genus *Willemetia*) willemetia 3. (genus *Phitosia*) phitosia

-RẔŇ- ‘SCOLYMINAE’ 1. (genus *Scolymus*) golden thistle / oyster thistle 2. (genus *Catananche*) Cupid’s dart 3. (genera *Gundelia, Hymenonema*) tumble thistle, other scolymine plant

-RẔH- ‘SCORZONERINAE’ 1. (genus *Tragopogon*) salsify, goatsbeard, oyster plant, jack-go-to-bed-at-noon 2. (genus *Scorzonera*) black salsify / Spanish salsify / black oyster plant / serpent root / viper’s herb, viper’s grass, scorzonera 3. (genus *Geropogon*) slender salsify / pasture goatsbeard

-RẔHW- ‘OTHER SCORZONERINAE & WARIONIINAE’ 1. (Scorzonerinae other than named genera; several genera) oother goatsbeard or salsify-like plant) 2. (Warioniinae; genus *Warionia*) warionia 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Campanulids > Asterales > Asteraceae > Cichorioideae [other than Cichorieae]

-**RMGW**- ‘VERNONIEAE’ I 1. (genus *Vernonia*) ironweed 2. (genus *Tarlmounia*) curtain creeper / vernonia creeper 3. (genus *Stokesia*) stokesia / Stokes’ aster

-**RMGY**- ‘VERNONIEAE’ II 1. (genus *Piptocarpha*) ashdaisy 2. (genus *Piptocoma*) velvetshrub 3. (genus *Gymnanthemum*) bitterleaf tree

-**RMGL**- ‘VERNONIEAE’ III 1. (genus *Centratherum*) lark daisy, Brazilian button 2. (genus *Elephantopus*) elephant’s foot, devil’s grandmother 3. (genus *Pseudelephantopus*) dog’s-tongue

-**RMGR**- ‘VERNONIEAE’ IV 1. (genus *Hesperomannia*) island-aster 2. (Vernonieae other than named genera; numerous genera) other vernoniine plant 3. [stem unassigned]

-**RMGŘ**- ‘GORTERIINAE’ 1. (genus *Gorteria*) beetle daisy 2. (genus *Berkheya*) thistle-thorn, yellow thistle, berkheya, Augusta thistle / Hamelin thistle 3. (genus *Gazania*) gazania, treasure flower

-**RMGV**- ‘ARCTOTIDINAE’ 1. (genus *Arctotis*) monarch-of-the-veld, African daisy, arctotis 2. (genus *Arctotheca*) Cape weed / creeping bear’s ear, capeweed / cape dandelion / cape marigold / plain treasureflower, beach daisy / beach pumpkin / sea pumpkin / dune cabbage / coast capeweed 3. (genus *Cymbonotus*) bear’s ears

-**RMGḌ**- ‘CICHORIOIDEAE’ 1. (Arctotidinae other than previously named genera; several genera) other arctotidine plant 2. (Gorteriinae other than previously named genera; several genera) other gorteriine plant 3. (Platycarpheae, Moquinieae, Liabeae, plus other unplaced cichorioid genera, e.g., *Distephanus, Trichospira, Heterolepis, Haplophyllum*) other cichorioid plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Campanulids > Asterales > Asteraceae > Carduoideae

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Campanulids > Asterales > Asteraceae > Carduoideae > Cardueae (a.k.a. Cynareae)

-FSPW- ‘CYNARA (ARTICHOKE)’ I 1. (*C. scolymus*) globe/French/green artichoke 2. (*C. cardunculus*) cardoon 3. (*C. cornigera*) white artichoke

-FSPY- ‘CYNARA (ARTICHOKE)’ II 1. (*C. humilis*) lesser wild artichoke 2. (*Cynara* species other than already named) various wild artichoke species

-**FSPL**- ‘CARDUINAE’ I 1. (genus *Carduus*) plumeless thistle, curled thistle, welted thistle, musk thistle, sheep thistle / shore thistle / slender thistle, Italian thistle 2. (genus *Cirsium*) [numerous types of] thistle 3. (genus *Onopordum*) cotton thistle, Scotch thistle, woolly thistle

-**FSPR**- ‘CARDUINAE’ II 1. (genus *Arctium*) burdock, beggar’s buttons / thorny burr / happy major, louse-bur / button-bur / cuckoo-button / wild rhubarb 2.(genus *Galactites*) milk thistle 3. (genus *Notobasis*) Syrian thistle

-**FSPŘ**- ‘CARDUINAE’ III 1. (genus *Saussurea*) saw-wort, snow lotus 2. (genus *Silybum*) silver milk thistle / elephant thistle / ivory thistle, blessed milk thistle / variegated thistle 3. (Carduinae other than named genera; numerous genera) other carduine thistle

-**FSPF**- ‘CARLININAE’ I 1. (genus *Carlina*) carline thistle 2. (genus *Atractylis*) atractylis 3. (Carlininae other than Stems 1 and 2; several genera) other carlinine thistle

-**FSPŢ**- ‘CENTAUREINAE’ I 1. (genus *Centaurea*) basketflower, knapweed, centaurea, centaury, starthistle, loggerhead, conrnflower / bachelor’s button / boutonniere flower / hurtsicle / bluebottle, yellow cockspur 2. (genus *Amberboa*) sweetsultan 3. (genus *Rhaponticum*) maral root, Russian knapweed / creeping knapweed / hardheads / bluweed, Austral cornflower / native thistle

-**FSPÇ**- ‘CENTAUREINAE’ II 1. (genus *Carthamus; C. tinctorius*) safflower 2. (genus *Carthamus;* species other than *C. tinctorius*) distaff thistle, saffron thistle / downy safflower, glaucous starthistle 3. (genus *Cheirolophus*) Maltese rock-centaury, cheirolophus

-**FSPĻ**- ‘CENTAUREINAE’ III 1. (genus *Crupina*) crupina, bearded-creeper / false saw-wort / starry scabious 2. (genus *Psephellus*) bachelor’s button 3. (genus *Serratula*) plumeless saw-wort

-**FSPH**- ‘CENTAUREINAE’ IV 1. (genus *Mantisalca*) dagger-flower 2. (Centaureinae other than named genera; numerous genera) other centaureine thistle plant 3. [stem unassigned]

-**FSPHW**- ‘OTHER CARDUEAE’ 1. (Echinopsinae; genus *Echinops*) globe-thistle 2. (Cardopatiinae; genera Cardopatium, Cousiniopsis) cardopatiine plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Campanulids > Asterales > Asteraceae > Carduoideae [other than Cardueae]

-**ŢBV**- ‘TARCHONANTHEAE & OTHER CARDUOIDEAE’ 1. (genus *Tarchonanthus*) camphor bush 2. (genus *Brachylaena*) silver oak, white alder 3. (Dicomeae, Oldenburgieae; seveal genera) other carduoid plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Campanulids > Asterales > Asteraceae > Mutisioideae & remaining Asteraceae

-**ŢBW**- ‘MUTISIOIDEAE’ I 1. (genus *Leibnitzia*) sunbonnet 2. (genus *Chaptalia*) silverpuff, pineland daisy, white sunbonnet 3. (genus *Gerbera*) African daisy, Barberton daiy / Transvaal daisy / gerbera daisy, Hilton daisy

-**ŢBY**- ‘MUTISIOIDEAE’ I 1. (genus *Adenocaulon*) trailplant, pathfinder 2. (genus *Acourtia*) desertpeony, sacapellote, desert holly, brownfoot, desert paeonia 3. (genus *Leucheria*) vanilla daisy

-**ŢBL**- ‘MUTISIOIDEAE’ III 1. (genus *Trixis*) threefold 2. (genus *Perezia*) perezia 3. (Mutisoideae other than named genera; numerous genera) other mutisioid plant

-**ŢBR**- ‘GOCHNATIEAE & HECASTOCLEIDOIDEAE’ 1. (genus *Gochnatia*) shrubby bullseye 2. (Gochnatieae other than Stem 1; several genera) other gochnatiid plant 3. (Hecastocleidoideae; genus *Hecastocleis*) prickleleaf

-**ŢBŘ**- ‘CORYMBIOIDEAE & OTHER ASTERACEAE’ 1. (Corymbioideae; genus *Corymbium*) plampers 2. (Famatinanthoideae, Stifftieae, Barnadesieae, Wunderlichioideae, Pertyeae, Gymnarrhenoideae; several genera) other mutisioid plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Campanulids > Asterales > Campanulaceae

-**LCPW**- ‘CAMPANULOIDEAE’ I 1. (genus *Campanula*) bellflower 2. (genus *Canarina*) Canary Island bellflower 3. (genus *Adenophora*) ladybell

-**LCPY**- ‘CAMPANULOIDEAE’ II 1. (genus *Asyneuma*) harebell 2. (genus *Edraianthus*) rockbell, dwarf harebell 3. (genus *Githopsis*) bluecup

-**LCPL**- ‘CAMPANULOIDEAE’ III 1. (genus *Heterocodon*) western pearlflower / rareflower heterocodon 2. (genus *Jasione*) sheep’s-bit / blue bonnet / blue button / blue daisy / iron flower 3. (genus *Triodanis*) Venus’ looking-glass

-**LCPR**- ‘CAMPANULOIDEAE’ IV 1. (genus *Legousia*) looking glass 2. (genus *Phyteuma*) rampion 3. (genus *Physoplexis*) tufted horned rampion

-**LCPŘ**- ‘CAMPANULOIDEAE’ V 1. (genus *Trachelium*) throatwort 2. (genus *Wahlenbergia*) small bellflower, flat bluebell, Cape bluebell, tufted bluebell, Roxburgh bellflower / dwarf cabbage tree, rock bluebell, fairy bluebell, river bluebell, Australian bluebell, royal bluebell, annual bluebell, austral bluebell, granite bluebell 3. (genus *Hanabusaya*) diamond bluebell

-**LCPF**- ‘CAMPANULOIDEAE’ VI 1. (genus *Codonopsis*) lance asiabell, poor man’s ginseng 2. (Campanuloideae other than named genera; numerous genera) other campanuloid plant 3. [stem unassigned]

-**LCPŢ**- ‘LOBELIOIDEAE’ I 1. (genus *Lobelia*) lobelia, cardinal flower, Indian tobacco, wild tobacco, pratia 2. (genera *Brighamia, Clermontia, Cyanea, Delissea, Trematolobelia*) Hawaiian lobelioid 3. (genus *Downingia*) calicoflower

-**LCPÇ**- ‘LOBELIOIDEAE’ II 1. (genus *Hippobroma*) Star of Bethlehem, madamfate 2. (genus *Hwellia*) water howellia 3. (genus *Isotoma*) isotome, blue star creeper, Woodbridge poison

-**LCPḶ**- ‘LOBELIOIDEAE’ III 1. (genus *Legenere*) false Venus’ looking-glass 2. (genus *Porterella*) fleshy porterella 3. (genus *Pratia*) poison pratia, white root

-**LCPH**- ‘LOBELIOIDEAE’ IV 1. (genus *Trimeris*) St. Helena lobelia / milkwood 2. (Lobelioideae other than named genera; numerous genera) other lobelioid plant 3. [stem unassigned]

-**LCPHW**- ‘CYPHIOIDEAE’ 1. (genus *Nemacladus*) threadplant, nemacladus 2. (Cyphioideae other than Stem 1; several genera) other cyphioid plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Campanulids > Asterales > families other than Campanulaceae

-**LČPW**- ‘ALSEUOSMIACEAE’ 1. (genus *Wittsteinia*) baw baw berry 2. (Alseuosmiaceae other than Stem 1; several genera) other alseuosmiacean plant 3. [stem unassigned]

-**LČPY**- ‘ARGOPHYLLACEAE’ 1. (genus *Argophyllum*) silver leaf 2. (genus *Corokia*) corokia 3. [stem unassigned]

-**LČPL**- ‘CALYCERACEAE’ 1. (genus *Nastanthus*) false-plantain, nastanthus 2. (Calyceraceae other than Stem 1; several genera) other calyceracean plant

-**LČPR**- ‘GOODENIACEAE’ I 1. (genus *Goodenia*) goodenia, native primrose 2. (genus *Brunonia*) blue pincushion / native cornflower 3. (genus *Dampiera*) dampiera

-**LČPŘ**- ‘GOODENIACEAE’ II 1. (genus *Lechenaultia*) lechenaultia 2. (genus *Scaevola*) scaevola, fan-flower, half-flower, naupaka 3. (genus *Selliera*) swampweed / bonking grass

-**LČPF**- ‘GOODENIACEAE’ III 1. (genus *Velleia*) pee-the-bed, velleia 2. (Goodeniaceae other than named genera; several genera) other goodeniacean plant 3. [stem unassigned]

-**LČPŢ**- ‘MENYANTHACEAE’ I 1. (genus *Menyanthes*) buckbean / bogbean 2. (genus *Liparophyllum*) marshwort 3. (genus *Nymphoides*) floatingheart, banana lily, water snowflake, fringed water lily, marshwort

-**LČPÇ**- ‘MENYANTHACEAE’ II 1. (genus *Nephrophyllidium*) deer cabbage 2. (Menyanthaceae other than named genera) other menyanthacean plant 3. [stem unassigned]

-**LČPḶ**- ‘ROUSSEACEAE’ 1. (genus *Cuttsia*) silver-leaved cuttsia / native elderberry / honey bush 2. (genus *Carpodetus*) marbleleaf / bucket-of-water -tree 3. (Rousseaceae other than Stems 1 and 2; several genera) other rousseacean plant

-**LČPH**- ‘STYLIDIACEAE’ 1. (genus *Stylidium*) triggerplant 2. (genus *Forstera*) forstera 3. (genus *Levenhookia*) stylewort

-**LČPHW**- ‘OTHER STYLIDIACEAE & OTHER ASTERALES FAMILIES’ 1. (genera *Phyllachne, Donatia*) cushionplant 2. Stylidiaceae other than named genera) other stylidiacean plant 3. (Pentaphragmataceae, Phellinaceae; several genera) other asterales plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Campanulids remaining families

-RZŅW- ‘OTHER CAMPANULID’ 1. (Paracryphiaceae, three genera) possumwood tree, paracryphiaceous plant 2. (Bruniaceae; several genera) brunia, button bush, redlegs 3. (Columelliaceae; genera *Collumelia, Desfontainia*) columelliaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Lamiids

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Lamiids > Lamiales

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Lamiids > Lamiales > Lamiaceae

-**FŠK**- ‘MENTHA’ I 1. (*Mentha spicata*) spearmint / common mint / garden mint / lamb mint 2. (*M. requienii*) Corsican mint 3. (*M.longifolia*) horse mint

-**FŠKW**- ‘MENTHA’ II 1. (*M. aquatica*) water mint / marsh mint 2. (*M. arvensis*) corn mint / field mint / banana mint 3. (*M. canadensis*) Canada mint, American wild mint

-**FŠKY**- ‘MENTHA’ III 1. (*M. laxiflora*) forest mint 2. (*M. satureioides*) creeping mint / native pennyroyal 3. (*M. suaveolens*) apple mint, pineapple mint

-**FŠKL**- ‘MENTHA’ IV (hybrids) 1. (*M. piperita*) peppermint 2. (*M. gracilis*) gingermint / redmint / Scotchmint / Scotch spearmint 3. (*M. villosa*) Cuban mint / large apple mint / foxtail mint / hairy mint / woolly mint / mojito mint

-**FŠKR**- ‘MENTHA’ V 1. (*M. laxiflora*) forest mint 2. (*M. satureioides*) creeping mint / native pennyroyal 3. (*M. suaveolens*) apple mint

-**FŠKŘ**- ‘MENTHA’ VI 1. (*M. pulegium*) pennyroyal / pennyrile / squaw mint / mosquito lant / pudding grass 2. (*M. australis, M. diemenica*) Australian mint / river mint / native mint / native peppermint, slender mint 3. (other *Mentha* species and hybrids) other mint plant

-**FŠKF**- ‘MENTHEAE’ I 1. (genus *Monarda*) beebalm, bergamot, Oswego tea, fragrantbalm, horse-mint, monarda 2. (genus *Monardella*) wildmint, coyote mint, monardella, mountain pennyroyal 3. (genus *Pycanthemum*) mountainmint, koellia

-**FŠKŢ**- ‘MENTHEAE’ II 1. (genus *Poliomintha*) rosemary mint 2. (genus *Pogogyne*) mesa mint 3. (genus *Piloblephis*) wild pennyroyal

-**FŠKÇ**- ‘MENTHEAE’ III 1. (genus *Stachydeoma*) mock pennyroyal 2. (genus *Hedeoma*) false pennyroyal 3. (genus *Dicerandra*) scrub mint, coastal plain mint, rose balm, Titusville mint, Radford’s balm

-**FŠKV**- ‘MENTHEAE’ IV 1. (genus *Conradina*) false rosemary, short-leaf rosemary 2. (genus *Blephilia*) pagoda plant, wood mint 3. (genus *Acanthomintha*) thornmint

-**FŠKM**- ‘MENTHEAE’ V 1. (genus *Hyssopus*) hyssop 2. (genus *Agastache*) giant hyssop, blue giant hyssop, anise hyssop, hummingbird mint 3. (genus *Dracocephalum*) dragonhead, dragonhead mint

-**FŠKN**- ‘MENTHEAE’ VI 1. (genus *Nepeta*) catnip, catmint 2. (genus *Cedronella*) Canary balm / Balm of Gilead 3. (genus *Lallemantia*) dragon’s head

-**FŠKH**- ‘MENTHEAE’ VII 1. (genus *Calamintha*) calamint 2. (genus *Prunella*) self-heal, heal-all, allheal 3. (genus *Horminum*) dragonmouth, Pyrenean dead-nettle

-**FŠKHW**- ‘MENTHEAE’ VIII 1. (genus *Melissa*) lemon balm / common balm / balm mint 2. (genus *Lepechinia*) pitcher sage, woodbalm 3. (genus *Rhododon*) sandmint

-**FŠGW**- ‘MENTHEAE’ IX 1. (genus *Lycopus*) water-horehound, gypsywort, bugleweed 2. (genus *Acinos*) rock thyme, basil thyme / spring savory 3. (genus *Clinopodium*) lesser calamint, Ozark calamint, Georgia calamint, Florida calamint, limestone calamint, wild basil, serpentine savory, monkeyflower savory, Browne’s savory, yerba buena

-**FŠGY**- ‘MENTHEAE’ X 1. (genus *Meehania*) Meehan’s mint / creeping mint 2. (genus *Glechoma*) ground-ivy / creeping charlie 3. (Mentheae other than named genera) other menthid plant

-**FŠP**- ‘THYMUS’ I 1. (*Thymus vulgaris*) [common] thyme 2. (*T. pulegiodes*) lemon thyme / broad-leaved thyme 3. (*T. citriodorus*) citrus thyme

-**FŠPW**- ‘THYMUS’ II 1. (*T. herba-barona*) caraway thyme 2. (*T. serpyllum*) wild thyme / creeping thyme / elfin thyme / Breckland thyme 3. (*T. praecox*) mother of thyme

-**FŠPY**- ‘THYMUS’ III 1. (*T. pseudolanuginosus*) woolly thyme 2. (*T. capitatus*) conehead thyme / Persian-hyssop / Spanish oregano 3. (other *Thymus* species) other thyme plant

-**FŠPL**- ‘ORIGANUM’ 1. (*Origanum vulgare*) oregano plant 2. (*O. majorana*) marjoram plant 3. (other *Origanum* species) other oregano or marjoram plant

-**FŠPR**- ‘SATUREJA’ 1. (*Satureja hortnesis*) summer savory 2. (*S. montana*) winter savory / mountain savory 3. (other *Satureja* species) other savory plant

-**FŠPŘ**- ‘SALVIA’ 1. (*Salvia officinalis*) sage 2. (*S. rosmarinus*) rosemary 3. (other *Salvia* species) other sage plant

-**FŠPF**- ‘OCIMEAE’ I 1. (genus *Ocimum*) basil 2. (genus *Platostoma*, a.k.a. *Mesona*) xiancao / cincau 3. (genus *Condea*) John Charles, desert lavender,

-**FŠPŢ**- ‘OCIMEAE’ II 1. (genus *Lavandula*) lavender 2. (genus *Hyptis*) bushmint, wild hops / false ironwort, Brazilian mint, bayamon 3. (genus *Plectranthus*, a.k.a. *Coleus*) spurflower, lobster bush / fly bush / mosquito bush, cockspur, coleus, scaredy cat plant, country potato / Chinese potato, coleus, Swedish ivy / Swedish begonia, Livingstone potato, Ethiopian potato

-**FŠPÇ**- ‘LAMIACEAE’ I 1. (genus *Lamiium*) dead-nettles 2. (genus *Sideritis*) ironwort / mountain tea / shepherd’s tea 3. (genus *Phlomis*) lampwick plant / Jerusalem sage

-**FŠPS**- ‘LAMIACEAE’ II 1. (genus *Holmskioldia*) Chinese hat plant / cup-and-saucer plant / mandarin’s hat 2. (genus *Scutellaria*) skullcaps 3. [stem unassigned]

-**FŠPŠ**- ‘LAMIACEAE’ III 1. (genus *Collinsonia*) richweed, stoneroot / whorled horse-balm 2. (genus *Elsholtzia*) Vietnamese balm 3. (genus *Perilla*) perilla, Japanese basil

-**FŠPĻ**- ‘LAMIACEAE’ IV 1. (genus *Ajuga*) bugleweed, bugle, carpet bugle, ground-pine, carpetweed, St. Lawrence plant 2. (genus *Teucrium*) germander 3. (genus *Clerodendrum*) glorybower, bagflower, bleeding-heart

-**FŠPFF**- ‘LAMIACEAE’ V 1. (genus *Tectona*) teak 2. (genus *Callicarpa*) beautyberry 3. (genus *Congea*) shower orchid, woolly congea

-**FŠPH**- ‘LAMIACEAE’ VI 1. (genus *Vitex*) chaste tree 2. (genus *Petitia*) bastard stopper 3. (genus *Premna*) firestick tree, fragrant premna

-**FŠPHW**- ‘LAMIACEAE’ VII 1. (genus *Prostanthera*) mintbush 2. (genus *Westringia*) westringia, white button bush, coastal rosemary 3. (Lamiacieae other than named genera; numerous genera) other lamiaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Lamiids > Lamiales > Acanthaceae

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Lamiids > Lamiales > Acanthaceae > Acanthoideae

-**LZFW**- ‘ACANTHOIDEAE’ I 1. (genus *Acanthus*) acanthus, bear’s breech, sea holly / holly mangrove 2. (genus *Aphelandra*) aphelandra, zebra plant 3. (genus *Andrographis*) creat / green cireta / king-of-bitters, false waterwillow

-**LZFY**- ‘ACANTHOIDEAE’ II 1. (genus *Barleria*) baleria, snake bush, bush violet, porcupine flower 2. (genus *Asystasia*) Chinese violet / coromandel / creeping foxglove, asystasia 3. (genus *Carlowrightia*) wrightwort

-**LZFL**- ‘ACANTHOIDEAE’ III 1. (genus *Odontonema*) toothedthread 2. (genus *Dyschoriste*) snakeherb 3. (genus *Hygrophila*) swampweed, temple plant / starhorn / giant hygro, glush weed, water wisteria, dwarf hygro, Indian waterweed

-**LZFR**- ‘ACANTHOIDEAE’ IV 1. (genus *Petalidium*) petal-bush 2. (genus *Ruellia*) ruellia, wild petunia 3. (Acanthoidiae other than named genera; numerous genera) other acanthoid plant

-**LZFŘ**- ‘ACANTHACEAE’ 1. (Avicennioideae; genus *Avicennia*) api api, black mangrove, grey mangrove / white mangrove, Indian mangrove 2. (Thungergioideae, Nelsonioideae; several genera) other acanthaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Lamiids > Lamiales > Oleaceae

-**LZM**- ‘OLEA’ 1. (*Olea europaea*) olive tree 2. (*O. capensis*) ironwood tree 3. (other *Olea* species) other oleid tree

-**LZMW**- ‘OLEACEAE’ I 1. (genus *Osmanthus*) osmanthus, sweet olive / tea olive / fragrant olive, holly olive, false holly 2. (genus *Phillyrea*) mock privet, green olive tree 3. (genus *Nestegis*) maire

-**LZMY**- ‘OLEACEAE’ II 1. (genus *Chionanthus*) fringetree 2. (genus *Forsythia*) forsythia, Easter tree, Korean goldenbell tree 3. (genus *Abeliophyllum*) white forsythia, Korean abelialeaf

-**LZML**- ‘OLEACEAE’ III 1. (genus *Jasminum*) jasmine 2. (genus *Menodora*) menodora, broom twinberry 3. (genus *Cartrema*) American olive / wild olive / devilwood

-**LZMR**- ‘OLEACEAE’ IV 1. (genus *Ligustrum*) privet 2. (genus *Forestiera*) swampprivet, stretchberry / desert olive/ tanglewood / devil’s elbow / spring herald / spring goldenglow / New Mexico privet / Texas forsythia, southern privet 3. [stem unassigned]

-**LZMŘ**- ‘OLEACEAE’ III 1. (genus *Syringa*) lilac 2. (genus *Fraxinus*) ash tree 3. (Oleaceae other than named genera) other oleaceous tree

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Lamiids > Lamiales

-**ŘŽP**- ‘LENTIBULARIACEAE’ 1. (genus *Genlislea*) corkscrew plant 2. (genus *Pinguicula*) butterwort 3. (genus *Utricularia*) bladderwort

-**ŘŽPW**- ‘LINDERNIACEAE’ 1. (genus *Lindernia*) false pimpernel, moist bank pimpernel 2. (genus *Torenia*) wishbone flower, bluewing, ola’a beauty 3. (Linderniaceae other than Stems 1 and 2; several genera) other linderniaceous flower plant

-**ŘŽPY**- ‘MARTYNIACEAE’ 1. (genus *Martynia*) cat’s claw / tiger’s claw 2. (genus *Proboscidea*) unicorn-plant, devil’s horn / devil’s claw, ram’s horn, doubleclaw 3. (Martyniaceae other than Stems 1 and 2; several genera) other martyniaceous plant, other unicorn plant

-**ŘŽPL**- ‘PEDALIACEAE’ 1. (genus *Sesamum*) sesame, benniseed 2. (genus *Harpagophytum*) devil’s claw / grapple plant / wood spider 3. (Pedaliaceae other than Stems 1 and 2; several genera) other pelaliaceous plant

-**ŘŽPR**- ‘PHRYMACEAE’ 1. (genus *Phryma*) lopseed 2. (genus *Glossostigma*) mudmat 3. (other Phymaceae; genera *Diplacus, Eythranthe, Mimulus, Thrydia*, five others) monkey-flower, musk flower, blood-drop-emlet

-**ŘŽPŘ**- ‘STILBACEAE’ 1. (genus *Nuxia*) forest elder / wild elder / forest nuxia 2. (genus *Bowkeria*) shell-flower 3. (Stilbaceae other than Stems 1 and 2; several genera) other stilbaceous plant

-**ŘŽPF**- ‘VERBENACEAE’ I 1. (genus *Verbena*) verbena / vervain 2. (genus *Phyla*) fogfruit / frogfruit 3. (genus *Petrea*) sandpaper vine / purple wreath / queen’s wreath

-**ŘŽPÇ**- ‘VERBENACEAE’ II 1. (genus *Citharexylum*) fiddlewood 2. (genus *Aloysia*) beebrush, whitebrush / whitebush, lemon verbena, almond verbena, mintbush lippia 3. (genus *Duranta*) golden dewdrop / pigeon berry / skyflower, espino

-**ŘŽPḶ**- ‘VERBENACEAE’ III 1. (genus *Lantana*) lantana, shrubverbena, tickberry, buttonsage, calico bush 2. (genus *Glandularia*) mock vervain, mock verbena 3. (genus *Stachytarpheta*) porterweed, velvetberry, rat tail, snakeweed, bastard vervain / Brazilian tea

-**ŘŽPH**- ‘VERBENACEAE’ IV 1. (genus *Lippia*) lippia, Mexican oregano / scented matgrass, licorice verbena, koseret 2. (genus *Nashia*) Bahamas berry / pineapple verbena / Moujean tea 3. (Verbenaceae other than named genera; numerous genera) other verbenaceous plant

-**ŘŽF**- ‘BIGNONIACEAE’ I 1. (genus *Bignonia*) crossvine 2. (genus *Adenoclymma*) garlic vine 3. (genus *Fridericia*) cricket-vine, chica

-**ŘŽFW**- ‘BIGNONIACEAE’ II 1. (genus *Jacaranda*) jacaranda tree, Brazilian caroba-tree, fern tree / black poui 2. (genus *Eccremocarpus*) Chilean glory-flower / Chilean glory creeper 3. (genus *Saritaea*) glowvine

-**ŘŽFY**- ‘BIGNONIACEAE’ III 1. (genus *Campsis*) trumpet creeper / trumpet vine 2. (genus *Deplanchea*) golden bouquet tree / wallaby wireless tree / yellow pagoda flower tree 3. (genus *Incarvillea*) Chinese trumpet flower

-**ŘŽFL**- ‘BIGNONIACEAE’ IV 1. (genus *Pandorea*) wonga vine, bower of beauty, Pandora’s vine, boat vine 2. (genus *Tecoma*) trumpetbush, Cape honeysuckle 3. (genus *Tecomanthe*) pink trumpet vine

-**ŘŽFR**- ‘BIGNONIACEAE’ V 1. (genus *Millingtonia*) tree jasmine / Indian cork tree 2. (genus *Oroxylum*) midnight horror / broken bones / tree of Damocles / Indian caper / Indian trumpet flower 3. (genus *Catalpa*) catalpa, catawba

-**ŘŽFŘ**- ‘BIGNONIACEAE’ VI 1. (genus *Chilopsis*) desert-willow 2. genus *Tabebuia*) roble, pink manjack, trumpet tree, pink poui, white ipê 3. (genus *Handroanthus*) epay / ipe / poui

-**ŘŽVL**- ‘BIGNONIACEAE’ VII 1. (genus *Spathodea*) African tulip tree / fountain tree / Nandi flame 2. (genus *Kigelia*) sausage tree / cucumber tree 3. (genus *Pajanella*) tender wild jack / pajanella

-**ŘŽVR**- ‘BIGNONIACEAE’ VIII 1. (genus *Pithecoctenium*) monkey’s comb 2. (Bignoniaceae other than named genera; numerous genera) other trumpet creeper/vine/bush/tree 3. [stem unassigned]

-**ŘŽG**- ‘GESNERIACEAE’ I 1. (genus *Fieldia*) fieldia 2. (genus *Sarmienta*) Chilean pitcher flower 3. (genus *Negria*) pumpkin tree

-**ŘŽGW**- ‘GESNERIACEAE’ II 1. (genus *Achimenes*) magic flowers, widow’s tears, Cupid’s bower, hot water plant 2. (genus *Episcia*) flame violet 3. (genus *Kohleria*) kohleria, tree gloxinia

-**ŘŽGY**- ‘GESNERIACEAE’ III 1. (genus *Haberlea*) Orpheus flower, resurrection plant 2. (Gesneriaceae other than named genera; numerous genera) other gesnerioid or cyrtandroid plant 3. [stem unassigned]

-**ŘŽGL**- ‘OROBANCHACEAE’ I 1. (genus *Orobanche*) boomrape 2. (genus *Castilleja*) Indian paintbrush, prairie-fire 3. (genus *Cistanche*) desert-boomrape

-**ŘŽGR**- ‘OROBANCHACEAE’ II 1. (genus *Conopholis*) cancer-root, squawroot 2. (genus *Agalinis*) false foxglove 3. (genus *Aureolaria*) yellow false foxglove, oak-leach, fernleaf false foxglove

-**ŘŽGŘ**- ‘OROBANCHACEAE’ III 1. (genus *Orthocarpus*) owl’s-clover 2. (genus *Triphysaria*) owel’s-clover, johnny-tuck / butter-and-eggs 3. (genus *Striga*) witchweed

-**ŘŽGV**- ‘OROBANCHACEAE’ IV 1. (genus *Melampyrum*) cow wheat 2. (genus *Lathraea*) toothwort 3. (genus *Rhinanthus*) rattle

-**ŘŽGḌ**- ‘OROBANCHACEAE’ V 1. (genus *Bartsia*) bartsia, velvetbells 2. (genus *Euphrasia*) eyebright 3. (genus *Parentucellia*) glandweed

-**ŘŽGM**- ‘OROBANCHACEAE’ VI 1. (genus *Boschniakia*) groundcone 2. (genus *Cordylanthus*) bird’s beak 3. (genus *Epifagus*) beech drops

-**ŘŽGN**- ‘OROBANCHACEAE’ VII 1. (genus *Schwalbea*) American chaffseed 2. (Orobanchaceae other than named genera; numerous genera) other orobanchaceous plant 3. [stem unassigned]

-**ŘŽV**- ‘ANTIRRHINEAE’ I 1. (genera *Antirrhinum, Misopates, Mohavea, Sairocarpus*) dragon flower, snapdragon 2. (genus *Asarina*) trailing snapdragon 3. (genus *Pseudorontium*) dog’s-mouth, Deep Canyon snapdragon

-**ŘŽVW**- ‘ANTIRRHINEAE’ II 1. (genus *Linaria*) Eurasian toadflax 2. (genus *Nuttallanthus*) Canada toadflax, Texas toadflax, Apalachicola toadflax 3. (genus *Cymbalaria*) ivy-leaved toadflax / coliseum ivy / mother of thousands / wandering sailor

-**ŘŽVY**- ‘ANTIRRHINEAE’ III 1. (genus *Kickxia*) cancerwort, fluellin 2. (genus *Maurandya*) angels trumpet / Mexican viper, climbing snapdragon / roving sailor 3. (Antirrhinines other than named genera) other antirrhinine snapdragon or toadflax-like plant

-**ŘŽM**- ‘CHELONEAE’ I 1. (genus *Chelone*) turtlehead 2. (genus *Collinsia*) blue eyed Mary, Chinese houses 3. (genus *Keckiella*) keckiella, beardtongue, penstemon

-**ŘŽMW**- ‘CHELONEAE’ II 1. (genus *Penstemon*) beardtongue 2. (genus *Nothocelone*) woodland beardtongue 3. (Cheloneae other than named genera; several genera) other cheloneid plant

-**ŘŽMY**- ‘GRATIOLEAE’ I 1. (genera *Gratiola*, *Sophronanthe*) hedge-hyssops 2. (genus *Bacopa*) waterhyssop 3. (genus *Capraria*) goatweed, Tamaulipan tea

-**ŘŽML**- ‘GRATIOLEAE’ II 1. (genus *Leucospora*) cliff conobea / cut-leaf conobea / much-cleft conobea / narrow-leaved paleseed / Obi-Wan conobea 2. (genus *Limnophila*) marshweed 3. (genus *Mecardonia*) axilflower

-**ŘŽMR**- ‘GRATIOLEAE’ III 1. (genus *Scoparia*) licorice weed, sweet broom, scoparia-weed 2. (genus *Stemodia*) twintip, stemodia, bluerod, pagurda 3. (Gratioleae other than named genera) other gratiolid plant

-**ŘŽMŘ**- ‘VERONICEAE’ I 1. (genus *Veronica*) speedwell, germander speedwell, bird’s eye, gypsyweed, brooklime 2. (genus *Veronicastrum*) Culver’s-root / Culver’s physic / Bowman’s root / black root, veronicastrum 3. (genus *Hebe*) hebe, New Zealand lilac, bushy veronica, mountain-box, koromiko, showy-speedwell

-**ŘŽX**- ‘VERONICEAE’ II 1. (genus *Wulfenia*) wulfenia 2. (Veroniceae other than named genera) other veronicid plant 3. [stem unassigned]

-**ŘŽXW**- ‘PLANTAGINACEAE’ I 1. (genus *Plantago*) plantain, fleawort 2. (genus *Digitalis*) foxglove 3. (genus *Erinus*) fairy foxglove / alpine balsam / starflower / liver balsam

-**ŘŽXL**- ‘PLANTAGINACEAE’ II 1. (genus *Callitriche*) water-starwort 2. (genus *Hippuris*) mare’s tail 3. (genus *Globularia*) globe daisy, globularia, ball flower

-**ŘŽXR**- ‘PLANTAGINACEAE’ III 1. (genus *Russelia*) firecracker plant, coralblow, russelia, fountainbush / coral plant 2. (Plantaginaceae other than named genera; numerous genera) other plantaginaceous plant 3. [stem unassigned]

-**ŘŽD**- ‘SCROPHULARIACEAE’ I 1. (genus *Scophularia*) figwort 2. (genus *Verbascum*) mullein 3. (genus *Buddleja*) butterfly bush

-**ŘŽDW**- ‘SCROPHULARIACEAE’ II 1. (genus *Limosella*) mudwort 2. (genus *Diascia*) twinspur 3. (genus *Leucophyllum*) barometer bush, Texas ranger / silverleaf

-**ŘŽDY**- ‘SCROPHULARIACEAE’ III 1. (genus *Myoporum*) boobialla, bastard ironwood / popwood, native myrtle, sugarwood, myoporum 2. (genus *Eremophila*) emu bush, turkey bush, poverty bush 3. (genus *Bontia*) wild olive, white alling

-**ŘŽDL**- ‘SCROPHULARIACEAE’ IV 1. (genus *Phygelius*) Cape fuchsia / Cape figwort 2. (Scorphulariaceae other than named genera; numerous genera) other scrophulariaceous plant, other figwort-like plant 3. [stem unassigned]

-**ŘŽŇ**- ‘LAMIALES’ I 1. (Byblidaceae; genus *Byblis*) rainbow plant 2. (Calceolariaceae; three genera) lady’s purse / slipper flower / pocketbook flower / slipperwort 3. (Mazaceae; two genera) mazus, swamp musk, lancea

-**ŘŽŇW**- ‘LAMIALES’ II 1. (Paulowniaceae; genus *Paulownia*) dragontree, empress tree, princess tree, foxglove-tree 2. (Tetrachondraceae; genera *Tetrachondra, Polypremum*) juniperleaf / rustweed 3. (Carlemanniaceae, Plocospermataceae, Schlegeliaceae, Thomandersaceae; several genera) other lamiales plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Lamiids > Gentianales

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Lamiids > Gentianales > Rubiaceae

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Lamiids > Gentianales > Rubiaceae > Rubioideae

-**NÇB**- ‘RUBIEAE’ I 1. (genus *Rubia*) madder 2. (genus *Kelloggia*) kelloggia 3. (genus *Gallium*) bedstraw

-**NÇBW**- ‘RUBIEAE’ II 1. (genus *Crucianella*) crosswort 2. (genus *Asperula*) woodruff 3. (Rubieae other than named genera; several genera) other rubiid plant

-**NÇBY**- ‘PSYCHOTRIEAE’ (genus *Psychotria*) psychotria, wild coffee 2. (genus *Myrmecodia*) ant-house plant 3. (Psychotrieae other than Stems 1 and 2) other psychotriid plant

-**NÇBL**- ‘SPERMACOCEAE’ I 1. (genus *Spermacoce*) false buttonweed 2. (genus *Diodella*) poorjoe / rough buttonweed 3. (genus *Diodia*) Virginia buttonweed

-**NÇBR**- ‘SPERMACOCEAE’ I 1. (genus *Ernodea*) beech creeper / cougbush 2. (genus *Hedyotis*) starviolet 3. (genus *Houstonia*) bluet

-**NÇBŘ**- ‘SPERMACOCEAE’ I 1. (genus *Mitracarpus*) girdlepod 2. (genus *Richardia*) Mexican-clover 3. (Spermacoceae other than named genera; numerous genera) other spermacoceous plant

-**NÇBV**- ‘ANTHOSPERMEAE’ 1. (genus *Opercularia*) stinkweed, dogweed 2. (genus *Nertera*) coral bead plant / coral moss / English baby tears 3. (Anthospermeae other than Stems 1 and 2; several genera) other anthosperm

-**NÇBḌ**- ‘RUBIOIDEAE’ 1. (Mitchelleae; genus *Mitchella*) partridge berry 2. (Paederieae; several genera) sewer vine, snowrose, tree of a thousand stars, Japanese boxthorn 3. (Palicoureeae, Coussareeae, Argostemmateae Colletoecemateae, Craterispermeae, Cyanoneuroneae, Danaideae, Dunnieae, Gaertnereae, Knoxieae, Lasiantheae, Morindeae, Ophiorrhizeae, Perameae, Prismatomerideae, Putorieae, Schizocoleeae, Theligoneae, Urophyleae; numerous genera) other rubioid plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Lamiids > Gentianales > Rubiaceae > Ixoroideae

-**ÇX**- ‘COFFEA’ 1. (*Coffea arabica*) arabica coffee plant 2. (*C. canephora*) robusta coffee plant 3. (all other *Coffea* species) other coffea plant

-**ÇXW**- ‘GARDENIEAE’ 1. (genus *Gardenia*) gardenia, cape jasmine, turpentine tree 2. (genus *Randia*) indigoberry 3. (Gardenieae other than Stems 1 and 2; numerous genera) other gardeniid plant

-**ÇXL**- ‘CONDAMINEEAE’ 1. (genus *Warszewiczia*) chaconia / wild poinsettia / pride of Trinidad and Tobago 2. (Condamineeae other than Stem 1; numerous genera) other condamineid plant 3. [stem unassigned]

-**ÇXR**- ‘IXOROIDEAE’ 1. (Ixoreae; genus *Ixora*) West Indian jasmine, jungle flame, jungle geranium 2. (Posoquerieae; genera *Posoqueria, Molopanthera*) needle flower 3. (Airospermeae, Alberteae, Aleisanthieae, Augusteae, Bertiereae, Coffeeae [other than genus *Coffea*], Cordiereae, Cremasporeae, Crossopterygeae, Greeneeae, Henriqueaieae, Jackieae, Mussaendeae, Octotropideae, Pavetteae, Retiniphylleae, Sabiceeae, Scyphiphoreae, Sherbournieae, Sipaneeae, Steenisieae, Trailliaedoxeae, Vanguerieae; numerous genera) other ixoroid plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Lamiids > Gentianales > Rubiaceae > Cinchonoideae

-**RGD**- ‘CINCHONEAE’ 1. (genus *Cinchona*) quinine, cinchona, Peruvian bark, Jesuit’s bark 2. (genus *Remijia*) remijia 3. (Cinchoneae other than Stem 1 and 2) other cinchonid plant

-**RGDW**- ‘CHIOCOCCEAE’ I 1. (genus *Chiococca*) milkberry, West Indian snowberry 2. (genus *Catesbaea*) lilythorn 3. (genus *Portlandia*) bell flower

-**RGDY**- ‘CHIOCOCCEAE’ II 1. (genus *Scolsanthus*) Puerto Rico devilbrush, maricao 2. (Chiococceae other than named genera; several genera) other chiococcid plant 3. [stem unassigned]

-**RGDL**- ‘GUETTARDEAE’ 1. (genus *Guettarda*) velvetseed 2. (genus *Bobea*) ‘ahakea 3. (Guettardeae other than Stems 1 and 2; several genera) other guettardid plant

-**RGDR**- ‘NAUCLEEAE’ 1. (genus *Cephalanthus*) buttonbrush 2. (genus *Uncaria*) gambier, cat’s claw 3. (Naucleeae other than Stems 1 and 2; numerous genera) other naucleid plant

-**RGDŘ**- ‘CINCHONOIDEAE’ 1. (Rondeletieae; numerous genera) rondeletia, Cleveland sunrise, Panama-rose, other rondeletiid plant 2. (Strumpfieae; genus *Strumpfia*) Pride of Big Pine 3. (Hamelieae, Hymenodictyeae, Hillieae, Isertieae; several genera) other cichonoid plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Lamiids > Gentianales > Rubiaceae > remaining sub-families/tribes/genera

-**RGDV**- ‘RUBIACEAE’ 1. (Coptosapelteae; genera *Coptosapelta*, *Acranthera*) coptosapelta, acranthera, coptosapeltid plant 2. (Luculieae; genus *Luculia*) luculia 3. (Rubiacieae other than named genera) other rubiaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Lamiids > Gentianales > Gentianaceae

-**GZVW**- ‘GENTIANACEAE’ I 1. (genera *Gentiana, Gentianella, Gentianopsis*) gentia, agueweed 2. (genus *Frasera*) frasera, American columbo, yellow gentian, green gentian, elkweed, deer’s ears / monument plant 3. (genus *Eustoma*) lisianthus, prairie gentian

-**GZVY**- ‘GENTIANACEAE’ II 1. (genus *Centaurium, Sebaea, Zeltnera*) centaury, sebaea, quinineweed 2. (genus *Cicendia*) slender cidendia / yellow centaury 3. (genus *Exacum*) Persian violet

-**GZVL**- ‘GENTIANACEAE’ III 1. (genus *Swertia*) felwort, swertia 2. (genus *Sabatia*) slender marsh-pink, rose gentian, prairie sabatia, meadow pink 3. (genus *Obolaria*) Virginia pennywort

-**GZVR**- ‘GENTIANACEAE’ IV 1. (genus *Voyria*) ghostplant 2. (genus *Bartonia*) screwstem 3. (genus *Fagraea*) kingwood, tembesu, perfume flower tree

-**GZVŘ**- ‘GENTIANACEAE’ V 1. (genus *Orphium*) sea rose 2. (Gentianaceae other than named genera; numerous genera) other gentianaceous plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Lamiids > Gentianales > Apocynaceae

-**RŢPW**- ‘APOCYNOIDEAE’ I 1. (genus *Apocynum*) dogbane, Indian hemp 2. (genus *Trachelospermum*) star jasmine, Confederate jasmine 3. (genus *Echites*) devil’s potato

-**RŢPY**- ‘APOCYNOIDEAE’ II 1. (genus *Nerium*) oleander, nerium 2. (genus *Mandevilla*) rocktrumpet, mandevilla, dipladenia, Brazilian jasmine, Chilean jasmine 3. (genus *Cycladenia*) Sacramento waxydogbane

-**RŢPL**- ‘APOCYNOIDEAE’ III 1. (genus *Angadenia*) pineland golden trumpet 2. (genus *Pentalinon*) hammock viper’s-tail / licebush / wild allamanda / wild wist yellow mandevilla / yellow dipladenia 3. (Apocynoideae other than named genera) other apocynoid plant

-**RŢPR**- ‘ASCLEPIADOIDEAE’ I 1. (genus *Asclepias*) milkweed, spider antelope horns, cow-cabbage, blood flower / cotton bush / Mexican / redhead, milkplant, milkflower, silkweed, solanoa, swallow-wort, butterfly flower, butterfly weed / pleurisy root, green antelopehorn 2. (genus *Cynanchum*) dog-strangling vine, climbing milkweed, caustic bush 3. (genus *Funastrum*) twinevine

-**RŢPŘ**- ‘ASCLEPIADOIDEAE’ II 1. (genus *Matelea*) milkvine 2. (genus *Marsdenia*) jungle netvine, waxflower / Madagascar jasmine / Hawaiian wedding flower / bridal wreath 3. (genus *Hoodia*) Bushman’s hat

-**RŢPF**- ‘ASCLEPIADOIDEAE’ III 1. (genus *Orbea*) star flower 2. (Stapeliads; numerous genera) stapeliad 3. (Ascelpiadoideae other than named genera; numerous genera) other asclepiadoid plant, other milkweed-like plant

-**RŢPŢ**- ‘PERIPLOCOIDEAE’ I 1. (genus *Periploca*) silkvine 2. (genus *Petopentia*) propeller vine 3. (genus *Cryptolepis*) milk rope

-**RŢPÇ**- ‘PERIPLOCOIDEAE’ II 1. (genus *Hemidesmus*) Indian sarsaparilla 2. (genus *Decalepis*) swallowroot 3. (Periplocoideae other than named genera; numerous genera) other periplocoid plant

-**RŢPH**- ‘RAUVOLFIOIDEAE’ I 1. (genera *Catharanthus, Vinca*) periwinkle 2. (genus *Rauvolfia*) devil pepper 3. (genus *Ochrosia*) yellowwood, bloodhorn / kopsia, ochrosia

-**RŢPV**- ‘RAUVOLFIOIDEAE’ II 1. (genus *Vallesia*) pearl bery / tearshrub, vallesia 2. (genus *Haplophyton*) cockroach plant 3. (genus *Aspidosperma*) wheel tree, quebracho

-**RŢPS**- ‘RAUVOLFIOIDEAE’ III 1. (genus *Alyxia*) chainfruit, sea box / dysentery bush, alyxia 2. (genus *Plumeria*) plumeria, frangipani, Singapore graveyard flower 3. (genus *Pteralyxia*) pteralyxia

-**RŢPŠ**- ‘RAUVOLFIOIDEAE’ IV 1. (genus *Cerbera*) cassowary plum / grey milkwood, sea mango, suicide plant 2. (genus *Carissa*) Bengal currant / Christ’s thorn, carandas plum, Natal plum, num-num, conkerberry / bush plum 3. (genus *Hancornia*) mangabeira / mangaba plant

-**RŢPL**- ‘RAUVOLFIOIDEAE’ V 1. (genus *Allamanda*) allamanda, trumpetvine, golden-trumpet 2. (genus *Melodinua*) melodinus 3. (Rauvolfioideae other than named genera; numerous genera) other rauvolfioid plant

-**RŢPHW**- ‘SECAMONOIDEAE’ 1. (genus *Secamone*) secamone, cork vine / corky milkvine 2. (Secamonoideae other than Stem 1; several genera) other secamonid plant 3. [stem unassigned]

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-**LZN**- ‘GELSEMIACEAE’ 1. (genus *Gelsemium*) heartbreak grass, swamp jessamine / Rankin’s trumpetflower, yellow jessamine / Carolina jessamine / evening trumpetfower 2. (genera *Mostuea* and *Pteleocarpa*) other gelsemiid plant 3. [stem unassigned]

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-**LZNW**- ‘LOGANIACEAE’ I 1. (genus *Logania*) logania 2. (genus *Labordia*) labordia 3. (genus *Mitreola*) hornpod

-**LZNY**- ‘LOGANIACEAE’ II 1. (genus *Strychnos*) strychnine tree / nux vomica / poison nut / quaker buttons, clearing-nut tree, threaded boxwood, black monkey orange 2. (genus *Spigelia*) pinkroot, Indian pink 3. (Loganiaceae other than named genera; several genera) other loganiaceous tree

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Lamiids > Boraginales

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Lamiids > Boraginales > Boraginaceae

-**FSTW**- ‘BORAGINOIDEAE’ I 1. (genus *Borago*) borage 2. (genus *Pulmonaria*) lungwort 3. (genus *Symphytum*) comfrey

-**FSTY**- ‘BORAGINOIDEAE’ II 1. (genus *Echium*) viper’s bugloss, Salvation Jane, Lady Campbell weed / Paterson’s curse, echium 2. (genus *Brunnera*) Siberian bugloss 3. [stem unassigned]

-**FSTL**- ‘BORAGINOIDEAE’ III 1. (genus *Anchusa*) bugloss / true alkanet / corn bugloss, Cape bugloss / Cape forget-me-not 2. (genus *Alkanna*) alkanet / dyer’s alkanet / dyer’s bugloss / orchanet / Spanish bugloss / Languedoc bugloss 3. (genus *Pentaglottis*) green alkanet, evergreen bugloss

-**FSTR**- ‘BORAGINOIDEAE’ IV 1. (genus *Nonea*) monkwort 2. (genus *Asperugo*) madwort 3. (genus *Cerinthe*) honeywort

-**FSTŘ**- ‘BORAGINOIDEAE’ V 1. (genus *Buglossoides*) gromwell, bastard alkanet 2. (genus *Lithodora*) purple gromwell, olive-leaved gromwell 3. (genus *Lithosperma*) gromwell, marbleseed, stoneseed, puccoon, Indian-paint, whiteseed

-**FSTF**- ‘BORAGINOIDEAE’ VI 1. (genus *Myosotis*) forget-me-not, scorpion grass 2. (genus *Eritrichium*) alpine forget-me-not 3. (genus *Myosotidium*) giant forget-me-not, chatham Islands forget-me-not

-**FSTV**- ‘BORAGINOIDEAE’ VII 1. (genus *Amsinckia*) fiddleneck 2. (genus *Carmona*) scorpionbush 3. (genus *Mertensia*) oysterplant, bluebells, mountain bell

-**FSTÇ**- ‘BORAGINOIDEAE’ VIII 1. (genus *Plagiobothrys*) popcorn flowers 2. (genus *Cryptantha*) cat’s eyes / popcorn flowers 3. (genus *Amebia*) Prophet’s flower

-**FSTĻ**- ‘BORAGINOIDEAE’ IX 1. (genus *Cynoglossum*) hounds’ tongue, wild comfrey 2. (genus *Hackelia*) stickseed, sticktight, beggar’s lice 3. (genus *Lappula*) stickseed, bluebur / bristly sheepbur / bur forget-me-not

-**FSTH**- ‘BORAGINOIDEAE’ X 1. (genus *Heliotropium*) heliotrope 2. (genus *Omphalodes*) navelwort 3. (genus *Pectocarya*) combseed

-**FSTHW**- ‘BORAGINOIDEAE’ XI 1. (genus *Cordia*) geigertree 2. (genus *Trichodesma*) camel bush / cattle bush 3. (Boraginoideae other than named genera; numerous genera) other boraginoid plant

-**FSMW**- ‘HYDROPHYLLOIDEAE’ I 1. (genus *Hydrophyllum*) waterleaf 2. (genus *Nemophila*) baby-blue-eyes, five-spot nemophila 3. (genus *Nama*) fiddeleaf

-**FSMY**- ‘HYDROPHYLLOIDEAE’ II 1. (genus *Eriodictyon*) yerba santa, Indian Knob mountainbalm, poodle-dog bush 2. (genus *Ellisia*) waterpod / false baby-blue-eyes / Aunt Lucy 3. (genus *Phacelia*) phacelia, scorpionweed, Whitlavia / wild Canterbury bells, Miami mist, lavender windows, wild heliotrope, California bluebell / desertbells

-**FSML**- ‘HYDROPHYLLOIDEAE’ III 1. (genus *Emmenanthe*) whispering bells 2. (genus *Romanzoffia*) mistmaid / mistmaiden 3. (genus *Pholistoma*) fiesta flower

-**FSMR**- ‘HYDROPHYLLOIDEAE’ IV 1. (genus *Draperia*) violet draperia 2. (genus *Eucrypta*) hideseed 3. (genus *Tricardia*) threeheart

-**FSMŘ**- ‘HYDROPHYLLOIDEAE’ V 1. (genus *Hesperochiron*) hesperochiron 2. (Hydrophylloideae other than named genera) other hydrophylloid plant 3. [stem unassigned]

-**FSNW**- ‘EHRETIOIDEAE’ I 1. (genus *Ehretia*) peace bush / native willow / peachwood, anacua, koda 2. (genus *Bourreria*) strongbark / strongback 3. (genus *Halgania*) halgania, blue bush

-**FSNY**- ‘EHRETIOIDEAE’ II 1. (genus *Rochefortia*) greenheart ebony 2. (genus *Tiquilia*) crinklemat, Peruvian sand flower 3. (Ehretioideae other than named genera; several genera) other ehretioid plant

-**FSN̆W**- ‘BORAGINACEAE’ 1. (Lennooideae; three genera) desert Christmas tree / scaly-stemmed sandplant, sandfood 2. (Cordioidiae; three genera) manjack, cordioid plant 3. (Boraginaceae other than named genera) other boraginaceous plant

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Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Lamiids > Solanales > Solanaceae

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Lamiids > Solanales > Solanaceae > Solanoideae > Solaneae

-ZP- ‘SOLANUM’ I 1. (*S. tuberosum*) potato plant 2. (*S. lycopersicum*) tomato plant 3. (*S. melongena*) eggplant plant / aubergine

-ZPL- ‘SOLANUM’ II 1. (tomato-producing *Solanum* species other than *S. lycospermum; S. pimpinellifolium, S. peruvianum, S. cheesmanii, S. galapagense, S. chilense,* etc.) wild tomato, currant tomato 2. (*S.aethiopicum*) Ethiopian eggplant / gilo 3. (*S. quitoense*) naranjilla / lulo

-ZPR- ‘SOLANUM’ III 1. (*S. torvum*) Turkey berry / wild eggplant / pea eggplant / shoo-shoo bush / prickly nightshade / susumber 2. (*S. muricatum*) pepino melon / melon pear 3. (*S. betaceum*) tamarillo / tree tomato / blood fruit

-ZPR̆- ‘SOLANUM’ IV 1. (*S. lycocarpum*) wolf apple 2. (*S. sessiliflorum*) cocona 3. (edible Australian wild tomato species, including *S. aviculare, S. centrale, S. chippendalei, S. diversiflorum, S. ellipticum, S. laciniatum, S. orbiculatum, S. phlomoides*) kangaroo apple, desert raisin / bush raisn / bush saltana, bush tomato, potato bush, round-leaved solanum, wild tomato

-ZPF- ‘SOLANUM’ V 1. (*S. retroflexum*) wonderberry / sunberry 2. (*S. scabrum*) garden huckleberry 3. (*S. ferox, a.k.a. S. lasiocarpum*) hiary-fruited eggplant / Indian nightshade

-**ZPŢ**- ‘SOLANUM’ VI 1. (*S. sisymbriifolium*) vila-vila / sticky nightshade / red buffalo-bur / fire-and-ice plant, litchi tomato 2. (*S. mauritanium*) earleaf nightshade /flannel weed / bugweed / tobacco weed / kerosene plant / wild tobacco 3. (*S. dulcamara*) bittersweet / bittersweet nightshade / blue bindweed / Amara Dulcis / climbing nightshade / fellenwort / felonwood / poisonberry / poisonflower / scarlet berry / trailing bittersweet / violet bloom / woody nightshade

-**ZPL̨**- ‘SOLANUM’ VII 1. (*S. mammosum*) nipplefruit / cow’s udder / fox head / apple of Sodom 2. (ornamental *Solanum* species, e.g., *S. capsicastrum, S. crispum, S. laxum, S. pseudocapsicum, S. rantonnetii, S. seaforthianum, S. wendlandii*) winter cherry, Jerusalem cherry, Chilean potato tree, potato vine, Christmas cherry, blue potato bush, Italian jasmine / St. Vincent lilac, paradise flower 3. (*Solanum* species not previously named) nightshade, soda apple, popolo, horsenettle, devil’s apple, porcupine tomato, devil’s-fig, wolfpeach, wild potato

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Lamiids > Solanales > Solanaceae > Solanoideae > Capsiceae

-ZPW- ‘CAPSICUM ANUUM (BELL/SWEET PEPPER)’ 1. bell pepper plant 2. sweet/Italian pepper, pepperoncino plant 3. paprika plant

-ZPY- ‘CAPSICUM ANUUM (HOT PEPPER)’ 1. serrano pepper plant 2. cayenne pepper plant 3. jalapeño pepper plant

-ZPH- ‘OTHER CULTIVATED CAPSICUM SPECIES’ 1. (*C. chinense*) habanero pepper plant [includes 7-pot, ají dulce, datil, Scotch bonnet, Trinidad scorpion varieties] 2. (*C. baccatum*) ají chili pepper plant 3. (other domesticated/cultivated *Capsicum* species; *C. frutescens, C. pubescens*) tabasco chili pepper / piri piri / Hawaiian pepper / xiaomila pepper, manzano pepper / rocoto pepper / locoto pepper

-**ZPHW**- ‘CAPSICEAE’ 1. (non-cultivated *Capsicum* species; numerous species) non-domesticated/ non-cultivated capsicum species 2. (genus *Lycianthes*) blue potato bush / Paraguay nightshade, various lycianthes nightshades 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Lamiids > Solanales > Solanaceae > Solanoideae > Physaleae

-ZPÇ- 'PHYSALIS' 1. (*P. philadelphica* a.k.a. *P. ixocarpa*) tomatillo / Mexican husk tomato plant 2. (*P. alkekengi*) Chinese/Japanese lantern, winter-cherry plant 3. (other *Physalis* species) groundcherry, strawberry-tomato plant, Inca berry plant, golden strawberry plant

-**ZPS**- 'PHYSALEAE' I 1. (genus *Quincula*) lobed groundcherry, purple groundcherry 2. (genus *Chamaesaracha*) five eyes, chamaesaracha 3. (genus *Mellissia*) Saint Helena boxwood

-**ZPŠ**- 'PHYSALEAE' II 1. (genus *Nothocestrum*) 'aiea 2. (genus *Withania*) Indian rennet / vegetable rennet, Indian ginseng / winter cherry 3. (Physaleae other than named genera; numerous genera) other physalid plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Lamiids > Solanales > Solanaceae > Solanoideae remaining families

-**ŇŢW**- 'HYOSCYAMEAE' 1. (genus *Hyoscyamus*) henbane 2. (genus *Atropa*) belladona, deadly nightshade 3. (Hyoscyameae other than Stems 1 and 2; several genera) scopolia, other hyoscyamid plant

-**ŇŢY**- 'LYCIEAE & MANDRAGOREAE' 1. (genus *Lycium*) box-thorn, desert-thorn, honey thorn, water-jacket, matrimony-vine / goji / goji-berry / Duke of Argyll's teaplant, wolfberry, Christmas berry, peachthorn, squawthorn 2. (other Lycieae; genus *Grabowskia*) grabowskia, other lyciid plant 3. (Mandragoreae; genus *Mandragora*) mandrake

-**ŇŢL**- 'DATUREAE' 1. (genus *Datura*) devil's trumpet 2. (genus *Brugmansia*) angel's trumpet 3. (other Datureae; genus *Trompettia*) trompettia, other daturid plant

-**ŇŢR**- 'SOLANDREAE' 1. (genus *Solandra*) chalice vine, Hawaiian lily 2. (genus *Juanulloa*) goldfinger 3. (Solandreae other than Stems 1 and 2; several genera) other solandrean plant

-**ŇŢŘ**- 'OTHER SOLANOIDEAE' 1. (Nicandreae; genus *Nicandra*) apple-of-Peru / shoo-fly plant 2. (Nolaneae; genus *Nolana*) Chilean bell flower 3. [stem unassigned]

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-**ŇŢM**- 'CESTROIDEAE' I 1. (genus *Cestrum*) cestrum, jessamine, lady of the night, green poisonberry 2. genus *Vestia*) Chilean box thorn 3. (genus *Salpiglossis*) painted tongue / scalloped tube tongue / velvet trumpet flower

-**ŇŢN**- 'CESTROIDEAE' II 1. (genus *Browallia*) bush-violet, amethyst flower 2. (genus *Streptosolen*) marmalade bush / firebush 3. (Cestroideae other than named genera; several genera) other cestroid flower

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-**ČVW**- 'PETUNIOIDEAE' I 1. (genus *Petunia*) petunia 2. (genus *Brunfelsia*) raintree, brunfelsia, lady-of-the-night 3. (genus *Fabiana*) false heath

-**ČVY**- 'PETUNIOIDEAE' II 1. (genus *Nierembergia*) cupflower, whitecup, nierembergia 2. (genus *Plowmania*) Chiapas red trumpet 3. (Petunioideae other than named genera; several genera) other pentunioid plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Lamiids > Solanales > Solanaceae remaining families

-**ČVL**- 'NICOTIANOIDEAE' I 1. (genus *Nicotiana*) tobacco plant 2. (genus *Anthocercis*) tailflower 3. (genus *Duboisia*) corkwood tree

-**ČVR**- 'NICOTIANOIDEAE' II 1. (genus *Symonanthus*) symonanthus, Bailey's symonanthus 2. (Nicotianoideae other than named genera; several genera) other nocotianoid plant 3. [stem unassigned]

-**ČVŘ**- 'OTHER SOLANACEAE' 1. (Schizanthaceae; genus *Schizanthus*) butterfly flower / fringeflower / poor-man's orchid 2. (Schwenckioideae, Goetzeoideae; several genera) other solanaceous plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Lamiids > Solanales > Convolvulaceae

-**PSVW**- 'CONVOLVULACEAE' I 1. (genus *Convolvulus*) morning glory, bindweed, convolvulus, rhodium / rhodium-wood, scammony, silverbush 2. (genus *Calystegia*) false bindweed, hearbind / hedgebell / hedge bindweed / large bindweed, plateau morning glory, beach morning glory 3. (genus *Evolvulus*) dwarf morning glory, Australian baby blue / sapphire blue, evovulus

-**PSVY**- 'CONVOLVULACEAE' II 1. (genus *Bonamia*) lady's nightcap 2. (genus *Cressa*) alkaliweed 3. (genus *Stylisma*) dawnflower

-**PSVL**- 'CONVOLVULACEAE' III 1. (*Ipomoea batatas*) sweet potato 2. (genus *Ipomoea* other than *I. batatas*) morning glory, moonflower, cypress vine / cardinal creeper / cardinal vine / star glory, man-of-the-earth / manroot / wild potato vine, moon vine, water spinach, railway creeper, spiderleaf, lilacbell, firevine / Spanish flag, whitestar / whitestar potato / pitted morning-glory 3. (genus *Argyreia*) Hawaiian baby woodrose

-**PSVR**- 'CONVOLVULACEAE' IV 1. (genus *Dichondra*) ponysfoot, dichondra 2. (genus *Jacquemontia*) clustervine, jacquemontia 3. (genus *Cuscuta*) dodder

-**PSVŘ**- 'CONVOLVULACEAE' V 1. (genus *Merremia*) woodrose, rock rosemary, Spanish arborvine, hogvine 2. (genus *Operculina*) turpeth / St. Thomas lidpod 3. (Convolvulaceae other than named genera; numerous genera) other convolvulaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Lamiids > Solanales remaining families

-**ŇŢŇ**- 'OTHER SOLONALES FAMILY' 1. (Hydroleaceae; genus *Hydrolea*) false fiddleleaf 2. (Montiniaceae; genera *Montinia, Grevea, Kaliphora*) montiniid plant 3. (Sphenocleaceae; genus *Sphenoclea*) sphenoclea

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Lamiids remaining orders: Vahliales, Garryales, Metteniusales, Icacinales

-**ŇŢMW**- 'GARRYALES' 1. (genus *Garrya*) silk tassel / tassel bush 2. (genus *Acuba*) spotted laurel / Japanese laurel / gold dust plant, acuba 3. (genus *Eucommia*) eucommia

-**ŇŢMY**- 'OTHER LAMIID ORDER' 1. (Icacinales; several genera) icacinacious plant, oncotheca plant 2. (Vahliales; genus *Vahlia*) vahlia 3. (Metteniusales; several genera) metteniusaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Ericales

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Ericales > Ericaceae

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Ericales > Ericaceae > Vaccinoideae

-ŇĎ- 'VACCINIUM SHRUB' I 1. (*V.* subgenus *Vaccinium*) blueberry, sparkleberry 2. (*V.* subgenus *Oxycoccus*) cranberry 3. (*V.* subgenus *Oxycoccus,* section *Myrtillus*) huckleberry, bilberry, whortleberry, grouseberry, blaeberry / fraughan / hurtleberry, Alaska blueberry

-ŇĎR- 'VACCINIUM SHRUB' II 1. (*V.* subgenus *Oxycoccus,* section *Vitis-idaea*) partridge berry / cowberry / redberry / lingonberry / foxberry / quailberry / bearberry / beaverberry / cougarberry 2. (*V.* subgenus *Oxycoccus,* section *Polycodium*) deerberry 3. (other *Vaccinium* species) other blueberry or whortleberry

-ŇĎL- 'GAULTHERIA' 1. (*Gaultheria procumbens*) eastern teaberry [source of wintergreen flavoring] 2. (*G. shallon*) salal / shallon 3. (other Gaultheria species) gaultheria

-ŇĎĎ- 'VACCINIOIDEAE' 1. (other Vaccinieae; numerous genera) other vaccinid plant 2. (Oxydendreae; genus *Oxydendrum*) sourwood / sorrel tree 3. (Andromedae, Gaultherieae [other than genus *Gaultheria*], Lyonieae; several genera) other vaccinioid plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Ericales > Ericaceae > Ericoideae

-ŇḌM- 'RHODODENDRON' (genus *Rhododendron*) 1. evergreen or vireya rhododendron plant 2. evergreen azalea plant 3. deciduous azalea plant

-ŇḌMW- 'ERICOIDEAE' I 1. (genus *Erica*) heath, heather 2. (genus *Phylloduce*) mountainheath, mountain heather 3. (genus *Ceratiola*) sand heath, sandhill-rosemary, Florida-rosemary

-ŇḌMY- 'ERICOIDEAE' II 1. (genus *Kalmia*) sheep-laurel, lambkill, calf-kill, kill-kid, sandmyrtle, whitewicky, mountain-laurel, bog-laurel / bob kalmia 2. (genus *Daboecia*) St. Dabeoc's heath 3. (Ericoideae other than named genera; several genera) other ericoid (heath-like) plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Ericales > Ericaceae > Arbutoideae

-ŇḌNW- 'ARBUTOIDEAE' I 1. (genus *Arbutus*) madrone, strawberry tree, cane apple tree 2. (genus *Arctostaphylos*) manzanita 3. (genus *Xylococcus*) mission manzanita

-ŇḌNY- 'ARBUTOIDEAE' II 1. (genus *Ornithostaphylos*) Baja birdbush 2. (genus *Comarostaphylis*) summer holly 3. (genus *Arctous*) bearberry

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Ericales > Ericaceae > Pyroloideae, Monotropoideae, Cassiopoideae, and Harrimanelloideae

-ŇḌN- 'ERICACEAE' 1. (Pyroloideae; several genera) wintergreen, shinleaf, shineleaf, prince's pine / pipsissewa, rheumatism root 2. (Monotropoideae; several genera) monotrope, gnome plant / cone plant, Indian pipe / ghost plant / corpse plant, Dutchman's pipe / pinesap / yellow bird's-nest, pygmy pipes, pinefoot, pinedrops / beechdrops / giant bird's nest, snow plant / snow flower 3. (Casiopoideae and Harrimanelloideae; genera *Cassiope, Harrimanella*) moss bell heather, other ericaceous heather

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Ericales > Actinidiaceae

-ŇḌW- 'ACTINIDIACEAE' 1. (*Actinidia deliciosa*) kiwi vine / Chinese gooseberry 2. (*A. arguta, A. kolomikta, A. polygama, A. chinensis*) kiwi berry / grape kiwi / dessert kiwi / cocktail kiwi vine, golden kiwi vine 3. (other Actinidiaceae; genera Clematoclethra, Saurauia, plus other Actinidia species) actinidia, actinidiaceous plant

-ŇḌV- 'CLETHRACEAE & CYRILLACEAE' 1. (Clethraceae; genera *Clethra, Purdiaea*) sweet pepper bush / summersweet, mountain pepper bush, lily of the valley tree, Japanese sweet shrub 2. (Cyrillaceae; genus *Cliftonia*) buck-wheat tree 3. (Cyrillaceae; genus *Cyrilla*) swamp cyrilla / swamp titi / leatherwood / ironwood

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Ericales > Ebenaceae

-ŇḌY- 'DIOSPYROS (PERSIMMON)' I 1. (*D. kaki*) Japanese persimmon tree 2. (*D. lotus*) date-plum tree 3. (*D. virginiana*) American persimmon

-ŇḌŇ- 'DIOSPYROS (EBONY WOOD)' 1. (black-wooded species of *Diospyros*, e.g., *D. ebenum*) ebony tree 2. (other timber species of *Diospyros*, e.g., *D. celebica, D. mun, D. mormorata, etc.*) calamander wood tree, Makassar ebony tree, marblewood tree 3. (other *Diospyros* species) other ebony/persimmon tree

-ŇḌŇW- 'EBENACEAE' 1. (genus *Euclea*) guarri 2. (other Ebenaceae; several genera) other ebenaceous tree/plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Ericales > remaining families

-CK- 'THEACEAE' I 1. (genus *Camellia*: *C. sinensis*) tea plant 2. (genus *Camellia* other than *C. sinensis*) camellia 3. (genus *Stewartia*) stewartia

-CKY- 'THEACEAE' II 1. (genus *Franklinia*) Franklin tree 2. (genus *Schima*) needlewood tree 3. (Theaceae other than named genera; several genera) other

-ŇZK- 'POLEMONIACEAE' 1. (genus *Phlox*) phlox 2. (genus *Linanthus*) linanthus, desert snow, desertbeauty, evening snow, sandblossom, granite prickly-phlox

3. (Polemoniaceae other than *Phlox* and *Linanthus*) Jacob's -ladder, gilyflower, gillia, trumpet flower, woollystars, gymnosteris, Arizona firecracker, ipomopsis, skyrocket, standing cypress, calico

-ŇZKW- 'PRIMULACEAE' I 1. (genus *Primula*) primrose, cowslip, auricula / bear's ear, oxlip, glory-of-the-marsh, shootingstar / mosquito-bill / sailor caps

Samolus) brookweed, water pimpernel, water cabbage, water rose 2. (genus *Ardisia*) ardisia, marlberry, coral bush / coralberry / spiceberry, duck's eye 3. (genus *Anagallis*) pimpernel, weather-glass, shepherd's clock

-ŇZKY- 'PRIMULACEAE' II 1. (genus *Trientalis*) starflower, chickweed-wintergreen 2. (genus *Androsace*) rock-jasmine, Northern fairy candelabra, androsace 3. (genus *Cyclamen*) sowbread, swinebread

-ŇZKL- 'PRIMULACEAE' III 1. (genus *Myrsine*) colicwood, red matipo, Cape myrtle / African boxwood 2. (genus *Lysimachia*) loosestrife, swamp candle 3. (Primulaceae other than previously named genera; numerous genera) snowbell, featherfoil, Joewood, other primulaceous plant

-ŇZKR- 'SAPOTACEAE' I 1. (genus *Manilkara*) bulletwood / balatá / ausubo / massaranduba / cow-tree, chicle, bully tree, nispero, sapotilla, sapota / sapodilla / naseberry, wild dilly 2. (genus *Pouteria*) pouteria, eggfruit, abiu, lúcuma, mamey sapote, canistel, green sapote, shiny-leaved condoo / black plum 3. (genus *Vitellaria*) shea tree

-ŇZKŘ- 'SAPOTACEAE' II 1. (genus *Synsepalum*) miracle berry / sweet berry plant 2. (genus *Chrysophyllum*) tar apple / star apple / golden leaf tree / milk fruit / cainito, Damson plum / wild star-apple / caimitillo, milkwood 3. (genus *Tieghemella*) makore wood tree

-ŇZKF- 'SAPOTACEAE' III 1. (genus *Palaquium*) gutta-percha tree 2. (genus *Sideroxylon*) bully tree 3. (Spotaceae other than named genera) other sapotaceous tree/plant

-ŇZKŢ- 'SARRACENIACEAE' 1. (genus *Sarracenia*) trumpet pitcher plant 2. (gneus *Darlingtonia*) California pitcher plant / cobra lily / cobra plant 3. (genus *Heliamphora*) sun pitcher plant

-ŇZKV- 'STYRACACEAE' 1. (genus *Halesia*) silverbell / snowdrop tree 2. (genus *Pterostyrax*) epaullete tree 3. (genus *Styrax* and several other genera) storax, snowbell, other styaracaceous plant

theaceous plant/tree

-ŇZKH- 'LECYTHIDACEAE' 1. (genus *Bertholletia*) brazil nut tree 2. (genus *Lecythis*) paradise nut tree / monkey pot tree / sapucaia 3. (genus *Barringtonia*) freshwater mangrove, mango-pine, itchytree, powder-puff tree

-ŇZKS- 'ERICALES' I 1. (Balsaminaceae; genera *Impatiens, Hydrocera*) balsam, jewelweed, touch-me-not, parrot flower, Congo cockatoo, impatiens, policeman's helmet, poor man's rhododendron, busy lizzie, patience, snapweed 2. (Diapensiaceae; several genera) wandplant / wandflower / beetleweed, pixiemoss, Oconee bells / acony bell, pincushion plant 3. (Fouquieriaceae; genus *Fouquieria*) ocotillo plant, boojum tree / cirio

-ŇZKŠ- 'ERICALES' II 1. (Roridulaceae; genus *Roridula*) dewstick / fly bush 2. (Symplocaceae; genus *Symplocos*) sweetleaf, sapphireberry 3. (Tetrameristaceae; three genera) tea mangrove

-ŇZKHW- 'ERICALES' III 1. (other Lecythidaceae genera) other lecythidaceous tree 2. (Sladeniaeceae, Marcgraviaceae, Mitrastemonaceae, Pentaphylacaceae; numerous genera) other ericales plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > asterids > Cornales

-**GZG**- 'HYDRANGEACEAE' I 1. (genus *Hydrangea*) hydrangea / hortensia 2. (genus *Decumaria*) woodvamp 3. (genus *Broussaisia*) kanawao

-**GZGW**- 'HYDRANGEACEAE' II 1. (genus *Philadelphus*) mock-orange 2. (genus *Whipplea*) whipplea / modesty 3. (genus *Carpenteria*) tree anemone / bush anemone

-**GZGY**- 'HYDRANGEACEAE' III 1. (genus *Fendlerella*) Utah fendlerbush 2. (genus *Deutzia*) deutzia 3. (genus *Kirengeshoma*) yellow waxbell, Korean kirengeshoma

-**GZGL**- ‘HYDRANGEACEAE’ IV 1. (genus *Fendlera*) fendlerbush 2. (genus *Jamesia*) jamesia / cliffbush / waxflower 3. (Hydrangeaceae other than named genera; several genera) other hydrangeaceous plant

-**GZGR**- ‘CORNACEAE’ 1. (genus *Cornus, subgenera Yinquania, Kraniopsis, Mesomora*) blue- or white-fruited dogwood 2. (genus *Cornus*, all other subgenera) other dogwood tree or shrub 3. (genus *Alangium*) alangium

-**GZGŘ**- ‘LOASACEAE’ I 1. (genus *Mentzelia*) blazingstar, stickleaf, yellowcomet, mentzelia 2. (genus *Petalonyx*) sandpaper plant 3. (genus *Eucnide*) stingbush, rock nettle

-**GZGV**- ‘LOASACEAE’ II 1. (genus *Blumenbachia*) electric shock plant 2. (genus *Caiophora*) carnation nettle / ortiga / ortiguilla 3. (Loasaceae other than named genera; several genera) other loasceous plant

-**GZGḌ**- ‘NYSSACEAE’ 1. (genus *Nyssa*) tupelo tree, black-gum tree 2. (genus *Davidia*) dove-tree / handkerchief tree / ghost tree 3. (genus *Camptotheca*) happy tree / cancer tree / tree of life

-**GZGZ**- ‘CORNALES’ 1. (Grubbiaceae; genera *Grubbia, Strobilocarpus*) sillyberry 2. (Curtisiaceae; genus *Curtisia*) assegai tree / Cape lancewood 3. (Hydrostachyaceae and Mastixiaceae; genera *Hydrostachys, Mastixia, Diplopanax*) other cornales plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales > Caryophyllaceae

-ŇZP- ‘CARYOPHYLLEAE’ I 1. (genus *Dianthus*) carnation, pink, sweet william 2. (genus *Acanthophyllum*) chickweed baby’s breath 3. (genus *Gypsophila*) baby’s-breath, gypsophila

-ŇZPW- ‘CARYOPHYLLEAE’ II 1. (genus *Petrorhagia*) tunica, tunic-flower / coat flower, pink 2. (genus *Psammophiliella*) cushion baby’s-breath / low baby’s-breath / annual gypsophila 3. (genus *Saponaria*) soapwort, bouncingbet / sweety Betty, pygmy pink, tumbling-Ted

-ŇZPY- ‘CARYOPHYLLEAE’ III 1. (genus *Vaccaria*) cowherb / cowcockle / cow basil / cow soapwort / prairie carnation 2. (Caryophylleae other than named genera; several genera) other caryophyllid plant 3. [stem unassigned]

-ŇZPL- ‘ALSINEAE’ I 1. (genus *Cerastium*) mouse-ear chickweed, chickweed, snow in summer 2. (genus *Holosteum*) jagged chickweed 3. (genus *Moenchia*) upright chickweed, erect chickweed

-ŇZPR- ‘ALSINEAE’ II 1. (genus *Stellaria*) starwort, chickweed, stitchwort 2. (genus *Pseudostellaria*) false starwort, tuber starwort, robust starwort, Sierra starwort 3. (Alsineae other than Stems 1 and 2; several genera) other alsinid plant

-ŇZPŘ- ‘PARONYCHIEAE’ 1. (genus *Paronychia*) nailwort, whitlow-wort, paronychia, chickweed 2. (genus *Herniaria*) rupturewort 3. (Paronychieae other than Stems 1 and 2; several genera) other paronychiid plant

-ŇZPF- ‘POLYCARPEAE’ I 1. (genus *Achyronychia*) onyxflower / frost-mat 2. (genus *Pollichia*) waxberry / barley sugar bush 3. (genus *Polycarpon*) manyseed

-ŇZPŢ- ‘POLYCARPEAE’ II 1. (genus *Scopulophila*) rockwort 2. (genus *Cardionema*) sandcarpet 3. (Polycarpeae other than named genera; several genera) other polycarpaeid plant

-ŇZPÇ- ‘SAGINEAE’ 1. (genera *Colobanthus*, *Sagina*) pearlwort 2. (genus *Minuartia*) sandwort 3. (Sagineae other than Stems 1 and 2; several genera) other saginid plant

-ŇZPĻ- SCLERANTHEAE’ I 1. (genus *Schiedea*) schiedea, alsinidendron, Nihoa carnation 2. (genus *Scleranthus*) knawel, cushion-bush, German-knotweed 3. (genus *Geocarpon*, a.k.a. *Mononeuria*) tinytim, earth-fruit

-ŇZPH- ‘SCLERANTHEAE’ II 1. (genus *Honckenya*) sea sandwort / seaside sandplant / sea chickweed / sea pimpernel / sea-beach sandwort / sea purslane 2. (genus *Alsinidendron*) alsinidendron 3. (Scleranthеae other than Stems 1 and 2; several genera) other scleranthis plant

-ŇZF- ‘SPERGULEAE’ 1. (genus *Spergula*) spurry / spurrey, stickwort 2. (genus *Spergularia*) sea-spurrey / sandspurry 3. (genera *Rhodalsine, Sanctambrosia*) other spergulid plant

-ŇZFW- SILENEAE' I 1. (genus *Silene*) catchfly, campion, gunpowder plant, silene, fire pink, Indian pink 2. (genus *Lychnis*) catchfly, white cockle, Maltese cross, ragged robin, flower-of-Jove, campion 3. (genus *Agrostemma*) corncockle, corn-pink

-ŇZFY- 'SILENEAE' II 1. (genus *Petrocoptis*) falguera 2. (genus *Eudianthe*) rose catchfly, rose of heaven 3. (Sileneae other than named genera; several genera) other sileneid plant

-ŇZPHW- 'CAROPHYLLACEAE' 1. (genus *Arenaria*) sandwort, golden moss / Irish moss 2. (genus *Corrigiola*) strapwort 3. (Carophyllaceae other than named genera; numerous genera) other carophyllaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales > Polygonaceae

-**RNTW**- 'POLYGONOIDEAE' I 1. (genus *Rheum*) rhubarb plant 2. (genus *Fagopyrum*) buckwheat plant 3. (genus *Rumex*) sorrel plant, dock

-**RNTY**- 'POLYGONOIDEAE' II 1. (genus *Polygonum*) knotweed, knotgrass, bindweed, birdweed / pigweed / lowgrass, wireweed, jointweed, tree hogweed, sandlace 2. (genus *Fallopia*) silver lace vine, russian vine / mile-a-minute vine / fleece vine / fleece flower, black-bindweed, copse bindweed / small-flower knotweed, climbing false buckwheat, Korean knotweed 3. (genus *Bistorta*) bistort, fleece flower, knotweed, mountain fleece

-**RNTL**- 'POLYGONOIDEAE' III 1. (genus *Muehlenbeckia*) maidenhair vine, lignum, wirevine, wire weed, shrubby tororaro / wiggy-wig bush, muehlenbeckia 2. (genus *Duma*) lignum 3. (genus *Koenigia*) knotweed, wild-rhubarb

-**RNTR**- 'POLYGONOIDEAE' IV 1. (genus *Oxyria*) mountain sorrel / wood sorrel / alpine sorrel 2. (genus *Persicaria*) smartweed, knotweed, willow-weed, tearthumb, amphibious bistort, water-pepper, kiss-me-over-the-garden-gate / prince's feather / princess-feather, arrowvine, Jesusplant / redshank, lady's thumb, jumpseed, Chinese indigo / polygonum-indigo, persicaria 3. (Polygonoideae other than named genera; several genera) other polygonoid plant

-**RNTŘ**- 'ERIOGONOIDEAE' I 1. (genus *Eriogonum*) wild buckwheat 2. (genus *Antigonon*) lovechain, Mexican coral vine / Mexican creeper . queen's wreath / queen's jewels / chain-of-love / coralvine / beebush / San Miguelito vine 3. (genus *Aristocapsa*) Indian Valley spineflower / Indian Valley spinecape

-**RNTF**- 'ERIOGONOIDEAE' II 1. (genus *Brunnichia*) redvine, Ladies' eardrops, buckwheat vine 2. (genus *Centrostegia*) Thurber's spineflower / red triangles 3. (genus *Chorizanthe*) spineflower, Turkish rugging

-**RNTM**- 'ERIOGONOIDEAE' III 1. (genus *Coccoloba*) pigeonplum, seagrape, false chiggergrape 2. (genus *Dedeckera*) July gold 3. (genus *Dodecahema*) slender-horned spineflower

-**RNTN**- 'ERIOGONOIDEAE' IV 1. (genus *Gilmania*) golden carpet 2. (genus *Goodmania*) yellow spinecap 3. (genus *Harfordia*) rabbit's purse

-**RNTÇ**- 'ERIOGONOIDEAE' V 1. (genus *Hollisteria*) false spikeflower 2. (genus *Lastarriaea*) leather spineflower 3. (genus *Mucronea*) spineflower

-**RNTH**- 'ERIOGONOIDEAE' VI 1. (genus *Nemacaulis*) woollyheads / cottonheads 2. (genus *Oxytheca*) oxytheca, round-leaf puncturebract 3. (genus *Pterostegia*) woodland threadstem / woodland pterostegia / fairy mist / fairy bowties

-**RNTHW**- 'ERIOGONOIDEAE' VII 1. (genus *Systenotheca*) Vortriede's spineflower 2. (genus *Triplaris*) ant tree, devil tree 3. (Eregionoideae other than named genera, plus Symmerioideae; genus *Symmeria*) other polygonaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales > Amaranthaceae

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales > Amaranthaceae > Amaranthoideae

-**RND**- 'AMARANTHOIDEAE' I 1. (genus *Amaranthus*) amaranth, greenstripe, pigweed, careless weed, Prince-of-Wales feather / prince's feather, Joseph's-coat, love-lies-bleeding / tassel flower 2. (genus *Celosia*) woolflower, cockscomb, Lagos spinach, quail grass, celosia 3. (genus *Chamissoa*) false chaff flower

-**RNDW**- 'AMARANTHOIDEAE' II 1. (genus *Aerva*) kapok bush / desert cotton, mountain knotgrass 2. (genus *Ptilotus*) foxtail, pussy tail, lamb's tail, mulla mulla 3. (genus *Achyranthes*) chaff flower

-**RNDY**- 'AMARANTHOIDEAE' III 1. (genus Nototrichium) rockwort 2. (Amaranthoideae other than named genera; numerous genera) other amaranthoid plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales > Amaranthaceae > Chenopodioideae

-**RMD**- 'CHENOPODIOIDEAE' I 1. (genus *Spinacia*) spinach plant 2. (edible species of genus *Chenopodium; C. album, C. pallidicaule, C. quinoa, C. giganteum*) quinoa, kañiwa, fat hen / white goosefoot plant, tree spinach 3. (genus *Chenopodium*; species other than Stem 2) goosefoot, bluebush, chualar, lambsquarters, huauzontle

-**RMDW**- 'CHENOPODIOIDEAE' II 1. (genus *Atriplex*) saltbush / orache plant 2. (genus *Extriplex*) California saltbush / California orache plant, San Joaquin saltbush / San Juaquin orache 3. (genus *Halimione*) sea purslane

-**RMDY**- 'CHENOPODIOIDEAE' III 1. (genus *Blitum*) goosefoot, blite, Good-King-Henry / poor-man's asparagus / Lincolnshire spinach / markery / English mercur, stawberry blite / strawberry spinach, poverty weed, Indian lettuce 2. (genus *Chenopodiastrum*) salt-green / sowbane / Australian-spinach 3. (genus *Oxybasis*) red goosefoot / saltmarsh goosefoot, upright goosefoot, coastblite goosefoot, oak-leaved goosefoot

-**RMDL**- 'CHENOPODIOIDEAE' IV 1. (genus *Stutzia*) Coville's orach, Suckley's orach 2. (genus *Grayia*) hop sage, siltbush 3. (genus *Lipandra*) manyseed goosefoot

-**RMDR**- 'CHENOPODIOIDEAE' V 1. (genus *Micromonolepis*) small povertyweed / red povertyweed 2. (genus *Proatriplex*) four-corners orach / Mancos shadscale 3. (genus *Krascheninnikovia*) winterfat

-**RMDŘ**- 'CHENOPODIOIDEAE' VI 1. (genus *Dysphania; D. ambrosioides*) epazote 2. (genus *Dysphania; D. botrys*) Jerusalem oak / sticky goosefoot / feathered geranium 3. (genus *Dysphania*; species other than Stems 1 and 2) wormseed, crumbwee, goosefoot

-**RMDV**- 'CHENOPODIOIDEAE' VII 1. (genus *Axyris*) axyris, Russian pigweed 2. (genus Cycloloma) winged pigweed / tumble ringwing / plains tumbleweed 3. (Chenopodioideae other than named genera; several genera) other chenopodioid plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales > Amaranthaceae > Betoideae

-**RMV**- 'BETA VULGARIS (BEET) 1. (*B. v. vulgaris 'Conditiva'*) beetroot plant (red beet) 2. (*B. v. vulgaris 'Altissima'*) sugar beet plant 3. (*B. v. vulgaris 'Cicla/Flavescens'*) [Swiss] chard plant

-**RMVW**- 'BETA VULGARIS (BEET) 1. (*B. v. vulgaris 'Crassa'*) mangelwurzel / mangold wurzel / mangold / mangel beet / fodder beet 2. (*B. v. maritima*) sea beet / wild spinach 3. (genus *Beta* other than *vulgaris* species) other beet species

-**RMVY**- 'BETOIDEAE' 1. (genus *Hablitzia*) Caucasian spinach 2. (genus *Aphanisma*) aphanisma, San Diego coastalcreeper 3. (other Betoideae genera; *Oreobliton, Patellifolia, Acroglochin*) other betoid plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales > Amaranthaceae > Camphorosmoideae

-**RMVL**- 'CAMPHOROSMOIDEAE' I 1. (genus *Bassia*) burningbush / belvedere / Mexican fireweed / Mexican firebush, kochia, prostrate summer-cypress, five-horn smotherweed / thorn orache, bassia 2. (genus *Neokochia*) molly 3. (genus *Spirobassia*) contorted smotherweed

-**RMVR**- 'CAMPHOROSMOIDEAE' II 1. (genus *Enchylaena*) barrier saltbush 2. (genus *Maireana*) bluebush 3. (genus *Sclerolaena*) goathead burr, galvanized burr, tall bindii

-**RMVŘ**- 'CAMPHOROSMOIDEAE' III 1. (genus *Threkeldia*) coast bonefruit 2. (Camphorosmoideae other than named genera; several genera) other camphorosmoid plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales > Amaranthaceae > Gomphrenoideae

-**RMF**- 'GOMPHRENOIDEAE' I 1. (genus *Gomphrena*) globe amaranth 2. (genus *Blutaparon*) silverhead / silverweed / saltweed / samphire, Galapagos amaranth 3. (genus *Froelichia*) snakecotton

-**RMFW**- 'GOMPHRENOIDEAE' II 1. (genus *Iresine*) bloodleaf, white snowplant, Texas shrub, Juba's bush 2. (genus *Alternanthera*) joyweed, ruby leaf, little ruby, calico-plant, chaff-flower, washerwoman, sanguinarea, alligatorweed, rabbit-meat, Joseph's coat 3. (genus *Tidestromia*) honeysweet, tidestromia

-**RMFY**- 'GOMPHRENOIDEAE' III 1. (genus *Guilleminea*) matweed 2. (genus *Pfaffia*) suma root / Brazilian ginseng 3. (Gomphrenoideae other than named genera; several genera) other gomphrenoid plant

-**RMFL**- 'POLYCNEMOIDEAE' 1. (genus *Polycnemum*) needleleaf 2. (genus *Nitrophila*) niterwort, boraxweed 3. (genera *Hemichroa* and *Surreya*) trailing saltstar / trailing jointweed / trailing hemichroa, surreya

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales > Amaranthaceae > Salicornoideae

-**RMFR**- 'SALICORNOIDEAE' I 1. (genus *Salicornia*) glasswort, saltwort, marsh samphire, samphire greens / sea asparagus / sea beans 2. (genus *Sarcocornia*) thick-headed glasswort, Australian samphire, Pacific swampfire / Pacific glasswort, Utah swampfire, chickenclaws / perennial glasswort 3. (genus *Tecticornia*) shrubby glasswort, shrubby samphire / grey glasswort, large-articled samphire, bead glasswort / bead samphire, blackseed glasswort / blackseed samphire, mat samphire

-**RMFŘ**- 'SALICORNOIDEAE' II 1. (genus *Allenrolfea*) iodine bush 2. (genus *Arthrocnemum*) Parish's glasswort 3. (Salicornoideae other than named genera) other salcornoid plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales > Amaranthaceae > Salsoloideae

-**RMFM**- 'SALSOLOIDEAE' I 1. (genus *Salsola*) saltwort, barilla plant 2. (genus *Kali*) Russian thistle, Tartar thistle, windwitch, tumbleweed, prickly saltwort / prickly glasswort 3. (genus *Halogeton*) saltlover, Aral barilla, halogeton

-**RMFN**- 'SALSOLOIDEAE' II 1. (genus *Haloxylon*) saxaul 2. (Salsoloideae other than named genera; several genera) other salsoloid plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales > Amaranthaceae remaining families

-**RMFŇ**- 'AMARANTHACEAE' 1. (Corispermoideae; genera *Corispermum, Agriophyllum, Anthochlamys*) bugseed, tickseed, tumbleweed, other corispermoid plant 2. (Suaedoideae; genera *Suaeda, Bienertia*) seepweed, seablite, iodine weed, bienertia 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales > Aizoaceae

-**RNX**- 'AIZOACEAE' I 1. (genus *Tetragonia*, edible species; *T. tetragonoides, T. decumbens*) New Zealand spinach, dune spinach 2. (genus *Tetragonia*; species other than Stem 1) bower spinach, tetragonia 3. (genus *Sceletium*) kanna / channa

-**RNXW**- 'AIZOACEAE' II 1. (genus *Conicosia*) narrow-leafed ice plant 2. (genus *Cleretum*) Livingstone daisy / Buck Bay vygie 3. (genus *Aloiopsis*) giant jewel plant

-**RNXL**- 'AIZOACEAE' III 1. (genus *Carpobrotus*) pigface, ice plant, sour fig, sea fig, beach banana 2. (genus *Delosperma*) pickle plant, ice plant 3. (genus *Disphyma*) round-leaved pigface / New Zealand iceplant / purple dewplant

-**RNXR**- 'AIZOACEAE' IV 1. (genus *Drosanthemum*) dew-flower , redondo creeper 2. (genus *Lapidaria*) Karoo rose 3. (genus *Malephora*) mesemb

-**RNXM**- 'AIZOACEAE' V 1. (genus *Muiria*) mouse-head 2. (genus *Pleiospilos*) split rock, mimicry plant 3. (genus *Zaleya*) African purslane

-**RNXN**- 'AIZOACEAE' VI 1. (genus *Sesuvium*) sea-purslane 2. (genus *Trianthema*) desert horse purslane / black pigweed / giant pigweed 3. (Aizoaceae other than named genera; numerous genera) other Aizoaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales > Molluginaceae

-**RNZV**- 'MOLLUGINACEAE' 1. (genus *Mollugo*) carpetweed 2. (genus *Glinus*) sweetjuice 3. (Molluginaceae other than Stems 1 and 2; several genera) other molluginaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales > Cactaceae

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales > Cactaceae > Cactoideae

-**FSK**- 'CACTEAE' I 1. (genus *Lophophora*) peyote, false peyote 2. (genus *Escobaria*) pincushion cactus / foxtail cactus 3. (genus *Epithelantha*) button cactus

-**FSKW**- 'CACTEAE' II 1. (genus *Echinocactus*) golden barrel cactus, devil's head / silverbell cactus / Turk's head cactus, horsecrippler / devil's pincushion, giant barrel cactus, cottontop cactus 2. (genus *Echinomastus*) redspine fishhook cactus / redspine butterfly cactus / needlespine pineapple cactus / red pineapple cactus / purple-spine viznagita / acuña cactus, Johnson's beehive cactus / Johnson's fishhook cactus, Lloyd's fishhok cactus / golfball cactus / silver column cactus / Mariposa cactus 3. (genus *Astrophytum*) sand dollar cactus / sea urchin cactus / star peyote, goat's horn cactus, bishop's ca cactus / bishop's hat / bishop's mitre cactus, monk's hood cactus

-**FSKY**- 'CACTEAE' III 1. (genus *Coryphantha*) beehive cactus, sea urchin cactus / hedghog cory cactus / rhinoceros cactus, nipple beehive cactus, Pima pineapple, bunched cory cactus / whiskerbush 2. (genus *Leuchtenbergia*) agave cactus / prism cactus 3. (genus *Mammillaria*) pincushion, finger cactus, powder-puff pincushion, green fishhook cactus, owl's eyes, mother of hundreds, strawberry cactus / California fishhook cactus, ladyfinger cactus, twin-spinded cactus, crested twin-spined cactus, Arizona fishhook cactus, nipple cactus, rattail cactus, feather cactus, red headed Irishman, bristle brush cactus, old lady cactus, mammillaria

-**FSKL**- 'CACTEAE' IV 1. (genus *Mammilloydia*) snowball cactus 2. (genus *Obregonia*) artichoke cactus 3. (genus *Pediocactus*) Brady's pincushion cactus / Brady's hedghog cactus / Marble Canyon cactus, Despain's cactus / San Rafael cactus, Knowlton's miniature cactus / Knowlton's minute cactus, Navajo pincushion cactus, Winkler's cactus, gypsum cactus

-**FSKR**- 'CACTEAE' V 1. (genus *Sclerocactus*) little barrel cactus 2. (genus *Stenocactus*) brain cactus 3. (Cacteae other than named genera; several genera) other cactid plant

-**FSKŘ**- 'PHYLLOCACTEAE' I 1. (genus *Carnegiea*) saguaro cactus 2. (genus *Cephalocereus*) old man cactus 3. (genus *Echinocereus*) hedgehog cactus, rainbow cactus, king cup cactus / claret cup cactus, green pitaya, Mojave mound cactus, ladyfinger cactus, dahlia cactus

-**FSKF**- 'PHYLLOCACTEAE' II 1. (genus *Bergerocactus*) snake cactus 2. (genus *Escontria*) chiotilla / jiotilla 3. (genus *Stenocereus*) octopus cactus, creeping devil caterpillar cactus, dagger cactus, sour pitaya, organ pipe cactus

-**FSKŢ**- 'PHYLLOCACTEAE' III 1. (genus *Myrtillocactus*) bilberry cactus / whortleberry cactus / blue candle, candelabra cactus 2. (genus *Pachycereus*) Indian comb, totem cactus / senita cactus 3. (genus *Peniocereus*) desert nightblooming cereus, gearstem cactus

-**FSKV**- 'PHYLLOCACTEAE' IV 1. (genus *Acanthocereus*) barbed-wire cactus 2. (genus *Brachycereus*) lava cactus 3. (genus *Jasminocereus*) candelabra cactus

-**FSKÇ**- 'PHYLLOCACTEAE' V 1. (genus *Epiphyllum*) climbing cactus, orchid cactus, leaf cactus 2.(genus *Selenicereus*) moonlight cactus 3. (Phyllocacteae other than named genera) other phyllocactid plant

-**FSKM**- 'CEREEAE' I 1. (genus *Cereus*) lady of the night cactus, giant club cactus / hedge cactus 2. (genus *Stetsonia*) toothpick cactus 3. (genus *Gymnocalycium*) chin cactus

-**FSKN**- 'CEREEAE' II 1. (genus *Melocactus*) Turk's cap cactus 2. (genus *Pilosocereus*) tree cactus, pipe organ cactus 3. (genus *Espestoa*) old Peruvian man

-**FSKH**- 'CEREEAE' III 1. (genus *Echinopsis*) hedgehog cactus, torch cactus, golden torch / golden column, Argentine saguaro, cardón, San Pedro cactus, sea-urchin cactus, Easter lily cactus, peanut cactus 2. (genus *Hylocereus*) dragon fruit 3. (Cereeae other than named genera) other cereid cactus

-**FSKHW**- 'RHIPSALIDEAE' 1. (genus *Rhipsalis*) mistletoe cactus 2. (genus *Schlumbergera*) Christmas cactus, Thanksgiving cactus, crab cactus, holiday cactus, Easter cactus / Whitsun cactus, claw cactus 3. (Rhipsalideae other than named genera; several genera) other rhipsalid cactus

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales > Cactaceae > Opuntioideae

-**FSGW**- 'OPUNTIEAE' 1. (genus *Opuntia*) prickly pear, old man's whiskers / cowboy's red whiskers, calico cactus, britttle cactus, Aaron's beard cactus / semaphore cactus, bunny ears cactus / polka-dot cactus, creeping cactus, red buttons opuntia 2. (genus *Consolea*) Florida semaphore cactus / semaphore pricklypear 3. (Opuntieae other than Stems 1 and 2; several genera) other opuntiid cactus

-**FSGY**- 'CYLINDROPUNTIEAE' 1. (genus *Cylindropuntia*) cholla bush 2. (genus *Grusonia*) matted cholla / Parish club cholla, sagebrush cholla 3. (Cylindropuntieae other than Stems 1 and 2; several genera) other cylindropuntiid cactus

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales > Cactaceae remaining sub-families

-**FSGL**- 'CACTACEAE' 1. (Pereskioideae; genus *Pereskia*) lemon vine, rose cactus, leaf cactus 2. (Notocacteae, Maihuenioideae, and Cactaceae of uncertain sub-family or tribe; several genera) other cactus 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales > Basellaceae

-**BVK**- 'BASELLACEAE' 1. (genus *Basella*) Malabar spinach / vine spinach / Ceylon spinach 2. (genus *Ullucus*) ulluco 3. (genera *Anredera, Tournonia*) Madeira vine, other basellaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales > Didiereaceae

-**BVKW**- 'DIDIEREACEAE' 1. (genus *Didierea*) octopus tree 2. (genus *Alluaudia*) Madagascar ocotillo 3. (Didiereaceae other than Stems 1 and 2) other didiereaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales > Droseraceae, Drosophyllaceae, Nepenthaceae

-**BVKY**- 'DROSERACEAE' 1. (genus *Drosera*) sundew 2. (genus *Dionaea*) Venus flytrap 3. (genus *Aldrovanda*) waterwheel plant

-**BVKL**- 'DROSOPHYLLACEAE & NEPENTHACEAE' 1. (Drosophyllaceae; genus Drosophyllum) Portuguese sundew / dewy pine 2. (Nepenthaceae; genus Nepenthes) tropical pitcher plant / monkey cup 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales > Montiaceae

-**BVKR**- 'MONTIACEAE' I 1. (genus *Montia*) miner's lettuce, tadlily, water-blinks, montia 2. (genus *Calandrinia*) purslane, redmaid, parakeelya 3. (genus *Cistanthe*) pussypaws, sandcress

-**BVKŘ**- 'MONTIACEAE' II 1. (genus *Claytonia*) spring beauty, winter purslane, Indian lettuce, grass-flower / fairy spud, claytonia 2. (genus *Lewisia*) lewisia, cliff maids, bitter-root 3. (genus *Lewisiopsis*) Tweedy's pussypaws / Tweedy's lewisia / Tweedy's bitterroot

-**BVKŢ**- 'MONTIACEAE' III 1. (genus *Lyallia*) lyallia cushion 2. (genus *Phemeranthus*) fameflower, rock rose , rock pink, sand pink, sunbright 3. (Montiaceae other than named genera; several genera) other montiid plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales > Nyctaginaceae

-**BVKF**- 'NYCTAGINACEAE' I 1. (genus *Bougainvillea*) bougainvillea, paperflower 2. (genus *Phaeoptilum*) brittle thorn 3. (genus *Tripterocalyx*) sandpuff, sand-verbena

-**BVKV**- 'NYCTAGINACEAE' II 1. (genus *Mirabilis*) four-o'clock, marvel of Peru, desert wishbone-bush, umbrella wort, maravilla 2. (genus *Abronia*) abronia, sand-verbena, heart's delight, prairie snowball / fragrant verbena, coastal 3. (genus *Acleisanthes*) trumpets, desert moonpod

-**BVKÇ**- 'NYCTAGINACEAE' III 1. (genus *Allionia*) windmills, trailing four o'clock 2. (genus *Anulocaulis*) ringstem 3. (genus *Boerhavia*) spiderling, alena, windflower, climbing wartclub / wishbone vine, alena, boerhavia

-**BVKH**- 'NYCTAGINACEAE' IV 1. (genus *Pisonia*) catchbirtree, pullback, devil's-claws, water mampoo, birdcatcher tree, birlime tree 2. (genus *Neea*) nia, neea, saltwood 3. (Nyctaginaceae other than named genera; several genera) other nyctaginaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales > Petiveriaceae

-**BVKM**- 'PETIVERIACEAE' 1. (genus *Rivina*) pigeonberry / rougeplant / baby peppers / bloodberry / coralito 2. (genus *Trichostigma*) hooopvine /black basket wythe / cooper's wythe / basket wiss / basket with / hoop with 3. (Petiveriaceae other than Stems 1 and 2; several genera) other petiveriaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales > Phytolaccaceae

-**BVKN**- 'PHYTOLACCACEAE' 1. (genus *Phytolacca*) pokeweed, poke, pokebush, pokeberry, pokeroot, poke sallet, red inkplant, inkberry 2. (genus *Agdestis*) rockroot 3. (Phytolaccaceae other than Stems 1 and 2; several genera) other phytolaccid plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales > Plumbaginaceae

-**BVM**- 'PLUMBAGINACEAE' I 1. (genus *Plumbago*) leadwort, plumbago, doctorbush, whorled plantain 2. (genus *Ceratostigma*) leadwort, plumbago 3. (genus *Armeria*) thrift, lady's cushion / sea pink

-**BVMW**- 'PLUMBAGINACEAE' II 1. (genus *Limonium*) sea-lavender, tree limonium, statice, caspia, marsh-rosemary 2. (Plumbaginaceae other than named genera; several genera) other plumbaginaceous plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Caryophyllales remaining families

-**BVN**- 'CARYOPHYLLALES' I 1. (Frankeniaceae; genus *Frankenia*) sea heath, alkali heath, Saint Helena tea, frankenia 2. (Microteaceae; genus *Microtea*) jumby pepper 2.(Portulacaceae; genus *Portulaca*) purslane, pigweed, 'ihi 3. (Sarcobataceae; genus *Sarcobatus*) greasewood / seepwood / saltbush

-**BVNW**- 'CARYOPHYLLALES' II 1. (Simmondsiaceae; genus Simmondsia) jojoba / goat nut / deer nut / pignut / wild hazel / quinine nut / coffeeberry / gray box bush 2. (Achatocarpaceae, Anacampserotaceae, Ancistrociadaceae, Asteropeiaceae, Barbeulaceae, Dioncophyllaceae, Gisekiaceae, Halophytaceae, Kewaceae, Limeaceae, Lophiocarpaceae, Macarthuriaceae, Physenaeceae, Rhabdodendron, Stegnospermataceae, Talinaceae, Tamaricaceae; numerous genera) other caryophyllales plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Berberidopsidales

-**CXN**- 'BERBERIDOPSIDALES' 1. (Berberidopsidaceae; genera *Berberidopsis,Streptothamnus*) montane tape vine, Chilean coral plant, streptothamnus 2. (Aextoxicaceae; genus *Aextoxicon*) olivillo / aceitunillo tree 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Santalales

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Santalales > Santalaceae

-**LFW**- 'SANTALACEAE' I 1. (genus *Santalum*) sandalwood, quandong, native peach, 'iliahi 2. (genus *Comandra*) bastard toadflax, comandra 3. (genus *Geocaulon*) false toadflax / northern comandra

-**LFY**- 'SANTALACEAE' II 1. (genus *Viscum*) mistletoe 2. (genus *Phoradendron*) mistletoe 3. (genus *Arceuthobium*) dwarf mistletoe

-**LFL**- 'SANTALACEAE' III 1. (genus *Buckleya*) piratebush 2. (genus *Choretrum*) sour bush 3. (genus *Exocarpos*) ballart, grass tree, dwarf cherry, slender cherry, broad-leaved cherry, cypress cherry, jointed cherry

-**LFR**- 'SANTALACEAE' IV 1. (genus *Korthalsella*) korthal mistletoe 2. (genus *Leptomeria*) currant bush 3. (genus *Nestronia*) leechbrush / conjurer's-nut / Indian olive / nestronia

-**LFŘ**- 'SANTALACEAE' V 1. (genus *Osyris*) osyris, Cape sumach, wild tea plant, African sandalwood 2. (genus *Thesium*) meadowflax 3. (Santalaceae other than named genera; numerous genera) other santalid plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Santalales > Loranthaceae

-**LFS**- 'LORANTHACEAE' I 1. (genus *Loranthus*) showy mistletoe 2. (genus *Muellerina*) Banksia mistletoe / coast mistletoe, Cypress-pine mistletoe, creeping mistletoe, hoop pine mistletoe myrtle-leaved mistletoe 3. (genus *Anyema*) pincushion mistletoe, stalked mistletoe, wireleaf mistletoe, drooping mistletoe

-**LFŠ**- 'LORANTHACEAE' II 1. (genus *Nuytsia*) moodjar, Australian Christmas tree 2. (genus *Atkinsonia*) Louisa's mistletoe 3. (genus *Psittacanthus*) parrot-flower

-**LFŢ**- 'LORANTHACEAE' III 1. (genus *Lysiana*) harlequin mistletoe, Mulga mistletoe, Northern mistletoe 2. (genus *Peraxilla*) red mistletoe 3. (genus *Macrosolen*) parasite honeysuckle

-**LFX**- 'LORANTHACEAE' IV 1. (genus *Dendrophthoe*) long-flowered mistletoe / apostle mistletoe, orang mistletoe 2. (genus *Agelanthus*) African mistletoe 3. (Loranthaceae other than named genera) other showy mistletoe

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Santalales > Balanophoraceae

-**LFM**- 'BALANOPHORACEAE' I 1. (genus *Balanophora*) coral plant, fungus root 2. (genus *Dactylanthus*) wood rose 3. (genus *Thonningia*) ground pineapple

-**LFN**- 'BALANOPHORACEAE' II 1. (genus *Hechettea*) hechettea 2. (Balanophoraceae other than named genera; several genera) other balanophorid plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > superasterids > Santalales remaining families

-**LFŇ**- 'SANTALALES' 1. (Misodendraceae; genus *Misodendrum*) feathery mistletoe 2. (Schoepfiaceae; genus *Schoepfia*) arana, schoepfia 3. (Olacaceae, Opiliaceae; several genera) other santalales plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > Saxifragales

-**LŠW**- 'SAXIFRAGACEAE' I 1. (genus *Saxifraga*) saxifrage, rockfoil, Dovedale moss, Madeira breakstone, fair maids of France, spider plant, Saint Patrick's cabbage, strawberry geranium, London pride 2. (genus *Micranthes*) saxifrage, brook lettuce 3. (genus *Heuchera*) alumroot, jill-of-the-rocks, jack-o'the-rocks, coral bells

-**LŠY**- 'SAXIFRAGACEAE' II 1. (genus *Chrysosplenum*) golden saxifrage 2. (genus *Astilbe*) false goat's beard, false spirea 3. (genus *Saxifragopsis*) strawberry saxifrage

-**LŠL**- 'SAXIFRAGACEAE' III 1. (genus *Tiarella*) foamflower 2. (genus *Tolmiea*) youth on age / pick-a-back-plant / piggyback plant / thousand mothers 3. (genus *Boykinia*) brookfoam, boykinia

-**LŠR**- 'SAXIFRAGACEAE' IV 1. (genus *Sullivantia*) coolwort 2. (genus *Bolandra*) false coolwort 3. (genus *Jepsonia*) jepsonia

-**LŠŘ**- 'SAXIFRAGACEAE' V 1. (genus *Lithophragma*) woodland star 2. (genus *Mitella*) miterwort, bishop's cap 3. (genus *Tellima*) fringecups

-**LŠF**- 'SAXIFRAGACEAE' VI 1. (genus *Bergenia*) elephant's ears / elephant-eared saxifrage, gergenia, pigsqueak / Siberian tea / Mongolian tea 2. (genus *Darmera*) Indian rhubarb / umbrella plant 3. (genus *Bensoniella*) Oregon bensoniella

-**LŠŢ**- 'SAXIFRAGACEAE' VII 1. (genus *Tankakaea*) Japanese foam flower 2. (Saxifragaceae other than named genera; several genera) other saxifragaceous plant 3. [stem unassigned]

-**LŠM**- 'GROSSULARIACEAE [genus RIBES]' I 1. (*Ribes rubrum*) redcurrant 2. (cultivar of *R. rubrum*) white currant 3. (*R. nigrum*) blackcurrant

-**LŠN**- 'GROSSULARIACEAE [genus RIBES]' II 1. (*R. uva-crispa / grossularia*) gooseberry 2. (*R. nidigrolaria*) jostaberry 3. (*R. aureum*) golden currant / clove currant / pruterberry / buffalo currant

-**LŠŇ**- 'GROSSULARIACEAE [genus RIBES]' III 1. (*Ribes* species cultivated as ornamental plants; several species) ornamental currant/gooseberry plant 2. (wild/non-cultivated *Ribes* species; numerous species) wild currant/gooseberry plant 3. [stem unassigned]

-**LŠV**- 'CRASSULACEAE' I 1. (genus *Crassula*) jade plant, pygmyweed, silver dollar plant, rattlesnake tail, stonecrop, silver beads, crassula, tailor's patch, watch chain / lizard's tail / zipper plant, string of buttons, red carpet, rosary plant / kebab bush / concertina plant 2. (genus *Sedum*) stonecrop, orpine, wallpepper, prick-madam / trip-madam 3. (genus *Sedella*) mock stonecrop

-**LŠX**- 'CRASSULACEAE' II 1. (genus *Kalanchoe*) velvet leaf / felt plant / felt bush, flaming katy / Madagascar widow's-thrill, devil's backbone / Mexican-hat plant, donkey ears / life plant, penwiper, paddle plant, silver teaspoons, pies from heaven, flapjacks / desert cabbage / white lady, chandelier plant / mother of millions 2. (genus *Andromischus*) andromischus, calico hearts 3. (genus *Cotyledon*) silver crown / silver ruffles, pig's ear / rounded-leafed navel-wort, bear's paw

-**LŠFW**- 'CRASSULACEAE' III 1. (genus *Hylotelephium*) stonecrop, live-forever, livelong / frog's-stomach / life-everlasting / midsummer-men / harping Johnny / orphan John / witch's moneybags 2. (genus *Rhodiola*) redpod stonecrop / queen's crown, golden root / rose root / Aaron's rod / Arctic root / king's crown / orpin rose, western roseroot / ledge stonecrop 3. (genus *Umbilicus*) Venus' navel, navelwort, gold drop, umbilicus

-**LŠFY**- 'CRASSULACEAE' IV 1. (genus *Sempervivum*) houseleek, liveforever, hen and chicks, Jupiter's beard 2. (genus *Jovibarba*) hen-and-chickens, rolling hen-and-chicks 3. (genus *Aeonium*) tree houseleek

-**LŠŢW**- 'CRASSULACEAE' V 1. (genus *Dudleya*) dudleya, live-forever, sea lettuce, fingertips, alabaster plant 2. (genus *Echeveria*) echeveria, Mexican hens-and-chicks, Mexican firecracker, desert savior, painted lady, Mexican snowball / Mexican gem / white Mexican rose 3. (genus *Graptopetalum*) leatherpetal, lavender pebbles / jewel-leaf plant, mother of pearl plant / ghost plant

-**LŠŢY**- 'CRASSULACEAE' V 1. (genus *Pachyphytum*) sugar-almond plant, moonstone, pachyphytum 2. (Crassulaceae other than named genera; numerous genera) other crassulaceous plant 3. [stem unassigned]

-**LŠŠ**- 'HALORAGACEAE' I 1. (genus *Haloragis*) seaberry, haloragis 2. (genus *Gonocarpus*) raspwort 3. (genus *Glischrocaryon*) golden pennants, popflower

-**LŠXW**- 'HALORAGACEAE' II 1. (genus *Myriophyllum*) water-milfoil, parrot's-feather 2. (genus *Proserpinaca*) mermaid-weed 3. (Haloragaceae other than Stems 1 and 2; several genera) other haloragid plant

-**LŠVW**- 'HAMAMELIDACEAE' I 1. (genus *Hamamelis*) witch-hazel, winterbloom 2. (genus *Fothergilla*) witch alder, fothergilla 3. (genus *Parrotia*) Persian ironwood, Chinese ironwood

-**LŠVY**- 'HAMAMELIDACEAE' II 1. (genus *Corylopsis*) winter-hazel 2. (genus *Rhodoleia*) Hong Kong rose 3. (Hamamelidaceae other than named genera) other hamamelid plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > Saxifragales remaining famlies

-**LŠMW**- 'SAXIFRAGALES' I 1. (Iteaceae; genera *Itea, Pterostemon*) sweetspire, pterostemon 2. (Aphanopetalaceae; genus *Aphanopetalum*) gum vine 3. (Paeoniaceae; genus *Paeonia*) peony

-**LŠMY**- ‘SAXIFRAGALES’ II 1. (Altingiaceae; genus *Liquidambar*) sweetgum / stargum, redgum, satin-walnut, americn storax 2. (Cercidiphyllaceae; genus *Cercidiphyllum*) katsura 3. (Tetracarpaea, Penthoraceae, Daphniphyllaceae, Peridiscaceae; several genera) other saxifragales plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > Vitales

-**ŢN**- ‘VITUS [GRAPE]’ 1. (*Vitus vinifera*) European grapevine 2. (culinary/cultivated *Vitus* species other than *V. vinifera*; several species, e.g., *V. labrusca, V. riparia, V. aestivalis, V. rotundifolia*) other grapevine [used for wine-making, table grapes, or preserve-making] 3. (wild or ornamental *Vitus* species; numerous species) wild grapevine, ornamental grapevine

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Brassicales

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Brassicales > Brassicaceae [a.k.a., Cruciferae]

-ŽK- ‘BRASSICA OLERACEA (BROCCOLI & RELATED SPECIES)’ I 1. (*B.o. Italica* group) broccoli plant 2. (*B.o. Bortrytis* group) cauliflower plant, romanesco plant, broccoflower hybrid plant 3. (*B.o. Alboglabra* group) Chinese broccoli plant / gai lan, broccolini hybrid

-ŽKW- ‘BRASSICA OLERACEA (BROCCOLI & RELATED SPECIES)’ II 1. (*B.o. Capitata* group) cabbage plant 2. (*B.o. Gemmifera* group) brussel sprout plant 3. (*B.o. Gongylodes* group) kohlrabi plant

-ŽKY- ‘BRASSICA OLERACEA (BROCCOLI & RELATED SPECIES)’ III 1. (*B.o. Acephala* group 1) kale plant 2. (*B.o. Acephala* group 2) collard plant 3. (*B.o. Acephala* group 3) spring greens

-ŽKL- ‘BRASSICA OLERACEA (BROCCOLI & RELATED SPECIES)’ IV 1. (*B.o. sabauda*) savoy cabbage plant 2. (*B.o.*hybrid with *B. rapa rapa*) rutabaga plant 3. (*B.o.* non-edible varieties) ornamental kale

-ŽKR- ‘BRASSICA RAPA’ I 1. (*B.r. rapa*) turnip plant 2. (*B.r. pekinensis*) Napa cabbage plant 3. (*B.r. chinensis*) bok choy

-ŽKŘ- ‘BRASSICA RAPA’ II 1. (*B.r. oleifera*) (field) mustard plant 2. (*B.r. rapifera*) rapini / broccoli rabe plant 3. (*B.r. perviridis*) komatsuna plant / Japanese mustard spinach

-ŽKŢ- ‘BRASSICA RAPA’ III 1. (*B.r. nipposinica*) mizuna 2. (*B.r. parachinensis*) choy sum / flowering cabbage 3. (*B.r. oleifera*) canola

-ŽKH- ‘BRASSICA (OTHER SPECIES)’ I 1. (*B. juncea*) brown mustard, wrapped heart mustard cabbage 2. (*B. hirta*) white mustard 3. (*B. nigra*) black mustard

-ŽKÇ- ‘BRASSICA (OTHER SPECIES)’ II 1. (*B. napus*) Siberian kale 2. (*B. rosularis*) tatsoi 3. [stem unassigned]

-ŽKF- ‘BRASSICEAE’ I 1. (genus *Raphanus*) radish plant, daikon 2. (genus *Nasturtium*) watercress plant 3. (genus *Eruca*) arugula/rocket plant

-ŽKV- ‘BRASSICEAE’ II 1. (genus *Wasabia/Eutremia*) wasabi plant 2. (genus *Lepidum*) garden cress, field cress / field pepperweed / field pepperwort, maca / Peruvian ginseng 3. (genus *Diplotaxus*) wild arugula / perennial wall-rocket

-ŽKM- ‘BRASSICEAE’ III 1. (genus *Erucastrum*) dogmustard 2. (genus *Coincya*) star mustard, Isle of Man cabbage, wallflower cabbage, Lundy cabbage 3. (genus *Hirschfeldia*) shortpod mustard / buchanweed / hoary mustard / Mediterranean mustard

-ŽKN- ‘BRASSICEAE’ IV 1. (genus *Rapistrum*) bastard cabbage 2. (genus *Cakile*) searocket 3. (Brassiceae other than named species; numerous genera) other brassicean plant

-ŽGW- ‘THELYPODIEAE’ I 1. (genus *Thelypodium*) thelypody 2. (genus *Sibara*) winged rockcress 3. (genus *Sibaropsis*) Hammitt’s clay-cress

-ŽGY- 'THELYPODIEAE' II 1. (genus *Streptanthus*) twistflower, jewelflower, shieldplant 2. (genus *Streptanthella*) longbeack streptanthella 3. (genus *Glaucocarpum*) waxfruit mustard / toad-flax cress /shrubby reed-mustard / Uinta Basin waxfruit

-ŽGL- 'THELYPODIEAE' III 1. (genus *Caulostramina*) cliffdweller 2. (genus *Caulanthus*) jewelflower, wild cabbage 3. (genus *Thysanocarpus*) fringepod, lacepod

-ŽGR- 'THELYPODIEAE' IV 1. (genus *Stanleya*) prince's plume 2. (genus *Phlebolobium*) Falkland rock-cress 3. (Thelypodieae other than named genera; numerous genera) other thelypodiid plant

-ŽGŘ- 'CARDAMINEAE' I 1. (genus *Barbarea*) winter cress / yellow rocket, land cress / American cress / bank cress / black wood cress / Bermuda cress / scurvy cress / upland cress / creasy greens, bittercress / herb barbara / rocketcress / winter rocket / wound rocket 2. (genus *Armoracia*) horseradish 3. (genus *Cardamine*) toothwort, bittercress

-ŽGV- 'CARDAMINEAE' II 1. (genus *Leavenworthia*) gladecress 2. (genus *Iodanthus*) purplerocket 3. (genus *Rorippa*) yellowcress

-ŽGḌ- 'CARDAMINEAE' III 1. (genus *Selenia*) selenia 2. (genus *Planodes*) planodes 3. (Cardamineae other than named genera; several genera) other cardamineid plant

-ŽFW- 'BRASSICACEAE' I 1. (genus *Isatis*) woad / dyer's woad / glastum / Asp of Jerusalem 2. (genus *Hugueninia*) tansy-leaved rocket 3. (genus *Thlaspi*) penny-cress

-ŽFY- 'BRASSICACEAE' II 1. (genus *Conringia*) hare's ear mustards 2. (genus *Lunaria*) honesty 3. (genus *Arabis*) rockcress

-ŽFL- 'BRASSICACEAE' III 1. (genus *Athysanis*) sandweed 2. (genus *Draba*) whitlow-grass 3. (genus *Heterodraba*) ladiestongue mustard

-ŽFR- 'BRASSICACEAE' IV 1. (genus *Cochlearia*) scurvy-grass, spoonwort 2. (genus *Hesperis*) dame's rocket / damask-violet / dame's-violet / dames-wort / gilliflower / summer lilac / mother-of-the-evening, hesperis 3. (genus *Erysimum*) wallflower

-ŽFŘ- 'BRASSICACEAE' V 1. (genus *Iberis*) candytuft 2. (genus *Teesdalia*) shepherdscress 3. (genus *Idahoa*) scalepod / oldstem idahoa

-ŽVW- 'BRASSICACEAE' VI 1. (genus *Coronopus*) swinecress, wartcress 2. (genus *Descurainia*) tansymustard 3. (genus *Subularia*) awlwort

-ŽVY- 'BRASSICACEAE' VII 1. (genus *Arabidopsis*) rockcress 2. (genus *Boechera*) rockcress 3. (genus *Cusickiella*) cusickiella

-ŽVL- 'BRASSICACEAE' VIII 1. (genus *Phoenicaulis*) daggerpod, phoenicaulis 2.(genus *Anelsonia*) daggerpod 3.(genus *Halimolobos*) fissurewort

-ŽVR- 'BRASSICACEAE' IX 1. (genus *Physaria*) twindpod, bladderpod 2. (genus *Dithyrea*) shieldpod 3. (genus *Paysonia*) bladderpod

-ŽVŘ- 'BRASSICACEAE' X 1. (genus *Aethionema*) stonecress 2. (genus *Berteroa*) false madwort 3. (Brassicaceae other than named genera; numerous genera) other brassicaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Brassicales > remaining families

-ŽŽP- 'CAPPARACEAE' I 1. (genus *Capparis*) caper bush / caper shrub, karir / kirir, wild orange, devil's guts, wild passionfruit, mabinlang 2. (genus *Boscia*) shepherd tree, hanza 3. (genus *Cadaba*) swartstorm, cadaba

-ŽŽPW- 'CAPPARACEAE' II 1. (genus *Apophyllum*) warrior bush / broom bush 2. (genus *Cladostemon*) three-finger bush 3. (genus *Crateva*) sacred barna, sacred garlic pear / temple plant

-ŽŽPY- 'CAPPARACEAE' II 1. (genus *Maerua*) rough-skinned bush cherry 2. (Capparaceae other than named genera; numerous genera) other capparaceous plant 3. [stem unassigned]

-ŽŽPL- 'CLEOMACEAE' I 1. (genus *Cleome*) spider flower, spider plant, spiderwisp / cat's whiskers / African cabbage, justago, cleome, mouse whiskers, bladderpod / burro-fat, bee plant, Navajo spinach / stinking-clover / skunk weed / beeweed / guaco, tick weed 2. (genus *Cleomella*) cleomella, stinkweed 3. (genus *Oxystylis*) spiny caper

-ŽŽPR- 'CLEOMACEAE' II 1. (genus *Polanisia*) clammyweed 2. (genus Wislizenia) spactacle fruit, jackass clover 3. (Cleomaceae other than named genera; several genera) other cleomaceous plant

-ŽŽPŘ- ‘RESEDACEAE’ 1. (genus *Reseda*) mignonette, weld, dyer’s rocket, bastard rocket 2. (Resedaceae other than Stem 1; several genera) other resedaceous plant 3. [stem unassigned]

-ŽŽPF- ‘GYROSTEMONACEAE’ 1. (genus *Gyrostemon*) camel poison, corkybark 2. (genus *Codonocarpus*) bell-fruit tree, camel poison 3. (Gyrostemonaceae other than Stems 1 and 2; several genera) button creeper, other gyrostemonaceous plant

-ŽŽPŢ- ‘CARICACEAE’ 1. (genus *Carica*) papaya / pawpaw 2. (genus *Vasconcellea* [other than *V. pubescens*]) babaco tree 3. (*V. pubescens*) mountain papaya / mountain pawpaw

-ŽŽPÇ- ‘LIMNANTHACEAE’ 1. (*Limnanthes alba*) white meadowfoam 2. (*Limnanthes* other than *L. alba*; several genera) meadowfoam 3. (genus *Floerkea*) false mermaid, floerkea

-ŽŽPĻ- ‘SALVADORACEAE’ 1. (genus *Salvadora*) toothbrush tree, mustard tree 2. (genus *Azima*) azima, bee sting bush 3. (genus *Dobera*) dobera

-ŽŽPS- ‘BRASSICALES’ I 1. (Akaniaceae; genera *Akania, Bretschneidera*) turnipwood, bretschneidera 2. (Bataceae; genus *Batis*) turtleweed / saltwort / beachwort / pickleweed 3. (Koeberliniaceae; genus *Koeberlinia*) crown of thorns / allthorn / crucifixion thorn

-ŽŽPŠ- ‘BRASSICALES’ II 1. (Moringaceae; genus *Moringa*) drumstick tree / horseradish tree / ben oil tree / benzolive tree, bottle tree, moringa 2. (Tropaeolaceae; genus *Tropaeolum*) nasturtium / nasturtian, canary creeper, flame flower, three-coloured Indian cress, mashua 3. (Tovariaceae, Pentadiplandraceae, Emblingiaceae, Setchellanthaceae; several genera) other brassicales plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Malvales

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Malvales > Malvaceae

-**LCK**- ‘MALVOIDEAE’ I 1. (genus *Malva*) mallow, cheeseweed / cheeseplant 2. (genus *Alcea*) hollyhock, alcea 3. (genus *Althaea*) marshmallow, hemp-leaved hollyhock

-**LCKW**- ‘MALVOIDEAE’ II 1. (genus *Lavatera*) tree-mallow 2. (genus *Iliamna*) globe-mallow, wild hollyhock, Peter’s mountain mallow 3. (genus *Malacothamnus*) bush-mallow, chaparral mallow

-**LCKY**- ‘MALVOIDEAE’ III 1. (genus *Abutilon*) Indian mallow, room maple / parlor maple / flowering maple, abutilon, Indian lantern-flower / monkeybush, desert Chinese-lantern, butterprint / abutilon- hemp / China-jute / velvetleaf 2. (genus *Wissadula*) velvetleaf 3. (genus *Allowissadula*) false Indianmallow

-**LCKL**- ‘MALVOIDEAE’ IV 1. (genus *Herissantia*) bladder mallow, curly abutilon 2. (genus *Akrosida*) bigleaf akrosida 3. (genus *Malvella*) scurfy mallow, alkali mallow, arrowleaf mallow

-**LCKR**- ‘MALVOIDEAE’ V 1. (genus *Sphaeralcea*) globemallow, desertmallow, false mallow 2. (genus *Eremalche*) white mallow, Parry’s mallow, desert five-spot 3. (genus *Napaea*) glade mallow

-**LCKŘ**- ‘MALVOIDEAE’ VI 1. (genus *Callirhoe*) poppy mallow, winecup 2. (genus *Sidalcea*) checkerbloom, checkermallow, prairie mallow, miniature hollyhock, sidalcea 3. (genus *Anoda*) anoda, violetta

-**LCKF**- ‘MALVOIDEAE’ VII 1. (genus *Hoheria*) houhere, lacebark, ribbonwood, New Zealand mallow 2. (genus *Lawrencia*) lawrencia, dunna dunna 3. (genus *Malope*) mallow-wort / purple Spanish mallow, malope

-**LCKŢ**- ‘MALVOIDEAE’ VIII 1. (genus *Modiola*) bristly-fruited mallow / Carolina bristlemallow / babosilla / redflower mallow 2. (genus *Gynatrix*) hemp bush, aboriginal hemp 3. (genus *Horsfordia*) velvetmallow

-**LCKÇ**- ‘MALVOIDEAE’ IX 1. (genus *Gossypium*) cotton plant 2. (genus *Kokia*) treecotton 3. (genus *Thespesia*) portia tree / Pacific rosewood / milo, maga

-**LCKM**- ‘MALVOIDEAE’ X 1. (genus *Abelmoschus*) okra, musk mallow, abelmosk / ambrette / musk okra / rose mallow 2. (genus *Hibiscus*) hibiscus, rosemallow, rosella, comfortroot, roselle, foulsapate marron, mahoe, mandrinette, pale face, flower-of-an-hour, rose of sharon 3. (genus *Kosteletzkya*) seashore mallow / sweat weed / saltmarsh mallow

-**LCKN**- ‘MALVOIDEAE’ XI 1. (genus *Malvaviscus*) wax mallow / Turkcap / Turk’s turban / ladies teardrop / Scotchman’s purse, Turk’s cap mallow, sleeping hibiscus, mazapan 2. (genus *Pavonia*) swampmallow, gingerbush, pavonia 3. (genus *Lagunaria*) pyramid tree / Norfolk Island hibiscus / Queensland white oak / sally wood / itchy bomb tree / cow itch tree

-**LCKH**- ‘MALVOIDEAE’ XII 1. (genus *Radyera*) desert rose mallow / bush hibiscus 2. (genus *Hibiscadelphus*) hau kuahiwi 3. [stem unassigned]

-**LCKHW**- ‘MALVOIDEAE’ XIII 1. (genus *Alyogyne*) coastal hibiscus, lilac hibiscus, sand hibiscus 2. (Malvoideae other than named genera; numerous genera) other malvoid plant 3. [stem unassigned]

-**LČK**- ‘BOMBACOIDEAE’ I 1. (genus *Bombax*) silk cotton tree, red cotton tree, kapok-tree, bombax 2. (genus *Chirantodendron*) devil’s hand tree / monkey’s hand tree / Mexican hand tree / hand-flower 3. (genus *Fremontodendron*) flannel bush, fremontia

-**LČKW**- ‘BOMBACOIDEAE’ II 1. (genus *Ochroma*) balsa tree / balsa wood tree 2. (genus *Pachira*) Malabar chestnut tree / French peanut tree / Provision tree, saba nut, pochote 3. (genus *Pseudobombax*) soroche, shaving brush tree

-**LČKY**- ‘BOMBACOIDEAE’ III 1. (genus *Quararibea*) south American sapote / chupa-chupa, wild palm, swizzlestick tree 2. (Bombacoideae other than named genera; several genera) other bombacoid plant 3. [stem unassigned]

-**LČKL**- ‘HELICTEROIDEAE’ I 1. (genus *Durio*) durian tree 2. (genus *Helicteres*) Indian screw tree 3. (genus *Kostermansia*) krepal, durian tuang

-**LČKR**- ‘HELICTEROIDEAE’ II 1. (genus *Triplochiton*) African whitewood tree 2. (Helicteroideae other than named genera) other helicteroid tree 3. [stem unassigned]

-**LČKŘ**- ‘BROWNLOWIOIDEAE’ 1. (genus *Berrya*) trincomalee wood tree 2. (Brownlowioideae other than Stem 1; several genera) other brownlowioid plant/tree 3. [stem unassigned]

-**LČKF**- ‘DOMBEYOIDEAE’ 1. (genus *Dombeya*) dombeya, pinkball / tropical hydrangea, dikba / South African wild pear, bois bete 2. (genus *Pentapetes*) noon flower 3. (Dombeyoideae other than Stems 1 and 2; several genera) other dombeyoid plant

-**LČKŢ**- ‘STERCULIOIDEAE’ I 1. (genus *Sterculia*) tropical chestnut, sterculia, Panama tree / manduvi tree, mopopaja tree, bastard poon tree / wild almond tree, China chestnut / seven sister’s fruit, lowveld chestnut, gorarbar, ulumbu tree, gulu, broad-leaved bottle tree 2. (genus *Cola*) kola tree / kola nut, hairy cola / Zulu coshwood 3 (genus *Brachychiton*) kurrajong, couramyn, illawarra flame tree, lacebark tree, bottletree

-**LČKÇ**- ‘STERCULIOIDEAE’ II 1. (genus *Franciscodendron*) tulip sterculia / tulip kurrajong / cabbage crowsfoot 2. (genus *Argyrodendron*) booyong, tulip oak, crowsfoot, blackjack 3. (genus *Pterygota*) buddha coconut, koto

-**LČKM**- ‘STERCULIOIDEAE’ III 1. (genus *Heritiera*) looking-glass mangrove, dungon 2. (genus *Scaphium*) malva nut tree / Taiwan sweet gum tree 3. (Sterculioideae other than named genera; several genera) other sterculioid plant/tree

-**LČKN**- ‘GREWIOIDEAE’ 1. (genus *Grewia*) falsa / phalsa, crossberry, emu-berry / dog’s balls / turkey bush / diddle diddle / dysentery bush 2. (genus *Triumfetta*) burbark, Chinese bur 3. (Grewioideae other than Stems 1 and 2; several genera) other grewioid plant

-**LČKH**- ‘TILIOIDEAE’ 1. (genus *Tilia*) linden / lime tree / limebush / basswood 2. (genera *Craigia, Mortoniodendron)* other tillioid plant/tree 3. [stem unassigned]

-**LČKHW**- ‘BYTTNERIOIDEAE’ I 1. (genus *Theobroma*) cacao tree / cocoa tree 2. (genus *Commersonia*) kerrawang 3. (genus *Abroma*) devil’s cotton

-**LČKS**- ‘BYTTNERIOIDEAE’ II 1. (genus *Waltheria*) sleepy morning 2. (genus *Hannafordia*) grey felt-bush 3. (genus *Seringia*) crinkle-leaved firebush

-**LČKŠ**- ‘BYTTNERIOIDEAE’ III 1. (genus *Lasiopetalum*) velvet bush, lasiopetalum 2. (genus *Ayenia*) ayenia 3. (Byttnerioideae other than named genera) other byttnerioid plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Malvales > remaining families

-**LCT**- ‘CISTACEAE’ I 1. (genus *Cistus*) rockrose, cistus 2. (genus *Halimium*) basil-leaved rock rose, woolly rock rose / false sun-rose, halimium 3. (genus *Fumana*) needle sunrose

-**LCTW**- 'CISTACEAE' II 1. (genus *Helianthemum*) rock rose / sunrose / rushrose / frostweed 2. (genus *Crocanthemum*) peak rockrose / peak rushrose, low rockrose / low frostweed, island rush-rose, bushy rockrose / bushy frostweed, rock frost / frostplant / frostwort / Canada frost weed / longbranch frostweed 3. (genus *Hudsonia*) goldenheather / povertygrass

-**LCTY**- 'CISTACEAE' III 1. (genus *Tuberaria*) spotted rock-rose / annual rock-rose 2. (genus *Lechea*) pinweed 3. [stem unassigned]

-**LCTL**- 'THYMELAEACEAE' I 1. (genus *Thymelaea*) sparrow-wort, spurge flax 2. (genus *Daphne*) daphne, garland flower, Nepalese paper plant, spurge-laurel, spurge olive / mezereum / mezereon 3. (genus *Lagetta*) lacebark tree / gauze tree

-**LCTR**- 'THYMELAEACEAE' II 1. (genus *Dirca*) leatherwood / moosewood / ropebark / wicopy 2. (genus *Wikstroemia*) false ohelo 3. (genus *Pimelea*) rice flower, bootlace bush, taranga, scrub kurrajong, pinatoro, New Zealand daphne / Strathmore weed, bunjong, banjine, Flinders poppy, pimelea

-**LCTŘ**- 'THYMELAEACEAE' III 1. (genus *Gonystylus*) ramin, melawis 2, (genus *Aetoxylon*) crocodile eaglewood / gaharu buaya 3. (genera *Aquilaria*, *Gyrinops*) lign-aloe tree

-**LCTF**- 'THYMELAEACEAE' IV 1. (genus *Phaleria*) God's crown / pau 2. (Thymelaeaceae other than named genera; numerous genera) other thymelaeaceous plant/tree 3. [stem unassigned]

-**LCTÇ**- 'DIPTEROCARPACEAE' I 1. (genus *Dipterocarpus*) keruing, gurjan 2. (genus *Neobalanocarpus*) chengal 3. (genus *Shorea*) lauan, Philippine mahogany

-**LCTḶ**- 'DIPTEROCARPACEAE' II 1. (genus *Parashorea*) white seraya / white lauan 2. (genus *Vateria*) white dammar 3. (Dipterocarpaceae other than named genera; several genera) other dipterocarpaceous plant/tree

-**LCTH**- 'BIXACEAE' 1. (genus *Bixa*) achiote, lipstick tree 2. (genus *Amoreuxia*) yellowshow, Santa Rita throwup weed 3. (genera *Cochlospermum, Diegodendron*) yellow cotton tree, kapok, diegodendron

-**LCTHW**- 'MALVALES' 1. (Muntingiaceae; genera *Muntingia, Neotessmannia, Dicraspidia*) calabur tree / capulin / Jamaica cherry / Panama berry / Singapore cherry / West Indian cherry / ornamental cherry / jamfruit tree, neotessmannia, dicraspidia 2.(Cytinaceae, Neuradaceae, Sarcolaenaceae, Sphaerosepalaceae; several genera) other malvales plant/tree 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Sapindales

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Sapindales > Sapindaceae

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Sapindales > Sapindaceae > Hippocastanoideae

-**VST**- 'HIPPOCASTANOIDEAE' 1. (genus *Acer*) maple tree, box elder tree, moosewood, sycamore 2. (genus *Aesculus*) horse chestnut, buckeye 3. (genera *Billia*, *Dipteronia*, *Handeliodendron*) other hippocastanoid plant/tree

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Sapindales > Sapindaceae > Sapindoideae

VSTW- 'SAPINDOIDEAE' I 1. (genus *Litchi*) lychee tree 2. (genus *Dimocarpus*) longan, alupag 3. (genus *Nephelium*) rambutan, korlan, pulasan

VSTY- 'SAPINDOIDEAE' II 1. (genus *Blighia*) blighia tree, ackee fruit tree 2. (genus *Allphylus*) titberry, bastard taaibos, dune false crowberry / dune false currant 3. (genus *Diploglottis*) native tamarind, small-leaved tamarind

VSTL- 'SAPINDOIDEAE' III 1 (genus *Melicoccus*) mamoncillo / Spanish lime / genip / quenette / limoncillo 2. (genus *Talisia*) pitomba tree 3. (genus *Alectryon*) hairy bird's eye / hairy alectryon, boonaree, titoki, beach bird's eye

VSTR- 'SAPINDOIDEAE' IV 1. (genus *Guioa*) glossy tamarind 2. (genus *Castanospora*) brown tamarind 3. (genus *Pappea*) jacket plum /indaba tree / bushveld cherry

VSTŘ- 'SAPINDOIDEAE' V 1. (genus *Sapindus*) soapberry / soapnut 2. (genus *Jagera*) foambark / ferntree / pink foambark / ferntop / pink tamarind, daintree foambark 3. (genus *Mischocarpus*) woolly bush apple

VSTF- ‘SAPINDOIDEAE’ VI 1. (genus *Sarcopteryx*) steelwood tree / corduroy 2. (genus *Toechima*) blunt-leaved steelwood, cape tamarind 3. (genus *Ungnadia*) Mexican buckeye

VSTÇ- ‘SAPINDOIDEAE’ VII 1. (genus *Arytera*) coogera 2. (Sapindoideae other than named genera; numerous genera) other sapindoid plant/tree 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Sapindales > Sapindaceae > Dodonaeoideae & Xanthoceroideae

-**PSXW**- ‘DODONAEOIDEAE’ I 1. (genus *Dodonaea*) hop-bush 2. (genus *Diplopeltis*) pepperflower 3. (genus *Harpulia*) tulipwood

-**PSXL**- ‘DODONAEOIDEAE’ II 1. (genus *Felicium*) ferntree / fern leaf tree 2. (genus *Ganophyllum*) scaly ash / scaly bark ash / Daintree hickory / honeywood 3. (genus *Hypelate*) inkwood / white ironwood

-**PSXR**- ‘XANTHOCEROIDEAE & OTHER DODONAEOIDEAE’ 1. (genus *Zanha*) velvet-fruited zanha 2. (Dodonaeoideae other than named genera; several genera) other dodonaeoid plant/tree 3. (Xanthoceroideae; genus Xanthoceras) yellowhorn / shiny leaf yellowhorn / goldenhorn / Chinse flowering chestnut

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Sapindales > Rutaceae

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Sapindales > Rutaceae > Rutoideae

-**MFKW**- ‘RUTOIDEAE’ I 1. (genus *Ruta*) rue 2. (genus *Cneoridium*) bushrue 3. (genus *Thamnosma*) desert rue, turpentinebroom, sandboegoe

-**MFKY**- ‘RUTOIDEAE’ II 1. (genus *Dictamnus*) burning bush / dittany / gas plant / fraxinella 2. (genus *Corymborkis*, a.k.a. *Macrostylis*) cinnamon orchid 3. (genus *Calodendrum*) Cape chestnut

-**MFKL**- ‘RUTOIDEAE’ III 1. (genus *Diosma*, a.k.a. *Coleonema*) diosma, confetti bush, buchu, boegoe, Cape May 2. (genus *Adenandra*) buchu, China flower 3. (genus *Agathosma*) buchu / boegoe / bucco / bookoo / diosma

-**MFKR**- ‘RUTOIDEAE’ IV 1. (genus *Angostura,* a.k.a. *Cusparia*) angostura 2. (genus *Esenbeckia*) esenbeckia, jopoy, limoncillo, crucecilla, coya / cuala-cuala, hueso de tigre, gasparillo 3. (genus Pilocarpus) jaborandi

-**MFKŘ**- ‘RUTOIDEAE’ V 1. (genus *Boronia*) boronia, native rose, granite rose 2. (genus *Acradenia*) whitey wood / wirewood, yellow satinheart / bonewood 3. (genus *Asterolasia*) starbush

-**MFKF**- ‘RUTOIDEAE’ VI 1. (genus *Chorilaena*) karri oak / chorilaena 2. (genus *Correa*) correa, Dunally correct, native fuschia 3. (genus *Crowea*) crowea, waxflower

-**MFKŢ**- ‘RUTOIDEAE’ VII 1. (genus *Diplolaena*) Yanchep rose, wild rose, diplolaena 2. (genus *Eriostemon*) pink wax flower 3. (genus *Microcybe*) microcybe

-**MFKÇ**- ‘RUTOIDEAE’ VIII 1. (genus *Nematolepis*) nematolepis, satinwood / lancewood / bobie bobie / satin box 2. (genus *Phebalium*) phebalium, St. Helens wax flower 3. (genus *Zieria*) zieria, twiggy midge bush, stinkbush

-**MFKM**- ‘RUTOIDEAE’ IX 1. (genus *Zanthoxylum*, including genus *Fagara*) prickly ash, toothache, yellowwood, Chinese pepper / Sichuan pepper, Hercules’ club / pepperbark, coco / cochucho / smelly sauco, forest knobwood, West Indian satinwood, yellow prickle, Japanese pepper / Korean pepper / chopi, Indian pepper / Indian ivy-rue, niaragato, zanthoxylum 2. (genus *Tetradium*) euodia / evodia / bee bee tree 3. (genus *Platydesma*) Hawai’i pilo kea

-**MFKN**- ‘RUTOIDEAE’ X 1. (genus *Pitavia*) pitao / pitran 2. (genus *Pentaceras*) penta ash / bastard crows ash / black teak 3. (genus *Orixa*) Japanese orixa / East Asian orixa

-**MFKS**- ‘RUTOIDEAE’ XI 1. (genus *Melicope*) doughwood / corkwood, melicope, pelea, wharangi, poataniwha, Maui ruta, catafaille, ‘alani, mokihana 2. (genus *Bosistoa*) bosistoa / bonewood 3. (genus *Choisya*) Mexican orange

-**MFKŠ**- ‘RUTOIDEAE’ XII 1. (genus *Geijera*) Australian willow / native willow / wilga, axebreaker 2. (genus *Medicosma*) pinkheart / bonewood 3. (Ruteae other than named genera; numerous genera) other rutoid plant/tree

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Sapindales > Rutaceae > Aurantioideae

-NL̥- 'GENUS *CITRUS* HYBRID' I 1. (*C. sinensis*) sweet orange 2. (*C. aurantium*) bitter orange / Seville orange 3. (*C. aurantiifola*) key lime

-NL̥N- 'GENUS *CITRUS* HYBRID' II 1. (*C. limon*) lemon 2. (*C. meyeri*) Meyer lemon 3. (*C. paradisi*) grapefruit

-NL̥NY- 'GENUS *CITRUS* HYBRID' III 1. (*C. clementina*) clementine 2. (*C. tangelo*) tangelo 3. (*Citrus* hybrid other than previously named) other citrus hybrid

-NL̥Ň- 'GENUS *CITRUS'* I 1. (*C. hystrix*) Kaffir lime / makrut 2. (*C. reticulata*) mandarin orange 3. (*C. medica*) citron

-NL̥NW- 'GENUS *CITRUS'* II 1. (*C. glauca, C. australasica, C. autralis, C. garrawayae, C. inodora, C. gracilis, C. warburgiana, C. wintersii*) Australian lime, finger lime, round lime, desert lime, wild lime 2. (*C. trifoliata*) trifoliate orange 3. (*C. platymamma*) byeonggyul

-NL̥ŇW- 'GENUS *CITRUS'* III 1. (*C. japonica*) kumquat 2. (*C. maxima*) pomelo / shaddock 3. (*Citrus* plant other than named species; numerous species) other citrus plant/tree

-**NL̥MW**- 'AURANTIOIDEAE' I 1. (genus *Citropsis*) African cherry orange 2. (genus *Burkillanthus*) Malay ghostlime 3. (genus *Clymenia*) clymenia, a-mulis

-**NL̥MY**- 'AURANTIOIDEAE' II 1. (genus *Pleiospermium*) Banguey Island orangeaster 2. (genus *Triphasia*) limeberry / sweet lime 3. (genus *Oxanthera*) oxanthera / false orange

-**NL̥ML**- 'AURANTIOIDEAE' III 1. (genus *Aegle*) bael / Bengal quince / golden apple / Japanese bitter orange / stone apple / wood apple 2. (genus *Limonia*) curd fruit / wood-apple / elephant-apple 3. (genus *Feroniella*) feroniella / ka sang / cra san / kawista-krikil / krasang / mak sang / canthan

-**NL̥MR**- 'AURANTIOIDEAE' IV 1. (genus *Monanthocitrus*) monanthocitrus / spotseed-lime 2. (genus *Merrillia*) flowering merrillia / katinga / Malay lemon 3. (genus *Murraya*) curry tree / curry leaf, orange jessamine / Hawaiian mock orange / bartree / cosmetic-bark-tree /Chinese box / Burmese-boxwood, murraya

-**NL̥MŘ**- 'AURANTIOIDEAE' V 1. (genus *Glycosmis*) orangeberry / gin berry, glycosmis, red-tangerine 2. (genus *Clausena*) horsewood, wampi / wampee, clausena 3. (Aurantioideae other than named genera; numerous genera) other aurantioid plant/tree

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Sapindales > Rutaceae > Spathelioideae

-**DVR**- 'SPATHELIOIDEAE' 1. (genus *Ptaeroxylon*) sneezewood tree 2. (genus *Cneorum*) spurge olive 3. (Spathelioideae other than named genera; several genera) other spathelioid plant/tree

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Sapindales > Rutaceae > Toddalioideae

-**PSMW**- 'TODDALIOIDEAE' I 1. (genus *Toddalia*) orange climber 2. (genus *Acronychia*) achronychia, silver aspen, hairy aspen, white aspen, doughwood, acid berry / plasticine tree, lemon aspen, hard aspen 3. (genus *Sarcomelicope*) yellowwood / yellow aspen / sarcomelicope

-**PSMY**- 'TODDALIOIDEAE' II 1. (genus *Ptelea*) hoptree 2. (genus *Phellodendron*) cork-tree 3. (genus *Skimmia*) skimmia

-**PSML**- 'TODDALIOIDEAE' III 1. (genus *Hortia*) bush orange 2. (genus *Casimiroa*) sapote, matasano 3. (genus *Amyris*) torchwood, chapotillo

-**PSMR**- 'TODDALIOIDEAE' IV 1. (genus *Vepris*) white ironwood 2. (Toddalioideae other than named genera; numerous genera) other toddalioid plant/tree 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Sapindales > Rutaceae > [remaining sub-families: Flindersioideae & Dictyolomatoideae]

-**PSMŘ**- 'FLINDERSIOIDEAE & DICTYOLOMATOIDEAE' 1. (Flindersioideae, genus *Flindersia*) silver silkwood / siver maple, Australian teak / crows ash, scented maple / rose ash, leopardwood / leopard tree, mountain silkwood, maple silkwood / rose silkwood, silver ash / cudgerie / bumpy ash, yellowwood / long Jack, hickory ash /

Cairns hickory, Queensland maple / maple silkwood, scented maple, Bennett's ash 2. (Flindersioideae, genus *Chloroxylon*) East Indian satinwood / Sri Lanka satinwood, Madagascar satinwood 3. (Dictyolomatoideae, genus *Dictyoloma*) dictyoloma

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Sapindales > Anacardiaceae

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Sapindales > Anacardiaceae > Anacardioideae

-NḶK- 'ANACARDIOIDEAE' I 1. (genus *Pistacia*) pistachio, American pistachio / Mexican pistache, mastic, Chinese pistache, betoum / wild pistachio / Persian turpentine tree / Mount Atlas mastic tree 2. (genus *Anacardium*) cashew, wild cashew 3. (genus *Mangifera*) mango tree

-**NḶKW**- 'ANACARDIOIDEAE' II 1. (genus *Rhus*) sumac 2. (genus *Malosma*) laurel sumac 3. (genus *Protorhus*) red beech

-**NḶKY**- 'ANACARDIOIDEAE' III 1. (genus *Comocladia*) maidenplum, poison ash / hogwood / thumbtack, guao 2. (genus *Cotinus*) smoketree / smokebush, chittamwood, Venetian sumach / dyer's sumach 3. (genus *Schinus*) pepper tree

-**NḶKL**- 'ANACARDIOIDEAE' IV 1. (genus *Amphipterygium*) cuachalalate 2. (genus *Toxicodendron*) poison oak, poison ivy, poison sumac, lacquer tree / varnish tree, wax tree 3. (genus *Metopium*) poisonwood / Florida poisontree

-**NḶKR**- 'ANACARDIOIDEAE' V 1. (genus *Astronium*) kingwood / locustwood / tigerwood /zebrawood, glassywood, urunday 2. (genus *Blepharocarya*) bollygum / rose butternut 3. (genus *Euroschinus*) pink poplar / ribbonwood / maiden's blush / blush cudgerie

-**NḶKŘ**- 'ANACARDIOIDEAE' VI 1. (genus *Buchanania*) chirauli-nut, green plum, satinwood / lightwood 2. (genus *Gluta*) rengas, ringas / Burmese lacquer / theetsee / thitsi 3. (genus *Pachycormus*) Baja elephant tree / torote blanco / copalquín

-**NḶKF**- 'ANACARDIOIDEAE' VII 1. (genus *Rhodosphaera*) deep yellowwood / yellow cedar / tulip satinwood 2. (genus *Schinopsis*) red quebracho 3. (genus *Searsia*) African sumac / willow rhus / karee

-**NḶKŢ**- 'ANACARDIOIDEAE' VIII 1. (genus *Semecarpus*) Australian cashew nut, marking nut tree / phobi nut tree / varnish tree, Chuuk poisontree, velvet badulla 2. (genus *Smodingium*) African poison ivy 3. (genus *Ozoroa*) ozoroa, resin tree

-**NḶKÇ**- 'ANACARDIOIDEAE' IX 1. (genus *Bouea*) plum mango / mango plum / gandaria 2. (Anacardioideae other than named genera; numerous genera) other anacardioid plant/tree 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Sapindales > Anacardiaceae > Spondiadoideae

-**NḶKM**- 'SPONDIADOIDEAE' I 1. (genus *Spondias*) hog plum / Spanish plum / golden apple, mombin, Tahitian apple, gully plum / ashanti plum / Java plum, Brazil plum 2. (genus *Choerospondias*) Nepali hog plum / lapsi 3. (Spondiadoideae other than Stems 1 and 2; numerous genera) other spondiadoid plant/tree

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Sapindales > Nitrariaceae

-**FXN**- 'NITRARIACEAE' 1. (genus *Nitraria*) nitre bush / dillon bush 2. (genus *Peganum*) wild rue / Syrian rue / African rue / esfand / harmel 3. (genus *Tetradiclis*) tetradiclis

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Sapindales > Simaroubaceae

-**NZXM**- 'SIMAROUBACEAE' I 1. (genus *Simarouba*) paradise-tree / dysentery-bark / bitterwood / Lakshmi Taru 2. (genus *Quassia*) quassia, amargo / bitter-ash / bitter-wood 3. (genus *Ailanthus*) ailanthus, tree of heaven

-**NZXN**- 'SIMAROUBACEAE' II 1. (genus *Leitneria*) corkwood 2. (genus *Eurycoma*) Malaysian ginseng / tongkat ali / pasak bumi 3. (Simaroubaceae other than named genera: numerous genera) other simaroubaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Sapindales > Meliaceae

-**MSFW**- 'MELIOIDEAE' I 1. (genus *Melia*) chinaberry tree / Persian lilac / white cedar / Pride of India / bread-tree / syringa berrtree 2. (genus *Azadirachta*) neem tree / nimtree / Indian lilac 3. (genus *Sandoricum*) santol tree / sentul / cotton tree

-**MSFY**- 'MELIOIDEAE' II 1. (genus *Guarea*) bossé, guarea, pink mahogany, cramantee, American muskwood 2. (genus *Turraeanthus*) African satinwood / African white mahogany 3. (genus *Aphanamixis*) pithraj tree

-**MSFL**- 'MELIOIDEAE' II 1. (genus *Aglaia*) aglaia, Pacific maple, droopy leaf / priyangu 2. (genus *Lansium*) langsat / lanzones / duku / dokong 3. (genus *Dysoxylum*) rosewood, ivory maghogany, pink mahogany, spicy mahogany, yellow mahogany, hairy rosewood, New Zealand mahogany

-**MSFR**- 'MELIOIDEAE' IV 1. (genus *Nymania*) Chinese lantern 2. (genus *Owenia*) emu apple, onion cedar / bog onion 3. (genus *Trichilia*) Natal-mahogany, bariaco

-**MSFŘ**- 'MELIOIDEAE' V 1. (genus *Synoum*) scentless rosewood 2. (Melioideae other than named genera; numerous genera) other melioid plant/tree 3. [stem unassigned]

-**MSXW**- 'CEDRELOIDEAE' I 1. (genus *Swietenia*) mahogany tree 2. (genus *Cedrela*) Argentine cedar, Spanish cedar / Cuban cedar 3. (genus *Toona*) redcedar / toon / toona

-**MSXL**- 'CEDRELOIDEAE' II 1. (genus *Carapa*) crabwood / andiroba, tangare 2. (genus *Khaya*) African mahogany 3. (genus *Chukasia*) Indian mahogany / bastard cedar / white cedar / Indian redwood / Burma almond wood / chickrassy / chittagong wood

-**MSXR**- 'CEDRELOIDEAE' III 1. (genus *Entandrophragma*) sapele mahogany / sapelli, sipo / utile, kosipo, tiama 2. (genus *Lovoa*) brown mahogany / Kilimanjaro mahogany 3. (Cedreloideae other than named genera; several genera) other cedreloid plant/tree

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Sapindales > Burseraceae and remaining Sapindales sub-families [i.e., Biebersteiniaceae & Kirkiacieae]

-**ŇSXW**- 'BURSERACEAE' I 1. (genus *Boswellia*) frankincense tree, luban / Yigaar tree, Indian oli-banum / Salai gugul 2. (genus *Commiphora*) myrrh plant, opopanax plant, bdellium plant, commiphora, rock corkwood, gugal / guggul / Mukul myrrh tree 3. (genus *Bursera*) torchwood copal / fragrant bursera, gumbo-limbo / copperwood / chaca / turpentine tree, elephant tree, palo santo

-**ŇSXL**- 'BURSERACEAE' II 1. (genus *Aucoumea*) gaboon / angouma / okoumé 2. (genus Canarium) Chinese white olive, mango bark / brown cudgerie / parsnip wood, scrub turpentine / carrot wood / Melville Island white beech, Chinese black olive, African canarium, black dhup / Raal dhup / black dammar 3. (genus Dacryodes) African pear / bush pear / bush plum / bush butter tree / butterfruit tree

-**ŇSXR**- 'BURSERACEAE' III & REMAINING SAPINDALES 1. (genus *Protium*) copal tree 2. (Burseraceae other than named genera; several genera) other burseraceous plant/tree 3. (Biebersteiniaceae and Kirkiacieae; several genera) other Sapindales plant/tree

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Huerteales

-**ŇSXM**- 'HUERTEALES' 1. (Dipentodontaceae; genera *Dipentodon, Perrottetia*) dipenodontaceous plant 2. (Tapisciaceae; genera *Tapiscia, Huertea*) tapisciaceous plant 3. (Gerrardinaceae and Petenaeaceae; genera *Gerrardina, Petenaea*) other huerteales plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Picramniales

-**ŇSXN**- 'PICRAMNIACEAE' 1. (genus *Picramnia*) bitterbush 2. (genus *Alvaradoa*) Mexican alvaradoa, alvaradoa 3. (genus *Nothotalisia*) nothotalisia

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Crossosomatales

-**RTĻM**- ‘CROSSOSOMATALES’ I 1. (genus *Crossosoma*) California rockflower, ragged rockflower 2. (genus *Glosspetalon*, a.k.a., *Forsellesia*) greasebush 3. (genus *Apacheria*) Chiricahua rock flower / cliff brittlebush / Apache bush

-**RTĻMW**- ‘CROSSOSOMATALES’ II 1. (genus *Staphylea*) bladdernut 2. (genus *Geissoloma*) guyalone 3. (genus *Ixerba*) tawari / whakou

-**RTĻMY**- ‘CROSSOSOMATALES’ III 1. (genus *Stachyurus*) stachyurus 2. (genera *Velascoa, Guamatela, Aphloia, Dalrympelea, Strasburgia*) other crossosomatales plant/tree 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Geraniales

-**TÇMW**- ‘GERANIACEAE’ I 1. (genus *Geranium*) geranium, cranesbill, giant herb robert 2. (genus *Erodium*) filaree / heron’s bill, storksbill / pinweed 3. (genus *California*) roundleaf stork’s bill

-**TÇMY**- ‘GERANIACEAE’ II 1. (genus *Pelargonium*) geranium, pelargonium, storksbill 2. (genus *Monsonia*) monsonia, sarcocaulon 3. (genus *Hypseocharis*) hypseocharis

-**TÇNW**- ‘FRANCOACEAE’ I 1. (genus *Francoa*) bridalwreath 2. (genus *Melianthus*) honey flower 3. (genus *Greyia*) Kei bottlebrush, Natal bottlebrush, woolly bottlebrush

-**TÇNY**- ‘FRANCOACEAE’ II 1. (genus *Viviana,* a.k.a. *Cissarobryon*) té de burro / oreganillo 2. (Francoaceae other than named genera; several genera) other francoaceous plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Myrtales

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Myrtales > Myrtaceae

-**ŘDK**- ‘MYRTEAE’ I 1. (genus *Psidium*) guava / yellow guava / lemon guava, Costa Rican guava / cas, Jamaican guava, Cattley guava / strawberry guava / cherry guava, mountain guava, Galápagos guava, little guava, purple guava, araçá 2. (genus *Pimenta*) allspice / myrtle pepper / pimenta / pimento, West Indian bay tree / bay rum tree / ciliment 3. (genus *Rhodomyrtus*) finger cherry / Cooktown loquat, native guava, rose myrtle

-**ŘDKW**- ‘MYRTEAE’ II 1. (genus *Acca*) feijoa / pineapple guava / guavasteen 2. (genus Hexachlamys) ubajay 3. (genus *Plinia*) Brazilian grapetree / jaboticaba

-**ŘDKY**- ‘MYRTEAE’ III 1. (genus *Myrtus*) myrtle 2. (genus *Archirhodomyrtus*) rose myrtle 3. (genus *Eugenia*) needle-leaf cherry, rainforest plum, dune myrtle, mountain cherry, cherry of the Rio Grande, cerrado pear, pitomba, guayabilla, bush cherry, pitanga, bush orange, mountain stopper / Cedar Bay cherry / beach cherry, agatelang

-**ŘDKL**- ‘MYRTEAE’ IV 1. (genus *Luma*) Chilean myrtle / arrayán, chequén / huillipeta 2. (genus *Pilidiostigma*) plum myrtle 3. (genus *Myrcianthes*) twinberry / twinberry stopper, arrayán, guabiyu, lucumillo

-**ŘDKR**- ‘MYRTEAE’ V 1. (genus *Austromyrtis*) midgen berry / midyim / silky myrtle, narrow-leaf myrtle 2. (genus *Calyptranthes*) lidflower, spicewood, mountainbay 3. (genus *Gossia*) scrub ironwood, python tree, Cape ironwood, sweet myrtle / small-leaved myrtle

-**ŘDKŘ**- ‘MYRTEAE’ VI 1. (genus *Lithomyrtus*) beach myrtella 2. (genus *Lophomyrtus*) ramarama, rohutu 3. (genus *Rhodamnia*) smooth scrub turpentine, brush turpentine, cliff malletwood

-**ŘDKF**- ‘MYRTEAE’ VII 1. (genus *Myrcia*) red rodwood, pedro hume de caá 2. (genus *Myrciaria*) rumberry / guavaberry, camu-camu, cabeludinha, blue grape 3. (Myrteae other than named genera; numerous genera) other myrtean plant / tree

-**ŘDKŢ**- ‘MYRTACEAE’ I 1. (genus *Syzygium*) clove tree, satinash, riberry, Malay apple / Malacca apple / Malay rose apple / Otaheite cashew / water apple, coolamon / durobby, lilly pilly, magenta cherry, onionwood, bellfruit, water cherry, hute / waterbessie, sour cherry, purple cherry / black water gum, jambul / jambolan / black plum Damson plum / Duhat plum / Portuguese plum, Indian bay leaf / Indonesian bay leaf / Salam leaf / Indonesian laurel, java apple / makopa / Samarang rose apple / wax jambu / wax apple, lady apple, Malabar plum / plum rose, sea apple 2. (genus *Waterhousea*) weeping lilli pilli 3. (genus *Syncarpia*) turpentine tree, satinay, yanderra

-**ŘDKÇ**- 'MYRTACEAE' II 1. (genus *Allosyncarpia*) an-binik 2. (genus *Eucalyptus*) eucalyptus, wattle-leaved peppermint, Badgingarra box, mallee, white mahogany, Lakefield coolibah, gum, stringybark, whitebark, blackbutt, apple-box, soak yate, mallet, ironbark 3. (genus *Angophora*) rough-barked apple, dwarf apple, Charmhaven apple, rusty gum, Coolabah apple, sandstone apple, broad-leaved apple, smudgee, red gum / smooth-barked apple, narrow-leaved apple

-**ŘDKS**- 'MYRTACEAE' III 1. (genus *Corymbia*) bloodwood tree, ghost gum tree, spotted gum tree, yellow jack / yellowjacket, range gum, snappy gum, apple gum / cabbage gum, mountain marri, Gilbert River box, rustyjacket, carbeen / Moreton Bay ash, cadaga / cadaghi 2. (genus *Stockwellia*) stockwellia, Vic Stockwell's puzzle 3. (genus *Backhousia*) curry myrtle, Johnstone river hardwood, lemon scented myrtle / sweet berbena tree / lemon scented verbena / lemon ironwood, stonewood / limewood / greay teak / stony backhousia, grey myrtle / carrol / neverbreak / iron myrtle / cinnamon myrtle / ironwood, shatterwood / boomerang tree, giant ironwood / scrub ironwood / lancewood / ironwood box

-**ŘDKŠ**- 'MYRTACEAE' IV 1. (genus *Tristania*) water gum 2. (genus *Xanthostemon*) golden penda, bridal tree / northern penda, Philippine ironwood, crimson penda / red penda 3. (genus *Metrosideros*) rātā, lehua, mountain rose, tepual

-**ŘDKM**- 'MYRTACEAE' V 1. (genus *Astartea*) astartea 2. (genus *Babingtonia*) camphor myrtle, broom baeckea, babingtonia 3. (genus *Baeckea*) heath-myrtle, baeckea

-**ŘDKN**- 'MYRTACEAE' VI 1. (genus *Balaustion*) native pomegranate 2. (genus *Calytrix*) starflower, snow myrtle, fringe-myrtle, Kimberley heather, calytrix 3. (genus *Chamelaucium*) waxflower

-**ŘDKV**- 'MYRTACEAE' VII 1. (genus *Cheyniana*) bush pomegranate 2. (genus *Darwinia*) darwinia, bell 3. (genus *Homoranthus*) homoranthus, fairy bells, mouse bush, eastern feather flower

-**ŘDKH**- 'MYRTACEAE' VIII 1. (genus *Hypocalymma*) white myrtle, large myrtle, Swan River myrtle 2. (genus *Micromyrtus*) heath-myrtle 3. (genus *Pileanthus*) coppercups

-**ŘDG**- 'MYRTACEAE' IX 1. (genus *Rinzia*) rinzia, desert rock myrtle 2. (genus *Sannantha*) tall baeckia, fern-leaf baeckia, sannantha 3. (genus *Scholtzia*) scholtzia

-**ŘDGW**- 'MYRTACEAE' X 1. (genus *Thryptomene*) thryptomene 2. (genus *Verticordia*) verticordia, feather flower, pixie ears, pink brownii / pink cauliflower, cauliflower bush, little chrysantha, landu, lambswool / native cauliflower / wild cauliflower, native tea, china cups, Morrison / Morrison-flower / Christmas Morrison, little grandiflora, Eurardy magenta 3. (genus *Lophostemon*) brush box / Queensland box / Brisbane box, Northern swamp box, milky box, swamp mahogany / swamp turpentine / swamp box

-**ŘDGY**- 'MYRTACEAE' XI 1. (genus *Agonis*) Western Australian peppermint / Swan River peppermint / willow myrtle, summer snowflakes 2. (genus *Asteromyrtus*) liniment-tree 3. (genus *Kunzea*) tick bush, rawiritoa, kunzea, spearwood / pondil, kanuka, manuka, muntries

-**ŘDGL**- 'MYRTACEAE' XII 1. (genus *Leptospermum*) tea-tree, tantoon / jellybush, turkey bush, manuka, wurunuru 2. (genus *Texandria*) wattie / native cedar / juniper myrtle, teatree, swamp peppermint 3. (genus *Melaleuca*) paperbark, honey-myrtle, teatree, mock olive, cajuput / white samet, umbrella bush, bottlebrush, karnbor, totem poles, hillock bush, moonah, gorada, robin redbreast bush, snow-in-summer / budjur, broom bush, banbar, boree, claw flower, mindiyed, melaleuca

-**ŘDGR**- 'MYRTACEAE' XIII 1. (genus *Callistemon*) bottlebrush 2. (genus *Calothamnus*) bottlebrush, claw flower, blood-flower 3. (genus *Beaufortia*) beaufortia, bottlebrush

-**ŘDGŘ**- 'MYRTACEAE' XIV 1. (genus *Lysicarpus*) brown hazelwood / budgeroo 2. (genus *Regelia*) regelia 3. (Myrtaceae other than named genera; numerous genera) other myrtaceous plant/tree

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Myrtales > Melastomataceae

-**MFX**- 'MELASTOMATACEAE' I 1. (genus Melastoma) blue tongue / native lasiandra, Asian melastome, Singapore rhododendron / Malabar melastome, Straits rhododendron, fox-gongued melastoma / blood-red melastoma / red melastome 2. (genus *Dissotis*) pink lady / Spanish shawl / rockrose 3. (genus *Medinilla*) medinilla, rose grape, tagimaucia, chandelier tree

-**MFXL**- 'MELASTOMATACEAE' II 1. (genus *Henriettea*) camasey 2. (genus *Rhexia*) meadow-beauty, handsome Harry 3. (genus *Tetrzygia*) clover ash, West Indian lilac

-**MFXR**- ‘MELASTOMATACEAE’ III 1. (genus *Clidemia*) soapbush / Koster’s curse 2. (Melastomataceae other than named genera; numerous genera) other melastomataceous plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Myrtales > Lythraceae

-**ŇŇP**- ‘LYTHRACEAE’ I 1. (genus *Punica*) pomegranate 2. (genus *Trapa*) water caltrop / buffalo nut / bat nut / devil pod / ling nut / mustache nut 3. (genus *Sonneratia*) mangrove apple, sonneratia, apple mangrove

-**ŇŇPW**- ‘LYTHRACEAE’ II 1. (genus *Lythrum*) loosestrife, lythrum, grass-poly, water purslane 2. (genus *Rotala*) Indian toothcup, rotala 3. (genus *Lawsonia*) henna tree / hina / mignonette tree / Egyptian privet

-**ŇŇPY**- ‘LYTHRACEAE’ III 1. (genus *Lagerstroemia*) crape myrtle / crêpe myrtle 2. (genus *Heimia*) sun opener / shrubby yellowcrest 3. (genus *Decodon*) swamp loosestrife / waterwillow

-**ŇŇPL**- ‘LYTHRACEAE’ IV 1. (genus *Cuphea*) cuphea, cigar plant, tarweed / blue waxweed, waxweed, false heather / Mexican heather 2. (genus Ammannia) redstem, ammannia 3. (Lythraceae other than named genera; numerous genera) other lythraceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Myrtales > Combretaceae

-**ŇŇPL**- ‘COMBRETACEAE’ I 1. (genus *Combretum*) bushwillow, leadwood tree, orange flame vine / chameleon vine, combretum 2. (genus *Conocarpus*) buttonwood / button mangrove 3. (genus *Terminalia*) white afara, white olive, crocodile tree, white arudah /arjuna / koha, palo amarillo / tanimbú, silver greywood, jalool, Indian almond / tropical almond / umbrella tree, myrobalan / inknut, pindang quondong, billygoat plum / kakadu plum / gubinge, idigbo / black afara /; blackbark / brimstone wood / shingle wood, okari nut, Madagascar almond tree, rosewood, marool, damson, limba, terminalia

-**ŇŇPR**- ‘COMBRETACEAE’ II 1. (genus *Laguncularia*) white mangrove 2. (genus *Lumnitzera*) black mangrove, Tonga mangrove 3. (genus *Bucida*) bullet tree, spiny black olive

-**ŇŇPŘ**- ‘COMBRETACEAE’ III 1. (genus *Buchenaviaia*) buchenavia, granadillo 2. (Combretaceae other than named genera; several genera) other combretaceous plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Myrtales > Onagraceae

-**ŇŇPF**- ‘ONAGRACEAE’ I 1. (genus *Fuchsia*) fuchsia 2. (genus *Circaea*) enchanter’s nightshade 3. (genus *Ludwigia*) primrose-willow, water-purslane, water-primrose, ludwigia, bushy seedbox / rattlebox, marsh seedbox, mosaic flower / false loosestrife

-**ŇŇPŢ**- ‘ONAGRACEAE’ II 1. (genus *Epilobium*) willowherb, spike-primrose, boisduvalia, hummingbird flower / hummingbird trumpet / California-fuchsia 2. (genus *Chamaenerion*) willowherb, fireweed, Saint Anthony’s laurel 3. (genus *Taraxia*) goldeneggs / sun cup

-**ŇŇPÇ**- ‘ONAGRACEAE’ III 1. (genus *Camissonia*) suncup, evening primrose 2. (genus *Calylophus*) sundrops, evening-primrose 3. (genus *Camissoniopsis*) suncup, evening primrose

-**ŇŇPS**- ‘ONAGRACEAE’ IV 1. (genus *Chylisma*) browneyes / brown-eyed primrose, yellow cups, suncup, evening primrose 2. (genus *Chylismiella*) wingfruit suncup 3. (genus *Oenothera* [including previous genus *Gaura*) suncup, sundrop, evening primrose, gaura, beeblossum

-**ŇŇPŠ**- ‘ONAGRACEAE’ V 1. (genus *Clarkia*) clarkia, godetia, mountain garland, farewell-to-spring, fairyfan, red ribbons, pink fairies / ragged robin 2. (genus *Gayophytum*) groundsmoke 3. (Onagraceae other than named genera; several genera) other onagraceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Myrtales > Vochysiaceae

-**ŇŇPH**- 'VOCHYSIACEAE' 1. (genus *Qualea*) jacaré, pau-terra 2. (Vochysiaceae other than Stem 1; several genera) other vochysiaceous plant/tree 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > malvids > Myrtales > remaining families: Alzateaceae, Crypteroniaceae & Penaeaceae [including Oliniaceae and Rhynchocalycaceae]

-**ŇŇPHW**- 'OTHER MYRTALES' 1. (Alzataceae; genus *Alzatea*) alzatea 2. (Crypteroniaceae; 3 genera) crypteroniaceous plant/tree 3. (Penaeaceae:several genera) penaeaceaous plant/tree

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Fabales

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Fabales > Fabaceae

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Fabales > Fabaceae > Faboideae

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Fabales > Fabaceae > Faboideae > Phaseoleae

-**JPW**- 'PHASEOLUS VULGARIS (COMMON BEAN)' I 1. kidney/red bean plant 2. white/navy bean plant 3. cannellini bean plant

-**JPY**- 'PHASEOLUS VULGARIS (COMMON BEAN)' II 1. pinto bean plant 2. black turtle bean plant 3. flageolet bean plant

-**JPL**- 'OTHER PHASEOLUS SPECIES' I 1. (*P. coccineus*) runner bean plant 2. (*P. acutifolius*) tepary bean plant 3. (*P. dumosus*) year bean plant

-**JPR**- 'OTHER PHASEOLUS SPECIES' II 1. (*P. lunatus*) lima/butter bean plant 2. (*P. maculatus*) spotted bean plant 3. (*Phaseolus* other than named species) slimjim bean plant, other Phaseolus bean plant

-**JPŘ**- 'VIGNA SPECIES' I 1. (*V. unguiculata*) cowpea, black-eyed pea, crowderpea, Chinese long-bean / pea-bean 2. (*V. angularis*) adzuki bean 3. (*V. radiata*) mung bean / green gram / golden gram / mash bean / green soybean / celera-bean / Jerusalem-pea

-**JPF**- 'VIGNA SPECIES' II 1. (*V. mungo*) black gram / black lentil / white lentil / urd-bean / urad bean 2. (*V. umbellata*) rice bean / red bean / climbing mountain bean / mambi bean / Oriental-bean 3. (*Vigna* other than named species; numerous species) moth bean / mat bean / Turkish gram, creole-bean, jungle mat bean / jungli-bean / African gram / three-lobe-leaved cowpea, zombi pea / wild cowpea, sarawak-bean, Dalrymple vigna, Bambara groundnut / Congo goober /hog-peanut / jugo bean, pencil yam / Maloga-bean / parsnip-bean, dune-bean / notched cowpea / sea-bean, Oahu cowpea, creeping vigna, other Vigna bean

-**JPÇ**- 'PHASEOLINAE' I 1. (genus *Cochliasanthus*) corkscrew vine / snail vine / snail creeper / snailflower / snail bean 2. (genus *Dipogon*) okie bean / Cape sweet-pea / dolichos pea / mile-a-minute vine 3. (genus *Lablab*) lablab-bean / hyacinth bean, bonavist bean / bonavist pea / Egyptian kidney bean / Indian bean / Australian pea / bataw

-**JPŢ**- 'PHASEOLINAE' II 1. (genus *Macroptilium*) purple bush-bean, supine bean 2. (genus *Macrotyloma*) ground bean / Kersting's groundnut, horse gram / kulthi 3. (genus *Psophocarpus*) winged bean / goa bean / four-angled bean / four-cornered bean / manila bean / princess bean / asparagus bean / dragon bean / cigarrillas

-**JPĻ**- 'PHASEOLINAE' III 1. (genus *Sphenostylis*) wild sweet-pea, African yam bean 2. (genus *Strophostyles*) wild bean, fuzzybean, amberique-bean / annual sand bean 3. (Phaseonlinae other than named genera/species [including *Phaseolus* other than named species]; numerous genera/species) other phaseoline bean

-**JFW**- 'PHASEOLEAE' I 1. (genus *Glycine*) soybean / soya bean, wild soybean, glycine, love creeper 2. (genus *Pachyrhizus*) jicama plant / yam bean, ahipa / Andean yam bean, goiteño / nupe / jacatupe / Amazonian yam bean 3. (genus *Pueraria*) kudzu / Japanese arrowroot, puero, East Asian arrowroot

-**JFY**- 'PHASEOLEAE' II 1. (genus *Clitoria*) pigeon wings, blue pea / butterfly pea / cordofan pea / Darwin bea / bluebellvine 2. (genus *Centrosema*) butterfly pea / wild blue vine / blue bell, centro 3. (genus *Amphicarpaea*) hogpeanut

-**JFL**- 'PHASEOLEAE' III 1. (genus *Cajanus*) pigeon pea / Congo pea 2. (genus *Rhynchosia*) snout-bean, jumby-bean / burn-mouth-vine 3. (genus *Flemingia*, a.k.a. *Thunbergia*) Bengal clockvine / Bengal trumpet / blue skyflower /blue thumbergia / blue trumpetvine / skyvine, luck plant / wild hops, sohphlang

-**JFR**- 'PHASEOLEAE' IV 1. *(genus Dioclea)* clusterpea 2. (genus *Canavalia*) jack-bean, barbicou-bean, giant stock-bean /gotani-bean / horse-bean / seaside-bean / wonder-bean, sword bean / scimitar-bean, Cathie's bean, bay bean / beach-bean / fire-bean / Mackenzie-bean 3. (genus *Galactia*) milk pea / beach pea / wild pea

-**JFŘ**- 'PHASEOLEAE' V 1. (genus *Kennedia*) kennedia, coral vine, black coral pea, purple running pea, running postman, dusky coral pea 2. (genus *Hardenbergia*) native lilac / wild sarsaparilla / wisteria climber, false sarsaparilla / purple coral pea / happy wanderer / waraburra 3. (genus *Butea*) flame-of-the-forest / bastard teak

-**JFM**- 'PHASEOLEAE' VI 1. (genus *Erythrina*) coral tree, flame tree, cock's spur, coral bean / Cherokee bean / red cardinal / cardinal spear, lucky bean tree, mulungu, mandara, tiger's claw / sunshine tree / roluos tree, ploughbreaker, erythrina 2. (genus *Mucuna*) deer-eye bean / donkey-eye bean / ox-eye bean / hamburger seed 3. (genus *Strongylodon*) jade vine / emerald vine

-**JFN**- 'PHASEOLEAE' VII 1. (genus *Apios*) potato bean / Indian potato / American groundnut / cinnamon vine / America-hoidomo / groundnut 2. (Phaseoleae other than named genera/species; numerous genera/species) other phaseolean bean 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Fabales > Fabaceae > Faboideae > tribes other than Phaseoleae

-**ŘPP**- 'PISUM SATIVUM (PEA)' 1. (*P. sativum*) common pea plant 2. (*P. s. macrocarpon*) snap pea plant 3. (*P. s. saccharatum*) snow pea plant

-**ŘPPW**- 'FABOIDEAE' I 1. (*Vicia faba*) fava bean plant 2. (*Vicia* species other than *V. faba*; numerous species) vetch 3. (genus *Lens*) lentil

-**ŘPPY**- 'FABOIDEAE' II 1. (genus *Lathyrus*) peavine, vetchling, Indian pea, red pea, tuberous pea 2. (genus *Cicer*) chickpea/garbanzo bean plant 3. (genus *Sesbania*) riverhemp, poisonbean, rattlebox / rattlebush, vegetable hummingbird, sesbania

-**ŘPPL**- 'FABOIDEAE' III 1. (genus *Arachis*) peanut plant 2. (genus *Inocarpus*) Tahitian chestnut 3. (genus *Stylosanthes*) pencilflower

-**ŘPPR**- 'FABOIDEAE' IV 1. (genus *Tipuana*) tipa / rosewood / pride of Bolivia 2. (genus *Centrolobium*) canarywood, amarillo de Guayaquil 3. (genus *Pterocarpus*) padauk, klaat, muninga / barwood, Indian kino / Malabar kino, mututi, red sandalwood / red sanders, African coralwood

-**ŘPPŘ**- 'FABOIDEAE' V 1. (genus *Dalbergia*) Madagascar rosewood, Brazilian tulipwood, Hluhluwe creeper, Bahia rosewod, climbing flat bean, frgrant rosewood, coin vine, kingwood, tracwood, Bombay blackwood / East Indian rosewood / Indian palisandre / Java palisandre, palisander, Siamese rosewood, Amazon rosewood, Honduras rosewood, Guatemalan rosewood, African blackwood / African ebony / African grenadio; zebrawood, moneybush 2. (genus *Aeschynomene*) jointvetch, sola / sola pith plant, shyleaf, ambatch 3. (genus *Nissolia*) yellowhood

-**ŘPPF**- 'FABOIDEAE' VI 1. (genus *Andira*) angelin / cabbagebark / cabbage tree, St. Martin rouge 2. (genus *Hovea*) purple pea, hovea, rusty pods, mountain beauty, devil's pins 3. (genus *Templetonia*) cockies tongues, centipede bush, templetonia

-**ŘPPÇ**- 'FABOIDEAE' VII 1. (genus *Calpurnia*) wild laburnum 2. (genus *Liparia*) orange nodding-head / mountain dahlia 3. (genus *Virgilia*) tree-in-a-hurry / cape lilack / blossom tree / pink blossom tree

-**ŘPPŢ**- 'FABOIDEAE' VIII 1. (genus *Ammodendron*) sand acacia 2. (genus *Baphia*) African sandalwood 3. (genus *Castanospermum*) blackbean / Moreton Bay chestnut

-**ŘPPĻ**- 'FABOIDEAE' IX 1. (genus *Cladrastis*) yellowwood 2. (genus *Dermatophyllum*) mescal bean 3. (genus *Haplormosia*) Liberian black grum

-**ŘPPS**- 'FABOIDEAE' X 1. (genus *Luetzelburgia*) sucupira / angelim 2. (genus *Ormosia*) horse-eye bean, ormosia 3. (genus *Pericopsis*) African teak, nandu wood

-**ŘPPŠ**- 'FABOIDEAE' XI 1. (genus *Sophora*) kohwhai, ku shen, mamane, sophora root, necklace pod, toromiro, mayo / mayú, pelú, lignum vitae 2. (genus *Styphnolobium*) necklacepod, coralbean / Eve's necklace 3. (genus *Xanthocercis*) Mashatu tree / Nyala tree

-**ŘPPH**- 'FABOIDEAE' XII 1. (genus *Genista*) broom, petty whin / needle furze, aulaga, talavera, greenweed, woodwaxen 2. (genus *Argyrocystisus*) pineapple broom / Moroccan broom 3. (genus *Calicotome*) thorny broom, spiny broom

-**ŘPPHW**- 'FABOIDEAE' XIII 1. (genus *Cytisus*) broom, Scotch broom, Spanish broom / French broom / Portuguese broom / hairy-fruited broom 2. (genus *Chamaecytisus*) Portuguese broom / white broom, big-flower broom / clustered broom, tagasaste, purple broom 3. (genus *Lembotropis*) black broom

-**ŘPK**- 'FABOIDEAE' XIV 1. (genus *Laburnum*) golden chain / golden rain / laburnum 2. (genus *Laburnocytisus*) groom laburnum / Adam's laburnum 3. (genus *Petteria*) Dalmatian laburnum

-**ŘPKW**- ‘FABOIDEAE’ XV 1. (genus *Lupinus*) lupin, lupine 2. (genus *Retama*) bridal broom 3. (genus *Spartium*) Spanish broom / rush broom / weaver’s broom

-**ŘPKY**- ‘FABOIDEAE’ XVI 1. (genus *Ulex*) gorse, furze, whin 2. (genus *Crotalaria*) rattlepod / rattlebox, chipilin, devil-bean / rattleweed / shack-shack, harebells / rabbit-bells, giant striata / chique chique, brown hemp / Indian hemp / Madras hemp / sunn hemp, green birdflower / regal birdflower 3. (genus *Cyclopia*) honeybush / honeybush tea plant

-**ŘPKL**- ‘FABOIDEAE’ XVII 1. (genus *Amorpha*) false indigo 2. (genus *Dalea*) prairie clover, purpletassels, summer farewell, dalea, white tassel-flower, whitetassels, Gentry’s indigobush 3. (genus *Eysenhardtia*) kidneywood tree

-**ŘPKR**- ‘FABOIDEAE’ XVIII 1. (genus *Marina*) false prairie-clover 2. (genus *Parryella*) common dunebroom 3. (genus *Psorothamnus*) indigo bush, dalea, smoketree / smokethorn

-**ŘPKŘ**- ‘FABOIDEAE’ XIX 1. (genus *Thermopsis*) goldenbanners / false-lupines 2. (genus *Baptisia*) wild indigo, galse indigo, catbells, scareweed, rattleweed, gopherweed, horseflyweed / indigo-broom / yellow broom 3. (genus *Wisteria*) wisteria

-**ŘPKF**- ‘FABOIDEAE’ XXI 1. (genus *Indigofera*) indigo plant 2. (genus *Cyamopsis*) guar / Lond bean 3. (genus *Lonchocarpus*) lancepod, philenoptera / apple leaf / rain tree -

-**ŘPKÇ**- ‘FABOIDEAE’ XXII 1. (genus *Millettia*) millettia, panga panga, Indian beeech / Pongam oiltree / pongamia 2. (genus *Piscidia*) fishpoison tree, Jamaican dogwood / fishfuddle 3. (genus *Tephrosia*) hoarypea, tephrosia, Flinders River poison, sandhill tippitoes, goat-rue / catgut / rabbit pea, fish-poison-bean

-**ŘPKŢ**- ‘FABOIDEAE’ XXIII 1. (genus *Alysicarpus*) moneywort, alys, alyce clover, buffalo clover / one-leaf clover 2. (genus *Desmodium*) tick-trefoil, tick clover, beggarweed, Kaimi clover, dollar leaf, silverleaf, hitchhikers, beggar lice 3. (genus *Pseudarthria*) pink velvet bean

-**ŘPKS**- ‘FABOIDEAE’ XXIV 1. (genus *Kummerowia*) Korean bushclover, Japanese bushclover 2. (genus *Lespedeza*) lespedeza, bush clover, Japanese clover 3. (genus *Abrus*) jequirity bean / rosary pea

-**ŘPKŠ**- ‘FABOIDEAE’ XXV 1. (genus *Bituminaria*) Arabian pea / pitch trefoil 2. (genus *Hoita*) scurfpea / leather-root 3. (genus *Orbexilum*) leather-root, French-grass / lanceleaf scurfpea, Sampson’s snakeroot

-**ŘPKV**- ‘FABOIDEAE’ XXVI 1. (genus *Pediomelum*) Indian breadroot, skunktop, turniproot, buckroot, buffalo pea / tall-bread scurf-pea 2. (genus *Psoralea*) white tumbleweed, prairie turnip / timpsula 3. (genus *Psoralidium*) lemon scurfpea / wild lemonweed / dune scurfpea, slimflower scurfpea

-**ŘPKH**- ‘FABOIDEAE’ XXVII 1. (genus *Rupertia*) California tea / rupertia, forest scurfpea 2. (genus *Syrmatium*) Bentham’s broom / Bentham’s deerweed, bird’s-foot trefoil, Haydon’s lotus / pygmy lotus, rush broom / rush deervetch, beach lotus / Nuttall’s lotus, island broom 3. (genus *Dipteryx*) tonka bean, almendro, baru, almendrillo

-**ŘPKHW**- ‘FABOIDEAE’ XXVIII 1. (genus *Swartzia*) Brazilian ebony / panococo 2. (genus *Cordyla*) wild mango 3. (genus *Pickeringia*) chaparral pea

-**ŘPT**- ‘FABOIDEAE’ XXIX 1. (genus *Astragalus*) milkvetch, locoweed, goat’s-thorn, cock’s-head, woollypod 2. (genus *Carmichaelia*) North Island broom, South Island broom, leafy broom, coral broom, dwarf broom, scented broom, large-flowered broom, pink broom, whip broom, wheeping broom / tree broom, giant-flowered broom 3. (genus *Galega*) goat’s rue

-**ŘPTW**- ‘FABOIDEAE’ XXX 1. (genus *Clianthus*) kakabeak / parrot’s beak / parrot’s bill / lobster claw 2. (genus *Colutea*) bladder senna 3. (genus *Montigena*) scree pea

-**ŘPTY**- ‘FABOIDEAE’ XXXI 1. (genus *Glycyrrhiza*) licorice / American licorice 2. (genus *Oxytropis*) locoweed, oxytrope, oxytropis 3. (genus *Sphaerophysa*) alkali swainsonpea / Austrian peawead / red bladdervetch

-**ŘPTL**- ‘FABOIDEAE’ XXXII 1. (genus *Sutherlandia*) cancer bush / balloon pea / sutherlandia 2. (genus *Swainsona*) swainson-pea, desert pea, darling pea 3. (genus *Sphinctospermum*) hourglass peaseed

-**ŘPTR**- ‘FABOIDEAE’ XXXIII 1. (genus *Lotus*) bird’s-foot trefol / bacon-and-eggs, deervetch, trefoil 2. (genus *Acmispon*) bird’s-foot trefoil, silver lotus, Spanish clover, deervetch, silky deerweed, Heermann’s lotus, Chile lotus 3. (genus *Ornithopus*) bird’s foot, yellow serradella

-**ŘPTŘ**- ‘FABOIDEAE’ XXXIV 1. (genus *Ottleya*) shrubby deervetch / desert rock-pea, stiff-haired lotus / strigose bird’s foot trefoil, Wright’s deervetch 2. (genus *Anthyllis*) kidneyvetch / woundwort 3. (genus *Coronilla* [including genus *Securigera*) crown vetch, scorpion vetch

-**ŘPTÇ**- 'FABOIDEAE' XXXV 1. (genus *Hippocrepis*) horseshoe vetch, scorpion senna 2. (genus *Hosackia*) big deervetch / broad-leafed lotus, bird's-foot trefoil, meadow lotus, thicket trefoil 3. (genus *Scorpiurus*) scorpion's-tail / caterpillar-plant

-**ŘPTĻ**- 'FABOIDEAE' XXXVI 1. (genus *Coursetia*) babybonnets 2. (genus *Genistidium*) brushpea 3. (genus *Gliricidia*) quickstick

-**ŘPC**- 'FABOIDEAE' XXXVII 1. (genus *Olneya*) ironwood / desert ironwood / palo fierro 2. (genus *Peteria*) spine-noded milkvetch / Thompson's peteria 3. (genus *Robinia*) locust, false acacia

-**ŘPTH**- 'FABOIDEAE' XXXVIII 1. (genus *Hedysarum*) sweetvetch, sweet-broom, alpine sainfoin 2. (genus *Alhagi*) camelthorn / camelthorn-bush, manna tree / mannaplant 3. (genus *Caragana*) Siberian peashrub / Siberian pea-tree / caragana

-**ŘTP**- 'FABOIDEAE' XXXIX 1. (genus *Ebenus*) Cretan ebony 2. (genus *Halimodendron*) common salt tree / Russian salt tree 3. (genus *Onobrychis*) sainfoin

-**ŘTPW**- 'FABOIDEAE' XL 1. (genus *Trifolium*) clover, trefoil 2. (genus *Melilotus*) melilot, sweet clover 3. (genus *Medicago*) alfalfa plant, medick, burclover, button clover, nonesuch, hop clover, moon trefoil, alfalfa arborea

-**ŘTPY**- 'FABOIDEAE' XLI 1. (genus *Ononis*) restharrow 2. (genus *Parochetus*) shamrock pea / blue oxalis 3. (genus *Trigonella*) blue fenugreek / blue melilot

-**ŘTPL**- 'FABOIDEAE' XLII 1. (genus *Aotus*) aotus, golden pea 2. (genus *Gompholobium*) glory pea / wedge-pea 3. (genus *Daviesia*) bitter-pea, bookleaf, marno

-**ŘTPR**- 'FABOIDEAE' XLIII 1. (genus *Oxylobium*) shaggy pea 2. (genus *Podolobium*) shaggy pea 3. (genus *Pultenaea*) bush-pea

-**ŘTPŘ**- 'FABOIDEAE' XLIV 1. (genus *Chorizema*) heart-leaf flame pea / Australian flame pea, holly flame pea 2. (genus *Dillwynia*) parrot-pea 3. (genus *Stonesiella*) clubmoss bush-pea

-**ŘTPÇ**- 'FABOIDEAE' XLV 1. (genus *Sphaerolobium*) globe-pea 2. (genus *Isotropis*) poison sage, granny bonnets, lamb poison 3. (genus *Gastrolobium*) poison plant, brother-brother, breelya, Swan River pea, mountain pea, cranbrook pea

-**ŘTPF**- 'FABOIDEAE' XLVI 1. (genus *Callistachys*) wonnich / native willow 2. (genus *Jacksonia*) stinkwood, dogwood, waldjumi 3. (genus *Viminaria*) native broom

-**ŘTPŢ**- 'FABOIDEAE' XLVII 1. (genus *Bossiaea*) bossiaea, water bush, cactus pea 2. (genus *Goodia*) golden-tip 3. (genus *Platylobium*) flat-pea, eggs-and-bacon pea

-**ŘTPĻ**- 'FABOIDEAE' XLVIII 1. (genus *Eutaxia*) eutaxia 2. (genus *Mirbelia*) mirbelia 3. (Faboideae other than named genera; numerous genera) other faboid plant/tree

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Fabales > Fabaceae > Cercidoideae

-**ŘTPS**- 'CERCIDOIDEAE' I 1. (genus *Cercis*) redbud, Judas tree 2. (genus *Barklya*) leather jacket / crown of gold tree 3. (genus *Piliostigma*) camel's foot tree / monkey bread / monkey biscuit tree, purple orchid tree

-**ŘTPŠ**- 'CERCIDOIDEAE' II 1. (genus *Bauhinia*) orchid tree, bauhinia, bidi leaf tree 2. (genus *Lysiphyllum*) bauhinia / jigal tree, mountain ebony / Queensland ebony, northern beantree 3. (genus *Schnella*) granny backbone, monkey ladder vine

-**ŘTPH**- 'CERCIDOIDEAE' III 1. (genus *Phanera* [including genus *Cheniella*]) bauhinia, butterfly tree, orchid tree / mountain ebony 2. (genus *Tylosema*) marama bean / gemsbok bean 3. (Cercidoideae other than named genera; several genera) other cercidoid plant/tree

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Fabales > Fabaceae > Caesalpinoideae

-**ŘPŢ**- 'CAESALPINEAE' I 1. (genus *Caesalpinia*) peacock flower / poinciana / red bird of paradise / Mexican bird of paradise / pride of Barbados / flos pavonis, nicker 2. (genus *Biancaea*) shoofly / Mauritius thorn / Mysore thorn / cat's claw, sappanwood / Indian redwood 3. (genus *Cordeauxia*) yeheb bush

-**ŘPŢW**- 'CAESALPINEAE' II 1. (genus *Denisophytum*) fewflower holdback 2. (genus *Erythrostemon*) tailed nicker, bird of paradise, Mexican holdback 3. (genus *Guilandina*) nicker, nicker nut

-**ŘPŢY**- ‘CAESALPINEAE’ III 1. (genus *Haematoxylum*) logwood, peachwood / brazilette 2. (genus *Hoffmannseggia*) rushpea, holdback, nicker, pig-nut / hog potato 3. (genus *Libidibia*) Brazilian ironwood / leopard tree, divi-divi, Argentinian brown ebony, quebrahacha

-**ŘPŢL**- ‘CAESALPINEAE’ IV 1. (genus *Moullava*) teri pod 2. (genus *Paubrasilia*) brazilwood / pernambuco wood 3. (genus *Pomaria*) holdback, nicker

-**ŘPŢR**- ‘CAESALPINEAE’ V 1. (genus *Pterolobium*) redwing, camp siege / bhoca 2. (genus *Tara*) tara 3. (Caesalpineae other than named genera; several genera) other caesalpine plant/vine

-**ŘTT**- ‘CAESALPINOIDEAE’ I 1. (genus *Cassia*) cassia, wishing-tree, gold medallion tree, sjambok pod, native laburnum, golden shower, Palawan cherry, drumstick tree 2. (genus *Senna*) senna, candlebush / Christmas candle / empress candle plant / ringworm shrub / candelabra bush, cassia, false sicklepod, avaram / Matara-tea, West Indian showertree, Christmas bush / money bush / yellow candlewood, coffee-weed / java-bean / American sicklepod, antbush / coffee-senna / septicweed / stinkingweed, monkey pod, Argentine wild sensitive plant, kassodtree / Thailand shower, arsenic-bush / Dooleyweed, New Mexico wild sensitive plant 3. (genus *Chamaecrista*) sensitive pea

-**ŘTTW**- ‘CAESALPINOIDEAE’ II 1. (genus *Burkea*) wild syringa tree 2. (genus *Erythrophleum*) Cooktown ironwood, ordeal tree 3. (genus *Melanoxylum*) brauna wood tree

-**ŘTTY**- ‘CAESALPINOIDEAE’ III 1. (genus *Delonix*) poinciana 2. (genus Parkinsonia, a.k.a. Cercidium) palo verde, brea 3. (genus Schizolobium) Brazilian firetree / Brazilian fern tree

-**ŘTL**- ‘CAESALPINOIDEAE’ IV 1. (genus Ceratonia) carob tree 2. (genus *Tachigali*) suicide tree 3. (genus Acrocarpus) pink cedar

-**ŘTTR**- ‘CAESALPINOIDEAE’ V 1. (genus Gleditsia) honey locust, water locust / swamp locust, Caspian locust 2. (genus Gymnocladus) coffee tree, dekang tree, soap tree / Chinese coffee tree 3. (Caesalpinoideae other than named genera) other caesalpinoid plant/tree

-**ŘKŢ**- ‘MIMOSOIDEAE’ I 1. (genus *Mimosa*) mimosa, sensitive plant, sensitive tree, sensitive-briar, catclaw briar 2. (genus *Anadenanthera*) calcium tree, yopo 3. (genus *Parkia*) African locust bean

-**ŘKŢW**- ‘MIMOSOIDEAE’ II 1. (genus *Acacia*) wattle, acacia, jam, raspberry jam / fine leaf jam, gidgee, mulga, salwood, ashburton willow / moondyne tree, creekline miniritchi, boobialla / native willow, yalgoo, prickly Moses, currawang, lancewood, Minni Ritchie, needlewood, myall, false bowgada, pin bush, bendee, motherumbah, soap bush, old man wodjil, brown spearwood / ringy rosewood, wantan / milhan, sugar brother, curracabah, granite bush, mountain hickory / broad-leaved hickory, scrub ironbark, New England hickory, dead finish / kurara / curara, wirilda, alumaru / nyalanyalara, gundabluey, waddy-wood / waddy, kangaroo thorn, kanji bush, broombush, camel bush / corky canji, bluebush, two-veined hickory, buchan blue, motherumbung, Broughton willow / cooba / Doolan, billy blue, umbrella bush wirra, wait-a-while / Bohemia, mamoose tree, gidyea, bowyakka, dwarf nealie, lightwood, yarran, brigalow / brigalow spearwood / orkor, boree / balaar / nilyah, bank catclaw, spine bush, desert oak / wirewood / dogwood, womel, eumong / dunthy / belalei / munumula / balkura / gurley / gooralee, catbush 2. (genus *Vachellia*) acacia, wattle, thorn tree 3. (genus *Senegalia*) acacia, catclaw, monkey thorn, cutch tree, visco / viscote / arca, catechu tree

-**ŘKŢY**- ‘MIMOSOIDEAE’ III 1. (genus *Ebenopsis*) Texas ebony, ebano 2. (genus *Pitecellobium*) blackbead, everfresh 3. (genus *Havardia*) chucum / cuisache, huajillo

-**ŘKŢL**- ‘MIMOSOIDEAE’ IV 1. (genus *Acaciella*) acacia, wattle, yellow tamarind 2. (genus *Albizia*) silk plant, silk tree, siris, acle / akle, albizia, musase, kalkora mimosa, lebbeck / lebbek tree / frywood / koko / woman’s-tongue tree, monkey’s earring, rain tree / saman / monkey pod 3. (genus *Archidendron*) lace flower, tulip siris, djenkol / jengkol / jering, bacon wood

-**ŘKŢR**- ‘MIMOSOIDEAE’ V 1. (genus *Calliandra*) powder-puff plant, fairy duster, soldierwood 2. (genus *Enterolobium*) monkey-ear tree / elephant-ear tree / guanacaste, pacara earpod tree 3. (genus *Faidherbia*) apple-ring acacia / winter thorn / ana tree

-**ŘKF**- ‘MIMOSOIDEAE’ VI 1. (genus *Lysiloma*) false tamarind, sabicu / horseflesh 2. (genus *Pararchidendron*) snow-wood / tulip siris / monkey’s earrings 3. (genus *Samanea*) rain tree

-**ŘKFW**- ‘MIMOSOIDEAE’ VII 1. (genus *Abarema*) abarema, shadbark, crabwood / jumbie head / soapy-soapy / wild tamarind 2. (genus *Entada*) sea bean / seaheart / monkey-ladder, St. Thomas bean , snuff box bean 3. (genus *Elephantorrhiza*) elephant root / sumach bean, eland’s wattle

-**ŘKFY**- ‘MIMOSOIDEAE’ VIII 1. (genus *Adenanthera*) red lucky seed / acacia coral / bead tree / Circassian seed, curly bean / jumbi-bead / red bead tree, peacock flower fence 2. (genus *Prosopis*) mesquite tree, algarrobo, alpataco, tamarugo, Argentine screwbean, American carob, tornillo 3. (genus *Desmanthus*) bundleflower, wild tantan, donkey bean

-**ŘKFL**- ‘MIMOSOIDEAE’ IX 1. (genus *Leucaena*) leadtree, leucania 2. (genus Schleinitzia) strand tangantangan 3. (genus *Dichrostachys*) sicklebush / Bell mimosa / Chinese lantern tree / Kalahari Christmas tree

-**ŘKFR**- ‘MIMOSOIDEAE’ X 1. (genus *Neptunia*) sensitive plant, yellow puff, water mimosa, neptunia 2. (Mimosoideae other than named genera; numerous genera) other mimosoid plant/tre 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Fabales > Fabaceae > Detarioideae

-**ŘTK**- ‘DETARIOIDEAE’ I 1. (genus *Detarium*) tallow tree, sweet detar / sweet dattock 2. (genus *Guibourtia*) Rhodesian teak, African rosewood, Rhodesian copalwood / false mopane, black hyedua, tiete rosewood / Patagonian cherry / sirari 3. (genus *Peltogyne*) purpleheart, violet wood

-**ŘTKW**- ‘DETARIOIDEAE’ II 1. (genus *Baikiaea*) African teak / Rhodesian teak / Zambian teak / Zambesi redwood 2. (genus *Colophospermum*) mopane / balsam tree / butterfly tree / turpentine tree 3. (genus *Daniellia*) West African copal tree / African copaiba balsam tree

-**ŘTKY**- ‘DETARIOIDEAE’ III 1. (genus *Copaifera*) diesel tree / kerosene tree, cabimo 2. (genus *Eperua*) wallaba tree 3. (genus *Hardwickia*) anjan tree

-**ŘTKL**- ‘DETARIOIDEAE’ IV 1. (genus *Saraca*) ashoka tree, saraca tree, sorrowless tree, asoca, ashok 2. (genus *Intsia*) Johnstone River teak / Pacific teak / scrub mahogany, Borneo teak / Malacca teak / Moluccan ironwood / merbau 3. (genus *Brachystegia*) miombo, mountain acacia, zebrawood

-**ŘTKR**- ‘DETARIOIDEAE’ V 1. (genus *Brownea*) mountain rose / scarlet flame bean / rose of Venezuela / copper hoop 2. (genus *Cynometra*) Uganda iron wood / muhimbi 3. (genus *Gilbertiodendron*) Liberian red oak

-**ŘTKŘ**- ‘DETARIOIDEAE’ VI 1. (genus *Tamarindus*) tamarind 2. (genus *Maniltoa*) handkerchief tree / pokok sapu tangan, cascading bean 3. (genus *Julbernardia*) congo zebrawood, mnondo, muchesa

-**ŘTKF**- ‘DETARIOIDEAE’ VII 1. (genus *Schotia*) weeping boerbean 2. (genus *Barnebydendron*) monkey-flower tree / fire of Pakistan 3. (Detarioideae other than named generap; numerous genera) other detarioid plant/tree

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Fabales > Fabaceae > remaining sub-families: Dialioideae and Duparquetia

-**ŘTKÇ**- ‘DIALIOIDEAE & DUPARQUETIOIDEAE’ 1. (genus *Dialium*) velvet tamarind, tamarind-plum 2. (Dialioideae other than Stem 1; numerous genera) other dialioid plant 3. (Duparquetioideae; genus *Duparquetia*) duparquetia

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Fabales > Polygalaceae

-**ŘTF**- ‘POLYGALACEAE’ I 1. (genus *Polygala*) milkwort, snakeroot, butterfly-bush / parrotbush / cascade curse, senega, senega-root 2. (genus *Comesperma*) pyramid flower, comesperma, love creeper, milkwort 3. (genus *Polygaloides*) gaywings / fringed polygala, shrubby milkwort

-**ŘTFW**- ‘POLYGALACEAE’ II 1. (genus *Habecarpa*) glandleaf milkwort, New Mexico milkwort, habecarpa 2. (genus *Rhinotropis*) desert polygala / thorn milkwort, California milkwort, Sierra milkwort, beaked spiny polygala / notch-beaked milkwort, intermountain milkwort, spiny milkwort 3. (Polygalaceae other than named genera) other polygalaceous plant, other milkwort

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Fabales > Surianaceae

-**ŘTFY**- 'SURIANACEAE' 1. (genus *Cadellia*) ooline 2. (genus *Suriana*) bay cedar 3. (Surianaceae other than Stems 1 and 2; genera *Recchia*, *Guilfoylia*, *Stylobasium*) other surianceous plant/tree

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Fabales > Quillajaceae

-**ŘTFL**- 'QUILLAJACEAE' 1. (*Quillaja saponaria*) soap bark tree 2. (*Quillaja brasiliensis*) Brazilian quillaja tree 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Celastrales

-**ÇFW**- 'CELASTRALES' I 1. (genus *Celastrus*) staff vine / staff tree, bittersweet, orange boxwood 2. (genus *Canotia*) crucifixion thorn, canotia 3. (genus *Cassine*) Cape saffron / bastard saffron / forest spoonwood, Bermuda olivewood

-**ÇFY**- 'CELASTRALES' II 1. (genus *Elaeodendron*) olive berry / red fruited olive plum / cassine 2. (genus *Catha*) khat / qat 3. (genus *Crossopetalum*) Christmas-berry, maiden berry

-**ÇFL**- 'CELASTRALES' III 1. (genus *Euonymus*) spindle, spindle tree, burning-bush, strawberry-bush, dingle-dangle tree, wahoo, wintercreeper, euonymus 2. (genus *Gymnosporia*) common spike-thorn 3. (genus *Hartogiella*) spoonwood

-**ÇFR**- 'CELASTRALES' IV 1. (genus *Lydenburgia*) bushman's tea 2. (genus *Maurocenia*) Khoi cherry / vulture-berry 3. (genus *Maytenus*) mayten, orangebark, koonkara, white cinnamon, orange bush, Indian's salt, maytenus, rock false candlewood, orange boxwood

-**ÇFŘ**- 'CELASTRALES' V 1. (genus *Mortonia*) saddlebush, sand paper bush, mortonia 2. (genus *Lepuropetalum*) petiteplant 3. (genus *Parnassia*) grass of Parnassus, bog-star

-**ÇFM**- 'CELASTRALES' VI 1. (genus *Paxistima*) Canby's mountain-lover / rat-stripper / cliff green, Oregon boxleaf / hedge / box / false box / myrtle box leaf 2. (genus *Pterocelastrus*) candlewood / cherrywood 3. (genus *Putterlickia*) spikethorn, false spikethorn, mock spike thorn, bastard spikethorn

-**ÇFN**- 'CELASTRALES' VII 1. (genus *Robsonodendron*) white silky bark 2. (genus *Salacia*) salacia, lolly berry 3.(genus *Schaefferia*) Florida boxwood, desert yaupon

-**ÇFŇ**- 'CELASTRALES' VIII 1. (genus *Stackhousia*) stackhousia, creamy candles, Gunn's mignonette / grasslands candles 2. (genus *Tripterygium*) threewingnut, thunder god vine 3. (Celestrales other than named genera; numerous genera) other celestrales plant/tree

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Malpighiales

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Malpighiales > Malpighiaceae

-**BZX**- 'MALPIGHIACEAE' I 1. (genus *Malpighia*) Singapore holly, Barbados cherry / acerola, bristly stingingbush, cowhage 2. (genus *Acridocarpus*) moth fruit 3. (genus *Banisteriopsis*) ayahuasca plant

-**BZXW**- 'MALPIGHIACEAE' II 1. (genus *Bunchosia*) peanut butter fruit, friar's plum, usama / bush tomato paste, yellow usama / yellow coffee of the bush 2. (genus *Brysonima*) locustberry, serret, nance / golden spoon, clam cherry / gooseberry 3. (genus *Galphimia*) gold shower / shower-of-gold / sslender goldshower / thryallis

-**BZXM**- 'MALPIGHIACEAE' III 1. (genus *Stigmaphyllon*) amazonvine 2. (Malpigiacieae other than named genera; numerous genera) other malpighiaceous plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Malpighiales > Elatinaceae & Centroplacaceae

-**BZXN**- 'ELATINACEAE & CENTROPLACACEAE' 1. (Elatinaceae; genus *Elatina*) waterwort 2. (Elatinaceae; genus *Bergia*) bergia 3. (Centroplacaceae; genera *Bhesa*, *Centroplacus*) centroplacaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Malpighiales > Euphorbioideae

-**LMÇ**- 'EUPHORBIACEAE' I 1. (genus *Euphorbia*) spurge, rattlesnake wed / white-margined sandmat, candelilla, sweet tabaiba, Hercules club, Medusa's head, copper tree, fire-on-the-mountain, African milk bush, desert poinsettia / fireplant / paint leaf, asthma-plant, African milk barrel, candelabra tree, frilled fan / elkhorn, gopher plant / mole plant, Madagascar jewel, snow on the mountain, crown-of-thorns / Christ plant, donkey tail, bonfire, Mexican flame leaf / Christmas star / winter rose, milk bush / pencil tree / firestick, devil's backbone / redbird cactus, African milk tree / cathedral cactus, poison tree, euphorbia 2. (genus *Homalanthus*) bleeding heart / native poplar / Queensland poplar 3. (genus *Ditrysinia*) Gulf Sebastian-bush

-**LMÇW**- 'EUPHORBIACEAE' II 1. (genus *Neoshirakia*) milktree, tallow tree 2. (genus *Triadica*) Chinese tallowtree / Florida aspen ; chicken tree / gray popcorn tree / candleberry tree 3. (genus *Sapium*) milktree / gumtree

-**LMÇL**- 'EUPHORBIACEAE' III 1. (genus *Sclerocroton*) duiker berry 2. (genus *Sebastiana*) "Mexican jumping bean" tree, sebastiana 3. (genus *Stillingia*) toothleaf, queen's-root, corkwood, queen's delight

-**LMÇR**- 'EUPHORBIACEAE' IV 1. (genus *Mercurialis*) mercury 2. (genus *Ricinus*) castor bean plant / castor oil plant 3. (genus *Argythamnia*) silverbush

-**LMÇF**- 'EUPHORBIACEAE' V 1. (genus *Chrosophora*) turnsole plant / dyer's croton / giradol 2. (genus *Omphalea*) Jamaican cobnut / popnut 3. (genus *Codiaeum*) fire croton / variegated croton

-**LMÇV**- 'EUPHORBIACEAE' VI 1. (genus *Croton*) croton, rushfoil, hogwort / goatweed, cascarilla, encinilla, grannybush, prairie-tea, dove weed / turkey mullein, forest fever-berry 2. (genus *Elateriospermum*) perah tree 3. (genus *Cnidoscolus*) tree spinach / chaya, mala mujer, bull nettle / spurge nettle / tread-softly / finger rot

-**LMÇM**- 'EUPHORBIACEAE' VII 1. (genus *Hevea*) rubber plant / rubber tree 2. (genus *Schinziophyton*) maongongo tree / mongongo nut tree / manketti tree 3. (genus *Ricinodendron*) njangsa / munguella / djansang

-**LMÇN**- 'EUPHORBIACEAE' VIII 1. (genus *Ricinocarpus*) wedding bush 2. (genus *Beyeria*) turpentine bush, wallaby bush, pinkwood 3. (Euphorbiaceae other than named genera; numerous genera) other euphorbiaceous plant/tree

-**RMÇ**- 'PERACEAE' 1. (genus *Clutia*) lightning bush 2. (genus *Chaetocarpus*) hedoke 3. (Peraceae other than Stems 1 and 2) other peraceous plant

-**RMÇW**- 'PICRODENDRACEAE' I 1. (genus *Androstachys*) lebombo ironwood 2. (genus *Austrobuxus*) pink cherry / hairybark 3. (genus *Micrantheum*) box micrantheum, western tridentbush

-**RMÇL**- 'PICRODENDRACEAE' II (genus *Oldfieldia*) African oak 2. (genus *Petalostigma*) quinine berry / quinine bush / quinine tree, long-leaved bitter bark 3. (genus *Piranhea*) three-leaf piranhea

-**RMÇR**- 'PICRODENDRACEAE' III 1. (genus *Picrodendron*) bitter plum, picrodendron 2. (genus *Tetracoccus*) scrubby-spurge, tetracoccus, hollybush 3. (Picrodenraceae other than named genera; numerous genera) other picrodendraceous plant

-**RMÇĞ**- 'PHYLLANTHACEAE' I 1. (genus *Phyllanthus*) gooseberry, leafflower, red root floater, scrubby spurge, gale of the wind / stonebreaker / seed-under-leaf, mousetail plant, brush sauropus, chamberbitter, sand riverchonia, gripeweed / shatterstone 2. (genus *Flueggea*) bushweed, mehamehame 3. (genus *Margaritaria*) pheasant-berry / egossa red pear / bushveld peacock-berry, bastard hogberry

-**RMÇF**- 'PHYLLANTHACEAE' II 1. (genus *Baccaurea*) Burmese grape, rambi / rambai, tampoi 2. (genus *Bridelia*) coastal golden-leaf / mitzeeri, brush ironbark / scrub ironbark 3. (genus *Pseudolachnostylis*) kudu berry

-**RMÇV**- 'PHYLLANTHACEAE' III 1. (genus *Leptopus*) maidenbush 2. (genus *Poranthera*) poranthera 3. (Phyllanthaceae other than named genera; numerous genera) other phyllantaceous plant

-**RMÇM**- 'LINACEAE' 1. (genus *Linum*) flax 2. (genus *Hesperolinon*) dwarf-flax / western flax 3. (genus *Reinwardtia*) yellow flax / pyoli

-**RMÇN**- 'RAFFLESIACEAE' 1. (genus *Rafflesia*) corpse lilly, rafflesia 2. (genus *Rhizanthes*) rhizanthes 3. (genus *Sapria*) sapria

-**RMÇŇ**- ‘OTHER LINACEAE & IXONANTHACEAE’ 1. (genus *Scerolinon*) northwestern yellowflax 2. (Linaceae other than named genera; several genera) other linaceous plant 3. (Ixonanthaceae; four genera) ixonanthaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Malpighiales > Salicaceae (including Samydaceae)

-**BZF**- ‘GENUS *POPULUS*’ 1. poplar tree 2. cottonwood tree 3. aspen tree

-**BZFW**- ‘SALICACEAE’ I 1. (genus *Salix*) willow tree, osier, sallow 2. (genus *Xylosma*) brushholly, logwood, xylosma 3. (genus *Oncoba*) snuff-box tree / fried egg tree / fried-egg flower

-**BZFY**- ‘SALICACEAE’ II 1. (genus *Azara*) azara 2. (genus *Dovyalis*) kei apple / kai apple / kau apple 3. (genus *Flacourtia*) Batoko plum, Indian coffee plum, rukam

-**BZFL**- ‘SALICACEAE’ III 1. (genus *Scolopia*) flintwood / mountain cherry / brown birch / scolopia, mountain saffron / red pear 2. (genus *Casearia*) sword-leaf 3. (Salicaceae other than named genera; numerous genera) other salicaceous plant/tree

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Malpighiales > Passifloraceae (including Turneraceae and Malesherbiaceae)

-**BZFR**- ‘PASSIFLORACEAE’ I 1. (genus *Passiflora*) passion flower, passion vine, grenadilla, passionfruit, maypop, water lemon / Jamaican honeysuckle, sweet calabash, bell apple, tumbo, goatsfoot 2. (genus *Adenia*) monkey rope / snake climber / wild granadilla, hondala, adenia 3. (genus *Turnera*) damiana, white buttercup sulphur alder / politician’s flower / dark-eyed turnera / white alder, turnera, ramgoat dashalong / yellow alder

-**BZFŘ**- ‘PASSIFLORACEAE’ II 1. (genus *Barteria*) ant tree 2. (Passifloraceae other than named genera; numerous genera) other passifloraceous plant/tree 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Malpighiales > Violaceae

-**BZFM**- ‘VIOLACEAE’ 1. (genus *Viola*) violet, pansy, heartsease 2. (genus *Melicytus*, a.k.a. *Hymenanthera*) porcupine shrub, mahoe, whitey-wood, tree violet 3. (Violaceae other than Stems 1 and 2; numerous genera) other violaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Malpighiales > Achariaceae and Humiriaceae

-**BZFN**- ‘ACHARIACEAE’ 1. (genus *Hydnocarpus*) chaulmoogra 2. (genus *Pangium*) keluak / keluwak / kepayang / payang 3. (genus *Kiggelaria*) wild peach

-**BZFŇ**- ‘OTHER ACHARIACEAE & HUMIRIACEAE’ 1. (genus *Capotroche*) pau de Lepra 2. (other Achariaceae; numerous genera) other achariaceous plant/tree 3. (Humiriaceae; several genera) humiriaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Malpighiales > Clusiaceae

-**GZFW**- ‘CLUSIACEAE’ I 1. (genus *Clusia*) autograph tree / copey / cupey / balsam apple / pitch-apple / Scotch attorney 2. (genus *Allanblackia*) tallow tree 3. (genus *Garcinia*) saptree, mangosteen, gambooge / brindleberry / brindall berry / Malabar tamarind, garcinia

-**GZFY**- ‘CLUSIACEAE’ II 1. (genus *Platonia*) bacuri / pakuri / maniballi / naranjillo 2. (genus *Symphonia*) chewstick / chestick / manni 3. (Clusiaceae other than named genera; several genera) other clusiaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Malpighiales > Calcophyllaceae

-**GZFL**- ‘CALOPHYLLACEAE’ I 1. (genus *Calophyllum*) mastwood / Alexandrian laurel / beauty leaf, Nicobar canoetree, pink touriga, poonspar / sirpoontree 2. (genus *Mammea*) mammee apple / mamey apple / Santo Domingo apricot / tropical apricot / South American apricot 3. (genus *Mesua*) Ceylon ironwood / Indian rose chestnut / cobra saffron

-**GZFR**- ‘CALOPHYLLACEAE’ II 1. (genus *Kielmeyera*) malva-do-campo, pau santo 2. (Calcophyllaceae other than named genera; several genera) other calcophyllaceous plant/tree 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Malpighiales > Hypericaceae & Podostemaceae

-**GZFŘ**- ‘HYPERICACEAE’ I 1. (genus *Hypericum*) St. John’s wort, goatweed, goldwire, sweet-amber, tutsan, pineweed / orangegrass, curry bush, sangrenaria, sanguinaria 2. (genus *Triadenum*) marsh St. John’s wort / bog St. John’s wort 3. (genus *Harungana*) dragon’s blood tree / orange-milk tree / haronga

-**GZFM**- ‘HYPERICACEAE’ II & PODOSTEMACEAE 1. (Hypericaceae other than genera named in Hypericaceae I; several genera) other hypericaceous plant 2. (Podostemaceae; numerous genera) riverweed, podostemaceous plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Malpighiales > Ochnaceae (including Quiinaceae & Medusagynaceae)

-**GZFN**- ‘OCHNACEAE (including QUIINACEAE & MEDUSAGYNACEAE)’ I 1. (genus *Ochna*) ochna, plane, yellow mai flower, Mickey Mouse bush, bird’s-eye bush 2. (genus *Lophira*) red ironwood tree 3. (genus *Brackenridgea*) yellow peeling plane

-**GZFŇ**- OCHNACEAE (including QUIINACEAE & MEDUSAGYNACEAE)’ II 1. (genus *Medusagyne*) jellyfish tree 2. (Ochnaceae other than named genera; several genera) other ochnaceous plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Malpighiales > Irvingiaceae and Pandaceae

-**MZVW**- ‘IRVINGIACEAE & PANDACEAE’ 1. (genus *Irvingia*) wild mango / African mango / bush mango, wild almond / barking deer’s mango 2. (Irvingiaceae other than Stem 1; genera *Allantospermum*, *Klainedoxa*) other irvingiaceous plant 3. (Pandaceae; genera *Panda*, *Microdesmis*, *Galearia*) pandaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Malpighiales > Erythroxylaceae, Ctenolophonaceae, and Rhizophoraceae

-**MZVY**- ‘ERYTHROXYLACEAE & CTENOLOPHONACEAE’ 1. (genus *Erythroxylum*) coca plant, catuaba plant, brown plum, kerosene wood / turpentine tree 2. (Erythroxylaceae other than Stem 1; other erythroxylaceous plant 3. (Ctenolophonaceae; genus Ctenolophon) ctenolophon)

-**MZVL**- ‘RHIZOPHORACEAE’ I 1. (genus *Rhizophora*) true mangrove, red mangrove 2. (genus *Ceriops*) yellow mangrove, spurred mangrove / Indian mangrove 3. (genus *Bruguiera*) orange mangrove, oriental mangrove / black mangrove / mangrove bean

-**MZVR**- ‘RHIZOPHORACEAE’ II 1. (genus *Carallia*) corkwood / butterfly plant 2. (Rhizophoraceae other than named genera; numerous genera) other rhizophoraceous plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Malpighiales > Dichapetalaceae

-**MZVŘ**- ‘DICHAPETALACEAE’ 1. (genus *Dichapetalum*) poison-leaf 2. (genus *Tapura*) tapura 3. (genus *Stephanopodium*) stephanopodium

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Malpighiales > Chrysobalanaceae

-**MŽVW**- 'CHRYSOBALANACEAE' I 1. (genus *Chrysobalanus*) coco plum / paradise plum 2. (genus *Couepia*) sweet angelim, pajurá 3. (genus *Licania*) gopher apple / ground oak, merecure, sansapote

-**MŽVY**- 'CHRYSOBALANACEAE' II 1. (genus *Parinari*) Annamese burada, mobola plum, nonda plum, guinea plum 2. (genus Atuna) tabon-tabon 3. (Chrysobalanaceae other than named genera) other chrysobalanaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Malpighiales > Caryocaraceae, Putranjivaceae, Lophopyxidaceae, Balanopaceae

-**MŽVL**- 'CARYOCARACEAE & LOPHOPYXIDACEAE' 1. (genus *Caryocar*) souari tree, butter-nut of Guiana / pekea-nut tree 2. (genus *Anthodiscus*) anthodiscus 3. (Lophopyxidaceae; genus *Lophopyxis*) lophopyxis

-**MŽVR**- 'PUTRANJIVACEAE & BALANOPACEAE' 1. (Putranjivaceae: genus *Drypetes*) ironplum, forest ironwood/ bastard white ironwood, yellow tulipwood / grey boxwood / white myrtle / grey bark 2. (Putranjivaceae: genus *Putranjiva*) putranjiva 3. (Balanopaceae; genus *Balanops*) balanops

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Malpighiales > remaining families

-**MŽVŘ**- 'OTHER MALPIGHIALES' 1. (Scyphostegiaceae, Lacistemataceae, Goupiaceae; several genera) scyphostegiaceous, lacistemataceous, or goupiaceous plant 2. (Bonnetiaceae; four genera) bonnetiaceous plant 3. (Trigoniaceae, Euphroniaceae; several genera) trigoniaceous or euphroniaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Oxilidales

-**LMSKW**- 'OXALIDACEAE' 1. (genus *Oxalis*) wood sorrel, yellow sorrel, pink sorrel, false shamrock, sourgrass, stabwort, Cape shamrock, lavender sorrel, alpine sorrel, sleeping beauty, finger-leaf, Mexican shamrock, redwood sorrel / Oregon sorrel, love plant, spiral sorrel, coamo, oca/ oka / New Zealand yam, volcanic sorrel, oxalis 2. (genus *Averrhoa*) starfruit tree / carambola / five-corner tree, bilimbi 3. (Oxalidaceae other than Stems 1 and 2; several genera) other oxalidaceous plant/tree

-**LMSKY**- 'CONNARACEAE & HUACEAE' 1. (genus *Connarus*) Indian zebrawood 2. (Connaraceae other than Stem 1; numerous genera) other connaraceous plant 3. (Huaceae; genera *Hua*, *Afrostyrax*) huaceous plant

-**LMSKL**- 'CUNONIACEAE' I 1. (genus *Cunonia*) butterspoon tree / butterknife tree / African red alder, cunonia 2. (genus *Vesselowskya*) marara 3. (genus *Weinmannia*) tawhero / towai, kamahi, encenillo, tineo

-**LMSKR**- 'CUNONIACEAE' II 1. (genus *Anodopetalum*) horizontal scrub 2. (genus *Ceratopetalum*) coachwood / scented satinwood / tarwood, New South Wales Christmas bush 3. (genus *Schizomeria*) Australian white birch / crab apple / white cherry / snowberry / humbug / squeaker

-**LMSKŘ**- 'CUNONIACEAE' III 1. (genus *Karrabina*) red carabeen / leather jacket /brush mahogany / red bean / pink marara / bush marara 2. (genus *Pseudoweinmannia*) rose marara / mararie / scrub rosewood / red carabeen 3. (genus *Caldcluvia*) soft corkwood / rose-leaf marara / brown alder / sugarbark, tiaca

-**LMSKF**- 'CUNONIACEAE' IV 1. (genus *Callicoma*) black wattle / butterwood / silver leaf / wild quince / callicoma 2. (genus *Pullea*) hard alder 3. (genus Bauera) bauera, dog rose / river rose

-**LMSKÇ**- 'CUNONIACEAE' V 1. (genus *Davidsonia*) Davidson plum, mullumbimby plum, ooray 2. (genus *Eucryphia*) leatherwood, pinkwood / plumwood 3. (Cunoniaceae other than named genera; several genera) other cunoniaceous plant

-**LMSKŢ**- 'ELAEOCARPACEAE' I 1. (genus *Elaeocarpus*) quandong, whitewood / pigeonberry ash, grey carabeen / freckled oliveberry, blueberry ash, yoga tree, blue marble tree / blue fig, kalia, rudraksha tree 2. (genus *Aceratium*) carabeen 3. (genus *Aristotelia*) mountain wineberry, Chilean wineberry / maqui, wineberry

-**LMSKS**- 'ELAEOCARPACEAE' II 1. (genus *Peripentadenia*) grey quandong / buff quandong 2. (genus Sloanea) maiden's blush, bullwood / montillo, acomat boucan, yellow carabeen 3. (genus Tetratheca) pink-bells, black-eyed susan, shy susan

-**LMSKŠ**- ‘ELAEOCARPACEAE’ III 1. (genus *Tremandra*) tremandra 2. (Elaeocarpaceae other than named genera; several genera) other elaeocarpaceous plant 3. [stem unassigned]

-**LMSKH**- ‘BRUNELLIACEAE & CEPHALOTACEAE’ 1. (genus *Brunellia*) West Indian sumac 2. (genus *Cephalotus*) Albany pitcher plant / Australian pitcher plant / fly-catcher plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Rosales

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Rosales > Rosaceae

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Rosales > Rosaceae > Rosoideae

-RŢM- ‘ROSOIDEAE’ I 1. (genus *Rosa*) rose 2. (genus *Agrimonia*) agrimony 3. (genus *Aremonia*) bastard-agrimony

-**RŢMW**- ‘ROSOIDEAE’ II 1. (genus *Hagenia*) African redwood / East African rosewood, hagenia 2. (genus *Leucosidea*) oldwood 3. (genus *Filipendula*) meadowsweet, mead wort, dropwort, queen-of-the-prairie

-**RŢMY**- ‘ROSOIDEAE’ III 1. (genus *Acaena*) bidibid, biddy-biddy, pirri-pirri-bur, sheep’s burr, New Zealand-bur, burnet, liliwai 2. (genus *Cliffortia*) caperose 3. (genus *Margyricarpus*) pearl-fruit

-**RŢML**- ‘ROSOIDEAE’ IV 1. (genus *Polylepis*) queñoa 2. (genus *Sanguisorba*) burnet, burnet bloodwort 3. (genus *Sarcopoterium*) prickly burnet / spiny burnet / thorny burnet

-**RŢMR**- ‘ROSOIDEAE’ V 1. (genus *Fallugia*) Apache plume / ponil 2. (genus *Geum*) avens, herb bennet, prairie smoke, Tasmanian snowrose 3. (genus *Waldsteinia*) barren strawberry

-**RŢMŘ**- ‘ROSOIDEAE’ VI 1. (genus *Alchemilla*) lady’s mantle 2. (genus *Aphanes*) parsley-piert, parsley breakstone, Australian piert, dew cup / lady’s mantle 3. (genus *Chamaerhodos*) little-rose

-**RŢNW**- ‘ROSOIDEAE’ VII 1. (genus *Comarum*) purple marshlocks / swamp cinquefoil / marsh cinquefoil 2. (genus *Dasiphora*) woody cinquefoil 3. (genus *Drymocallis*) sticky cinquefoil, tall cinquefoil / cream cinquefoil, rock cinquefoil

-**RŢNY**- ‘ROSOIDEAE’ VIII 1. (genus *Argentina*) silverweed 2. (genus *Horkelia*) horkelia 3. (genus *Ivesia*) mousetail, rock whitefeather, ivesia

-**RŢŇ**- ‘ROSOIDEAE’ IX 1. (genus *Sibbaldia*) sibbaldia, creeping sibbaldia 2. (genus *Potentilla*) cinquefoil, barren strawberry, tormentil, fan-foil, old-field five-fingers 3. (genus *Sibbaldiopsis*) three-toothed cinquefoil / shrubby fivefingers, wineleaf

-**RŢŇW**- ‘ROSOIDEAE’ X 1. (genus *Duchesnea*) mock strawberry / Indian-strawberry / false strawberry 2. (Rosoideae other than named genera or *Fragaria* or *Rubus* below) other rosoid plant 3. [stem unassigned]

-ŘŽ- ‘STRAWBERRY (genus *Fragaria*)’ 1. (wild) strawberry plant 2. strawberry cultivar 3. strawberry hybrid

-ŘT- ‘BERRY (genus *Rubus*)’ I 1. raspberry plant 2. blackberry plant 3. dewberry plant

-ŘTW- ‘BERRY (genus *Rubus*)’ II 1. cloudberry plant 2. salmonberry plant 3. thimbleberry plant

-ŘTY- ‘BERRY (genus *Rubus* [HYBRID])’ I 1. loganberry plant 2. boysenberry plant 3. tayberry plant

-ŘTL- ‘BERRY (genus *Rubus* [HYBRID])’ II 1. marionberry plant 2. youngberry plant 3. olallieberry plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Rosales > Rosaceae > Amygdaloideae

-NŽ- ‘MALEAE’ I 1. (*Malus domestica*) apple 2. (genus *Malus* except *M. domestica*) crabapple 3. (genus *Pyrus*) pear

-NŽW- ‘MALEAE’ II 1. (genus *Cydonia*) quince 2. (genus *Pseudocydonia*) Chinese quince 3. (genus *Chaenomeles*) Japanese quince, Chinese flowering quince, Tibetan quince

-**NŽY**- ‘MALEAE’ III 1. (genus *Sorbus*) rowan / mountain-ash 2. (genus *Chamaemespilus*) false medlar / dwarf whitebeam 3. (genus *Amelanchier*) serviceberry, juneberry, shadbush, snowy mespilus, saskatoon berry, shadbush, shadwood, shadblow, sarvisberry, sugarplum / wild-plum, chuckley pear

-**NŽL**- ‘MALEAE’ IV 1. (genus *Aria* [including *Micromeles*]) whitebeam 2. (genus *Cormus*) [true] service-tree / sorb tree 3. (genus *Torminalis*) wild service-tree, chequers / checker tree

-**NŽR**- ‘MALEAE’ V 1. (genus *Crataegus*) hawthorn, quickthorn, thornapple, May-tree, whitethorn, hawberry 2. (genus *Rhaphiolepis*) India hawthorn / Hong Kong hawthorn 3. (genus *Eriobotrya*) loquat

-**NŽŘ**- ‘MALEAE’ VI 1. (genus *Mespilus*) medlar 2. (genus *Malacomeles*) false serviceberry 3. (genus *Peraphyllum*) squaw apple / wild crab apple

-**NŽV**- ‘MALEAE’ VII 1. (genus *Aronia*) chokeberry 2. (genus *Heteromeles*) toyon / Christmas berry / California holly 3. (genus *Cotoneaster*) cotoneaster

-NŽT- ‘GENUS PRUNUS’ I (subgenus *P. Amygdalus*) 1. almond tree 2. peach tree 3. peach tree bearing nectarine fruit / “nectarine tree”

-NŽTW- ‘GENUS PRUNUS’ II (subgenus *P. Prunus*) 1. plum tree 2. apricot tree 3. plum-apricot hybrid (pluot, plumcot, apriplum)

-NŽTY- ‘GENUS PRUNUS’ III 1. (subgenus *P. Cerasus*) cherry tree 2. (subgenus *P. Lithocerasus*) sand cherry shrub 3. (subgenera *P. Padus* and *P. Laurocerasus*) bird cherry, cherry-laurel, bitter-berry, chokeberry

-**NŽTL**- ‘AMYGDALOIDEAE’ I 1. (genus *Kerria*) Japanese marigold bush / miracle marigold bush / kerria 2. (genus *Rhodotypos*) jetbead, rhodotypos 3. (genus *Coleogyne*) blackbrush

-**NŽTR**- ‘AMYGDALOIDEAE’ II 1. (genus *Neviusia*) snow-wreath 2. (genus *Osteomeles*) Hawaiian rose / Hawaiian hawthorn 3. (genus *Pyracantha*) firethorn, pyracantha

-**NŽTŘ**- ‘AMYGDALOIDEAE’ III 1. (genus *Photinia*) photinia 2. (genus *Stranvaesia*) Chinese photinia 3. (genus *Vauquelinia*) Arizona rosewood, slimleaf rosewood

-**NŽTÇ**- ‘AMYGDALOIDEAE’ IV 1. (genus *Neillia*) lace shrub 2. (genus *Physocarpus*) ninebark 3. (genus *Gillenia*) Indian physic, Bowman’s root

-**NŽTḶ**- ‘AMYGDALOIDEAE’ V 1. (genus *Lyonothamnus*) Catalina ironwood 2. (genus *Exochorda*) pearlbush 3. (genus *Oemleria*) osoberry / Indian plum

-**NŽD**- ‘AMYGDALOIDEAE’ VI 1. (genus *Sorbaria*) false spiraea / false goat’s beard 2. genus *Chamaebatiaria*) fern bush / desert sweet 3. (genus *Adenostoma*) chamise, redshanks / ribbonwood / ribbon bush

-**NŽDW**- ‘AMYGDALOIDEAE’ VII 1. (genus *Spiraea*) spiraea, bridewort, meadowsweet, hardhack / steeplebush 2. (genus *Petrophytum*) rockmat, mat rock spiraea 3. (genus *Luetkea*) partridgefoot / luetkea

-**NŽDY**- ‘AMYGDALOIDEAE’ VIII 1. (genus *Holodiscus*) ocean spray / creambush, mountain spray / rock-spiraea 2. (genus *Aruncus*) goatsbeard 3. (Amygdaloideae other than named genera or genus *Prunus* below; numerous genera) other amygdaloid plant/tree/shrub

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Rosales > Rosaceae > Dryadoideae

-**NŽMW**- ‘DRYADOIDEAE’ 1. (genus *Dryas*) avens 2. (genus *Chamaebatia*) mountain misery, bearclover / kit-kit-dizze 3. (genus *Cercocarpus*) mountain mahogany

-**NŽMY**- ‘DRYADOIDEAE’ 1. (genus *Purshia*) cliffrose, bitterbrush 2. (genus *Cowania*) evergreen cliffrose 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Rosales > Moraceae

-ŘÇ- ‘MORACEAE’ I 1. (genus *Ficus*) fig tree 2. (genus *Morus*) mulberry tree 3. (genus *Treculia*) African breadfruit tree

-ŘÇW- ‘MORACEAE’ II – (genus *Artocarpus*) 1. (*A. altilis*) breadfruit 2. (*A. heterophyllus*) jackfruit 3. (*A. lacucha*) monkey fruit / lakoocha

-ŘÇL- ‘MORACEAE’ III 1. (genus *Milicia*) iroko / African teak 2. (genus *Bagassa*) cow wood 3. (genus *Streblus*) milk tree, Siamese rough bush

-ŘÇR- ‘MORACEAE’ IV 1. (genus *Antiaris*) bark cloth tree, antiaris, false iroko, upas tree 2. (genus *Trilepisium*) urnfig / false-fig 3. (genus *Maclura*, a.k.a. *Cudrania*) Osage orange, old fustic / dyer’s mulberry, cudrang / mandarin melon berry / silkworm thorn / Chinese mulberry, cockspur thorn

-ŘÇŘ- ‘MORACEAE’ V 1. (genus *Brosimum*) snakewood, breadnut / Maya nut, satine bloodwood, mama-cadela 2. (genus *Broussonetia*) paper mulberry 3. (genus *Fatoua*) mulberry weed / crabweed

-ŘÇÇ- ‘MORACEAE’ VI 1. (genus *Dorstenia*) dorstenia 2. (Moraceae other than named genera; numerous genera) other moraceous plant/tree

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Rosales > Urticaceae

-**NŽK**- ‘URTICACEAE’ I 1. (genus *Urtica*) nettle, stinging nettle, creeping nettle, swamp nettle 2. (genus *Dendrocnide*) nettle tree, stinging tree, bympie-bympie, lip tree 3. (genus *Hesperocnide*) Hawaii nettle, western nettle

-**NŽKW**- ‘URTICACEAE’ II 1. (genus *Laportea*) woodnettle 2. (genus *Urera*) flameberry, hopue, scratchbush, climbing-nettle 3. (genus *Elatostema*) New Zealand begonia / parataniwha

-**NŽKY**- ‘URTICACEAE’ III 1. (genus *Soleirolia*) baby’s tears / angel’s tears / bits and pieces / bread and cheese / Corsican creeper / Corsican curse / friendship plant ; mind-your-own-business / mother of thousands / Paddy’s wig / Pollyanna vine 2. (genus *Parietaria*) pellitory, lichwort, asthma 3. (genus *Pilea*) silver springkles , friendship plant, artillery plant / gunpowder plant, Moon Valley plant / creeping Charlie , chinese money plant / missionary plant, Canadian clearweed, black-leaf panamiga

-**NŽKL**- ‘URTICACEAE’ IV 1. (genus *Pourouma*) Amazon grape / Amazon tree-grape / uvilla 2. (genus *Myrianthus*) giant yellow mulberry / monkey fruit 3. (genus *Cecropia*) pumpwood, cecropia, trumpet tree, guarumo

-**NŽKR**- ‘URTICACEAE’ V 1. (genus *Pipturis*) mamaki, Waimea pipturus 2. (genus *Neraudia*) ma’oloa 3. (Urticaceae other than named genera) other urticaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Rosales > Cannabaceae

-**NŽKŘ**- ‘CANNABACEAE’ I 1. (genus *Cannabis*) cannabis, hemp, marijuana plant 2. (genus *Humulus*) hop plant 3. (genus *Celtis*) hackberry, nettle tree, lote tree, white stinkwood, cottonwood, tala, almex

-**NŽKF**- ‘CANNABACEAE’ II 1. (genus *Pteroceltis*) blue sandalwood / Tara wingceltis 2. (genus *Chaetachme*) thorny elm 3. Cannabaceae other than named genera; several genera) other cannabaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Rosales > Ulmaceae

-**NŽKÇ**- ‘ULMACEAE’ I 1. (genus *Ulmus*) elm 2. (genus *Zelkova*) zelkova 3. (genus *Hemiptelea*) thorn-elm

-**NŽKŢ**- ‘ULMACEAE’ II 1. (genus *Holoptelea*) Indian elm / jungle cork tree 2. (genus *Planera*) planertree / water elm 3. (Ulmaceae other than named genera; genera *Ampelocera* and *Phyllostylon*) other ulmaceous tree

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Rosales > Rhamnaceae

-**NŽP**- ‘RHAMNACEAE’ I 1. (genus *Rhamnus*) buckthorn, redberry, Avignon berry, staddo 2. (genus *Frangula*) buckthorn, coffeeberry, Indian cherry, black dogwood 3. (genus *Berchemia*) bird plum, supplejack, rattan vine, pink ivory

-**NŽPW**- ‘RHAMNACEAE’ II 1. (genus *Condalia*) bluewood, snakewood, purple haw, logwood, yana yana, condalia 2. (genus Karwinskia) coyotillo / cacachila 3. (genus Krugiodendron) black ironwood / leadwood

-**NŽPY**- ‘RHAMNACEAE’ III 1. (genus *Reynosia*) darlingplum 2. (genus *Sageretia*) mock buckthorn, sageretia 3. (genus *Scutia*) cat-thorn

-**NŽPL**- ‘RHAMNACEAE’ IV 1. (genus *Ventilago*) vine tree, whip vine, supplejack 2. (genus *Maesopsis*) umbrella tree 3. (genus Gouania) chewstick, toothbrush tree

-**NŽPR**- ‘RHAMNACEAE’ V 1. (genus *Adolphia*) California prickbush / California pineshrub, junco 2. (genus *Colletia*) crucifixion thorn, thorn of the creoss / anchor plant 3. (genus *Discaria*) hairy anchor plant / Australian anchor plant, matagouri

-**NŽPŘ**- ‘RHAMNACEAE’ VI 1. (genus *Hovenia*) Japanese raisin tree / oriental raisin tree 2. (genus *Pailurus*) Jerusalem thorn / garland thorn / Christ’s thorn / crown of thorns 3. (genus *Ziziphus*) jujube, buffalo thorn, lotebush, hinap, red date / Chinese date, Chinese apple / Indian plum / dunks, mistol, graythorn / gumdrop tree / Texas buckthorn, zunna berry, ziziphus

-**NŽPF**- ‘RHAMNACEAE’ VII 1. (genus *Noltea*) soapbush / soap dogwood 2. (genus *Phylica*) Saint Helena rosemary 3. (genus *Cryptandra*) cryptandra

-**NŽPÇ**- ‘RHAMNACEAE’ VIII 1. (genus *Pomaderris*) pomaderris, kumarahou / gum-digger’s soap, dogwood 2. (genus *Spyridium*) basket bush, dusty miller, spyridium 3. (genus Trymalium) karri hazel

-**NŽPŢ**- ‘RHAMNACEAE’ IX 1. (genus *Alphitonia*) soap bush / red ash, pink ash / white ash, kauila, sarsaparilla tree 2. (genus *Ceanothus*) ceanothus, Jersey tea, New Jersey tea / red root, buckbrush, pinemat, blueblossom, chaparral whitethorn, Barranca brush, coast whitethorn, California lilac 3. (genus *Colubrina*) nakedwood, snakewood, greenheart, Texan hogplum, mabi / soldierwood, Pondo weeping thorn

-**NŽPĻ**- ‘RHAMNACEAE’ IX 1. (genus *Lasiodiscus*) red-hair bush 2. (genus Emmenosperma) yellow ash / bonewood 3. (Rhamnaceae other than named genera; numerous genera) other rhamnaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Rosales > Elaeagnaceae

-**NŽPS**- ‘ELAEAGNACEAE’ 1. (genus *Elaeagnus*) silverberry, oleaster, wolf-willow, Russian olive, lingaro berry, silverthorn, autumn olive 2. (genus *Shepherdia*) buffaloberry / bullberry 3. (genus *Hippophae*) sea-buckthorn / seaberry / sallowthorn / sandthorn

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Rosales > remaining families: Barbeyaceae & Dirachmaceae

-**NŽPŠ**- ‘BARBEYACEAE & DIRACHMACEAE’ 1. (Barbeyaceae; genus *Barbeya*) barbeya 2. (Direchmaceae; genus *Dirachma*) dirachma 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Fagales

-**ŘNTW**- ‘FAGACEAE’ I 1. (genus *Fagus*) beech tree 2. (genus *Quercus*) oak tree 3. (genus *Castanea*) chestnut tree

-**ŘNTY**- ‘FAGACEAE’ II 1. (genus *Castanopsis*) chinquapin / chinkapin 2. (genus *Chrysolepis*) golden chinquapin 3. (genus *Trigonobalanus*) trigaonobalanus

-**ŘNTL**- ‘FAGACEAE’ III 1. (genus *Lithocarpus*) stone oak 2. (genus *Notholithocarpus*) tanoak / tanbark oak 3. [stem unassigned]

-**ŘNTR**- ‘NOTHOFAGACEAE’ 1. (genus *Nothofagus*, subgenera *Nothofagus* and *Brassospora*) Antarctic beech, Magellan’s beech, Dombey’s beech, lenga beech, New Guinea beech, New Caledonian beech 2. (genus *Nothofagus*, subgenus *Fuscospora*) ruil, mountain beech, red beech, tanglefoot-beech, black beech, hard beech 3. (genus *Nothofagus*, subgenus *Lophozonia*) rauli, myrtle beech, hualo, Santiago’s oak, silver beech, Antarctic beech, Patagonian oak / roble beech

-**ŘNTŘ**- ‘MYRICACEAE’ 1. (genus *Myrica* [including *Morella*]) bayberry / bay-rum tree, wax-myrtle, sweet gale / bog-myrtle, yumberry, candleberry 2. (genus *Comptonia*) sweet-fern 3. (genus *Canacomyrica*) canacomyrica

-**ŘNTF**- ‘JUGLANDACEAE’ I 1. (genus *Juglans*) walnut tree, butternut tree 2. (genus *Pterocarya*) wingnut tree 3. (genus *Cyclocarya*) wheel wingnut tree

-**ŘNTÇ**- ‘JUGLANDACEAE’ II 1.(genus *Carya* other than section *Apocarya*) hickory tree, pignut, kingnut, shellbark 2. (genus *Carya*, section *Apocarya*) pecan tree, bitternut hickory, bitter pecan / water hickory, Mexican hickory 3. (genus *Annamocarya*) Chinese hickory / beaked hickory

-**ŘNTḶ**- ‘JUGLANDACEAE’ III 1. (genus *Oreomunnea*) gavilán, guayaba amarillo 2. (genus *Rhoiptelea*) horsetail tree 3. (Juglandaceae other than named genera; several genera) other juglandaceous tree

-**ŘNTM**- ‘CASUARINACEAE’ 1. (genus *Causarina*) she-oak, swamp oak, grey oak / river oak, common ironwood / Australian pine tree / whistling pine tree, belah 2. (genus *Allocasuarina*) she-oak, bull-oak, tamma, desert oak, karri oak, hairy oak 3. (Casuarinaceae other than Stems 1 and 2: genera *Gymnostoma, Ceuthostoma*) other casuarinaceous tree

-**ŘNTN**- ‘BETULACEAE’ I 1. (genus *Betula*) birch tree 2. (genus *Alnus*) alder tree 3. (genus *Corylus*) hazel tree, filbert tree

-**ŘNTŇ**- ‘BETULACEAE’ II 1. (genus *Carpinus*) hornbeam tree 2. (genus *Ostrya*) hop-hornbeam, ironwood 3. (genus *Ostryopsis* and Ticodendraceae; genus *Ticodendron*) hazel-hornbeam, ticodendron

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Cucurbitales

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Cucurbitales > Cucurbitaceae

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Cucurbitales > Cucurbitaceae > Benincaseae

-BḌ- ‘GENUS CUCUMIS’ I – MUSKMELON & ASSOCIATED CULTIVARS 1. European cantaloupe plant 2. North American cantaloupe plant 3. muskmelon plant

-BḌW- ‘GENUS CUCUMIS’ II – MUSKMELON & ASSOCIATED CULTIVARS 1. honeydew melon plant 2. Crenshaw melon plant 3. casaba melon plant

-BḌY- ‘GENUS CUCUMIS’ III 1. cucumber vine 2. cackrey / West Indian gherkin vine 3. aardvark cucumber / aardvark pumpkin

-BḌN- ‘BENINCASEAE’ I 1. (genus *Citrullus*) watermelon 2. (genus *Benincasa*) winter melon / wax gourd / ash gourd / white gourd / tallow gourd / Chinese preserving melon 3. (genus *Praecitrullus*) tinda / Indian squash / round melon

-BḌM- ‘BENINCASEAE’ II 1. (genus *Acanthosicyos*) nara melon, gemsbok cucumber 2. (genus *Coccinia*) scarlet gourd, ivy gourd / tindora / kowai fruit 3. (genus *Ctenolepis*) Garcin’s bur cucumber

-BḌŇ- ‘BENINCASEAE’ III 1. (genus *Diplocyclos*) native bryony / striped cucumber 2. (genus *Lagenaria*) calabash / bottle gourd / long melon / New Guinea bean / Tasmania bean / white-flowered gourd 3. (genus *Zehneria*) native cucumber / giant cucumber

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Cucurbitales > Cucurbitaceae > Cucurbiteae

-**GḌM**- ‘CUCURBITA (SQUASH) – Species *C. Pepo*’ I 1. zucchini squash vine 2. scallop summer squash 3. crookneck squash vine

-**GḌN**- ‘CUCURBITA (SQUASH) – Species *C. Pepo*’ II 1. pumpkin vine 2. acorn squash vine 3. spaghetti squash vine

-**GḌŇ**- ‘CUCURBITEAE’ 1. (genus *Abobra*) cranberry gourd 2. (genus *Cayaponia*) melonleaf, São Caetano melon, tayuya 3. (genus *Sicana*) cassabanana / casbanan / sikana /puttigel / musk cucumber

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Cucurbitales > Cucurbitaceae > remaining tribes

-**GḌG**- ‘CUCURBITACEAE’ I 1. (genus *Dendrosicyos*) Socatran cucumber tree 2. (genus *Tumamoca*) tumamoc globeberry 3. (genus *Sicyos*) burr cucumber

-**GḌGW**- ‘CUCURBITACEAE’ II 1. (genus *Luffa*) luffa / loofah 2. (genus *Trichosantes*) serpent gourd, snake gourd 3. (genus *Cyclanthera*) exploding cucumber, stuffing cucmber / slipper gourd / caigua

-**GḌGY**- ‘CUCURBITACEAE’ III 1. (genus *Sechium*) chayote / mirliton / choko / christophine 2. (genus *Marah*) manroot / wild cucumber / cucumber gourd / old man in the ground 3. (genus *Echinocystis*) wild cucmber / prickly cucumber / bur cucumber

-**GḌGL**- ‘CUCURBITACEAE’ IV 1. (genus *Bryonia*) bryony 2. (genus *Ecballium*) squirting cucumber / exploding cucumber 3. (genus *Telfairia*) fluted pumpkin, oyster nut, queen’s nut / Zanzibar oilvine

-**GḌGR**- ‘CUCURBITACEAE’ V 1. (genus *Momordica*) bitter melon / bitter apple / bitter gourd / bitter squash / balsam-pear, gac, spiny gourd / spine gourd / teasle gourd / kantola 2. (genus *Siraitia*) monk fruit / Buddha fruit 3. (genus *Thladiantha*) Manchu tubergourd / goldencreeper / wild potato

-**GḌGǍ**- ‘CUCURBITACEAE’ VI 1. (genus *Gerrardanthus*) bigfoot 2. (genus *Fevillea*) antidote vine 3. (genus *Momordica*) African cucumber / balsam apple / balsam pear

-**GḌGV**- ‘CUCURBITACEAE’ VII 1. (genus *Alsomitra*) Javan cucumber 2. (genus *Gynostemma*) jiaogulan / five-leaf ginsent / poor man’s ginseng / miracle grass / fairy herb / sweet tea vine / gospel herb / southern ginseng 3. (Cucurbitaceae other than named genera, including unnamed Benincaseae and Cucurbiteae) other curcurbitaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Cucurbitales > remaining families: Begoniacea, Datiscaceae, Tetramelaceae, Corynocarpaceae, Coriariaceae, Apodanthaceae, and Anisophyllaceae

-**GḌMW**- 1. (Begoniaceae; genera *Begonia, Hellebrandia*) begonia, aka ‘aka ‘awa 2. (Datiscaceae; genus *Datisca*) Durango root 3. (Tetramelaceae; genera *Tetrameles, Octomeles*) tetrameles tree, octomeles tree

-**GḌNW**- 1. (Carynocarpaceae; genus *Corynocarpus*) New Zealand laurel, Glenugie karaka 2. (Coriariaceae; genus *Coriaria*) redoul, tutu, deu / huique 3. [stem unassigned]

-**GḌǸW**- 1. (Apodanthaceae; genera *Apodanthes, Pilstyles*) stemsucker, apodanthes 2. (Anisophylleaceae; several genera) anisophylleaceous plant 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Eudicots > supperrosids > rosids > eurosids > fabids > Zygophyllales

-BḌBW- ‘ZYGOPHYLLACEAE’ I 1. (genus *Zygophyllum*) Syrian bean-caper 2. (genus Fagonia) fagonbush 3. (genus *Roepera*) coast twin-leaf 3. (genus *Tetraena*) simple-leaved bean caper / hureim

-BḌBY- ‘ZYGOPHYLLACEAE’ II 1. (genus *Kallstroemia*) California caltrop, Arizona poppy, hairy caltrop, big caltrop, warty caltrop, perennial caltrip, Caribbean caltrop 2. (genus *Tribulus*) Jamaican feverplant / puncture vine, cork hopbush, goat’s-head / bullhead, cat’s-head / devil’s eyelashes / tackweed 3. (genus *Balanites*) Egyptian balsam, Angolan greenthorn, small torchwood

-BḌBL- ‘ZYGOPHYLLACEAE’ III 1. (genus *Guaiacum*) lignum-vitae 2. (genus *Porlieria*) Texas gualacum / Texas lignum-vitae / soapbush / huayacán 3. (genus *Bulnesia*) maracaibo lignum-vitae, verawood, Argentine lignum-vitae / Paraguay lignum-vitae

-BḌBR- ‘OTHER ZYGOPHYLLACEAE & KRAMERIACEAE’ 1. (genus *Larrea*) chaparral, creosote bush / greasewood 2. (Zygophyllaceae other than named genera; several genera) other zygophyllaceous plant 3. (Krameriaceae; genus *Krameria*) rhatany / rattany, krameria, abrojo colorado

Tracheophytes > Spermatophytes > Angiosperms > Magnolids

Tracheophytes > Spermatophytes > Angiosperms > Magnolids > Magnoliales

-**CFW**- ‘MAGNOLIACEAE’ 1. (genus *Magnolia*) magnolia, sweetbay, bull bay 2. (genus *Liriodendron*) tulip tree / tulipwood / tulip poplar / whitewood / fiddletree / yellow-poplar 3. (Magnolliaceae other than Stems 1 and 2; genera *Pachylarnax, Kmeria, Elmerillia, Michelia*) champak, champaca, banana shrub, other magnoliaceous plant/tree

-**CFY**- ‘MYRISTICACEAE’ 1. (genus *Myristica*) nutmeg tree 2. (genus *Virola*) baboonwood / ucuhuba / ucuuba 3. (Myristicaceae other than Stems 1 and 2; numerous genera) other myristicaceous plant/tree

-**CFL**- ‘ANNONACEAE’ I 1. (genus *Annona*) custard apple, soursop / graviola, pond apple / alligator apple, sugar apple / sweetsop, cherimoya, guanabana 2. (genus *Anonidium*) junglesop 3. (genus *Asimina*) pawpaw

-**CFR**- ‘ANNONACEAE’ II 1. (genus *Cananga*) cananga tree, ylang-ylang 2. (genus *Artabotrys*) climbing lang-lang / ylang ylang vine 3. (genus *Uvaria*) finger-root / bush banana

-**CFŘ**- ‘OTHER ANNONACEAE’ 1. (genus *Monodora*) calabash nutmeg 2. (genus *Xylopia*) xylopia, grains of Selim tree / Kani pepper tree / Senegal pepper tree / Ethiopian pepper tree 3. (Annonaceae other than named genera; numerous genera) other annonaceous plant

-**CFF**- ‘HIMANTANDRACEAE & DEGENERIACEAE & EUPOMATIACEAE’ 1. (Himantandraceae; genus *Galbulimima*) northern pigeonberry ash, white magnolia 2. (Degeneriaceae; genus *Degeneria*) karawa, masiratu 3. (Eupomatiaceae; genus *Eupomatia*) bolwarra, copper laurel

Tracheophytes > Spermatophytes > Angiosperms > Magnolids > Laurales

Tracheophytes > Spermatophytes > Angiosperms > Magnolids > Laurales > Lauraceae

-**CVW**- ‘LAURACEAE’ I 1. (genus *Laurus*) laurel / bay laurel / sweet bay 2. (genus *Umbellularia*) Oregon myrtle / California bay laurel / California bay / cinnamon bush / peppernut tree / headache tree / mountain lurel / balm of heaven 3. (genus *Cassytha*) dodder laurel / laurel dodder, devil’s twine, love vine

-**CVY**- ‘LAURACEAE’ II 1. (genus *Cinnamomum*) cinnamon tree, camphor laurel / camphorwood, Selasian wood / Martaban camphor wood / saffrol laurel, kalingag tree, Indian bay leaf, malabathrum, red barked sassafras, stout camphor tree, false camphor tree / Nepal camphor tree 2. (genus *Sassafras*) sassafras 3. (genus *Lindera*) spicebush, spicewood, Benjamin bush

-**CVL**- ‘LAURACEAE’ III 1. (genus *Litsea*) bollywood / bollygum, brown beech, bolly beech, Mexican bay leaf, litsea, may chang, mangeao 2. (genus *Neolitsea*) green bolly gum, hairy bolly gum 3. (genus *Beilschmiedia*) blush walnut / hard bolly gum / nut wood, yellow walnut / yellow nut / canary ash, northern acorn tree, tawa tree, taraire tree

-**CVR**- ‘LAURACEAE’ IV 1. (genus *Persea;* subgenus *Persea; P. american* and *P. schiedeana*) avocado tree, coyo 2. (genus *Persea*, subgenera *Eriodaphne* and *Machilus*; numerous species) redbay, swampbay, lingue, large-flowered bay tree 3. (genus *Nectandra*) sweetwood

-**CVŘ**- ‘LAURACEAE’ V 1. (genus *Ocotea*) stinkwood, Brazilian sassafras, cape laurel, sword laurel 2. (genus *Chlorocardium*) greenheart 3. (Lauraceae other than named genera; numerous genera) other lauraceous plant/tree

Tracheophytes > Spermatophytes > Angiosperms > Magnolids > Laurales > Monimiaceae

-**ŠŢW**- ‘MONIMIACEAE’ I 1. (genus *Hedycarya*) Australian mulberry, pigeonwood / porokaiwhiri 2. (genus *Xymalos*) lemonwood 3. (genus *Tambourissa*) bois tambour

-**ŠŢY**- ‘MONIMIACEAE’ II 1. (genus *Peumus*) boldo 2. (genus *Wilkiea*) tetra beech, wilkiea 3. (Monimiaceae other than named genera: numerous genera) other monimiaceous plant/tree

Tracheophytes > Spermatophytes > Angiosperms > Magnolids > Laurales > Hernandiaceae

-**ŠŢL**- ‘HERNANDIACEAE’ 1. (genus *Hernandia*) lantern tree 2. (genus *Gyrocarpus*) helicopter tree / propeller tree / whirly whirly tree / stinkwood / shitwood 3. (Hernandiaceae other than Stems 1 and 2: several genera) other hernandiaceous plant/tree

Tracheophytes > Spermatophytes > Angiosperms > Magnolids > Laurales > Atherospermataceae

-**ŠŢR**- ‘ATHEROSPERMATACEAE’ I 1. (genus *Atherosperma*) southern sassafrass / blackheart sassafras 2. (genus *Daphnandra*) socketwood, light yellowwood 3. (genus *Doryphora*) yellow sassafras / canary sassafras / golden sassafras / golden deal

-**ŠŢŘ**- ‘ATHEROSPERMATACEAE’ II 1. (genus *Laurelia*) Peruvian nutmeg / Chilean laurel / Chilean sassafras, pukatea 2. (genus *Laureliopsis*) tepa 3. (Atherospermataceae other than named genera; genera *Dryadodaphne, Nemuaron*) other atherospermataceous plant

Tracheophytes > Spermatophytes > Angiosperms > Magnolids > Laurales > Calycanthaceae

-**ŠŢM**- 'CALYCANTHACEAE' 1. (genus *Calycanthus* [including *Sinocalycanthus*]) sweetshrub, spicebush 2. (genus *Chimonanthes*) wintersweet, Japanese allspice 3. (genus *Idiospermum*) ribbonwood / idiot fruit

Tracheophytes > Spermatophytes > Angiosperms > Magnolids > Laurales > remaining families: Siparunaceae & Gomortegaceae

-**ŠŢN**- 'SIPARUNACEAE & GOMORTEGACEAE' 1. (Siparunaceae; genere *Siparuna, Glossocalyx*) siparunaceous plant 2. (Gomortegaceae; genus *Gomortega*) gomortega, keule / queule 3. [stem unassigned]

Tracheophytes > Spermatophytes > Angiosperms > Magnolids > Piperales

-**SSW**- 'PIPERACEAE' 1. (genus *Piper*) pepper, pepper vine, pepperleaf, peppercorn, root bear plant, yerba santa, betel leaf vine, long pepper, ant plant / ant piper, kawakawa, cubeb, kava, pariparoba 2. (genus *Peperomia*) radiator plant, peperomia, watermelon begonia, baby rubberplant / pepper face, jade necklace / trailing jade / creeping buttons 3. (Piperaceae other than Stems 1 and 2; genera *Verhuellia, Zippelia, Manekia*) other piperales plant

-**SSY**- 'SAURURACEAE' 1. (genus *Saururus*) lizard's tail / water-dragon / dragon's tail / swamp root 2. (genus *Anemopsis*) yerba mansa / lizard tail 3. (genera *Houttuynia* and *Gymnotheca*) fish mint / fish leaf / rainbow plant / chameleon plant / fish wort / bishop's weed / Chinese lizard tail

-**SSL**- 'ARISTOLOCHIACEAE' I 1. (genus *Aristolochia*) birthwort, Dutchman's pipe, fox's ears, work killer, pipevine, swanflower, pelican flower, rooster flower, calico flower, sarrasine, smearwort, snakeroot, aristolochia 2. (genus *Isotrema*) large-leaved Dutchman's pipe 3. (genus *Hydnora*) jackal food

-**SSR-** 'ARISTOLOCHIACEAE' II 1. (genus *Asarum*) wild ginger, hazelwort / wild spikenard / asarabacca, Sichuan ginger 2. (genus *Hexastylis*) heartleaf, little brown jug 3. (Aristolochiaceae other than named genera) other aristolochiaceous plant

Tracheophytes > Spermatophytes > Angiosperms > Magnolids > Canellales

-**SSŘ**- 'CANELLACEAE' 1. (genus *Canella*) wild cinnamon / white cinnamon / cinnamon bark 2. (genus *Pleodendron*) chupacallos 3. (Canellaceae other than Stems 1 and 2: genera *Cinnamodendron, Cinnamosma, Warburgia*) other canellaceous plant/tree

-**SSM**- 'WINTERACEAE' I 1. (genus *Drymis*) winter's bark / canelo 2. (genus *Pseudowintera*) horopito, pepperwood 3. (genus *Tasmannia*) pepperbush, pepper tree

-**SSN**- 'WINTERACEAE' II 1. (genus *Zygogynum*) hotbark 2. (genus *Takhtajania*) takhtajania 3. [stem unassigned]

7.3.3 Holomycota (Fungi, Yeast, and Cristidiscoidea)

Holomycota > Cristidiscoidea

-**RJX**- 'CRISTIDISCOIDEA' 1. (Fonticulaceae; genus *Fonticula*) fonticula slime mold 2. (Nucleariidae; genus *Nuclearia*) nuclearia amoeba 3. (Nucleariidae; genus *Parvularia*) parvularia amoeba

Holomycota > Opisthosporidia (a.k.a. Fungi) other than Dikarya

-**RJG**- 'NON-EUMYCOTOID FUNGUS' 1. (Rozellidea; genera *Rozella, Nucleophaga, Paramicrosporidium*) rozellid fungus 2. (Microsporidea; numerous genera) microsporidian fungus 3. (Aphelidomyceta; several genera) aphelidiaceous fungus

-**RJGW**- 'CHYTRIDIOMYCETA' 1. (Neocallimastigomycota; several genera) neocallimastigaceous fungus 2. (Monoblepharomycota; several genera) monoblepharid fungus 3. (Chytridiomycotina; numerous genera) chytridiomycotine fungus

-**RJGY**- 'BLASTOCLADIOMYCETA' 1. (Physodermatales; genera *Physoderma, Paraphysoderma, Urophlyctis*) pysodermataceous fungus 2. (Blastocladiales; several genera) blastocladiaceous fungus 3. (Entomophthoromycota [excluding Basidiobolomycota]; numerous genera) entomophthoromycote fungus other than basidiobolomycote fungus

-**RJGL**- 'BASIDIOBOLOMYCOTA' 1. (Basidiobolales; three genera) basidiobolaceous fungus 2. (Neozygitales; genera *Apterivorax, Neozygites*) neozygitaceous fungus 3. (Kickxellomycota; numerous genera) kickxellomycotefungus

-**RJGR**- 'NON-ZOOPAGOMYCETIC FUNGI OTHER THAN MUCOROMYCOTA' 1. (Mortierellomycota; several genera) mortierellomycote fungus 2. (Calcarisporiellomycota; genera *Calcarisporiella, Echinochlamydosporium*) calcarisporiellomycote fungus 3. (Glomeromycota; numerous genera) glomeromycote fungus

-**RJGV**- 'MUCOROMYCOTA' 1. (Mucorales genera *Mucor, Rhizomucor, Rhizopus, Absidia, Apophysomyces, Cunninghamella*) mucormycosis-causing fungus 2. (Mucorales fungus other than previous stem; several genera) other mucorales fungus 3. (Endogonomycetes & Umbelopsidomycetes) other mucoromycotine fungus

Holomycota > Opisthosporidia > Dikarya

Holomycota > Opisthosporidia > Dikarya > Ascomycota

Holomycota > Opisthosporidia > Dikarya > Ascomycota > Pezizomycotina

-**RJT**- 'LICHENIZED ASCOMYCETE FUNGUS' 1. (Lecanoromycetes; numerous genera) lecanoromycetes fungus 2. (Lichinales; numerous genera) lichinales fungus 3. (Arthoniomycetes; numerous genera) arthoniomycetes fungus

-**RJTW**- 'PATHOGENIC DOTHIDEOMYCETES SPECIES' 1. (*Hortea wernecki*) hortea wernecki 2. (genus *Pyrenochaeta*) pyrenochaeta 3. (genus *Alternaria*) alternaria

-**RJTY**- 'DOTHIDEOMYCETES' 1. (Dothideomycetidae; numerous genera) capnodiales, dothideales, or myriangiales fungus/mold 2. (Pleosporomycetidae including Pleosporales other than genera named in previous root; numerous genera) jahnulales, hysteriales, mytilinidiales, or pleosporales fungus/mold 3. (remaining Dothideomycetes families; numerous genera) botryosphaeriales, microthyriales, patellariales, or trypetheliales fungus/mold

-**RJTL**- 'EUROTIOMYCETES' I 1. (genus *Cladophialophora*) cladophialophora 2. (genus *Rhinocladiella*) rhinocladiella 3. (genus *Exophiala*) exophiala [all three are causes of phaeohyphomycosis]

-**RJTR**- 'EUROTIOMYCETES' II 1. (genus *Coccidioides*) coccidioides 2. (genus *Aspergillus*) aspergillus 3. (genus *Talaromyces*) talaromyces

-**RJTŘ**- 'EUROTIOMYCETES' III 1. (genus *Trichophyton*) trichophyton 2. (genus *Microsporum*) microsporum 3. (genus *Epidermophyton*) epidermophyton [all 3 are causes of ringworm/athlete's foot]

-**RJTÇ**- 'EUROTIOMYCETES' IV 1. (genus *Histoplasma*) histoplasma 2. (genus *Blastomyces*) blastomyces 3. (genus *Paracoccidioides*) paracoccidioides

-**RJTF**- 'EUROTIOMYCETES' V 1. (Mycocaliciomycetidae; several genera) mycocaliciomycetid fungus 2. (Chaetothyriomycetidae other than *Exophiala*) other chaetothryiomycetid fungus 3. (Eurotimycetidae other than named genera; numerous genera) other eurotimycetid fungus

-**RJTH**- 'SORDARIOMYCETES' I 1. (genus *Fusarium*) fusarium 2. (genus *Acremonium*) acremonium [cause of hyalohyphomycosis] 3. (genus *Madurella*) - madurella

-**RJTĻ**- 'SORDARIOMYCETES' II 1. (genus *Pseudallescheria*) pseudallescheria 2. (genus *Sporothrix*) sporothrix [cause of rose handler's disease] 3. (genus *Claviceps*) claviceps, ergot

-**RJTHW**- 'SORDARIOMYCETES' III 1. (genus *Xylaria*) xylaria 2. (genus *Colletotrichum*) colletotrichum 3. (Sordariomycetes other than named genera; numerous genera) other sordariomycetes fungus

RJD- 'LEOTIOMYCETES' I 1. (Cyttariales; genera *Cyttaria, Cyttariella*) cytariales fungus 2. (Erysiphales; numerous genera) erysiphaceous fungus, powdery mildew 3. (Helotiales; numerous genera) helotiales fungus

-**RJDV**- 'LEOTIOMYCETES' II 1. (Leotiales; several genera) leotiales fungus 2. (Rhytismatales; numerous genera) rhytismatales fungus 3. (Thelebolales; numerous genera) thelobolales fungus

-**RJB**- 'PEZIZOMYCETES' I 1. (Pezizaceae; numerous genera) cup fungus, pezizaceous fungus 2. (Pyronemataceae; numerous genera) cup fungus, orange peel fungus, pyronemataceous fungus 3. (Sarcoscyphaceae; several genera) cup fungus, sarcoscyphaceous fungus

-**RJBW**- 'PEZIZOMYCETES' II 1. (genus *Morchella*) morel 2. (Morchellaceae other than Stem 1; several genera) other morchellaceous fungus 3. (Chorioactidaceae; several genera) cup fungus, chorioactidaceous fungus

-**RJBY**- 'PEZIZOMYCETES' III 1. (Discinaceae; genus *Gyromitra*) false morel 2. (Discinaceae other than Stem 1; several genera) other discinaceous fungus 3. (Helvellaceae; several genera) elfin saddle, helvellaceous fungus

-**RJBL**- 'PEZIZOMYCETES' IV 1. (Tuberaceae; genus *Tuber*) truffle 2. (Tuberaceae other than Stem 1; several genera) other tuberaceous fungus 3. (remaining Pezizomycetes families: Sarcosomataceae, Ascobolaceae, Ascodesmidaceae, Rhizinaceae, Glaziellaceae, Carbomycetaceae, Caloscyphaceae; several genera) other pezizomycetes fungus

-**RJBR**- 'PEZIZOMYCOTINA' I 1. (Coniocybomycetes; three genera) coniocybaceous fungus 2. (Xylonomycetes; three genera) xylonaceous or symbiotaphrinacous fungus 3. (Geoglossomycetes; several genera) geoglossaceous fungus

-**RJBŘ**- 'PEZIZOMYCOTINA' II 1. (Laboulbeniomycetes; numerous genera) laboulbeniomycetes fungus 2. (Orbiliomycetes: several genera) orbiliaceous fungus 3. (other Pezizomycotine fungus; several genera) lahmia, medeolaria, triblidiaceous fungus

Holomycota > Opisthosporidia > Dikarya > Ascomycota > Saccharomycotina

-**RJXM**- 'SACCHAROMYCOTINE YEAST' 1. (genus *Saccharomyces*) baker's yeast 2. (genus *Candida*) candida yeast 3. (Saccharomycotina other than Stems 1 and 2) other saccharomycotine yeast

Dikarya > Ascomycota > Taphrinomycotina

-**RJXN**- 'TAPHRINOMYCOTINE YEAST' 1. (Schizosaccharomycetes) fission yeast 2. (genus *Pneumocystis*) pneumocystis yeast 3. (Taphrinomycotina other than Stems 1 and 2) other taphrinomycontine yeast

Holomycota > Opisthosporidia > Dikarya > Ascomycota > remaining families

-**RJXW**- 'OTHER ASCOMYCETES' 1. (Thelocarpaceae; 3 genera) thelocarpaceous fungus 2. (Vezdaeaceae; genus *Vezdaea*) vezdaea fungus 3. (Acomycetes other than from above-named orders or classes; numerous genera) other ascomycete fungus

Holomycota > Opisthosporidia > Dikarya > Basidiomycota

Holomycota > Opisthosporidia > Dikarya > Basidiomycota > Agaricomycotina

Holomycota > Opisthosporidia > Dikarya > Basidiomycota > Agaricomycotina > Agaricomycetes

-**RŢK**- 'CANTHARELLALES' I 1. (genus *Cantherellus*) chanterelle / yellow chanterelle, 2. (genus *Craterellus*) tube chanterelle / yellowfoot chanterelle / yellow-leg, trompette de la mort / horn of plenty 3. (Cantharellaceae; several genera) other cantharellaceous mushroom

-**RŢKW**- 'CANTHARELLALES' II 1. (Hydnaceae; genus *Hydnum*) sweet tooth fungus / hedgehog mushroom / urchin of the woods, pied de mouton 2. (Hydnaceae other than Stem 1; several genera) hydnoid fungus / tooth fungus 3. (Botryobasidiaceae; several genera) corticioid fungus / patch fungus / crust fungus

-**RŢKY**- 'CANTHARELLALES' III 1. (Clavulnaceae; several genera) clavarioid fungus 2. (Aphelariaceae; several genera) club fungus / coral fungus 3. (Ceratobasidaceae and Tullasnellaceae; several genera) other cantharellales fungus

-**RŢKL**- ‘POLYPORACEAE’ 1. (genus *Laetiporus*) sulpher shelf / sulphur polypore / chicken-of-the-woods / crab-of-the-woods / lobster-of-the-woods 2. (genus *Laccocephalum*, a.k.a. *Polyporus*) native bread / blackfellow’s bread 3. (Polyporaceae; genus *Cerioporus*) dryad’s saddle / pheasant’s back mushroom

-**RŢKR**- ‘POLYPORALES’ 1. (Meriplilaceae; genus *Grifola*) maitake mushroom / ram’s head mushroom / sheep’s head mushroom 2. (Sparassidaceae; genus *Sparassis*) cauliflower fungus / cauliflower mushroom 3. (Polyporales other than named genera, including remaining Polyporaceae; numerous genera) other polyporales fungus

-**RŢKŘ**- ‘AGARICOMYCETES’ I 1. (Auriculariales; genus *Auricularia*) wood ear / Judas’ ear / jelly ear, cloud ear 2. (other Auriculariales; numerous genera other than *Auricularia*) other auriculariales fungus 3. (Sebacinales; several genera) sebacinaceous fungus, jelly fungus

-**RŢKF**- ‘AGARICOMYCETES’ II 1. (Stereopsidales; genera *Stereopsis, Clavulicium*) stereopsidales fungus 2. (Geastraceae; several genera) earthstar fungus 3. (Hysterangiales; numerous genera) hysterangiales fungus

-**RŢKŢ**- ‘AGARICOMYCETES’ III 1. (Gomphaceae; genus *Chroogomphus*) pine-spike / spike-cap 2. (genus Ramaria) coral fungus, ramaria, 3. (Gomphaceae other than Stem 1; numerous genera) pig’s ear fungus, gomphaceous fungus

-**RŢKÇ**- ‘AGARICOMYCETES’ IV 1. (Trechisporales/Hydnodontaceae; numerous genera) hydnodontaceous fungus 2. (Hymenochaetales; numerous genera) hymenochaetales fungus 3. (Phallaceae; numerous genera) stinkhorn fungus

-**RŢKS**- ‘AGARICOMYCETES’ V 1. (Bankeraceae; genus *Sarcodon*) shingled hedgehog fungus / scaly hedgehog fungus 2. (Bankeraceae other than Stem 1; several genera) bankeraceous fungus 3. (Thelephoraceae; several genera) leathery earthfan fungus

-**RŢKŠ**- ‘AGARICOMYCETES’ VI 1. (Corticiales; numerous genera) corticiales fungus 2. (Jaapiales; genus *Jaapia*) jaapia fungus 3 (Gloeophyllales; several genera) gloeophyllaceous fungus

-**RŢKM**- ‘RUSSULACEAE’ 1. (genus *Russula*) russula, brittlegill, the Charcoal Burner, the Sickener, erupting mushroom, shrimp mushroom, the Flirt 2. (genus *Lactarius*) milk-cap 3. (genus *Lactifluus*) milky, milk-cap

-**RŢKN**- ‘RUSSULALES’ 1. (Hericiaceae; genus *Hericium*) lion’s mane / monkey head / bearded tooth / satyr’s bear / bearded hedgehog / pom pom mushroom 2. (Bondarzewiaceae; genus *Bondarzewia*) Berkeley’s polypore / stum blossom 3. (Russulales including remaining Russulaceae; numerous genera) other russulales fungus

-**RŢKH**- ‘AGARICOMYCETIDAE’ 1. (Lepidostromataceae; several genera) lepidostramataceous fungus 2. (Atheliaceae; numerous genera) atheliaceous fungus 3. (Amylocorticiaceae; numerous genera) amylocorticiaceous fungus

-**RŢX**- ‘BOLETALES’ I 1. (Boletaceae; genus *Boletus*) cep mushroom / penny bun / porcini mushroom / bolete, old man of the woods 2. (Boletaceae; genus *Leccinum*) red-capped scaber stalk, northern roughstem, birch bolete, Aspen bolete, slate bolete, manzanita bolete 3. (Suillaceae; genus *Suillus*) weeping bolete / granulated bolete, Slippery Jack, woolly-capped suillus, Jersey cow mushroom / bovine bolete

-**RŢXW**- ‘BOLETALES’ II 1. (Rhizopogonaceae; genus *Rhizopogon*) shoro 2. (Sclerodermataceae; several genera) earthball fungus, hard-skinned puffball, prettymouth, dyemaker’s puffball 3. (Boletales remaining families; numerous genera) other boletales fungus

Holomycota > Opisthosporidia > Dikarya > Basidiomycota > Agaricomycotina >Agaricomycetes >Agaricomycetidae > Agaricales

Holomycota > Opisthosporidia > Dikarya > Basidiomycota > Agaricomycotina >Agaricomycetes > >Agaricomycetidae Agaricales > Agaricineae

-**PŢP-** ‘AGARICACEAE’ I 1. (genus *Agaricus*) common mushroom, agaricus mushroom, button mushroom, crimini mushroom, champignon, baby bella, portabella 2. (genus *Coprinus*) shaggy ink cap / shaggy mane / lawyer’s wig 3. (genus *Leucocoprinus*) yellow pot-plant mushroom

-**PŢPW-** ‘AGARICACEAE’ II 1. (genus *Bovista*) true puffball 2. (genus *Calbovista*) sculptured puffball / sculptured giant puffball / warted giant puffball 3. (genus *Calvatia*) puffball, giant puffball

-**PŢPY-** ‘AGARICACEAE’ III 1. (genus *Lycoperdon*) common puffball / warted puffball / gem-studded puffball / wolf farts / devil’s snuff-box, pear-shaped puffball, meadow puffball, grassland puffball / spring puffball, peeling puffball, smooth puffball / soft puffball, dusky puffball, umber-brown puffball 2. (genus *Tulostoma*) stalkball / stalked puffball 3. (genus *Lepiota)* lepiota

-**PŢPL-** ‘AGARICACEAE’ IV 1. (genus *Macrolepiota*) parasol mushroom 2. (genus *Chlorophyllum*) shaggy parasol mushroom 3. (remaining Agaricaceae; numerous genera) other agaricaceous mushroom

-**PŢPR-** ‘HYMENOGASTRACEAE’ I 1. (genus *Psilocybe*) magic mushroom, psychedelic mushroom 2. (genus *Deconica*) dung-loving Psilocybe, mountain moss Psilocybe 3. (genus *Gymnopilus*) magic blue gym, laughing gym / spectacular rustgill, common rustgill, scaly restgill, western jumbo gym, gymnopilus

-**PŢPŘ-** ‘HYMENOGASTRACEAE’ II 1. (genus *Galerina*) autumn skullcap, turf bell 2. (genus *Hebeloma*) poison pie / fairy cake, ghoul fungus 3. remaining Hymenogastraceae; several genera) other hymenogastraceous mushroom

-**PŢPF-** ‘STROPHARIACEAE’ I 1. (genus *Stropharia*) winecap / burgundy / garden giant / king stropharia mushroom, roundhead mushroom 2. (Strophariaceae; genus *Agrocybe*, a.k.a., *Cyclocybe*) pioppino / velvet pioppino / poplar mushroom / black poplar mushroom 3. (Strophariaceae; genus *Pholiota*) nameko mushroom

-**PŢPÇ-** ‘STROPHARIACEAE’ II 1. (genus *Hypholoma*) sulphur tuft / clustered woodlover, brick cap / chestnut mushroom / cinnamon cap / brick top / red woodlover / kuritake 2. (genus *Kuehneromyces*) sheathed woodtuft 3. (remaining Strophariaceae; several genera) other strophariaceous mushroom

-**PŢPĻ-** ‘AGARICINEAE’ I 1. (Nidulariaceae; several genera) nidulariaceous fungus / bird’s nest fungus 2. (Cystodermateae, a.k.a., Squamanitaceae; several genera) cystodermateous fungus 3. (Crassisporiaceae; genera *Crassisporium, Romagnesiella*) crassisporiaceous mushroom

-**PŢPS-** ‘AGARICINEAE’ II 1. (Cortinariaceae; genus *Cortinarius*) cortinar / webcap, Gypsy mushroom 2. (remaining Cortinariaceae; several genera) other cortinariaceous mushroom 3. (Bolbitiaceae; numerous genera) bolbitiaceous mushroom

-**PŢPŠ-** ‘AGARICINEAE’ III 1. (Hydnangiaceae; genus *Laccaria*) laccaria, the deceiver mushroom 2. (remaining Hydnangiaceae; several genera) other hydnangiaceous mushroom 3. (Psathyrellaceae; several genera) psathyrellaceous mushroom

-**PŢPH-** ‘AGARICINEAE’ IV 1. (Crepidotaceae; several genera) crepidotaceous fungus 2. (Inocybaceae; several genera) inocybaceous fungus 3. (Tubariaceae; several genera) tubariaceous fungus

-**PŢPHW-** ‘AGARICINEAE’ V 1. (Mythicomycetaceae; 2 genera) mythicomycetaceous fungus 2. (Gimnopileae; several genera) gimnopilean fungus 3. (remaining Agaracineae; several genera) oher agaricine fungus

Holomycota > Opisthosporidia > Dikarya > Basidiomycota > Agaricomycotina >Agaricomycetes >Agaricomycetidae > Agaricales > Tricholamatineae

-**PŠXW**- ‘TRICHOLAMATINEAE’ I 1. (Mycenaceae; numerous genera) mycenaceous mushroom 2. (Catathelasmataceae/Biannulariaceae; numerous genera) catathelasmataceous mushroom 3. (Entolomataceae; several genera) entolomataceous mushroom

-**PŠXL**- ‘TRICHOLAMATINEAE’ II 1. (Lyophyllaceae; genus *Hypsizygus*) beech mushroom, bunapi-shimeji, buna-shimeji 2. (Lyophyllaceae; genus *Calocybe*) St. George’s mushroom / marzolino 3. (remaining Lyophyllaceae; several genera) other lyophyllaceous mushroom

-**PŠXR**- ‘TRICHOLAMATINEAE’ III 1. (Tricholomataceae; genus *Clitocybe*) wood blewit / blewitt muschroom 2. (Tricholomataceae; genus *Tricholoma*) matsutake mushroom, ponderosa mushroom / pine mushroom / American matsutake mushroom, grey knight / dirty tricholoma 3. (remaining Tricholamatineae; several genera) other tricholamatine mushroom

Holomycota > Opisthosporidia > Dikarya > Basidiomycota > Agaricomycotina >Agaricomycetes >Agaricomycetidae > Agaricales > Marasmiineae

-**LVM**- ‘MARASMIACEAE’ 1. (genus *Merasmius*) parachute mushroom, fairy ring mushroom, pinwheel mushroom 2. (genus *Clitocybula*) coincap mushroom 3. (remaining Marasmiaceae; numerous genera) other marasmiaceous mushroom

-**LVMW**- ‘OMPHALOTACEAE’ 1. (genus *Lentinula*) shiitake mushroom 2. (genus *Gymnopus*) wood woolly-foot mushroom, spindle-shank mushroom 3. (remaining Omphalotaceae; several genera) other omphalotaceous mushroom

-**LVMY**- ‘PHYSALACRIACEAE’ 1. (genus *Flammulina*) enoki / golden needle / seafood mushroom / lily mushroom / winter mushroom / velvet foot / velvet shank / velvet stem / winter fungus 2. (genus *Amillaria*) honey mushroom / honey fungus 3. (remaining Physalacriaceae; numerous genera) other physalacriaceous mushroom

-**LVMR**- ‘MARASMIINEAE’ 1. (Cyphellaceae; numerous genera) cyphellaceous mushroom 2. (Niaceae/Lachnellaceae; several genera) niaceous/lachnellaceous fungus 3. (remaining Marasmiineae; numerous genera) other marasmiine mushroom

Holomycota > Opisthosporidia > Dikarya > Basidiomycota > Agaricomycotina >Agaricomycetes >Agaricomycetidae > Agaricales > remaining classes/families

-**BZVW**- ‘SCHIZOPHYLLINEAE’ 1. (Schizophyllaceae; two genera) schizophyllaceous mushroom 2. (Fistulinaceae; genus *Fistulina*) beefsteak mushroom / ox tongue mushroom 3. (remaining Fistulinaceae; several genera) other fistulinaceous mushroom

-**BZVY**- ‘HYGROPHORACEAE’ 1. (genus *Hygrophorus*) woodwax / waxy cap, herald of winter 2. (genus *Hygrocybe*) waxcap mushroom 3. (remaining Hygrophoraceae; numerous genera) other hygrophoraceous mushroom

-**BZVL**- ‘HYGROPHORIINEAE’ 1. (Clavariaceae; several genera) clavariaceous fungus, coral fungus / antler fungus / finger fungus / worm mold / spaghetti mushroom 2. (Typhulaceae; several genera) typhulaceous mushroom 3. (remaining Hygrophoriineae; numerous genera) other hygrophoriine fungus or mushroom

-**NZGW**- ‘AMANITACEAE’ 1. (*Amanita caesaria*) Caesar’s mushroom 2. (genus *Amanita* other than *A. caesaria*) fly agaric, blusher mushroom, panther cap, death cap, orange spring amanita, destroying angel 3. (remaining Amanitaceae; several genera) other amanitaceous mushroom

-**NZGY**- ‘PLUTEACEAE’ 1. (genus *Pluteus; P. cervinus*) deer mushroom / deer shield / fawn mushroom 2. (*P. salicinus*) knackers crumpet 3. (genus *Volvariella*) straw mushroom

-**NZGL**- ‘PLUTEINEAE’ 1. (remaining Pluteaceae; several genera) other pluteaceous mushroom 2. (Pterulaceae; numerous genera) pterulaceous mushroom 3. (Radulomytaceae; several genera) radulomytaceous mushroom

-**NZGR**- ‘PLEUROTINEAE’ 1. (Pleurotaceae; genus *Pleurotus*) oyster mushroom, abalone mushroom, king tuber mushroom 2. (remaining Pleurotaceae; several genera) other pleurotaceous mushroom 3. (remaining Agaricales fungi; several genera) other agaricales fungus

Holomycota > Opisthosporidia > Dikarya > Basidiomycota > Agaricomycotina > Tremellomycetes

-**Ẓ BV**- ‘TREMELLOMYCETES’ I 1. (Tremellaceae; genus *Tremella*) snow fungus / snow ear / silver ear fungus / white jelly mushroom 2. (genus *Cryptococcus*) cryptococcus 3. (genus *Trichosporon*) trichosporon

-**Ẓ DV**- ‘TREMELLOMYCETES’ II 1. (remaining Tremellales families: remaining Tremellaceae species; Carcinomycetaceae, Cuniculitremaceae, Phragmoxenidaceae, Rhynchogastremataceae, Sirobasidiaceae, Tetragoniomycetaceae, Trichosporonaceae; numerous genera) other tremellales fungus/yeast 2. (Filobasidiales; genus *Filobasidium*) filobasidium 3. (Cystofilobasidiales; several genera) cystofilobasidiaceous fungus

Holomycota > Opisthosporidia > Dikarya > Basidiomycota > Agaricomycotina > remaining classes: Dacrymycetes and Bartheletiomycetes

-**Ẓ GV**- ‘DACRYMYCETES & BARTHELETIOMYCETES’ 1. (Dacrymycetaceae; several genera) dacrymycetaceous fungus 2. (Bartheletiaceae; genus Bartheletia) bartheletia fungus 3. [stem unassigned]

Holomycota > Opisthosporidia > Dikarya > Basidiomycota > remaining subdivisions and classes: Pucciniomycotina, Ustilagomycotina, Wallemiomycetes

-**BZDV**- ‘OTHER BASIDIOMYCOTA’ 1. (Pucciniomycotina; numerous genera) pucciniomycotine fungus / rust fungus 2. (Ustilaginomycotina; numerous genera) ustilaginomycotine fungus / smut fungus 3. (Wallemiaceae; genus *Wallemia*) wallemia fungus

Holomycota > Opisthosporidia > Dikarya > Entorrhizomycota

-**RZDV**- 'ENTORRHIZOMYCOTA' 1. (genus *Talbotiomyces*) talbotiomyces 2. (genus *Juncorrhiza*) juncorrhiza 3. (genus *Entorrhiza*) entorrhiza

7.3.4 Unicellular Eukaryotes & Multicellular Algae

-**MḶM**- 'HAPTISTA & OTHER PRIMITIVE UNICELLULAR EUKARYOTE' I 1. (Glaucophyta; several genera) glaucophyte alga / glaucocystid alga 2. (Alveidea; genus *Ancoracysta*) ancoracysta 3. (Hemimastigophora; several genera) spironemid

-**MḶMW**- 'HAPTISTA & OTHER PRIMITIVE UNICELLULAR EUKARYOTE' II 1. (Pavlovophyceae; several genera) pavlovophyceous alga 2. (Prymnesiophyceae; numerous genera) prymnesiophyceous alga 3. (Rappephyceae; genera unknown) rappephyceous alga / rappemonid alga

-**MḶMY**- 'CENTROHELIOZOA' 1. (Pterocystida; several genera) pterocystid heliozoan 2. (Acanthocystida; several genera) acanthocystid heliozoan 2. (genus *Spiculophrys*) spiculophrys

-**MḶML**- 'CORBIHELIA' 1. (Endohelea; genera Microheliella, Heliomorpha) microhelid or heliomorphid alga 2. (Picomonadea; genus *Picomonas*) picozoan / Picomonas 3. (Telonemea; genera *Telonema, Lateronema*) telonemid alga

-**MḶMR**- 'CRYPTISTA' 1. (Cryptophyta; numerous genera) cryptophytic alga 2. (Leucocrypta, a.k.a. Katablepharidophyta, several genera) leucocryptid or katablepharid alga 3. (Palpitophyta; genus Palpitomonas) palpitomonad alga

Unicellular Rhodophyta (Red Alga)

-**BZGV**- 'UNICELLULAR RED ALGA' 1. (Cyanidiales; several genera) cyanidiales alga 2. (Rhodellales; several genera) rhodellales alga 3. (Porphyridiophyceae; several genera) porphyridiales alga

Multicellular Rhodophyta (Red Alga)

-**BZGW**- 'NEMALIOPHYCIDAE' 1. (genus *Palmaria*) dulse / dillisk / dilsk / red dulse / sea lettuce flakes 2. (genus *Audouinella*) black algae , black brush algae / black beard algae 3. (remaining Nemaliophycidae; numerous genera) other nemaliophycid alga/seaweed

-**BZGY**- 'GIGARTINACEAE' 1. (genus *Chondrus*) Irish moss 2. (genus *Chondracanthus*) Turkish towel seaweed 3. (remaining Gigartinaceae; several genera) other gigartinaceous alga/seaweed

-**BZGL**- 'PHYLLOPHORACEAE' 1. (genus *Mastocarpus*) carragheen / false Irish moss, Turkish washcloth / black tar spot / grapestone 2. (genus *Phyllophora*) leaf bearer 3. (remaining Phyllophoraceae; numerous genera) other phyllophoraceous alga/seaweed

-**BZGR**- 'SOLIERIACEAE' 1. (genus *Euchuma*) guso 2. (genus *Meristotheca*) tosaka-nori 3. (genus *Kappaphycus*) elkhorn sea moss

-**BZGŘ**- 'GIGARTINALES' 1. (genus *Callophyllis*) carola seaweed 2. (genus *Dilsea*) sea belt, poor man's weather glass 3. (remaining Gigartinales including remaining Solieriaceae; numerous genera) other gigartinales seaweed

-**BZGM**- 'RHODYMENIOPHYCIDAE' 1. (genus *Polysiphonia*) banded siphon weed, lobster horns 2. (genus *Gracilaria*) ogo / ogonori / sea moss 3. (genus *Gelidium*) umutgasari, tengusa, makusa, genso

-**BZGN**- 'FLORIDEOPHYCEAE (REMAINING FAMILIES)' 1. (Corallinophycidae; numerous genera) coralline alga 2. (remaining Rhodymeniophycidae; numerous genera) other rhodymeniophycid alga/seaweed 3. (Hildenbrandiophycidae &Ahnfeltiophycidae; several genera) other florideophyceous alga

-**BZGZ**- ‘BANGIACIEAE’ 1. (genus *Porphyra*) nori / purple laver seaweed 2. (genus Pyropia) southern laver, gim / nori, karengo 3. (Bangiaceae other than Stems 1 and 2; numerous genera) other bangiacieous alga/seaweed

-**BZGḌ**- ‘METARHODOPHYTINA ‘ 1. (Compsopogonophyceae; several genera) compsopogonophyceous alga/seaweed 2. (Stylonemataceae; numerous genera) stylonemataceous alga/seaweed 3. (Rufusiaceae; genus *Rufusia*) rufusia alga

Stramenopiles

-**LMḌM**- ‘HETEROPTROPHIC STRAMENOPILES/HETEROKONT’ I 1. (Oomycetes, a.k.a., Peronosporomycetes; numerous genera) water mold, downy mildew 2. (Hyphochytridiomycota; several genera) hyphochytrid 3. (Bigyromonadea; genus *Developayella*) developayella

-**LMḌN**- ‘HETEROPTROPHIC STRAMENOPILES/HETEROKONT’ II 1. (Bicosecia, including Cyathobodoniae; several genera) bicosoecid 2. (Sagenista, including Labyrinthulomycetes and Eogyrea; several genera) slime net, thraustochytrid, amphitrematid 3. (Placidozoa, including Nanomonadea, Proteromonadida; several genera) placidian protist

-**LMḌŇ**- ‘HETEROPTROPHIC STRAMENOPILES/HETEROKONT’ III 1. (Opalinata; several genera) opaline protist 2. (Nucleohelea; genera Actinophrys, Actinosphaerium) actinophryid protist 3. (Opalozoa other than Stems 1 and 2) other opalozoan protist

-**LMḌḌ**- ‘HETEROPTROPHIC STRAMENOPILES/HETEROKONT’ IV 1. (Pirsoniaceae; genus *Pirsonia*) pirsonia 2. (Platysulcida; genus *Platysulcus*) platysulcus 3. (Blastocystea; genus *Blastocystis*) blastocystis

Ochrophyta/Stamenochromes

-**GḌD**- ‘BACILLARIPHYTA (DIATOM)’ 1. (Bacillariophyceae; numerous genera) raphid diatom 2. (Fragilariophyceae; numerous genera) araphid diatom 3. (Coscinodiscophyceae; numerous genera) centric diatom

-**GḌDW**- ‘HYPOGYRISTA’ 1. (Bolidophyceae; several genera) bolidophyte, bolidophyceous plankton 2. (Dictyochophyceae; numerous genera) dictyochophyte, dictyochophyceous alga 3. (Pelagophyceae; numerous genera) pelagophyte, pelagophyceous alga

-**GḌDY**- ‘CHRYSISTA’ I 1. (Pinguiophyceae; several genera) pinguiophyte, pinguiophyceous alga 2. (Eustigmatophyceae; numerous genera) eustigmatophyte, eustigmatophyceous alga 3. (Picophgales; genus *Picophagus*) picophagus

-**GḌDL**- ‘CHRYSISTA’ II 1. (Synchromophyceae; two genera) synchromophyte / synchromophyceous alga 2. (Chrysophyceae, including Synurophyceae; numerous genera) desmid / golden alga / golden-brown alga / chrysophyte / chrysomonad, synurid 3. (genus *Leukarachnion*) leukarachnion

-**GḌDR**- ‘CHRYSISTA’ III 1. (Raphidophyceae; several genera) raphidophyte alga 2. (Chrysomerophyceae; several genera) chrysomerphyte / chrysomerophyceous alga 3. (Phaeothamniophyceae; several genera) phaeothamniophyte, phaeothamniophyceous alga

-**GḌDŘ**- ‘CHRYSISTA’ IV 1. (Xanthophyceae; numerous genera) yellow-green algae 2. (Chrysista algae other than named genera) other chrysista alga 3. [stem unassigned]

-**ŇŽG**- ‘EDIBLE BROWN ALGA (KELP)’ I 1. (genus *Laminaria*) kombu, haidai seaweed, oarweed 2. (genus *Saccharina*) kombu 3. (genus *Undaria*) wakame, hiromi

-**ŇŽGW**- ‘EDIBLE BROWN ALGA (KELP)’ II 1. (genus *Eisenia*) arame / sea oak 2. (genus *Alaria*) winged kelp, badderlocks 3. (genus *Ecklonia*) paddle weed / kajime / noro-kajime

-**ŇŽGY**- ‘EDIBLE BROWN ALGA (KELP)’ III 1. (genus *Durvillaea*) rimurapa / cochayuyo 2. (genus *Postelsia*) sea palm / palm seaweed 3. (genus *Nereocystis*) bull kelp / bullwhip kelp / ribbon kelp

-**ŇŽGL**- ‘EDIBLE BROWN ALGA (KELP)’ IV 1. (genus *Kjellmaniella*) sugar kelp / sea belt / Devil’s apron 2. (genus *Fucus*) bladderwrack, spiral wrack 3. (genus *Pelvetia*) channelled wrack

-**ŇŽGR**- ‘EDIBLE BROWN ALGA (KELP)’ V 1. (genus *Himanthalia*) thongweed 2. (genus *Hizikia*) hiziki 3. (genus *Cladosiphon*) mozuku

-**ŇŽGŘ**- ‘EDIBLE BROWN ALGA (KELP)’ VI 1. (genus *Nemacystus*) ito-mozuku 2. (genus *Tinocladia*) futo mozuku 3. (genus *Sphaerotrichia*) ishi-mozuku

-**ŇŽGV**- ‘OTHER BROWN ALGA (KELP)’ 1. (genus *Sargassum*) sargassum, sargasso kelp 2. (Chrysoparadoxales; genus *Chrysoparadoxa*) chrysoparadoxa 3. (Phaeophyceae other than previously named genera) brown alga, paeophyte

Alveolata

-**NŢK**- ‘HETEROTRICHEA’ I 1. (genus *Blepharisma*) blepharisma 2. (genus *Climacostomum*) climacostomum 3. (genus *Fabrea*) fabrea

-**NŢKW**- ‘HETEROTRICHEA’ II 1. (genus *Phacodinium*) phacodinium 2. (genus *Protocruzia*) protocruzia 3. (genus *Condylostoma*) condylostoma

-**NŢKY**- ‘HETEROTRICHEA’ III 1. (genus *Folliculina*) folliculina 2. (genus *Spirostomum*) spirostomum 3. (genus *Stentor*) stentor

-**NŢKL**- ‘HETEROTRICHEA’ IV 1. (genus *Maristentor*) maristentor 2. (genus *Peritromus*) peritromus 3. (Heterotrichea other than named genera; several genera) other heterotrich ciliate

-**NŢKR**- ‘KARYORELICTEA’ I 1. (genus *Loxodes*) loxodes 2. (genus *Cryptopharynx*) cryptopharynx 3. (genus *Geleia*) geleia

-**NŢKŘ**- ‘KARYORELICTEA’ II 1. (genus *Kentrophorus*) kentrophorus 2. (genus *Tracheloraphis*) tracheloraphis 3. (Karyorelictea other than named genera; several genera) other karyorelict ciliate

-**NŢKF**- ‘COLPODEA’ I 1. (genus *Colpoda*) colpoda 2. (genus *Bresslaua*) bresslaua 3. (genus *Tillina*) tillina

-**NŢKÇ**- ‘COLPODEA’ II 1. (genus *Woodruffia*) woodruffia 2. (genus *Bryophrya*) bryophrya 3. (Colpodea other than named genera; several genera) other colpodid ciliate

-**NŢKS**- ‘NASSOPHOREA’ 1. (genus *Nassula*) nassula 2. (genus *Discotricha*) discotricha 3. (Nassophorea other than Stems 1 and 2; several genera) other nassophorid ciliate

-**NŢKŠ**- ‘PERITRICHIA’ 1. (genus *Vorticella*) vorticella 2. (genus *Cothurnia*) cothurnia 3. (Peritrichia other than Stems 1 and 2; numerous genera) other peritrich ciliate

-**NŢKM**- ‘OLIGOHYMENOPHOREA’ I 1. (genus *Paramecium*) paramecium 2. (genus *Frontonia*) frontonia 3. (genus *Urocentrum*) urocentrum

-**NŢKN**- ‘OLIGOHYMENOPHOREA’ II 1. (genus *Tetrahymena*) tetrahymena 2. (genus *Glaucoma*) glaucoma 3. (genus *Colpidium*) colpidium

-**NŢKH**- ‘OLIGOHYMENOPHOREA’ III 1. (genus *Ichthyophthirius*) ichthyophthirius 2. (genus *Ophryoglena*) ophryoglena 3. (Oligohymenophorea other than named genera; numerous genera) other oligohymenophorean ciliate

-**NŢTW**- ‘PHYLLOPHARYNGEA’ I 1. (Cryptogemmida; several genera) cryptogemmid ciliate 2. (Exogemmida; several genera) exogemmid ciliate 3. (Chlamydodontida; several genera) chlamydodontid ciliate

-**NŢTY**- ‘PHYLLOPHARYNGEA’ II 1. (Dysteriida; several genera) dysteriid ciliate 2. (Hypocomatida; several genera) hypocomatid ciliate 3. (Rhynchodida; several genera) rhynchodid ciliate

-**NŢTL**- ‘PHYLLOPHARYNGEA’ III [SUCTORIA] 1. (Endogenida; several genera) endogenid suctorian 2. (Exogenida; several genera) exogenid suctorian 3. (Evaginogenida and other Suctoria; several genera) evagionogenid or other suctorian

-**NŢTR**- ‘PROSTOMATEA’ I 1. (genus *Coleps*) coleps 2. (genus *Prorodon*) prorodon 3. (genus *Placus*) placus

-**NŢTŘ**- ‘PROSTOMATEA’ II 1. (genus *Urotricha*) urotricha 2. (genus *Chilophrya*) chilophrya 3. (genus *Balanion*) balanion

-**NŢTÇ**- ‘PROSTOMATEA’ III 1. (genus *Bursella*) bursella 2. (genus *Holophrya*) holophrya 3. (Prostomatea other than named genera; several genera) other prostomate ciliate

-**NŢTF**- ‘PLAGIOPYLEA’ I 1. (genus *Plagiopyla*) plagiopyla 2. (genus *Sonderia*) sonderia 3. (genus *Discomorphella*) discomorphella

-**NŢTĻ**- ‘PLAGIOPYLEA’ II 1. (genus *Epalxella*) epalxella 2. (genus *Saprodinium*) saprodinium 3. (Plagiopylea other than named generra; several genera) other plagiopylean ciliate

-**NŢC**- ‘LITOSTOMATEA’ I 1. (genus *Chaenia*) chaenia 2. (genus *Didinium*) didinium 3. (genus *Homalozoon*) homolozoon

-**NŢCW**- ‘LITOSTOMATEA’ II 1. (genus *Lacrymaria*) lachrymaria 2. (genus *Dileptus*) dileptus 3. (genus *Trachelius*) trachelius

-**NŢČ**- ‘LITOSTOMATEA’ III 1. (genus *Spathidium*) spathidium 2. (genus *Enchyelys*) enchelys 3. (genus *Mesodinium*) mesodinium

-**NŢČW**- ‘LITOSTOMATEA’ IV 1. (genus *Amphileptus*) amphileptus 2. (genus *Loxophyllum*) loxophyllum 3. (genus *Litonotus*) litonotus

-**NŢTH**- ‘LITOSTOMATEA’ V 1. (genus *Trachelophyllum*) trachelophyllum 2. (genus *Balantidium*) balantidium 3. (Litostomatea other than named genera; several genera) other litostomate ciliate

-**NŢP**- ‘SPIROTRICHEA’ I 1. (genus *Euplotes*) euplotes 2. (genus *Paraeuplotes*) paraeuplotes 3. (genus *Diophrys*) diophrys

-**NŢPW**- ‘SPIROTRICHEA’ II 1. (genus *Euplotidium*) euplotidium 2. (genus *Aspidisca*) aspidisca 3. (genus *Uronychia*) uronychia

-**NŢPY**- ‘SPIROTRICHEA’ III 1. (genus *Stylonychia*) stylonychia 2. (genus *Steinia*) steinia 3. (genus *Pleurotricha*) pleurotricha

-**NŢPL**- ‘SPIROTRICHEA’ IV 1. (genus *Oxytricha*) oxytricha 2. (genus *Urosoma*) urosoma 3. (genus *Kahliella*) kahliella

-**NŢPR**- ‘SPIROTRICHEA’ V 1. (genus *Urostyla*) urostyla 2. (genus *Uroleptus*) uroleptus 3. (genus *Tachysoma)* tachysoma

-**NŢPŘ**- ‘SPIROTRICHEA’ VI 1. (genus *Amphisiella*) amphisiella 2. (genus *Uroleptoides*) uroleptoides 3. (genus *Tetrastyla)* tetrastyla

-**NŢPÇ**- ‘SPIROTRICHEA’ VII 1. (genus *Gonostomum*) gonostomum 2. (genus *Holosticha*) holosticha 3. (genus *Epiclintes*) epiclintes

-**NŢPF**- ‘SPIROTRICHEA’ VIII 1. (genus *Strongylidium*) strongylidium 2. (genus *Kerona*) kerona 3. (genus *Trachelostyla)* trachelostyla

-**NŢPĻ**- ‘SPIROTRICHEA’ IX 1. (genus *Hypotrichidium*) hypotrichidium 2. (genus *Stichotricha*) stichotricha 3. (Spirotrichea other than named genera; several genera*)* other spirotrich ciliate

-**NŢPS**- ‘OLIGOTRICHEA’ I 1. (genus *Halteria*) halteria 2. (genus *Strombidium*) strombidium 3. (genus *Tontonia*) tontonia

-**NŢPŠ**- ‘OLIGOTRICHEA’ II 1. (genus *Tintinnidium*) tintinnidium 2. (genus *Strombidinopsis*) strombidinopsis 3. (Oligotrichea other than named genera; numerous genera) other oligotrich ciliate

-**NŢPH**- ‘ARMOPHOREA’ 1. (genus *Metopus*) metopus 2. (genus *Chaenomorpha*) chaenomorpha 3. (Armophorea other than Stems 1 and 2) other armophorean ciliate

-**NŢPHW**- ‘OTHER CILIOPHORA’ 1. (genus *Muranothrix*) muranothrix 2. (genus *Cariacothrix*) cariacothrix 3. (remaining Ciliophora; numerous genera) other unicellular ciliate

-**NŢFW**- ‘APICOMPLEXA / SPOROZOA’ I 1. (genus *Plasmodium*) plasmodium 2. (genus *Toxoplasma*) toxoplasma 3. (genus *Cryptosporidium*) cryptosporidium

-**NŢFY**- ‘APICOMPLEXA / SPOROZOA’ II 1. (genus *Cyclospora*) cyclospora 2. (genus *Cystoisospora*) cystoisospora / isospora 3. (genus *Hepatocystis*) hepatocystis

-**NŢFL**- ‘APICOMPLEXA / SPOROZOA’ III 1. (genus *Babesia*) babesia 2. (genus *Entopolypoides*) entopolypoides 3. (genus *Theileria*) theileria [cause of equine piroplasmosis]

-**NŢFR**- ‘APICOMPLEXA / SPOROZOA’ IV 1. (genus *Nycteria*) nycteria 2. (genus *Rayella*) rayella 3. (remaining apicomplexa; numerous genera) other apicomplexan/sporozoan parasite

-**NŢVW**- ‘DINOFLAGELLATA’ I 1. (genus *Tripos*) tripos 2. (genus *Ceratium*) ceratium 3. (genus *Gonyaulax*) gonyaulax

-**NŢVY**- ‘DINOFLAGELLATA’ II 1. (genus *Peridinium*) peridinium 2. (genus *Gymnodinium*) gymnodinium 3. (genus *Oxyrrhis*) oxyrrhis

-**NŢVL**- ‘DINOFLAGELLATA’ III 1. (genus *Amoebophrya*) amoebophrya 2. (genus *Noctiluca*) noctiluca’ 3. (Dinoflagellata other than named genera) other dinoflagellate

-**NŢVR**- ‘OTHER MYZOZOA’ 1. (Chromerida; several genera) chromerid myzozoan 2. (Perkinsozoa; several genera) perkinsean parasite 3. (Myzozoa other than named genera, including Protalveloata; numerous genera) other myzozoan

Rhizaria

-**RMT**- ‘MONADOFILOSE CERCOZOAN’ I 1. (genus *Euglypha*) euglypha 2. (genus *Trinema*) trinema 3. (Thaumatomonadida; several genera) thamatomonad cercozoan

-**RMTW**- ‘MONADOFILOSE CERCOZOAN’ II 1. (Phaeodarea; numerous genera) phaeodarian cercozoan 2. (Tectofilosida; several genera) tectofilosid cercozoan 3. (Thecofilosea other than Stems 1 and 2; several genera) other thecofilose cercozoan

-**RMTY**- ‘MONADOFILOSE CERCOZOAN’ III 1. (genus *Cercomonas*) cercomonas 2. (genus *Kraken*) kraken [amoebid cercozoan] 3. (Monadofilosa other than named genera; numerous genera) other monadofilose cercozoan

-**RMTL**- ‘RETICULOFILOSE CERCOZOAN’ 1. (genus *Clathrulina*) clathrulina 2. (genus *Hedriocystis*) hedriocystis 3. (Reticulofilosa other than Stems 1 and 2; numerous genera) other reticulose cercozoan

-**RMTR**- ‘ENDOMYXAN CERCOZOAN’ I 1. (genus *Vampyrella*) vampyrella 2. (genus *Pseudospora*) pseudospora 3. (genus *Gromia*) gromia

-**RMTŘ**- ‘ENDOMYXAN CERCOZOAN’ II 1. (Plasmodiophorida; numerous genera) plasmodiophorid parasite 2. (Phagomyxida; genus Phagomyxa) phagomyxa 3. (Endomyxa other than Stems 1 and 2 and previous root; several genera) other endomyxan cercozoan

-**RMTÇ**- ‘MONOTHALAMEAN FORAMINIFERA’ 1. (Allogromiida; numerous genera) allogromiid foraminiferan 2. (Astrorhizida; numerous genera) astrorhizid foraminiferan 3. (Monothalamea other than Stems 1 and 2, including Xenophyophorea and genus *Reticulomyxa*; several genera) other monothalamean foraminiferan

-**RMTF**- ‘TUBOTHALAMEAN FORAMINIFERA’ 1. (Miliolida; numerous genera) miliolid foraminiferan 2. (Spirillinida; numerous genera) spirillinid foraminiferan 3. (Silicoloculinida; genus *Miliammellus*) miliammellus

-**RMTL̦**- ‘GLOBOTHALAMEAN FORAMINIFERA’ 1. (Rotaliida; numerous genera) rotaliid foraminiferan 2. (Textulariida; numerous genera) texturlariid formaminiferan 3. (Globothalamea other than Stems 1 and 2) other globothalamean foraminiferan

-**RMTH**- ‘OTHER FORAMINIFERA’ 1. (Nodosariidae; numerous genera) nodosariid foraminiferan 2. (Lagenidae; numerous genera) lagenid foraminifera 3. (Foraminifera other than named genera, classes, or orders; numerous genera) other foraminiferan

-**RMTHW**- ‘RADIOZOAN’ 1. (Polycystina, including Collodaria; numerous genera) polycystine radiolarian 2. (Acantharea; numerous genera) acantharean radiolarian 3. (Taxopodida; genus *Sticholonche*) sticholonche

Discoba (Excavata)

-**RSK**- ‘EUGLENOIDA’ I 1. (genus *Euglena*) euglena 2. (genus *Astasia*) astasia 3. (genus *Khawkinea*) khawkinea

-**RSKW**- ‘EUGLENOIDA’ II 1. (genus *Ascoglena*) ascoglena 2. (genus *Trachelomonas*) trachelomonas 3. (genus *Strombomonas*) strombomonas

-**RSKY**- ‘EUGLENOIDA’ III 1. (genus *Phacus*) phacus 2. (genus *Lepocinclus*) lepocinclus 3. (genus *Eutreptia*) eutreptia

-**RSKL**- ‘EUGLENOIDA’ IV 1. (genus *Peranema*) peranema 2. (genus *Anisonema*) anisonema 3. (genus *Urceolus*) urceolus

-**RSKR**- ‘EUGLENOIDA’ V 1. (genus *Heteronema*) heteronema 2. (genus *Distigma*) distigma 3. (genus *Neometanema*) neometanema

-**RSKŘ**- ‘EUGLENOIDA’ VI 1. (genus *Rhabdomonas*) rhabdomonas 2. (genus *Lentomonas*) lentomonas 3. (genus *Ploetia*) ploetia

-**RSKF**- ‘EUGLENOIDA’ VII 1. (genus *Rapaza*) rapaza 2. (genus *Keelungia*) keelungia 3. (genus *Petalomonas*) petalomonas

-**RSKŢ**- ‘EUGLENOIDA’ VIII 1. (genus *Entosiphon*) entosiphon 2. (genus *Colacium*) colacium 3. (Euglenoida other than named genera) other euglenoid flagellate

-**RSKÇ**- ‘KINETOPLASTEA’ 1. (genus *Trypanosoma*) trypanosoma 2. (genus *Leishmania*) leishmania 3. (Kinetoplastea other than Stems 1 and 2; several genera) other kinetoplast flagellate

-**RSKS**- ‘DIPLONEMEA’ 1. (genus *Diplonema*) diplonema 2. (genus *Rhynchopus*) rhynchopus 3. (Diplonemidae other than Stems 1 and 2; several genera) other diplonemid flagellate

-**RSKŠ**- ‘OTHER EUGLENOZOAN’ 1. (genus *Calkinsia*) calkinsia 2. (genus *Postgaardi*) postgaardi 3. (Euglenozoan other than named genera; numerous genera) other euglenozoan flagellate

-**RSKM**- ‘DISCOBA’ I 1. (Jakobea; several genera) jakobid flagellate 2. (Tsukubea; genus *Tsukubamonas*) tsukubamonas 3. (Percolatea; two genera) percolate flagellate

-**RSKN**- ‘DISCOBA’ II 1. (Acrasida; several genera) non-mycetozoan cellular slime mold 2. (Lyromonadida; genus *Lyromonas*) lyromonas 3. (Gruberellidae; several genera) gruberellid flagellate

-**RSKH**- ‘DISCOBA’ III 1. (genus *Naegleria*) naegleria 2. (genus *Vahlkampfia*) vahlkampfia 3. (Vahlkampfiidae other than Stems 1 and 2; several genera) other vahlkampfiid amoeboid

Loukazoa

-**SKMW**- ‘TRICHOZOA’ 1. (genus *Trichomonas*) trichomonas 2. (genus *Dientamoeba*) dientamoeba 3. (genus *Giardia*) giardia

-**SKMY**- ‘OTHER LOUKAZOA’ 1. (Anaeromonadea; three genera) anaeromonad metamonad flagellate 2. (Malawimonadidae; genera *Malawimonas, Gefionella*) malawimonad glagellate 3. (Ancyromonadida, a.k.a., Planomonadida, plus remaining Trichozoa; several genera) ancyromonad flagellate, other trichozoan metamonad flagellate

-**SKM**- ‘DIPHYLLATEA & RIGIFILIDA & MANTAMONADIDA’ 1. (Diphylleida, a.k.a. Collodictyonidae; three genera) collodictyonid flagellate 2. (Rigifilida; two genera) rigifilid 3. (Mantamonadida; genus *Mantamonas*) mantamonas

Amoebozoa

-**ŇŠK**- ‘AMOEBOZOAN’ I 1. (genus *Amoeba*) amoeba 2. (genus *Chaos*) chaos 3. (genus *Metachaos*) metachaos

-**ŇŠKW**- ‘AMOEBOZOAN’ II 1. (genus *Polychaos*) polychaos 2. (genus *Parachaos*) parachaos 3. (genus *Metamoeba*) metamoeba

-**ŇŠKY**- ‘AMOEBOZOAN’ III 1. (genus *Flabellula*) flabellula 2. (genus *Harmannella*) harmannella 3. (genus *Echinameoba*) echinamoeba

-**ŇŠKL**- ‘AMOEBOZOAN’ IV 1. (genus *Arcella*) arcella 2. (genus *Nebela*) nebela 3. (genus *Difflugia*) difflugia

-**ŇŠKR**- ‘AMOEBOZOAN’ V 1. (genus *Acanthamoeba*) acanthamoeba 2. (genus *Balamuthia*) balamuthia 3. (genus *Cochliopodium*) cochliopodium

-**ŇŠKŘ**- ‘AMOEBOZOAN’ VI 1. (genus *Thecamoeba*) thecamoeba 2. (genus *Dinamoeba*) dinamoeba 3. (genus *Mayorella*) mayorella

-**ŇŠKF**- ‘AMOEBOZOAN’ VII 1. (genus *Vexillifera*) vexillefera 2. (genus *Oscillosignum*) oscillosignum 3. (genus *Subulamoeba*) subulamoeba

-**ŇŠKŢ**- ‘AMOEBOZOAN’ VIII 1. (genus *Clydonella*) clydonella 2. (genus *Vanella*) vanella 3. (genus *Pellita*) pellita

-**ŇŠKÇ**- ‘AMOEBOZOAN’ IX 1. (genus *Discamoeba*) discamoeba 2. (genus *Unda*) unda 3. (genus *Paramoeba*) paramoeba

-**ŇŠKM**- ‘AMOEBOZOAN’ X 1. (genus *Mastigamoeba*) mastigamoeba 2. (genus *Iodamoeba*) iodamoeba 3. (genus *Endolimax*) endolimax

-**ŇŠKN**- ‘AMOEBOZOAN’ XI 1. (genus *Entamoeba*) entamoeba 2. (genus *Pelomyxa*) pelomyxa 3. (Amoebozoan other than named genera; numerous genera) other amoeba

-**ŇŠKV**- ‘MYCETOZOAN (AMOEBOID SLIME MOLD)’ 1. (Myxogastria/Myxomycetes; numerous genera) syncytial, plasmodial, or acellular slime mold 2. (Dictyosteliida; several genera) cellular slime mold 3. (Protostelida; several genera) protostelid slime mold

Non-Opisthokont Obazoa

-**LMVL**- 'BREVIATEA & APUSOMONADIDA (i.e., NON-OPISTHOKONT OBAZOAN)' 1. (Breviatea; several genera) breviatid amoeboid flagellate 2. (Apusomonadida; several genera) apusomonad flagellate 3. [stem unassigned]

Non-Animalia Holozoa

-**LMVR**- 'ICHTHYOSPOREA, a.k.a., MESOMYCETOZOA' 1. (Ichthyophonida, including Amoebidiales genus *Amoebidium*; several genera) ichtyophonid mesomycetozoan 2. (Dermocystida; several genera) dermocystid (or rhinosporidiaceous) mesomycetozoan 3. (genus *Rhinosporidium*) rhinosporidium

-**LMVW**- 'NON-CHOANOZOAN UNICELLULAR HOLOZOAN FLAGELLATE' 1. (Pluriformea; genera *Syssomonas, Corallochytrium*) pluriform flagellate 2. (Filasterea; genera *Ministeria, Capsaspora, Pigoraptor*) filasterean flagellate 3. (genus *Tunicaraptor*) tunicaraptor

-**LMVY**- 'CHOANOFLAGELLATEA' 1. (Craspedida, a.k.a., Salpingoecidae; numerous genera) craspedid or salpingoecid flagellate 2. (Acanthoecidae; several genera) acanthoecid flagellate 3. (Stephanoecidae; numerous genera) stephanoecid flagellate

7.3.5 Bacteria & Archaea

Terrabacteria

-**PFTW**- 'CYANOBACTERIA' I 1. (genus *Arthrospira*) spirulina 2. (genus *Aphanizomenon*) aphanizomenon 3. (genus *Nostoc*) nostoc

- **PFTY**- 'CYANOBACTERIA' II 1. (genus *Cyanothece*) cyanothece 2. (genus *Synechocystis*) synechocystis 3. (Cyanobacteria other than Stems 1 and 2; numerous genera) other cyanobacterium

-**PFTL**- 'ACTINOBACTERIA' I 1. (genus *Actinomyces*) actinomyces 2. (genus *Corynebacterium*) corynebacterium 3. (genus *Mycobacterium*) mycobacterium

-**PFTR**- 'ACTINOBACTERIA' II 1. (genus *Nocardia*) nocardia 2. (genus *Streptomyces*) streptomyces 3. (genus *Gardnerella*) gardnerella

-**PFTŘ**- 'ACTINOBACTERIA' III 1. (genus *Atopobium*) atopobium 2. (genus *Olsenella*) olsenella 3. (genus *Cryptobacterium*) cryptobacterium

-**PFTÇ**- 'ACTINOBACTERIA' IV 1. (genus *Eggerthella*) eggerthella 2. (genus *Frankia*) frankia 3. (genus *Bifidobacterium*) bifidobacterium

-**PFTĻ**- 'ACTINOBACTERIA' V 1. (genus *Mobiluncus*) mobiluncus 2. (genus *Actinomadura*) actinomadura 3. (Actinobacteria other than named genera; numerous genera) other actinobacterial bacterium

-**PFTH**- 'DEINOCOCCUS-THERMUS' 1. (Deinococcaceae; genera *Deinococcus, Denobacterium*) deinococcaceous bacterium 2. (Trueperaceae; genus *Truepera*) truepera 3. (Thermaceae; several genera) thermaceous bacterium

-**PFP**- 'FIRMICUTES' I 1. (genus *Bacillus*) bacillus 2. (genus *Listeria*) listeria 3. (genus *Staphylococcus*) staphylococcus

-**PFPW**- 'FIRMICUTES' II 1. (genus *Enterococcus*) enterococcus 2. (genus *Streptococcus*) streptococcus 3. (Lactobacillaceae; numerous genera) lactobacillaceous bacterium [formerly all included in genus *Lactobacillus*]

-**PFPY**- 'FIRMICUTES' III 1. (genus *Leuconostoc*) leuconostoc 2. (genus *Pediococcus*) pediococcus 3. (genus *Aerococcus*) aerococcus

-**PFPL**- 'FIRMICUTES' IV 1. (genus *Clostridium*) clostridium 2. (genus *Clostridioides*) clostridioides 3. (Heliobacteria; several genera) heliobacterial bacterium

-**PFPR**- 'FIRMICUTES' V 1. (genus *Acetobacterium*) acetobacterium 2. (genus *Anaerococcus*) anaerococcus 3. (genus *Peptococcus*) peptococcus

-**PFPŘ**- 'FIRMICUTES' VI 1. (genus *Peptostreptococcus*) peptostreptococcus 2. (genus *Peptoniphilus*) peptoniphilus 3. (genus *Finegoldia*) finegoldia

-**PFPÇ**- ‘FIRMICUTES’ VII 1. (genus *Veillonella*) veillonella 2. (genus *Erysipelothrix*) erysipelothrix 3. (Firmicutes other than named genera; numerous genera) other firmicutes bacterium

-**PFPĻ**- ‘TENERICUTES (MOLLICUTES)’ 1. (genus *Mycoplasma*) mycoplasma 2. (genus *Ureaplasma*) ureaplasma 3. (Mollicutes other than Stems 1 and 2, including Haloplasmatales) other mollicutes bacterium

-**PFPH**- ‘OTHER TERRABACTERIUM’ 1. (Chloroflexi; numerous genera) chloroflexian bacterium 2. (Armatimonadetes; several genera) armatimonadetes bacterium 3. (Abditibacteriota; several genera) abditibacterial bacterium

Proteobacteria

-**KSKW**- ‘PROTEOBACTERIA’ I 1. (genus *Salmonella*) salmonella 2. (genus *Shigella*) shigella 3. (genus *Escherichia*) escherichia

-**KSKY**- ‘PROTEOBACTERIA’ II 1. (genus *Yersinia*) yersinia 2. (genus *Haemophilus*) haemophilus 3. (genus *Vibrio*) vibrio

-**KSKL**- ‘PROTEOBACTERIA’ III 1. (genus *Pseudomonas*) pseudomonas 2. (genus *Francisella*) francisella 3. (genus *Xanthomonas*) xanthomonas

-**KSKR**- ‘PROTEOBACTERIA’ IV 1. (genus *Legionella*) legionella 2. (genus *Coxiella*) coxiella 3. (genus *Klebsiella*) klebsiella

-**KSKŘ**- ‘PROTEOBACTERIA’ V 1. (genus *Aggregatibacter*) aggregatibacter 2. (genus *Azotobacter*) azotobacter 3. (genus *Acidithiobacillus*) acidithiobacillus

-**KSKF**- ‘PROTEOBACTERIA’ VI 1. (genus *Neisseria*) neisseria 2. (genus *Bordetella*) bordetella 3. (genus *Burkholderia*) burkholderia

-**KSKŢ**- ‘PROTEOBACTERIA’ VII 1. (genus *Chromobacterium*) chromobacterium 2. (genus *Eikenella*) eikenella 3. (genus *Ralstonia*) ralstonia

-**KSKÇ**- ‘PROTEOBACTERIA’ VIII 1. (genus *Brucella*) brucella 2. (genus *Bartonella*) bartonella 3. (genus *Rhizobium*) rhizobium

-**KSKM**- ‘PROTEOBACTERIA’ IX 1. (genus *Rickettsia*) rickettsia 2. (genus Orientia) orientia 3. (genus *Ehrlichia*) ehrlichia

-**KSKN**- ‘PROTEOBACTERIA’ X 1. (genus *Helicobacter*) helicobacter 2. (genus *Campylobacter*) campylobacter 3. (genus *Wolinella*) wolinella

-**KSKH**- ‘PROTEOBACTERIA’ XI 1. (genus *Myxococcus*) myxococcus 2. (genus *Disulfovibrio*) disulfovibrio 3. (genus *Geobacter*) geobacter

-**KSKHW**- ‘PROTEOBACTERIA’ XII 1. (genus *Cardiobacterium*) cardiobacterium 2. (genus *Stenotrophomonas*) stenotrophomonas 3. (Proteobacteria other than named genera; numerous genera) other proteobacterium

Sphingobacteria / FCB Group

-**PFMW**- ‘BACTEROIDETES’ I 1. (genus *Bacteroides*) bacteroides 2. (genus *Prevotella*) prevotella 3. (genus *Porphyromonas*) porphyromonas

-**PFMY**- ‘BACTEROIDETES’ II 1. (genus *Tannerella*) tannerella 2. (genus *Sphingobacterium*) sphingobacterium 3. (Bacteroidetes other than named genera) other bacteroidetes bacterium

-**PFML**- ‘CHLOROBI’ 1. (genus *Chlorobium*) chlorobium 2. (Chlorobiaceae other than Stem 1; several genera) chlorobiaceous bacterium / green sulfur bacterium 3. (Ignavibacteriaceae; several genera) ignavibacterial bacterium

-**PFMR**- ‘OTHER FCB GROUP BACTERIUM’ 1. (Fibrobacteres; several genera) fibrobacterial bacterium 2. (Gemmatimonadetes; several genera) gemmatimonad bacterium 3. (FCB Group Bacteria other than Stems 1 and 2, e.g. Fermentibacteria, Delphibacteria; several genera) other FCB group bacterium

Planctobacteria / PVC Group

-**VZXW**- ‘CHLAMYDIAE’ 1. (genus *Chlamydia*) chlamydia 2. (genus *Chlamydophila*) chlamydophila 3. (Chlamydiae other than Stems 1 and 2) other chlamydian bacterium

-**VZXL**- ‘PLANCTOBACTERIA’ (Lentisphaerae; several genera) lentisphaeran bacterium (Planctomycetes; numerous genera) plactomycetic bacterium (Verrucomicrobia; numerous genera) verrucomicrobian bacterium

-**VZXR**- ‘OTHER PVC GROUP BACTERIUM’ 1. (Poribacteria) poribacterial bacterium 2. (Omnitrophica) omnitrophic bacterium 3. (PVC Group bacteria other than named classes/orders/genera) other PVC group bacterium

Other Bacteria

-**LMSW**- ‘SPIROCHAETES’ I 1. (genus *Borrelia*) borrelia 2. (genus *Leptospira*) leptospira 3. (genus *Treponema*) treponema

-**LMSY**- ‘SPIROCHAETES’ II 1. (genus *Brachyspira*) brachyspira 2. (genus *Brevinema*) brevinema 3. (Spirochaetes other than named genera;) other spirochaete

-**LMSL**- ‘ACIDOBACTERIA’ I 1. (genus *Acidobacterium*) acidobacterium 2. (genus *Holophaga*) holophaga 3. (genus *Geothrix*) geothrix

-**LMSR**- ‘ACIDOBACTERIA’ II 1. (genus *Acanthopleuribacter*) acanthopleuribacter 2. (genus *Bryobacter*) bryobacter 3. (remaining Acidobacteria; several genera) other acidobacterium

-**LMŠL**- ‘AQUIFICAE’ 1. (Aquificales; several genera) aquificales bacterium 2. (Desulfurobacteriaceae; three genera) desulfurobacterium 3. (genus *Thermosulfidibacter*) thermosulfidibacter

-**LMŠ**- ‘FUSOBACTERIA’ 1. (genus *Fusobacterium*) fusobacterium 2. (genus *Leptotrichia*) leptotrichia 3. (Fusobacteria other than Stems 1 and 2; several genera) other fusobacterial bacterium

-**LMŠM**- ‘CALDISERICA / ‘CRYOSERICOTA’ 1. (genus *Caldisericum*) caldisericum 2. (genus *Cryosericum*) cryosericum 3. (Caldiserica/Cryosericota other than stems 1 and 2) other caldiseric/cyroseric bacterium

-**LMŠN**- ‘THERMOTAGAE’ 1. (Thermotogaceae; two genera) thermotogaceous bacterium 2. (Fervidobateriaceae; two genera) vervidobateriaceous bacterium 3. (Kosmotogaceae and Petrotogaceae; several genera) kosmotogaceous or petrotogaceous bacterium

-**LMŠR**- ‘OTHER BACTERIUM’ I 1. (Calditrichaeota; genus *Caldithrix* and other possible genera) calditrichaeotic bacterium 2. (Chrysiogenetes; three genera) chrysiogenaceous bacterium 3. (Coprothermobacteria; genus *Coprothermobacter*) coprothermobacter

-**LMŠW**- ‘OTHER BACTERIUM’ II 1. (Deferribacteres; several genera) deferribacterium 2. (Dictyoglomi; genus *Dictyoglomus*) dictyoglomus 3. (Elusimicrobia; several genera) elusimicrobial bacterium

-**LMŠY**- ‘OTHER BACTERIUM’ III 1. (Nitrospirae; several genera) nitrospiraceous bacterium 2. (Synergistetes; numerous genera) synergistaceous bacterium 3. (Thermodesulfobacteria; several genera) thermodesulfobacteriaceous bacterium

Archaea

-**ŠŠPL**- ‘ARCHAEON’ 1. (“DPANN” Archaea; numerous genera) DPANN archaeon 2. (Euryarchaeota; numerous genera) euryarchaeotic archaeon 3. (Proteoarchaeota; numerous genera) proteoarchaeotic archaeon

7.3.6 Viruses & Viroids

Duplodnaviria

-**LFMW**- ‘ALPHAHERPESVIRUS’ 1. (genus *Varicellovirus*) varicella zoster / human alphaherpesvirus 3 / VZV, bovine alphaherpesvirus 1 & 5, bubaline alphaherpresvirus 1, canid alphaherpesvirus 1, caprine alphaherpesvirus 1, cercopithecine alphaherpesvirus 9, cervid alphaherpesvirus 1 & 2, equid alphaherpesvirus 1/3/4/8/9,

felid alphaherpesvirus 1, monodontid alphaherpesvirus 1, phocid alphaherpesvirus 1, suid alphaherpesvirus 1 2. (genus *Simplexvirus*) herpes simplex 1 / human alphaherpesvirus 1 / HHV-1, herpes simplex 2 / human alphaherpesvirus 2 / HHV-2, ateline alphaherpesvirus 1, bovine alphaherpesvirus 2, cercopithecine alphaherpesvirus 2, leporid alphaherpesvirus 4, macacine alphaherpesvirus1, macropodid alphaherpesvirus 1 & 2, panine alphaherpesvirus 2 & 3, pteropodid alphaherpesvirus 1, saimiriine alphaherpesvirus 1 3. (Alphaherpesvirinae other than Stems 1 and 2; genera Iltovirus, Mardivirus, Scutavirus) other alphaherpesvirus

-**LFMY**- 'BETAHERPESVIRUS' 1. (genus *Cytomegalovirus*) cytomegalovirus / CMV / human betaherpesvirus 5, aotine betaherpesvirus 1, cebine betaherpesvirus 1, cercopithecine betaherpesvirus 5, macacine betaherpesvirus 3 & 8, mandrilline betaherpesvirus 1, papiine betaherpesvirus 3/4, saimiriine betaherpesvirus 4 2. (genus Roseolovirus) roseolovirus / human betaherpesvirus 6A/6B/7, macacine betaherpesvirus 9, murid betaherpesvirus 3, suid betaherpesvirus 2 3. (Betaherpesvirinae other than Stems 1 and 2; genera *Muromegalovirus, Proboscivirus*) other betaherpesvirus

-**LFML**- 'GAMMAHERPESVIRUS' 1. (genus *Rhadinovirus*) rhadinovirus / gamma-2 herpesvirus, human gammaherpesvirus 8, ateline gammaherpesvirus 2& 3, bovine gammaherpesvirus 4, cricetid gammaherpesvirus 2, mcacine gammaherpesvirus 5/8/11/12, murid gammaherpesvirus 4 & 7, saimiriine gammaherpesvirus 2 2. (genus *Lymphocryptovirus*) lymphocryptovirus / human gammaherpesvirus 4 / Epstein-Barr virus, callitrichine gammaherpesvirus 3, cercopithecine gammaherpesvirus 14, gorilline gammaherpesvirus 1, macacine gammaherpesvirus 4 & 10, panine gammaherpesvirus 1, papiine gammaherpesvirus 1, pongine gammaherpesvirus 2 3. (Gammaherpesvirinae other than Stems 1 & 2: genera *Macavirus, Percavirus*) other gammaherpesvirus

-**LFMR**- 'OTHER HERPES VIRUS & CAUDOVIRALES' 1. (Alloherpesviridae; genera *Batrachovirus, Cyprinivirus, Ictalurivirus, Salmonivirus*) alloherpesvirus 2. (Malacoherpesviridae; genera *Aurivirus, Ostreavirus*) malacoherpesvirus 3. (Caudovirales; numerous genera) caudovirales virus / tailed bacteriophage

Monodnaviria

-**RŠTW**- 'PARVOVIRUS' 1. (Parvovirinae; several genera) parvovirine parvovirus, primate erythroparvovirus 1 2. (Densovirinae; several genera) densovirine parvovirus 3. (Hamaparvovirinae; several genera) hamaparvovirine parvovirus

-**RŠTY**- 'PAPILLOMAVIRUS' 1. (Human Papillomaviruses; genera *Alphapapillomavirus, Betapapillomavirus, Gammapapillomavirus, Mupapillomavirus, Nupapillomavirus*) human papillomavirus / HPV 2. (genus *Kappapapillomavirus 2*) Shope papilloma virus / SPV / cottontail rabbit papillomavirus / CRPV 3. (Papillomavirus other than Stems 1 and 2; numerous genera) other papillomavirus

-**RŠTL**- 'HUMAN POLYOMAVIRUS' I 1. (genus *Betapolyomavirus*; species: human polyomavirus 1) BK polyomavirus 2. (genus *Betapolyomavirus*; species: human polyomavirus 2) JC polyomavirus 3. (genus *Alphapolyomavirus*; species: human polyomavirus 5) Merkel cell polyomavirus

-**RŠTR**- 'HUMAN POLYOMAVIRUS' II 1. (genus *Deltapolyomavirus*; species: human polyomavirus 6) human polyomavirus 6 2. (genus *Deltapolyomavirus*; species: human polyomavirus 7) human polyomavirus 7 3. (genus *Alphapolyomavirus*; species: human polyomavirus 8) trichodysplasia spinulosa polyomavirus

-**RŠTŘ**- 'POLYOMAVIRUS' I 1. (genus *Alphapolyomavirus* other than named species above [species with non-human hosts or non-symptomatic in humans]) alphapolyomavirus 2. (genus *Betapolyomavirus* other than named species above [species with non-human hosts or non-symptomatic in humans]) betapolyomavirus 3. (genus *Deltapolyomavirus* other than named species above [species with non-human hosts or non-symptomatic in humans]) deltapolyomavirus

-**RŠTÇ**- 'POLYOMAVIRUS II & OTHER COSSAVIRICOTA' 1. (genus *Gammapolyomavirus* other) gammapolyomavirus 2. (Polyomaviridae species as yet unassigned to a genus) other polyomavirus 3. (Mouvericetes; genus *Bidensovirus*) *Bombyx mori* bidensovirus / silkworm bidensovirus

-**RŠTF**- 'CRESSDNAVIRICOTA' 1. (genus *Cyclovirus*) cyclovirus 2. (Geminiviridae; several genera) geminivirus 3. (Cressdnaviricota other than Stems 1 & 2; Genomoviridae; several genera and Arfiviricetes; numerous genera) other cressdnaviricote virus

-**RŠTḶ**- 'OTHER MONADNAVIRIA' 1. (Trapavirae; several genera) pleolipovirid virus 2. (Sangervirae; several genera) microvirid bacteriophage 3. (Loebvirae; numerous genera) tubulavirales bacteriophage

Riboviria

Riboviria > Orthonavirae

Riboviria > Orthonavirae > Pisuviricota

Riboviria > Orthonavirae > Pisuviricota > Pisoniviricetes

Riboviria > Orthonavirae > Pisuviricota > Pisoniviricetes > Nidovirales

-**ŠŠT**- 'BETACORONAVIRUS' I 1. (genus *Merbecovirus*) Middle East respiratory syndrome-related coronavirus (MERS-CoV), hedgehog coronavirus 1, pipistrellus bat coronavirus HKU5, Tylonycteris bat coronavirus HKU4 2. (genus *Sarbecovirus*, species *SARS-CoV*) severe acute respiratory syndrome coronavirus (SARS-CoV) 3. (genus *Sarbecovirus*, species *SARS-CoV-2*) severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2)

-**ŠŠTW**- 'BETACORONAVIRUS' II 1. (genus *Embecovirus*) human coronavirus OC43, human coronavirus HKU1, murine coronavirus 2. (genus *Nobecovirus*) rousettus bat coronavirus HKU9 3. (genus *Hibecovirus*) bat Hp-betacoronavirus Zhejiang2013

-**ŠŠTY**- 'ALPHACORONAVIRUS' I 1. (genus *Alphacoronavirus* 1) alphacoronavirus 1, feline coronavirus, canine coronavirus, transmissible gastroenteritis virus 2. (genus *Pedacovirus*) porcine epidemic diarrhea virus, scotophilus bat coronavirus 512 3. (genus *Minacovirus*) mink coronavirus 1

-**ŠŠTL**- 'ALPHACORONAVIRUS' II 1. (genus *Duvinacovirus*) human coronavirus 229E 2. (genus *Setracovirus*) human coronavirus NL63 3. (Alphacoronavirus other than named genera; several genera) other alphacoronavirus

-**ŠŠTR**- 'GAMMACORONAVIRUS & DELTACORONAVIRUS' 1. (genus *Igacovirus*) avian coronavirus, duck coronavirus 2. (genus *Cegacovirus*) beluga whale coronavirus SW1 (Gammacoronavirus and Deltacoronavirus other than Stems 1 and 2) other gamma- or delta- coronavirus

-**ŠŠTŘ**- 'TOBANIVIRUS' 1. (genus *Torovirus*) torovirus 2. (genus *Bafinivirus*) bafinivirus 3. (Tobanivirus other than Stems 1 & 2; several genera) other tobanivirus

-**ŠŠTH**- 'OTHER NIDOVIRALES' 1. (genus *Alphaarterivirus*) equine arteritis virus 2. (genus *Okavirus*) okavirus 1, gill-associated virus, yellow head virus 3. (remaining nidovirales viruses; several genera) other nidovirales virus

Riboviria > Orthonavirae > Pisuviricota > Pisoniviricetes > Picornavirales

-**MÇP**- 'GENUS ENTEROVIRUS' I 1. (species *enterovirus C*) poliovirus, enterovirus C 2. (species *enterovirus D*) enterovirus D, enterovirus 68, enterovirus 70, enterovirus 94, enterovirus 111, enterovirus 120 3. (species *enterovirus E*) enterovirus E / bovine enterovirus / BEV / enteric cytopathic bovine orphan virus (ECBO)

-**MÇPW**- 'GENUS ENTEROVIRUS' II 1. (species *enterovirus G*) enterovirus G, porcine enterovirus B 2. (species enterovirus H) enterovirus H, simian enterovirus A 3. (other enterovirus species) enterovirus A, enterovirus B, enterovirus F, enterovirus I, enterovirus J, enterovirus K, enterovirus L

-**MÇPY**- 'GENUS ENTEROVIRUS' III 1. (species *rhinovirus A*) rhinovirus A, human rhinovirus A 2. (species *rhinovirus B*) rhinovirus B, human rhinovirus B (species *rhinovirus C*) rhinovirus C, human rhinovirus C

-**MÇPL**- 'GENUS CARDIOVIRUS' 1. (species *cardiovirus B*) cardiovirus B, Saffold virus (SAFV), Theiler's murine encephalomyelitis virus (TMEV) 2. (species *cardiovirus A*) cardiovirus A, encephalomyocarditis virus 3. (*Cardiovirus* species other than Stems 1 & 2) cardiovirus C/D/E/F

-**MÇPR**- 'GENUS KOBUVIRUS' 1. (*Kobuvirus*, species *aichivirus A*) aichi virus, canine kobuvirus 2. (*Kobuvirus*, species *aichivirus B*) bovine kobuvirus 3. (*Kobuvirus*, species *aichivirus C*) porcine kobuvirus

-**MÇPŘ**- 'GENUS PARECHOVIRUS' 1. (species *parechovirus A*) parechovirus A /human parechovirus 1 / HPeV1, human parechovirus 2 2. (species *parechovirus D*) parechovirus D / ferret parechovirus 3. (species other than Stems 1 & 2) other parechovirus

-**MÇPÇ**- 'GENERA HEPATOVIRUS & SALIVIRUS' 1. (*Hepatovirus*, species *hepatovirus A*) hepatovirus A, hepatitis A virus 2. (*Hepatovirus*, species other than Stem 1) hepatovirus B/C/D/E/F/G/H/I 3. (*Salivirus*, species *salivirus A*) salivirus A

-**MÇPF**- 'GENUS APHTHOVIRUS' 1. (species *FMDV*) foot-and-mouth disease virus (FMDV) 2. (species *ERAV*) equine rhinitis A virus 3. (species *BRAV* and *BRBV*) bovine rhinitis A virus, bovine rhinitis B virus

-**MÇPŢ**- 'OTHER PICORNAVIRUS' 1. (genus *Avihepatovirus*) avihepatovirus A 2. (genus *Erbovirus*) erbovirus A 3. (remaining Picornavirales; numerous genera) other picornavirus

-**MÇPĻ**- 'CALICIVIRUS' 1. (genus *Norovirus*) norovirus, winter vomiting bug 2. (genus *Sapovirus*) Sapporo virus 3. (Calicivirus other than Stems 1 & 2; several genera) other calicivirus

-**MÇPH**- ‘DICISTROVIRIDAE’ 1. (genus *Aparavirus*) acute bee paralysis virus, Israeli acute paralysis virus, Kashmir bee virus, Taura syndrome virus, mud crab virus 2. (genus *Cripavirus*) aphid lethal paralysis virus, cricket paralysis virus, *Drosophila* C virus, *Rhopalosiphum padi* virus 3. (genus *Triatovirus*) black queen cell virus, *Himetobi* P virus, *Homalodisca coagulata* virus-1, *Plautia stali* intestine virus, triatoma virus

-**MÇPHW**- ‘OTHER PICORNAVIRALES’ 1. (genus *Iflavirus*) iflavirus 2. (Secoviridae; several genera) secovirus 3. (remaining Pircornavirales; Marnaviridae, Polycipiviridae, Solinviviridae; several genera) other picornovirales virus

Riboviria > Orthonavirae > Pisuviricota > Pisoniviricetes > Sobelivirales

-**NSŢW**- ‘SOBELIVIRALES’ 1. (Alvernaviridae; genus *Dinornavirus*) heterocapsa circularisquama RNA virus 01 2. (Barnaviridae; genus *Barnavirus*) mushroom bacilliform virus 3. (Solemoviridae; genera *Polemovirus, Sobemovirus*) poinsettia latent virus, blueberry shoestring virus, cocksfoot mottle virus, lucerne transient streak virus, papaya lethal yellowing virus, rice yellow mottle virus, ryegrass mottle virus, sesbania mosaic virus, solanum nodiflorum mottle virus, southern bean mosaic virus, southern cowpea mosaic virus, sowbane mosaic virus, subterranean clover mottle virus, turnip rosette virus, velvet tobacco mottle virus, imperata yellow mottle virus

Riboviria > Orthonavirae > Pisuviricota > Dupolopiviricetes

-**NSŢR**- ‘DURNAVIRALES’ 1. (Picobirnaviridae; genus *Picobirnavirus*) human picobirnavirus, equine picobirnavirus, beihai picobirnavirus 2. (Amalgviridae; genera *Amalgavirus, Zybavirus*) *Allium cepa* amalgavirus 1 & 2, blueberry latent virus, rhododendron virus A, southern tomato virus, spinach amalgavirus 1, vicia cryptic virus M, *Zoostera marina* amalgavirus 1 & 2, *Zygosaccharomyces bailii* virus Z 3. (remaining Durnavirales families: Hypoviridae, Partitiviridae; several genera and unassigned species) cryphonectria hypovirus, partitivirus

Riboviria > Orthonavirae > Pisuviricota > Stelpaviricetes

-**ŇSŢ**- ‘POTYVIRUS’ I 1. (genus *Potyvirus*; numerous species) [various plant mosaic viruses, mottle viruses, dwarf viruses, shoestring viruses, distortion viruses, ringspot viruses, stunt viruses, break viruses, streak viruses] 2. (genus *Ipomovirus*) sweet potato mild mottle virus 3. (genus *Tritimovirus*) wheat streak mosaic virus

-**ŇSŢW**- ‘POTYVIRUS’ II 1. (genus *Rymovirus*) ryegrass mosaic virus 2. (genus *Poacevirus*) triticum mosaic virus 3. (remaining Potyvirus; several genera) other potyvirus

-**ŇSŢR**- ‘ASTROVIRUS’ 1. (genus *Mamastrovirus*; species HAstV, HAstV-VA, HAstV-MLB) human astrovirus 2. (genus *Mamastrovirus*; species other than Stem 1) ovine astrovirus, bovine astrovirus, feline astrovirus, sea lion astrovirus, murine astrovirus, porcine astrovirus, mink astrovirus 3. (genus *Avastrovirus*) chicken astrovirus, pigeon astrovirus, avian nephritis astrovirus

Riboviria > Orthonavirae > Duplornaviricota

-**MSŢ**- ‘REOVIRUS’ I 1. (genus *Rotavirus*) rotavirus 2. (genus *Orbivirus*) orbivirus, bluetongue virus, African horse sickness virus, epizootic hemorrhagic disease virus 3. (genus *Seadornavirus*) banna virus (BAV), kadipiro virus, liao ning virus

-**MSŢW**- ‘REOVIRUS’ II 1. (genus *Coltivirus*) Colorado tick fever virus, European Eyach virus 2. (genus *Phytoreovirus*) wound tumour virus (WTV), rice dwarf virus (RDV), rice gall dwarf virus (RGDV), tobacco leaf enation virus (TLEF), rice buncy stunt virus (ICTV) sweet potato virus (VIDE) 3. (Reovirus other than named genera) other reovirus

-**MSŢR**- ‘CHRYMOTIVIRICETES & VIDAVERVIRICETES’ 1. (Ghabrivirales; Totiviridae; several genera) totivirus 2. (Ghabrivirales other than Stem 1; Chrysoviridae, Megabirnaviridae, Quadriviridae; several genera) ghabrivirales virus other than totivirus 3. (Mindivirales, Cystoviridae; genus *Cystovirus*) pseudomonas virus

Riboviria > Orthonavirae > Kitrinoviricota > Flaviviridae

-**LFSK**- ‘GENUS FLAVIVIRUS’ I 1. (species *yellow fever virus*) yellow fever virus 2. (species *Dengue virus*) dengue virus 3. (species *Zika virus*) zika virus

-**LFST**- ‘GENUS FLAVIVIRUS’ II 1. (species *West Nile virus*) West Nile virus, Kunjin virus 2. (species *Murray Valley encephalitis virus*) Murray Valley encephalitis virus (MVEV) 3. (species *Tick-borne encephalitis virus*) tick-borne encephalitis virus (TBEV)

-**LFSP**- ‘GENUS FLAVIVIRUS’ III 1. (species *Omsk hemorrhagic fever virus*) Omsk hemorrhagic fever virus 2. (species *Kyansur Forest disease virus*) Kyansur Forest disease virus 3. (species *Powassan virus*) Powassan virus (POWV)

-**LFSC**- ‘GENUS FLAVIVIRUS’ IV 1. (species *Louping ill virus*) Louping ill virus 2. (species *Langat virus*) Langat virus 3. (species *Alkhurma virus*) Alkhurma virus

-**LFSČ**- ‘GENUS FLAVIVIRUS’ V 1. (species *Japanese encephalitis virus*) Japanese encephalitis virus 2. (species *Saint Louis encephalitis virus*) Saint Louis encephalitis virus 3. (species other than named species) other flavivirus

-**LFSW**- ‘GENUS HEPACIVIRUS’ 1. (species *Hepacivirus C*) human hepatitis C virus 2. (species *Hepacivirus A*) canine hepacivirus, equine hepacivirus 3. (*Hepacivirus* species other than stems of previous root above) other hepacivirus

-**LFSY**- ‘OTHER FLAVIVIRIDAE’ 1. (genus *Pegivirus*) pegivirus 2. (genus *Pestivirus*) pestivirus 3. [stem unassigned]

-**LFSL**- ‘HEPELIVIRALES’ 1. (genus *Rubivirus*) rubella virus (RuV) 2. (genus *Orthohepevirus*) orthohepevirus, human hepatitis E virus 3. (Hepelivirales other than Stems 1 & 2; genera *Omegatetravirus, Betatetravirus, Benyvirus, Piscihepevirus*) other hepelivirales virus

-**LFSR**- ‘TYMOVIRALES’ 1. (Tymoviridae; genera *Tymovirus, Marafivirus, Maculavirus*) tymovirid virus 2. (Betaflexiviridae; numerous genera) betaflexivirus, quinvirine or triverine virus 3. (Alphaflexiviridae, Deltaflexiviridae, Gammaflexiviridae; numerous genera) other tymovirales virus

-**LFSŘ**- ‘GENUS ALPHAVIRUS’ I 1. (species *Chikungunya virus*) Chikungunya virus 2. (species *Mayaro virus*) Mayaro virus, Una virus 3. (species *Ross River virus*) Ross River virus (RRV)

-**LFSM**- ‘GENUS ALPHAVIRUS’ II 1. (species *Eastern equine encephalitis virus*) Eastern equine encephalitis virus (EEEV) 2. (species *Venezuelan equine encephalitis virus*) Venezuelan equine encephalitis virus 3. (species *Western equine encephalitis virus*) Western equine encephalitis virus

-**LFSN**- ‘GENUS ALPHAVIRUS’ III 1. (species *Middelburg virus*) Middelburg virus (MIDV) 2. (species *Barmah Forest virus*) Barmah Forest virus (BFV) 3. (Togaviridae other than Stems 1 & 2 or *Alphavirus* species) other togavirus

-**LFSF**- ‘MARTELLIVIRALES (other than Togaviridae)’ I 1. (Bromoviridae; several genera) bromovirid virus 2. (Closteroviridae; several genera) closterovirid virus 3. (Endornaviridae; numerous genera) endornavirid virus

-**LFSX**- ‘MARTELLIVIRALES (other than Togaviridae)’ II 1. (Kitaraviridae; several genera) kitaravirid virus 2. (Mayoviridae; genera *Idaeovirus, Pteridovirus*) mayovirid virus 3. (Virgaviridae; numerous genera) virgavirid virus

-**LFSV**- ‘OTHER KITRINOVIRICOTA’ 1. (Nodamuvirales; genera *Alphanodavirus*, *Betanodavirus, Sinaivirus*) nodavirus, sainaivirus 2. (Tolivirales; several genera) carmotetravirus, luteovirus, umbravirus, procedovirine virus 3. (Kitrinoviricota other than named orders/classes/families/genera) other kitrinoviricote virus

Riboviria > Orthonavirae > Negarnaviricota

-**NXP**- ‘ORTHOMYXOVIRUS (INFLUENZA VIRUS)’ 1. (genus *Alphainfluenzavirus*) influenza A virus, avian influenza virus / “bird flu” virus 2. (genus *Betainfluenzavirus*) influenza B virus 3. (genera *Gammainfluenzavirus, Deltainfluenzavirus*) influenza C virus, influenza D virus

-**NXPW**- ‘OTHER ORTHOMYXOVIRUS’ 1. (genus *Quaranjavirus*) quaranjavirus 2. (genus *Thogotovirus*) thogotovirus (THOV), dhori virus (DHOV) 3. (genus *Isavirus*) salmon isavirus

-**NXPY**- ‘BUNYAVIRALES’ I 1. (genus *Orthohantavirus*) hantavirus 2. (genus *Orthobunyavirus*) California encephalitis virus, La Crosse virus (LACV), Jamestown Canyon virus, Cache Valley orthobunyavirus (CVV) 3. (genus *Orthonairovirus*) Crimean-congo hemorhagic fever orthonairovirus (CCHF)

-**NXPL**- ‘BUNYAVIRALES’ II 1. (genus *Mammarenavirus-Old World Complex*) Lassa virus (LASV), lymphocytic choriomeningitis virus (LCMV), Lujo virus (LUJV) 2. (genus *Mammarenavirus-New World Complex*) Whitewater Arroyo virus (WWAV), Machupo virus (MACV), Sabiá virus (SBAV), Chapare virus (CHAPV), Guanarito virus (GTOV), Junín virus (JUNV) 3. (genus *Phlebovirus*) Rift Valley fever phlebovirus, Punta Toro phlebovirus, Sicilian phlebovirus

-**NXPR**- ‘BUNYAVIRALES’ III 1. (genus *Orthotospovirus*) tomato spottel wilt orthotspovirus, soybean vein necrosis orthotospovirus, impatiens necrotic spotted orthotospovirus 2. (genus Emaravirus) High Plains wheat mosaic emaravirus, fig mosaic emaravirus, blackberry leaf mottle associated emaravirus, raspberry leaf bloth emaravirus 3. (Bunyavirales other than named gernera; several genera) other bunyavirales virus

-**NXPŘ**- ‘FILOVIRIDAE 1. (genus *Ebolavirus*) ebolavirus 2. (genus *Marburgvirus*) Marburg virus, Ravn virus 3. (Filoviridae other than Stems 1 & 2; several genera) other filovirus

-**NXPÇ**- ‘RHABDOVIRIDAE’ I 1. (genus *Lyssavirus*; species *rabies lyssavirus*) rabies virus (RABV) 2. (genus *Lyssavirus*; species other than Stem 1) other lyssavirus 3. (genus *Vesiculovirus*) Chandipura virus (CHPV), vesicular stomatitis virus

-**NXPḶ**- ‘RHABDOVIRIDAE’ II 1. (genus *Tibrovirus*) tibrovirus, Bas-Congo virus, Ekpoma virus 1 & 2 2. (genus *Novirhabdovirus*) novirhabdovirus) 3. (Rhabdoviridae other than named genera; numerous genera) other rhabdovirus

-**NXPŢ**- ‘PARAMYXOVIRIDAE’ I 1. (genus *Morbillivirus*; species *Measles morbillivirus*) measles virus (MeV) 2. (genus *Morbillivirus*; species other than Stem 1) canine distemper virus, feline morbillivirus, cetacean morbillivirus, rinderpest virus, phocine distemper virus 3. (genus *Respirovirus*) human respirovirus 1 & 3, bovine respirovirus 3, caprine respirovirus 3, porcine respirovirus 1, squirrel respirovirus, murine respirovirus

-**NXPH**- ‘PARAMYXOVIRIDAE’ II 1. (genus *Aquaparammyxovirus*) aquaparamyxovirus 2. (genus *Ferlavirus*) reptilian ferlavirus / Fer-de-Lance virus 3. Paramyxoviridae other than Stems 1 & 2; several genera) other paramyxovirus

-**NXFW**- ‘OTHER NEGARNAVIRICOTA’ I 1. (Rubulavirinae genus *Orthorubulavirus*) mumps virus (MuV) 2. (Rubulavirinae other than Stem 1) other rubulavirus 3. (Pneumoviridae; genera *Orthopneumovirus, Metapneumovirus*) human respiratory syncytial virus A2 & B1 (HRSV), human metapneumovirus (HMPV), avian metapneumovirus (AMPV), bovine repiratory syncytial virus (BRSV), murine pneumonia virus (MPV)

-**NXFY**- ‘OTHER NEGARNAVIRICOTA’ II 1. (Avulavirinae; several genera) avulavirus, Newcastle disease virus 2. (Bornaviridae; genus *Orthobornavirus*) Borna disease virus 1 & 2, bornavirus 3. (Bornaviridae other than Stem 1; genera *Carbovirus, Cultervirus*) python virus, carbovirus, cultervirus

-**NXFL**- ‘OTHER NEGARNAVIRICOTA’ III 1. (Qinviridae; genus *Yingvirus*) yingvirus 2. (Aspiviridae; genus *Ophiovirus*) ophiovirus 3. (Yueviridae; genus *Yuyuevirus*) yuyuevirus

-**NXFR**- ‘OTHER NEGARNAVIRICOTA’ IV 1. (Sunviridae; genus *Sunshinevirus*) reptile sunshinevirus 1 2. (Nyamviridae; several genera) Midway virus, Nyamanini virus, Sierra Nevada virus, soybean cyst nematode virus 1 3. (Articulavirales and Negarnaviricota other than named families/genera; several genera) other negarnaviricote virus

Riboviria > Orthonavirae > Lenarviricota

-**TÇB**- ‘LENARVIRICOTA’ 1. (Botourmiaviridae; genus Ourmiavirus) ourmia melon virus, epirus cherry virus, cassava virus C 2. (Levivirales; genera *Levivirus, Allolevivirus*) levivirales virus 3. (Narnaviridae & Mitoviridae; genera *Narnavirus, Mitovirus*) narnavirus, mitovirus

Riboviria > Orthonavirae > other orthonavirid viruses

-**TÇG**- ‘BIRNAVIRIDAE’ 1. (genus *Avibirnavirus*) infectious bursal disease virus 2. (genus *Aquabirnavirus*) infectious pancreatic necrosis virus, Tellina virus, yellowtail ascites virus 3. (genus *Blosnavirus*) blotched snakehead virus

-**TÇD**- ‘OTHER ORTHONAVIRUS’ 1. (Birnaviridae other than stems of previous root above; several genera) other birnavirus 2. (Permutotetraviridae; genus *Alphapermutotetravirus*) alphapermutotetravirus 3. (genus *Botybirnavirus*) botybirnavirus

Riboviria > Pararnavirae

-**RTÇP**- ‘HEPADNAVIRUS’ 1. (genus *Orthohepadnavirus*) hepatitis B virus 2. (genus *Avihepadnavirus*) duck hepatitis B virus, heron hepatitis B virus, parrot hepatitis B virus 3. (Hepadnavirus other than Stems 1 & 2; several genera) other hepadnavirus

-**RTÇT**- ‘GENUS LENTIVIRUS’ I 1. (species *Human immunodeficiency virus 1*) human immunodeficiency virus 1 / HIV-1 2. (species *Human immunodeficiency virus 2*) human immunodeficiency virus 2 / HIV-2 3. (species *Simian immunodeficiency virus*) simian immunodeficiency virus

-**RTÇK**- ‘GENUS LENTIVIRUS’ II 1. (species *Feline immunodeficiency virus*) feline immunodeficiency virus 2. (species *Bovine immunodeficiency virus*) bovine immunodeficiency virus 3. (species *Equine infectious anemia virus*) equine infectious anemia virus

-**RTÇW**- ‘GENUS LENTIVIRUS’ III 1. (species *Visna-maedi virus*) visna-maedi virus 2. (species *Jembrana disease virus*) jembrana disease virus 3. (Lentivirus species other than named) caprine arthritis encephalitis virus, puma lentivirus

-**RTÇL**- ‘GENUS DELTARETROVIRUS’ 1. (species *Human T-cell lymphotropic virus*) human T-cell lymphotropic virus (HTLV-1, HTLV-2, HTLV-3, HTLV-4) 2. (species *Simian T-cell lymphotropic virus*) simian T-cell lymphotropic virus (STLV-1, STLV-2, STLV-3, STLV-4) 3. (species *Bovine leukemia virus*) bovine leukemia virus (BLV)

-**RTÇR**- ‘OTHER RETROVIRIDAE’ I 1. (genus *Alpharetrovirus*) avian leukosis virus, Rous sarcoma virus 2. (genus *Betaretrovirus*) mouse mammary tumour virus 3. (genus *Gammaretrovirus*) murine leukemia virus, feline leukemia virus

-**RTÇM**- ‘OTHER RETROVIRIDAE’ II 1. (genus *Deltaretrovirus*) bovine leukemia virus 2. (genus *Epsilonretrovirus*) walleye dermal sarcoma virus 3. (Spumaretrovirinae; several genera) bovine foamy virus, feline foamy virus, equine foamy virus, eastern chimpanzee simian foamy virus, brown greater galago prosimian foamy virus

-**RTÇN**- ‘OTHER RETROVIRUS’ 1. (Caulimoviridae; numerous genera) cauliflower mosaic virus, commelina yellow mottle virus, cassava vein mosaic virus, petunia vein clearing virus, tobacco vein clearing virus, soybean chlorotic mottle virus, rice tungro bacilliform virus 2. (Belpaoviridae; genus *Semotivirus*) *Anopheles gambiae* moose virus, *Ascaris lumbricoides* Tas virus, *Bombyx mori* Pao virus, *Caenorhabditis elegans* Cer13 virus, *Drosophila melanogaster* Bel virus, *Drosophila melanogaster* Roo virus 3. (Metaviridae, Pseudoviridae; several genera) other retrovirus

Varidnaviria

Varidnaviria > Bamfordvirae

Varidnaviria > Bamfordvirae > Nucleocytoviricota

-**PSF**- ‘GENUS ORTHOPOXVIRUS’ I 1. (species *Variola virus*) smallpox virus / variola virus 2. (species *vaccinia virus*) vaccinia virus 3. (species *cowpox virus*) cowpox virus

-**PSFW**- ‘GENUS ORTHOPOXVIRUS’ II 1. (species *monkeypox virus*) monkeypox virus 2. (species *camelpox virus*) camelpox virus 3. (species *raccoonpox virus*) raccoonpox virus

-**PSFY**- ‘GENUS ORTHOPOXVIRUS’ III 1. (species *skukpox virus*) skunkpox virus 2. (species *volepox virus*) volepox virus 3. (species other than named) other orthopox virus

-**PSFL**- ‘GENUS PARAPOXVIRUS’ 1. (species *orf virus*) orf virus 2. (species *bovine papular stomatitis virus*) bovine papular stomatitis virus 3. (species other than Stems 1 & 2) other parapox virus

-**PSFR**- ‘OTHER POXVIRUS’ I 1. (genus *Yatapox virus*) tanapox virus, monkey tumor virus 2. (genus *Molluscipox virus*) molluscum contagiosum virus (MCV) 3. (genus *Avipoxvirus*) avian pox virus

-**PSFŘ**- ‘OTHER POXVIRUS’ II 1. (genus *Suipoxvirus*) swinepox virus 2. (genus *Leporipoxvirus*) myxoma virus, hare fibroma virus, rabbit fibroma virus, squirrel fibroma virus 3. (remaining poxvirus genera; several genera) other poxvirus

-**PSFM**- ‘OTHER NUCLEOCYTOVIRICOTA’ I 1. (Asfarviridae; genus *Asfivirus*) African swine fever virus (ASFV) 2. (Ascoviridae; genera *Ascovirus, Toursvirus*) ascovirid virus 3. (Iridoviridae; several genera) iridovirid virus

-**PSFN**- ‘OTHER NUCLEOCYTOVIRICOTA’ II 1. (Mimiviridae; several genera) mimivirid virus 2. (Phycodnaviridae; several genera) phycodnavirid virus 3. (Nucleocytoviricota other than named families/genera; several genera) other nucleocytovirus

Varidnaviria > Bamfordvirae > Preplasmiviricota

Varidnaviria > Bamfordvirae > Preplasmiviricota > Tectiliviricetes

-**GZXL**- 'GENUS MASTADENOVIRUS' I 1. (species *human mastadenovirus A/B/C/D/E/F/G*) human mastadenovirus A/B/C/D/E/F/G 2. (*simian mastadenovirus A/B/C/D/E/F/G/H/I*) simian mastadenovirus A/B/C/D/E/F/G/H/I 3. (species canine mastadenovirus A) canine mastadenovirus A

-**GZXR**- 'GENUS MASTADENOVIRUS' II 1. (species *bovine mastadenovirus A/B/C*) bovine mastadenovirus A/B/C 2. (species *porcine mastadenovirus A/B/C*) porcine mastadenovirus A/B/C 3. (species other than named; numerous species) other mastadenovirus

-**GZXW**- 'OTHER ADENOVIRIDAE' 1. (genus *Siadenovirus*) frog siadenovirus 2. (genus *Aviadenovirus*) avian adenovirus 3. (Adenoviridae other than named genera; genera *Atadenovirus, Ichtadenovirus*) other adenovirus

-**GZXM**- 'OTHER TECTILIVIRICETES' 1. (Corticoviridae; genus *Corticovirus*) corticovirus / psedoalteromonas phage PM2 2. (Turriviridae; genus *Alphaturrivirus*) Alphaturrivirus / sulfolobus turreted icosahedral virus 1 & 2 3. (Tectiviridae; several genera) tectivirus

Varidnaviria > Bamfordvirae > Preplasmiviricota > Maveriviricetes

-**LCXW**- 'LAVIDAVIRIDAE (VIROPHAGE)' 1. (genus *Mavirus*, species *Cafeteriavirus-dependent mavirus*) Cafeteriavirus-dependent mavirus 2. (genus *Sputnikvirus*, species *Mimivirus-dependent virus Sputnik*) Mimivirus-dependent virus Sputnik 3. (genus *Sputnikvirus*, species *Mimivirus-dependent virus Zamilon*) Mimivirus-dependent virus Zamilon

Varidnaviria > Helvetiavirae > Dividoviricota

-**RCXW**- 'SPHAEROLIPOVIRIDAE' 1. Alphasphaerolipovirus 2. Yingchengvirus 3. Hukuchivirus

Miscellaneous Viruses, Viroids, Satellites

-**FŠT**- 'AVSUNVIROIDAE' 1. (genus *Avsunviroid*) avocado sunblotch viroid 2. (genus *Elaviroid*) eggplant latent viroid 3. (genus *Pelamoviroid*) peach latent mosaic viroid, appler hammerhead viroid, chrysanthemum chlorotic mottle viroid

-**FŠTW**- 'VIROID SATELLITE' 1. (Alphasatellitidae; several genera) alphasatellite 2. (Tolecusatellitidae; genera *Betasatellite, Deltasatellite*) Ageratum yellow vein betasatellite, tomato leaf curl deltasatellite 3. viroid satellite other than Stems 1 & 2

-**FŠTY**- 'OTHER VIRUS' I 1. (Ligamenvirales; several genera) lipothrixvirus, rudivirus 2. (Baculoviridae; several genera) baculovirus 3. (Nudiviridae; genera *Alphanudivirus, Betanudivirus*) Oryctes rhinoceros nudivirus, Gryllus bimaculatus nudivirus, Heliothis zea nudivirus

-**FŠTL**- 'OTHER VIRUS' II 1. (Ampullaviridae; genus *Ampullavirus*) Acidianus bottle-shaped virus 2. (Bicaudaviridae; genus *Bicaudavirus*) Acidianus two-tailed virus 3. (Clavaviridae; genus *Clavavirus*) Aeropyrum pernix bacilliform virus 1

-**FŠTR**- 'OTHER VIRUS' III 1. (Fuselloviridae; genera *Alphafusellovirus, Betafusellovirus*) Sufolobus spindle-shaped virus 1 & 6, acidianus spindle-shaped virus 1 2. (Globuloviridae; genus *Globulovirus*) Pyrobaculum spherical virus, Thermoproteus tenax spherical virus 1 3. (Guttaviridae; genera *Alphaguttavirus, Betaguttavirus*) Sulfolobus newzealandicus droplet-shaped virus, Aeropyrum pernix ovoid virus 1

-**FŠTŘ**- 'OTHER VIRUS' IV 1. (Ovaliviridae; genus *Alphaovalivirus*) Sulfolobus ellipsoid virus 1 2. (Plasmaviridae; genus *Plasmavirus*) Archoleplasma phage L2 3. (Portogloboviridae; genus *Alphaportoglobovirus*) Sulfolobus alphaportoglobovirus 1

-**FŠTÇ**- 'OTHER VIRUS' V 1. (Hytrosaviridae; genera *Glossinavirus, Muscavirus*) Glossina hytrosavirus, Musca hytrosavirus 2. (Nimaviridae; genus *Whispovirus*) white spot syndrome virus 3. (Polydnaviridae; genera *Ichnovirus, Bracovirus*) ichnovirus, bracovirus

-**FŠTF**- ‘OTHER VIRUS’ VI 1. (Anelloviridae; numerous genera) anellovirus, torque teno virus (TTV) 2. (Spiraviridae; genus *Alphaspiravirus*) Aeropyrum coil-shaped virus (ACV) 3. (Pospiviroidae; several genera) potato spindle tuber viroid, hop stunt viroid, coconut cadang-cadang viroid, apple scar skin viroid, Coleus blumei viroid 1

-**FŠTV**- ‘OTHER VIRUS’ VII 1. (Halspiviridae; genus *Salterprovirus*) salterprovirus His1 2. (Tristromaviridae; genus *Alphatristromavirus*) Pyrobaculum filamentous virus 1 3. (Finnlakeviridae; genus *Finnlakevirus*) Flavobacterium virus FLiP

-**FŠTĻ**- ‘OTHER VIRUS’ VIII 1. (genus *Pithovirus*) pithovirus 2. (genus *Dinodnavirus*) dinodnavirus 3. (genus *Rhizidiovirus*) rhizidiovirus

-**FŠTH**- ‘OTHER VIRUS’ IX 1. (genus *Deltavirus*) deltavirus 2. (genus *Pandoravirus*) pandoravirus 3.(genus *Papanivirus*) panicum papanivirus 1

-**FŠTM**- ‘OTHER VIRUS’ X 1. (genus *Albetovirus*) tobacco albetovirus 1/2/3 2. (genus *Virtovirus*) tobacco virtovirus 1 3. (virus other than named order/class/family/genus) other virus